Training IndraMotion MLC

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Daily Schedule

Morning Session

08:00 –	Training
–	Coffee break
– 11:45	Training

11:45 – 12:45	Lunch break

Afternoon Session

12:45 –	Training
–	Coffee break
– 16:00	Training

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Learning Targets

- Understanding the architecture and functions of IndraMotion MLC
- Commissioning of the overall system with IndraWorks (control, drives, IOs)
- Diagnosis of the overall system with IndraWorks
- Motion programming according to IEC 61131
  - based on PLCopen
  - based on the AxisInterface
- Maintenance procedures
  - data backup and restore
  - drive exchange

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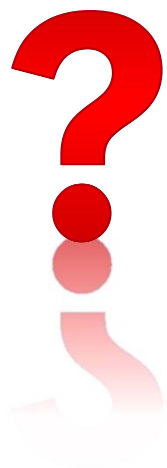
Seminar contents

- Overview of IndraMotion MLC architecture and features
- Motion and logic functionalities
- Engineering with IndraWorks
- Data archiving
- Glimpse on Motion Programming with PLCopen
- Motion Programming with AxisInterface and IMC
- Synchronized Motion
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- Creating CAMs with CamBuilder
- Generic Application Template
- Using IMST – IndraMotion Service Tool

Round of introduction

- Name
- Company / Department
- Activity
- Experience with Rexroth products
  - IndraWorks
  - IndraDrive
  - IEC programming
  - PLCopen
  - IndraMotion MLC / IndraMotion MLD
- Your expectations



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Agenda

▪ Introduction to IndraMotion MLC ▪ System topology and system components ▪ IndraWorks licenses and installation	Introduction
▪ First steps with IndraWorks ▪ Motion Programming Basics ▪ MLC Diagnosis system ▪ Data backup and restore ▪ Task System	Working with IndraWorks
▪ Synchronized Motion with PLCopen ▪ Electronic CAMs: Point table – MotionProfile – FlexProfile ▪ CamBuilder	Synchronized Motion
▪ AxisInterface and IMC ▪ Generic Application Template ▪ IMST ▪ Additional sources of information	Advanced Topics

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Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation

Introduction

- First steps with IndraWorks
- Motion Programming Basics
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
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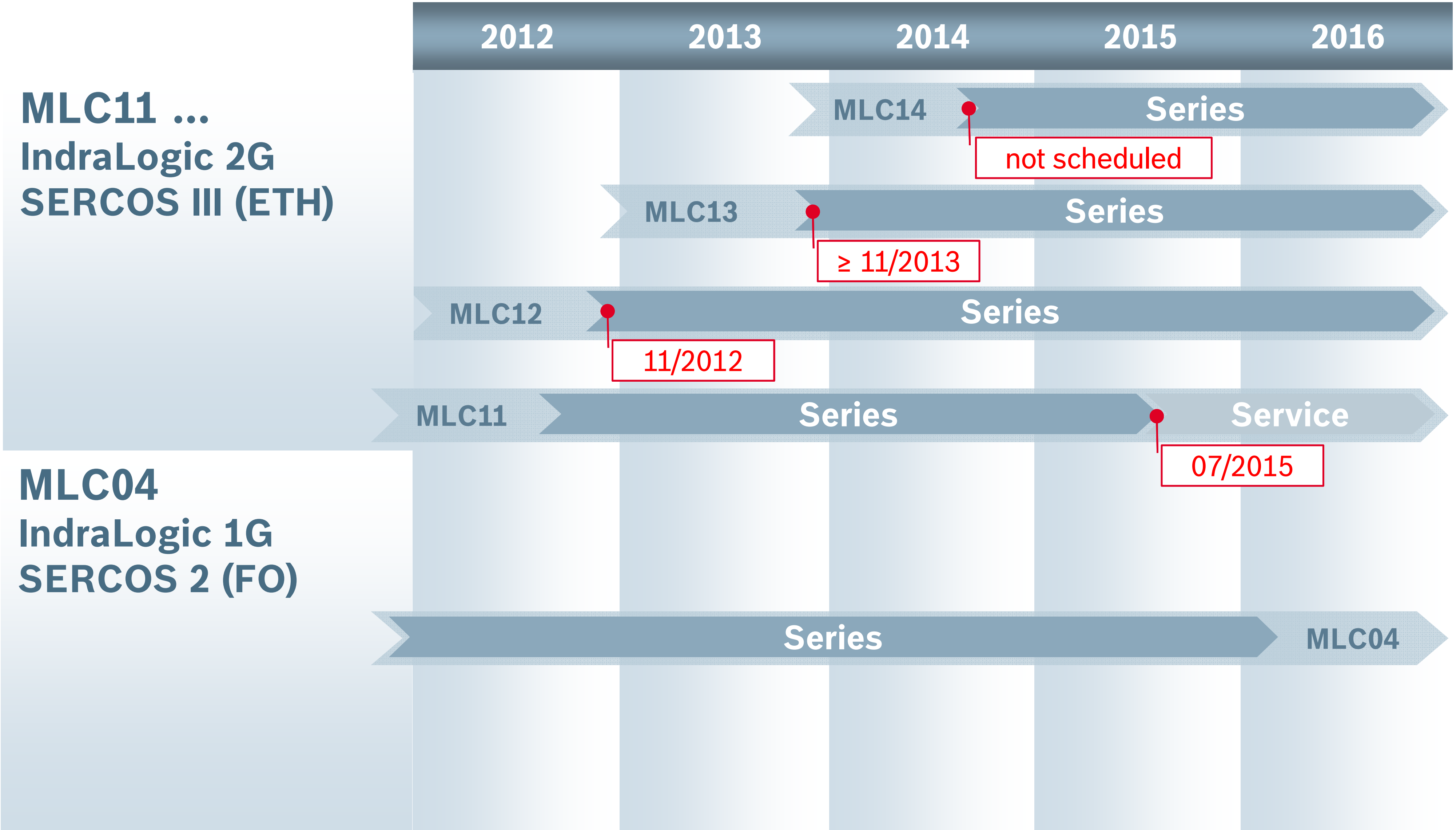
Product status



Integrated Motion-Logic System

- Integrated system for all tasks with motion and logic
- Motion kernel with hard real-time (<1 µs jitter) through SERCOS III
- Seamless networking with SERCOS III drives, I/O, periphery and cross communication
- Comprehensive functions for
  - PLC
  - axis positioning
  - axis synchronization
- Standardized function blocks compliant to PLCopen
- Pre-engineered technology function blocks
- Robotics

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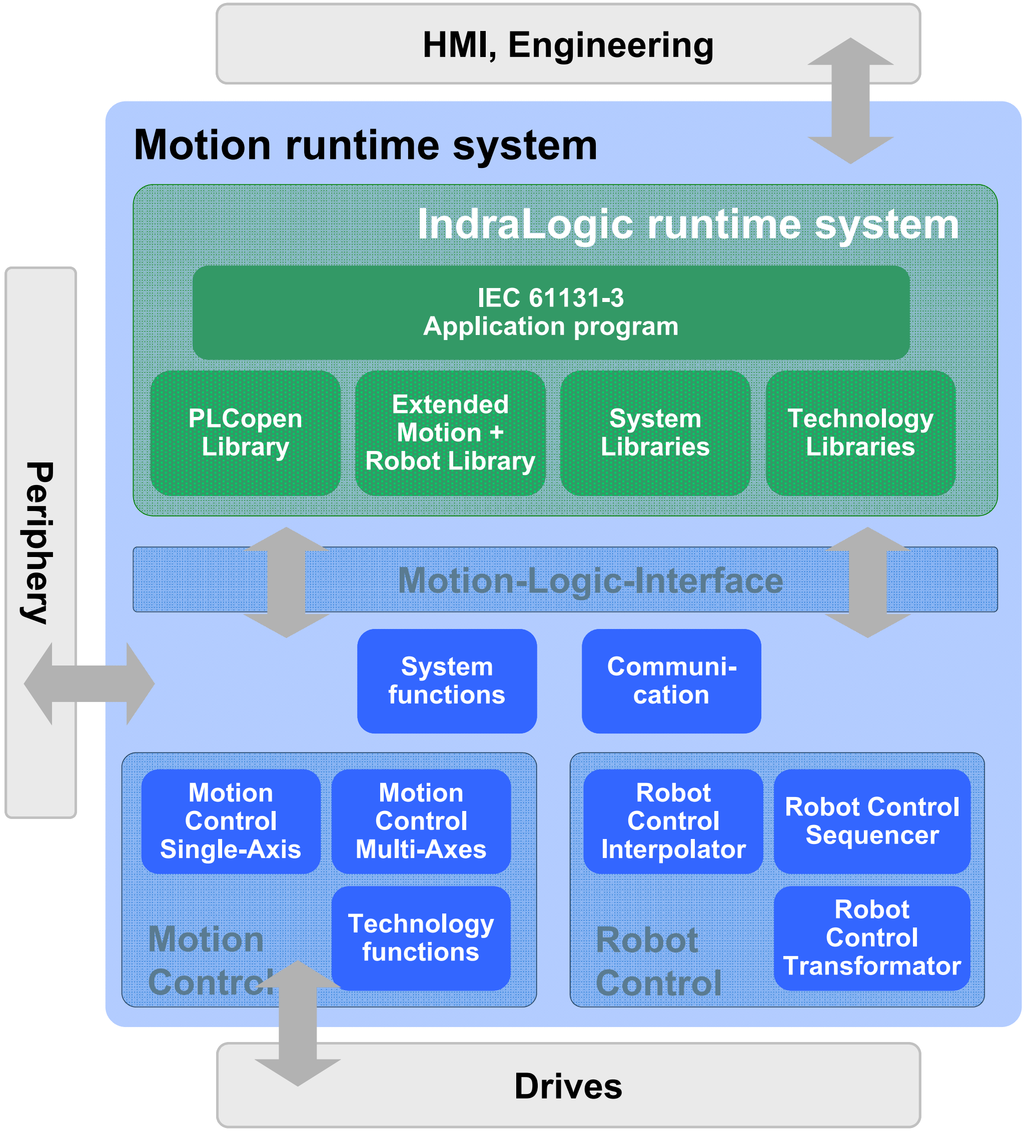
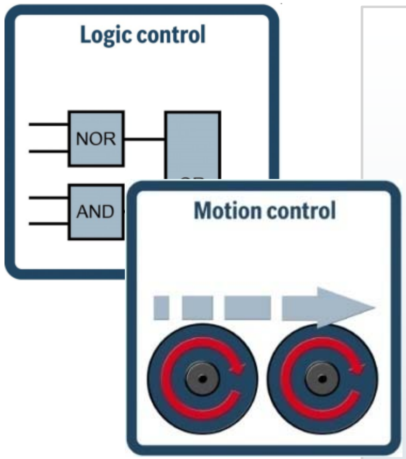
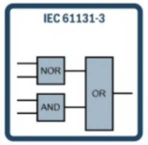
Integration of Motion and Logic



- **Integrated Motion-Logic runtime system**
- **Motion runtime system**
  - Synchronization of 64 axes
  - Virtual, real, encoder or link axes
  - IndraDrive C/M, IndraDrive Cs SERCOS drives (Pack Profile)
- **Logic runtime system**
  - Embedded in motion kernel
  - Standardized application interface according to IEC-61131-3
  - With transparent interface to motion kernel

Functions for Motion Control (selection)

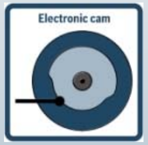




Logic control
according to IEC 61131-3 with extensions

PLCopen
Standardized function blocks for Motion Control

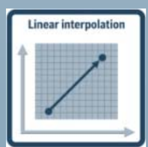
Electronic gears



Electronic cams

Multi-axes synchronization
Electronic Line Shafting

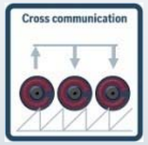
Positioning



Linear interpolation

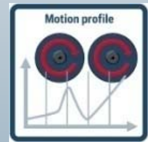
Real master axes
Encoder axes

Virtual master axes



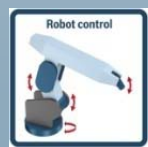
Cross communication
Link axes

Master axes cascading



MotionProfile
Master-axes related motion profiles

FlexProfile
Master-axes or time related motion profiles



Robot control
Tools and functions for coordinated motion

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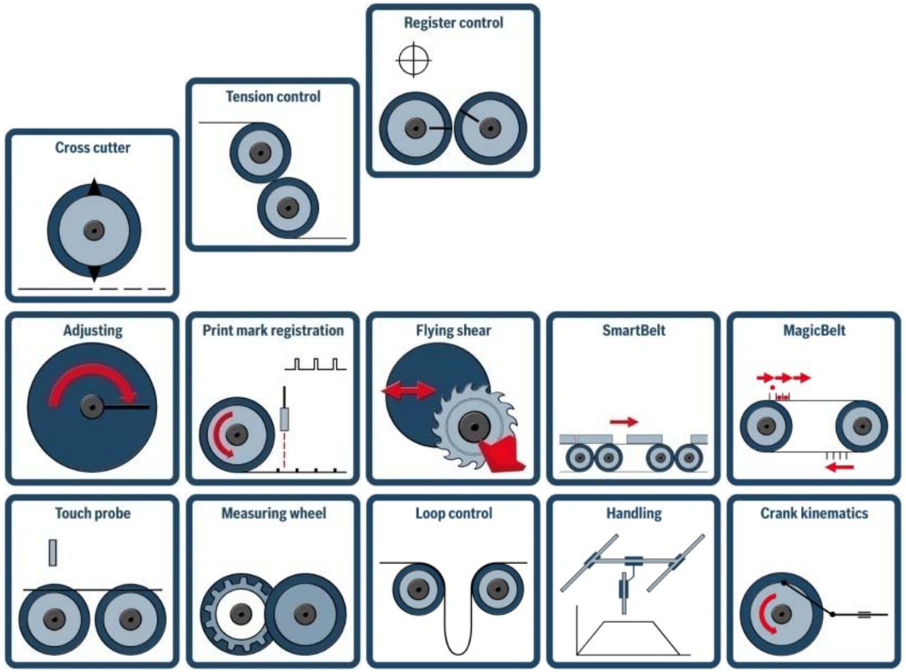
Technology function blocks



- **Pre-engineered function blocks** for easy implementation of complex processes
- Basis for **modular machine and control designs**
- **Faster reaction** to market and customer requirements

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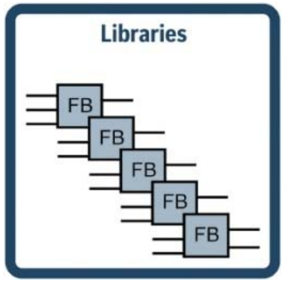
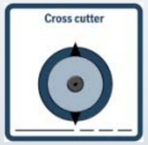
Technology function blocks (selection)



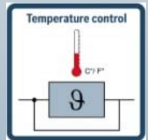
Sag Control

Tension Control



Cross Cutter

Handling
Interpreter for PTP multi-axes coordination



Temperature Control

High Speed Cam Switches

Winder Toolbox
Functions for dancer, diameter, tension

Flying Shear

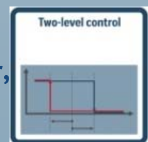
Print Mark Registration

Color Register Control

Manual Axis Positioning (Jog)
Incremental, continuous

Measuring Wheel

Touch Probe

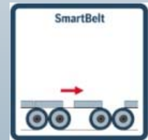


Two-point Control

PID Control

Crank Cinematic

Cross Sealing



Smart Belt

Magic Belt

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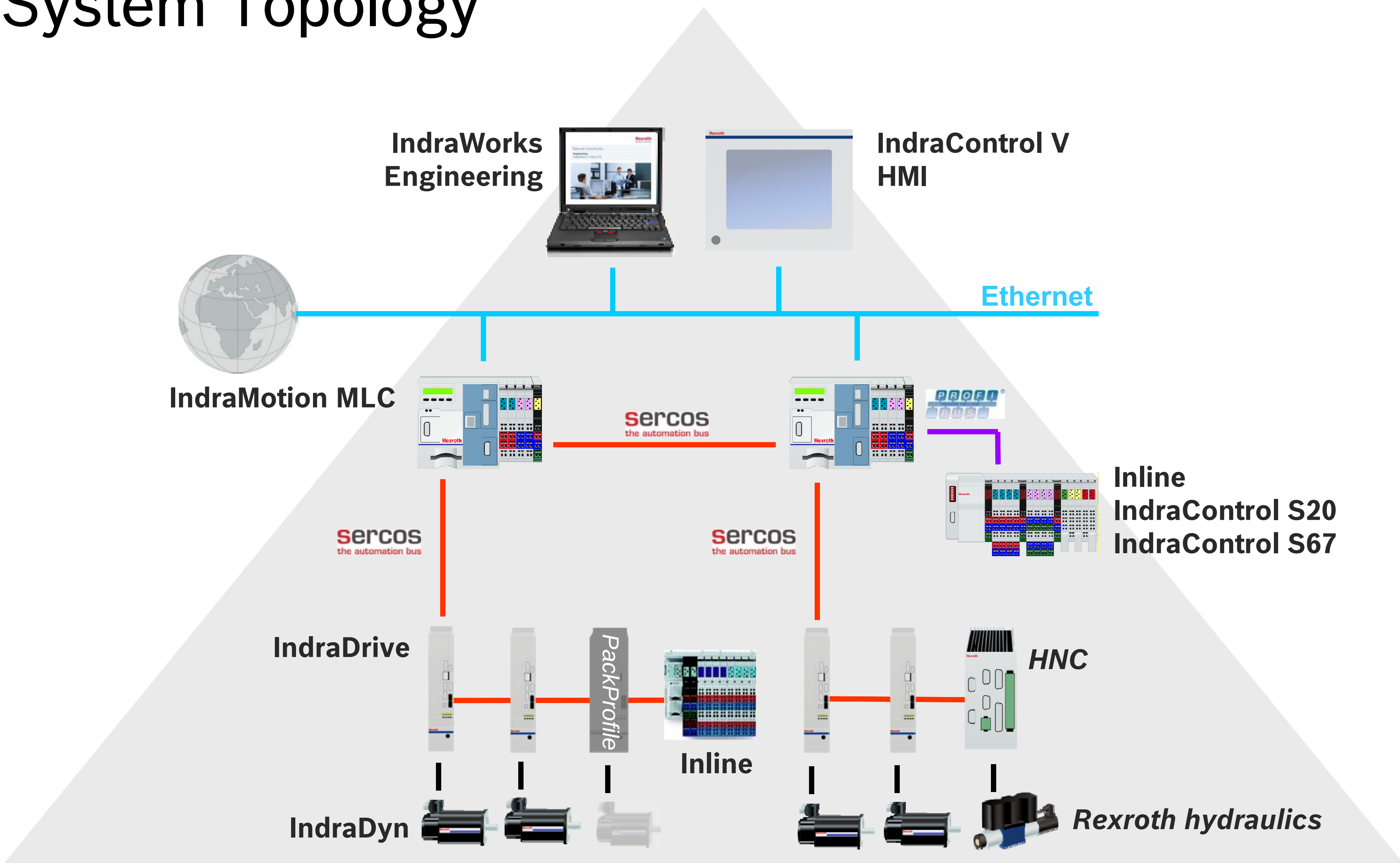
Introduction

System Topology



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Control Portfolio

				
Platform	**L25**	**L45**	**L65**	**L40.2**
Drive interface	SERCOS III	SERCOS III	SERCOS III	SERCOS 2
Max. axis number	16	32	64	32
Cycle time	2 ms / 1 ms	1 ms / 1 ms	0.5 / 0.25 ms	2 ms
Max. number of function modules	2	4	4	4
Onboard IO	☒	8I/8O	8I/8O	8I/8O
C2C	FM SERCOS III	FM SERCOS III	FM SERCOS III	FM SERCOS 2
EtherNet/IP Scanner/Adapter	with FM	≥ MLC14 / ☑	≥ MLC14 / ☑	☒ / ☑
PROFINET IO Contr./Device	with FM	☑ / ☑	☑ / ☑	☒
PROFIBUS Master/Slave	with FM	☑ / ☑	☑ / ☑	☑ / ☑
Robot Control	☑	☑	☑	☑

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IndraControl L25

Display

FM bus for Function modules

Ethernet interface

Ready contact with status LED

1 GByte CF
3 Partitions

SERCOS III
Drives + I/O

Connector for supply voltage

Local Inline IOs

Up to 16 axes

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IndraControl L45/L65

- Display
- Option: 8 MByte SRAM
- FM bus for function modules
- Ethernet interface
- Ready contact with status LED
- 1 GByte CF 3 Partitions
- PROFINET IO EtherNet/IP
- SERCOS III Drives + I/O
- Onboard I/O 8I/8O
- Local Inline IOs
- Connector for supply voltage
- PROFIBUS DP Master/Slave

Up to 32/64 axes

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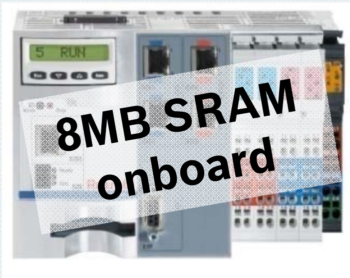
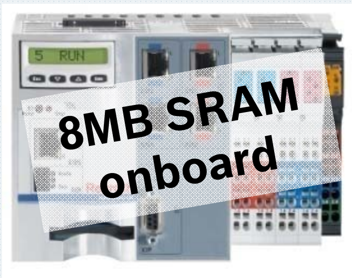
Hardware Platforms

IndraMotion MLC



CML45.1-3P-504...



CML65.1-3P-504...

IndraMotion MLC + IndraLogic XLC

CML25.1-3N-400...

CML45.1-3P-500...

CML65.1-3P-500...

IndraLogic XLC

CML25.1-PN-400...

CML45.1-NP-500...

CML65.1-NP-500...

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IndraMotion MLC versus IndraLogic XLC

Focus	PLC and general automation	Motion Control and technology specific solutions
Hardware platform	optional SERCOS interface	SERCOS interface required
Synchronous operation	☑	☑
Branch specific Technology functions	☒	☑
C2C (Cross comm)	☒	☑
Hydraulic drives	☒	☑
Robot Control	☒	☑
Function modules	Real-time Ethernet Fast IO	all
GAT support	$\text{GAT}^{\text{compact}}$	$\text{GAT}^{\text{compact}}$, $\text{GAT}^{\text{central}}$, $\text{GAT}^{\text{decentral}}$
IMST	☑	☑

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Function modules (1)

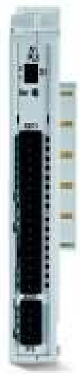
Functional expansion by up to 4 function modules

Functional expansion by up to 2 function modules

IndraControl L45/65

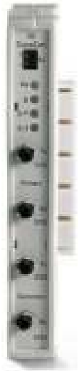
IndraControl L25

PLS



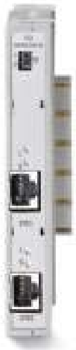
Fast I/O

SERCOS 2



SRAM → *option: 8MB SRAM onboard!*

SERCOS III



Real-Time Ethernet

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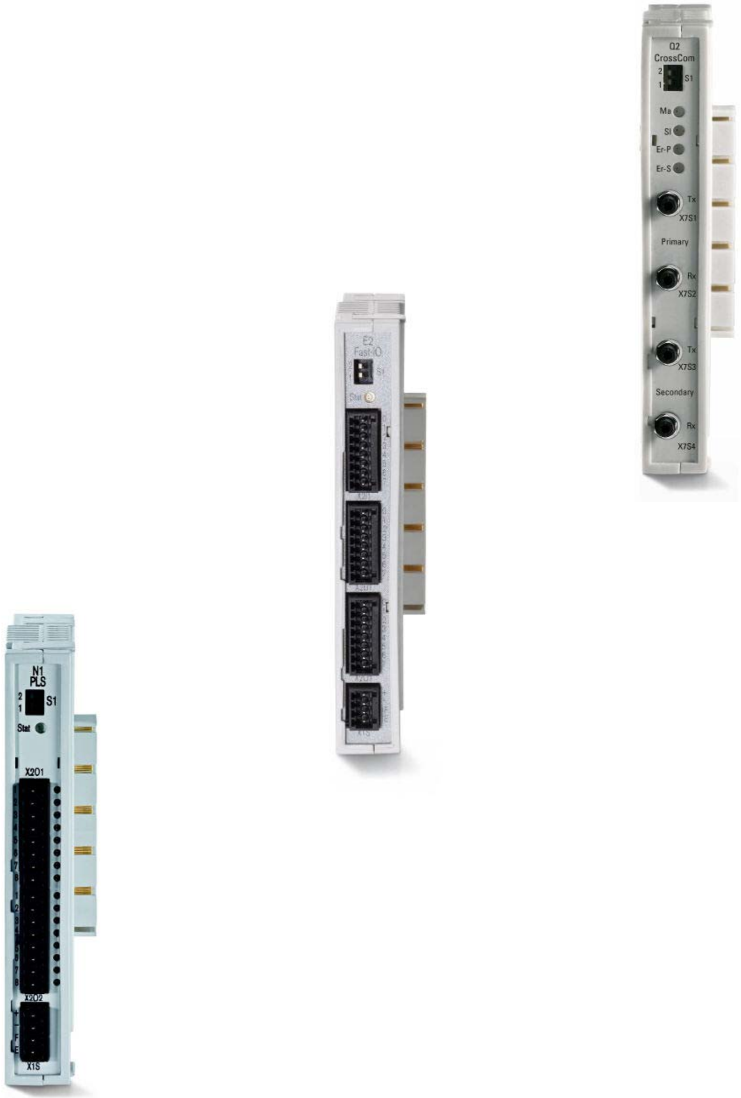
Function modules (2)



- **SERCOS 2** [CFL01.1-Q2]
  - Medium: FO
  - MLC-link with up to 64 MLCs
  - Drive link (e.g. for IndraDrive Mi)
- **Digital Cam Switch** [CFL01.1-N1]
  - 16 outputs
  - Scanning rate 125 µs
- **Fast I/O** [CFL01.1-E2]
  - 8 inputs, 8 outputs
  - 8 freely programmable IOs

Function modules (3)

- **SERCOS III** [CFL01.1-R3]
  - Medium: Ethernet
  - Cross communication
- **SRAM** [CFL01.1-Y1]
  - 8 MB SRAM battery-buffered
  - *Robot Control on L25*
- **Real-time Ethernet** [CFL01.1-TP]
  - PROFINET IO
  - EtherNet/IP Adapter
  - EtherNet/IP Scanner (≥ MLC13)
  - PROFIBUS DP Master
  - PROFIBUS DP Slave

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Function modules SERCOS 2 and SERCOS III

FM SERCOS 2
FM SERCOS III

IndraMotion MLC

Mounting		Function		
FM1	FM2	SERCOS III onboard	FM1	FM2
-	-	Master comm	-	-
SERCOS III	-	Master comm	C2C	-
SERCOS 2	-	Master comm	C2C	
SERCOS 2	-	inactive	Master comm	-
SERCOS 2	SERCOS 2	inactive	Master comm	C2C

- C2C is available only with L45/L65, availability for L25 with MLC13!
- Mixed operation of SERCOS 2 and SERCOS III drives is not supported!

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Interoperability with IndraDrive

IndraDrive C/M

Firmware MPx07/08

IndraDrive Cs

Firmware MPx16/17

- Economy:
- ☞ Interpolation on MLC
- ☞ Restrictions with regards to technology functions

IndraDrive Mi SERCOS 2

Firmware MPB07/08

- ☞ Function module SERCOS 2 required (CFL01.1-Q2)

IndraDrive Mi SERCOS III

Firmware MPB17

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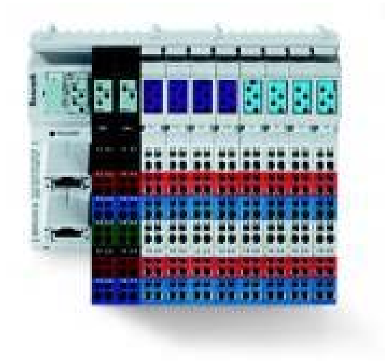
SERCOS III IOs

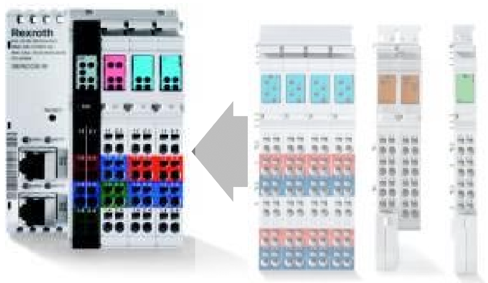


SERCOS III Block I/O Digital [R-ILB S3 24 DI16 DIO16]

- 16 digital inputs
- 16 combined inputs and outputs
- Short circuit and overload safe outputs
- 2/3-wire technology
- Mode “Fast Response” (not SERCOS III synchronized)
- Reaction time approx. 200 µs

SERCOS III Block I/O Analog [R-ILB S3 AI4 AO2]

- 4 analog, configurable differential inputs (current/voltage/temperature)
- 2 analog, configurable outputs (current/voltage)
- Short circuit and over-current safe
- Flexible measurement operation with adjustable filter times
- 2/3/4-wire technology
- Mode “Fast Response” (not SERCOS III synchronized)
- Max. reaction time 4 ms/channel



SERCOS III Bus coupler [R-IL S3 BK DI8 DO4-PAC]

- 8 digital inputs onboard (2/3-wire technology)
- 4 digital outputs (0.5 A) onboard
- Up to 63 modules can be mounted, max. 244 Bytes I/O
- Automatic recognition of Inline module baud rate

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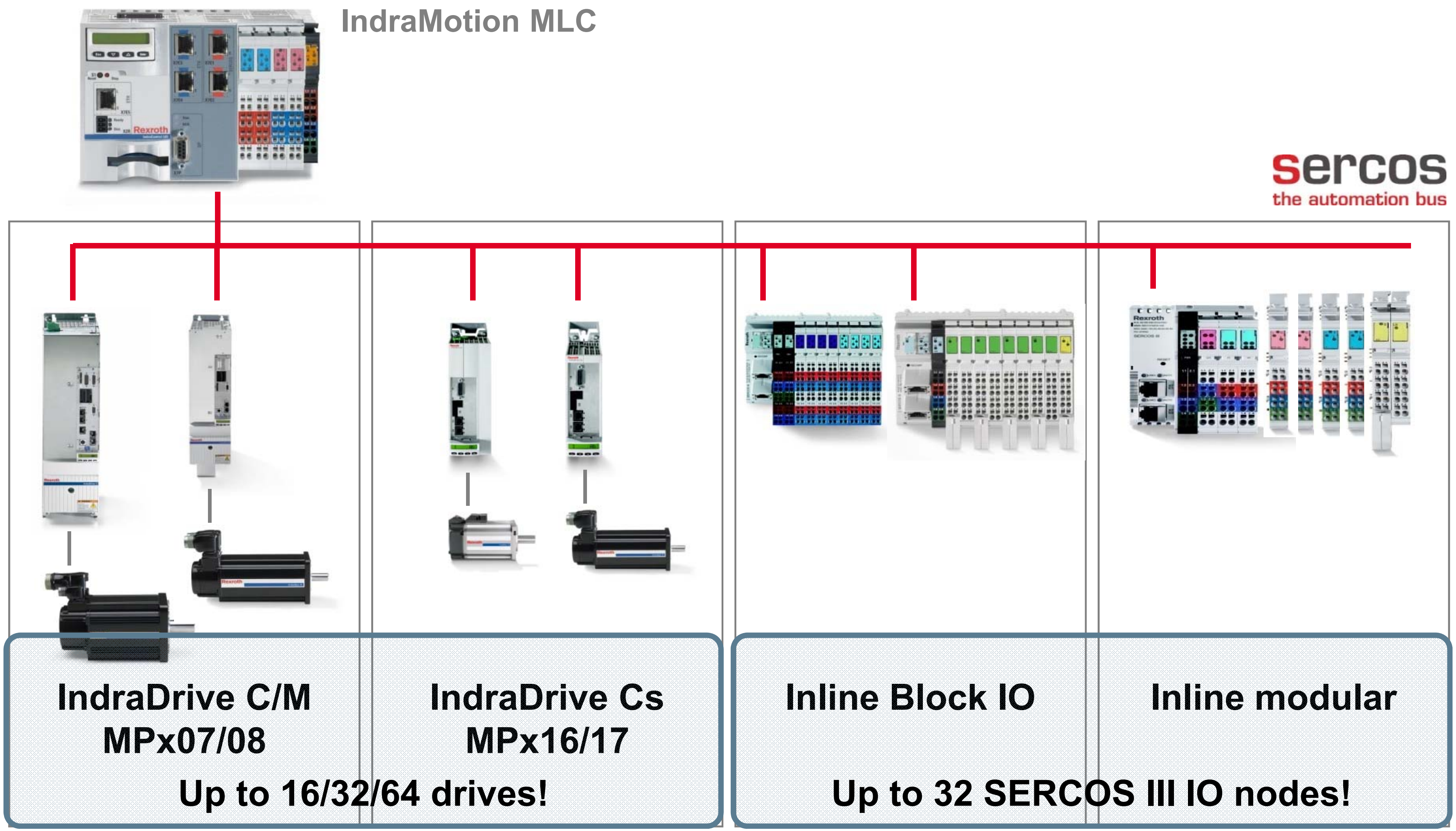
Connectivity – SERCOS III



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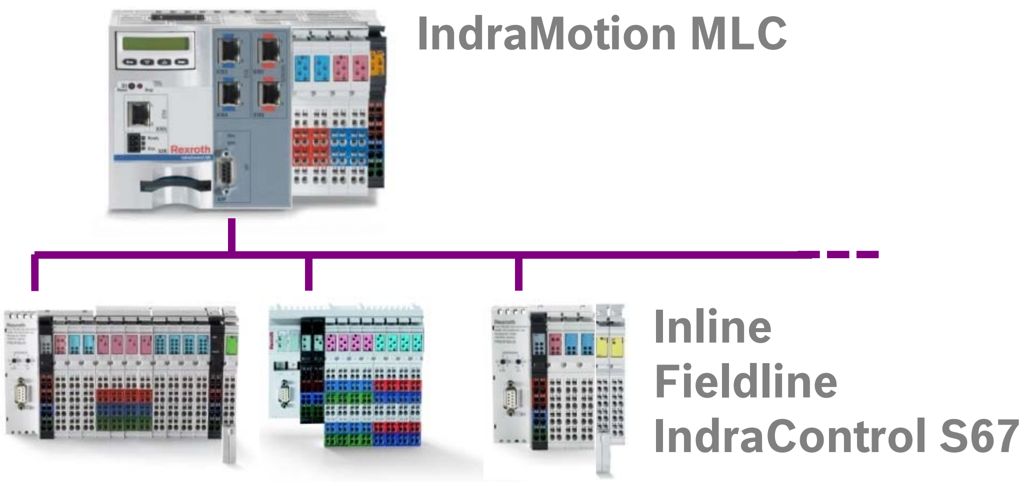
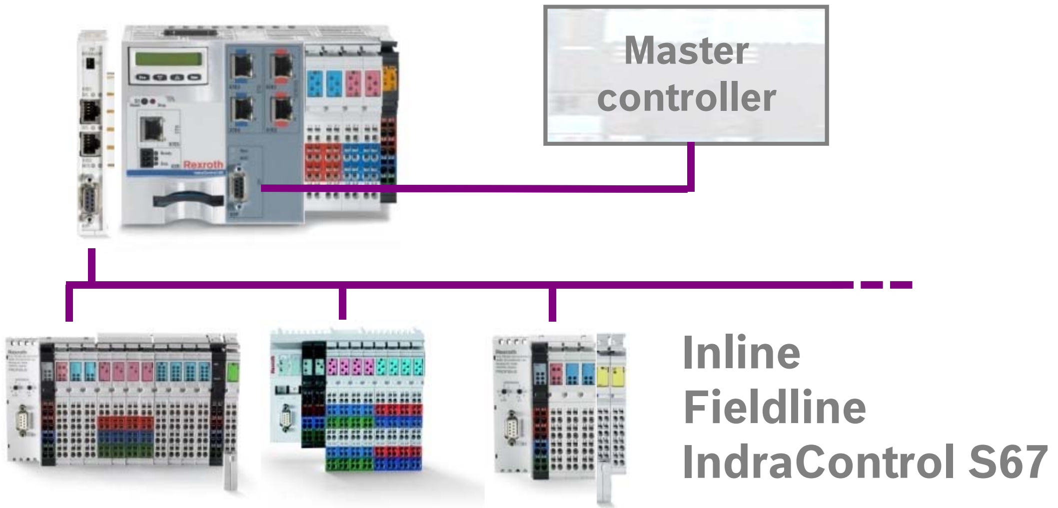
Connectivity – PROFIBUS



- **Onboard interface as DP master**
  - For connection of Rexroth Inline, Rexroth Fieldline and IndraControl S67



- **Onboard interface as DP slave**
  - Connection to master controller
- Function module Real-time Ethernet
  - For connection of Rexroth Inline, Rexroth Fieldline and IndraControl S67

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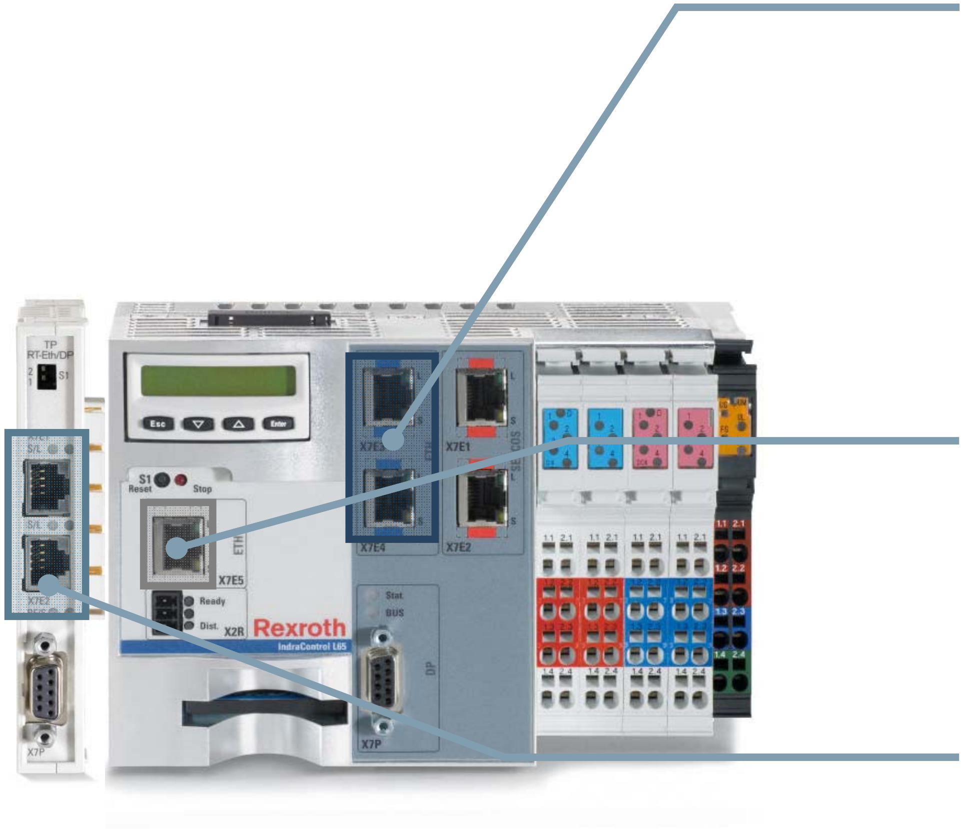
Connectivity – PROFINET & EtherNet/IP



- PROFINET IO Controller
- PROFINET IO Device
- EtherNet/IP Adapter
- EtherNet/IP Scanner ≥ ***MLC14***
- Standard TCP/IP

- EtherNet/IP Adapter
- Standard TCP/IP
  (e.g. Engineering, HMI)

- PROFINET IO Controller
- PROFINET IO Device
- EtherNet/IP Adapter
- EtherNet/IP Scanner ≥ ***MLC14***
- Standard TCP/IP

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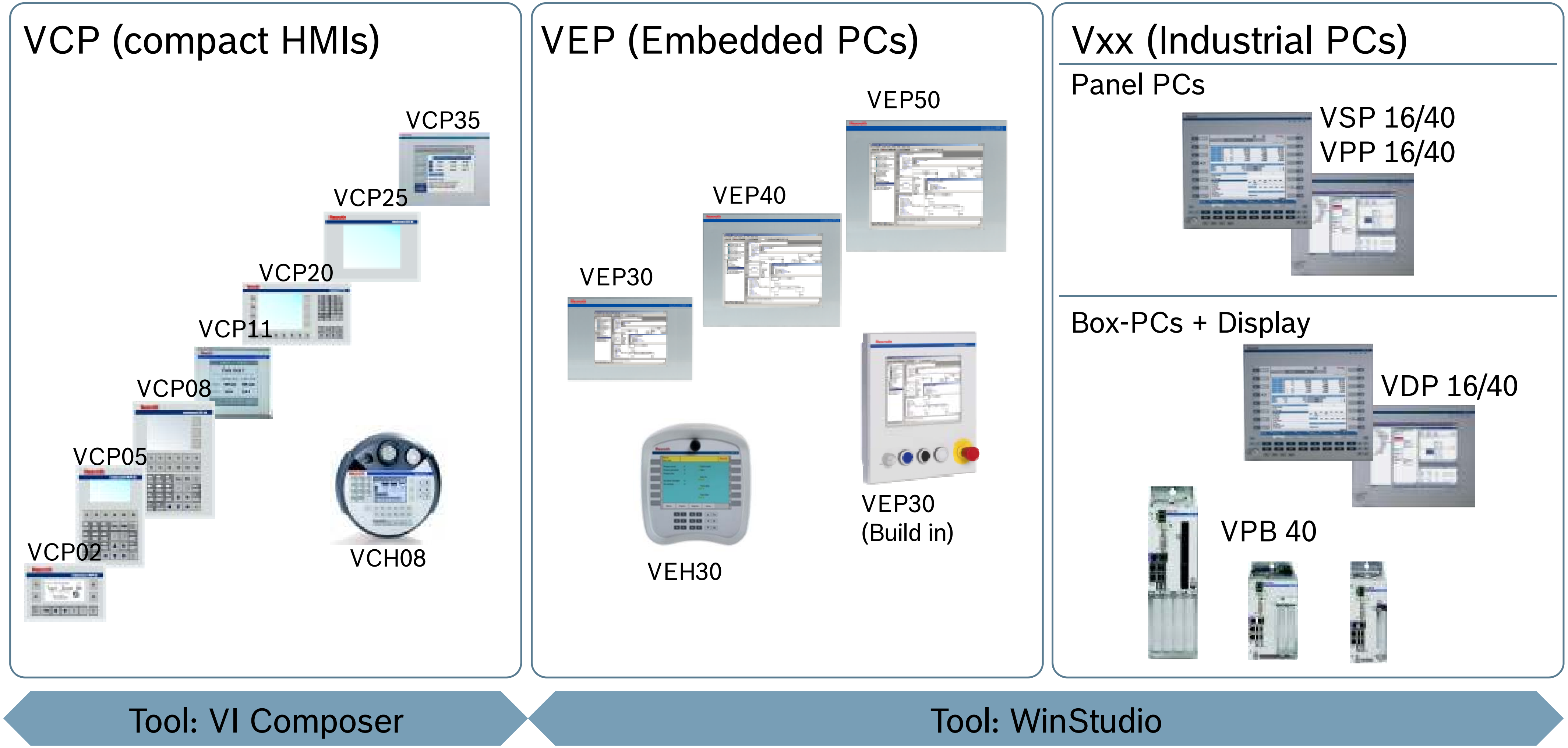
IndraControl V – HMIs



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Display on IndraControl Lx5 (1)

- 8-digits display with 4 keys
- Display of diagnosis and status information
- Additional information with regards to hardware and installed firmware:
  - Material number
  - Type code
  - Hardware index
  - Serial number
  - Firmware version
  - etc.
- Network Settings
- Load base parameters

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Display on IndraControl Lx5 (1)

Exercise

- How to load base parameters of MLC via display
- How to set IP parameters via display
- How to check the Ethernet connection

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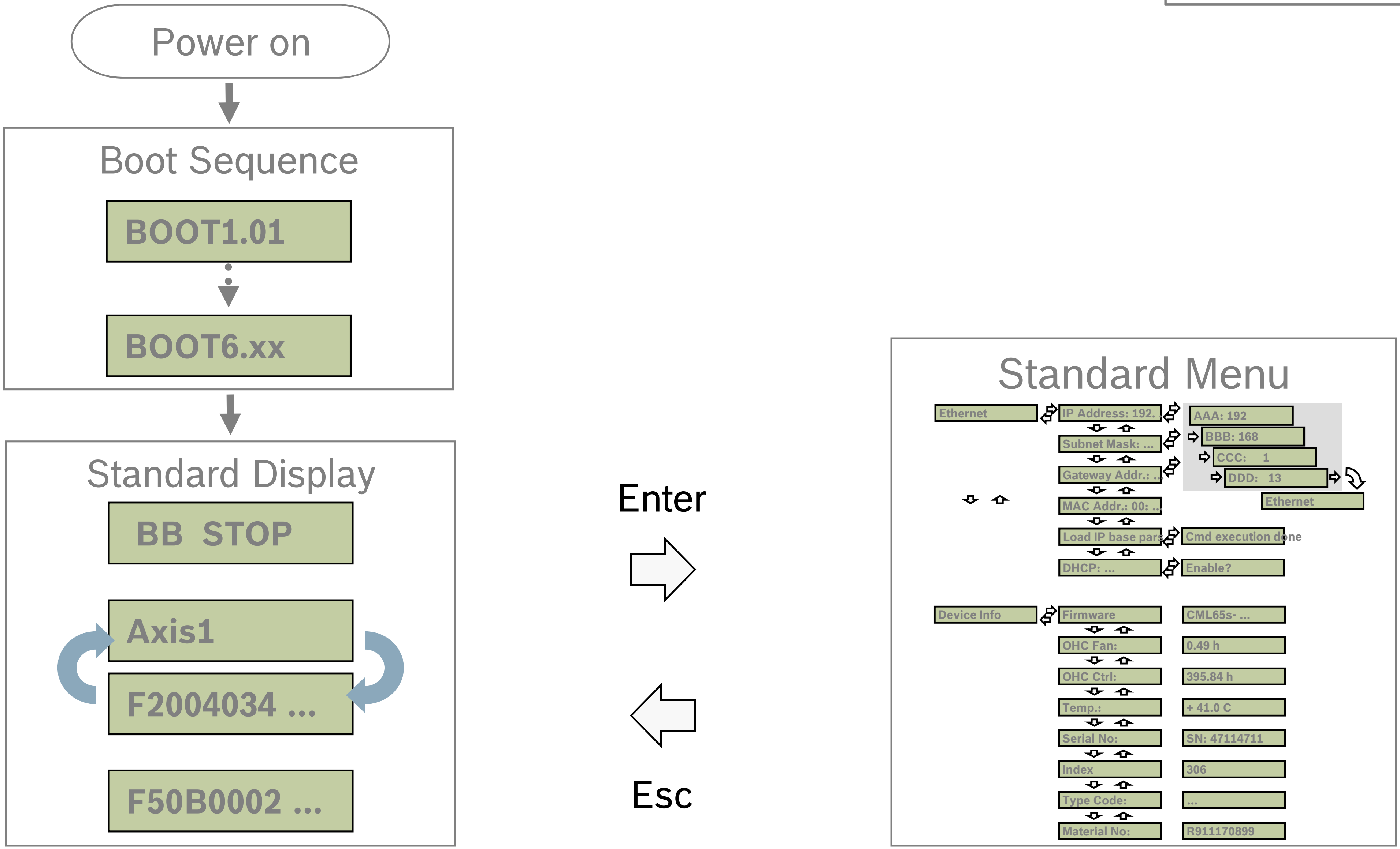
Display on IndraControl Lx5 (2)

Exercise



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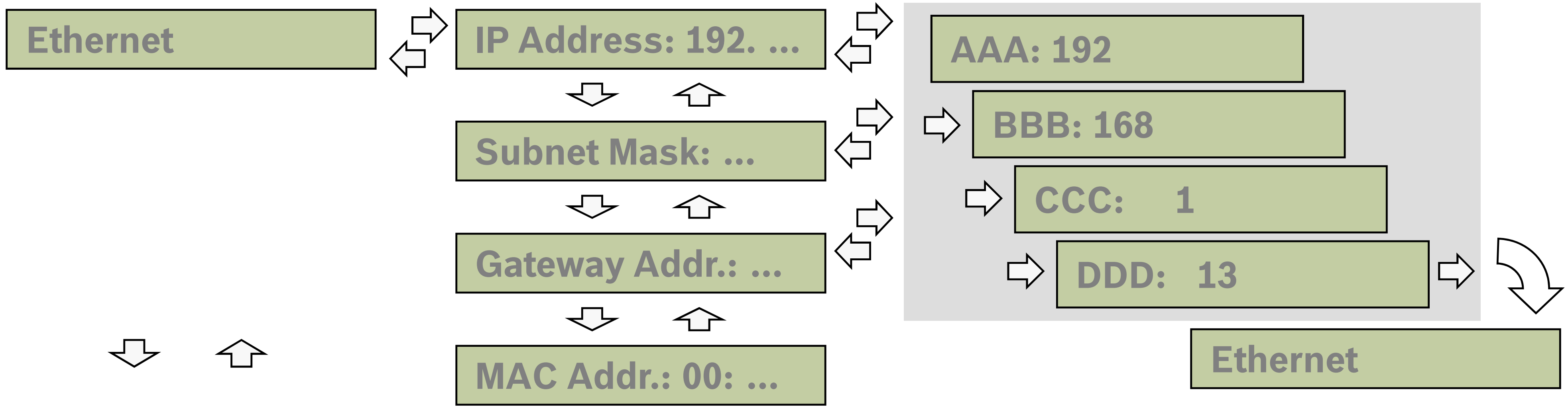
Display on IndraControl Lx5 (3)

Exercise



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Display on IndraControl Lx5 (4)

Exercise

Device Info

Firmware	CML65s-MLC ...
OHC Fan:	0.49 h
OHC Ctrl:	395.84 h
Temp.:	+ 41.0 C
Serial No:	SN: 47114711
Index	306
Type Code:	CML65.1-3P-504 ...
Material No:	R911170899

Enter

Esc

Down

Up

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Display on IndraControl Lx5 (5)

Exercise

Power on

Boot Sequence

BOOT1.01

BOOT6.xx

Standard display

BB RUN

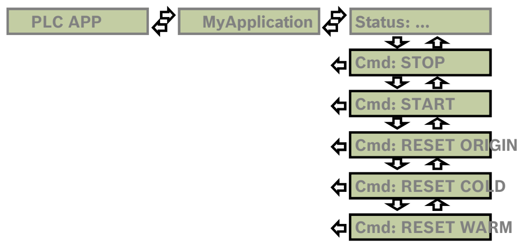
Axis1

F2004034 ...

F50B0002 ...

Press Esc and Enter for 8 seconds!

Advanced Menu



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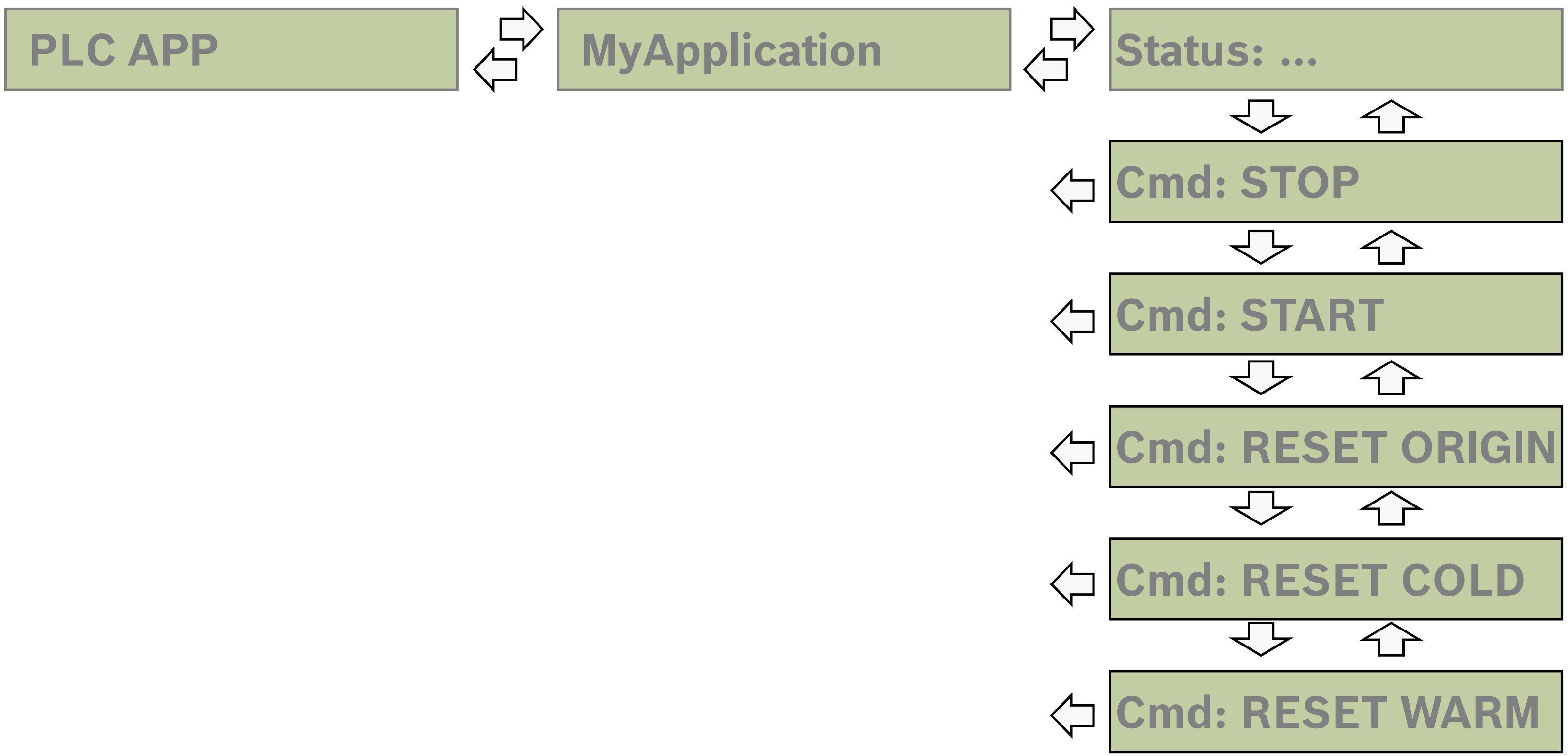
Display on IndraControl Lx5 (6)

Exercise



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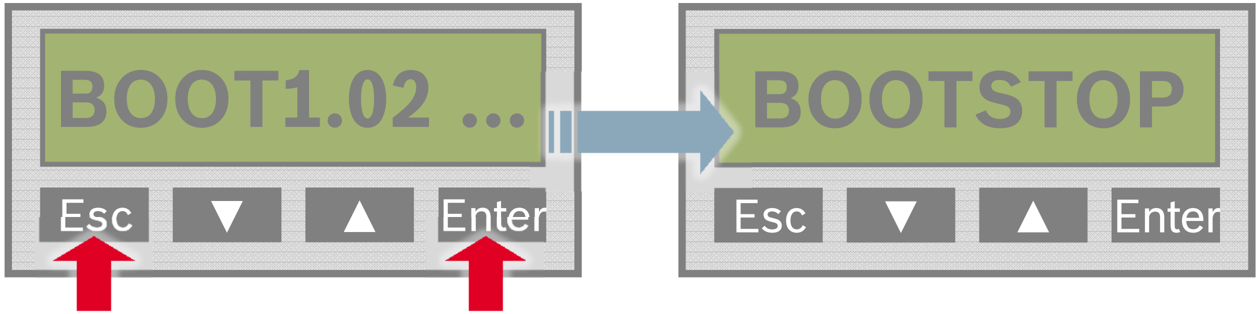
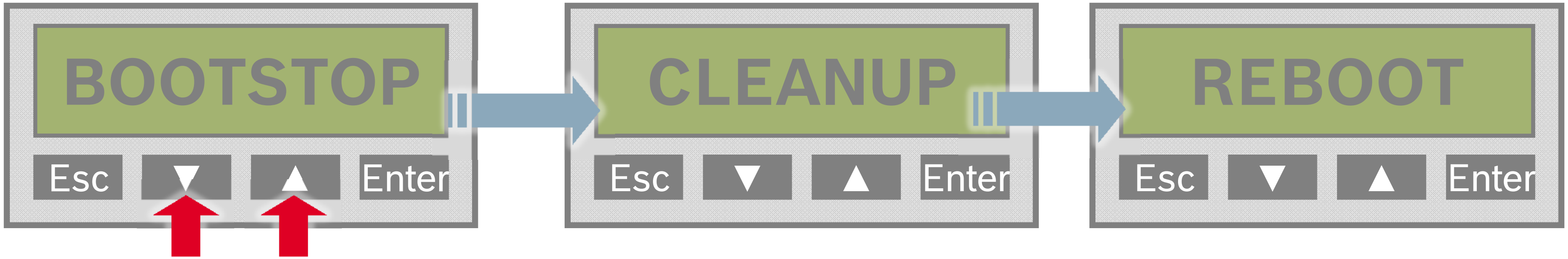
Display on IndraControl Lx5 (7)

Exercise

- How to load MLC base parameters?
- Press Esc and Enter simultaneously during the boot sequence until “BOOTSTOP” is displayed:



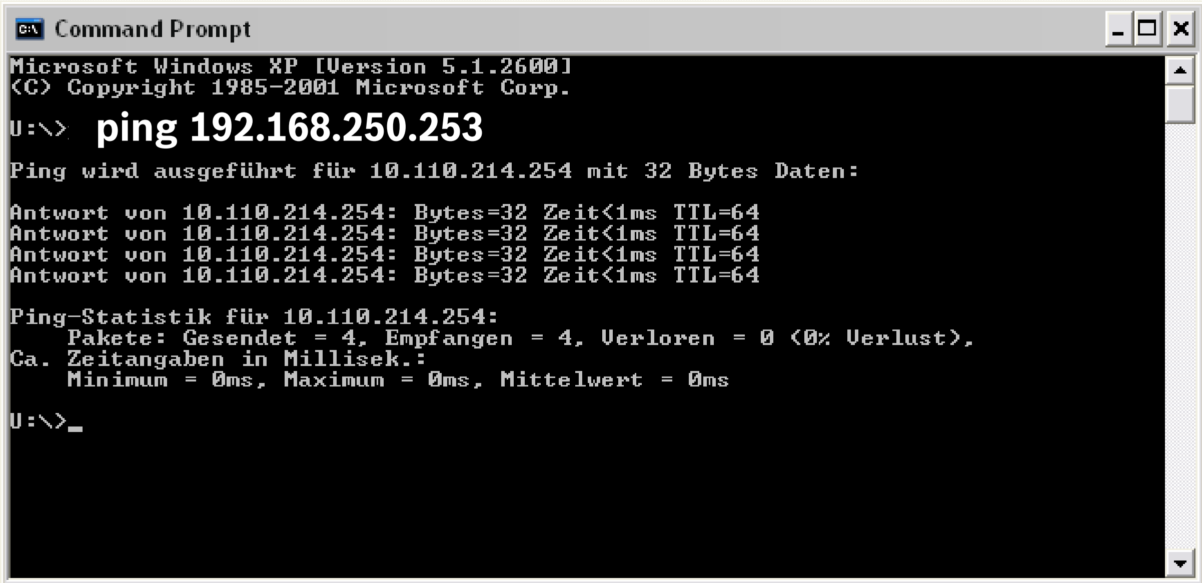
- Press ▼ and ▲ simultaneously to clear all data on MLC!



- MLC reboots automatically!

Display on IndraControl Lx5 (8)

Exercise

- Load MLC base parameters using the display
- Set IP address of your PC to **192.168.123....**
- Set IP Address of your MLC to **192.168.123.141**
- Set Gateway Address of your MLC to **192.168.123.254**
- Set Subnet Mask to 255.255.255.0
- Check Firmware version of MLC
- 

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- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- Motion Programming Basics
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Introduction

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IndraWorks Licenses for MLC[1]

IndraWorks ML* Data medium — **SWA-IWORKS-ML*-12VRS-D0-DVD**

version dependent	**SWL-IWORKS-ML*-12VRS-D0-ENG**	Single License MLC Engineering
	SWL-IWORKS-ML*-12VRS-D0-ENG*M25	Volume License MLC Engineering
	SWL-IWORKS-ML*-12VRS-D0-OPD	Single License Operation Desktop for MLC
	SWL-IWORKS-ML*-12VRS-D0-OPD*M25	Volume License Operation Desktop for MLC
	SWL-IWORKS-ML*-12VRS-D0-TEAMSERVER	Team Server License for MLC
version independent	**SWL-IWORKS-ML*-NNVRS-D0-ENG**	Single License MLC Engineering
	SWL-IWORKS-ML*-NNVRS-D0-ENG*M25	Volume License MLC Engineering
	SWL-IWORKS-ML*-NNVRS-D0-COM	Single License MLC Communication
	SWL-IWORKS-ML*-NNVRS-D0-COM*M25	Volume License MLC Communication
	SWL-IWORKS-ML*-NNVRS-D0-OPD	Single License Operation Desktop for MLC
	SWL-IWORKS-ML*-NNVRS-D0-OPD*M25	Volume License Operation Desktop for MLC

[1]) Without Campus licenses and Upgrade licenses

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IndraWorks – Supported operation systems

Microsoft Windows xp SP3

Windows Vista

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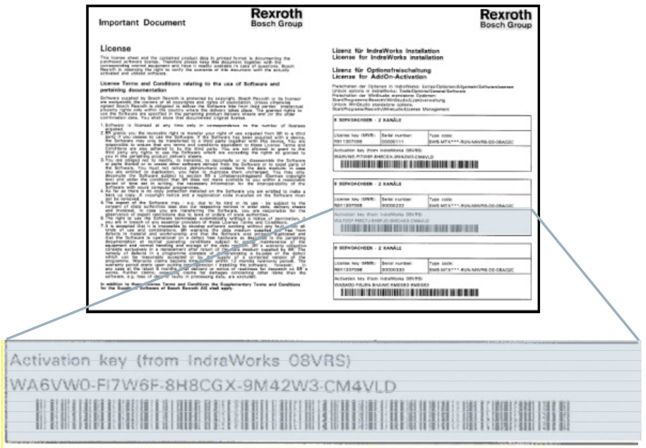
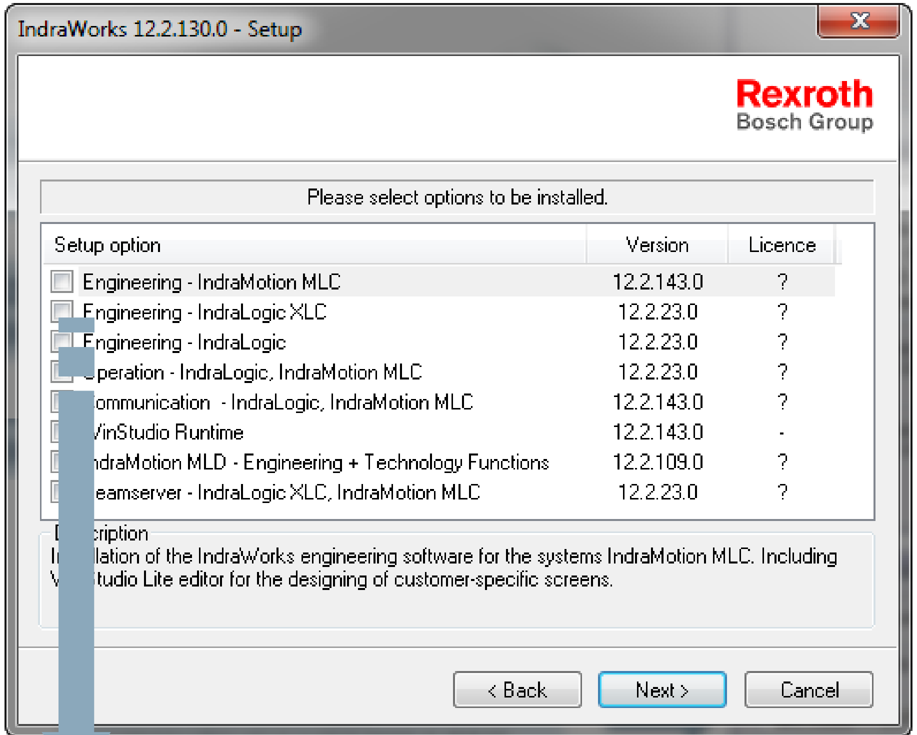
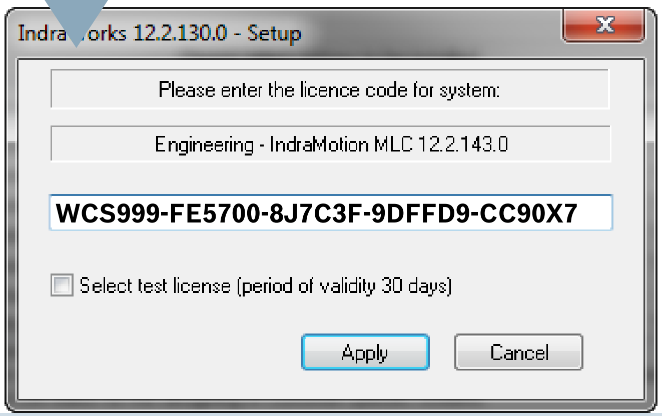
IndraWorks ML* – Installation Procedure

R911334632
SWA-IWORKS-ML*-12VRS-D0-DVD**

R911332861
SWL-IWORKS-ML*-NNVRS-D0-ENG







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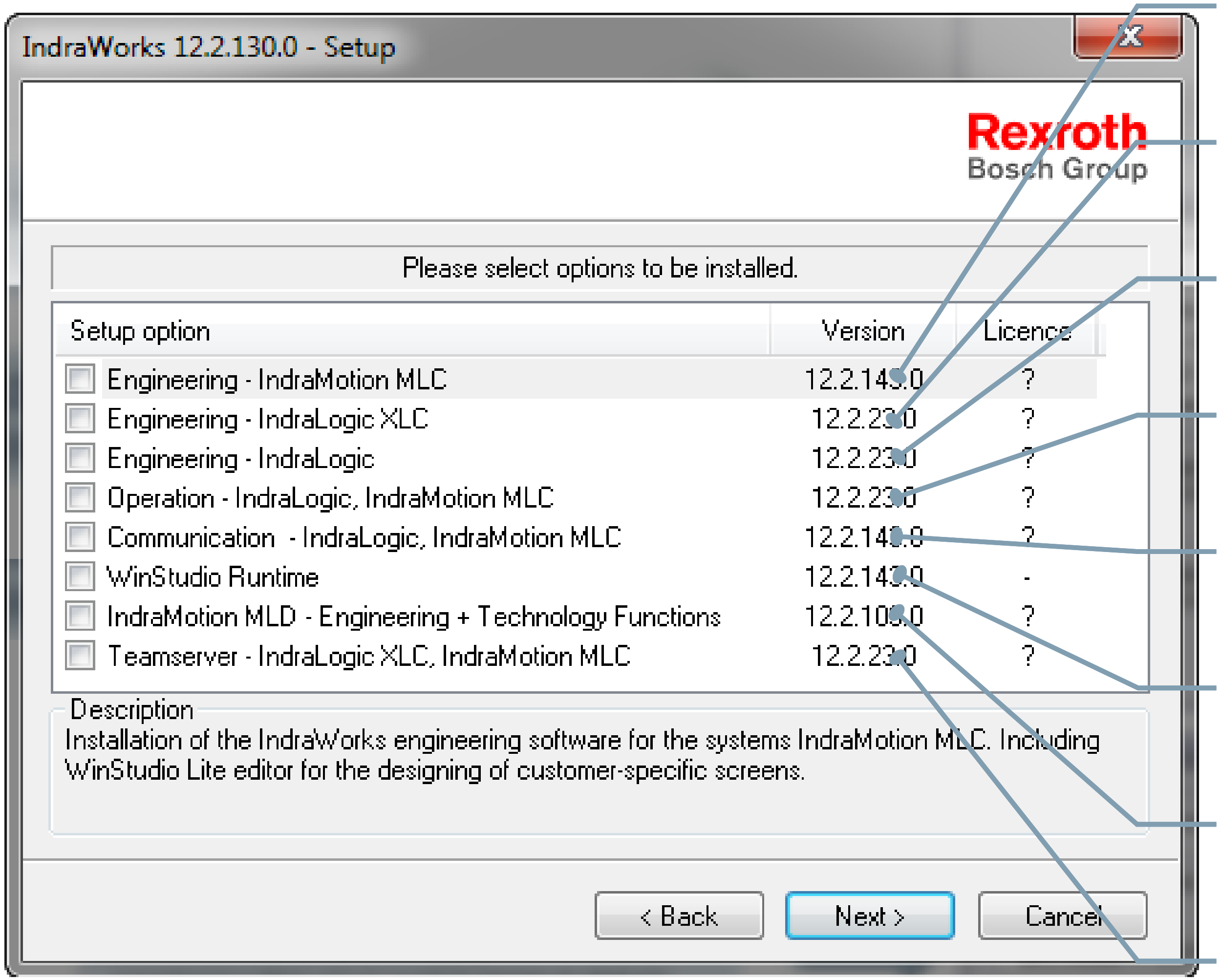
IndraWorks ML* – Installation varieties



- IndraMotion MLC Engineering
- IndraLogic XLC Engineering
- IndraLogic L/V Engineering (1G)
- Operation Desktop for MLC/XLC
- OPC Server for MLC/XLC
- WinStudio Runtime
- MLD Engineering
- Team server for MLC/XLC (VCS)

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IndraWorks Options for MLC[1]

version dependent	**SWS-IWORKS-VCS-12VRS-D0**	Single License VCS[2]
	SWS-IWORKS-VCS-12VRS-D0-M10	Volume License VCS[2] (10 Single Licenses)
	SWS-IWORKS-VCS-12VRS-D0-M25	Volume License VCS[2] (25 Single Licenses)
	SWS-IWORKS-CAM-12VRS-D0	Single License CAM Builder
	SWS-IWORKS-CAM-12VRS-D0-M25	Volume License CAM Builder
version independent	**SWS-IWORKS-VCS-NNVRS-D0**	Single License VCS[2]
	SWS-IWORKS-VCS-NNVRS-D0-M10	Volume License VCS[2] (10 Single Licenses)
	SWS-IWORKS-VCS-NNVRS-D0-M25	Volume License VCS[2] (25 Single Licenses)
	SWS-IWORKS-CAM-NNVRS-D0	Single License CAM Builder
	SWS-IWORKS-CAM-NNVRS-D0-M25	Volume License CAM Builder

[1]) Without Campus licenses and Upgrade licenses [2]) Version Control System for Team Engineering

Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- Motion Programming Basics
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Working with IndraWorks

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Engineering Workflow

1. Create new IndraWorks project
2. add control to project
3. scan axes and add to project
4. configure axes and control
5. configure IOs
6. motion programming

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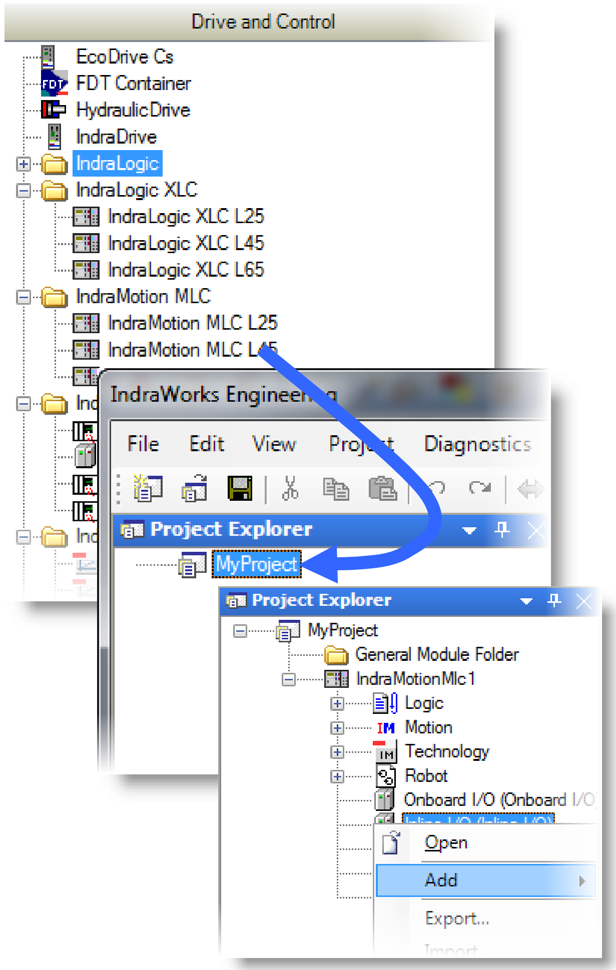
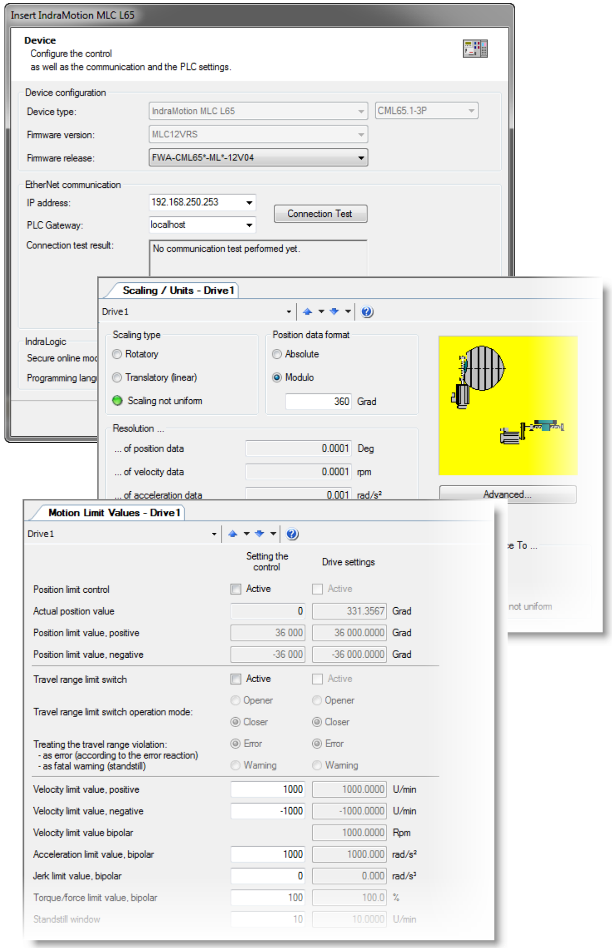
Engineering with IndraWorks

Selection of devices by drag & drop

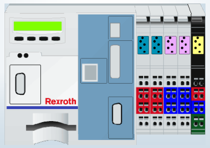


Parameterization through wizards



Automatic generation of program data

control parameters



drive parameters

PLC program template and data structures

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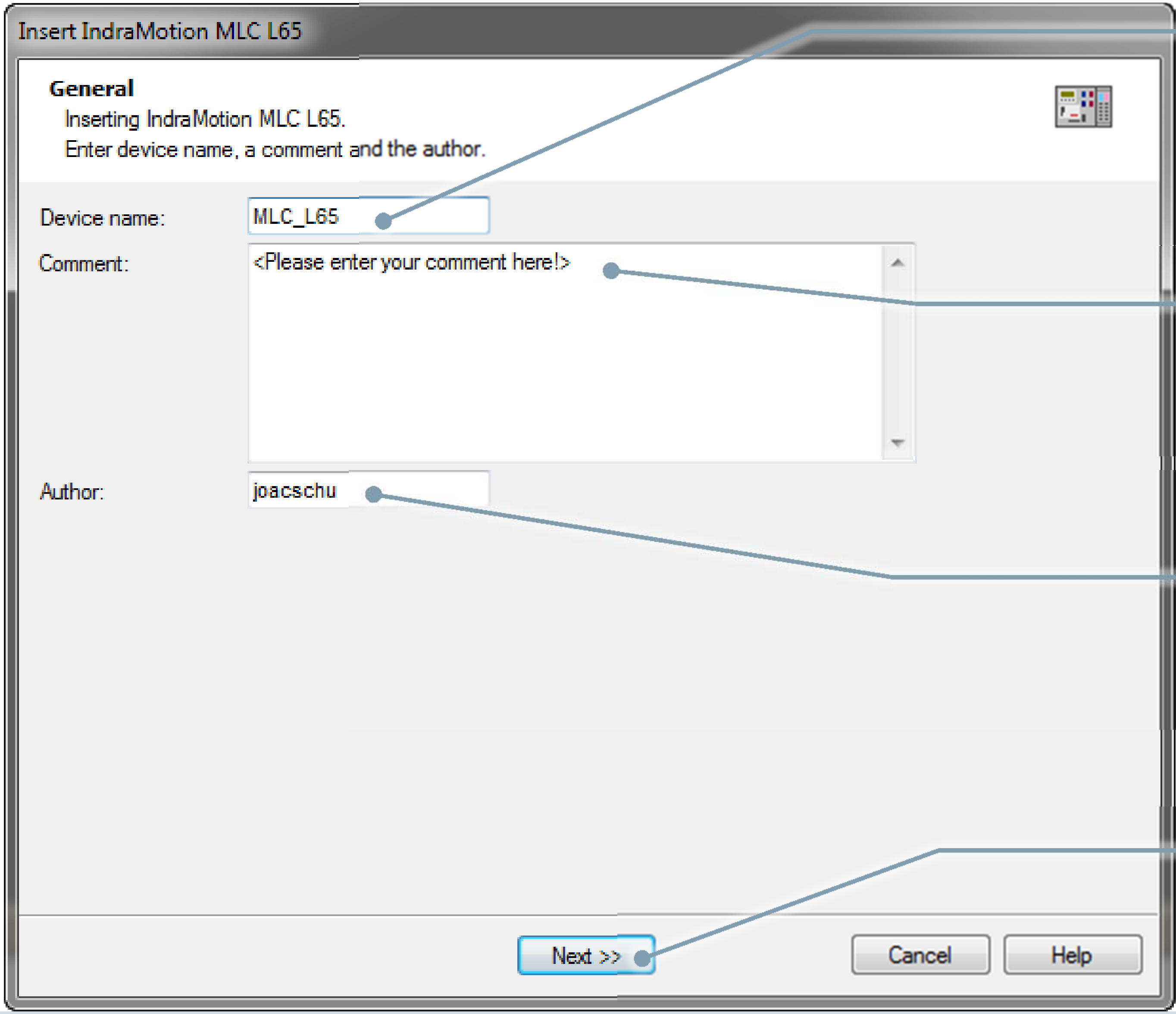
Parameterization through wizards



Enter device name

Comment
(tooltip in project explorer)

User name can be modified

Push “Next” button

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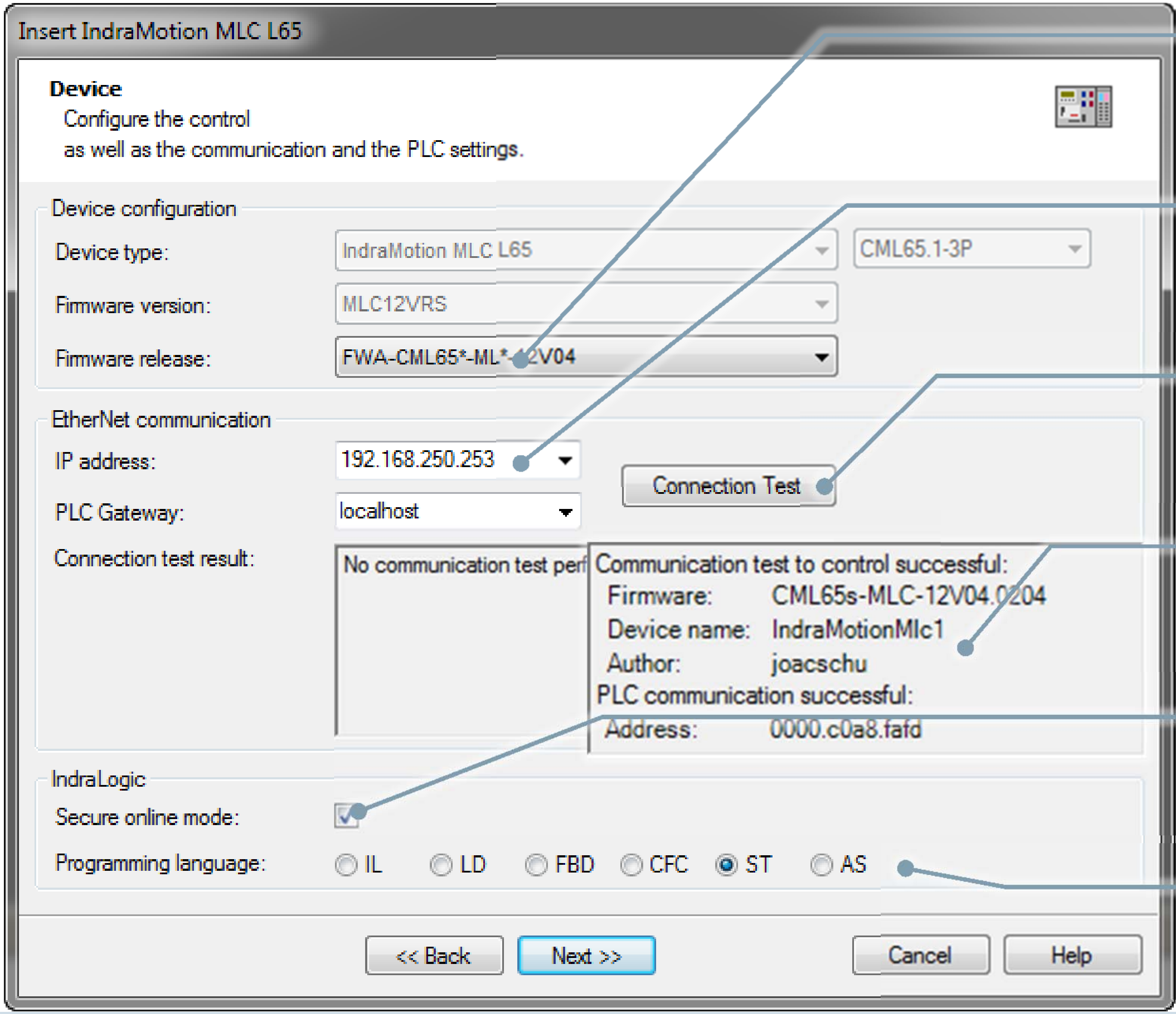
Parameterization through wizards



Select appropriate firmware version

Enter IP address

Execute Connection Test

Connection has been established

Secure online mode can be deselected!

Select favorite IEC language

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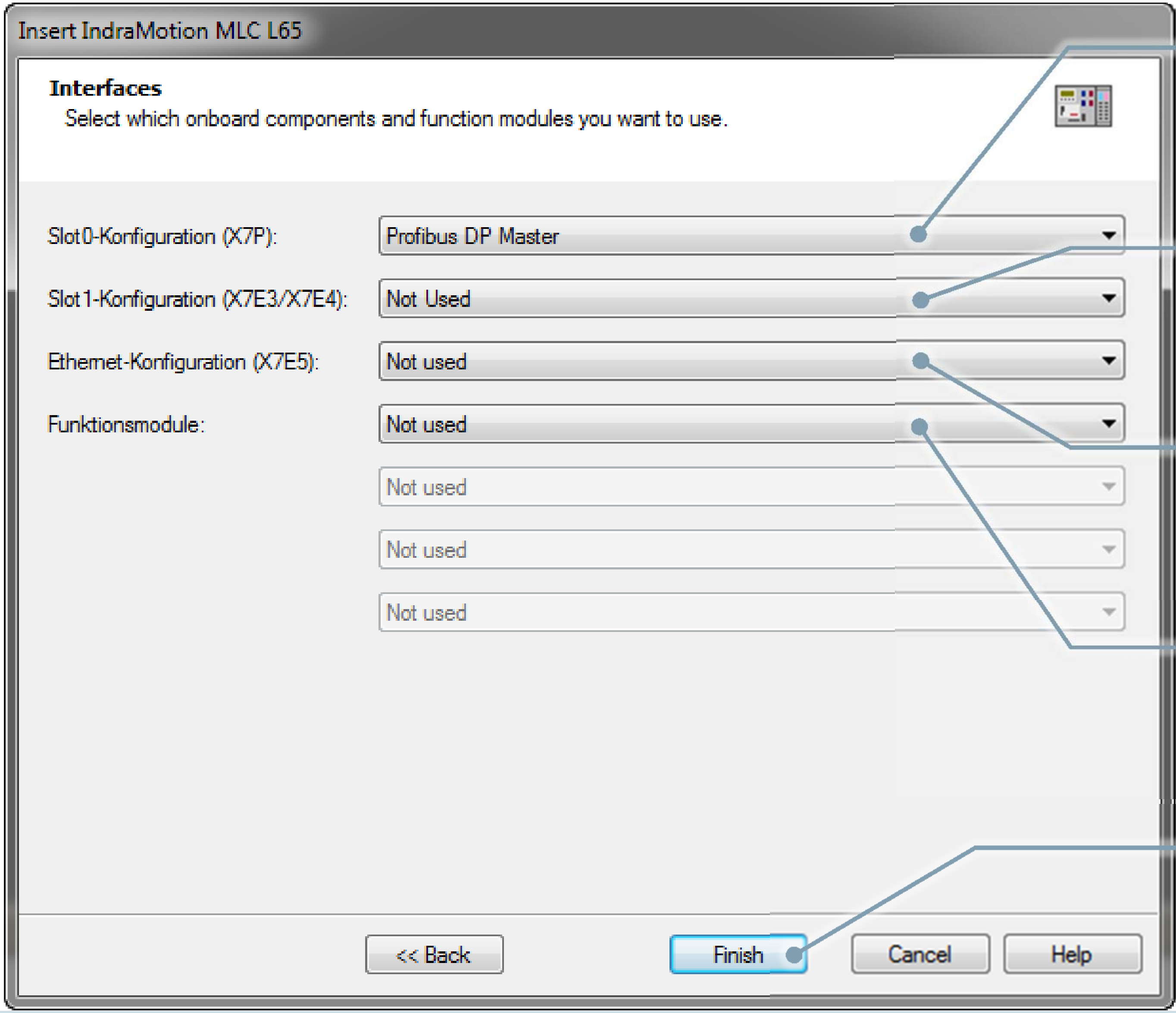
Parameterization through wizards



Select mode of PROFIBUS interface

Select protocol on real time Ethernet ports

Optionally enable EtherNet/IP Adapter on Engineering Port

Function modules

Press "Finish" button to add MLC to project

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1st Steps with IndraWorks

Exercise

- Create an empty IndraWorks project
- Add an IndraMotion MLC Lx5 to your project
- Select Firmware version
- Enter correct IP address
- Check Ethernet connection
- Upgrade MLC Firmware

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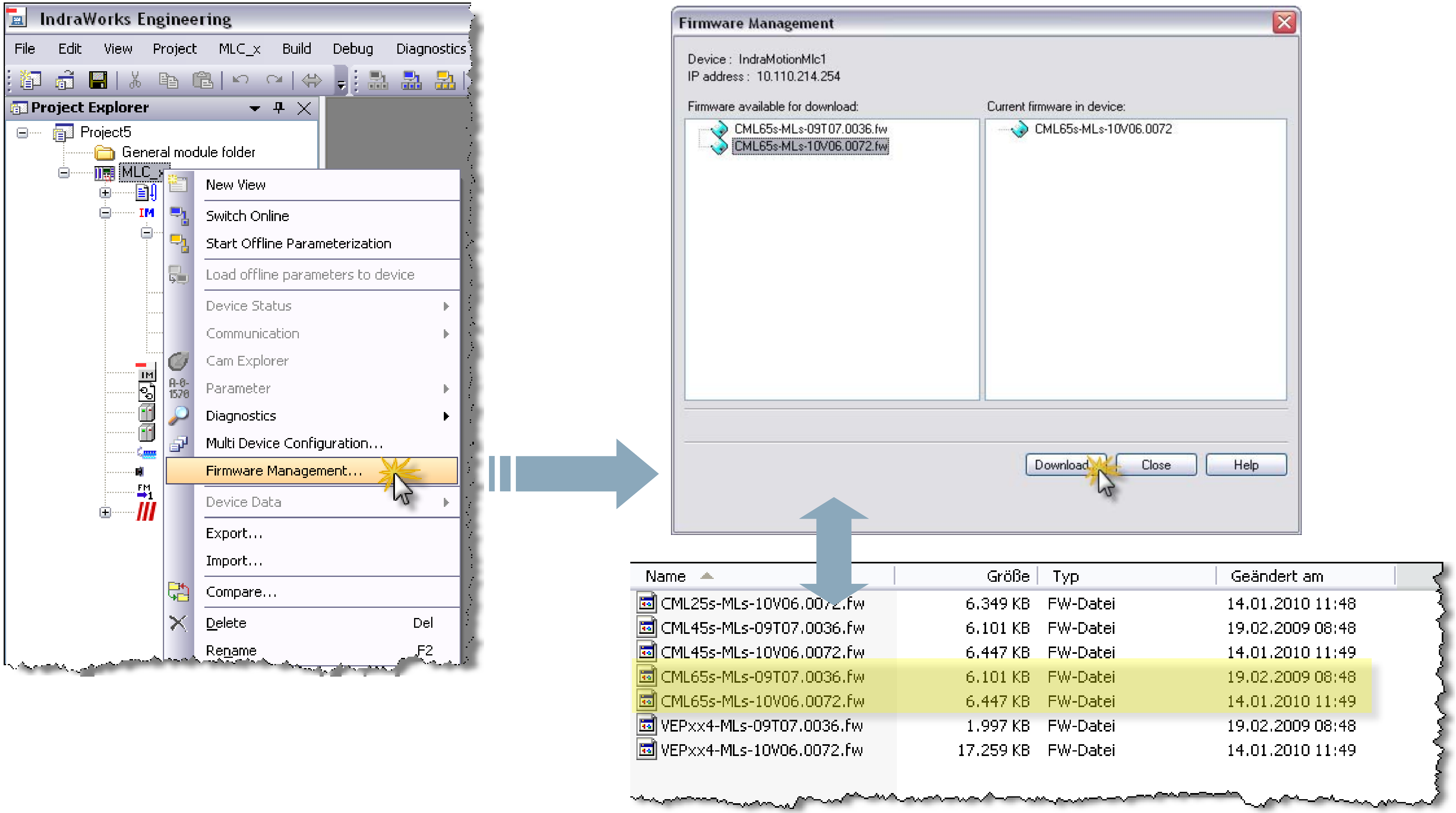
MLC Firmware Download

Exercise



Firmware files are located in
C:\Program Files\Rexroth\IndraWorks\MLC\Firmware

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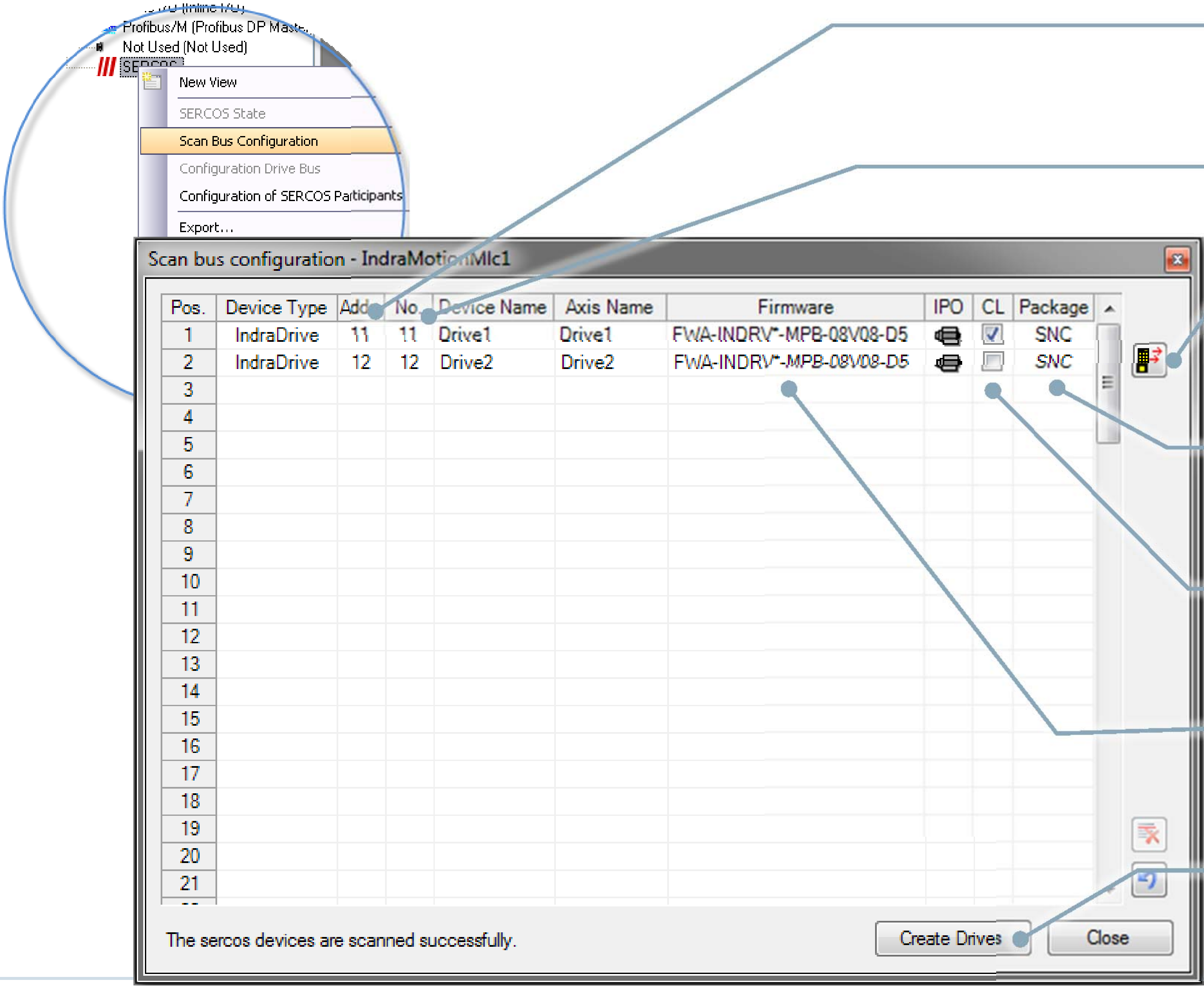
Scanning and Remote address assignment



SERCOS address

Unique axis number

Remote address assignment

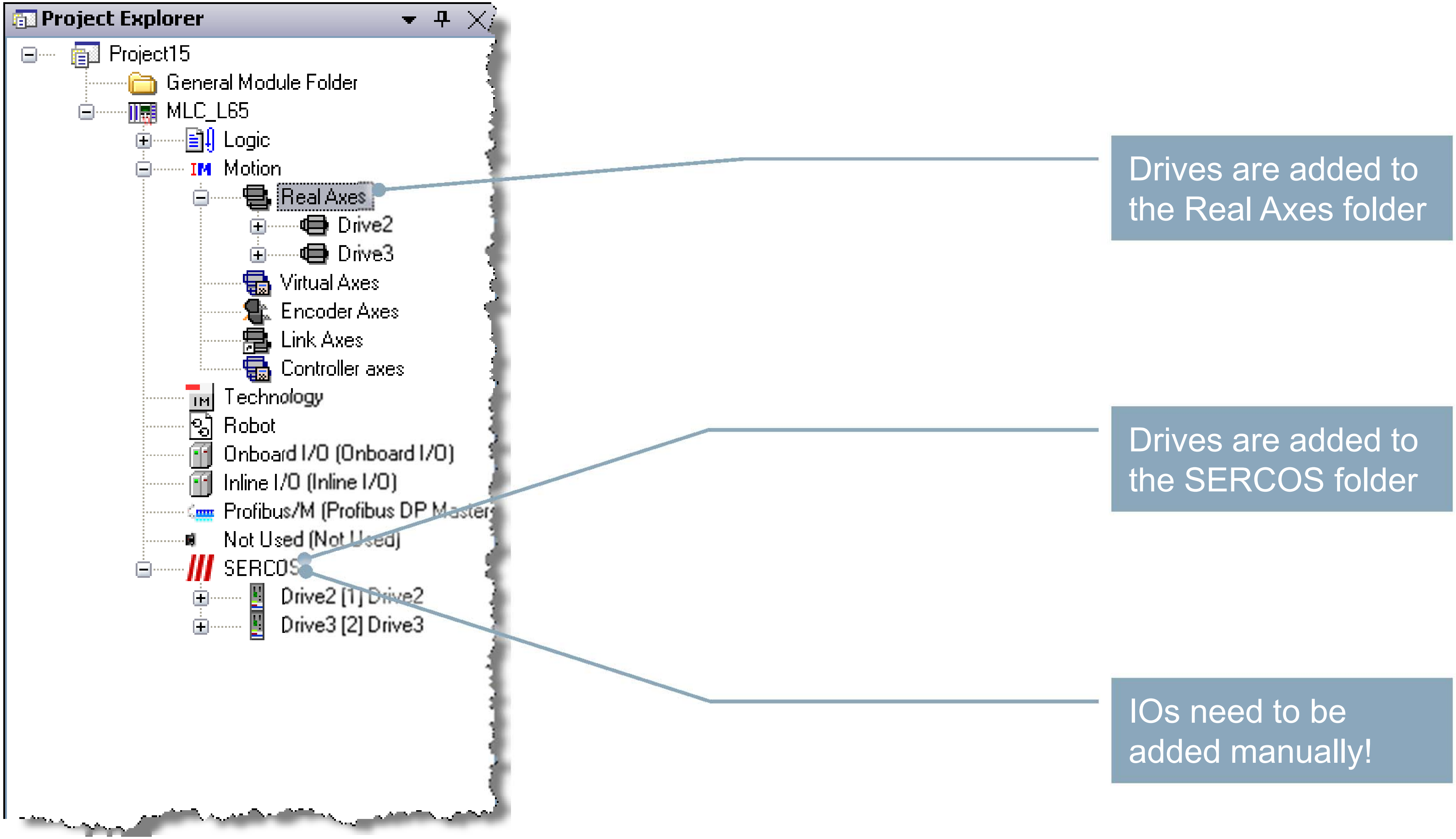
Interpolation on IndraDrive: Yes/No

IndraDrive firmware version

Axis name (can be modified)

Add drives to IndraWorks project

Scanning and Remote address assignment



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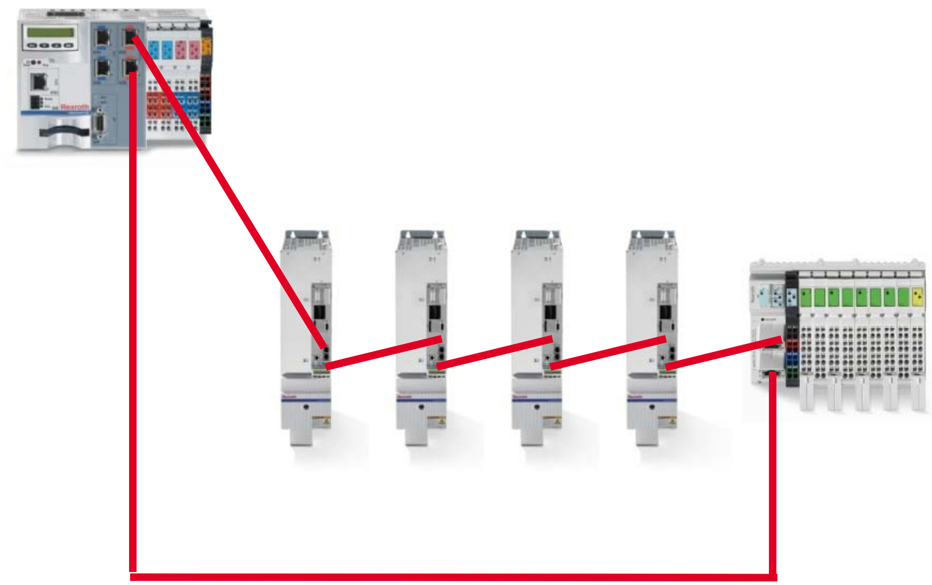
Remote address assignment

- Motivation:
  - Address assignment for devices without address switch (e.g. IO modules)
  - Better usability by centralized assignment from IndraWorks

- The topology address corresponds to the position in the ring
- Visualization of all devices in ascending order in a table form
- The SERCOS address can be modified



Topology Address	1	2	3	4	5
SERCOS Address	1	2	3	4	20

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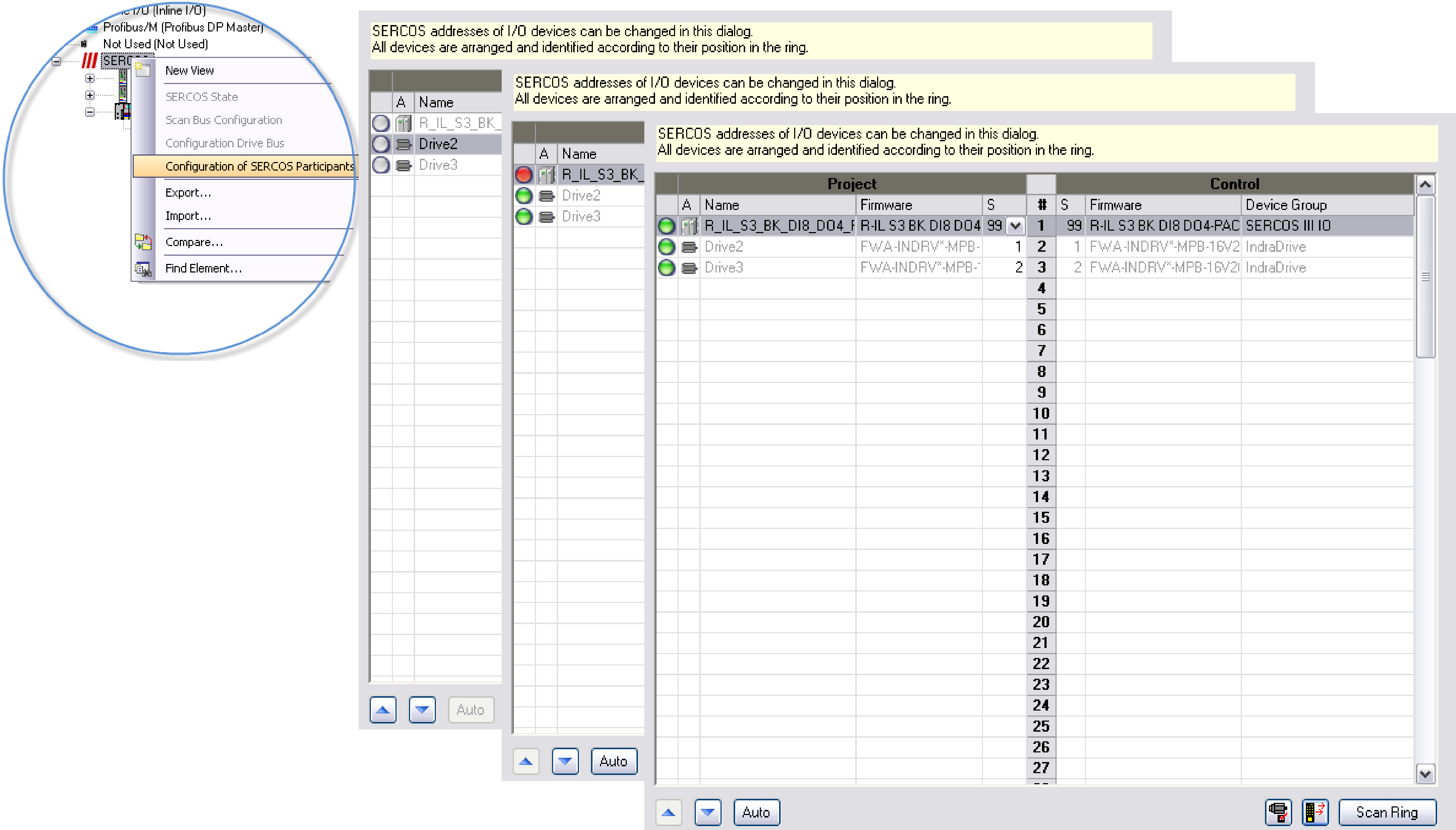
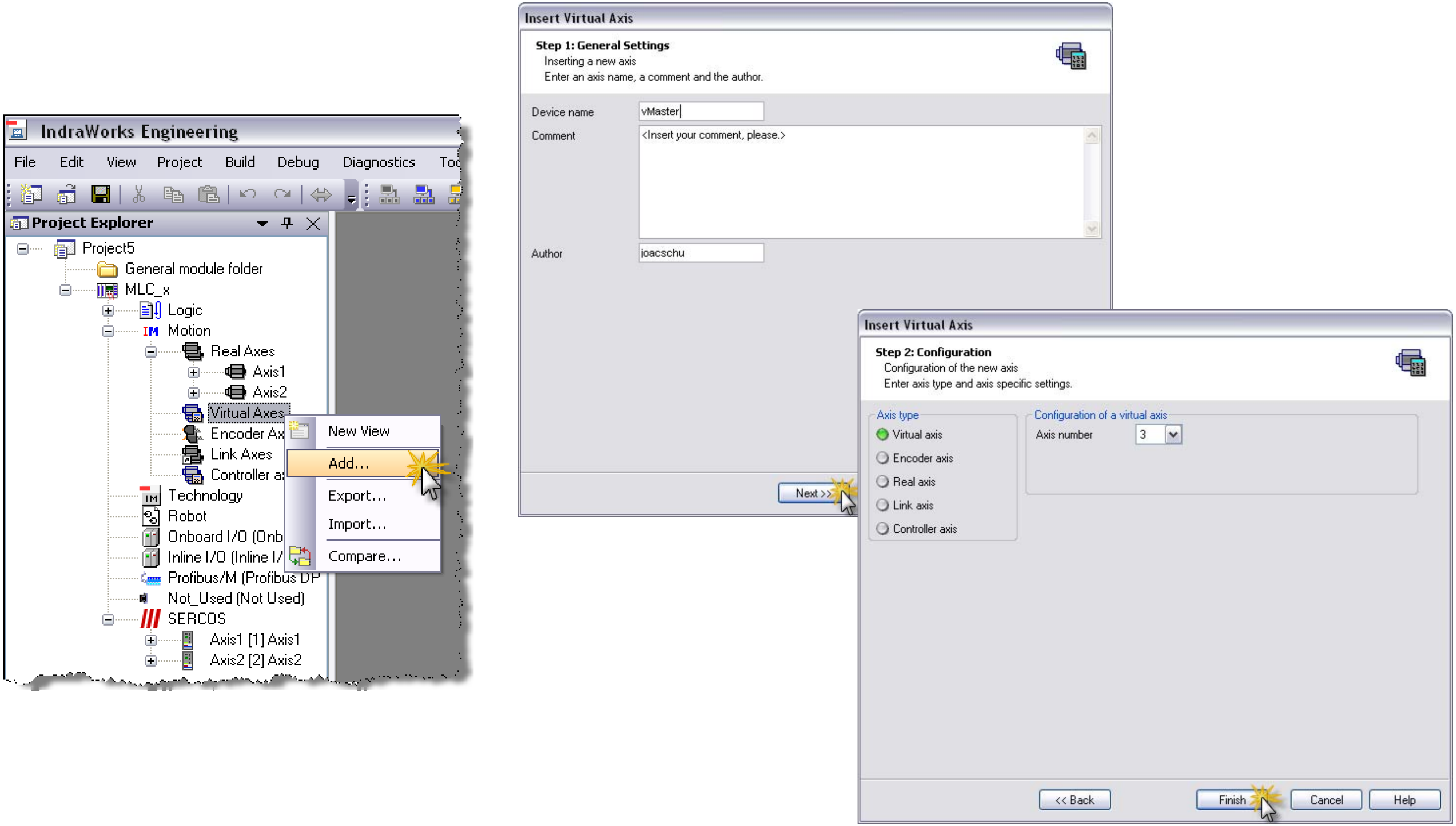
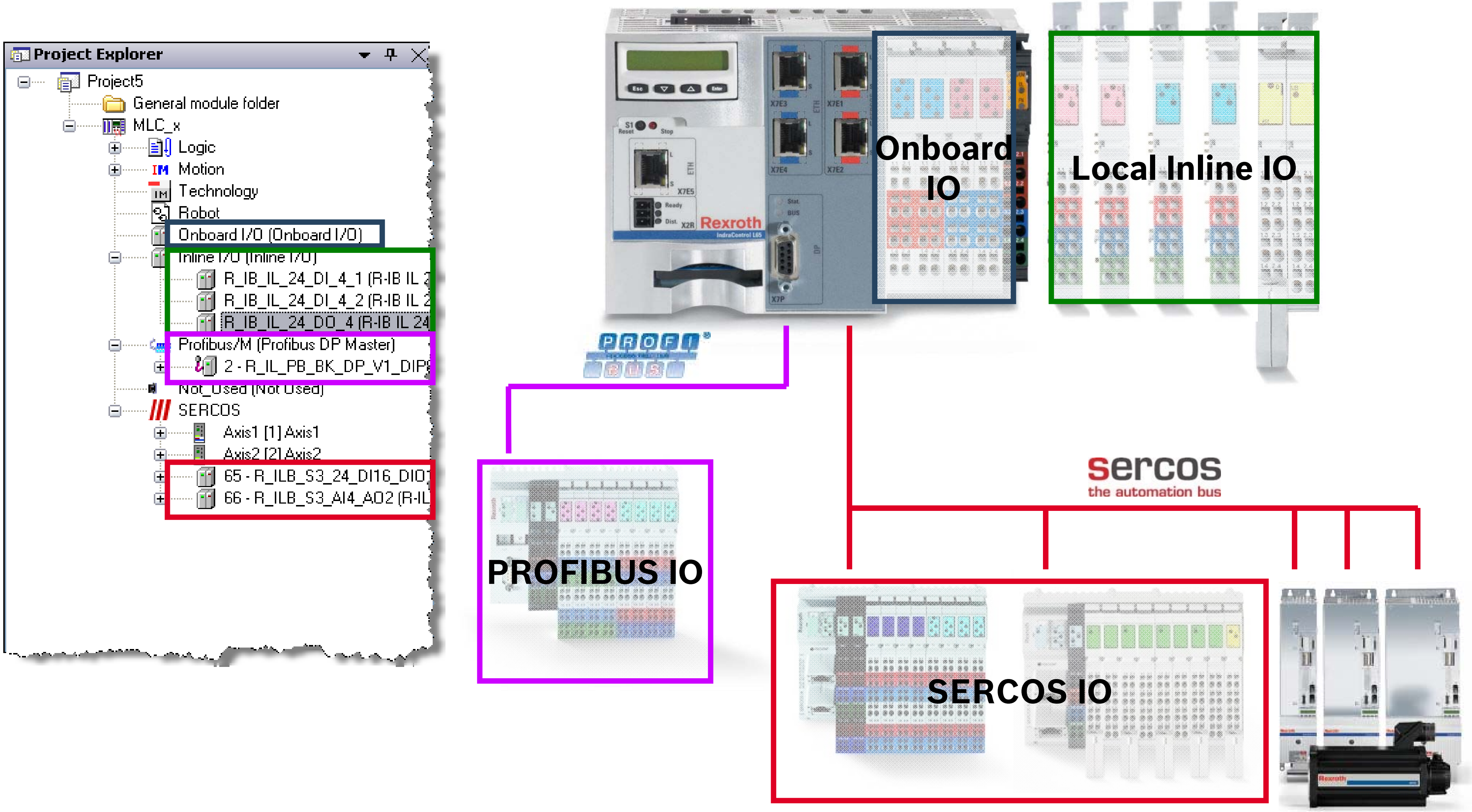
Remote address assignment



IndraWorks – Adding axes to the project



Consistent IO configuration



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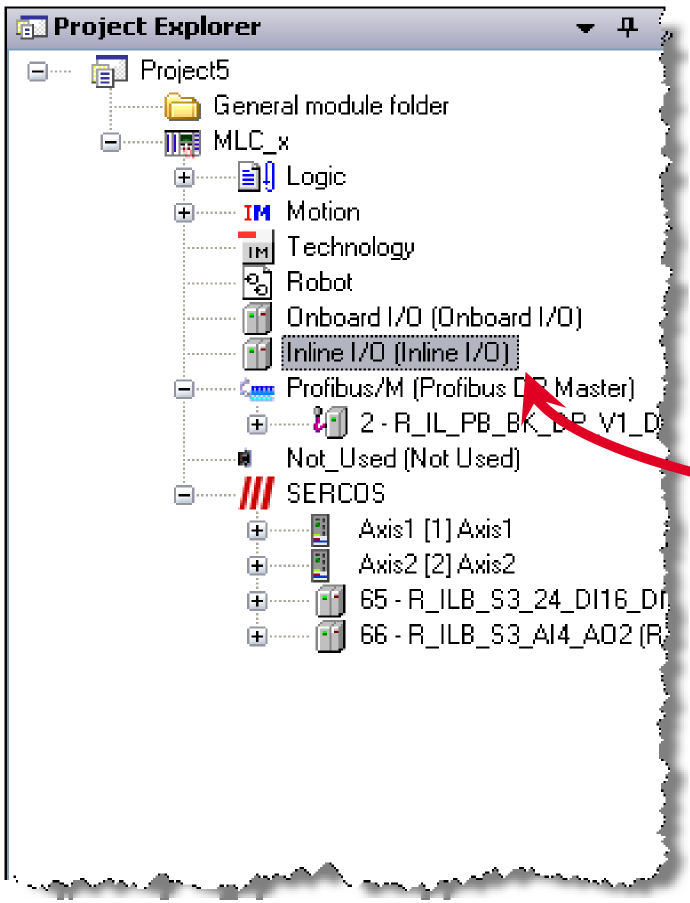
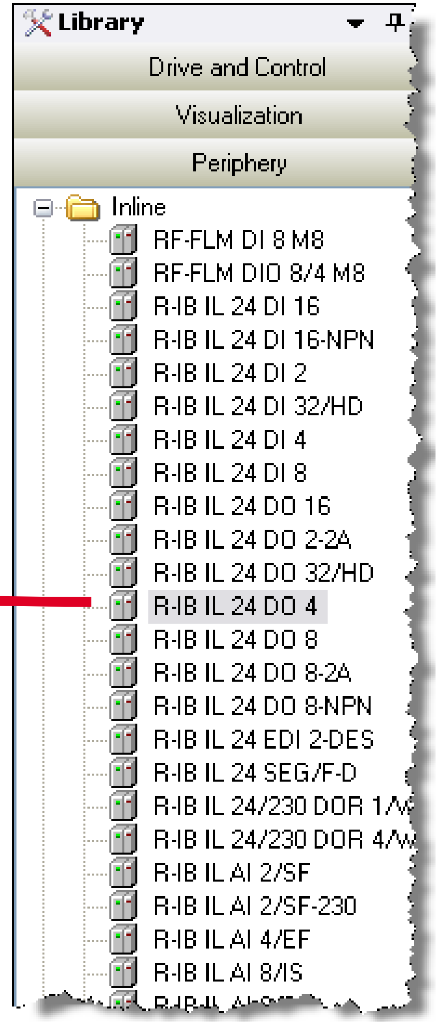
IO configuration – Inline IOs



Add appropriate Inline modules to the node Inline IO

Drag & Drop



The IO configuration in IndraWorks is uniform.
The configuration of PROFIBUS IOs and SERCOS III IOs follows the same philosophy.

Open **Periphery** in Device Library
Select category **Inline**

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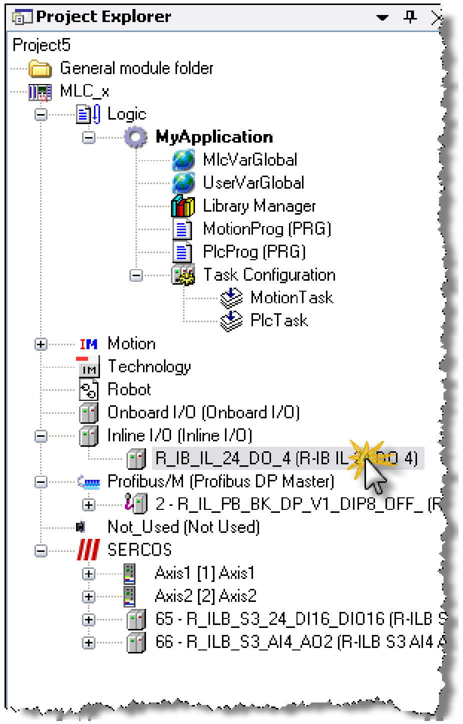
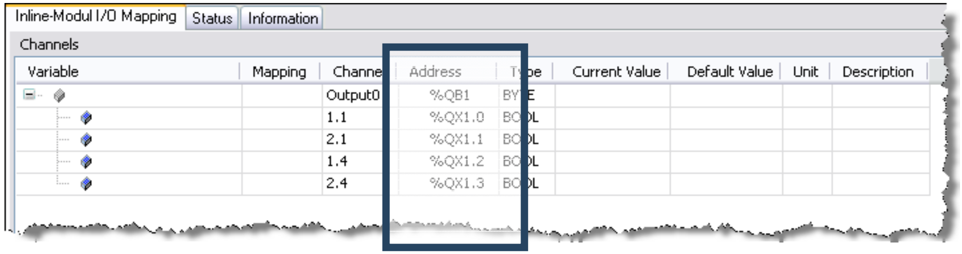
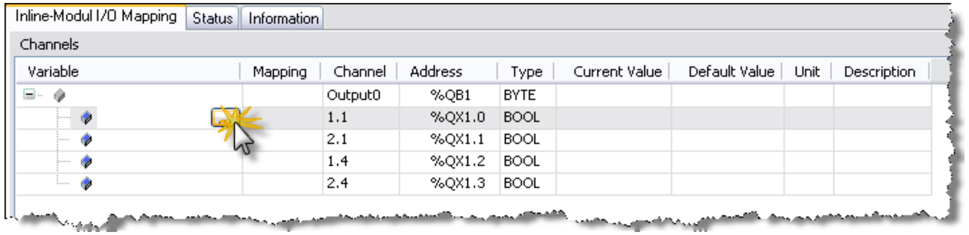
Address assignment & Mapping

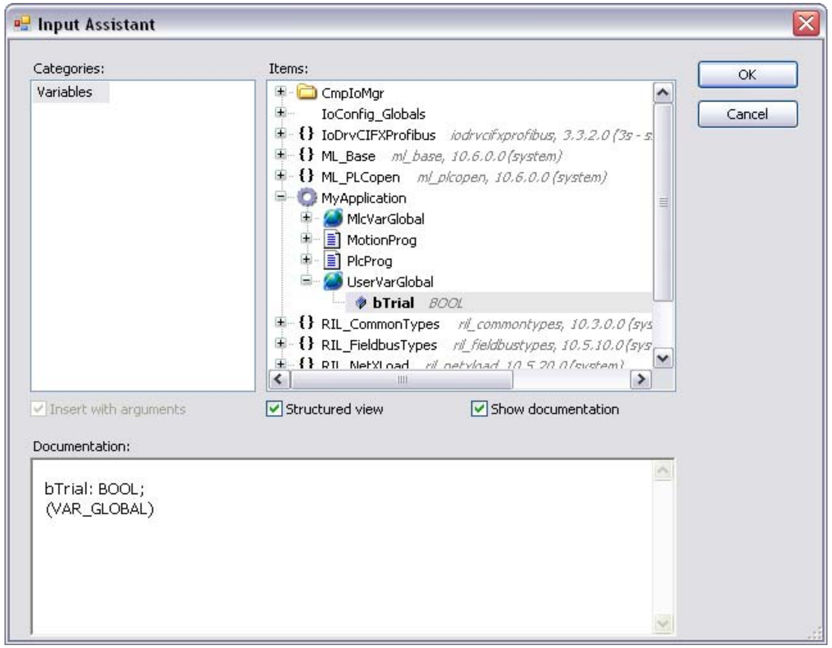




Automatic assignment of PLC addresses



The IOs can be mapped to PLC variables



Browse the PLC variables and select the appropriate variable

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IO configuration – Symbolic access to the IOs

```
VAR_GLOBAL
    // Bitadressen
    bSwitch_0    AT    %IX4.0  :  BOOL   ;  // Switch 0
    bSwitch_1    AT    %IX4.1  :  BOOL   ;  // Switch 1
    bSwitch_2    AT    %IX4.2  :  BOOL   ;  // Switch 2
    bSwitch_3    AT    %IX4.3  :  BOOL   ;  // Switch 3
    bSwitch_4    AT    %IX4.4  :  BOOL   ;  // Switch 4
    bSwitch_5    AT    %IX4.5  :  BOOL   ;  // Switch 5
    bSwitch_6    AT    %IX4.6  :  BOOL   ;  // Switch 6
    bSwitch_7    AT    %IX4.7  :  BOOL   ;  // Switch 7

    bSwitch_8    AT    %IX5.0  :  BOOL   ;  // Switch 8
    bSwitch_9    AT    %IX5.1  :  BOOL   ;  // Switch 9
    bSwitch_10   AT    %IX5.2  :  BOOL   ;  // Switch 10
    bSwitch_11   AT    %IX5.3  :  BOOL   ;  // Switch 11
    bSwitch_12   AT    %IX5.4  :  BOOL   ;  // Switch 12
    bSwitch_13   AT    %IX5.5  :  BOOL   ;  // Switch 13
    bSwitch_14   AT    %IX5.6  :  BOOL   ;  //
    bSwitch_15   AT    %IX5.7  :  BOOL   ;  //

    // Byte addresses
    byS0_S7      AT    %IB4    :  BYTE   ;  //
    byS8_S15     AT    %IB5    :  BYTE   ;  //

    // Word addresses
    wS0_S15      AT    %IW4    :  WORD   ;  //

    // Double word addresses
    dwS0_S31     AT    %ID4    :  DWORD  ;  //

    // Long word addresses
    dwS0_S63     AT    %IL4    :  LWORD  ;  //
END_VAR
```

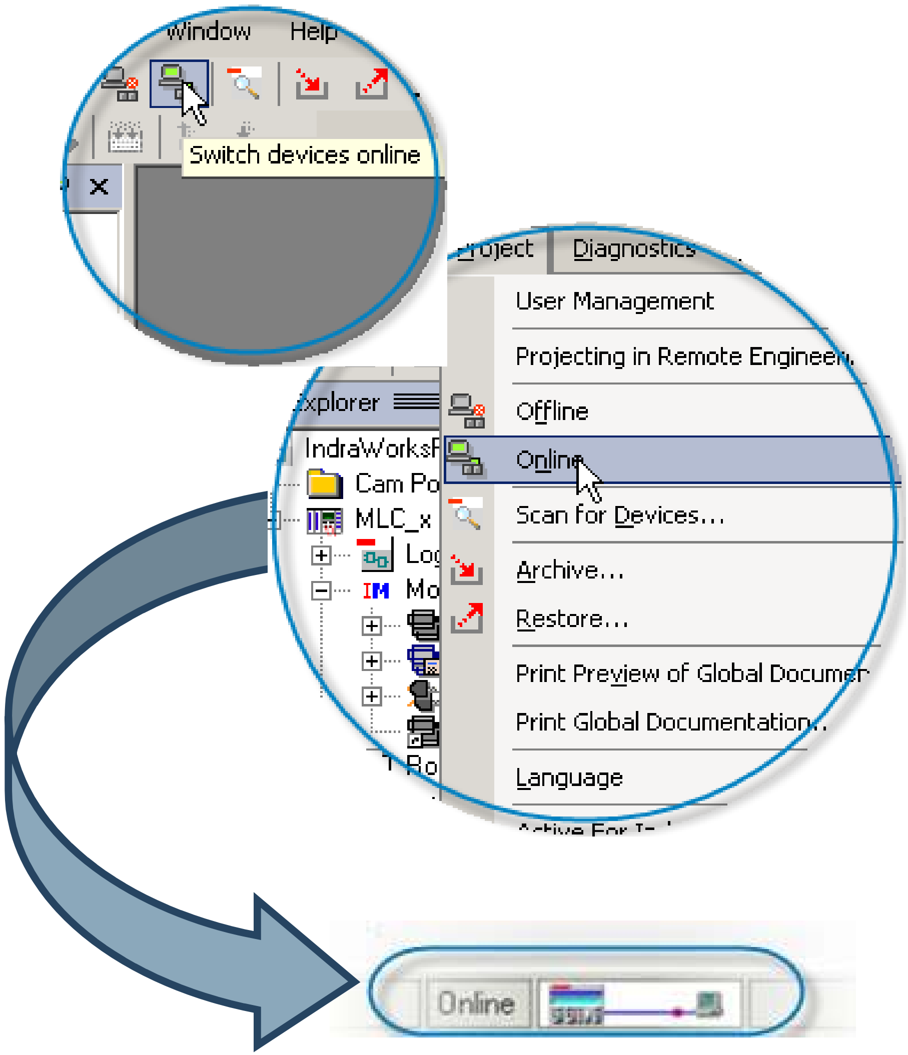
Variables can be mapped to IOs in IndraLogic using the keyword AT followed by the PLC address

```
VAR_GLOBAL
    // Single comment line

(* Bit addresses *)
    bValve_0     AT    %QX4.0   :   BOOL   ;   // Valve 0
    bValve_1     AT    %QX4.1   :   BOOL   ;   // Valve 1
    bValve_2     AT    %QX4.2   :   BOOL   ;   // Valve 2
    bValve_3     AT    %QX4.3   :   BOOL   ;   // Valve 3
    bValve_4     AT    %QX4.4   :   BOOL   ;   // Valve 4
    bValve_5     AT    %QX4.5   :   BOOL   ;   // Valve 5
    bValve_6     AT    %QX4.6   :   BOOL   ;   // Valve 6
    bValve_7     AT    %QX4.7   :   BOOL   ;   // Valve 7)

    bValve_8     AT    %QX5.0   :   BOOL   ;   // Valve 8
```

Switching to Online Mode



- Establish the Ethernet connection to the control
- Comparison of motion configuration with real devices
- Download motion configuration to MLC
- Online data exchange possible
- Switch from parameter mode (P2) to operating mode (BB)
- Start up of SERCOS, with all drives showing "bb" or "Ab"!

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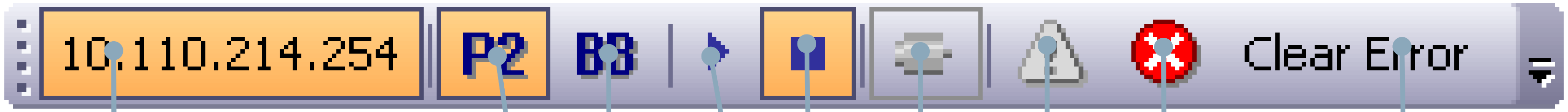
Toolbar MLC Info



Clear all errors

Show error log

Show warning log

Show device status (MLC and axis state information)

Switch PLC from STOP to RUN and vice versa

Switch SERCOS link from parameter mode to operating mode and vice versa

IP address of MLC

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IO configuration

Exercise

- Complete your IndraWorks project
- Scan drives and add them to your project
- Add virtual axis to your project
- Generate correct configuration of Local Inline IOs
- Generate correct configuration of SERCOS III IOs

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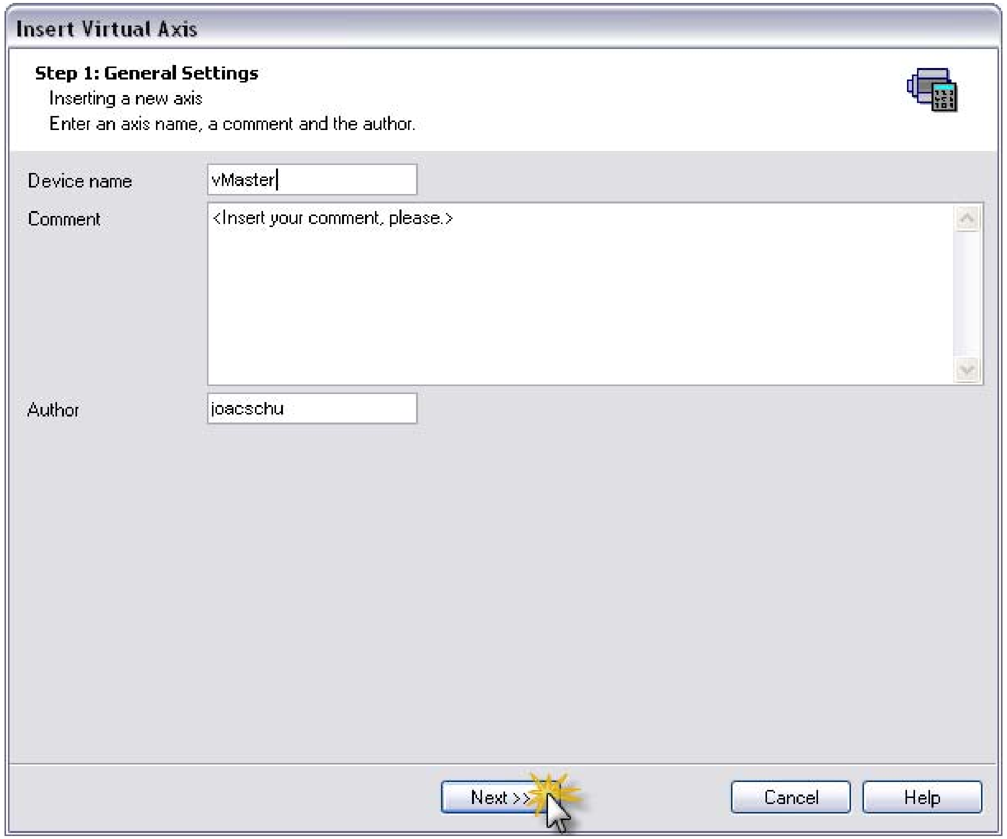
Add virtual axis

Exercise



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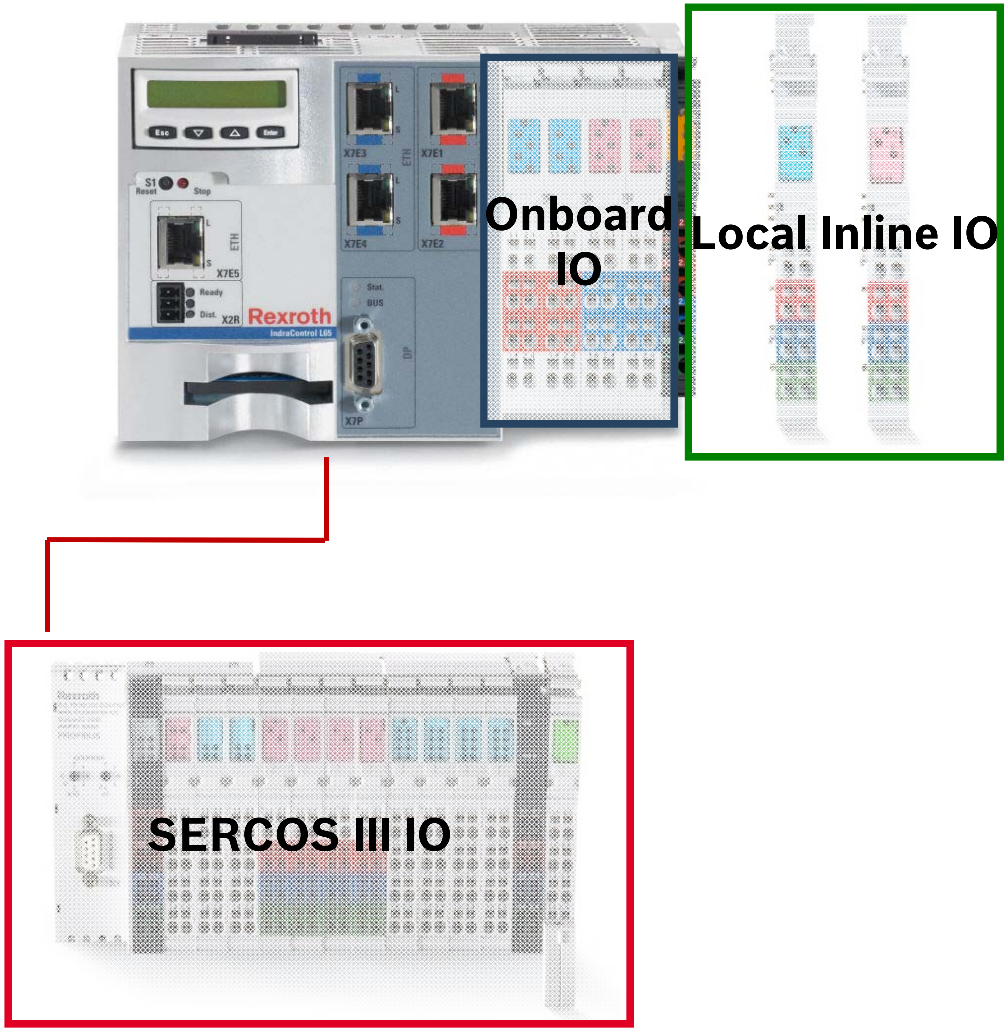
IO configuration

Exercise

- Onboard IOs need not to be configured
- Configure Local Inline IOs for your demo system
- Configure SERCOS III IOs for you demo system
- Check the SERCOS III address of the SERCOS IO module and eventually adjust it (remote addressing)
- No mapping of IOs to PLC variables required!
- At which point of time a bad configuration is detected?
- How to detect a bad configuration?



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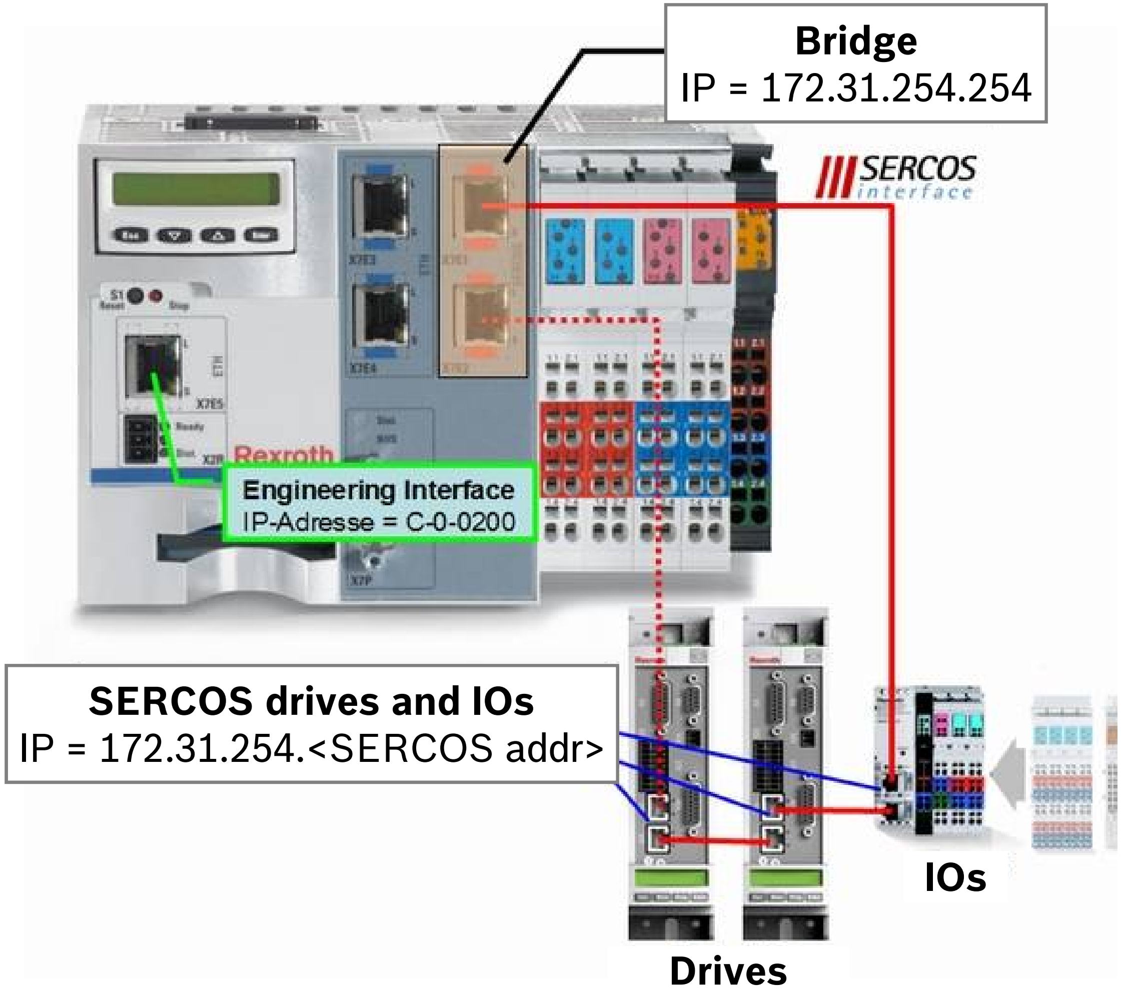
Using the IP channel in SERCOS III (1)

- IP communication with devices in the SERCOS III link
- Required parameters
  - IP settings Engineering port
  - SERCOS addresses of drives and IO
- Automatic setting of all IP parameters for drives and IOs
- Modification of drive's Gateway address is effective only after switching on and off the IndraDrive (bug!)



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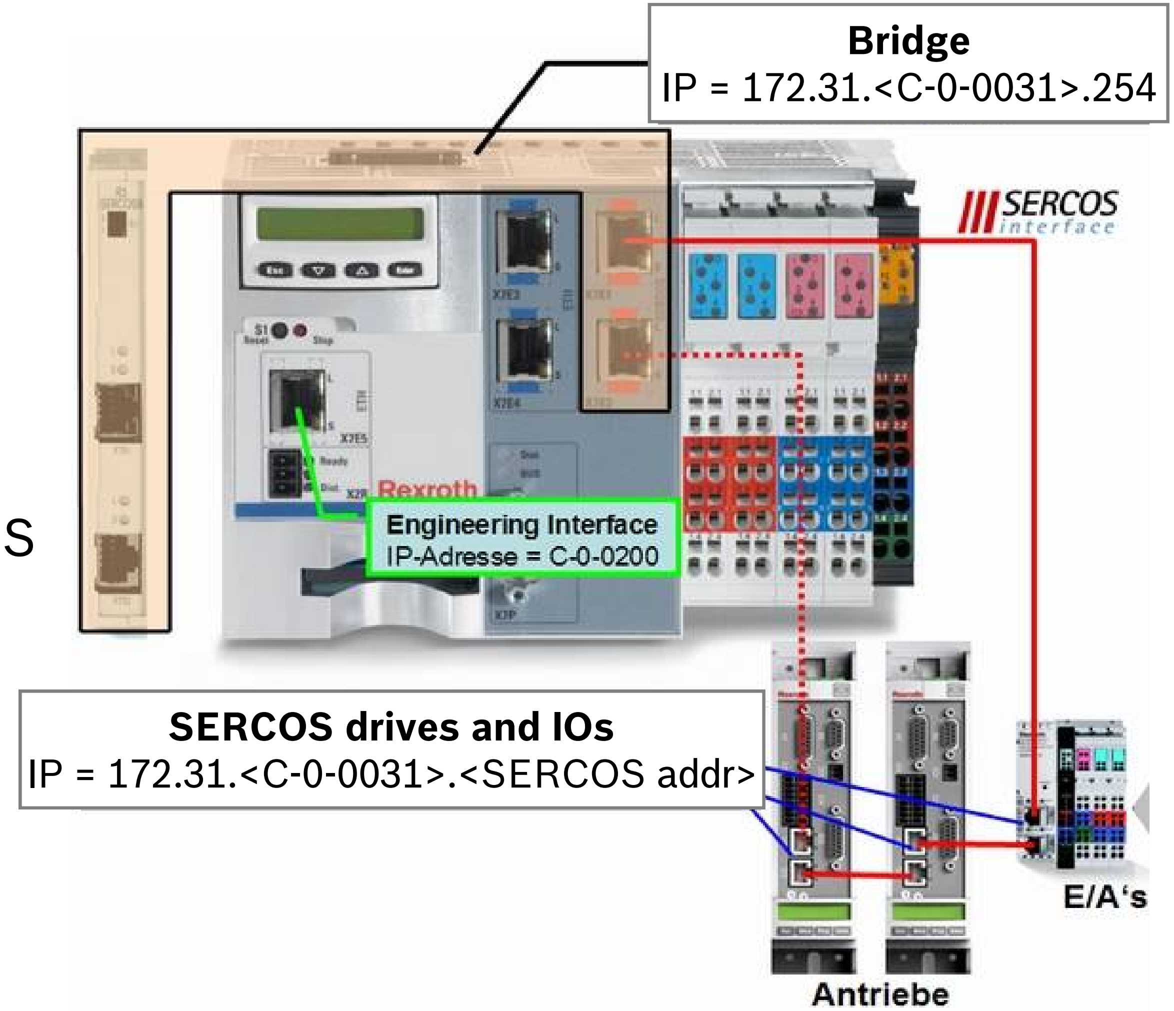
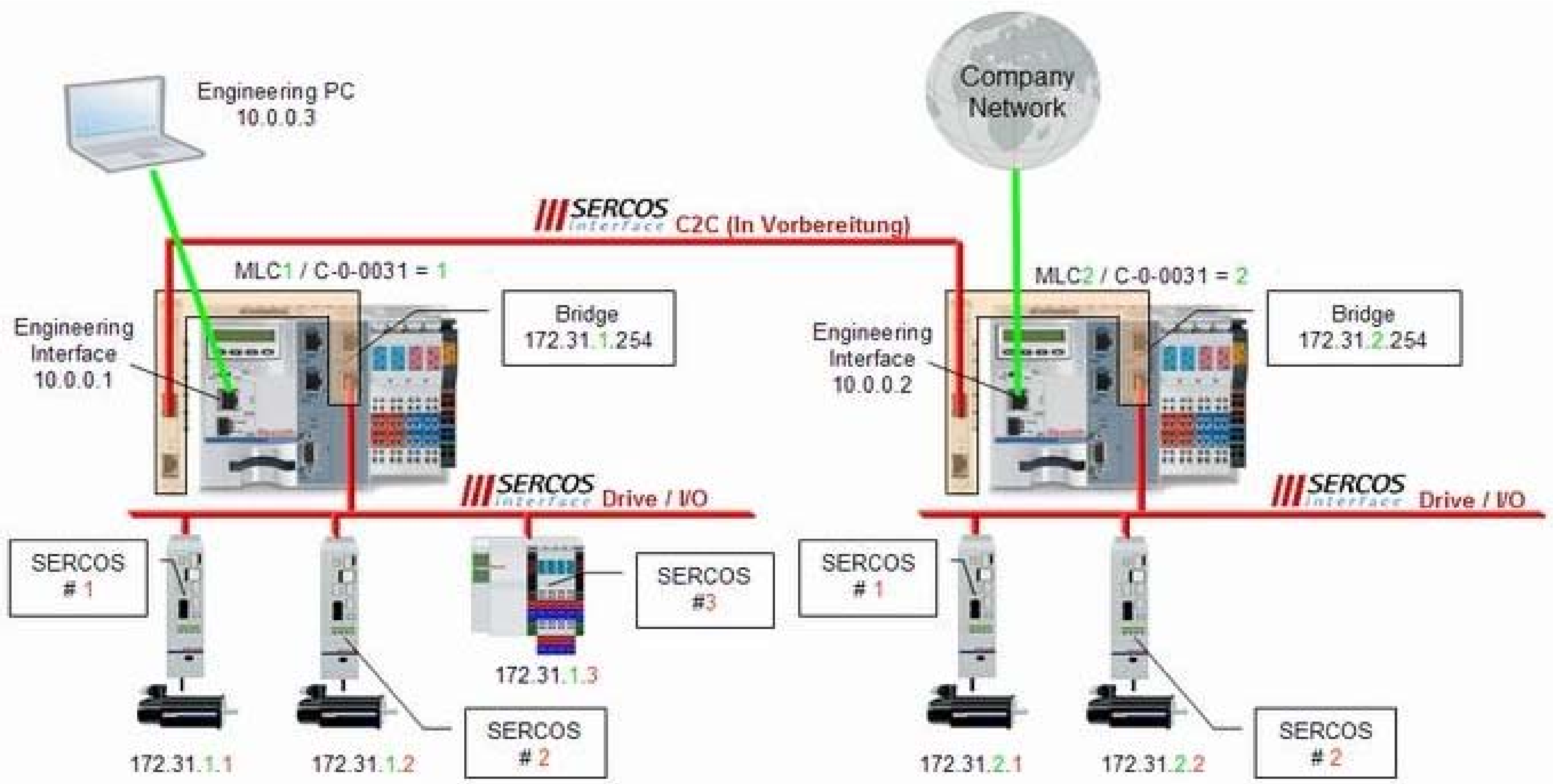
Using the IP channel in SERCOS III (2)

- SERCOS III interfaces of MLC are “bridged”
- Bridging means that different physical networks are combined to one logical network
- Common IP address for all SERCOS III interfaces of MLC
- All drives and IOs in the SERCOS drive link are accessible via Engineering port
- ... and also remote MLCs in the SERCOS C2C link



Using the IP channel in SERCOS III (3)



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IndraDrive FW download

Exercise

- How to upgrade IndraDrive firmware via SERCOS III link

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IndraDrive FW download

Exercise

```
U:\>ping 172.31.254.1

Ping wird ausgeführt für 172.31.254.1 mit 32 Bytes Daten:

Zeitüberschreitung der Anforderung.
Zeitüberschreitung der Anforderung.
Zeitüberschreitung der Anforderung.
Zeitüberschreitung der Anforderung.

Ping-Statistik für 172.31.254.1:
    Pakete: Gesendet = 4, Empfangen = 0, Verloren = 4 (100% Verlust),

U:\>_
```

```
U:\>route print
===========================================================================
Schnittstellenliste
0x1 ........................... MS TCP Loopback interface
0x2 ...00 50 56 c0 00 08 ...... VMware Virtual Ethernet Adapter for VMnet8
0x3 ...00 50 56 c0 00 01 ...... VMware Virtual Ethernet Adapter for VMnet1
0x4 ...00 19 d2 57 dd e2 ...... Intel(R) PRO/Wireless 3945ABG Network Connection
 - Deterministic Network Enhancer Miniport
0x5 ...00 17 a4 d5 c9 39 ...... Broadcom NetXtreme Gigabit Ethernet - Determinis
tic Network Enhancer Miniport
===========================================================================
===========================================================================
Aktive Routen:
     Netzwerkziel    Netzwerkmaske          Gateway    Schnittstelle  Anzahl
          0.0.0.0          0.0.0.0   10.110.214.129   10.110.214.137      20
   10.110.214.128  255.255.255.128   10.110.214.137   10.110.214.137      20
   10.110.214.137  255.255.255.255        127.0.0.1        127.0.0.1      20
   10.255.255.255  255.255.255.255   10.110.214.137   10.110.214.137      20
        127.0.0.0        255.0.0.0        127.0.0.1        127.0.0.1      1
     192.168.52.0    255.255.255.0     192.168.52.1     192.168.52.1      20
     192.168.52.1  255.255.255.255        127.0.0.1        127.0.0.1      20
   192.168.52.255  255.255.255.255     192.168.52.1     192.168.52.1      20
    192.168.199.0    255.255.255.0    192.168.199.1    192.168.199.1      20
    192.168.199.1  255.255.255.255        127.0.0.1        127.0.0.1      20
  192.168.199.255  255.255.255.255    192.168.199.1    192.168.199.1      20
        224.0.0.0        240.0.0.0   10.110.214.137   10.110.214.137      20
        224.0.0.0        240.0.0.0     192.168.52.1     192.168.52.1      20
        224.0.0.0        240.0.0.0    192.168.199.1    192.168.199.1      20
  255.255.255.255  255.255.255.255   10.110.214.137   10.110.214.137      1
  255.255.255.255  255.255.255.255     192.168.52.1     192.168.52.1      1
  255.255.255.255  255.255.255.255    192.168.199.1    192.168.199.1      1
  255.255.255.255  255.255.255.255    192.168.199.1                4      1
Standardgateway:    10.110.214.129
===========================================================================
Ständige Routen:
  Keine

U:\>_
```

- Use the ***ping*** command to check IP communication to the IndraDrive
- Check the routing table on your notebook using the command ***route print***
- IP communication doesn't work, because a routing entry for the IndraDrive is missing!

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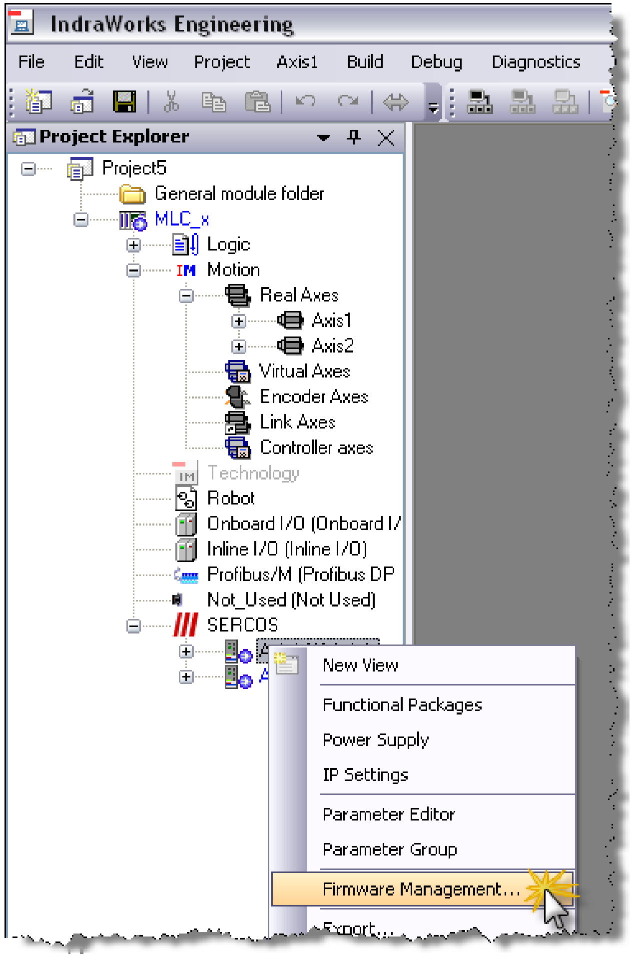
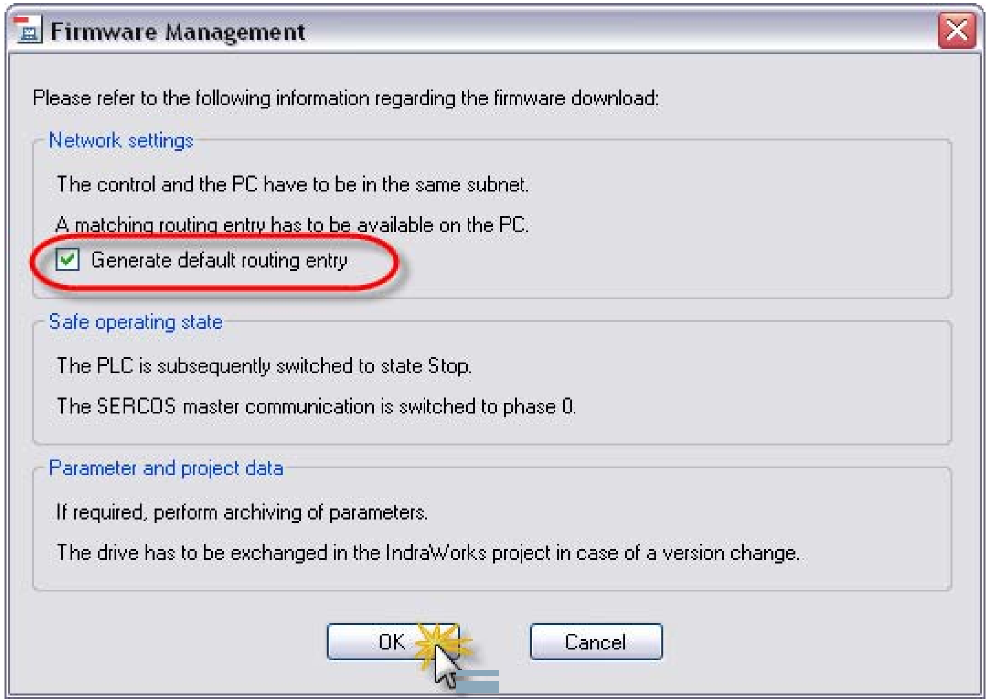
IndraDrive FW download

Exercise





To download the drive firmware select the corresponding axis and open the context menu to call the Firmware Management

A routing entry for this IndraDrive is generated!

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IndraDrive FW download

Exercise

```
U:\>route print
===========================================================================
Schnittstellenliste
0x1 ........................... MS TCP Loopback interface
0x2 ...00 50 56 c0 00 08 ...... VMware Virtual Ethernet Adapter for VMnet8
0x3 ...00 50 56 c0 00 01 ...... VMware Virtual Ethernet Adapter for VMnet1
0x4 ...00 19 d2 57 dd e2 ...... Intel(R) PRO/Wireless 3945ABG Network Connection
 - Deterministic Network Enhancer Miniport
0x5 ...00 17 a4 d5 c9 39 ...... Broadcom NetXtreme Gigabit Ethernet - Determinis
tic Network Enhancer Miniport
===========================================================================
===========================================================================
Aktive Routen:
     Netzwerkziel    Netzwerkmaske          Gateway    Schnittstelle   Anzahl
          0.0.0.0          0.0.0.0   10.110.214.129   10.110.214.137       20
   10.110.214.128  255.255.255.128   10.110.214.137   10.110.214.137       20
   10.110.214.137  255.255.255.255        127.0.0.1        127.0.0.1       20
   10.255.255.255  255.255.255.255   10.110.214.137   10.110.214.137       20
        127.0.0.0        255.0.0.0        127.0.0.1        127.0.0.1       1
     172.31.254.1  255.255.255.255   10.110.214.254   10.110.214.137       1
     192.168.52.0    255.255.255.0     192.168.52.1     192.168.52.1       20
     192.168.52.1  255.255.255.255        127.0.0.1        127.0.0.1       20
   192.168.52.255  255.255.255.255     192.168.52.1     192.168.52.1       20
    192.168.199.0    255.255.255.0    192.168.199.1    192.168.199.1       20
    192.168.199.1  255.255.255.255        127.0.0.1        127.0.0.1       20
  192.168.199.255  255.255.255.255    192.168.199.1    192.168.199.1       20
        224.0.0.0        240.0.0.0   10.110.214.137   10.110.214.137       20
        224.0.0.0        240.0.0.0     192.168.52.1     192.168.52.1       20
        224.0.0.0        240.0.0.0    192.168.199.1    192.168.199.1       20
  255.255.255.255  255.255.255.255   10.110.214.137   10.110.214.137       1
  255.255.255.255  255.255.255.255     192.168.52.1     192.168.52.1       1
  255.255.255.255  255.255.255.255    192.168.199.1    192.168.199.1       1
  255.255.255.255  255.255.255.255    192.168.199.1                4       1
Standardgateway:    10.110.214.129
===========================================================================
Ständige Routen:
  Keine

U:\>_
```

```
U:\>ping 172.31.254.1

Ping wird ausgeführt für 172.31.254.1 mit 32 Bytes Daten:

Antwort von 172.31.254.1: Bytes=32 Zeit=4ms TTL=63
Antwort von 172.31.254.1: Bytes=32 Zeit=2ms TTL=63
Antwort von 172.31.254.1: Bytes=32 Zeit=8ms TTL=63
Antwort von 172.31.254.1: Bytes=32 Zeit=2ms TTL=63

Ping-Statistik für 172.31.254.1:
    Pakete: Gesendet = 4, Empfangen = 4, Verloren = 0 (0% Verlust),
Ca. Zeitangaben in Millisek.:
    Minimum = 2ms, Maximum = 8ms, Mittelwert = 4ms

U:\>
```

- Check the routing table using the command ***route print***
- A new entry has been created
- Use the ***ping*** command to check IP communication
- IP communication works!

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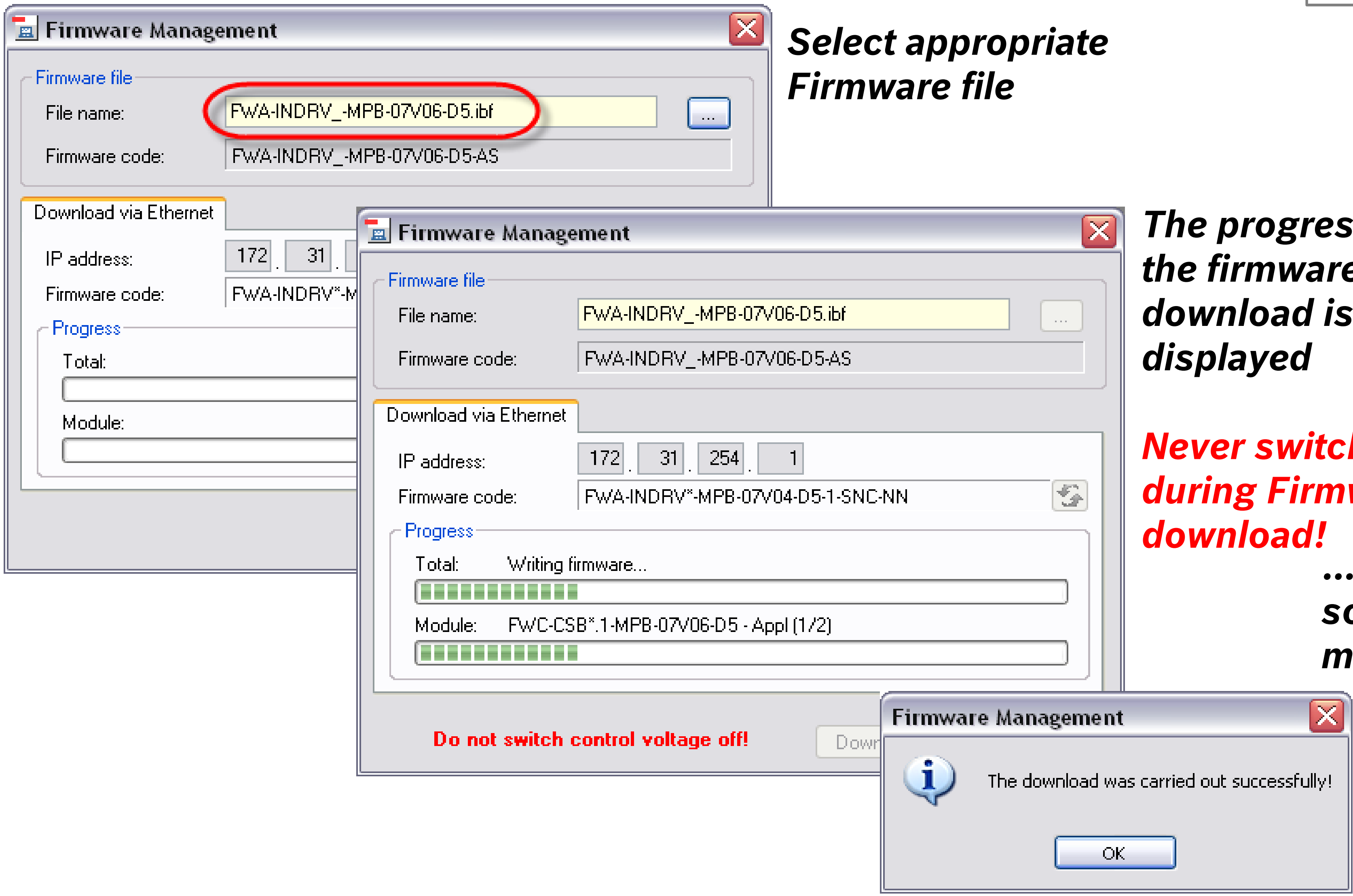
IndraDrive FW download

Exercise



Select appropriate Firmware file

The progress of the firmware download is displayed

Never switch off during Firmware download!

… after some minutes …

Download completed successfully!

Check the correct result (display of IndraDrive) and do the same procedure for the 2nd drive!

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IndraDrive FW download

Exercise


```
U:\>route delete 172.31.254.1

U:\>route delete 172.31.254.2

U:\>route add 172.31.254.0 mask 255.255.255.0 10.110.214.254

U:\>
```

- You also can add the routing entry yourself:
- Deselect the option *Generate default routing entry*
- ... before downloading
- Create the routing entry with DOS command ***route add***
- Delete the previous routes
- Call ***route add*** to set routing entry

Syntax: route add 172.31.254.0 mask 255.255.255.0 192.168.123.141

IP address of SERCOS III subnet **IP address of MLC**

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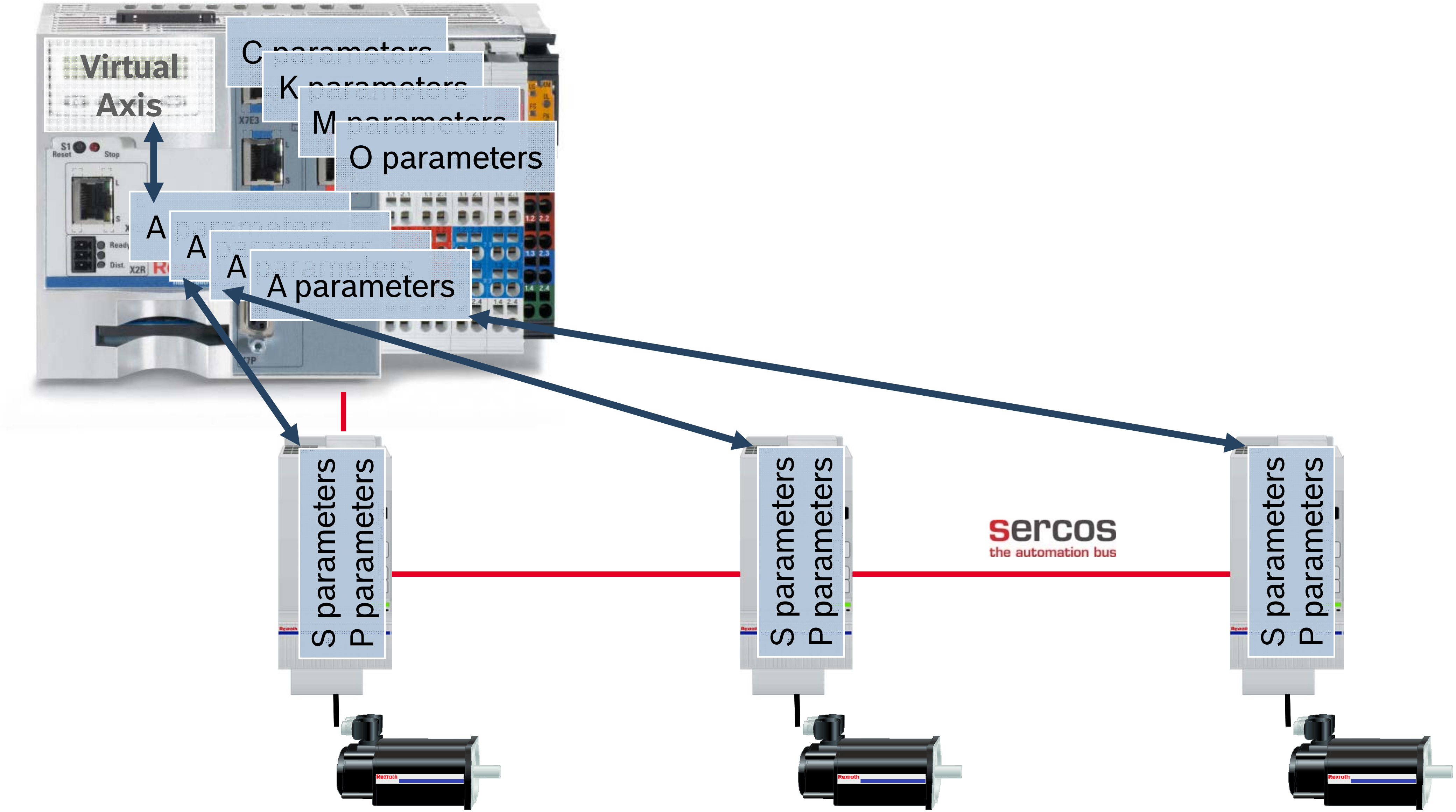
MLC Parameter System



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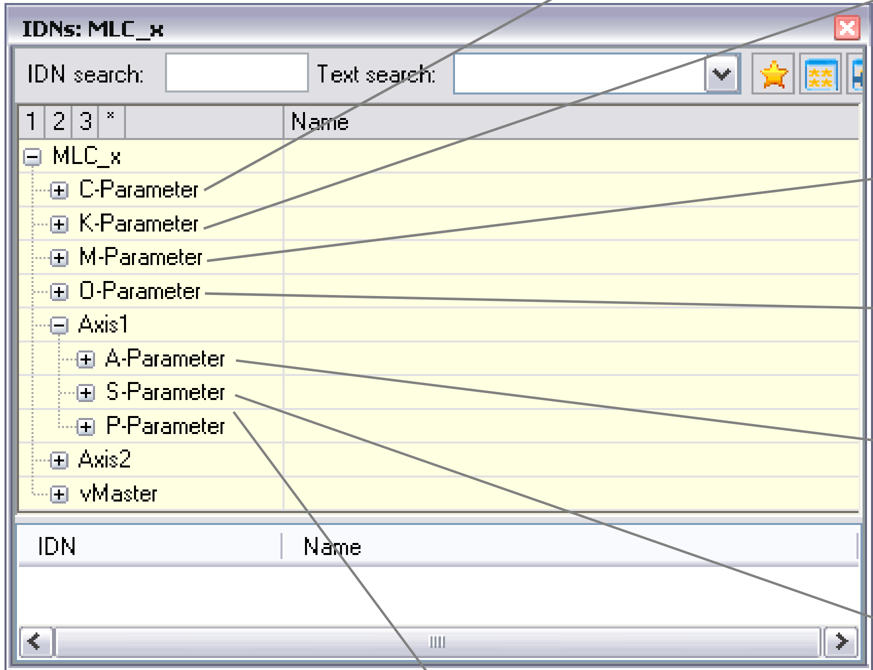
MLC Parameter System



Control parameters
Configuration data of the control (or information)
One set of C parameters per IndraMotion MLC

Kinematics parameters
Configuration data for Robot Control
One set of K parameters per Kinematics

Probe parameters
Configuration data for probes (M001 – M100)

Oscilloscope parameters
Configuration data for the oscilloscope

Axis parameters
Configuration data of the axes (or information)
One set of A parameters per axis resides in the control

Standard parameters defined in the SERCOS standard
Configuration data of the drive (or information)
Stored in the IndraDrive (Flash memory of control section)

Manufacturer/product-specific parameters in terms of the SERCOS standard
Configuration data of the drive (or information)
Stored in the IndraDrive (Flash memory of control section)

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MLC Parameter System

MLC/MLP-Parameter 10VRS
A/C Parameters Cover S/P Parameters

IndraDrive: A/C Paramete
SercosDrive: A/C Parame
HNC Devices: A/C Param

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MLC/MLP-Parameter 10VRS
IndraDrive: A/C Parameters Cover S/P Parameters

The following table provides an overview for IndraDrive on how the information of A or C parameters of the MLC are transferred to the S or P parameters:

> The calculation of the motion is carried out on the IndraDrive.
>
> If the calculation is carried out on the control, see SercosDrive: A/C Parameters cover S/P Parameters.

- Ident_IDV means that the copy is identical,
- Used_IDV means that at least one bit is influenced.

IDN	Parameter name	Ident_IDV	Used_IDV
A-0-0002	Axis name	S-0-0142	S-0-0142
A-0-0007	[illegible]	[illegible]	S-[illegible]-0033, S-0-0034, S-0-[illegible]035, S-0-[illegible]5, S-0-[illegible]7
A-0-0021	Axis sta[illegible]		[illegible]figuration
A-0-0022	Extended axis status		AT configuration
A-0-0023	Axis diagnostic number		S-0-0390
A-0-0024	Axis condition		S-0-0139
A-0-0028	Travel range limit switch	P-0-0090	P-0-0090
A-0-0029	Position polarities	S-0-0055	S-0-0055
A-0-0030	Positive position limit value	S-0-0049	S-0-0049
A-0-0031	Negative position limit value	S-0-0050	S-0-0050
A-0-0032	Positive velocity limit value	S-0-0038, S-0-0091	S-0-0038, S-0-0091
A-0-0033	Negative velocity limit value	S-0-0039, S-0-0091	S-0-0039, S-0-0091
A-0-0034	Bipolar acceleration limit value	S-0-0138	S-0-0138
A-0-0036	Bipolar jerk limit value	S-0-0349	S-0-0349

During SERCOS startup (transition from P2 to BB) the information of A or C parameters is sent to the drives. As a result some S or P parameters are overlapped by A and C parameters!

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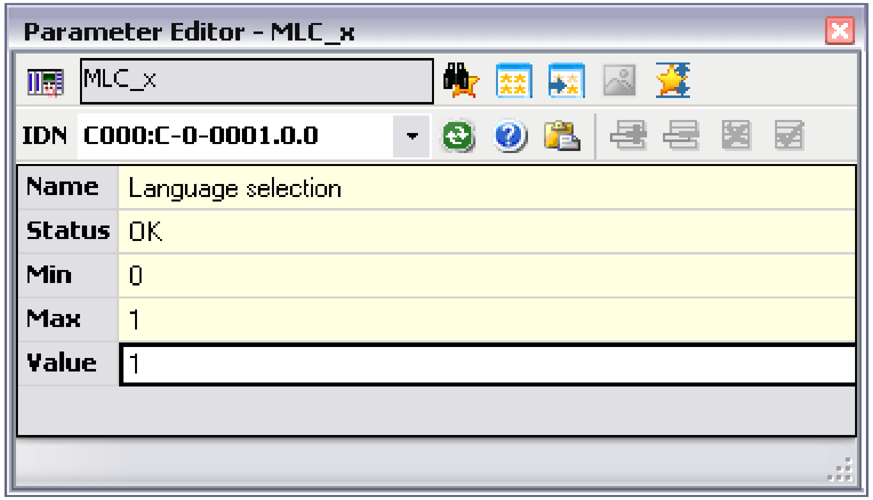
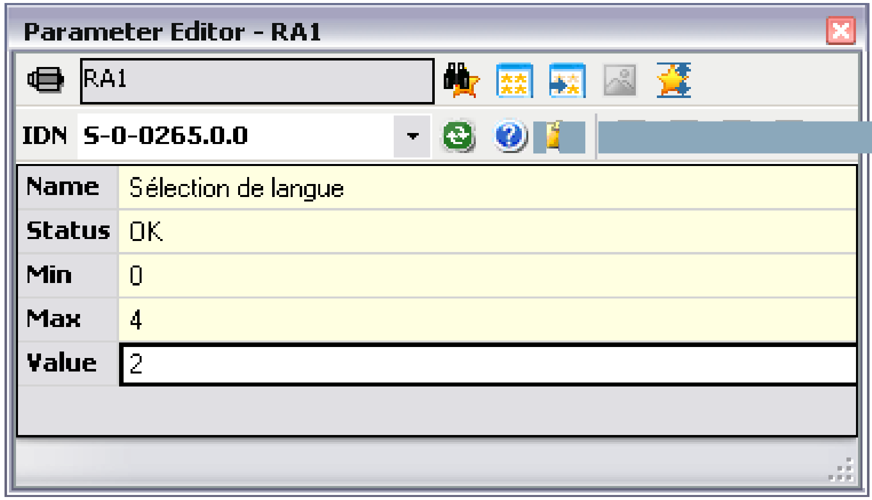
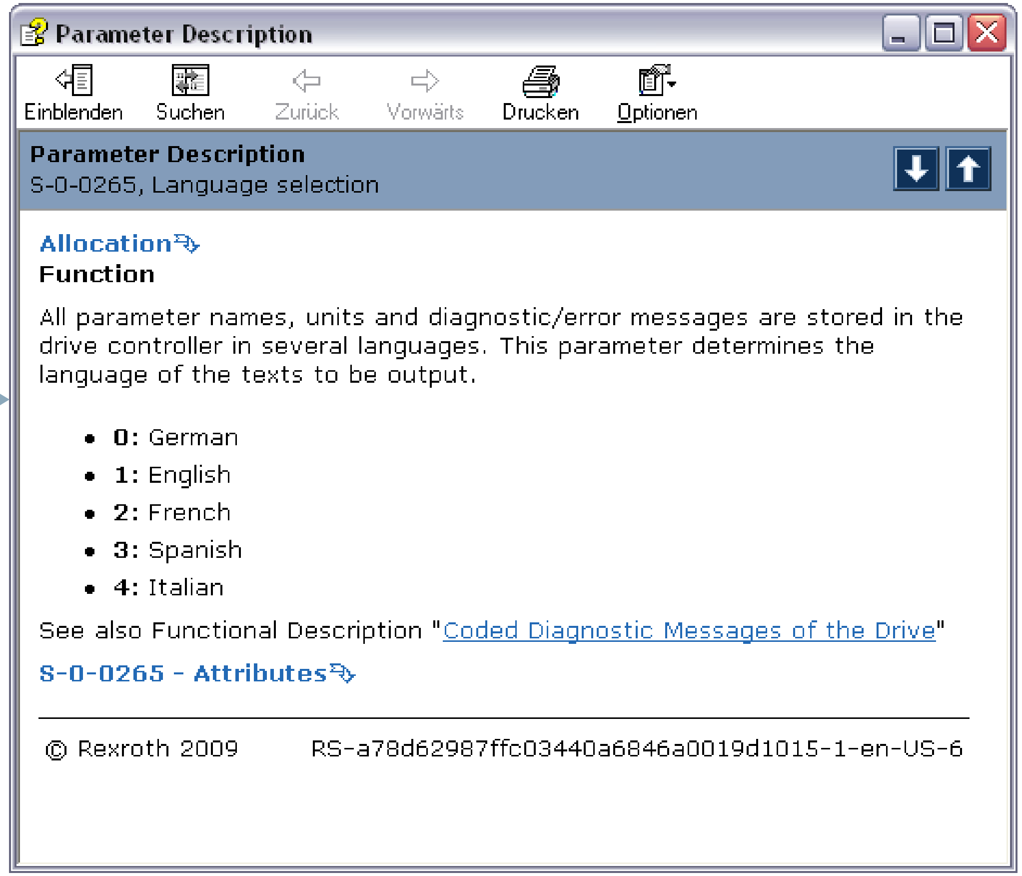
MLC Parameter System – Parameter Editor



Multiple instances of the Parameter Editor can be opened. Provides direct access to the parameter system of MLC and the drives.





Invoke the Online Help directly from the Parameter Editor

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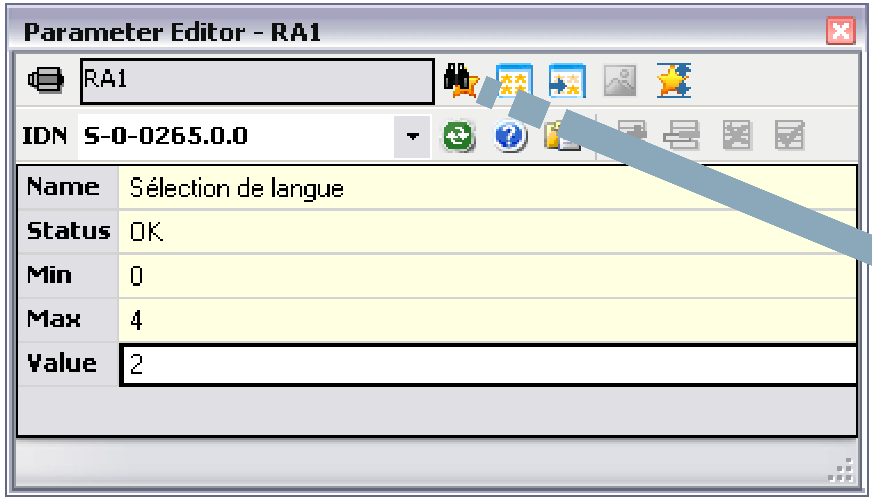
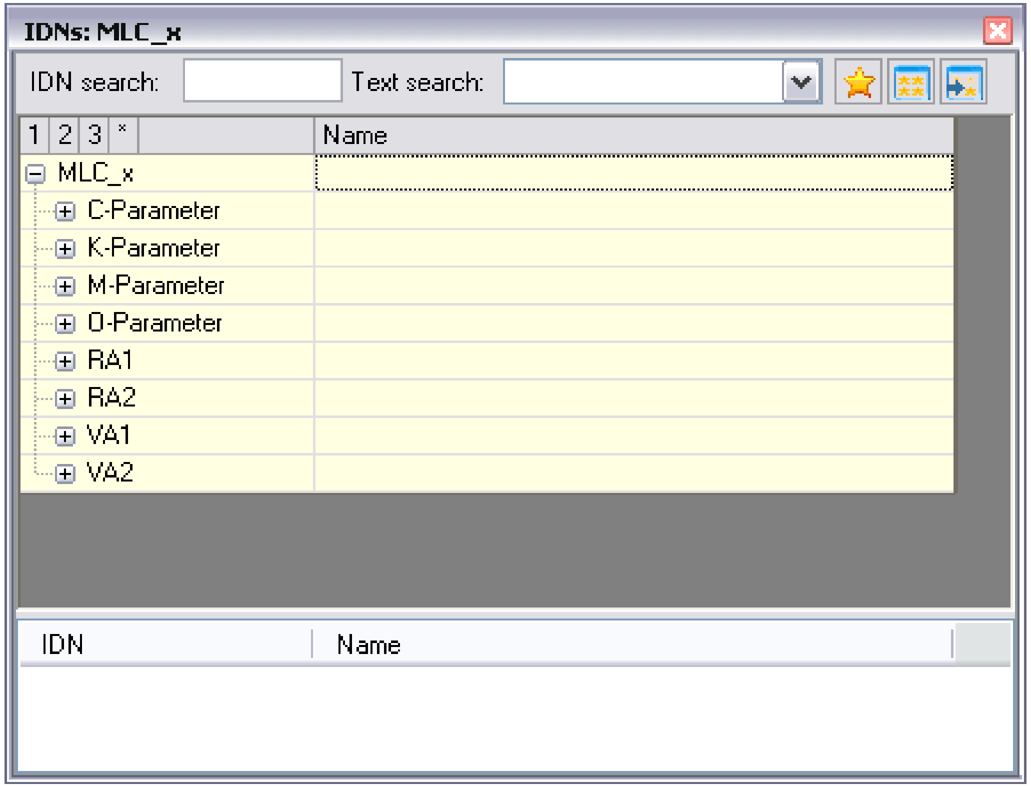
MLC Parameter System – Parameter Editor



With IDN search you get all the control and drive parameters in a tree structure and you can browse the parameter system



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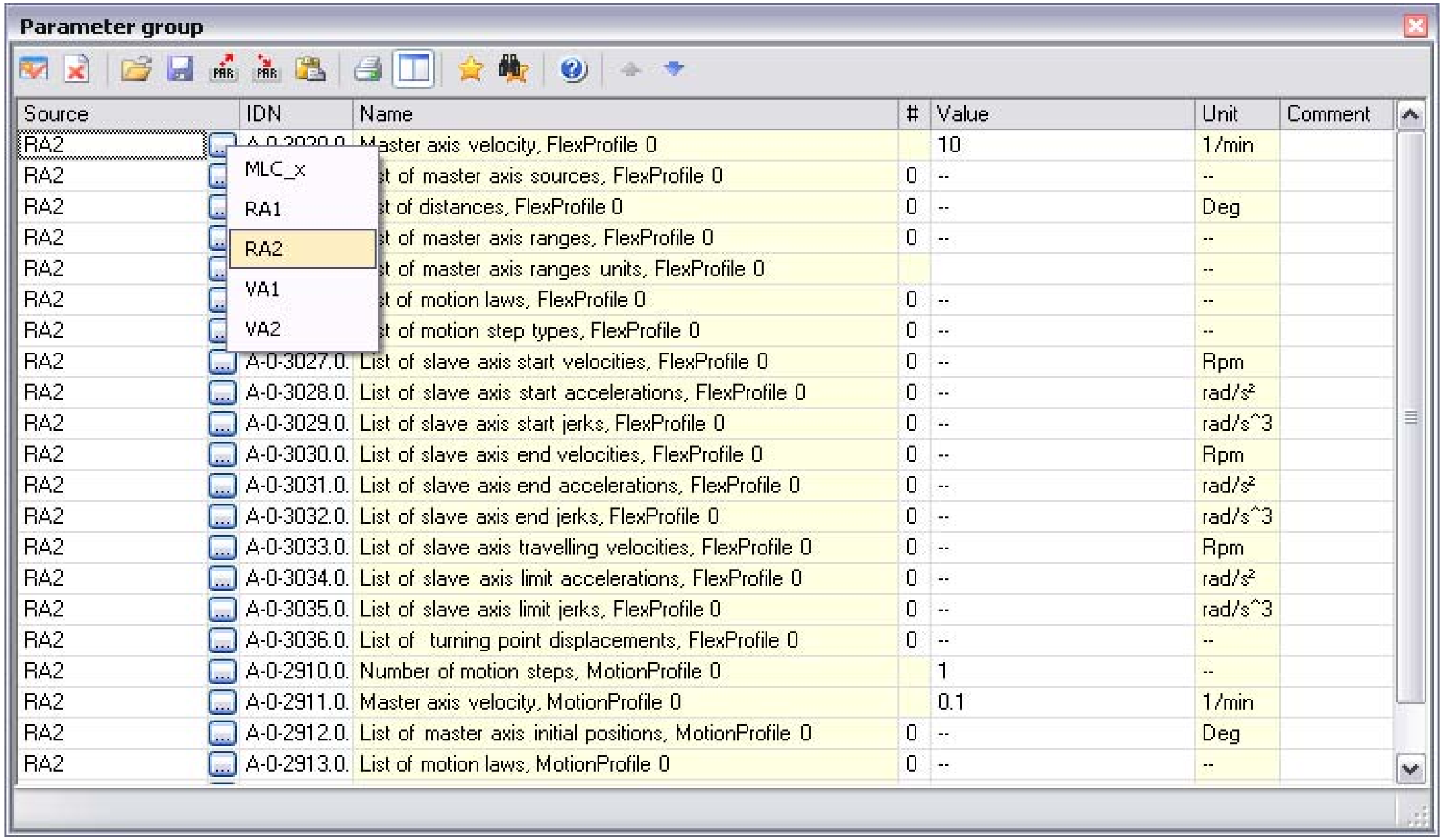
MLC Parameter System – Parameter Groups



Source	IDN	Name	#	Value	Unit	Comment
RA2	A-0-3020.0	Master axis velocity, FlexProfile 0		10	1/min	
RA2		st of master axis sources, FlexProfile 0	0	--	--	
RA2		st of distances, FlexProfile 0	0	--	Deg	
RA2		st of master axis ranges, FlexProfile 0	0	--	--	
RA2		st of master axis ranges units, FlexProfile 0			--	
RA2		st of motion laws, FlexProfile 0	0	--	--	
RA2		st of motion step types, FlexProfile 0	0	--	--	
RA2	A-0-3027.0.	List of slave axis start velocities, FlexProfile 0	0	--	Rpm	
RA2	A-0-3028.0.	List of slave axis start accelerations, FlexProfile 0	0	--	rad/s²	
RA2	A-0-3029.0.	List of slave axis start jerks, FlexProfile 0	0	--	rad/s^3	
RA2	A-0-3030.0.	List of slave axis end velocities, FlexProfile 0	0	--	Rpm	
RA2	A-0-3031.0.	List of slave axis end accelerations, FlexProfile 0	0	--	rad/s²	
RA2	A-0-3032.0.	List of slave axis end jerks, FlexProfile 0	0	--	rad/s^3	
RA2	A-0-3033.0.	List of slave axis travelling velocities, FlexProfile 0	0	--	Rpm	
RA2	A-0-3034.0.	List of slave axis limit accelerations, FlexProfile 0	0	--	rad/s²	
RA2	A-0-3035.0.	List of slave axis limit jerks, FlexProfile 0	0	--	rad/s^3	
RA2	A-0-3036.0.	List of turning point displacements, FlexProfile 0	0	--	--	
RA2	A-0-2910.0.	Number of motion steps, MotionProfile 0		1	--	
RA2	A-0-2911.0.	Master axis velocity, MotionProfile 0		0.1	1/min	
RA2	A-0-2912.0.	List of master axis initial positions, MotionProfile 0	0	--	Deg	
RA2	A-0-2913.0.	List of motion laws, MotionProfile 0	0	--	--	

Use Parameter groups to group all parameters relating to a specific MLC function, e.g. FlexProfile, MotionProfile, Phase synchronous operation etc.

The axis for which the parameters are displayed can easily be switched!
For only one entry as shown above ...

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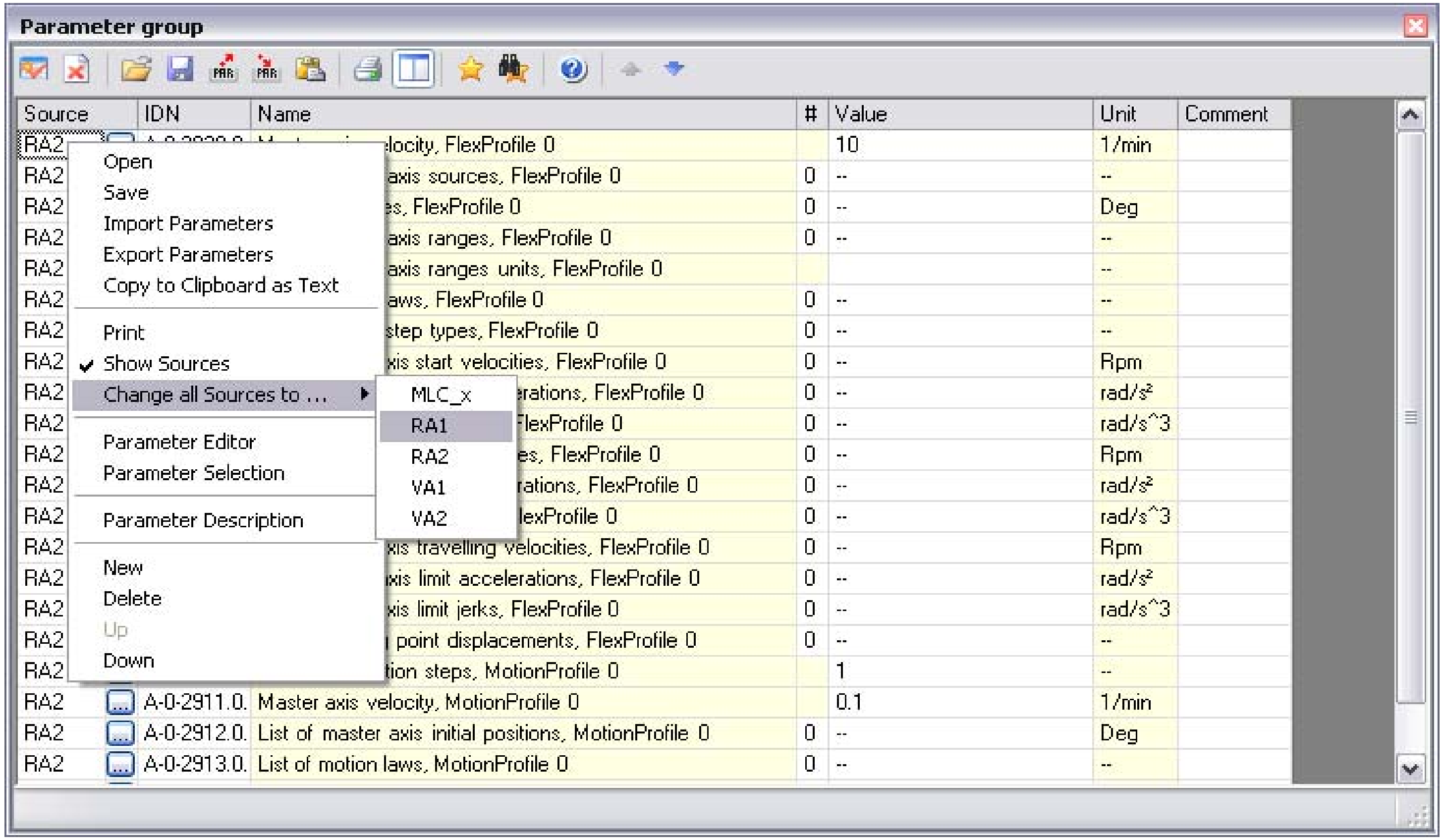
MLC Parameter System – Parameter Groups



... or for all group entries simultaneously!

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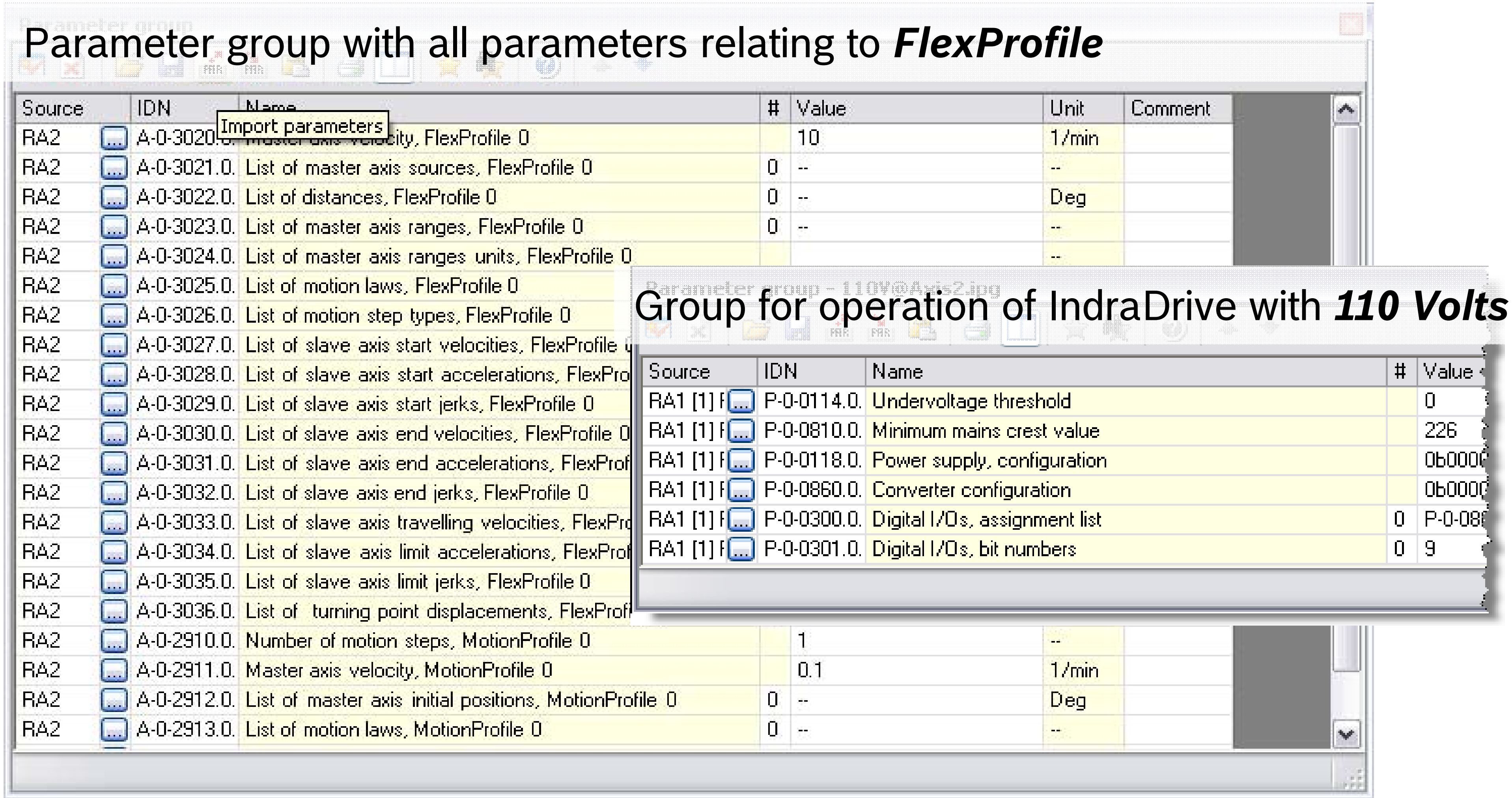
MLC Parameter System – Parameter Groups



Parameter group with all parameters relating to ***FlexProfile***

Import parameters

Source	IDN	Name	#	Value	Unit	Comment
RA2	A-0-3020.	...city, FlexProfile 0		10	1/min	
RA2	A-0-3021.0.	List of master axis sources, FlexProfile 0	0	--	--	
RA2	A-0-3022.0.	List of distances, FlexProfile 0	0	--	Deg	
RA2	A-0-3023.0.	List of master axis ranges, FlexProfile 0	0	--	--	
RA2	A-0-3024.0.	List of master axis ranges units, FlexProfile 0			--	
RA2	A-0-3025.0.	List of motion laws, FlexProfile 0				
RA2	A-0-3026.0.	List of motion step types, FlexProfile 0				
RA2	A-0-3027.0.	List of slave axis start velocities, FlexProfile 0				
RA2	A-0-3028.0.	List of slave axis start accelerations, FlexPro				
RA2	A-0-3029.0.	List of slave axis start jerks, FlexProfile 0				
RA2	A-0-3030.0.	List of slave axis end velocities, FlexProfile 0				
RA2	A-0-3031.0.	List of slave axis end accelerations, FlexProf				
RA2	A-0-3032.0.	List of slave axis end jerks, FlexProfile 0				
RA2	A-0-3033.0.	List of slave axis travelling velocities, FlexPro				
RA2	A-0-3034.0.	List of slave axis limit accelerations, FlexProf				
RA2	A-0-3035.0.	List of slave axis limit jerks, FlexProfile 0				
RA2	A-0-3036.0.	List of turning point displacements, FlexProf				
RA2	A-0-2910.0.	Number of motion steps, MotionProfile 0		1	--	
RA2	A-0-2911.0.	Master axis velocity, MotionProfile 0		0.1	1/min	
RA2	A-0-2912.0.	List of master axis initial positions, MotionProfile 0	0	--	Deg	
RA2	A-0-2913.0.	List of motion laws, MotionProfile 0	0	--	--	

Group for operation of IndraDrive with ***110 Volts***

Source	IDN	Name	#	Value
RA1 [1] F	P-0-0114.0.	Undervoltage threshold		0
RA1 [1] F	P-0-0810.0.	Minimum mains crest value		226
RA1 [1] F	P-0-0118.0.	Power supply, configuration		0b000(
RA1 [1] F	P-0-0860.0.	Converter configuration		0b000(
RA1 [1] F	P-0-0300.0.	Digital I/Os, assignment list	0	P-0-08
RA1 [1] F	P-0-0301.0.	Digital I/Os, bit numbers	0	9

Import/Export as well as Load/Save of parameter groups is supported!

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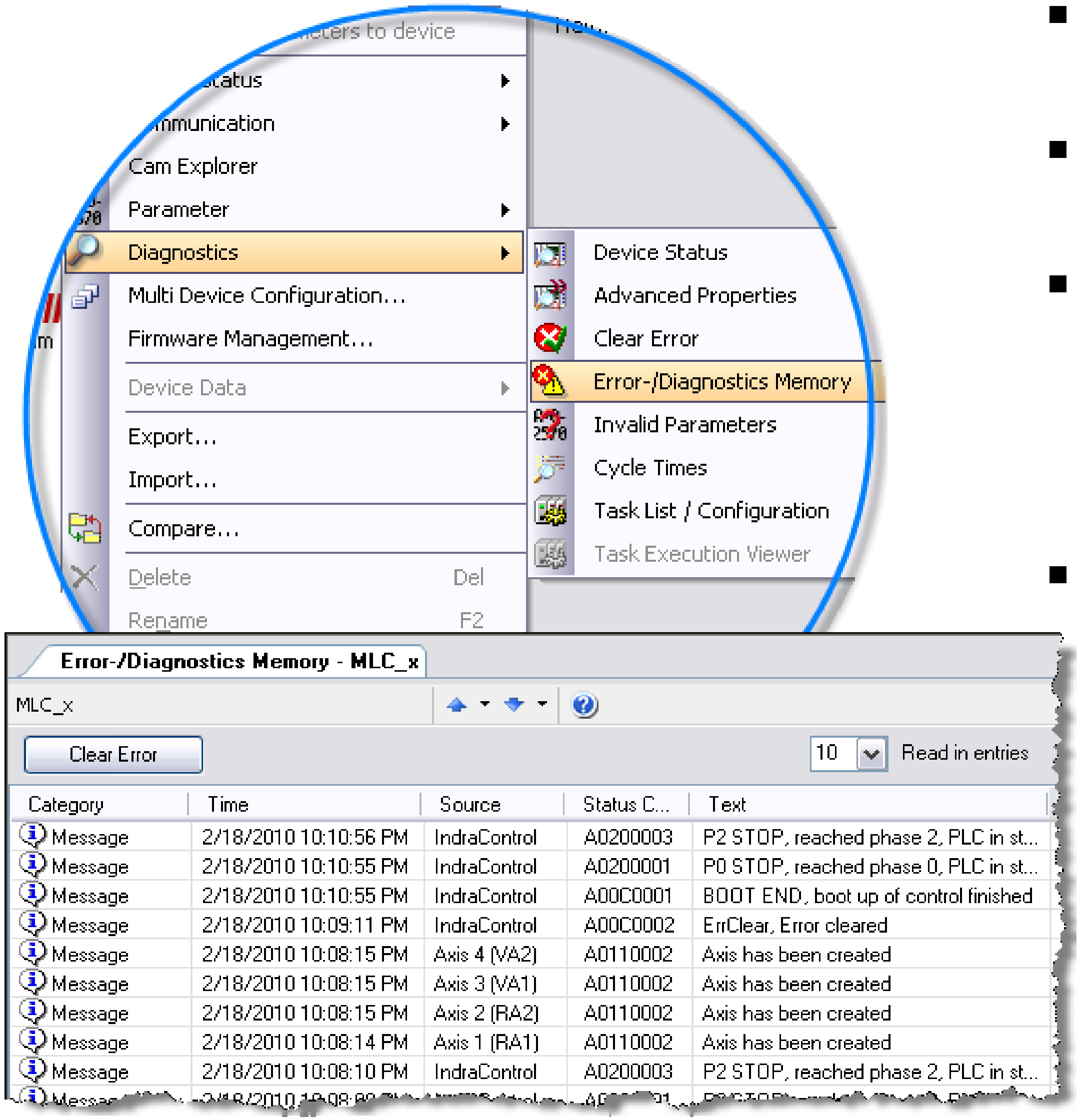
System Diagnostics



- Display of **device status**
- Function for **clearing errors**
- **Diagnosis / error memory** for the entire system (max. 1000 entries)
- List of invalid parameters

Display of **cycle times**
Time balance of the tasks for motion, logic and communication

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System Messages

- Message categories

Categories	Note
Message	Status information of message type
Error	Error reaches, the error reaction is triggered, the error is active
Error	Error passive, the error cause still exists, but the error was reset
Error	Error leaves, the error cause does not exist any longer **and** the error was reset
Warning	Warning reached, the warning reaction is triggered, the warning is active
Warning	Warning passive, the warning cause still exists, the warning was reset
Warning	Warning leaves, the warning cause does not exist any longer **and** the warning was reset

- Display of error/diagnostic memory

Error/Diagnostic Logbook - MLC...

Clear Error

max. 10 most recent entries

Category	Time	Source	Status Code	Text
Message	4/11/2007 5:23:01 PM	IndraControl	A0200003	P2 STOP, reached phase 2, PLC in stop
Error	4/11/2007 5:23:00 PM	IndraControl	F0160021	RTOS error (Real Time Operating System)
Message	4/11/2007 5:22:59 PM	IndraControl	A00C0001	BOOT END, boot up of control finished
Message	4/11/2007 5:22:59 PM	IndraControl	A0200001	P0 STOP, reached phase 0, PLC in stop
Message	4/11/2007 4:09:39 PM	IndraControl	A0200003	P2 STOP, reached phase 2, PLC in stop
Message	4/11/2007 4:09:39 PM	IndraControl	A0200001	P0 STOP, reached phase 0, PLC in stop
Message	4/11/2007 4:09:38 PM	IndraControl	A0200005	BB STOP, Motion ready, PLC in stop
Error	4/11/2007 4:09:38 PM	IndraControl	F0160021	RTOS error (Real Time Operating System)
Message	4/11/2007 4:09:37 PM	IndraControl	A0200004	P3 STOP, reached phase 3, PLC in stop
Message	4/11/2007 4:09:37 PM	IndraControl	A0200003	P2 STOP, reached phase 2, PLC in stop

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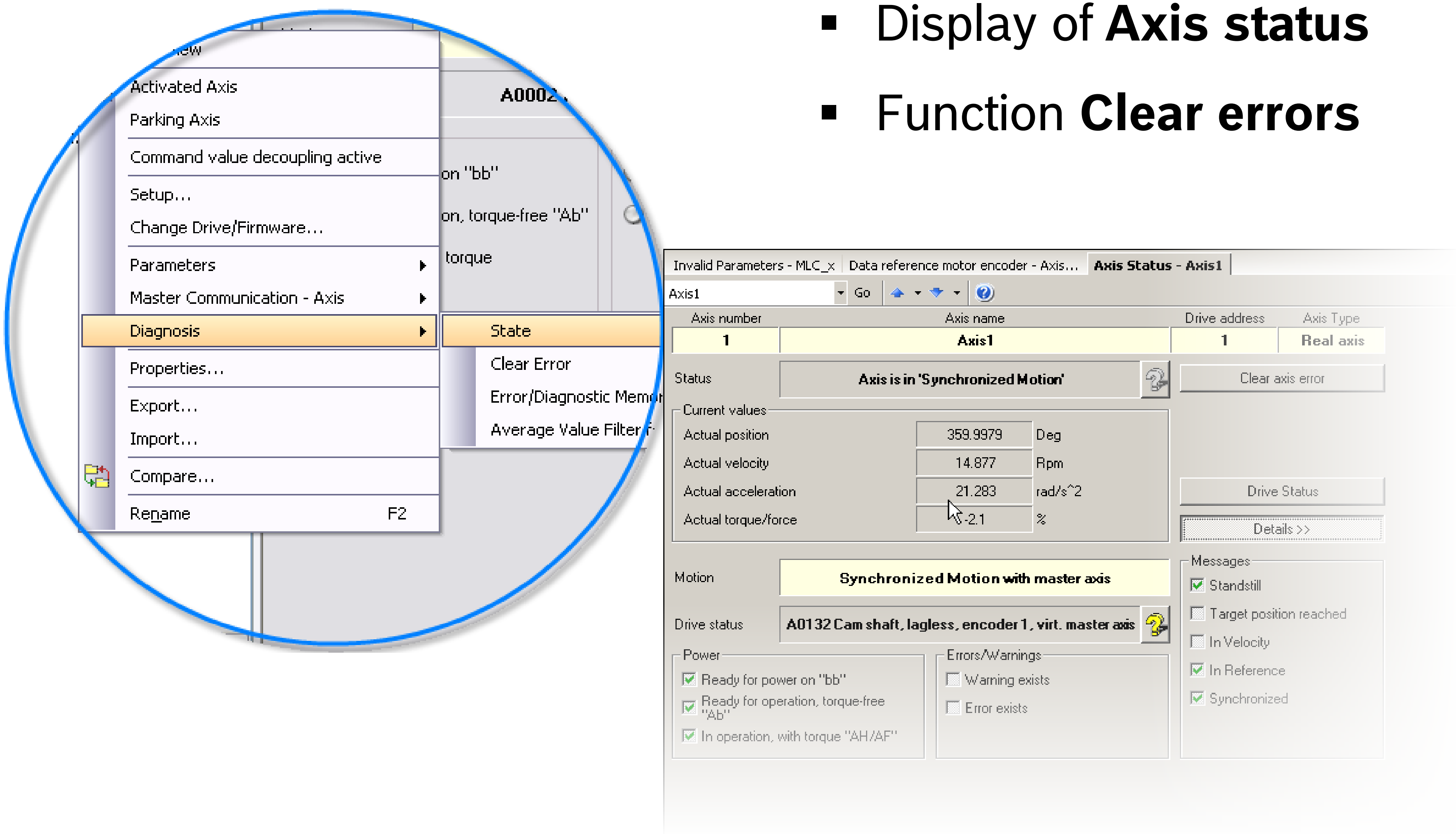
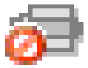
Axis Diagnosis

- Display of **Axis status**
- Function **Clear errors**



Activate/Deactivate

- The real axis is created in the IndraMotion MLC project
- The “deactivated" status can be assigned via the context menu and is then marked with  in the Project Explorer
- The A parameters can be accessed. The drive must not exist physically
- The drive is always in P0, even if the SERCOS ring is in P4 (applies for SERCOS 2)
- Virtual axes and encoder axes cannot be "deactivated"
- If the drive electronics is taken away from the encoder axis by "deactivating" a real axis, it is also marked as "deactivated"

Usage:

- During the commissioning, only certain drives should be moved. The other drives are configured and also programmed in the PLC program, but they do not yet exist physically.

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Parking Axis

- The real axis is created in the IndraMotion MLC project
- The “parking" property can be assigned via its context menu and is then marked with in the Project Explorer.
- The drive must exist physically. The motor does not need to exist. The drive command "Parking Axis" is used
- The A parameters as well as S/P parameters are accessible
- The drive follows the SERCOS phase. No error messages are generated by the drive
- Virtual axes and encoder axes cannot be "parked“

Usage:

- Prevent drive from generating an error with the transition to P4 without a connected motor

Command value decoupling

- Available for real axes with interpolation on the control
- Enabled via context menu and marked with
- The axis is subject to the regular PLCopen state machine in case of an active command value decoupling.
- This means that the MC_Power has to be executed before executing a motion command. This does not cause the drive to be switched to **AF**. The **AB** mode in the drive is not required for the execution of the FB.
- In case of an active command value decoupling, the position command value calculated by the control will still be written cyclically into the parameter S-0-0047. Since the drive is not in **AF**, the command values are not processed.
- The calculated command value of the axis is reflected by parameter A-0-0151 „Interpolated position of the control".

Usage:

- Test motion program without axis movements

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Command value decoupling

- The command value decoupling allows the execution of traveling commands without a motion carried out in the drive.
- The command value decoupling is activated via A-0-0024, „Axis state".
- The previous states
  - 0 - axis activated
  - 1 - axis parked
  - 2 - axis deactivated
- ... were extended by the following states
  - 4 - axis is activated without command value processing in the drive
  - 5 - axis is activated with command value calculation. Drive is parked
  - 6 - axis activated with command value calculation. Drive not connected

Axis status	Without command value decoupling	Command value decoupling activated
Activated		
Parked		
Deactivated		

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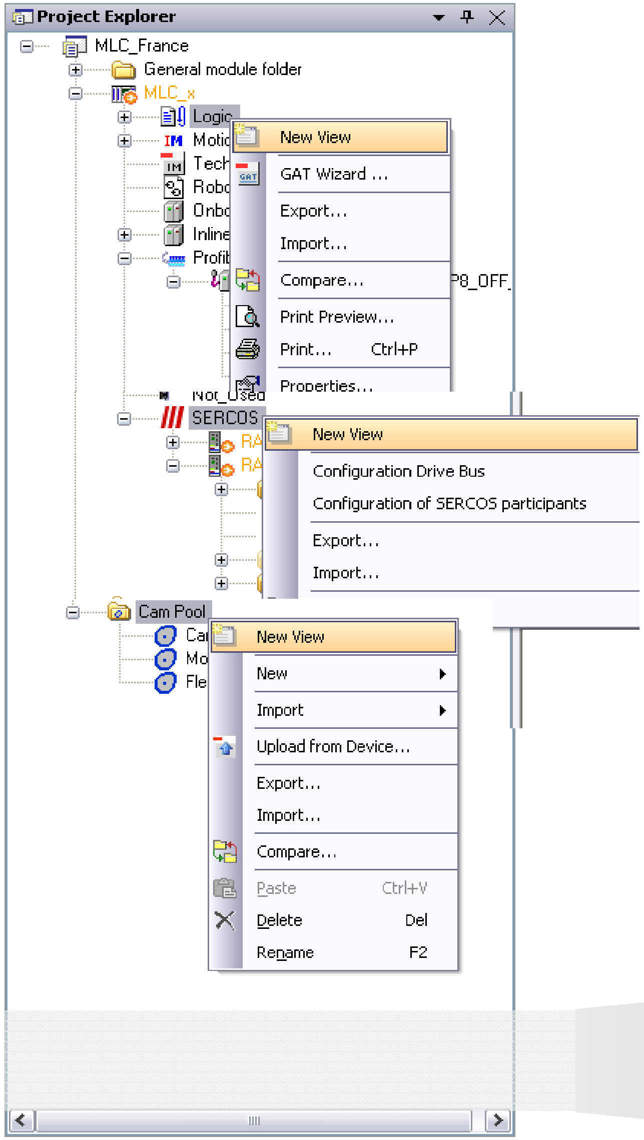
Tabbed views in Project Explorer



Create your own tabs for faster navigation in the Project Explorer!

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Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- **Motion Programming Basics**
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Working with IndraWorks

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IEC 61131 – What is it?

- The international standard IEC 61131 was created as a basis for uniform PLC programming, where modern software technology designs are considered
- The standard comprises 7 parts:
  - IEC 61131-1 General overview, definitions
  - IEC 61131-2 Hardware
  - IEC 61131-3 Programming languages
  - IEC 61131-4 User guidelines
  - IEC 61131-5 Messaging service specification
  - IEC 61131-7 Fuzzy logic
  - IEC 61131-8 Technical report
- MLC is a IEC 61131 compliant system, i.e. Motion programming is based on this standard
- All IEC programming languages are supported

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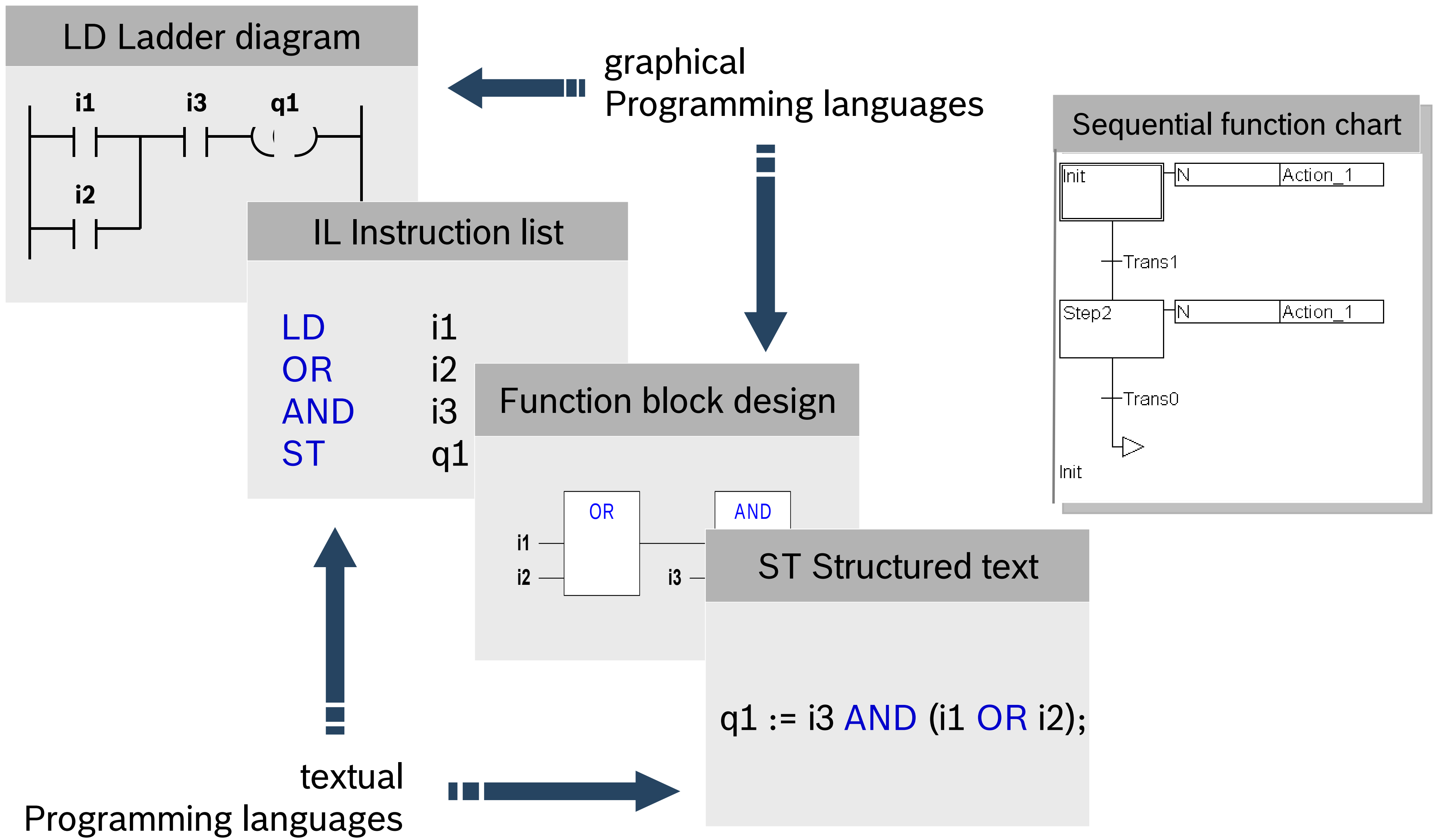
IEC 61131 – Programming languages



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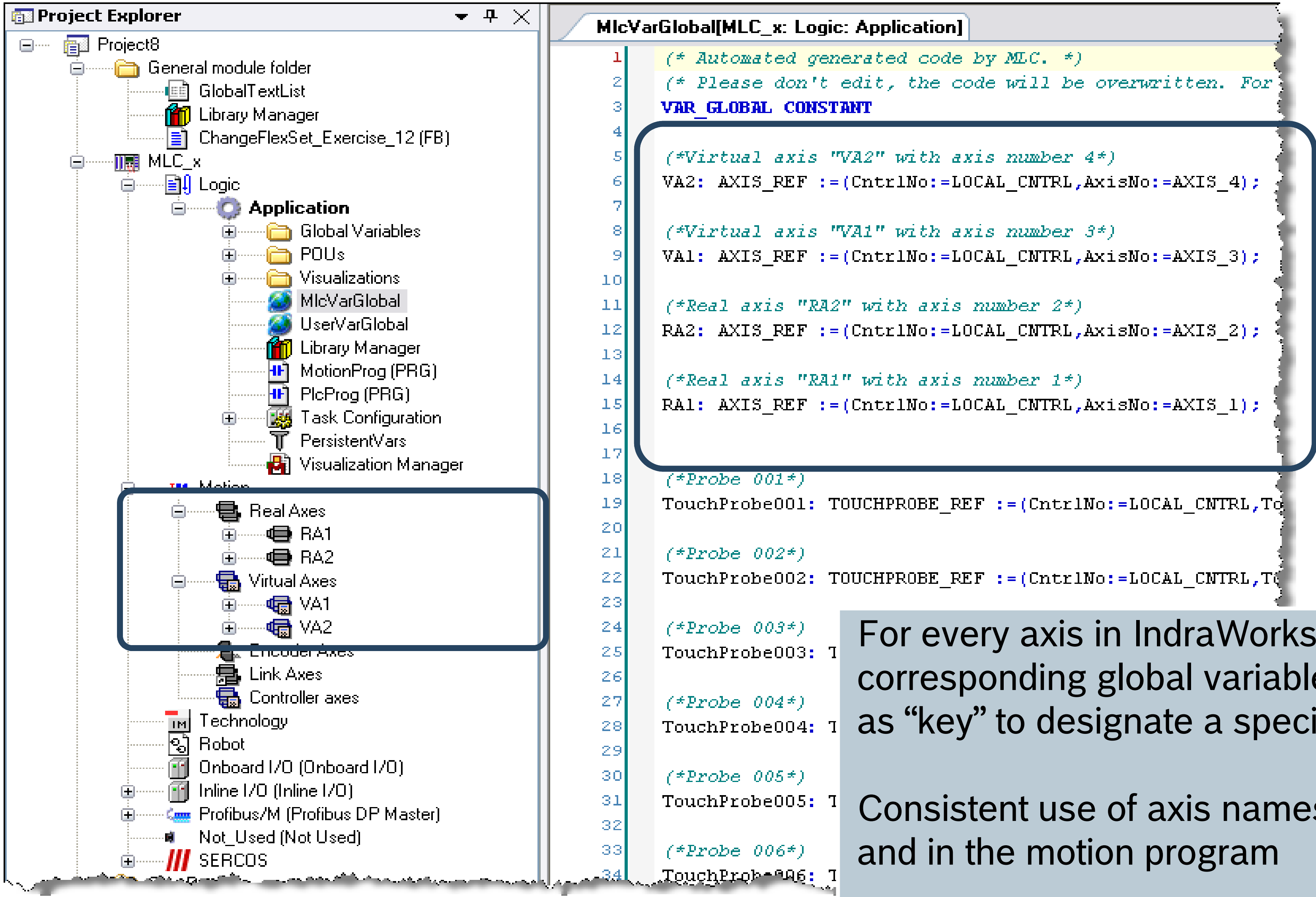
Relation IndraWorks ↔ IndraLogic



For every axis in IndraWorks there is a corresponding global variable which is used as “key” to designate a specific axis

Consistent use of axis names in IndraWorks and in the motion program

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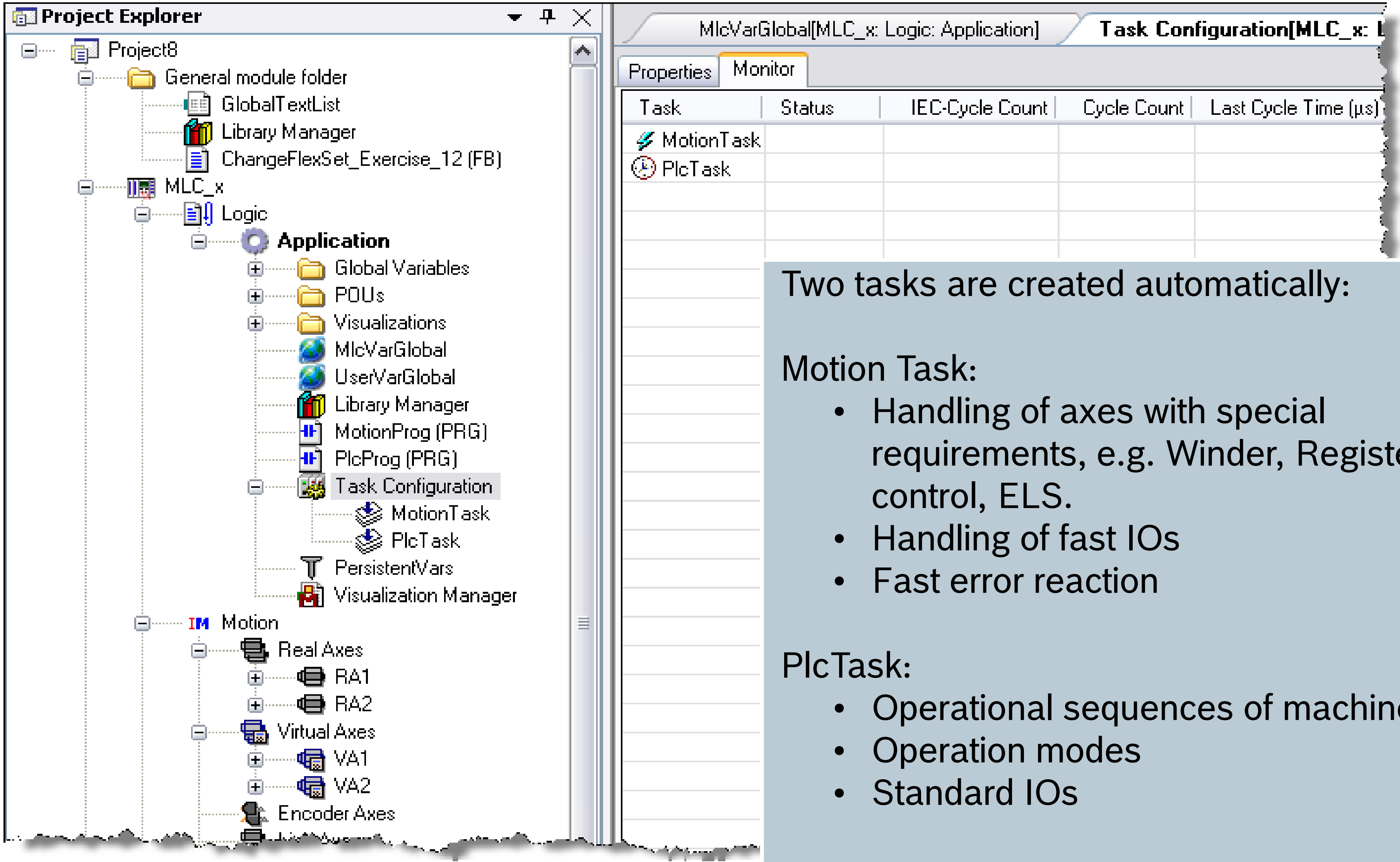
Relation IndraWorks ↔ IndraLogic



Two tasks are created automatically:

Motion Task:

- Handling of axes with special requirements, e.g. Winder, Register control, ELS.
- Handling of fast IOs
- Fast error reaction

PlcTask:

- Operational sequences of machine
- Operation modes
- Standard IOs

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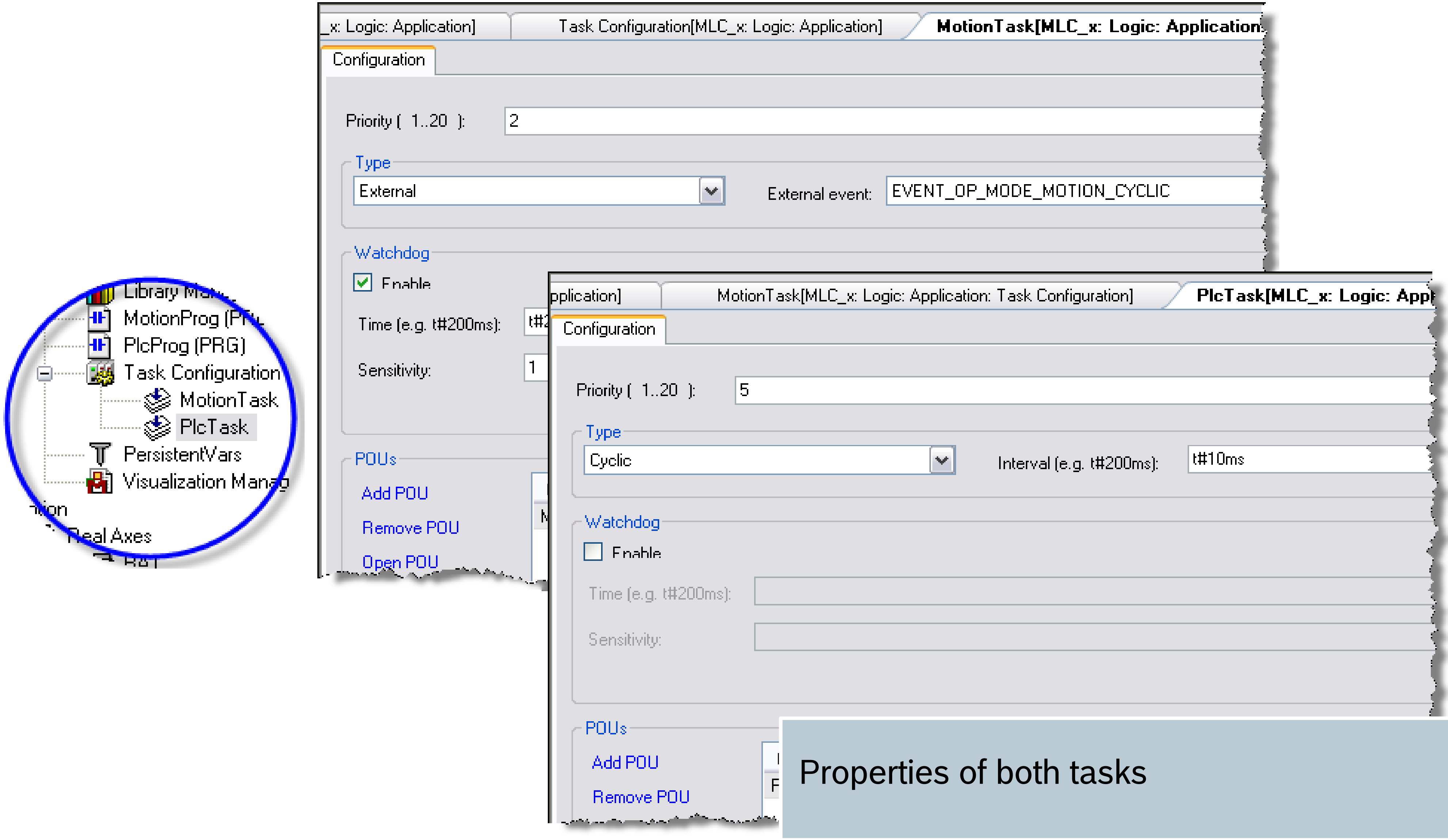
Relation IndraWorks ↔ IndraLogic



Properties of both tasks

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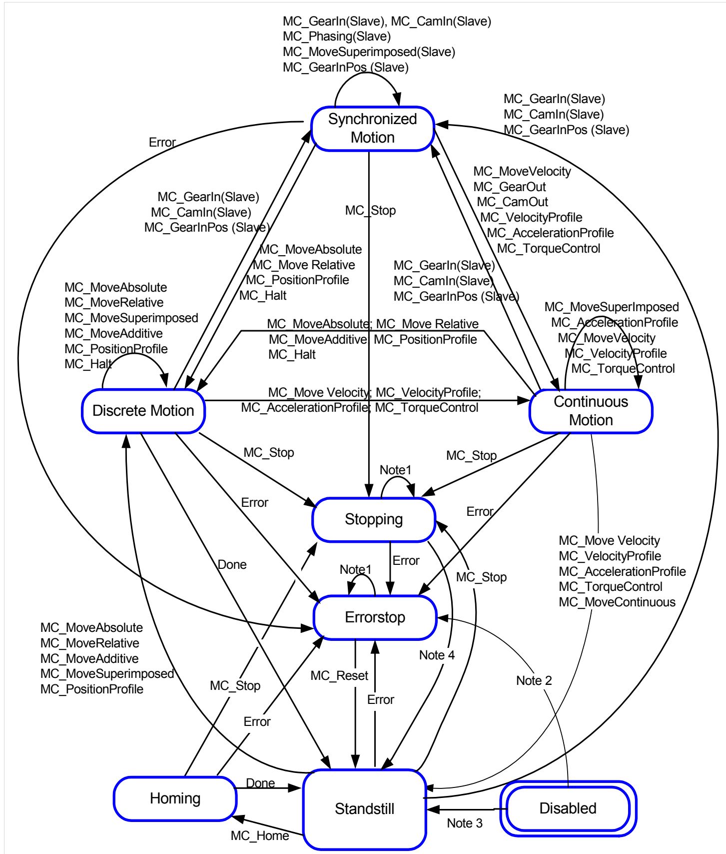
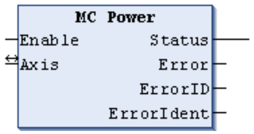
What is PLCopen?

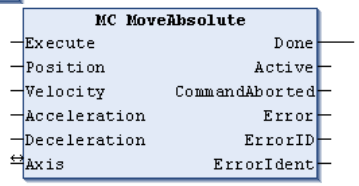
- Interest group, independent of controller manufacturers and users
- founded in 1992
- Ambition:
  promotion, development and use of IEC 61131-3 compatible software
- Advantages:
  cost reduction e. g. for
  - Training,
  - Development,
  - or Service

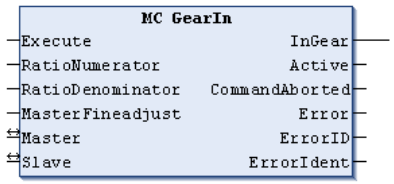
PLCopen and Motion Control

- First Library for standardized motion control functions specified in November 2001
- Main subjects:
  - State Machine
  - Function Blocks
  - Directives for use of Function Blocks









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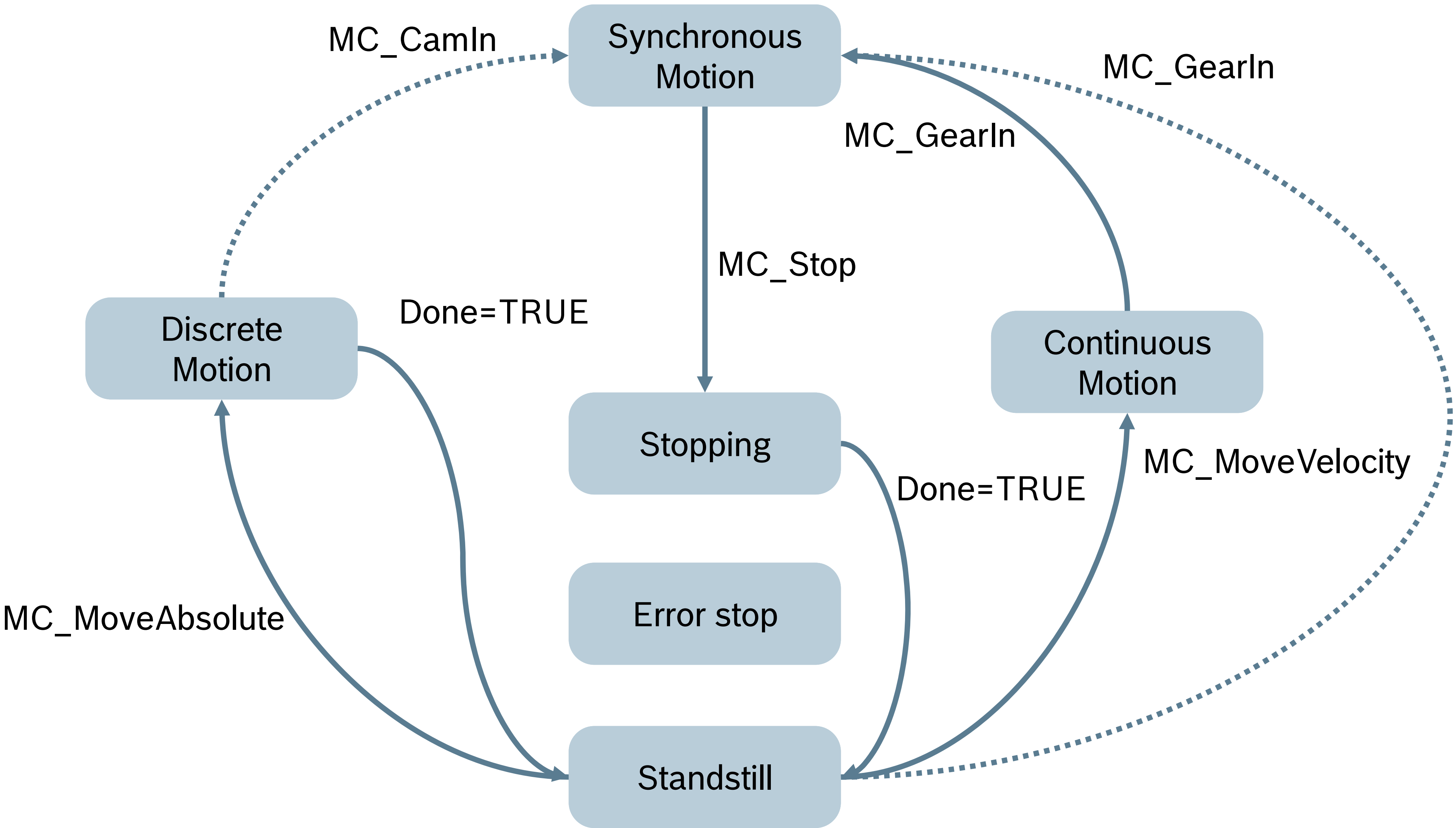
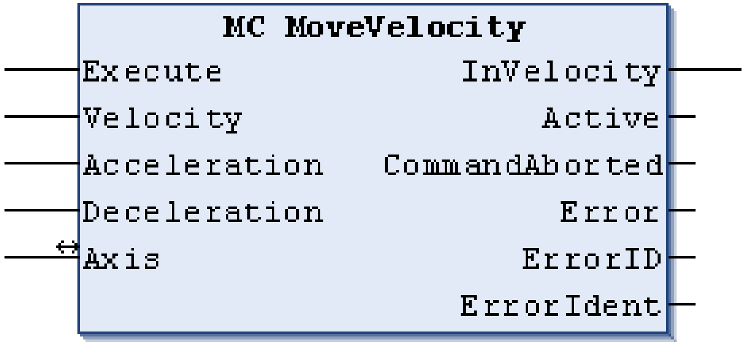
State Diagram – simplified



Example of a PLCopen Function block



	Name	Type	Comment [1]
VAR_IN_OUT	Axis	AXIS_REF	Contains information regarding the actual axis
VAR_INPUT	Execute	BOOL	Starts the motion if there is a rising edge
	Velocity	REAL	Maximum velocity value (does not necessarily have to be reached).
	Acceleration	REAL	Acceleration (always +).
	Deceleration	REAL	Deceleration (always +).
VAR_OUTPUT	InVelocity	BOOL	Velocity reached (for the first time)
	Active	BOOL	Processing of data runs after preprocessing is completed
	CommandAborted	BOOL	Command aborted by the following command.
	Error	BOOL	Indicates that an error has occurred in the FB instance.
	ErrorID	ERROR_CODE	Brief indication of the cause for error
	ErrorIdent	ERROR_STRUCT	Detailed information regarding the error

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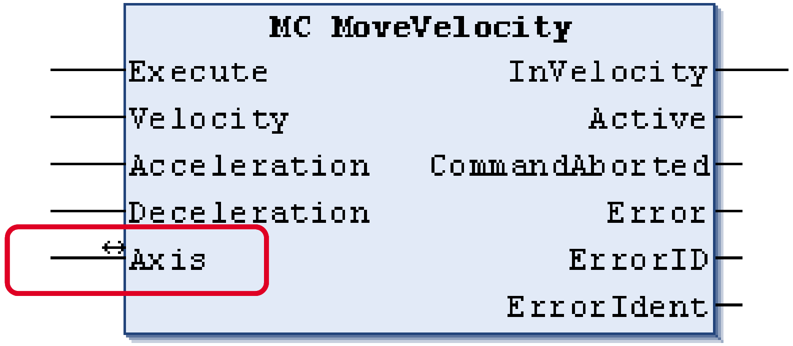
PLCopen – AXIS_REF

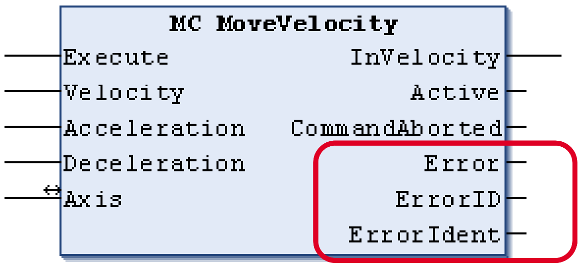


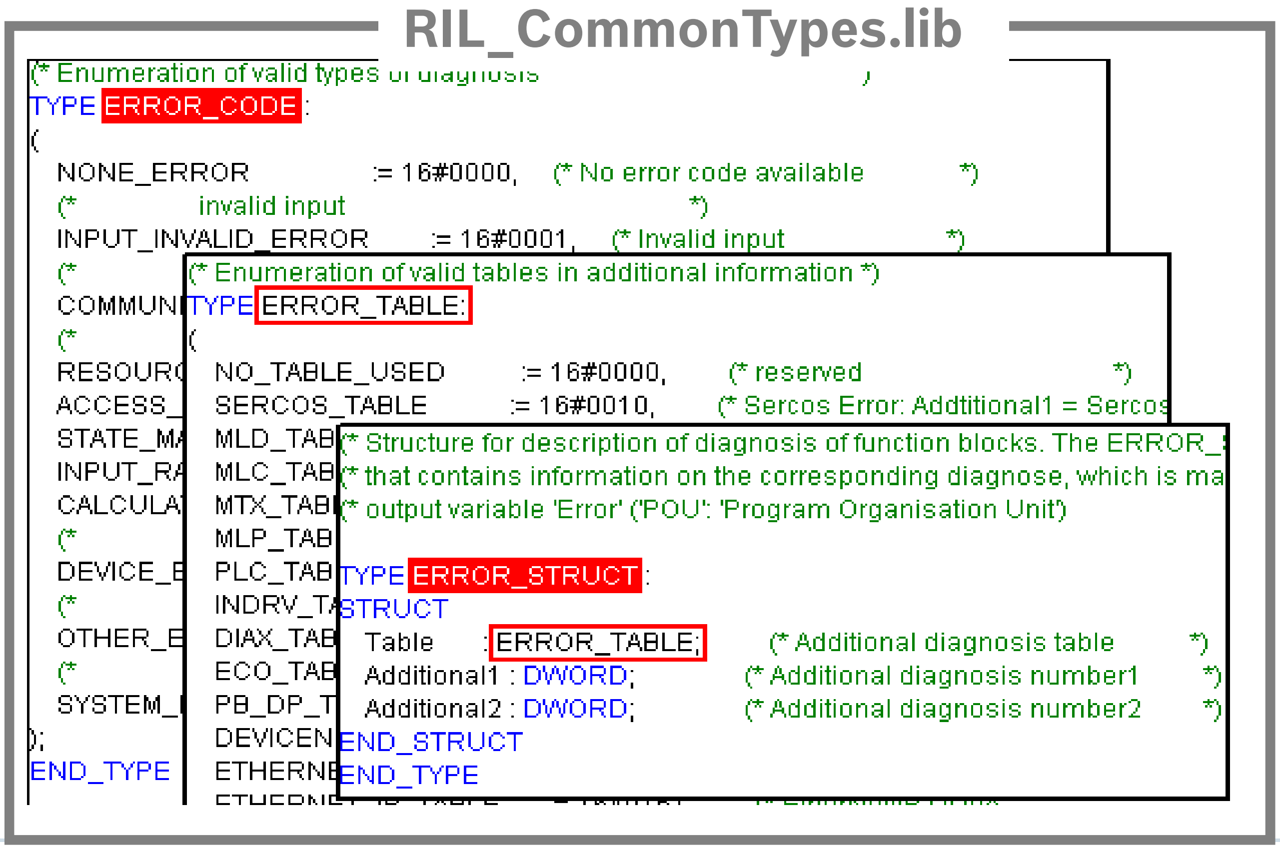

```
(* Structure of PLCopen element AXIS_REF. The AXIS_REF is a structure that contains *)
(* information on the corresponding object.                                          *)
(* It is used as a VAR_IN_OUT in all Motion Control Function Blocks.                  *)
TYPE AXIS_REF :
STRUCT
    CntrlNo: CONTROLS := LOCAL_CNTRL;  (* Control number init: local control          *)
    AxisNo:  OBJECTS  := NO_OBJECT;    (* Axis reference number, init: no object      *)
END_STRUCT
END_TYPE
```

- AXIS_REF
  - ... is used as a “key” to designate a specific axis
  - ... unique reference to an axis of the system
  - VAR_IN_OUT of all PLCopen Function blocks
  - Definition is manufacturer-specific

PLCopen Function blocks – Error handling

- Every Function block has three output parameters for the error handling:
  - **Error** Boolean output parameter. If it is TRUE an error has occurred during the processing of the function block
  - **ErrorID** Error number
  - **ErrorIdent** Rexroth-specific detailed error information





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Function Blocks according to PLCopen

Parameter access

- MC_ReadActualPosition
- MC_ReadStatus
- MC_ReadAxisError
- MB_Command
- MB_ReadLintParameter
- MB_ReadListParameter
- MB_ReadLrealParameter
- MB_ReadLrealListParameter
- MB_ReadParameter
- MB_ReadRealParameter
- MB_ReadSercosAttribute
- MB_ReadSercosDataStatus
- MB_ReadStringParameter
- MB_WriteLintParameter
- MB_WriteListParameter
- MB_WriteLrealParameter
- MB_WriteParameter
- MB_WriteRealParameter
- MB_WriteStringParameter

Single axis

- MC_Power
- MC_MoveAbsolute
- MC_MoveRelative
- MC_MoveAdditive
- MC_MoveVelocity
- MB_Home
- MC_Stop
- MC_Reset
- MC_TorqueControl
- MB_ChangeCamData
- ML_OpenCyclicPositionChannel
- ML_OpenCyclicAnalogChannel
- ML_OpenCyclicTorqueChannel
- ML_OpenCyclicVelocityChannel
- ML_WriteCyclicPosition
- ML_WriteCyclicAnalogl
- ML_WriteCyclicTorquel
- ML_WriteCyclicVelocity

Multi axis

- MC_CamIn
- MC_CamOut
- MC_GearIn
- MC_GearOut
- MB_GearInPos
- MB_Phasing
- MB_PhasingSlave
- MB_PhasingSlave

- MB_MotionProfile
- ML_FlexProfile
- MB_ChangeCamData
- MB_ChangeFlexEventSet
- MB_ChangeFlexProfileSet
- MB_ChangeProfileSet
- MB_ChangeProfileStep

Motion Profile / FlexProfile

Robot Control

- MC_AddAxisToGroup
- MC_RemAxisFromGroup
- MC_GroupReset
- MB_GroupReadListParameter
- MB_GroupReadParameter
- MB_GroupRealRealParameter
- MB_GroupReadSercosAttribute
- MB_GroupReadSercosDataStatus
- MB_GroupReadStringParameter
- MB_GroupWriteListParameter
- MB_GroupWriteParameter
- MB_GroupWriteRealParameter
- MB_GroupWriteStringParam.
- MB_IRDProgSelect
- MB_IRDProgStart
- MB_IRDProgStop

PLCopen motion control

MC_xxx specified according to PLCopen **MB_xxx** Bosch Rexroth standard **ML_xxx** MLC specific

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Standard and System-specific FBs, Portability

	Scope	Prefix	Example
Standard	**PLCopen V1.0**	**MC_**	**MC_**MoveVelocity
Standard	DCC Standard	**MB_**	**MB_**ReadParameter
System-specific	MLC	**ML_**	**ML_**PhasingSlave
System-specific	MLD	**MX_**	**MX_**SetDeviceMode
System-specific	VisualMotion	**MV_**	**MV_**Hysteresis
System-specific	Synax	**MS_**	**MS_**ReadSingleParameter
System-specific	MTX	**MT_**	**MT_**Hysteresis
System-specific	IndraLogic	**MP_**	**MP_**...
System-specific	Synax and VisualMotion	**MSV_**	**MSV_**ReadMaxValue

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Global data as interface to the motion kernel

- Access to axis information via global variables AxisData and DV_Axis
- AxisData is an array with elements of type ML_AXISDATA_SM
- This axis-specific structure holds „per axis" information like:
  - **Axis state information, e.g.**
    `AxisData[1].Axis_InVelocity`
  - **Actual values for Position, Velocity and Force/Torque, e.g.**
    `AxisData[1].rActualPosition_i`
    `AxisData[1].rActualVelocity_i`
  - **Container for customized command values and actual values**
    `AxisData[1].dwUserCmdDataA_q`
    `AxisData[1].dwUserActualDataC_i`
- DV_Axis is an array with elements of type ML_DirectVarAxis
  - **Access to A-Parameters**
    ```
    DV_Axis[1].A_0_0100        // actual position
    DV_Axis[1].A_0_0102        // actual velocity
    ```
- DV_Control is an array with elements of type ML_DirectVarControl
  - **Access to C-Parameters**
    ```
    DV_Control[LOCAL_CNTRL].C_0_0023 // MC system status
    ```


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Libraries

▪ ML_Base	structures, data types, AxisData etc.
▪ ML_PLCopen	PLCopen Function Blocks
▪ ML_TechInterface	AxisInterface and IMC Interface
▪ ML_*	Technology Function Blocks etc.
▪ RIL_Utilities	Data conversion, byte-swapping, etc.
▪ RIL_CommonTypes	Common data types and constants
▪ RIL_ProfibusDP	Fieldbus diagnosis and DP-V1 services
▪ RIL_ProfibusDP_02	Fieldbus diagnosis and DP-V1 services
▪ Standard	IEC standard functions and function blocks
▪ Syslibxxx	Miscellaneous system functions

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Programming Guidelines for IndraLogic

- A cross-system working group has elaborated IndraLogic programming guidelines
- Aim: unify all functions, function blocks etc. across systems
- Definitions and recommendations for different subjects:
  - Standardized headers for function blocks
  - History
  - Type identifiers
  - Variable identifiers
  - Constant identifiers
  - Interfaces of function blocks
  - Error handling
  - Names of libraries
  - Versioning of libraries

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IL Guidelines – Standardized header

```
(*##########################################################################################*)
(*##########################################################################################*)

(* General Header *)
(*------------------------------------------------------------------------------------------*)
(* Shortdescription         : This function block provide the communication between device xyz *)
(*                            via Profibus and the programmable logic controller *)
(* Version                  : 1.3 *)
(* Name                     : Max Mustermann *)
(* Date                     : 2004-02-02 *)
(* Company                  : Bosch Rexroth AG *)
(* Target                   : SYNAX200-MotionLogic; VisualMotion *)
(* Functional description : A communication with device xyz is only possible over a special *)
(*                            multiplex process. The function block decodes and encodes the *)
(*                            telegram and makes sure that ... *)
(* Handling specials        : Attend the following points: *)
(*                            - It's essential that the device xyz is connected with the *)
(*                              control unit. *)
(*                            - Connect the device to the Profibus and provide that the bus is *)
(*                              running without any error. *)
(*------------------------------------------------------------------------------------------*)

(* Additional Header *)
(*------------------------------------------------------------------------------------------*)
(* Customer                 : Koenig & Bauer AG *)
(* Machine                  : FA0815STX *)
(*------------------------------------------------------------------------------------------*)

(*##########################################################################################*)
(*##########################################################################################*)
PROGRAM ZZMainProgram
VAR
```


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IL Guidelines – Prefix for type & variable identifiers

Data type	Prefix	Instance example	Type example
Function block	fb	fbJogAxisX1	MT_Jogging
Structure	st	stDeviceCommunication	MX_COM_DATA
Array	ar	arStateControlUnit	MV_STATE_INFO
Enumerator	en	enDeviceDiagnosis	ML_DIAG_DATA

Data type	Prefix	Example	Memory allocation	Data type designation	Data type description
BOOL	b	bVar	1 Bit	Boole	Bit oriented boolean format
BYTE	by	byVar	8 Bit	Byte	Bit oriented Byte format
WORD	w	wVar	16 Bit	Word	Bit oriented format with simple word length
DWORD	dw	dwVar	32 Bit	Double Word	Bit oriented format with double word length
LWORD	lw	lwVar	64 Bit	Long Word	Bit oriented format with fourfold word length
SINT	si	siVar	8 Bit	Short Integer	integer signed format with shortened length
INT	i	iVar	16 Bit	Integer	integer signed format with simple length
DINT	di	diVar	32 Bit	Double Integer	integer signed format with double length
LINT	li	liVar	64 Bit	Long Integer	integer signed format with fourfold length
USINT	usi	usiVar	8 Bit	UnsignedShort Integer	integer unsigned format with shortened length
UINT	ui	uiVar	16 Bit	Unsigned Integer	integer unsigned format with simple length
UDINT	udi	udiVar	32 Bit	Unsigned Double Integer	integer unsigned format with double length
ULINT	uli	uliVar	64 Bit	Unsigned Long Integer	integer unsigned format with fourfold length
REAL	r	rVar	32 Bit	Real	real number with simple length

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IL Guidelines – Prefix for type & variable identifiers

LREAL	lr	lrVar	64 Bit	Long Real	real number with double length
STRING	str	strVar	8 Bit per character	String	string of 1-255 characters (ANSI code possible)
WSTRING	wstr	wstrVar	16 Bit per character	Wide String	string of 1-65535 characters (UNI code possible)
TIME	t	tVar	32 Bit	Time	Time format
DATE	d	dVar	32 Bit	Date	Date format
TIME_OF_DAY	tod	todVar	32 Bit	Time Of Day	Time of day format
DATE_AND_TIME	dat	datVar	32 Bit	Date And Time	Date and time format
POINTER TO ???	p???	p???Var	32 Bit	Pointer To ???	Pointer / address of a variable with special data type
POINTER TO DWORD	pdw	pdwVar	32 Bit	Pointer To Double Word	Example: Pointer to a double word variable

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Agenda

- Introduction to IndraMotion MLC
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- MLC Diagnosis system
- Data backup and restore
- Task System
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- Additional sources of information

Working with IndraWorks

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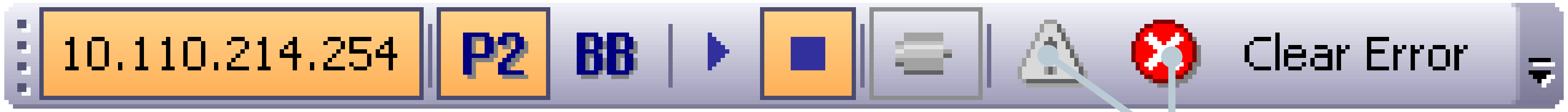
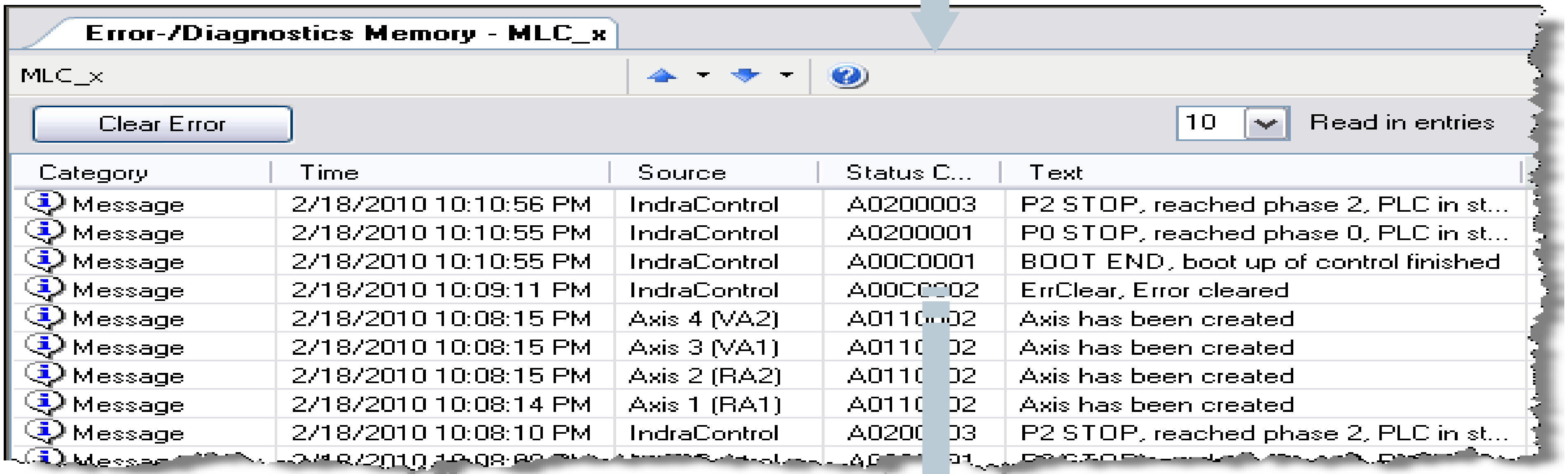
Display of pending errors, warnings, and infos



Colored icon
indicates pending error or warning

By pushing the button the internal **diagnosis / error memory** is displayed.
It holds diagnostic information of the overall system (max. 1000 entries)



Category	Time	Source	Status C...	Text
Message	2/18/2010 10:10:56 PM	IndraControl	A0200003	P2 STOP, reached phase 2, PLC in st...
Message	2/18/2010 10:10:55 PM	IndraControl	A0200001	P0 STOP, reached phase 0, PLC in st...
Message	2/18/2010 10:10:55 PM	IndraControl	A00C0001	BOOT END, boot up of control finished
Message	2/18/2010 10:09:11 PM	IndraControl	A00C[illegible]02	ErrClear, Error cleared
Message	2/18/2010 10:08:15 PM	Axis 4 (VA2)	A0110[illegible]02	Axis has been created
Message	2/18/2010 10:08:15 PM	Axis 3 (VA1)	A0110[illegible]02	Axis has been created
Message	2/18/2010 10:08:15 PM	Axis 2 (RA2)	A0110[illegible]02	Axis has been created
Message	2/18/2010 10:08:14 PM	Axis 1 (RA1)	A0110[illegible]02	Axis has been created
Message	2/18/2010 10:08:10 PM	IndraControl	A0200[illegible]03	P2 STOP, reached phase 2, PLC in st...

After a double click on an entry the Online Help opens and you get more information on the occurred error or warning!

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Diagnosis concept

- MLC has a consistent diagnostic system
- Every MLC diagnosis info has a 8-digit diagnosis code as a unique identifier
- The 8-digit diagnosis code is displayed as well as a plain text message in German or English language (for MLC)
- The IndraDrive has some more languages available …
- Diagnosis infos are displayed
  - in IndraWorks (Device status, Axis-/Drive status)
  - in the MLC-internal diagnosis memory (in chronological order)
  - on the display of IndraControl L25/45/65
  - Error output of PLCopen function blocks, technology function blocks etc.

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Diagnosis codes

8-digit diagnosis code

F 5 0B 0002

plain text message

SERCOS ring interrupted

4-digit code number

Diagnosis group (PLC, drive, ...)

Severity of diagnosis

Diagnosis identifier (F, E, C, A)

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Diagnosis codes – Identifier & Severity

Diagnostic identifier

- Currently, 4 different categories are supported:
  - F - Error
  - C - Command error
  - E - Warning
  - A - Message
- Priority is defined as follows: F > C > E > A

Severity of the diagnostic

- Currently ten degrees of severity are supported for the MLC:
- F9 > ... >F1 >F0 ; C9 > ... >C1 >C0 ; E0 ; A0.

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Diagnosis codes – Severity

Severity	Description /Designation	Error reaction
0	Non-fatal error	Logbook entry is generated, message is shown on the display, no error reaction.
1	Axis group error	Logbook entry is generated, message is shown on the display, If an axis belonging to a group activates an error of severity F2, Disengage the axis, then this axis is currently lost to the group and the group reacts with an F1 error, disengaging **all** axes in the group.
2	Axis error	A logbook entry is generated, a message is shown on the display, Axis (or drive) is disengaged as best as possible, all other axes are unaffected by this.
3/4	Reserved	-
5	Controller error	A logbook entry is generated, a message is shown on the display, **All** axes are disengaged as best as possible.
6	Reserved	-
7	Reserved	-
8	Fatal controller error	A logbook entry is generated, a message is shown on the display, **All** axes are disengaged as best as possible.
9	Fatal system error Exception, undefined system status	A logbook entry is generated, a message is shown on the display, Firmware no longer working, request for FatalSystemErrorHandler (), no error reaction to the drive, torque disable.

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Diagnosis codes – Group

Diagnosis group

- ... indicates which component has caused an error etc.

Group	Cause / Error Table
00	IndraDrive drive/ INDRV_TABLE (see IndraDrive Drive Errors, IndraDrive Drive Warning)
01...A9	MLC firmware, e.g. a virtual axis error Virtual Axis, Error Message
2D	SERCOS error/ SERCOS_TABLE SERCOS error message
2E	Generic axis / generic drive Generic Axis/SercosDrive, Error Message
AE	User program (to be provided by the user)

- Drive errors are translated as follows:
  - IndraDrive error ***F4034 Emergency-Stop***
  - Expansion to 8 digits → ***F2004034***
  - Severity 2, no other axis is affected
  - Diagnosis group 00 → IndraDrive
  - **S-0-0390** F4034
  - **A-0-0020** “Drive Error”
  - **A-0-0023** F2004034
  - Lx5 display: “**Axis1 F2004034 Drive error**“

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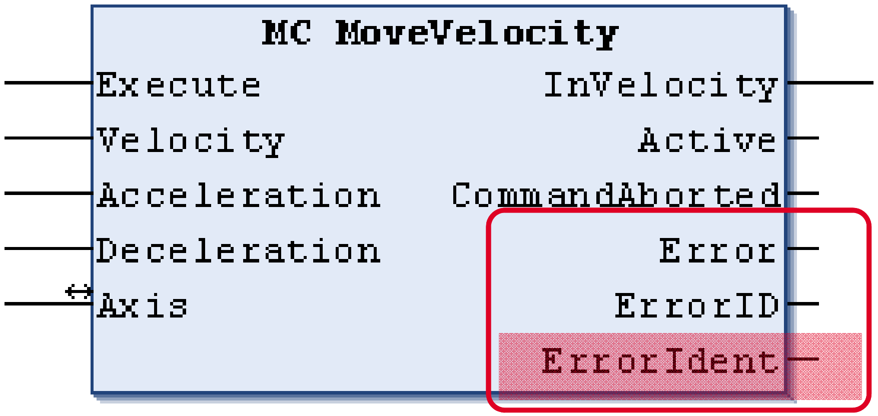
Error information in the motion program




```
(* Structure to implement the diagnostics of the function block *)
TYPE ERROR_STRUCT :
STRUCT
    Table       : ERROR_TABLE;  (* Additional diagnosis table          *)
    Additional1 : DWORD;        (* Contains the error number diagnostics *)
                                (* corresponding to the table.         *)
    Additional2 : DWORD;        (* eventually supplements to Additional1 *)
END_STRUCT
END_TYPE
```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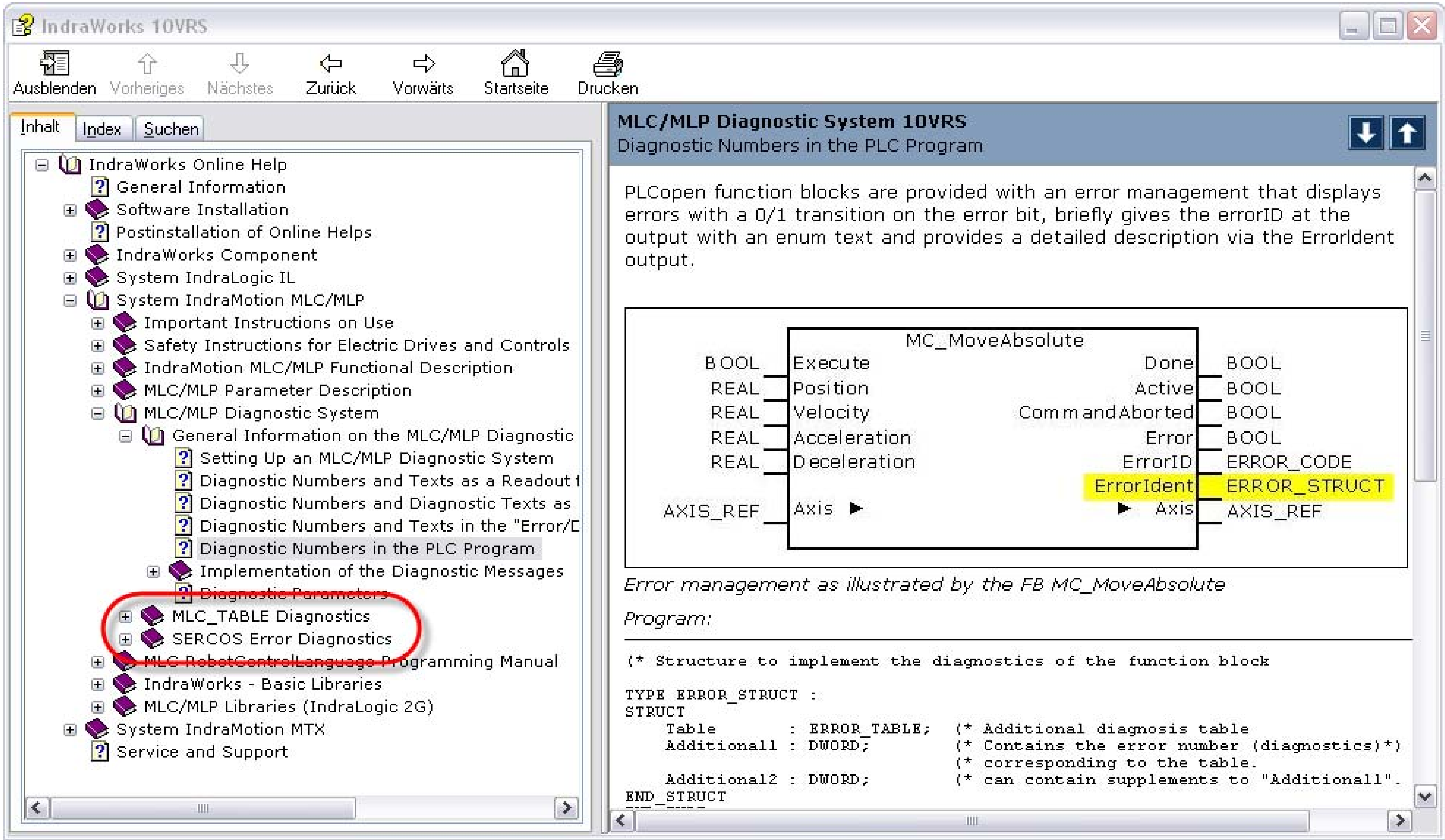
Error information in the motion program

```
(* Enumeration of valid tables in additional information *)
TYPE ERROR_TABLE:
(
  NO_TABLE_USED      := 16#0000, (* reserved     *)
  SERCOS_TABLE       := 16#0010, (* SERCOS : Addtitional1 = SERCOS code  *)
  MLD_TABLE          := 16#0020, (* Drive-based Motion – Logic   *)
  MLC_TABLE          := 16#0030, (* Controller-based Motion Logic         *)
  MTX_TABLE          := 16#0040, (* CNC  *)
  MLP_TABLE          := 16#0050, (* PC-based Motion Logic          *)
  PLC_TABLE          := 16#0060, (* PLC  *)
  INDRV_TABLE        := 16#0070, (* IndraDrive   *)
  …      (* DIAX, EcoDrive, Profibus, DeviceNet, *)
         (* Ethernet, EtherNet/IP, Interbus,     *)
         (* function related, CAN        *)
  INLINEIO_TABLE     := 16#0190, (* Inline IO bus        *)
  USER1_TABLE        := 16#1000, (* free User Table      *)
                                 (* ...  *)
  USER10_TABLE       := 16#1009  (* free User Table      *)
);
END_TYPE
```

Error information in the motion program



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Working with IndraWorks

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Archiving IndraWorks projects

For which purpose you need an archived IndraWorks project?

- Service
  For diagnosis purposes, drive replacement, and other service tasks a valid IndraWorks project is required
- Support
  A qualified support can be done based on the complete IndraWorks project. For this purpose the exchange of a complete IndraWorks archive is very useful
- Serial machines
  The IndraWorks project is required to duplicate machines

Archiving IndraWorks projects

Engineering PC

IndraWorks project
- Control, axis & IO configuration
- Offline parameters
- IndraLogic project (sources)
- CAMs
- FlexProfiles
- …

MLC

CF

System partition
- Firmware

OEM partition
- PLC boot project
- IndraWorks project

User partition
- … (other)

unpartioned area
- C parameters
- A parameters
- K, M, O parameters
- remanent PLC data

IndraDrive

Internal Flash / MMC
- Firmware
- S parameters
- P parameters

IndraWorks archive (on HDD and/or on CF of MLC)

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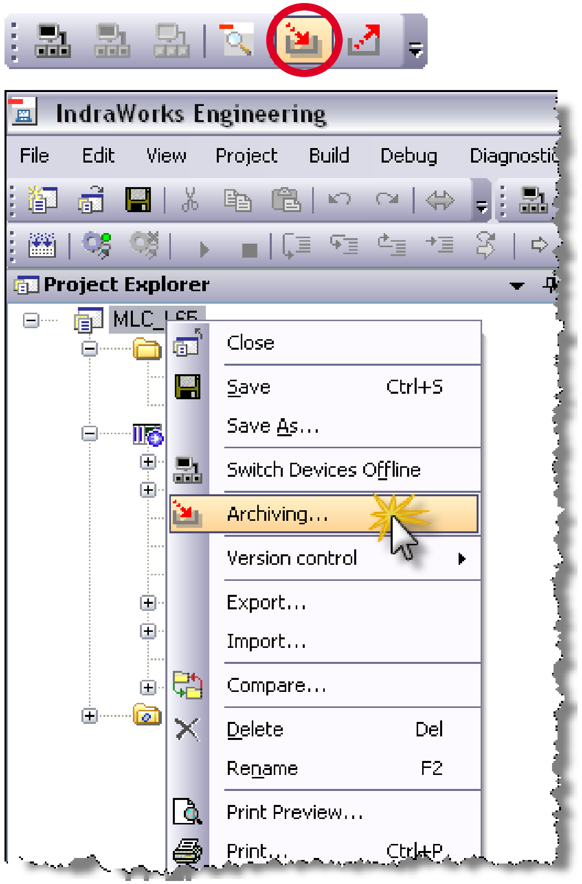
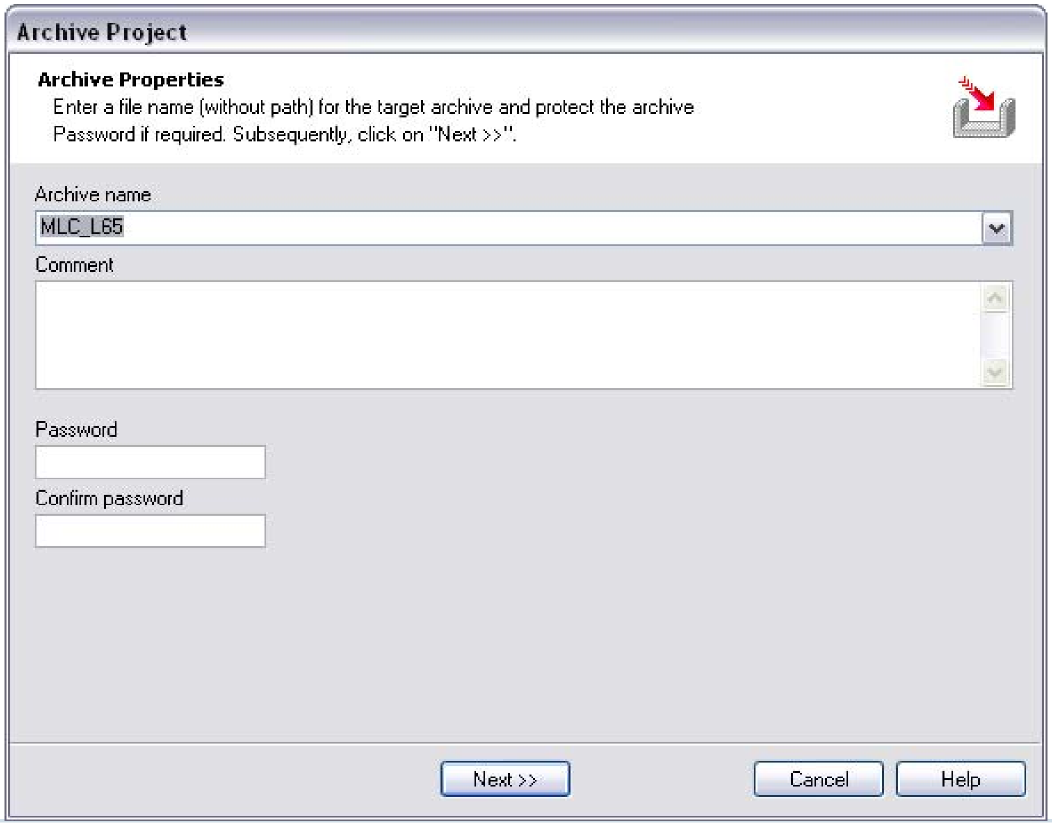
Archiving IndraWorks projects



- Invoke the Archiving function via menu or toolbar
- ... IndraWorks is in online mode!
- ☞ For later Offline Parameterization update the offline parameters!



- Enter an archive name
- Optionally you can protect the archive by a password

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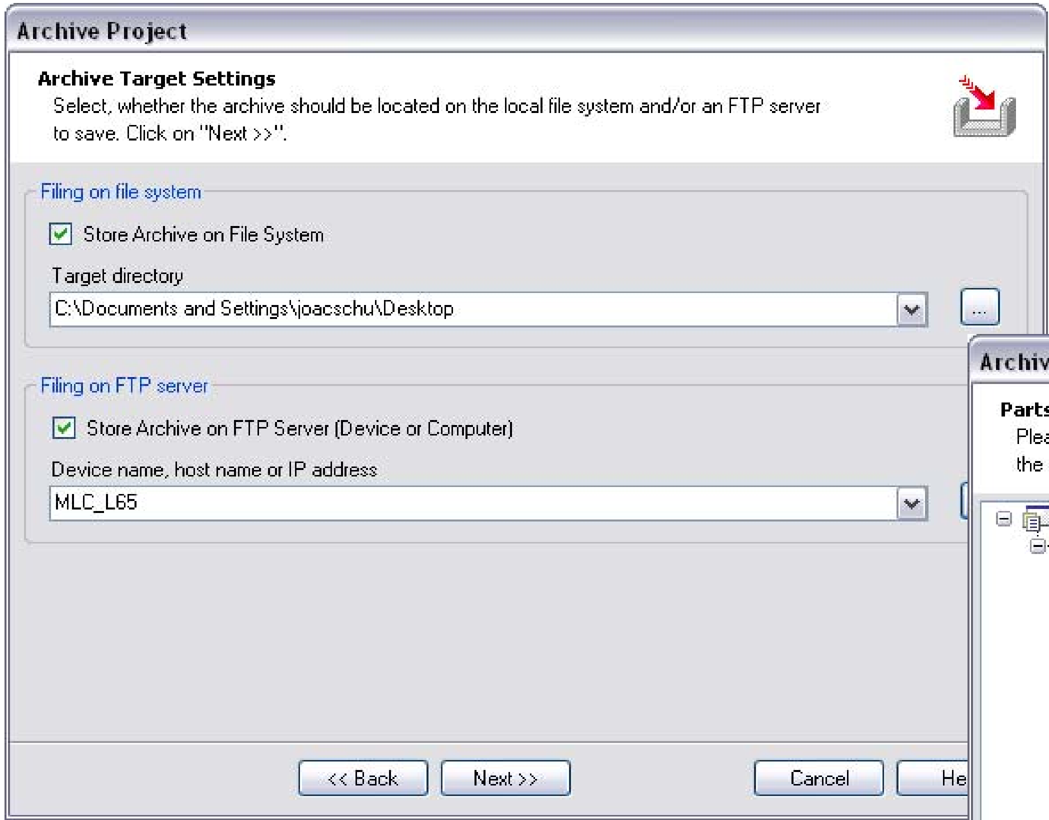
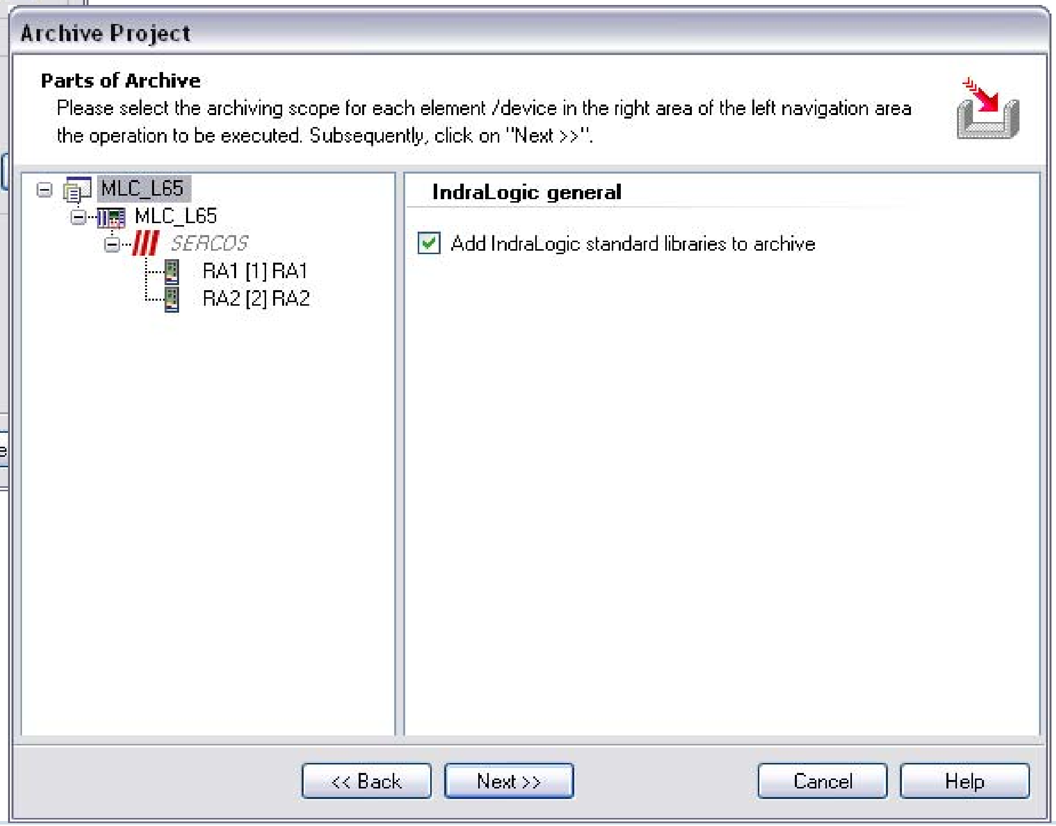
Archiving IndraWorks projects



- Specify the target directory and/or
- the device name or IP address of the MLC
- (storage on a FTP server is also supported)



- Add IndraLogic libraries to archive

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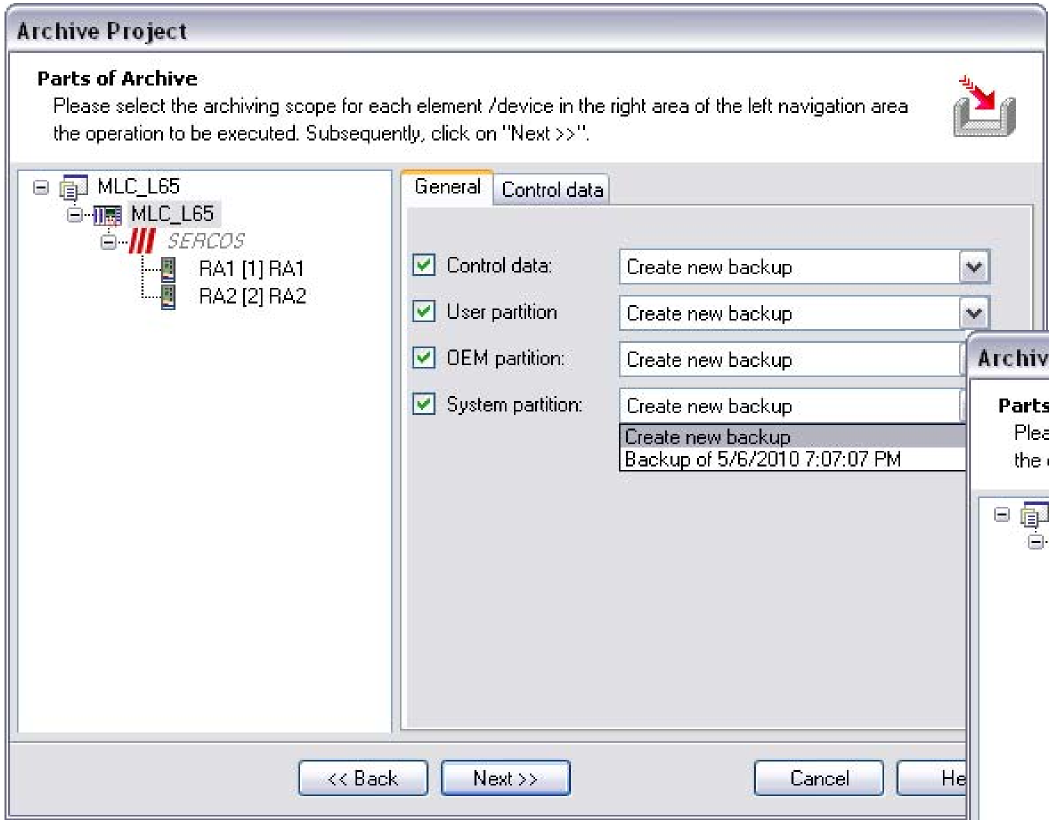
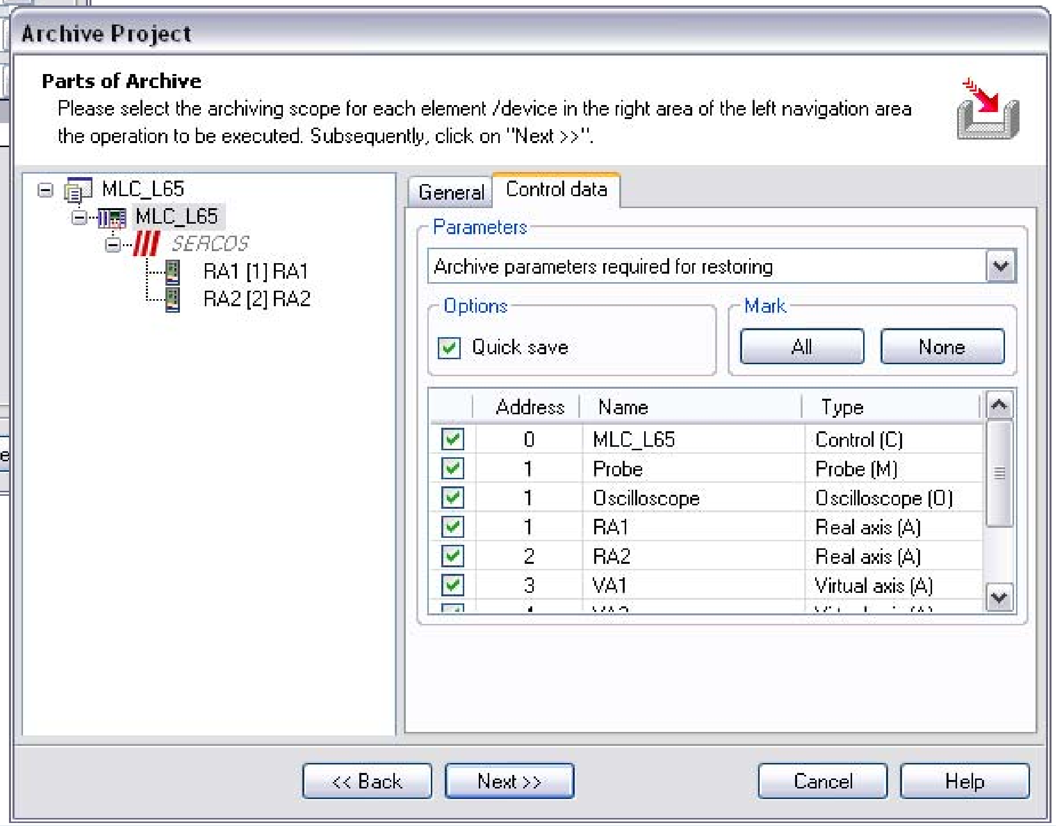
Archiving IndraWorks projects



- **Tab "General":**
- Select control data (includes drive data)
- Select User and OEM partition
- Select new backup or an existing backup, which is still up-to-date



- **Tab "Control data":**
- Mark all parameters
- Select option "Archive parameters required for restoring"

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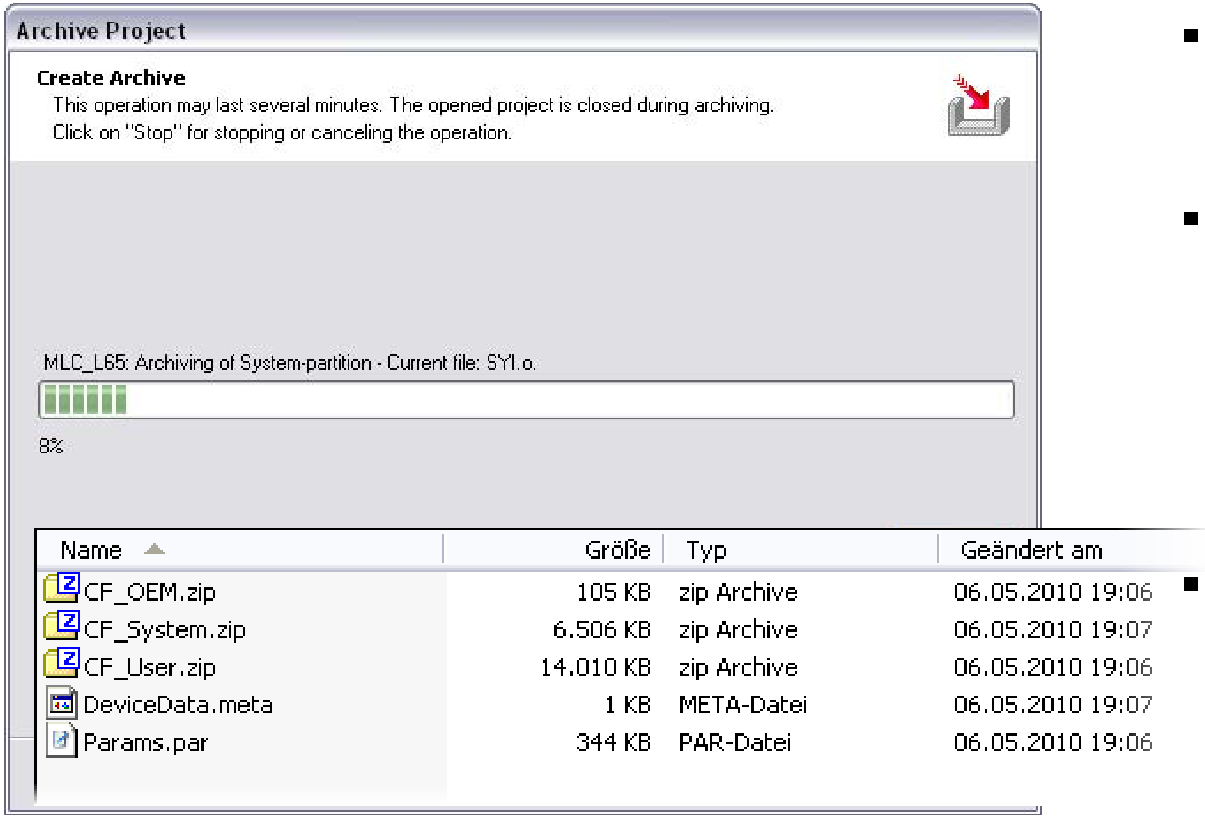
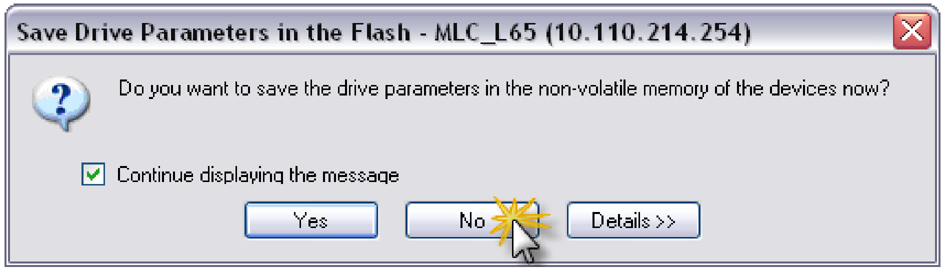
Archiving IndraWorks projects



- During the archive operation the selected data is transferred to the engineering PC
- … and stored in a subfolder of the current IndraWorks project

- Subfolder "ArchivedData" in current IndraWorks project

- Afterwards IndraWorks switches to offline mode
- Acknowledge this operation, normally no save of drive parameters is necessary



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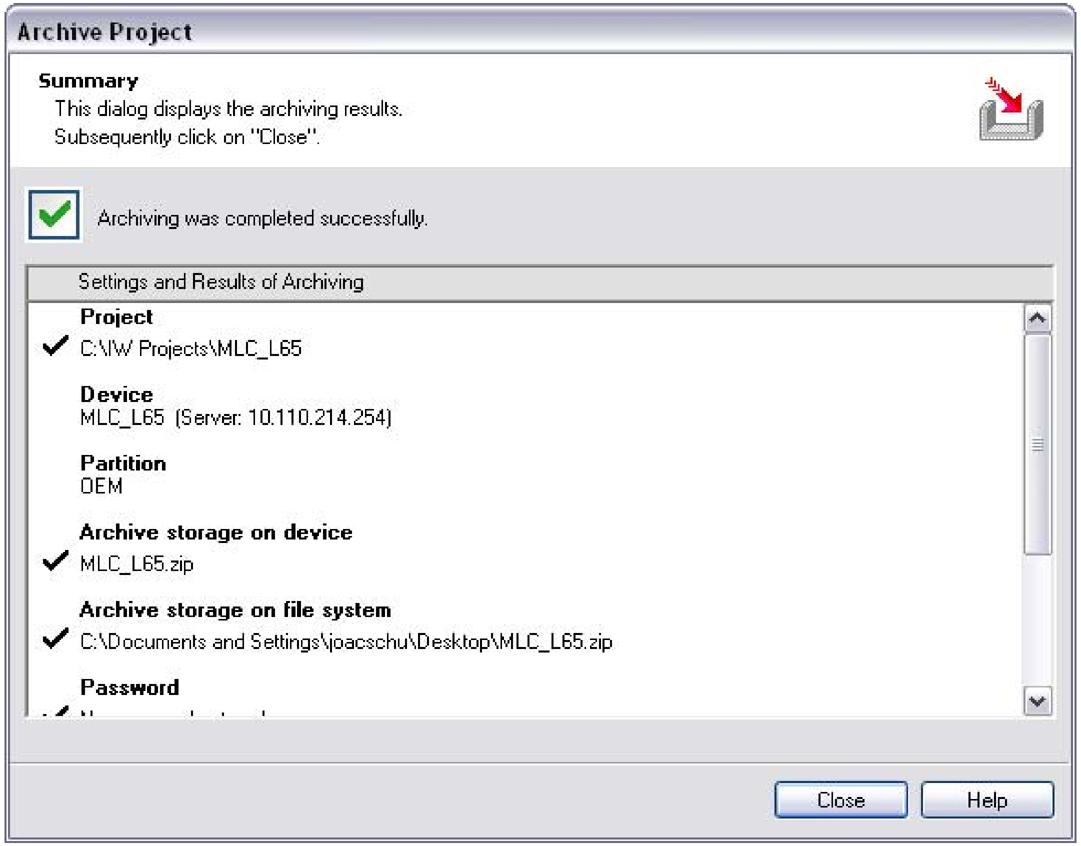
Archiving IndraWorks projects

- Finally the complete IndraWorks project is zipped and stored as an archive in the specified target directory
- ... and/or on the CF card of the MLC (partition OEM)
- The successful result is displayed:



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Restoring IndraWorks projects

- Restoring an MLC project archive may occur in different scenarios:
  - Service – diagnosis
    A service engineer restores an IndraWorks project to his own engineering notebook to perform service tasks on a running machine. After restoring the project direct switching to online mode is possible without downloading anything to MLC
  - Service – exchange of control or drive components
    In case of drive replacement etc. parameters have to be downloaded to the new drive.
  - Duplicating machines – 1st time installation
    The IndraWorks project is used to duplicate an existing machine. In this case all device data (control & drive parameters, CF partitions) have to be restored
  - Support of a customer
    During the commissioning phase of a machine the complete IndraWorks archive can be forwarded to Bosch Rexroth for further processing. After fixing a problem the revised project is returned to the customer

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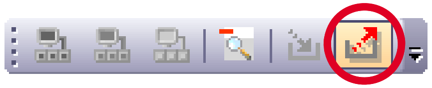
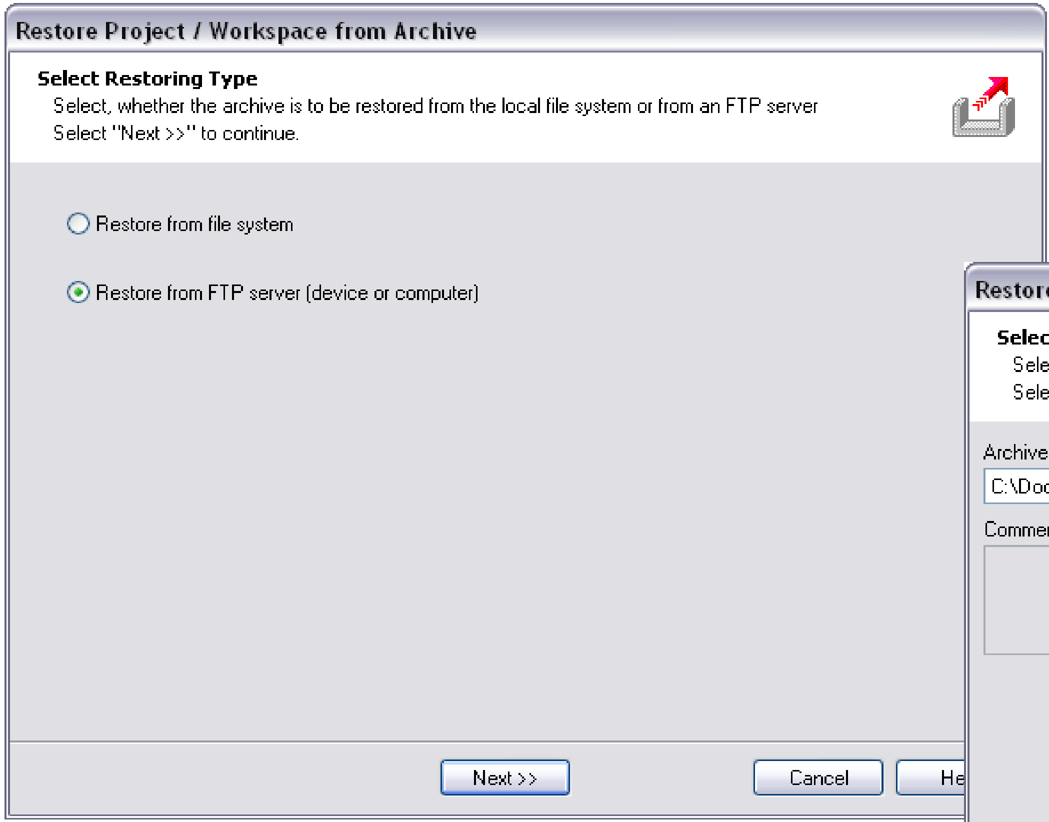
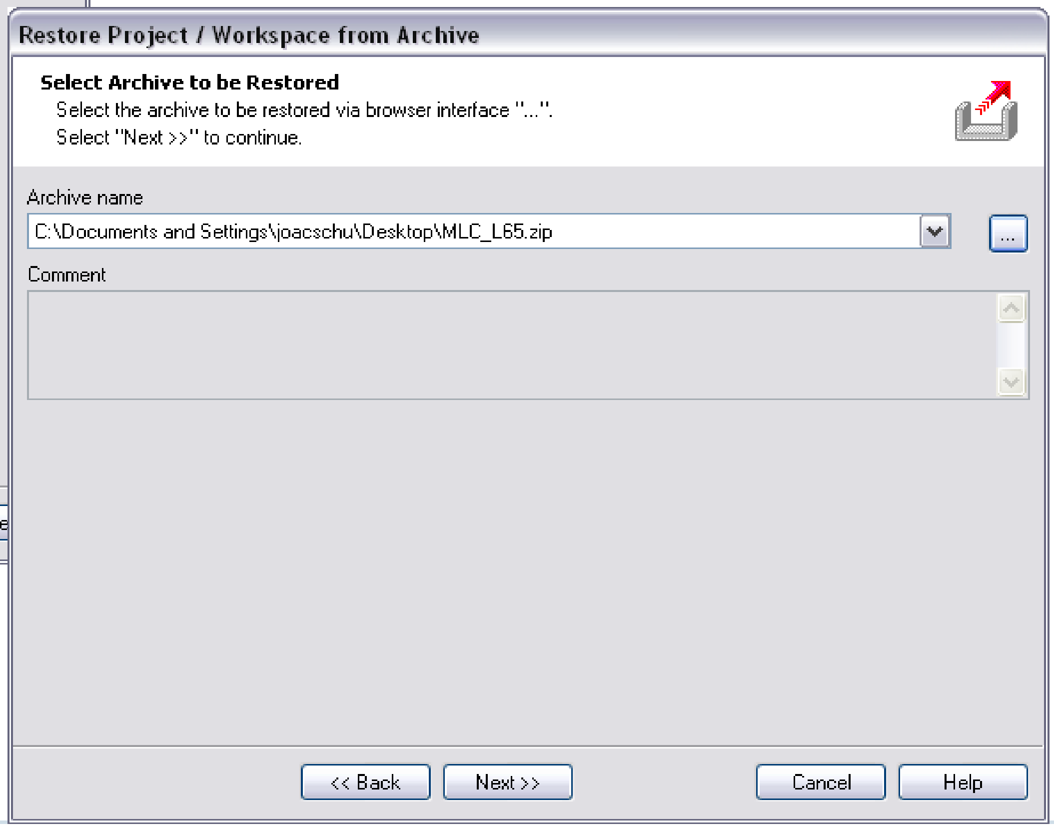
Restoring IndraWorks projects





- To restore push the button "Restore project"
- You can restore from the HDD
- … or from IndraMotion MLC



- If you restore from the file system specify the path and name of the archive

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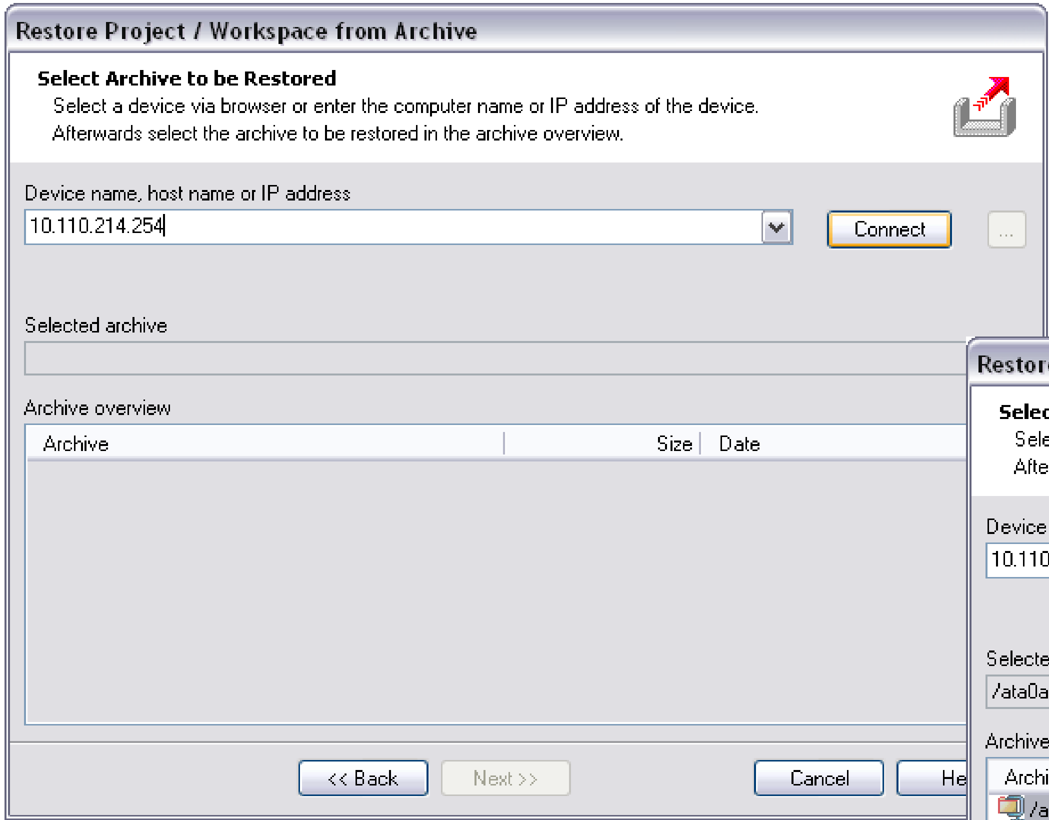
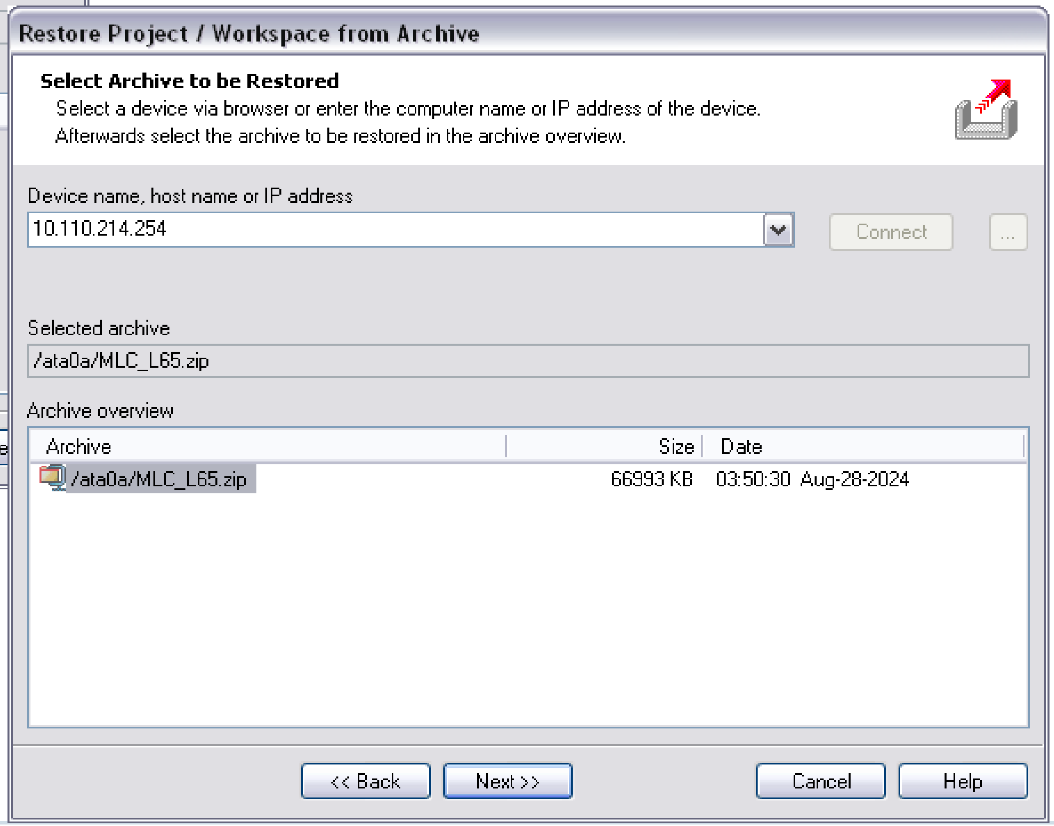
Restoring IndraWorks projects



- If you want to restore from the CF card
- … enter an IP address or host name
- … and push “Connect” button



- A list of the archived project(s) on the CF is displayed
- Select the appropriate entry and push the “Next” button

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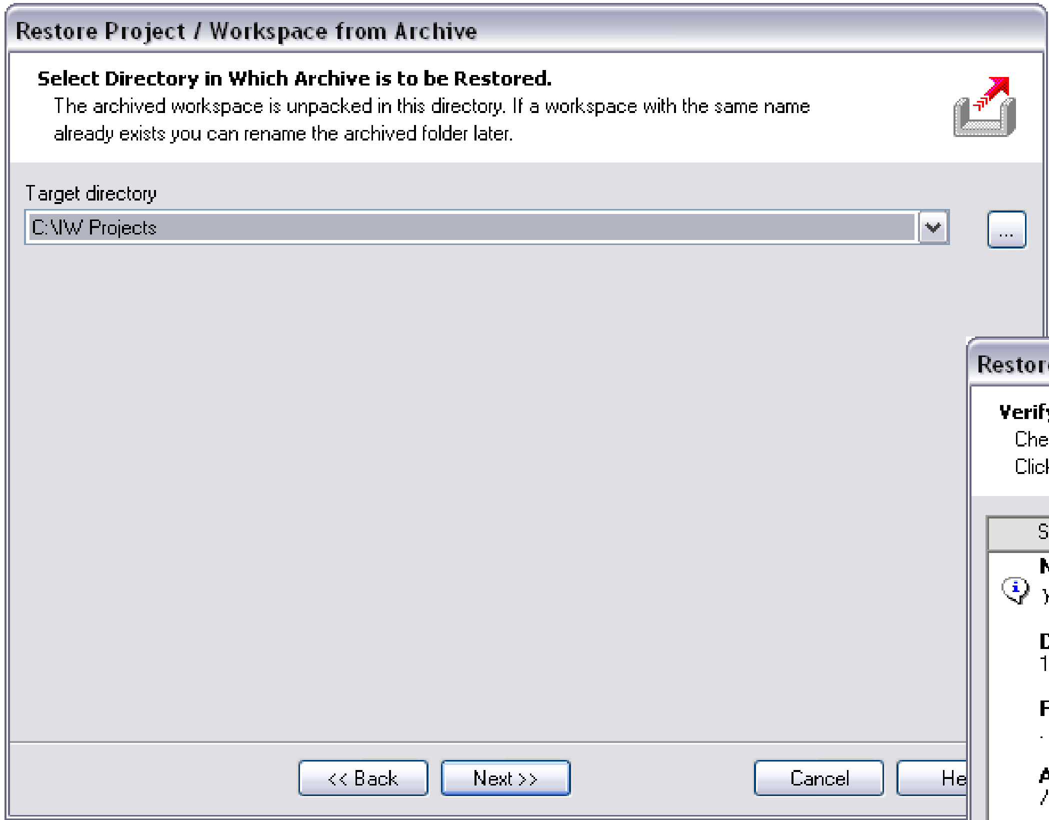
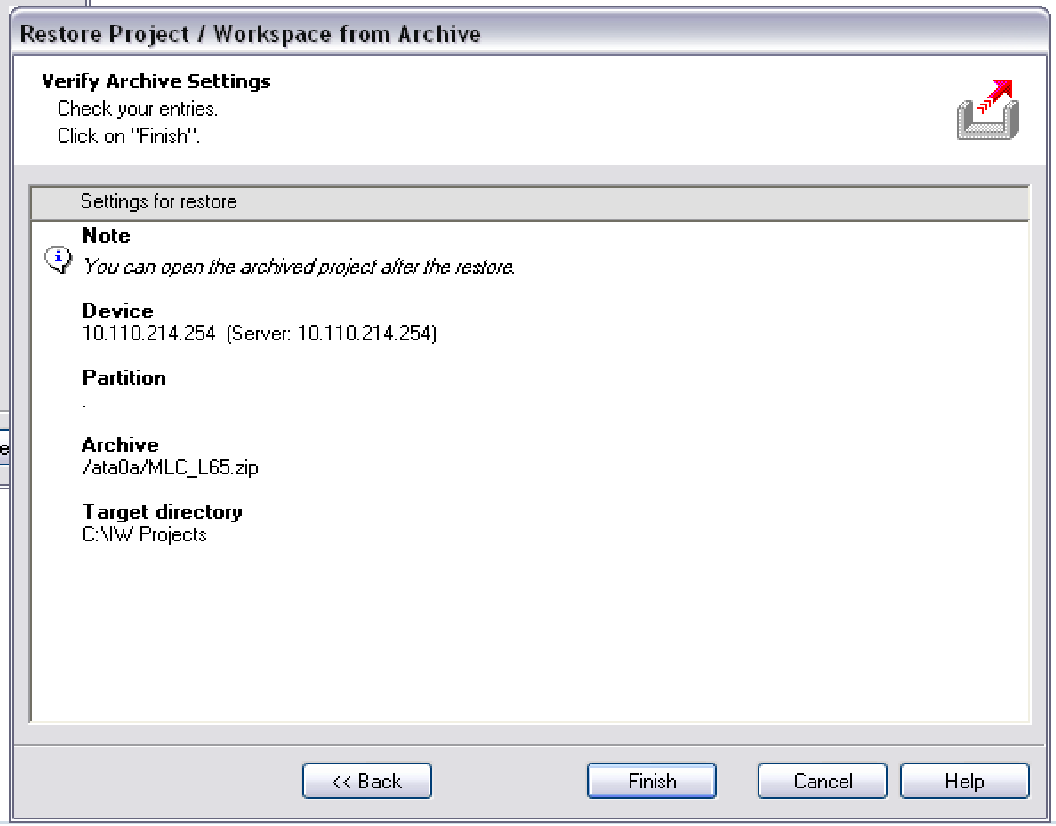
Restoring IndraWorks projects



- Enter the target directory on your HDD to which you want to restore to



- A summary of the operations to be performed is displayed
- Push the “Finish” button to acknowledge the restore procedure

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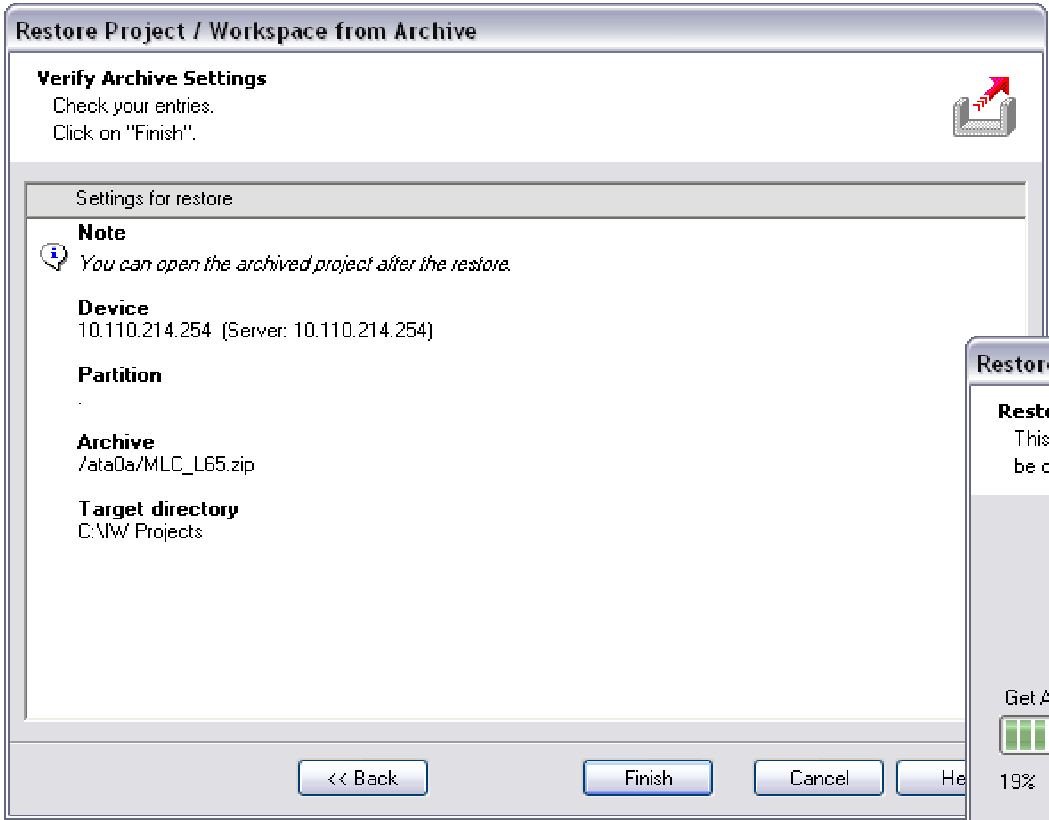
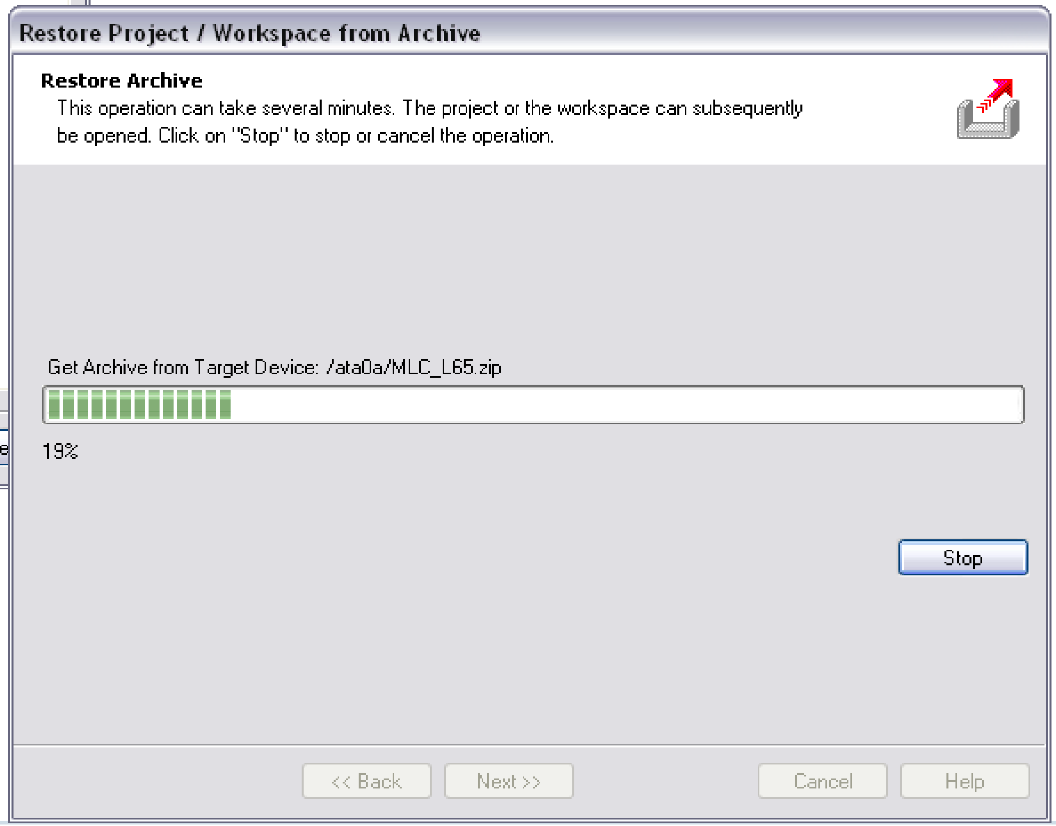
Restoring IndraWorks projects



- A summary of the procedure to be performed is displayed
- Acknowledge this operation by pushing the “Finish” button



- As a result is transferred from the MLC and unzipped
- ... or directly unzipped from the HDD

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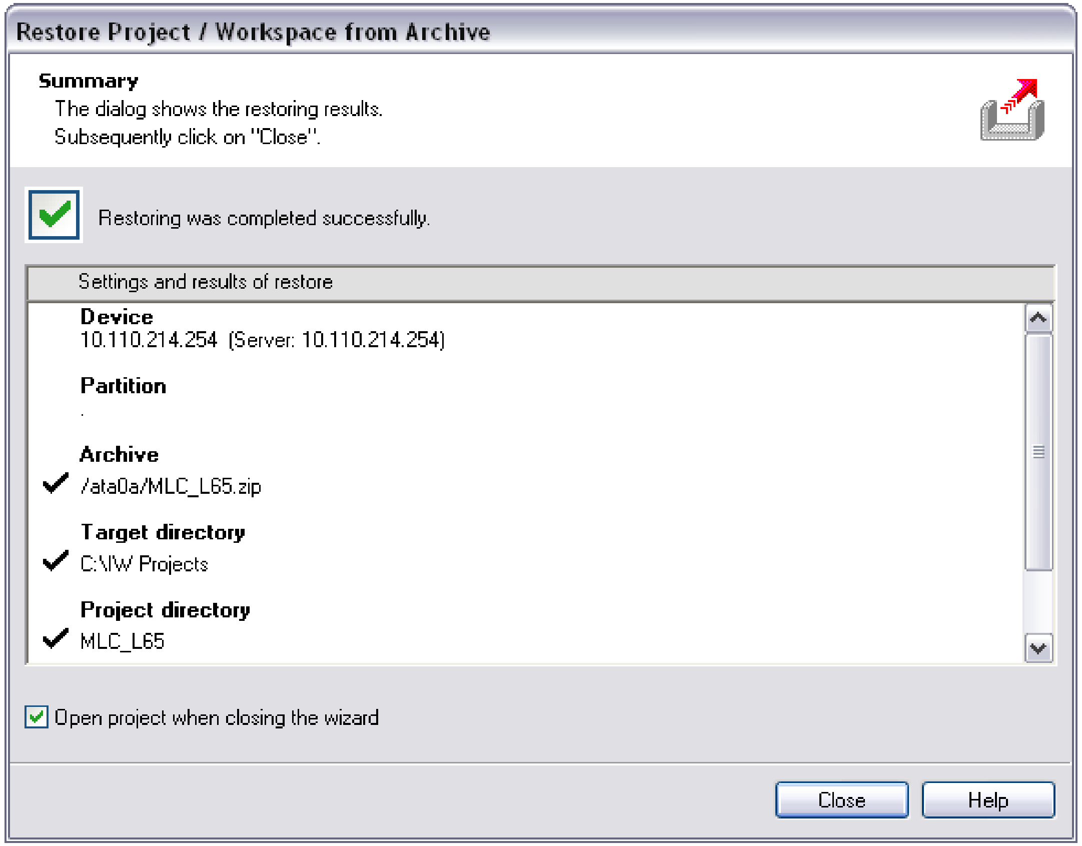
Restoring IndraWorks projects



- On completion the result of the restore operation is displayed
- ... and you can immediately open the restored project
- ... by pressing the “Close” button
- Finally a reminder is displayed to restore also the device data

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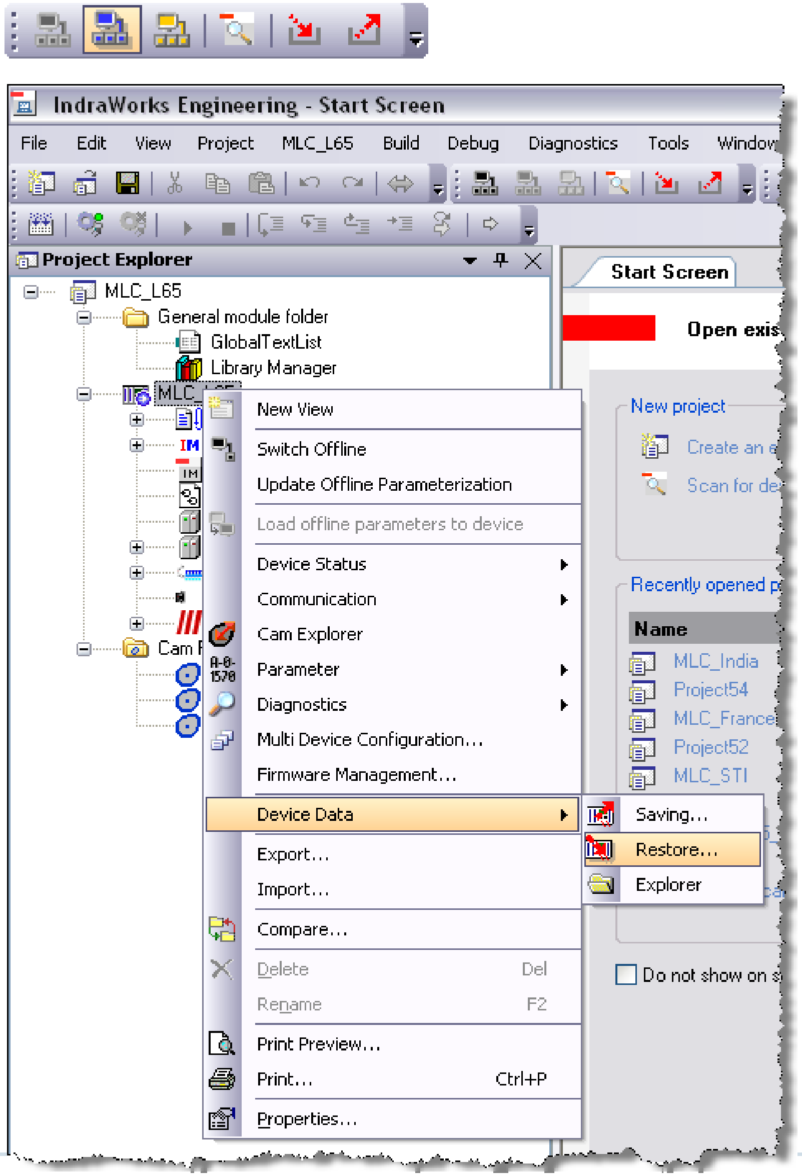
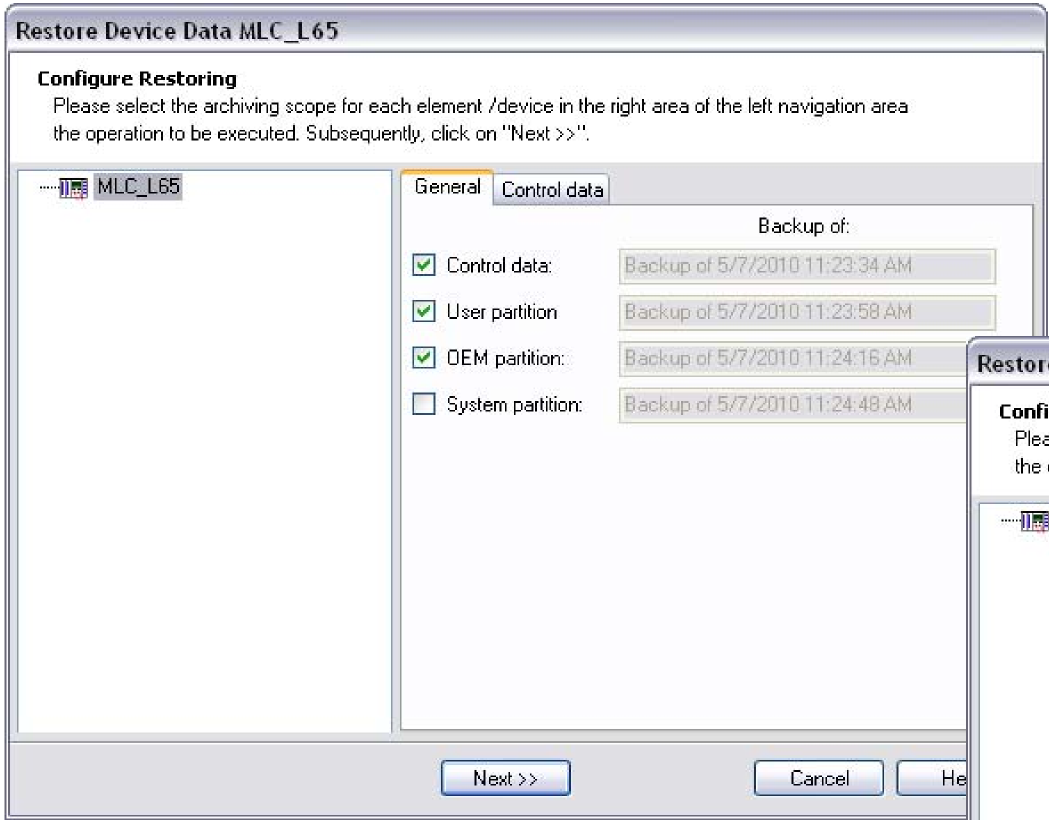
Restoring IndraWorks projects

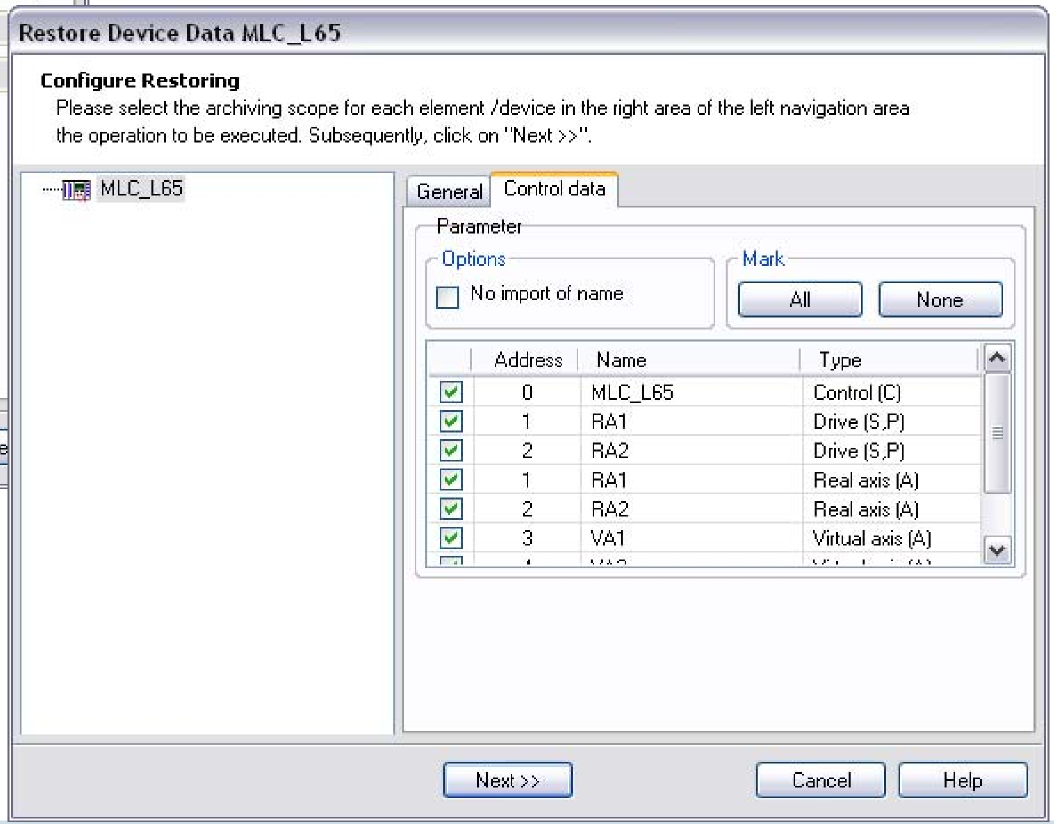


- To restore the Device Data switch to Online mode
- … and select the submenu entry “Device Data/Restore”

Restoring IndraWorks projects



- Select what you want to restore
- In most cases you will restore the complete control data (control & drive parameters)
- ... as well as User & OEM partition



- On the tab “Control data” select the items you want to restore
- In this case restore the complete parameter set

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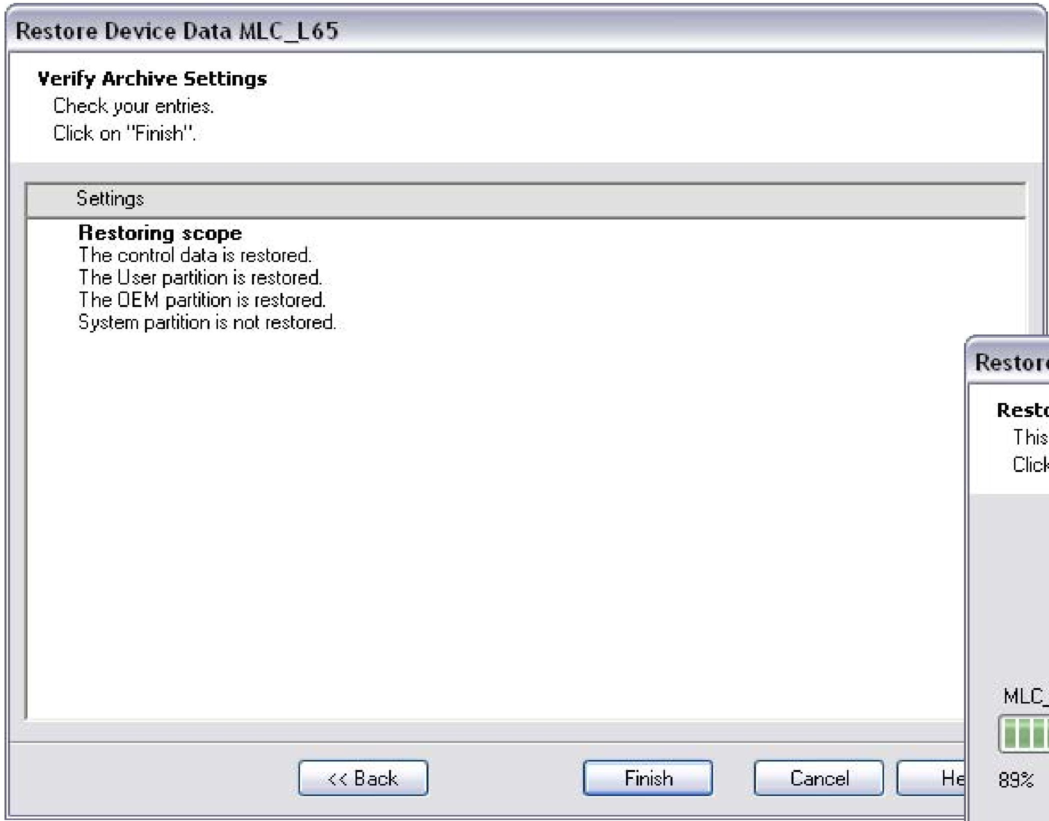
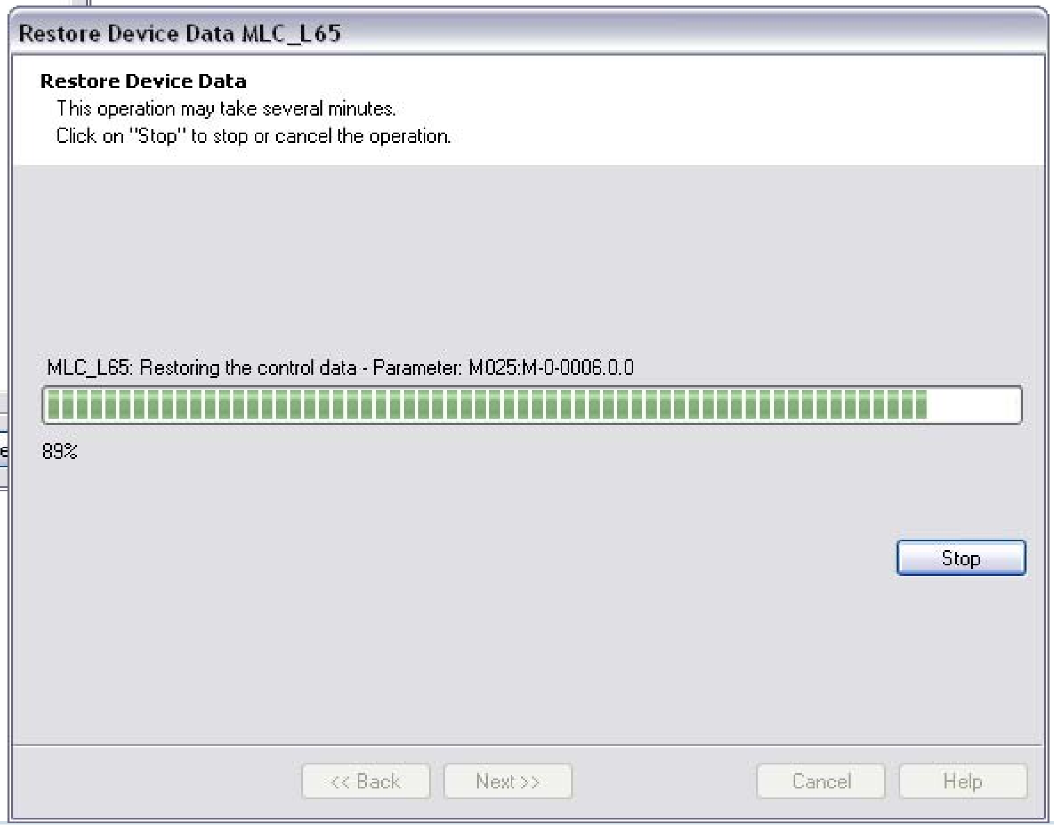
Restoring IndraWorks projects



- A summary of the procedure to be performed is displayed
- Acknowledge this operation by pushing the “Finish” button



- As a result the selected partitions
- ... and the selected control and drive parameters are restored

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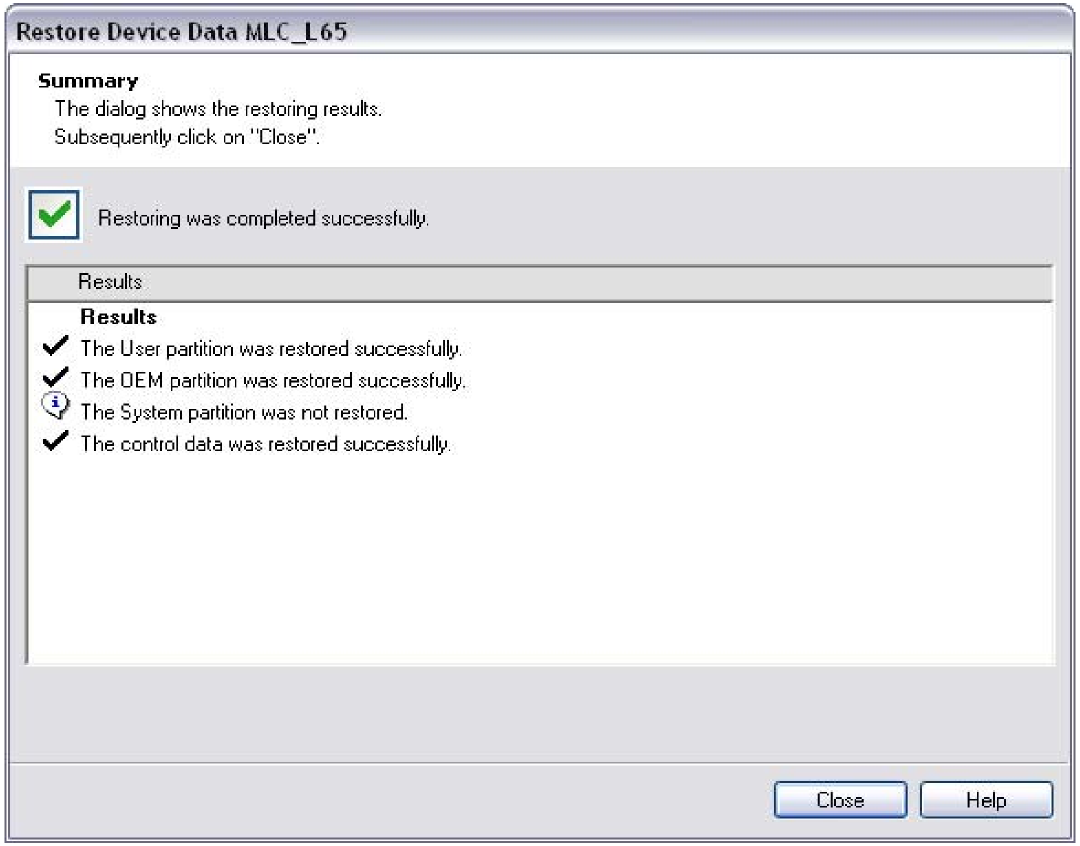
Restoring IndraWorks projects

- After all data has been downloaded to the MLC
- ... the successful result is displayed:



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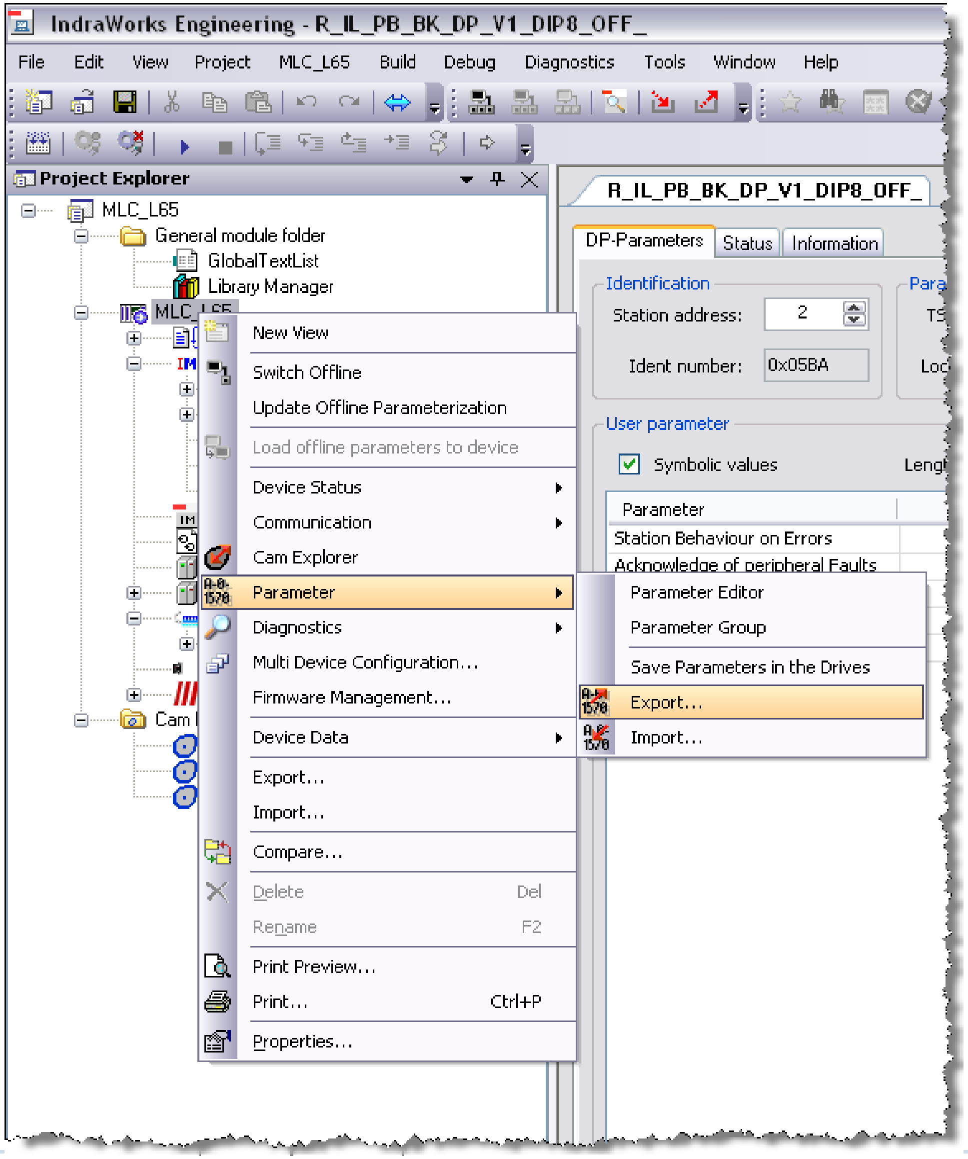
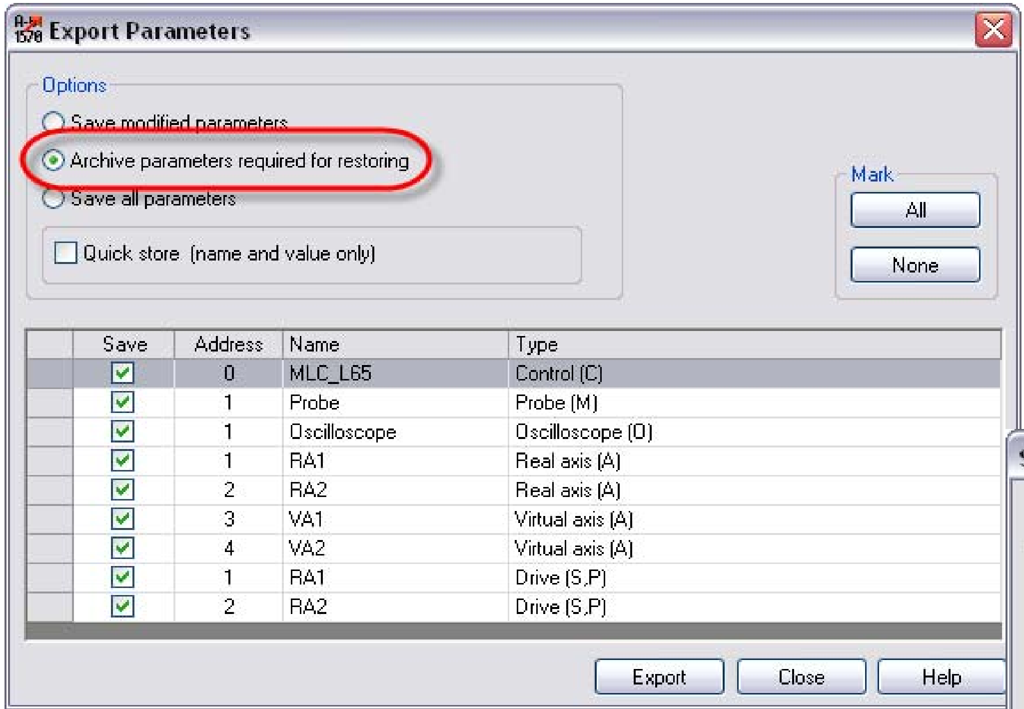
Export of control or drive parameters

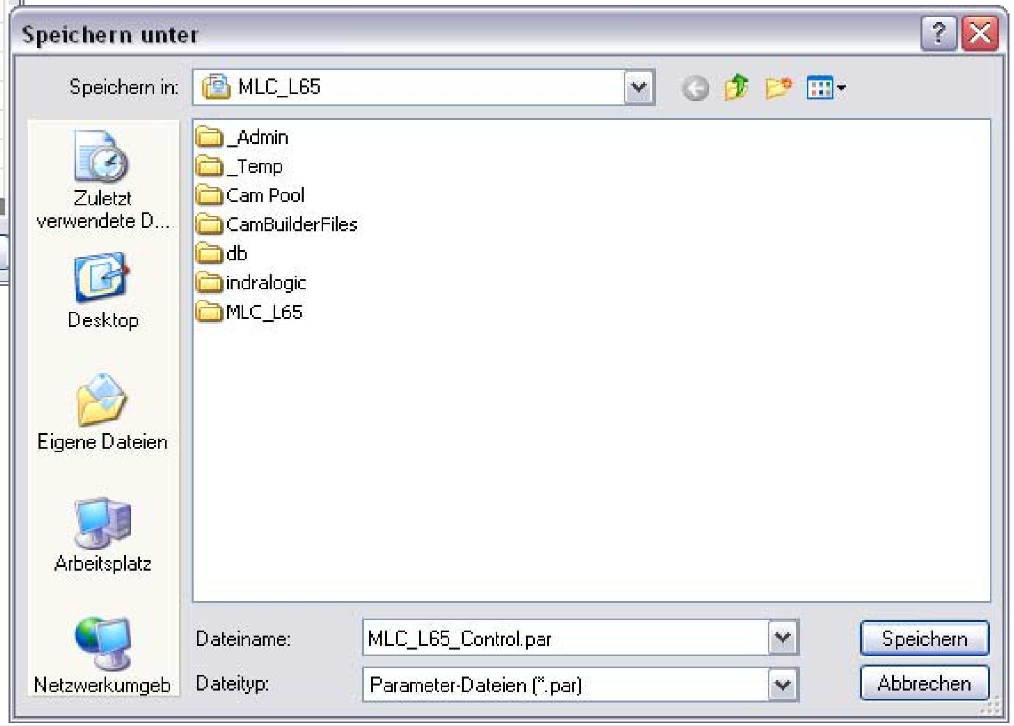


- **Parameter export**
- If you want to save the actual parameter values of your control and / or drives
- … select the submenu entry “Parameter/Export”
- This is necessary for instance before downloading a new firmware

Export of control or drive parameters



- Select “Archive parameters required for restoring”
- Export of all parameters or
- ... only of selected parameters is supported
- Push “All” to export the complete parameter set



- The parameter file is stored within the IndraWorks project folder
- If needed a subfolder for the parameter file can be created
- Enter a filename for the parameter file

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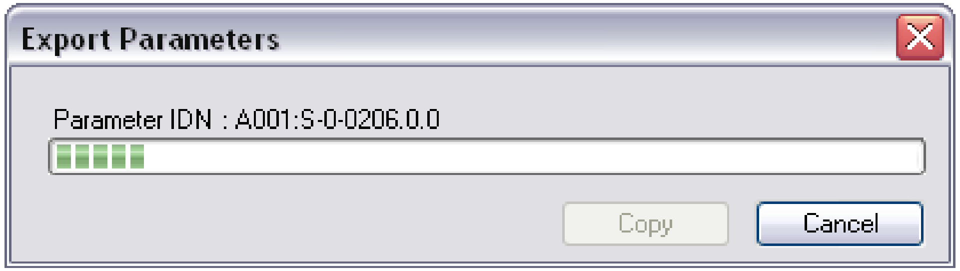
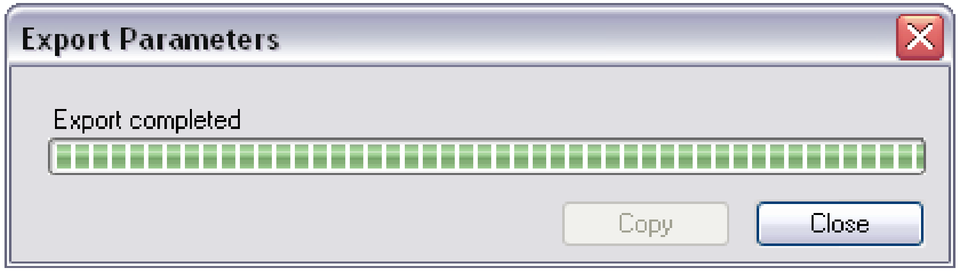
Export of control or drive parameters

- The parameters are uploaded from MLC and stored in a parameter file
- During this procedure the actual progress is displayed



- On completion this information is displayed



- As a result the parameters are saved in the parameter file which can be imported later on

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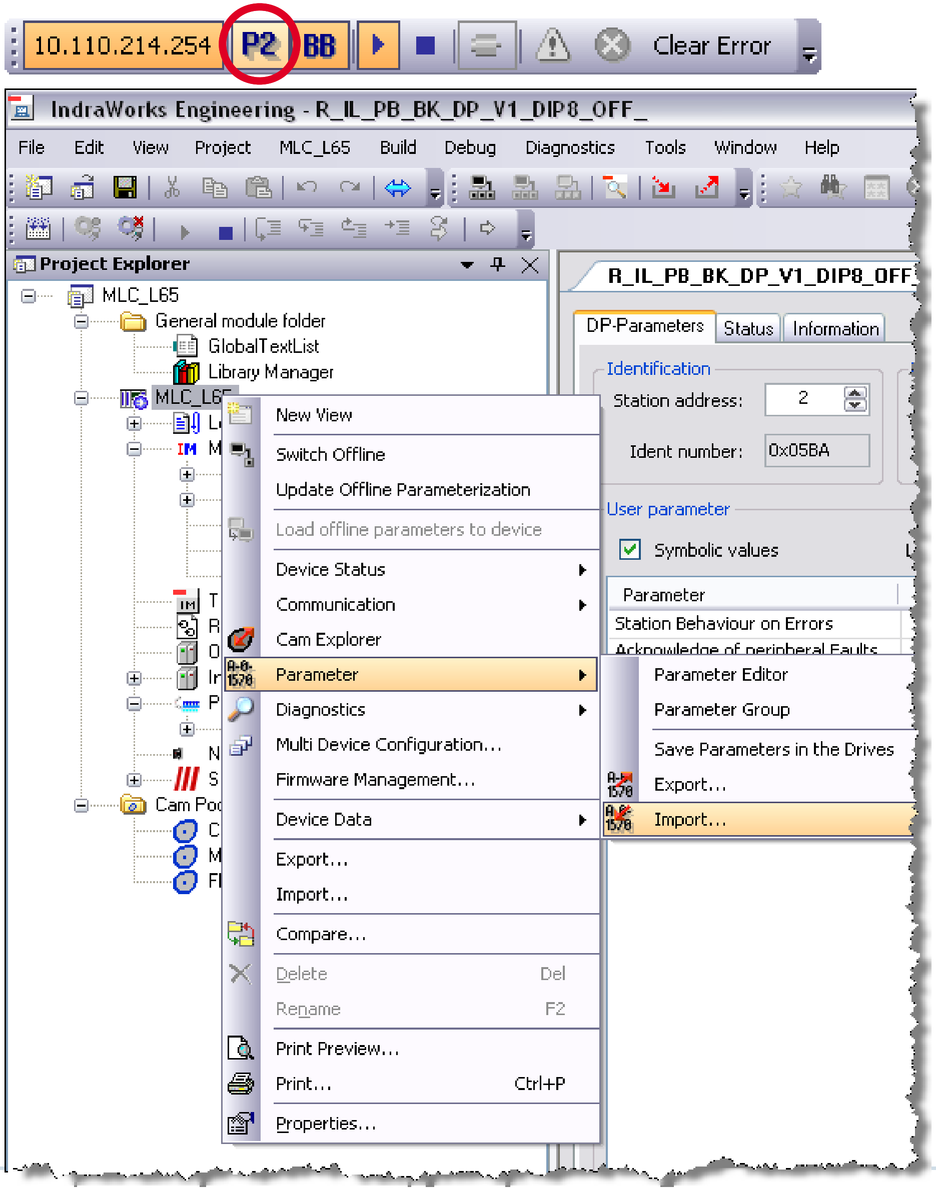
Import of control or drive parameters



- **Parameter import**
- If you have a parameter file you can download the parameters to the MLC and in this way restore a prior state
- Before importing parameters switch the control to parameter mode
- … then select the submenu entry “Parameter/Import”
- This procedure is necessary for instance after a firmware upgrade

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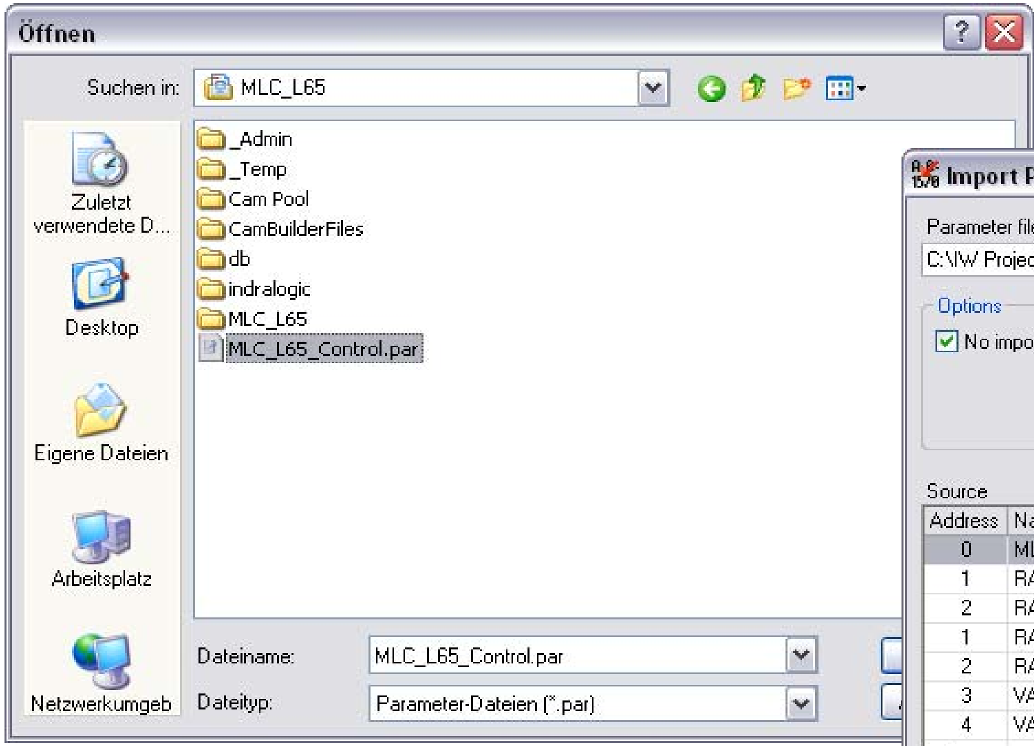
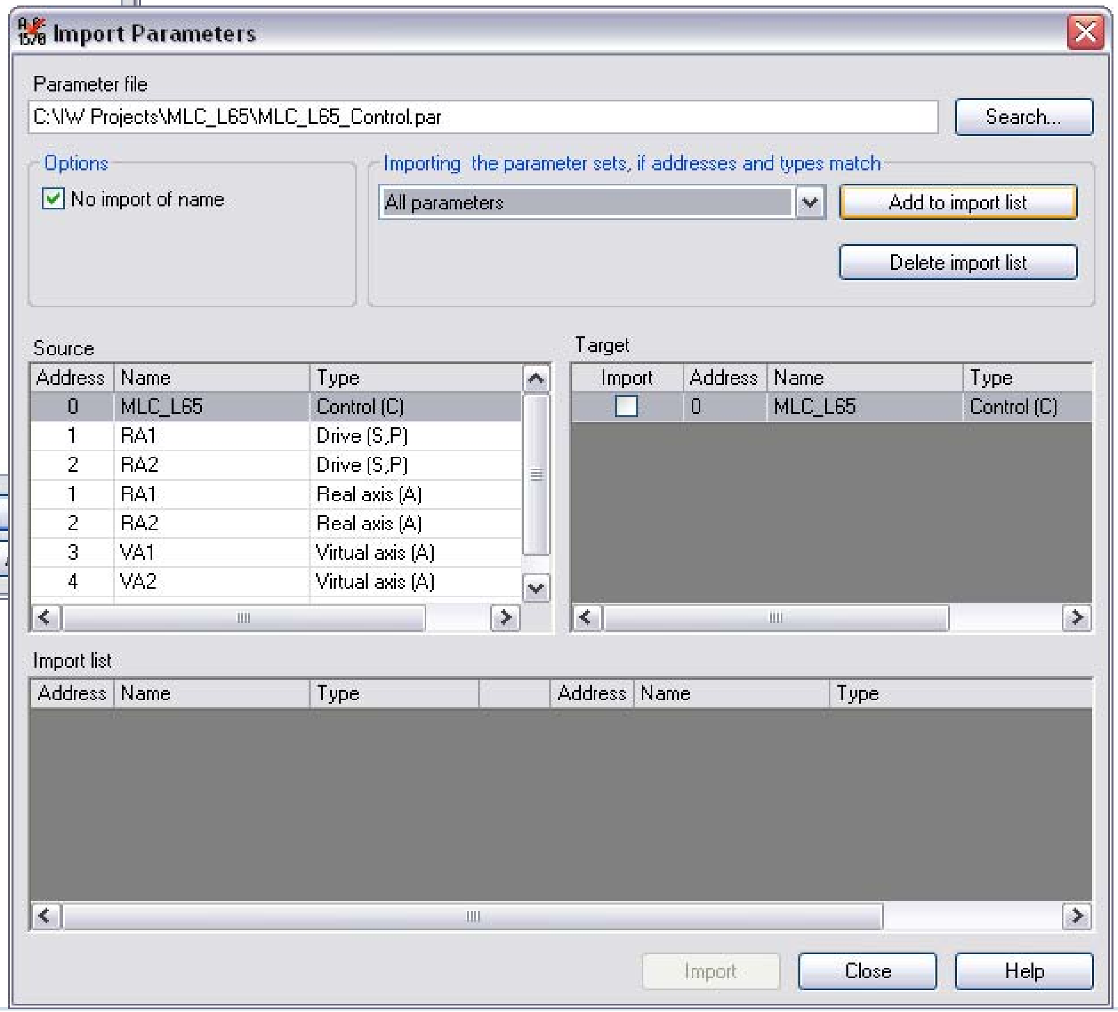
Import of control or drive parameters



- Select the appropriate parameter file



- The components within the parameter file are displayed
- A selective import
- ... as well as a full import is supported

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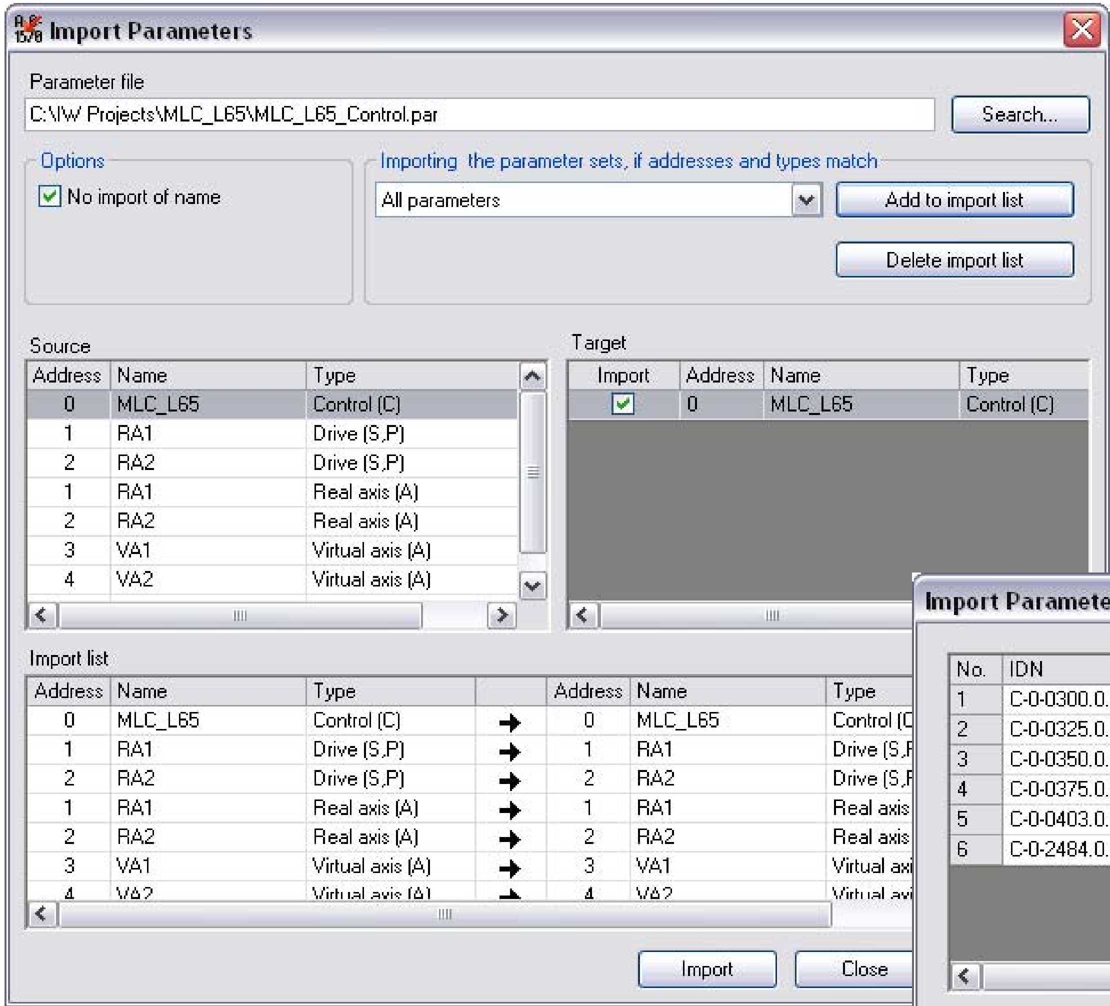
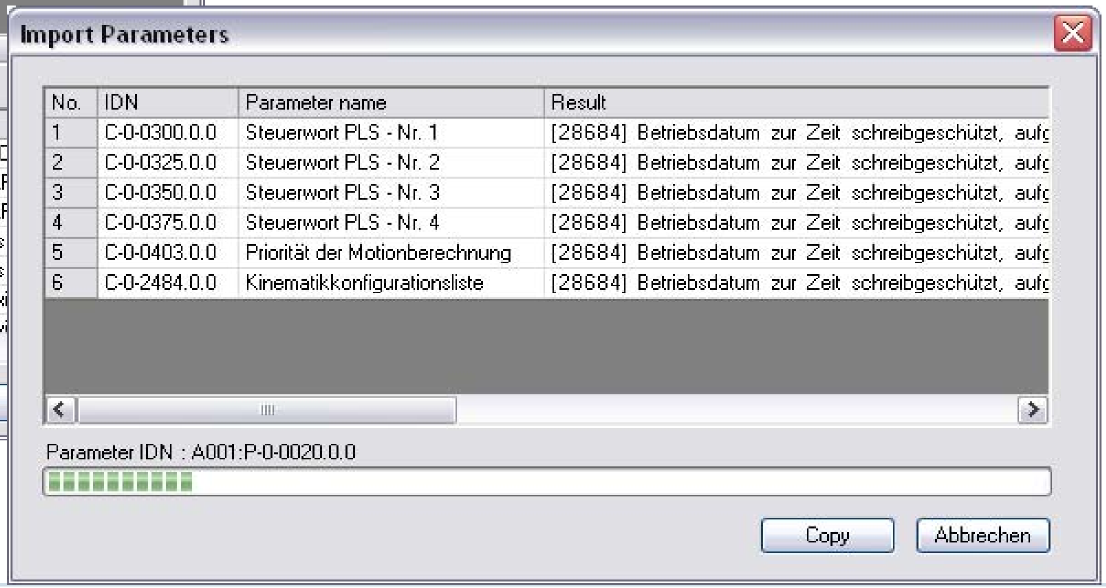
Import of control or drive parameters



- If you press “Add to import list” a full import list is generated
- … and after pressing the button “Import” the parameters are downloaded to the MLC and/or drives



- The progress of this operation is displayed as well as errors during the import

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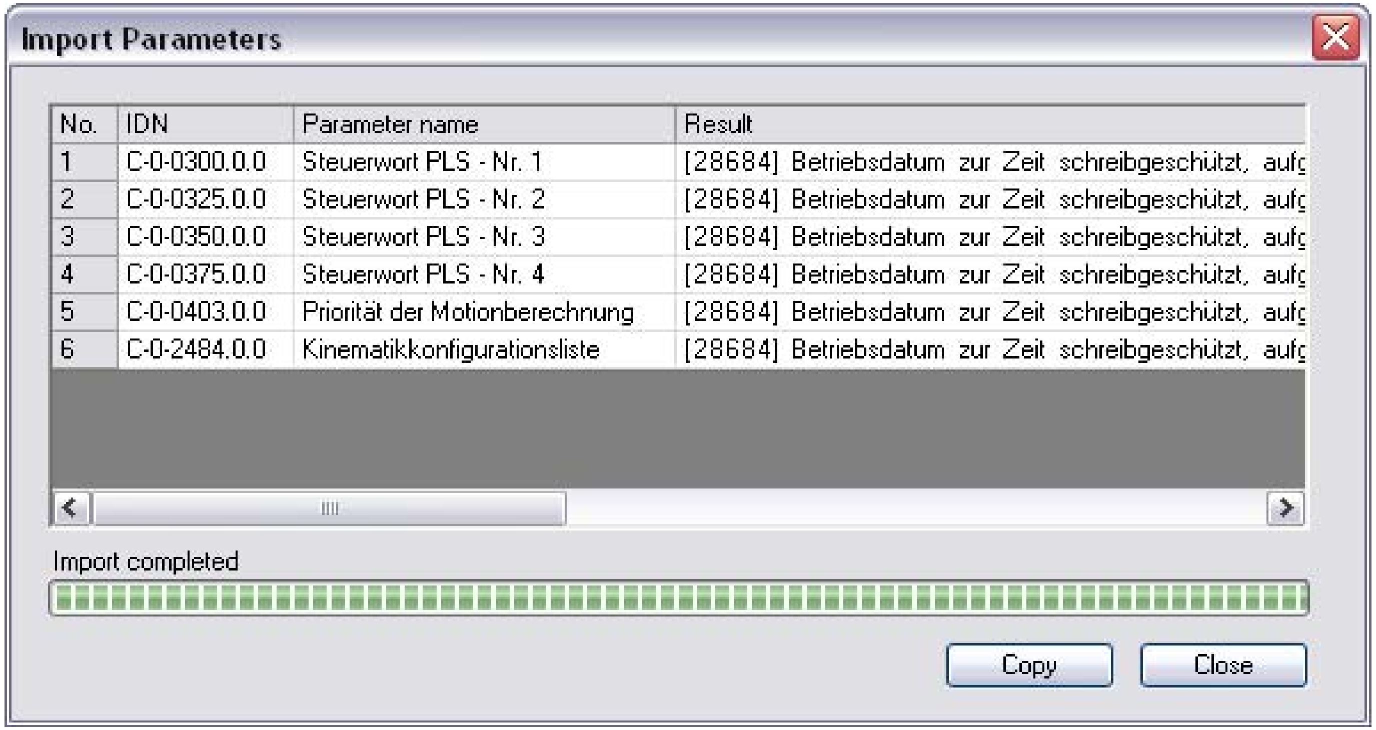
Import of control or drive parameters

- On completion the following information is displayed:



No.	IDN	Parameter name	Result
1	C-0-0300.0.0	Steuerwort PLS - Nr. 1	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg
2	C-0-0325.0.0	Steuerwort PLS - Nr. 2	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg
3	C-0-0350.0.0	Steuerwort PLS - Nr. 3	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg
4	C-0-0375.0.0	Steuerwort PLS - Nr. 4	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg
5	C-0-0403.0.0	Priorität der Motionberechnung	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg
6	C-0-2484.0.0	Kinematikkonfigurationsliste	[28684] Betriebsdatum zur Zeit schreibgeschützt, aufg

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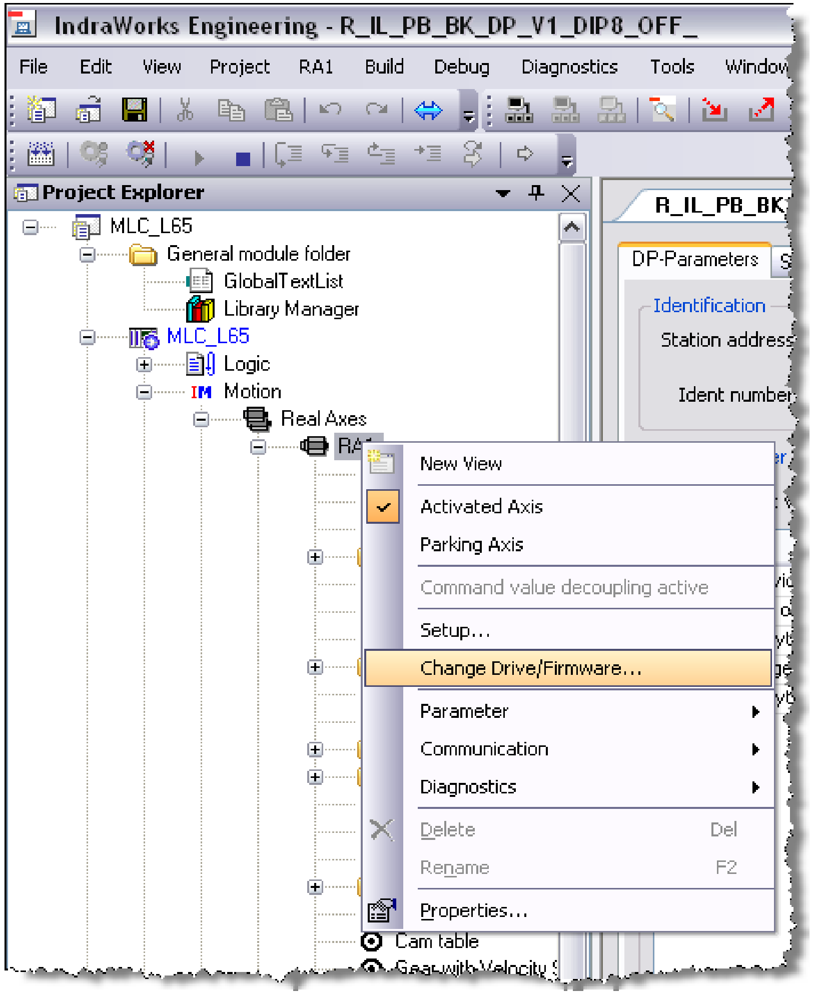
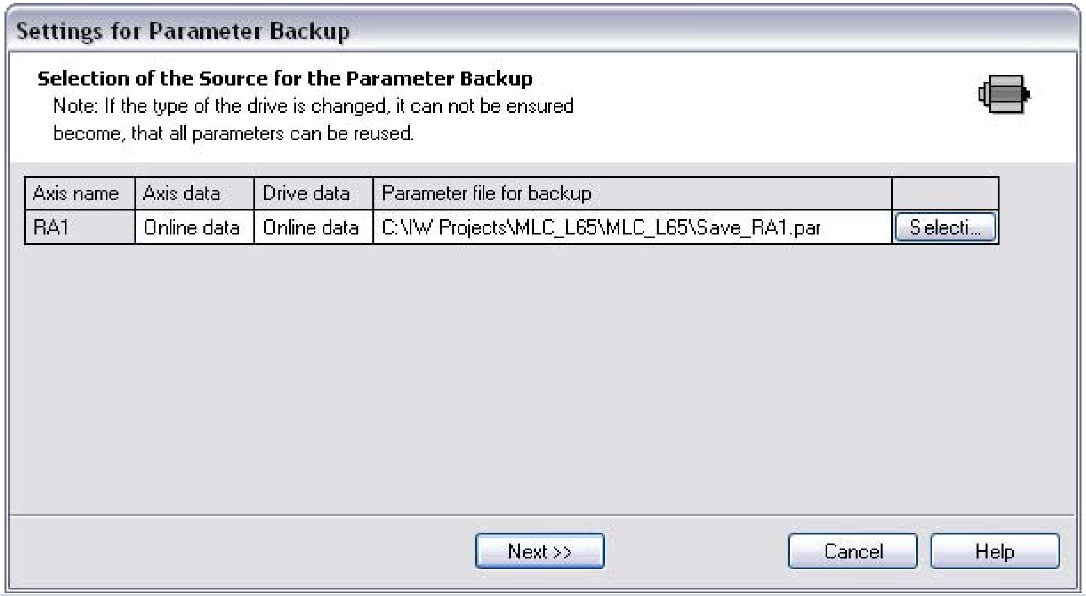
Drive exchange



- For drive exchange there is a special wizard which simplifies this task
- Select the appropriate axis you want to exchange
- … and select the submenu entry “Change Drive/Firmware”



- In the 1st step enter a file name for the parameter backup of the current drive

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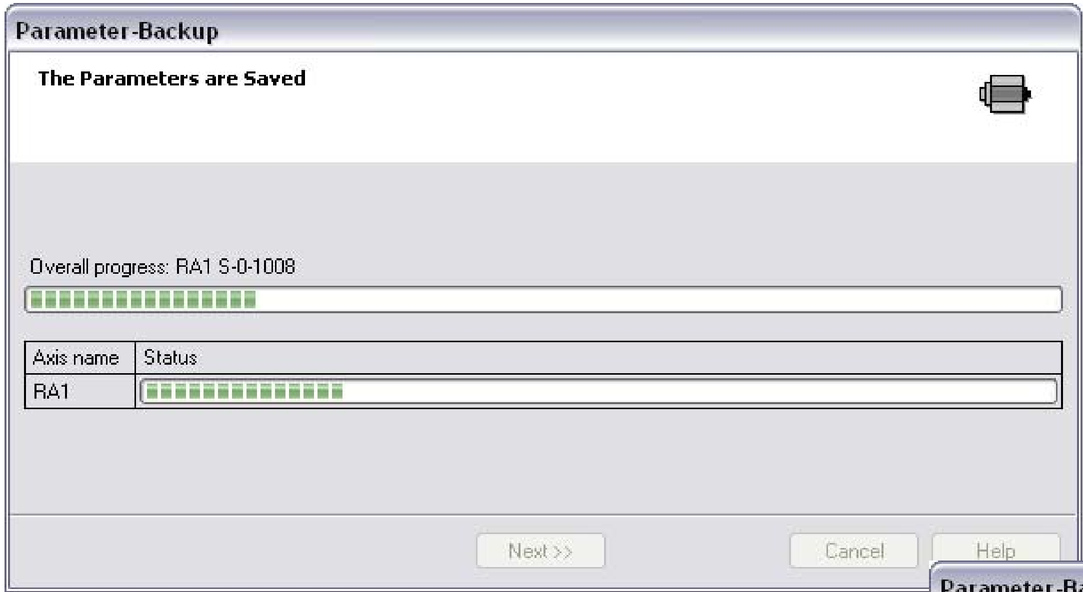
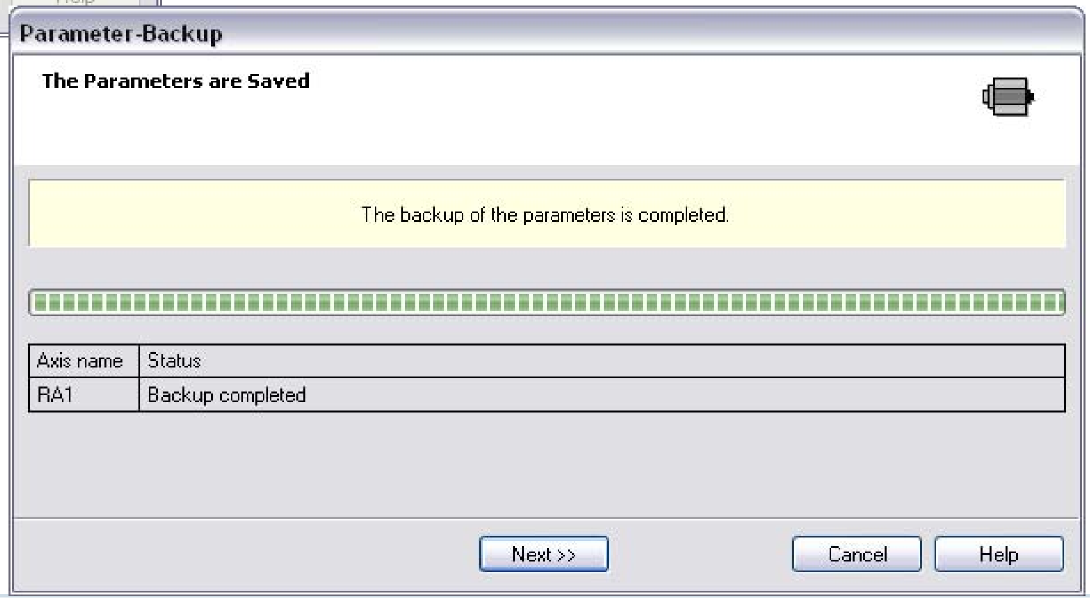
Drive exchange

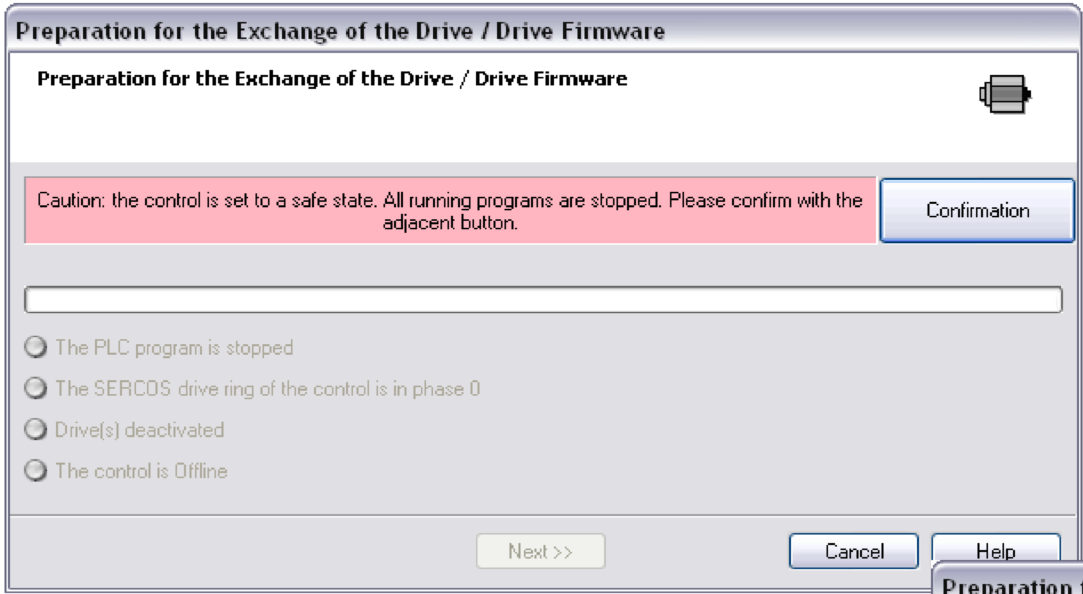


- All parameters of the original drive are saved

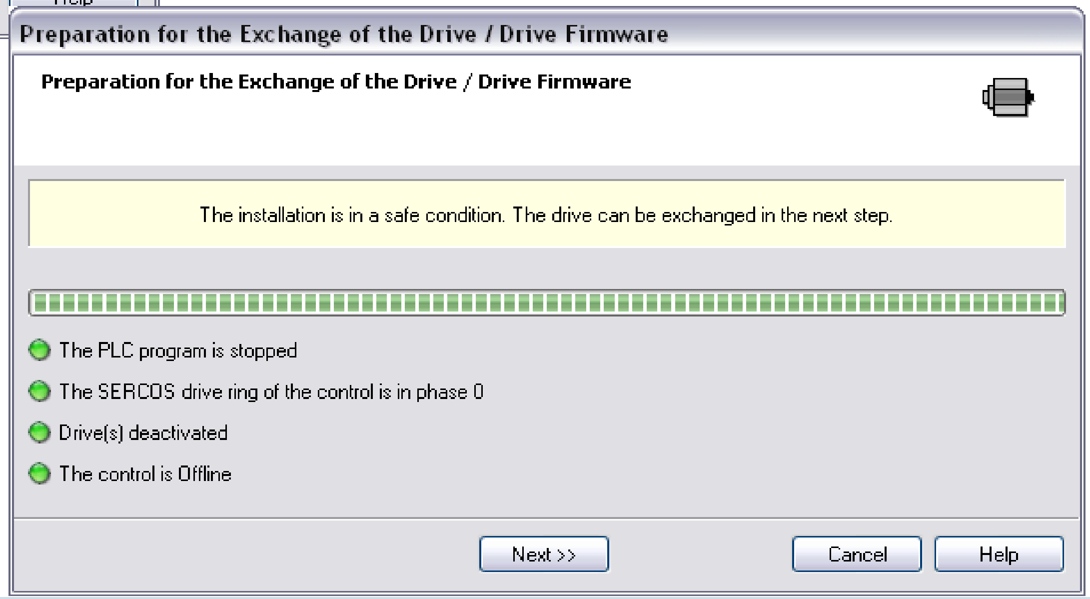


- After the backup is complete the specified file holds all parameters of the original drive

Drive exchange



- Before exchanging the drive some preparations have to be done (stop PLC, switch to parameter mode etc.)
- Press the button "Confirmation" to enable IndraWorks to perform these preparations



- The control is now in a safe condition
- ... and IndraWorks is offline

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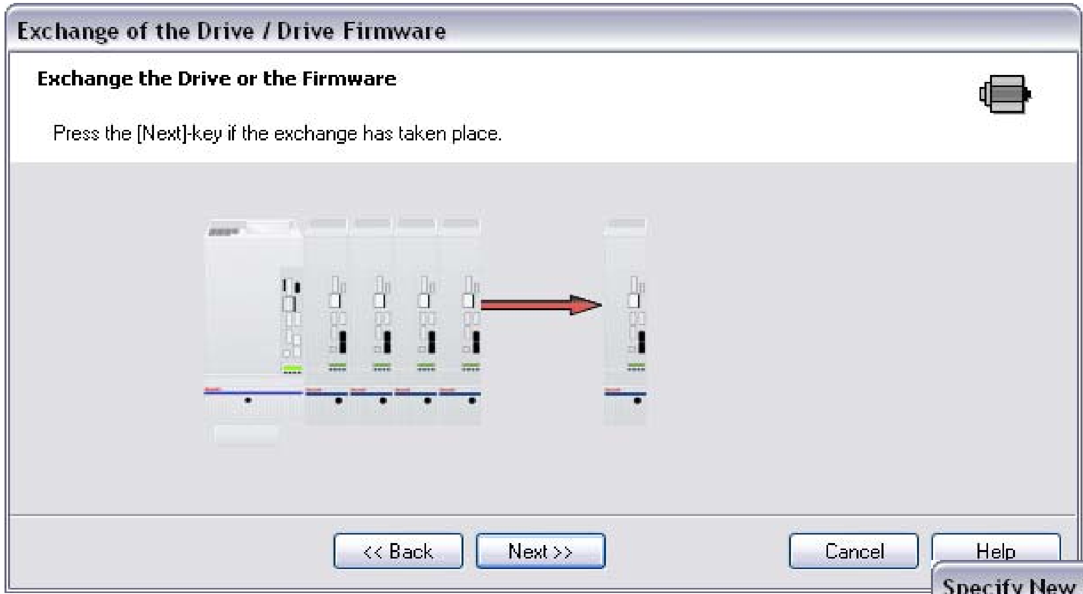
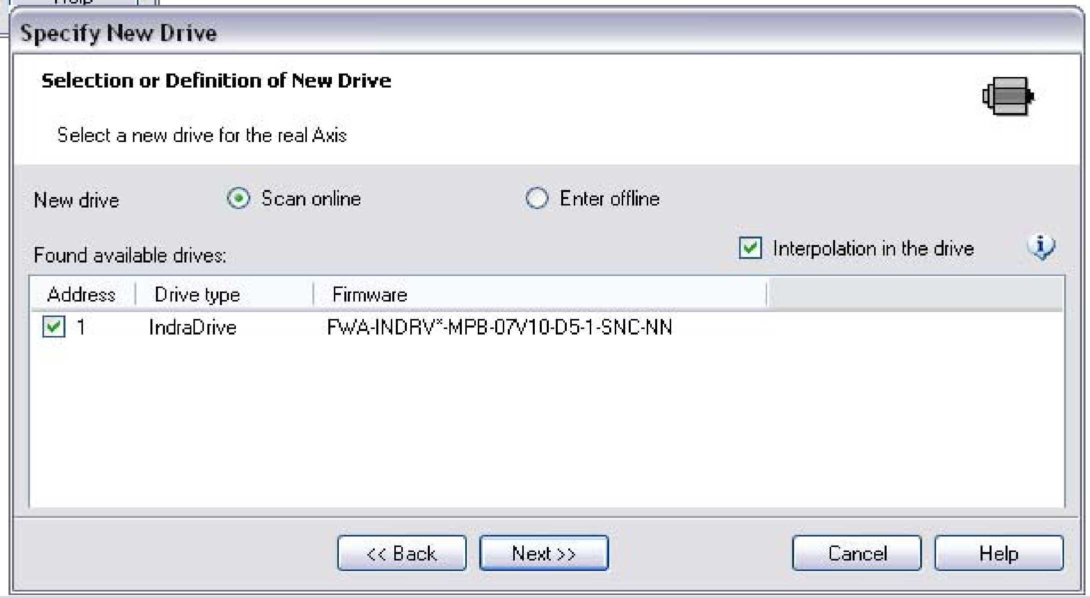
Drive exchange



- Exchange the drive now or
- … upgrade the firmware of the existing drive



- The new drive can be scanned or can be added manually

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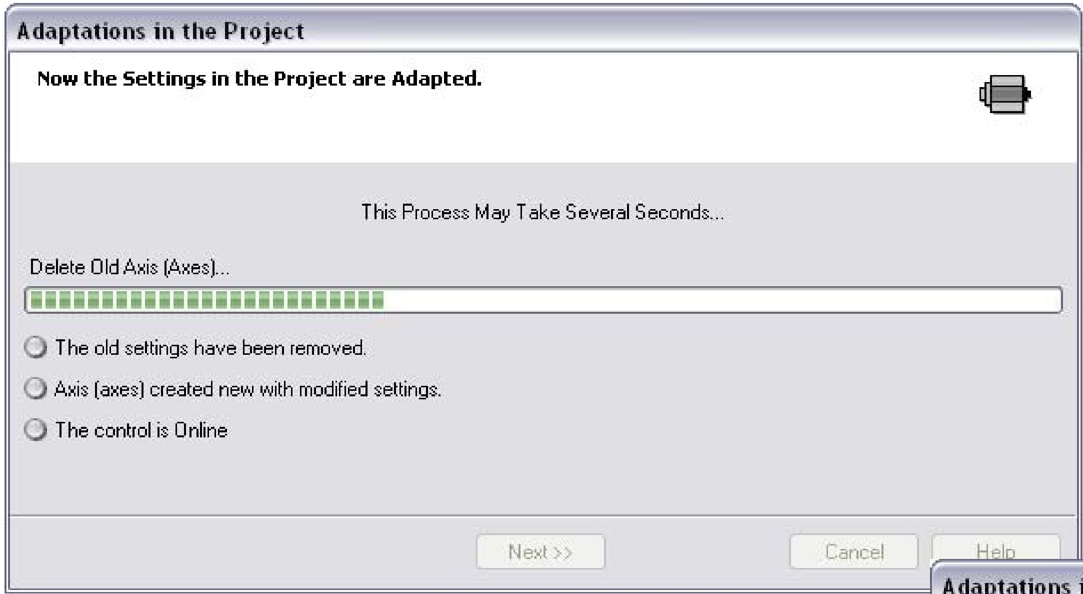
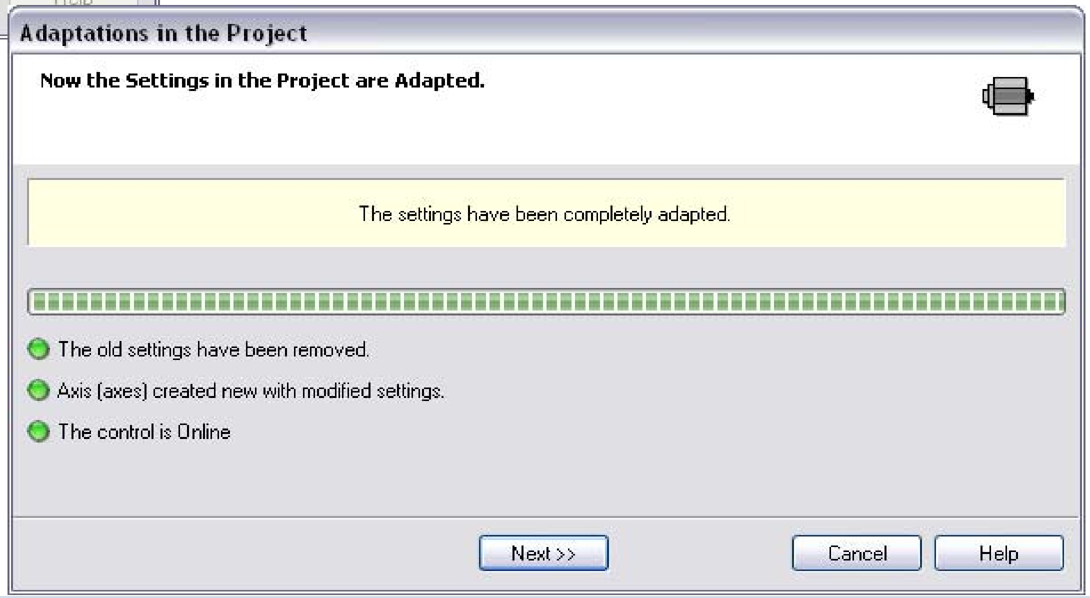
Drive exchange



- The IndraWorks project needs to be adjusted
- The old drive is deleted from the project



- ... a new axis based on the scanned drive is inserted
- ... and IndraWorks switches to online mode

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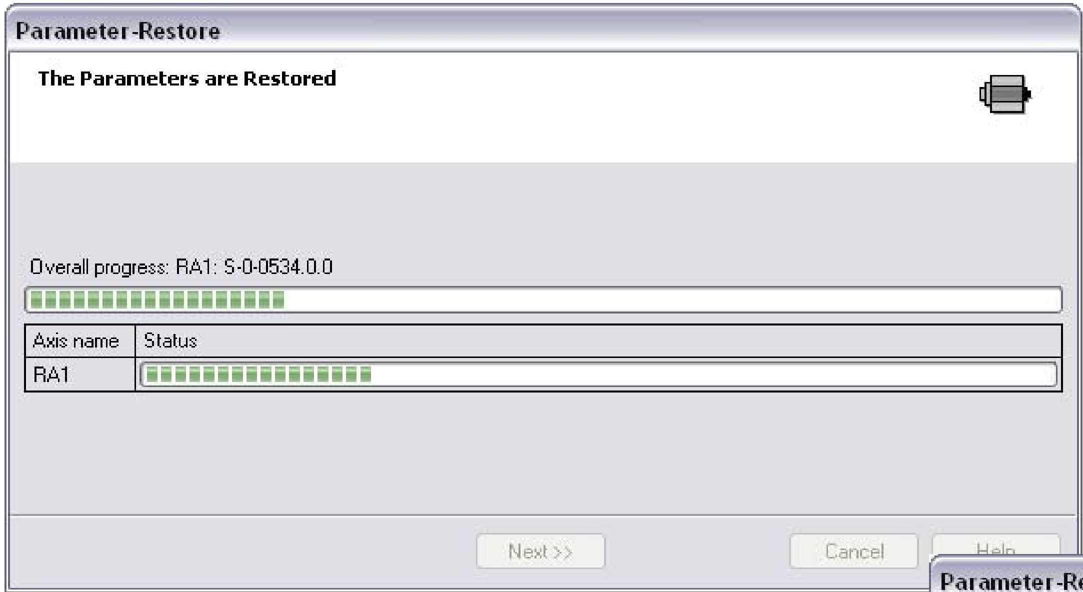
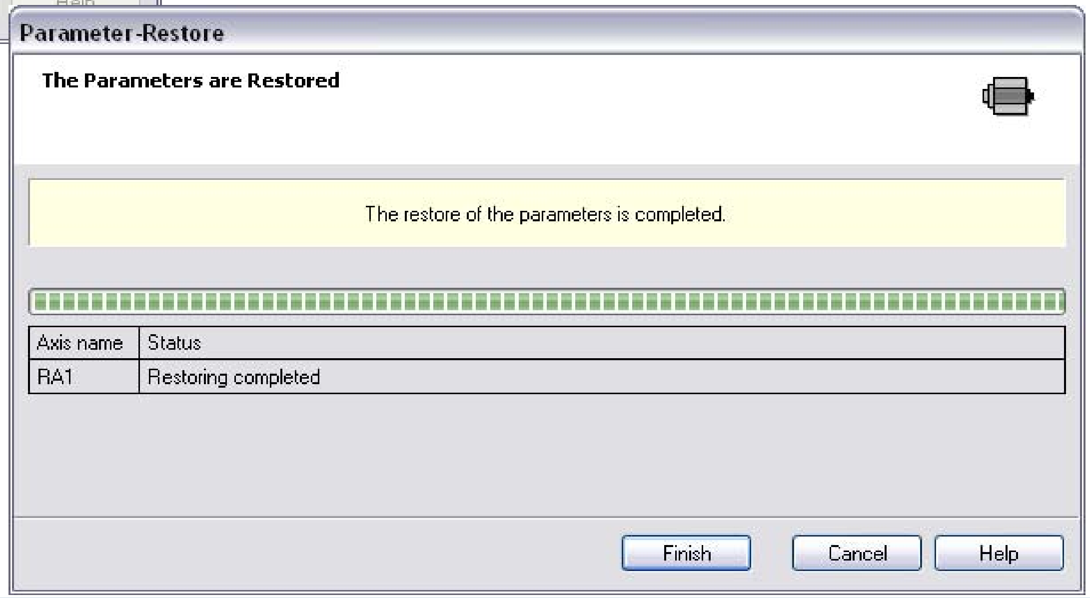
Drive exchange



- Now the parameters of the original drive are restored using the parameter file generated beforehand



- ... as a result all parameters are restored to the new drive
- ... and the drive exchange is completed

Agenda

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- Data backup and restore
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- IMST
- Additional sources of information

Working with IndraWorks

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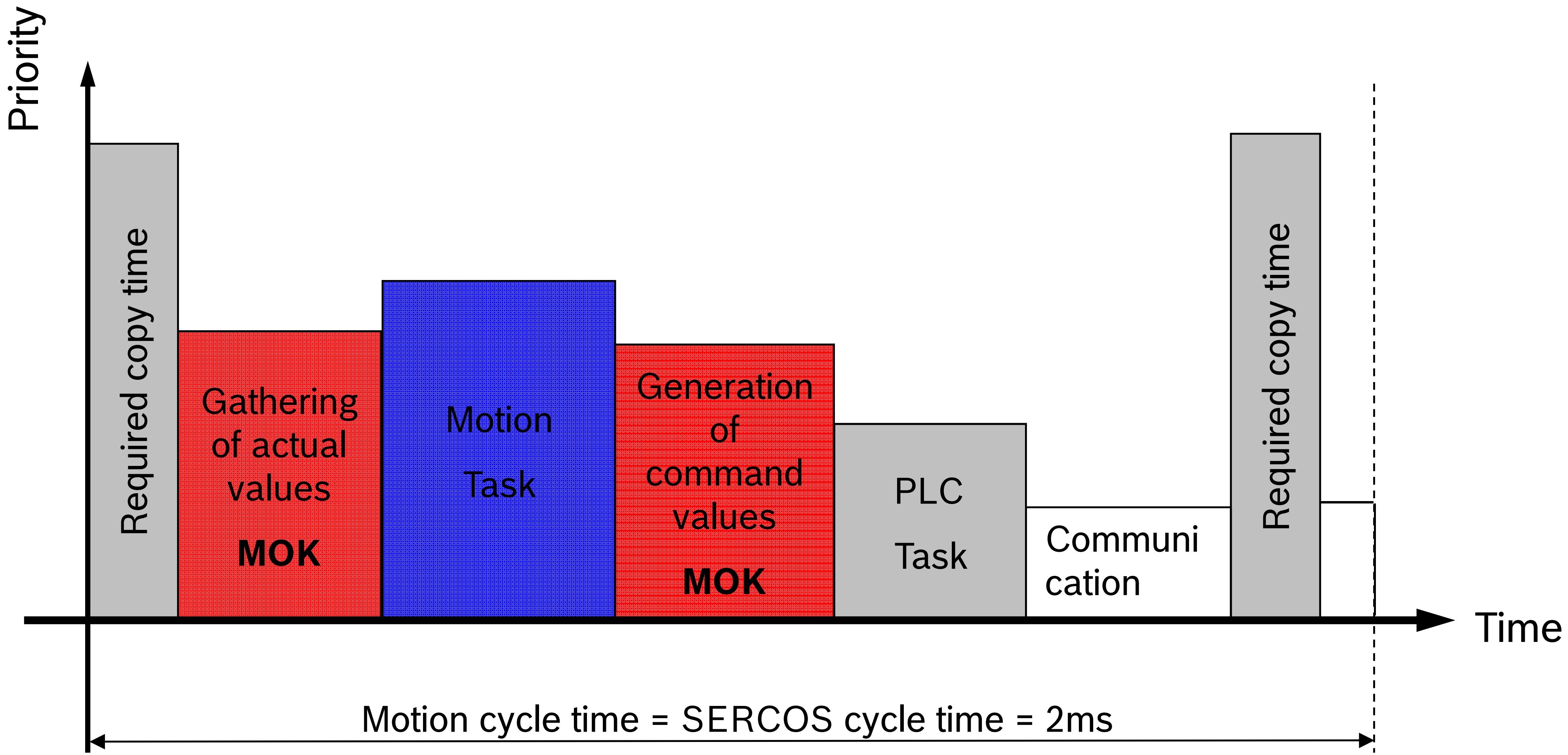
MLC task system

- One motion cycle with default settings



MLC task system

- SERCOS cycle time and Motion cycle time are adjustable independently
  → Allows short SERCOS cycle time (L65: minimum 250µs)
- Adjustment of the priority of the motion kernel
  → Motion kernel can be interrupted by high prior PLC task
  → Fast processing of IO data
- All tasks are priority controlled
  - At a given point in time only one task can be processing (“active”)
  - The RTOS scheduler activates always the “ready” task with the highest priority
- The “behavior” of the overall system can be manipulated by adjusting priority, start condition, and cycle time of the individual tasks
- Visualization of the task scheduling and runtime in the task viewer

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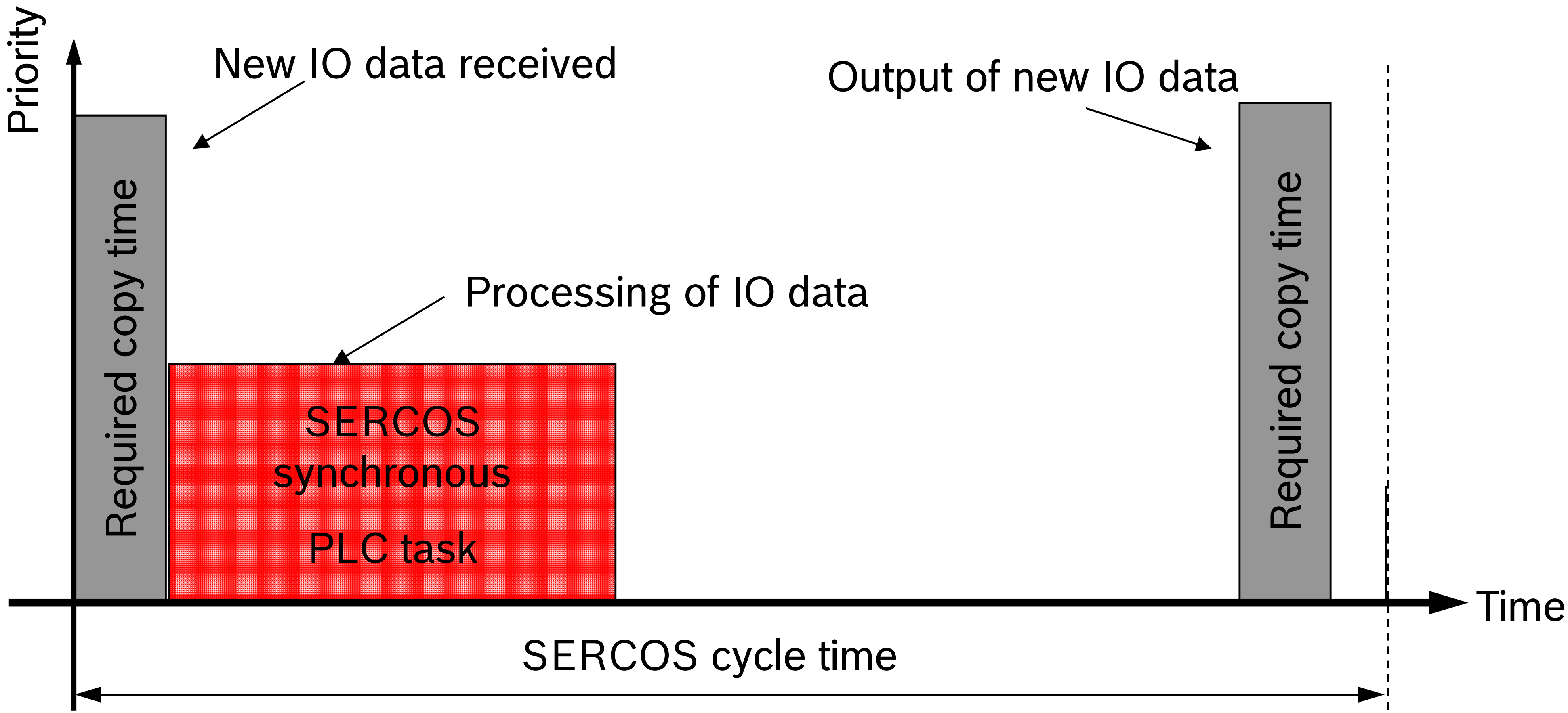
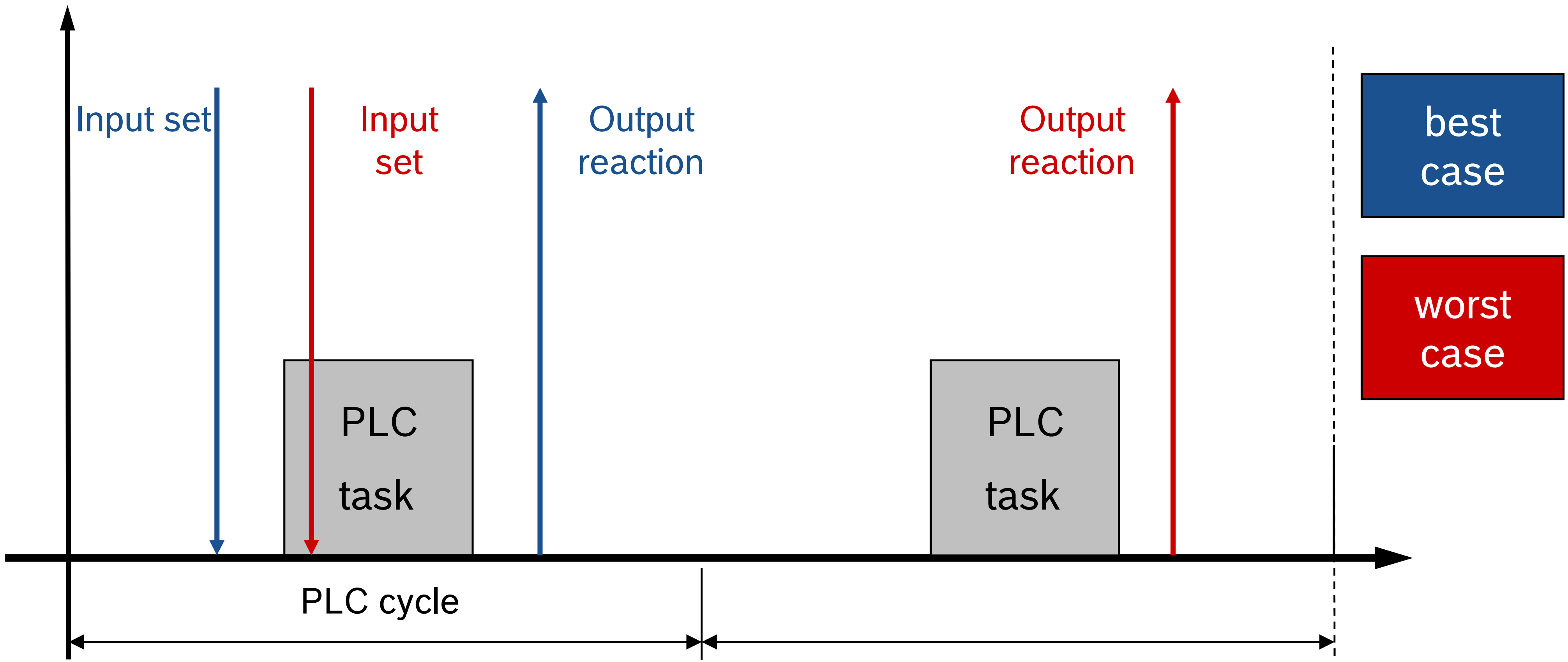
Fast IO turn-around-times – SERCOS IOs

- Fast IO turn-around-times can be achieved
  - ... by adjustment of the SERCOS cycle time to the desired value
  - ... and creating a SERCOS synchronous PLC task with high priority



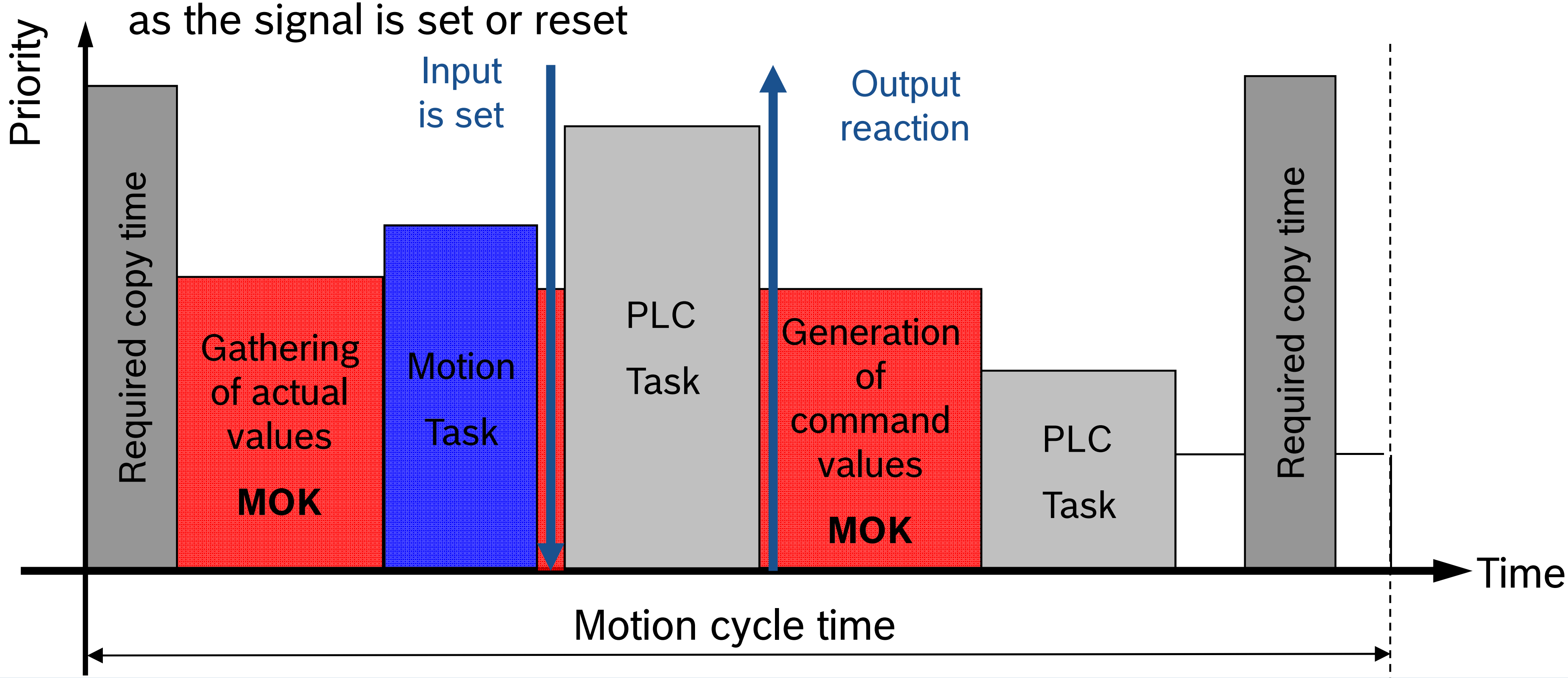
Fast IO turn-around-times – Onboard & Fast IO

- Fast IO turn-around-times can be achieved
  - IO data are processed by a cyclic PLC task
  - A shorter cycle time of this task results in faster turn-around-time

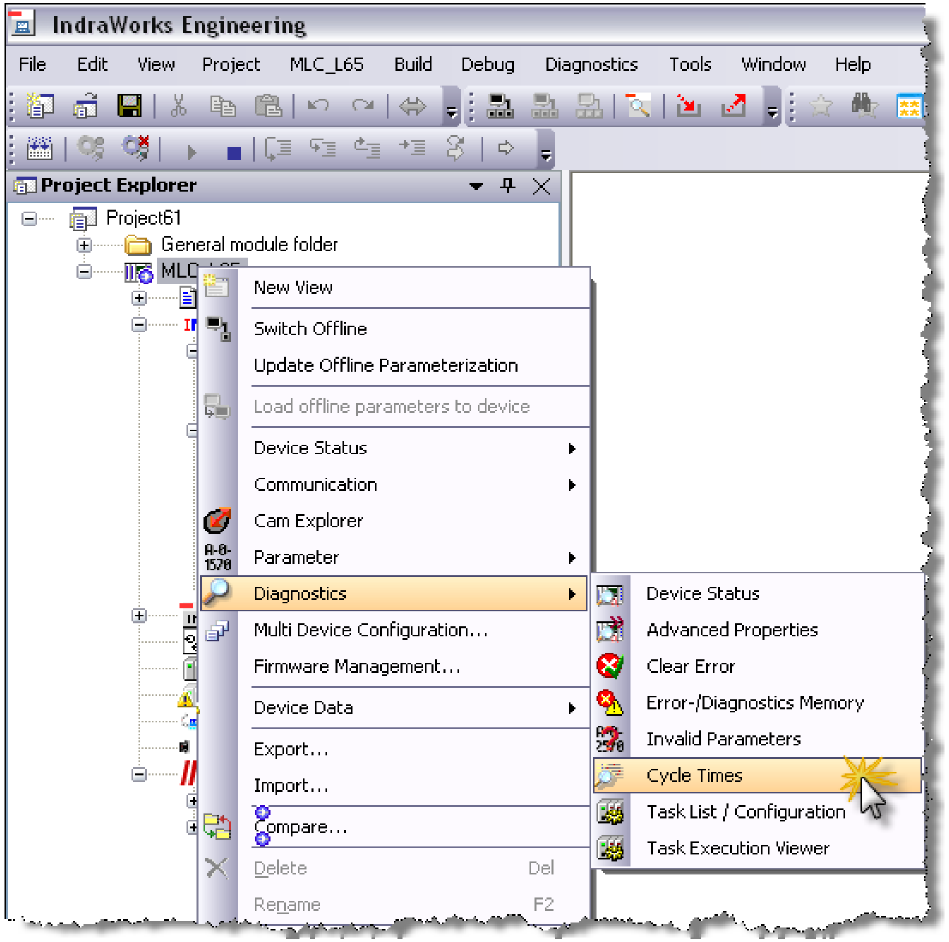


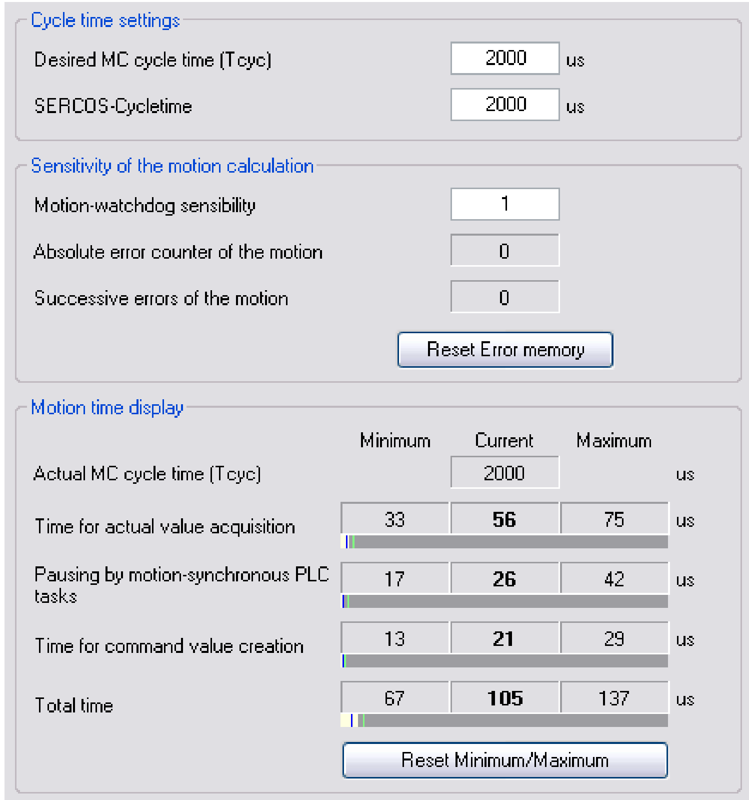
Fast IO turn-around-times – Onboard IO

- Fast IO turn-around-times can be achieved
  - Creation of a task triggered by an external event → input signal of Onboard IOs or complete input byte
  - This task has to be defined with high priority so that it is activated as soon as the signal is set or reset

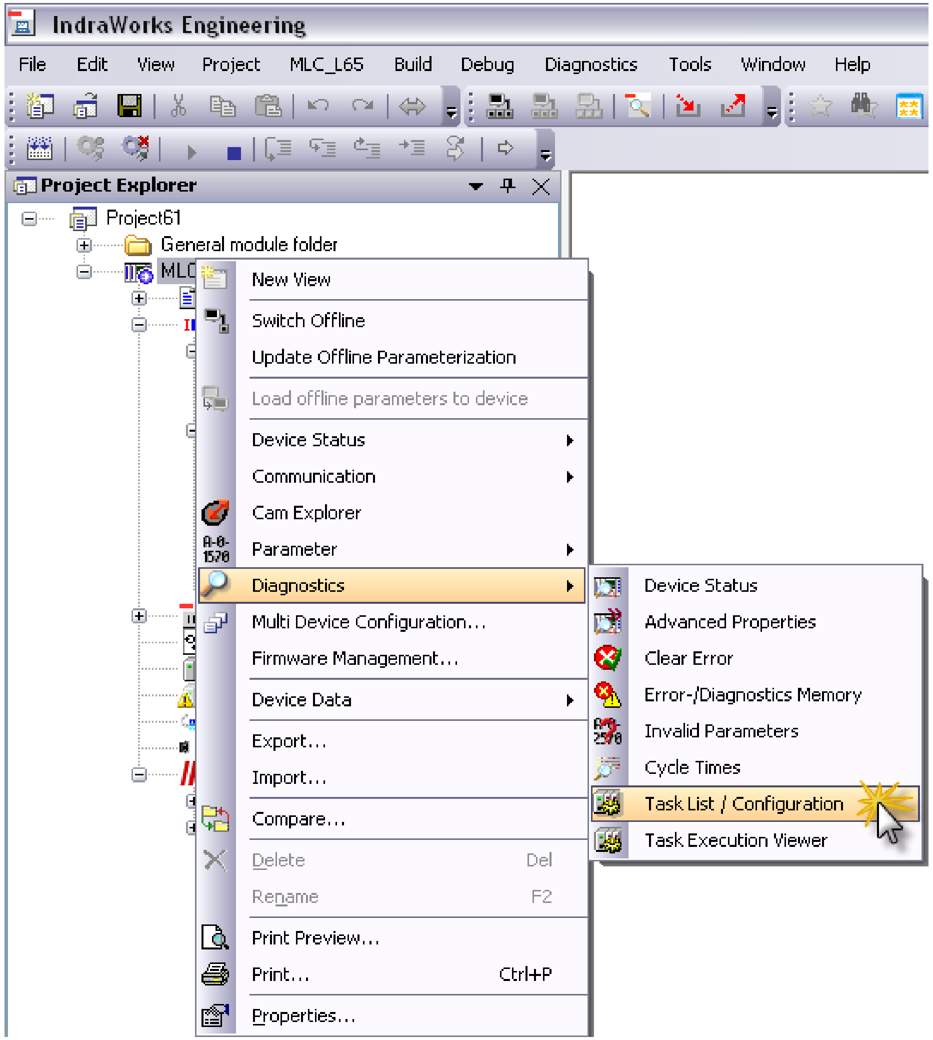


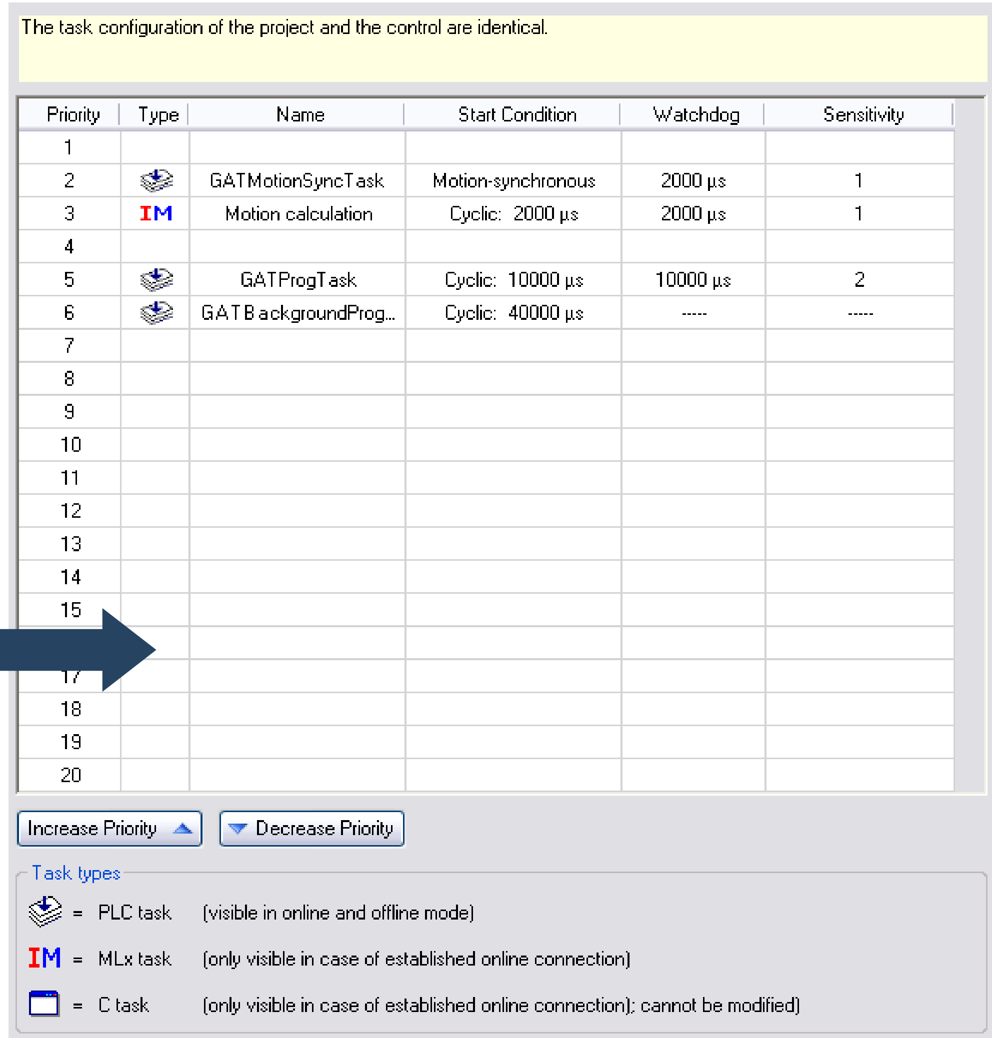
Diagnosing the task system





Diagnosing the task system





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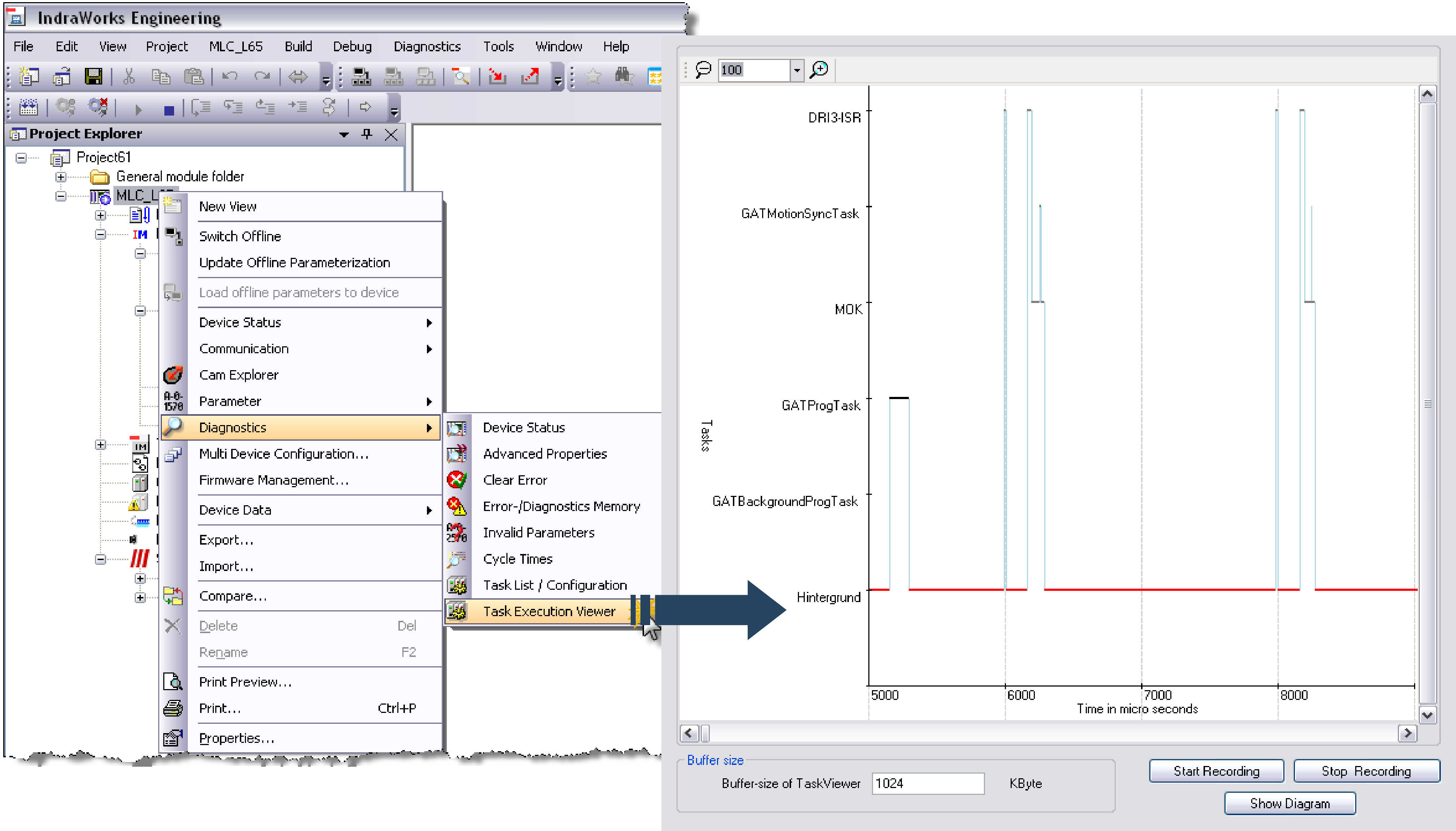
Diagnosing the task system



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Synchronized Motion

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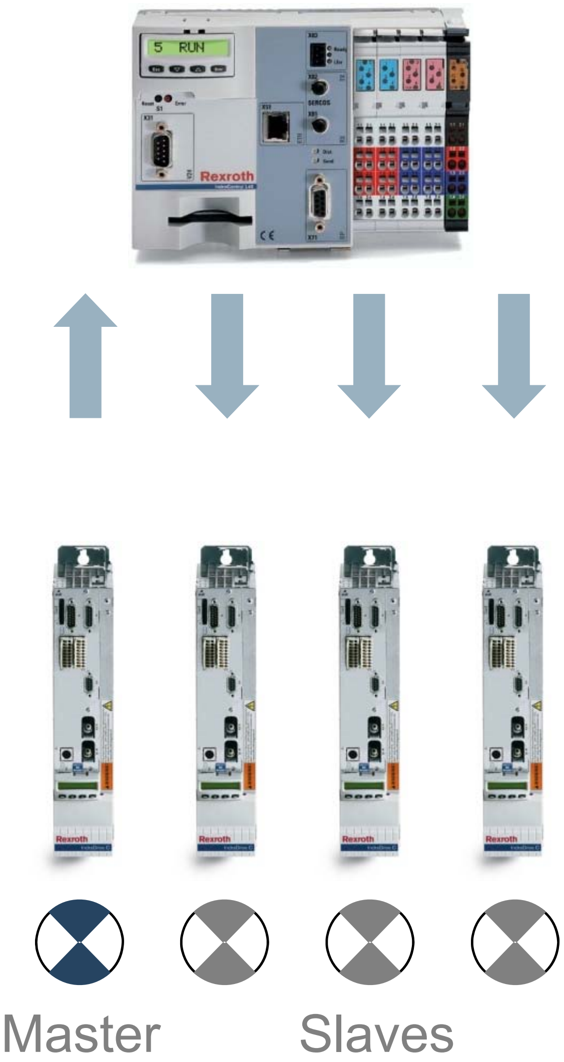
What is synchronized motion?

- In synchronized motion one or more slave axes follow a master axis
- The rotational angle φ of the master axis (master axis position) is sent to the slave axis
- On the basis of this master axis position a synchronous command position is calculated:

  $\psi = f(\varphi)$ rotary axis

  $s = f(\varphi)$ linear axis

- As a result the slave axis follows the master axis synchronously



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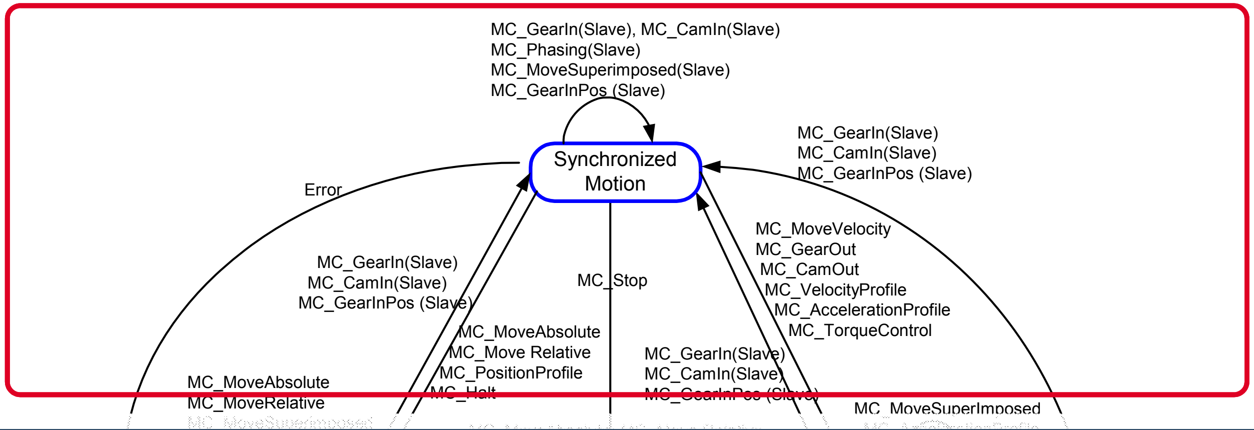
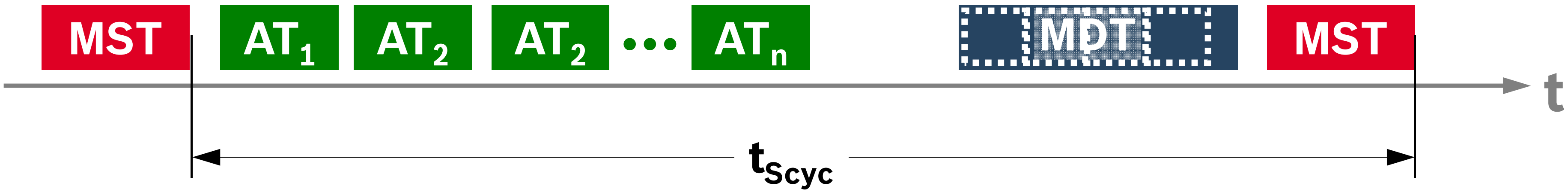
Synchronized motion and bus systems





- A prerequisite of synchronous operation is an adequate communication system
- SERCOS 2 and SERCOS III offer equidistance
- ... based on TDMA → Time Division Multiple Access
- Fieldbus systems like PROFIBUS DP and DeviceNet don't meet this assumption
- Synchronous operation is not available for PROFIBUS drives!

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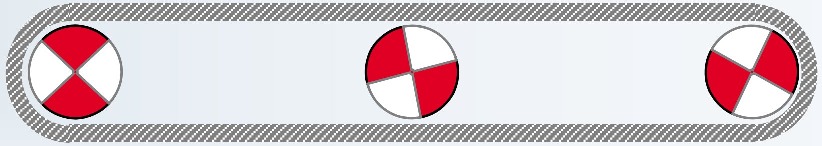
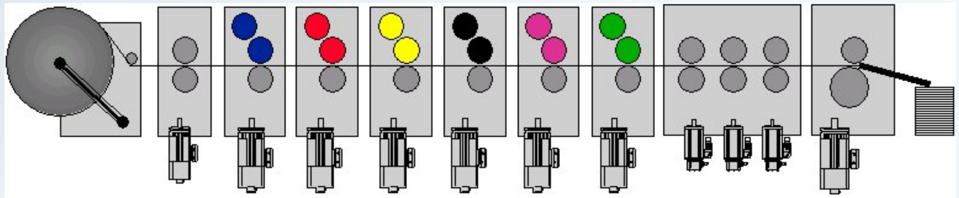
Available synchronization modes

Velocity synchronization



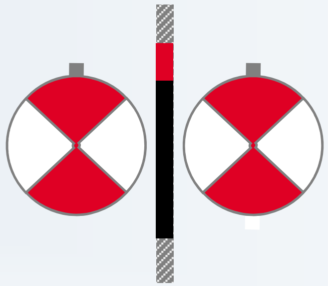
Conveyor belt

Phase synchronization



Offset printing machine

Electronic CAM



Cross seal

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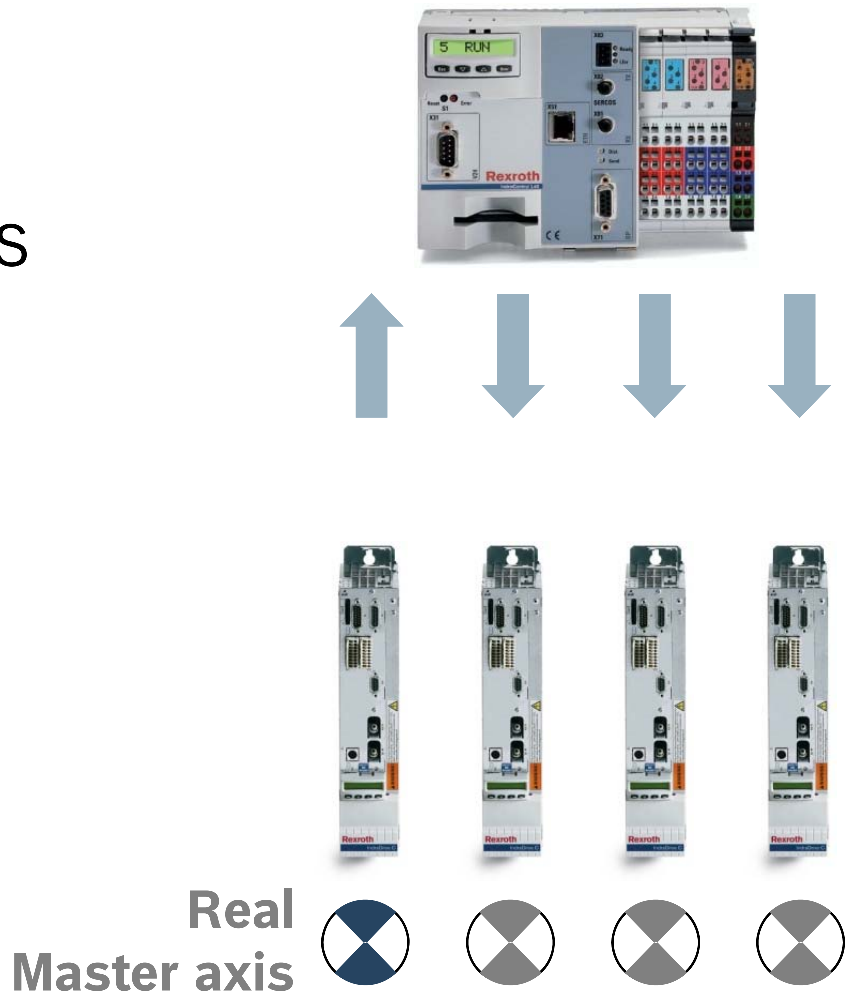
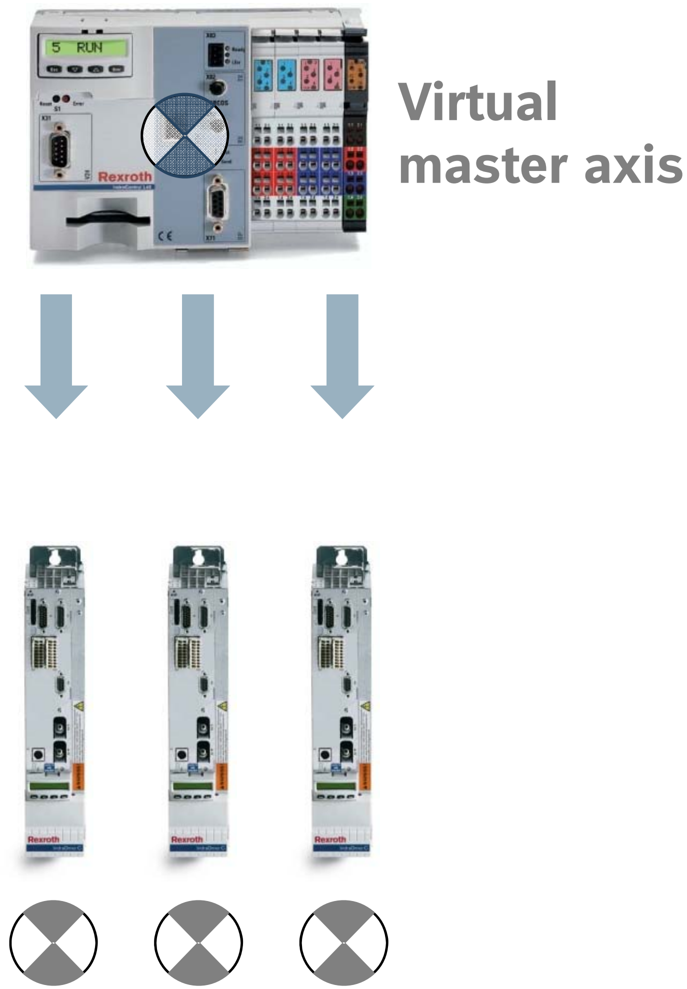
Real master axis

- The master axis position is based on an encoder signal
- For example encoder signal of an IndraDrive connected by SERCOS to an MLC



Virtual master axis

- Master axis positions are generated internally on controller level
- ... and transmitted cyclically to the slave axis
- As a result the real slave axes follow this virtual master
- A virtual axis is an imaginary object which exists only in software
- ... and can be controlled with standard PLCopen function blocks or the AxisInterface



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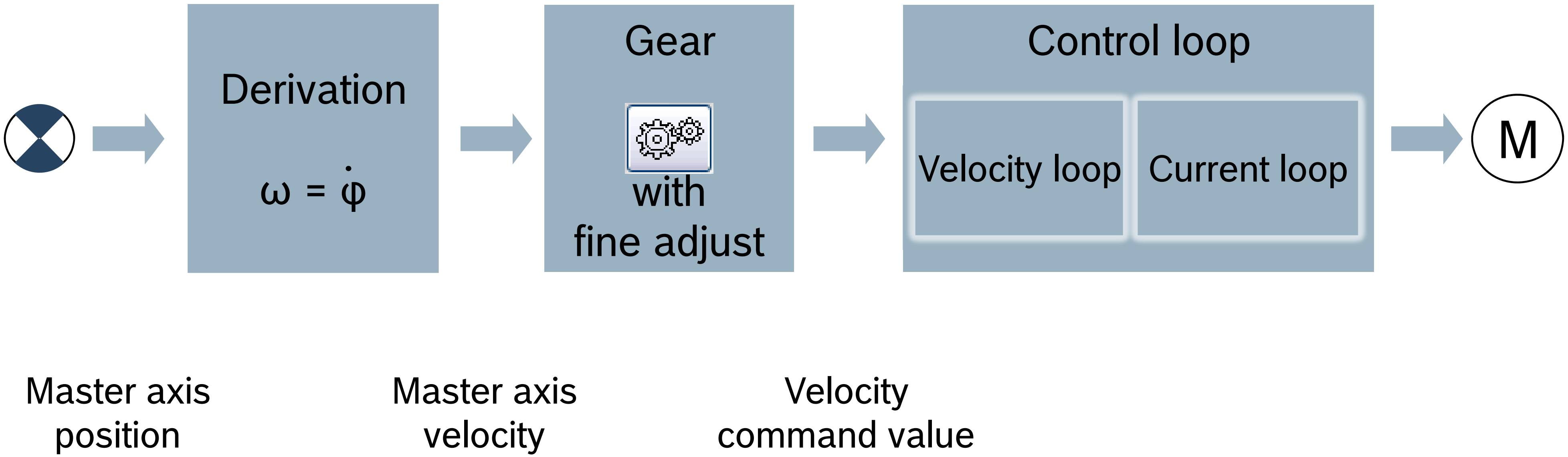
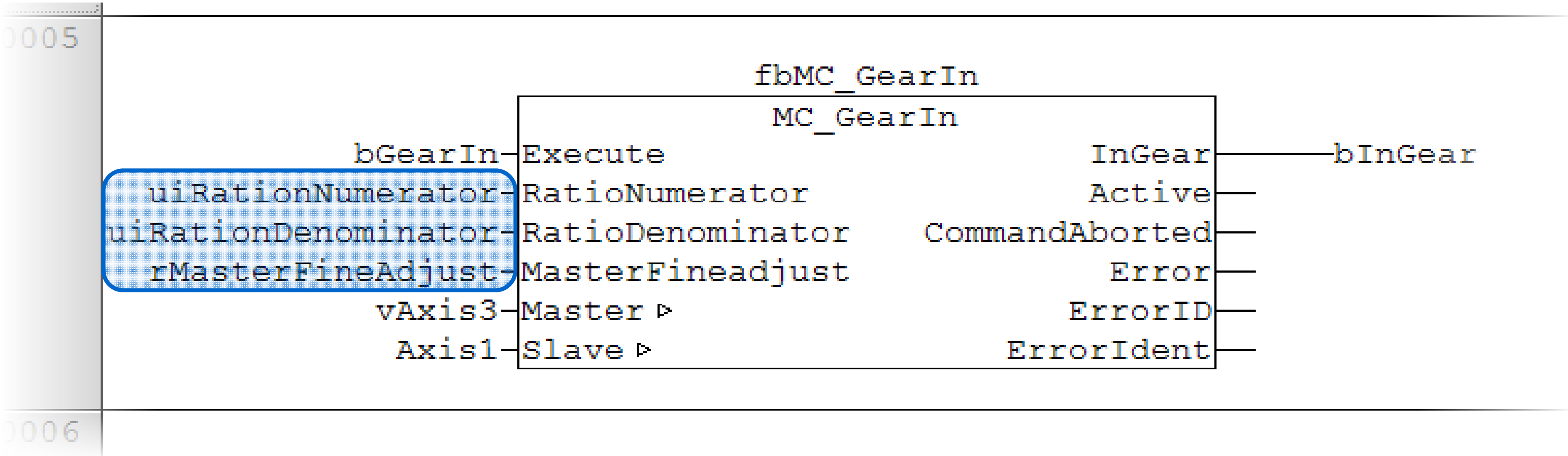
Velocity synchronization (1)



Velocity synchronization (2)



Execute	*→ refer to general information on PLCopen*
RationNumerator,	
RatioDenominator	*Gear ratio*
MasterFineadjust	*Master fine adjust (e.g. tension control)*

$$n_{Slave} = n_{Master} \times \frac{nRatioNumerator}{RatioDenominator} \times (1 + MasterFineadjust)$$

Master	*Master axis*
Slave	*Slave axis*
InGear	*Slave axis in velocity synchronization*

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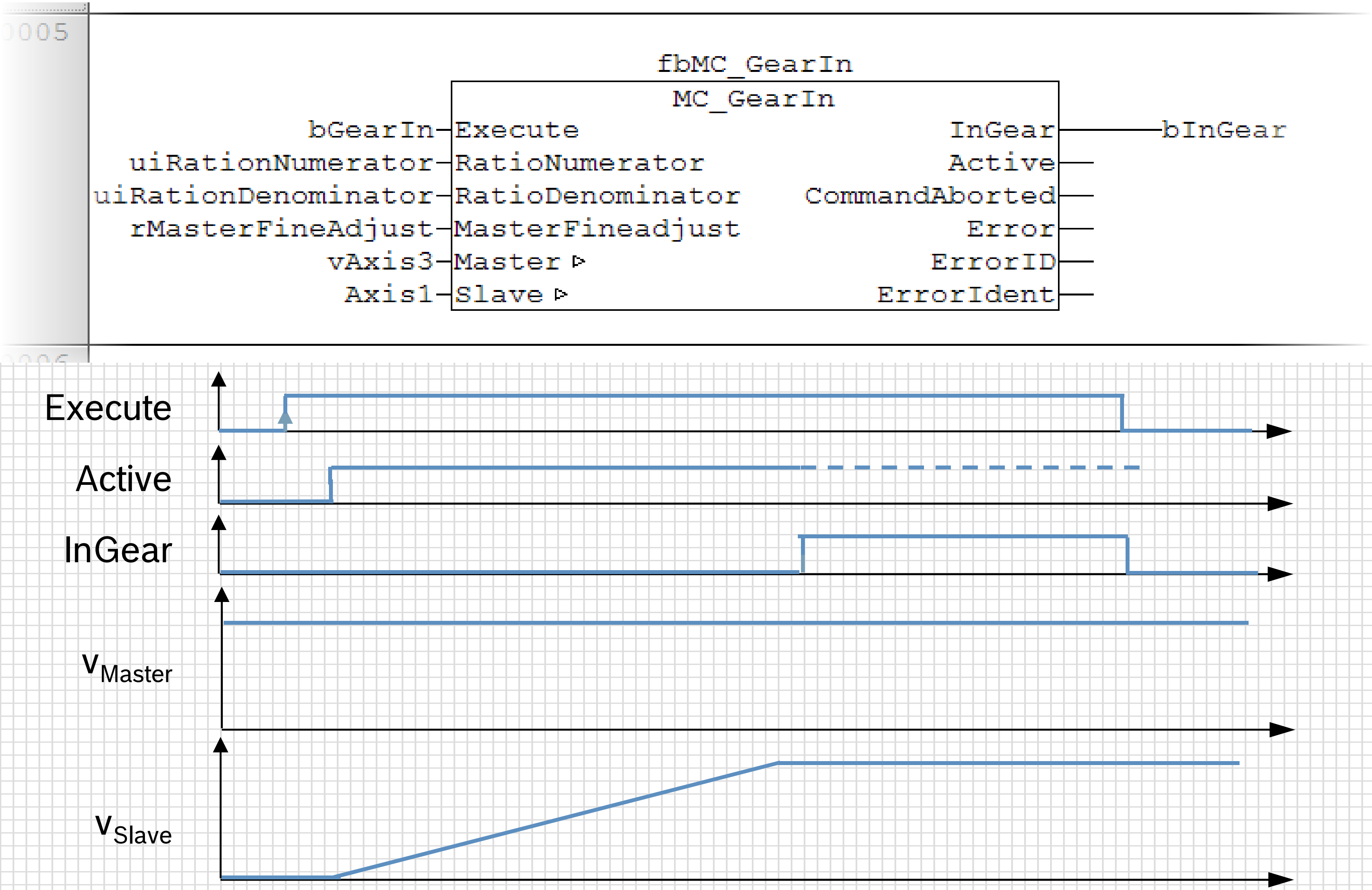
Velocity synchronization (3)



Velocity synchronization (4)

ML_TechInterface.library
Operation Mode "Velocity Synchronization"

When this operation mode is activated, the drive is operated using an electronic gear with fine adjustment. This functionality causes a velocity synchronization between the master axis and the selected slave axis.

The following command activates the operation mode:

```
arAxisCtrl_gb[].Admin._OpMode.en: = ModeSyncVel;
```

or

```
arAxisCtrl_gb[].Admin._OpMode.b.MODE_SYNC_VEL: = TRUE;
```

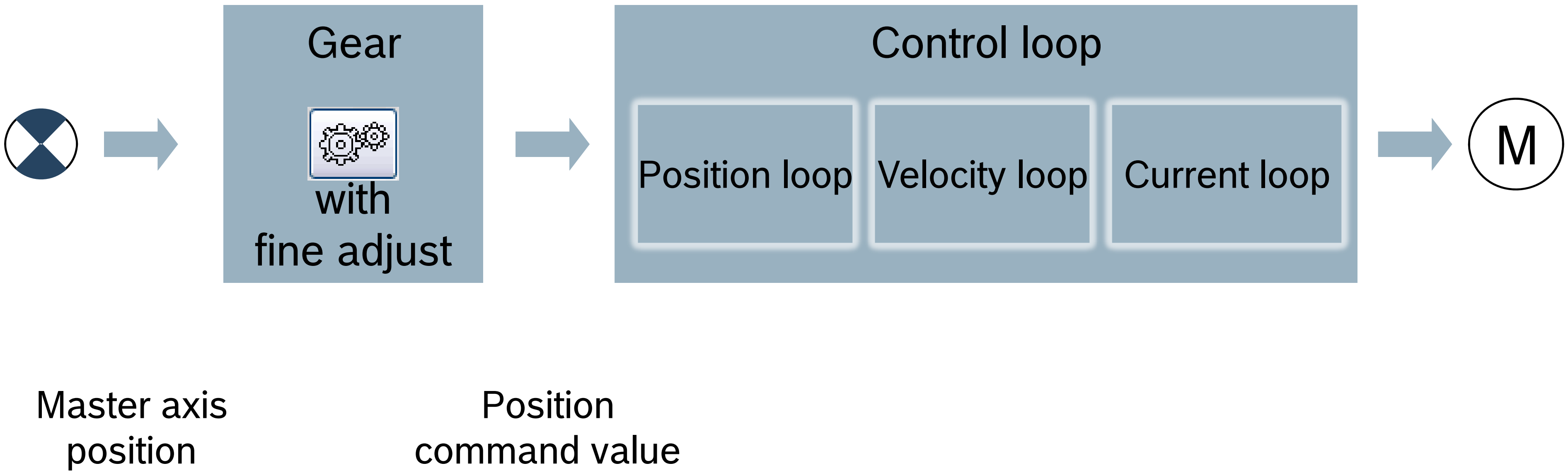
The AxisInterface uses the MC_Power, MC_GearIn and MC_GearOut PLCopen FBs internally to carry out the switchover.

The following table contains the attributes supported by this operation mode:

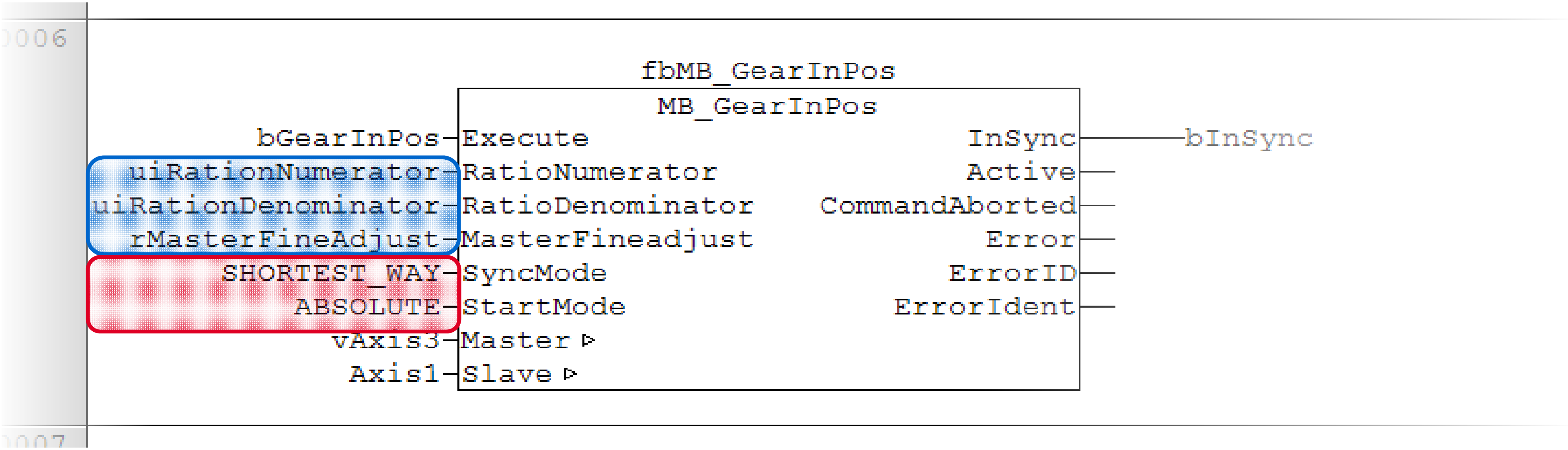
Element	Name	Type	Default	Cyclically scanned
arAxisCtrl_gb[ ]	SyncMode.OutputRevolution	UINT	1	Yes
	SyncMode.InputRevolution	UINT	1	Yes
	SyncMode.Fineadjust	REAL	0.0	Yes
	SyncMode.Master	AXIS_REF		Yes
	Admin.Axis	AXIS_REF		No
arAxisStatus_gb[]	Admin.MODE_SYNC_VEL	BOOL		Not applicable

Attributes, operation mode "velocity synchronization"

Phase synchronization (1)



Phase synchronization (2)



Execute	*→ refer to general information on PLCopen*
RationNumerator,	
RatioDenominator,	*→ refer to MC_GearIn*
MasterFineadjust	
SyncMode	*Synchronization direction*
StartMode	*Synchronization type (absolute, relative)*
Master	*Master axis*
Slave	*Slave axis*
InSync	*Slave axis in phase synchronization*

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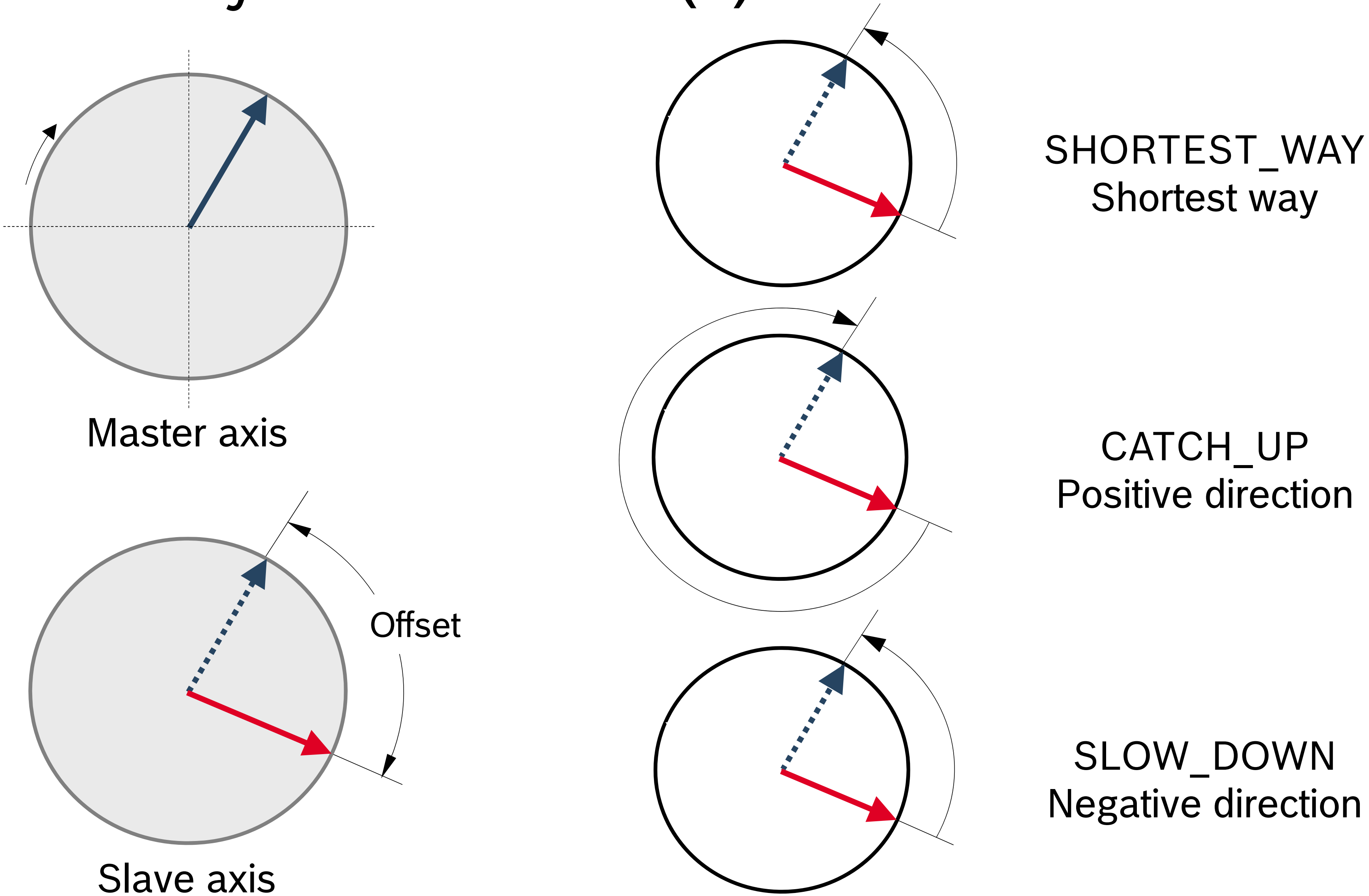
Phase synchronization (3)



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Phase synchronization (4)

ML_TechInterface.library
Operation Mode "Phase Synchronization"

When this operation mode is activated, the drive is operated using an electronic gear with fine adjustment. This functionality causes a phase synchronization between the master axis and the selected slave axis.

The following command activates the operation mode:

```
arAxisCtrl_gb[].Admin._OpMode.en: = ModeSyncPhase;
```

or

```
arAxisCtrl_gb[].Admin._OpMode.b.MODE_SYNC_PHASE: = TRUE;
```

☞ The AxisInterface uses the MC_Power, MB_GearInPos, MC_GearOut and MB_PhasingSlave PLCopen FBs internally to carry out the switchover.

The following table contains the attributes supported by this operation mode:

Element	Name	Type	Default	Cyclically scanned
arAxisCtrl_gb[ ]	SyncMode.OutputRevolution	UINT	1	Yes
	SyncMode.InputRevolution	UINT	1	Yes
	SyncMode.Fineadjust	REAL	0.0	Yes
	SyncMode.SyncDirection	MC_SYNC_DIRECTION	SHORTEST_WAY	Yes
	SyncMode.StartMode	MC_START_MODE	ABSOLUTE	No
	SyncMode.Master	AXIS_REF		Yes
	Admin.Axis	AXIS_REF		No
	SyncMode.PhaseOffset	REAL	0.0	Yes
	SyncMode.PhaseOffsetVel	REAL	1.0	No
	SyncMode.PhaseOffsetAcc	REAL	100.0	No
arAxisStatus_gb[]	Admin.MODE_SYNC_PHASE	BOOL		Not applicable
	SyncMode.PhasingSlaveDone	BOOL		Not applicable

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Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- Motion Programming Basics
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Synchronized Motion

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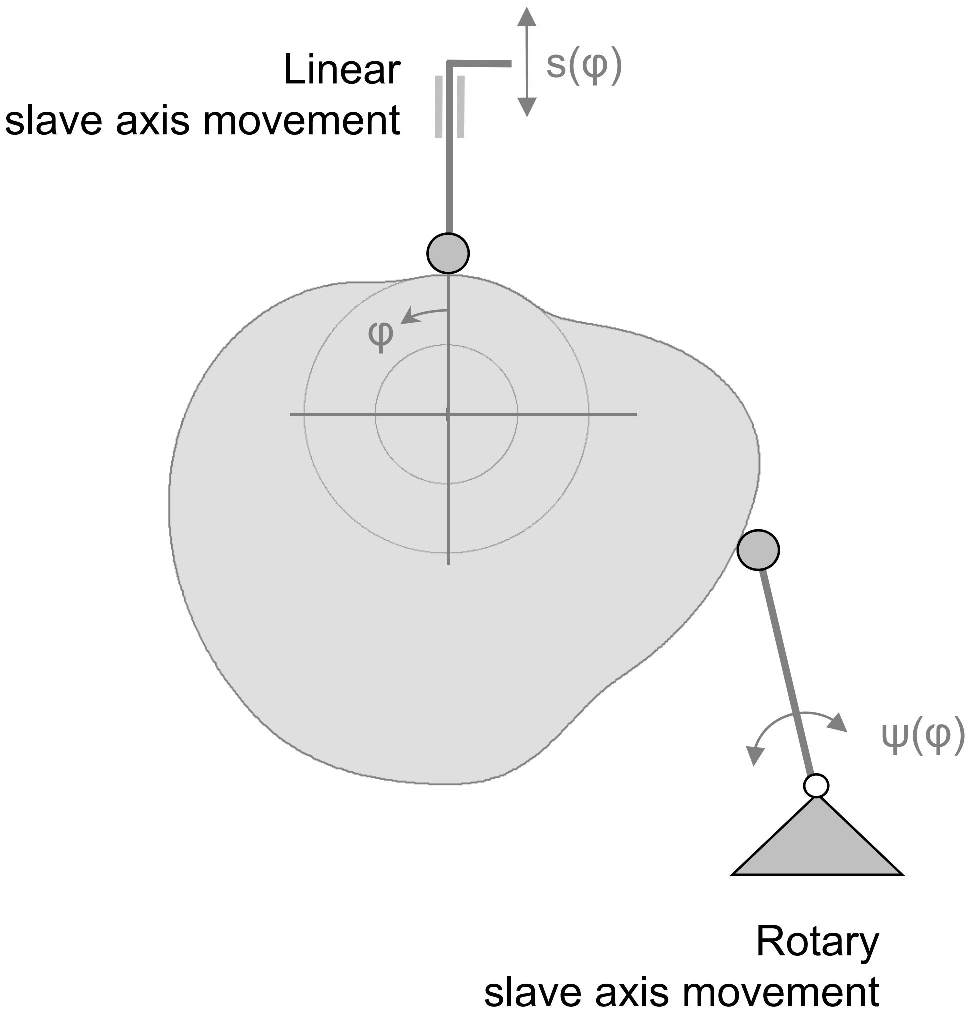
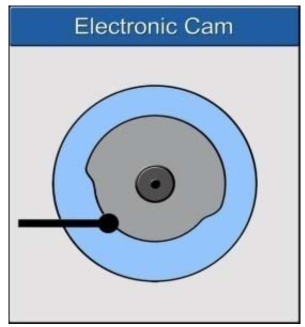


Cams

- For complex non-linear motion profiles
- Generate rotary or linear slave axis movement
- … depending on a master axis
- Master signal is rotary angle φ (generally)



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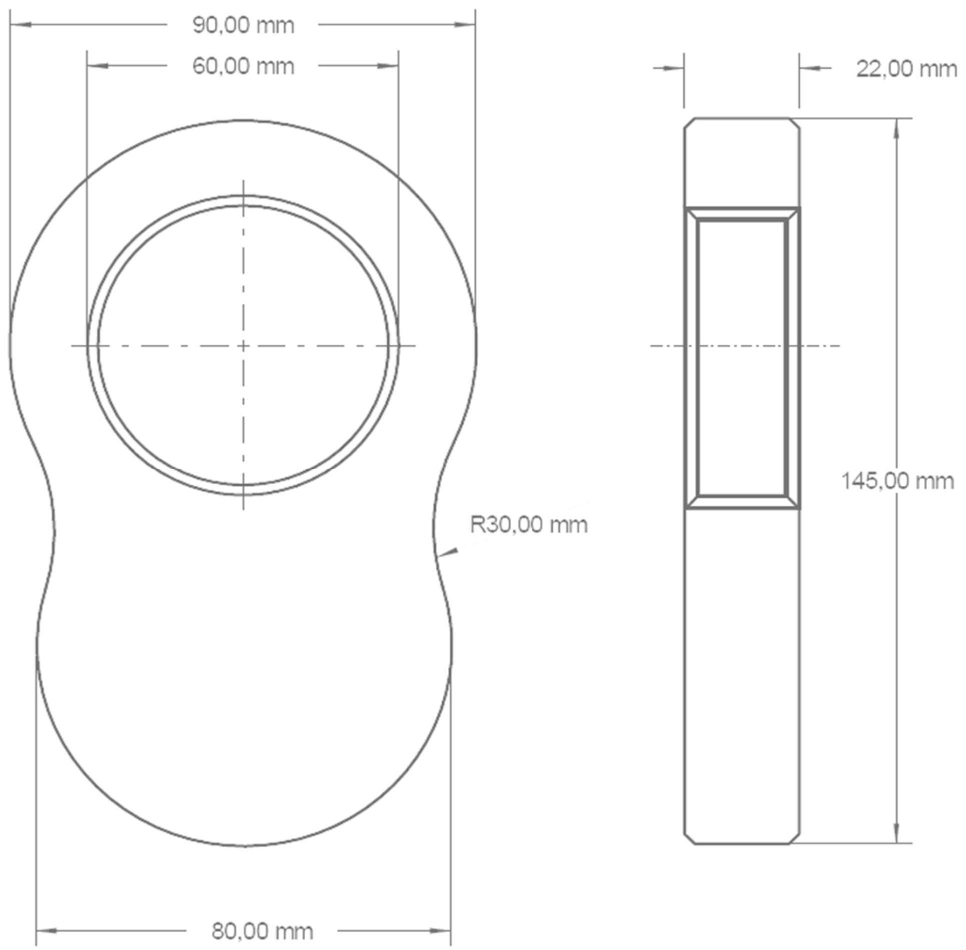
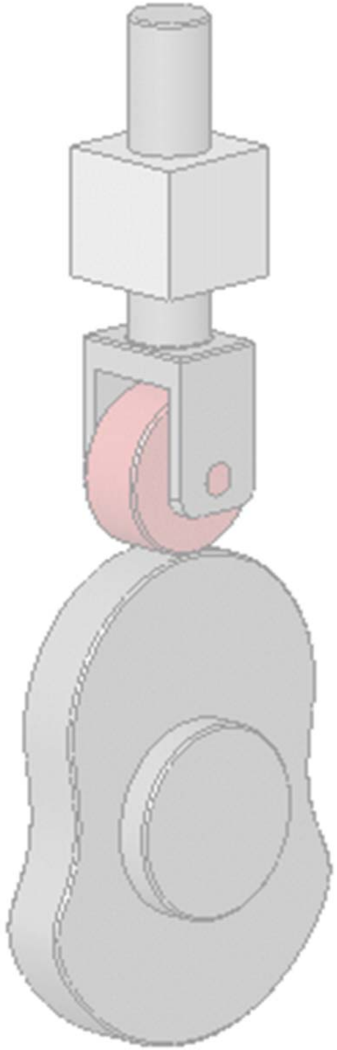
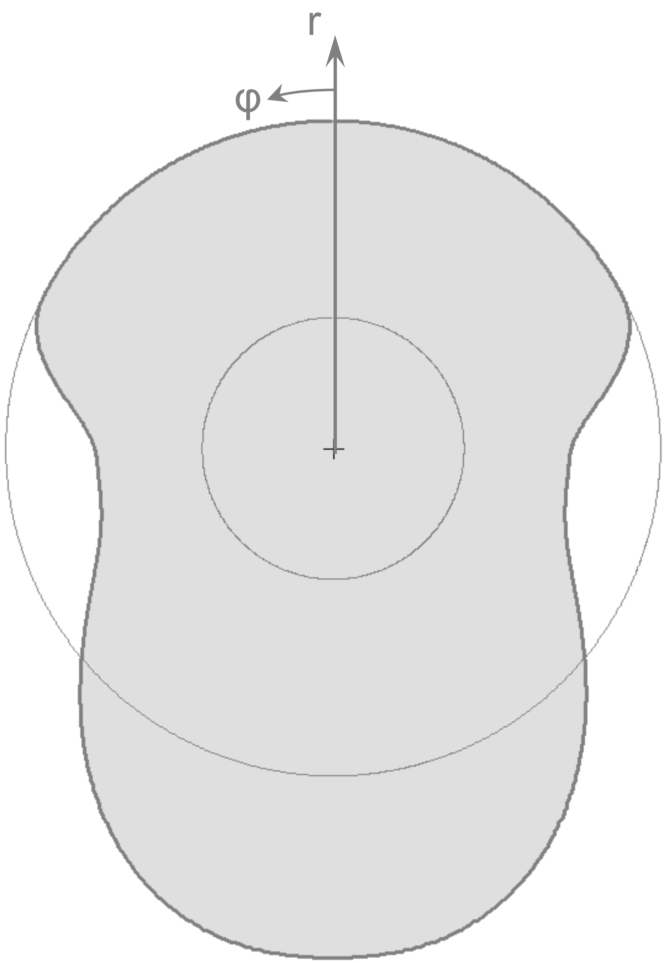
Cams – Examples (1)

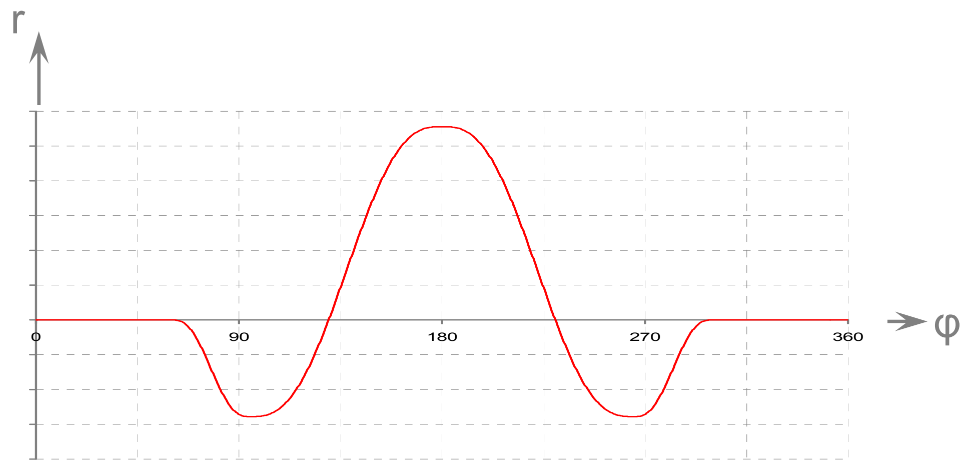




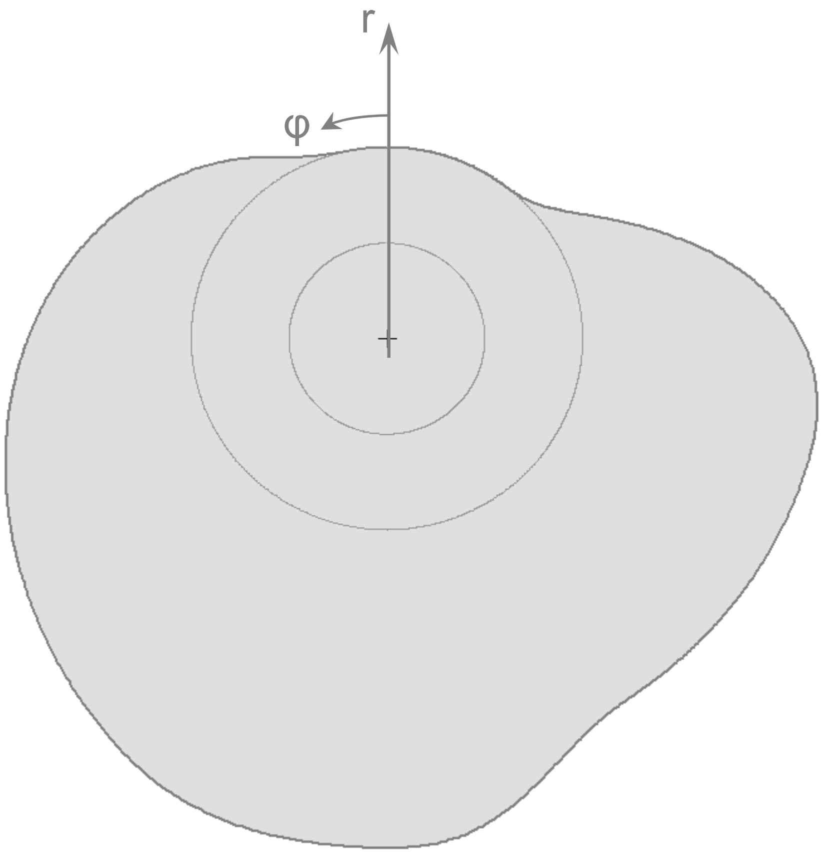
Source: www.wikipedia.de

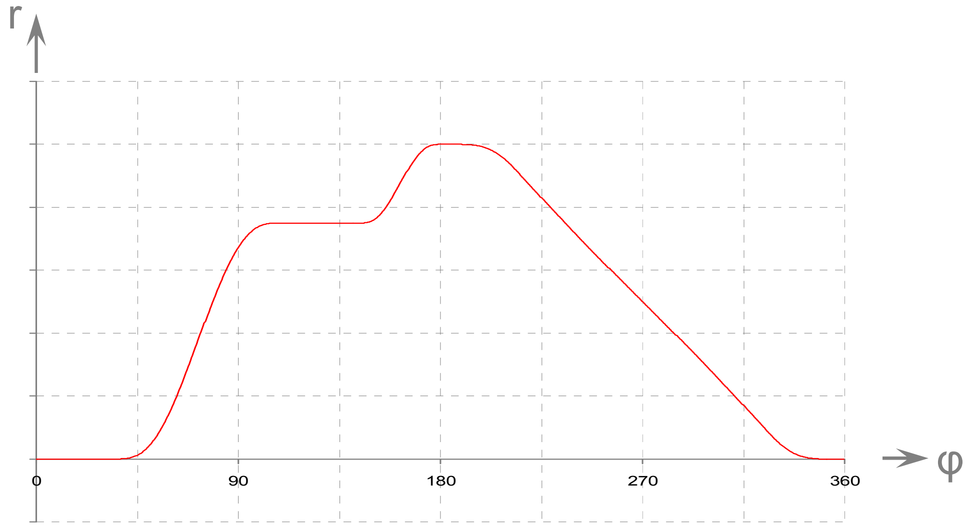
Cams – Examples (2)





Cams – Examples (3)

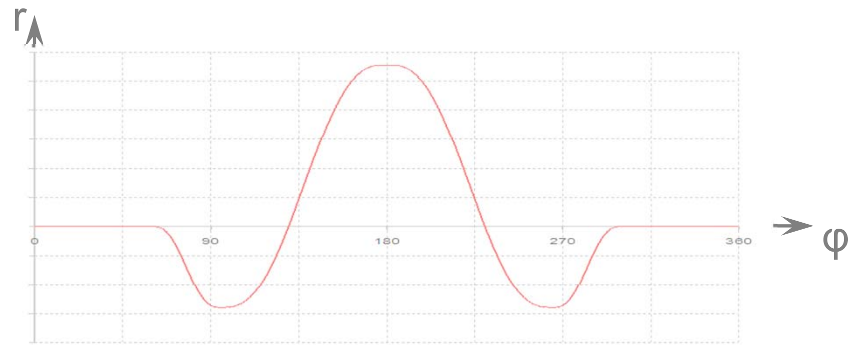




Cams – Overview

- Cam
  - Mechanical Cam
  - Electronic Cam
    - Point table
    - Analytic description *Electronic Motion profile*
    - FlexProfile

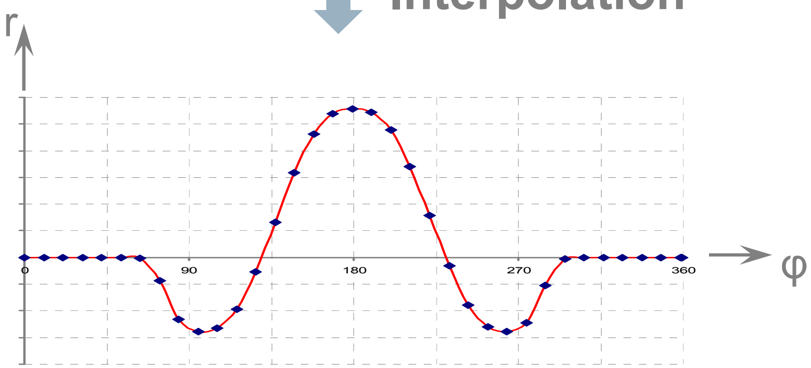
Point table



Point table

φ [Grad]	0	10	358,59375	358,945313	359,296875	360
r [%]	0		0	0	0	0

Interpolation



- Point table describes coupling of master axis and slave axis
- Calculation of slave axis positions is determined by interpolation between discrete table positions

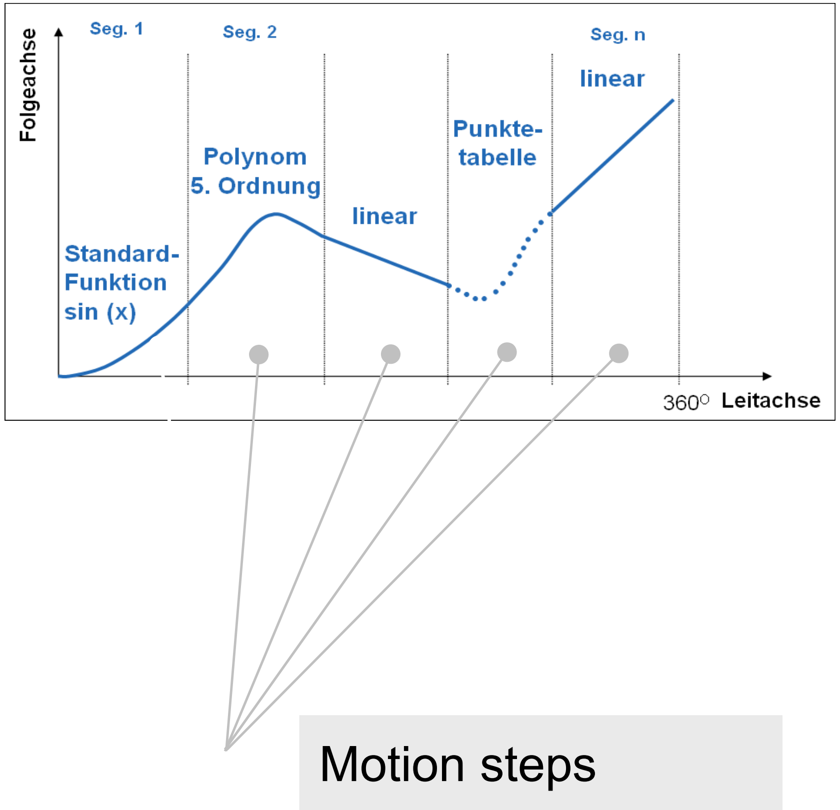
IndraDrive

- 4 tables with 1024 points
- 4 tables with 128 points
- Cubic spline interpolation is used to calculate intermediate slave axis positions
- Values are normalized (Percentage values)
- Cam shaft distance as additional parameter

Point table

- ☑ Simple and clear function
- ☑ Low computing effort
  All the coefficients required for spline interpolation are determined in a calculation that is made before the time of execution
- ☑ The slave axis position can then be determined very easily during runtime
- ☑ Suitable if no online modification is required
- ☑ Easy data exchange with 3rd party tools
- ☒ All polynomial coefficients must be recalculated after a support point is changed
- ☒ Large data amounts

Motion Profile



- Mathematical function describes motion curve („Formula“)
- These motion rules describe the coupling of slave axis and master axis as an analytical function
- The complete motion profile is defined step-by-step using motion rules
- A step with the associated motion rule is called “motion step”
- For more details on motion rules refer to VDI 2143
- Mainly polynomial functions of the 5th order or higher are used

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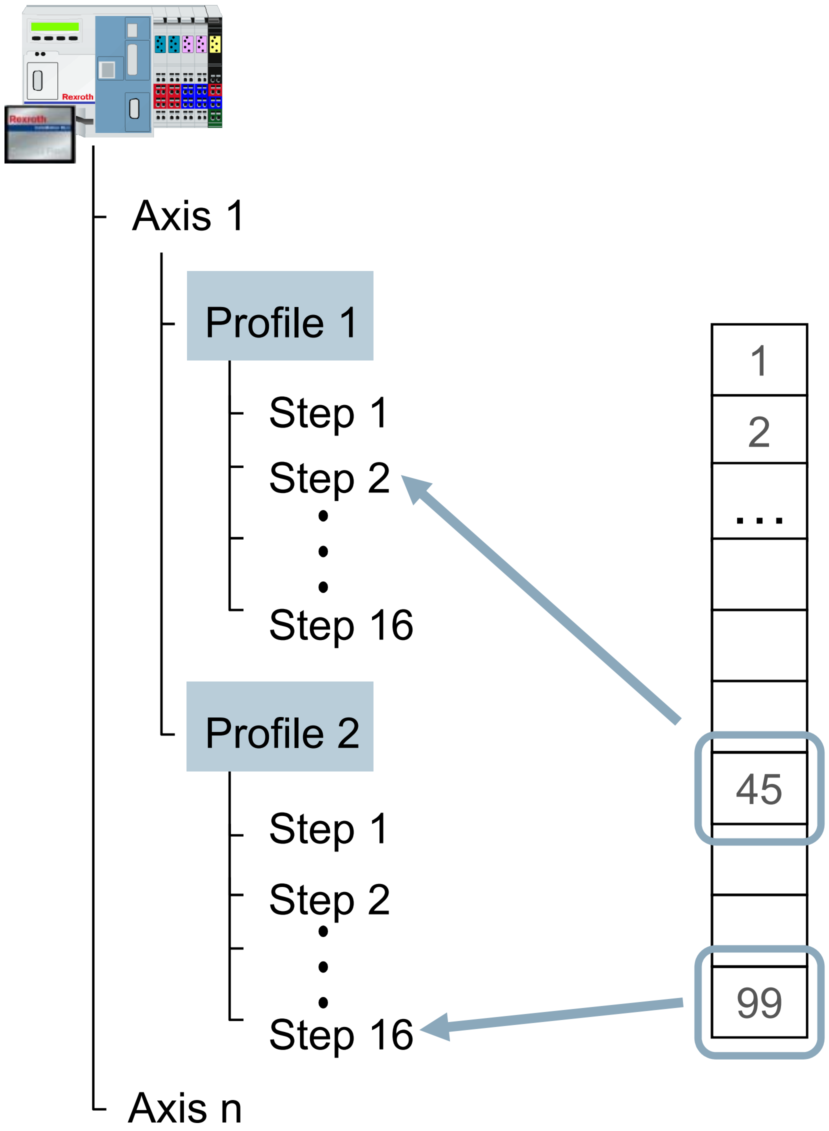
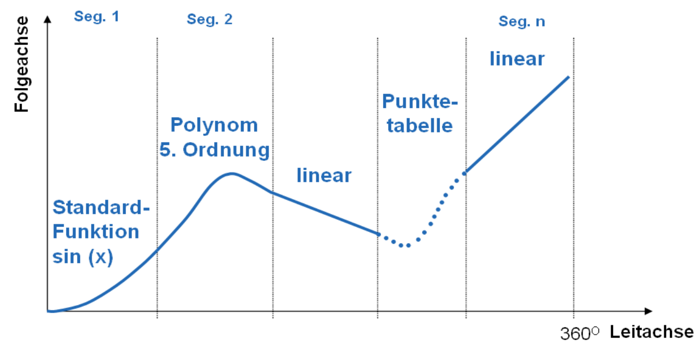
Motion Profile



- 2 **Motion profiles** per axis with up to 16 steps
- Point tables still can be used within a motion profile
- 99 Point tables with 1024 points
- **Processing in PLC program** by function blocks
  - MB_MotionProfile
  - MB_ChangeProfileSet
  - MB_ChangeProfileStep
  - MB_ChangeCamData



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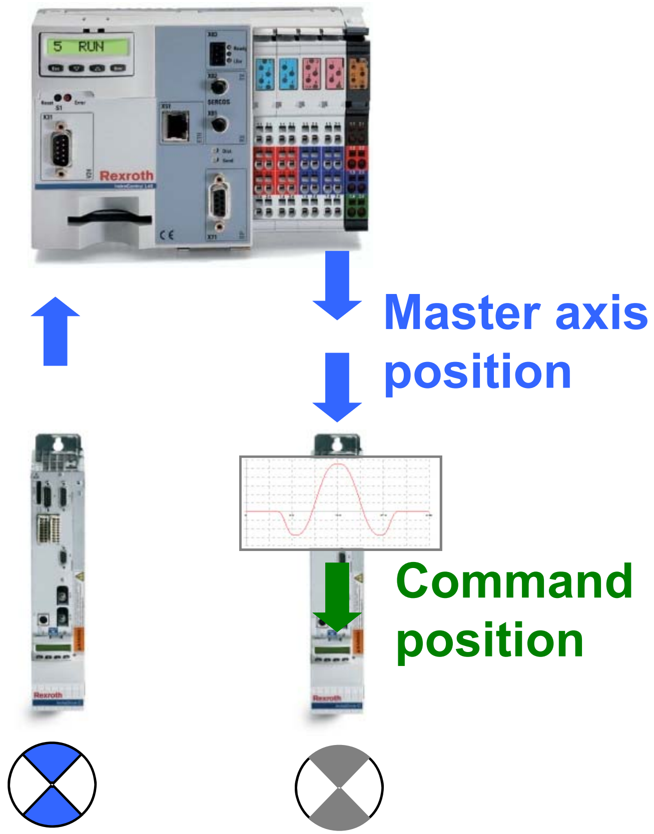
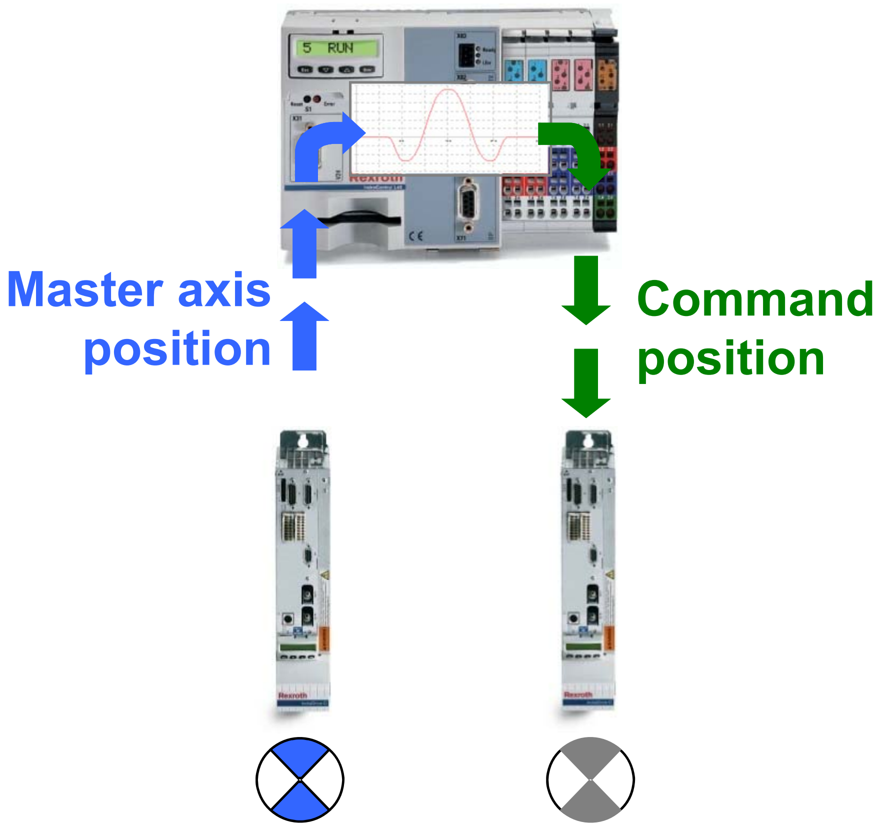
Centralized and decentralized Cam concept

Decentralized Concept



Centralized Concept



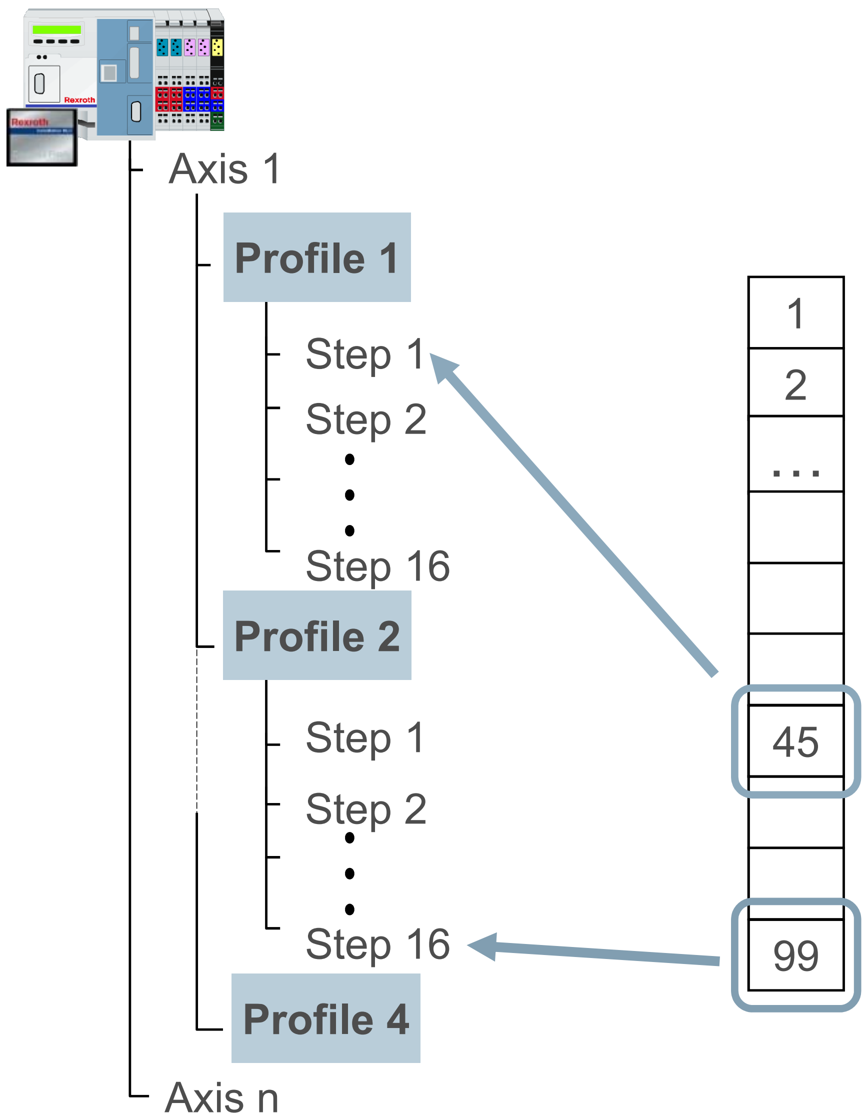
- ☑ Drives with basic operation sufficient
- ☒ Increased Computing effort in MC

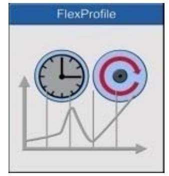
Motion Profile

- ☑ Low amounts of data
  The required information is limited to only the specification of the boundary values of the individual motion steps
- ☑ Easy online modification
  The individual motion steps easily can be modified independent of other motion steps
- ☑ Only drives with basic operation required
  Drives with PackProfile are adequate

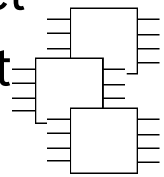
- ☒ Increased computing effort at runtime

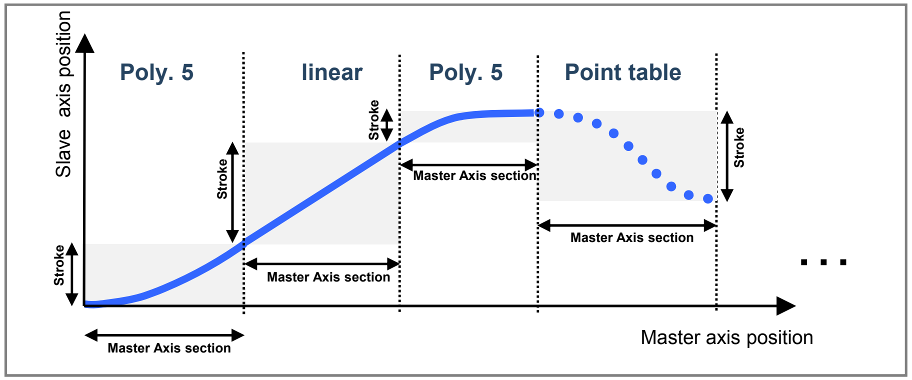
FlexProfile





- **4 FlexProfiles** per axis
- Max. 16 steps described by motion rules or point tables with max. 1024 points
- **Processing in PLC program** by function blocks
  - ML_FlexProfile
  - MB_ChangeFlexProfileSet
  - MB_ChangeFlexEventSet
  - MB_ChangeCamData





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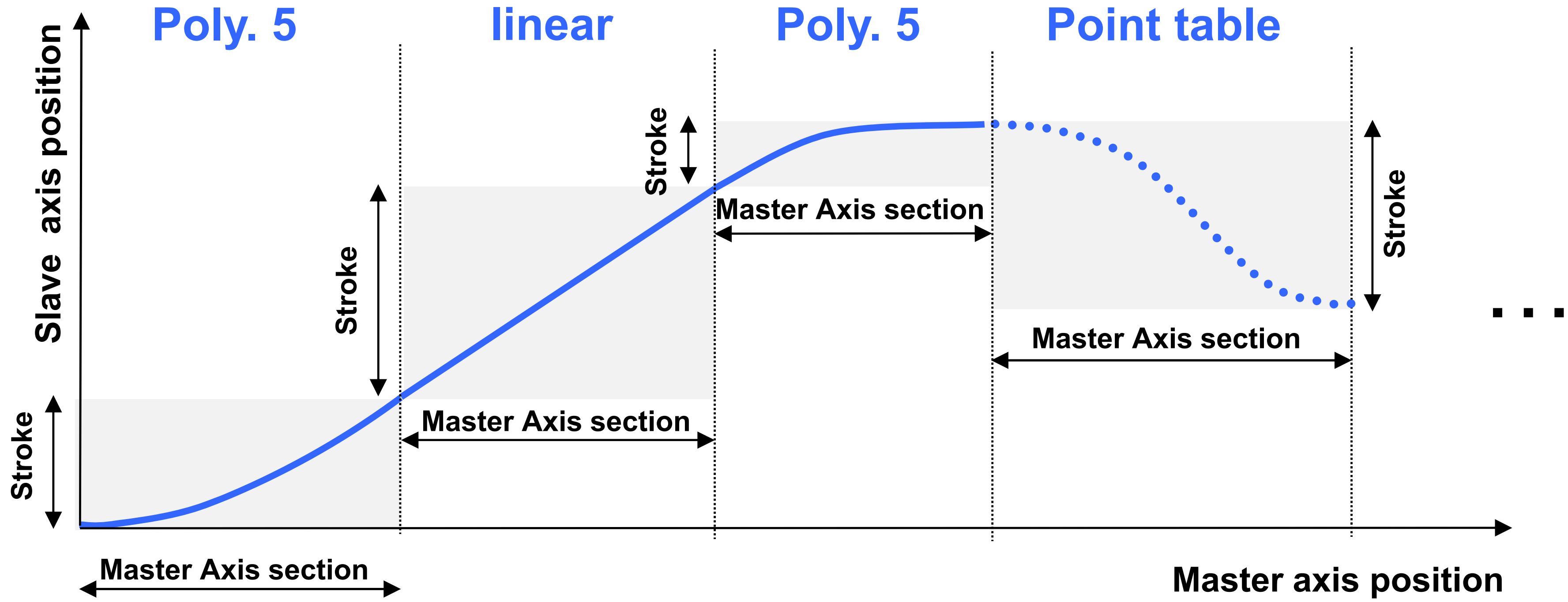
FlexProfile – Basic Features



- Compose up to 16 motion steps to a FlexProfile
- Free scaling of master axis (no restriction to 360°)
- Free scaling of slave axis (no restriction to 360°)
- Relative definition of master axis section and stroke
- Time controlled motion supported
- Rich set of motion rules (cf. VDI 2143 and extended motion rules)

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FlexProfile – Standardized Motion rules (VDI 2143)

Rest → Rest
- Standstill *
- Simple sinoid (simple sine curve) *
- Bestehorn sinoid (offset sine curve)
- Acceleration-optimized offset sine curve *
- Torque-optimized offset sine curve *
- Gutman sinoid *
- Modified sinoid *
- Modified acceleration trapezoid *
- 5^{th}-degree polynomial
- 7^{th}-degree polynomial *

Velocity → Velocity
- Linear interpolation
- 5^{th}-degree polynomial
- 7^{th}-degree polynomial *
- Modified sinoid *

Rest → Velocity
- 5^{th}-degree polynomial
- 7^{th}-degree polynomial *

Velocity → Rest
- 5^{th}-degree polynomial
- 7^{th}-degree polynomial *

General motion
- 5^{th}-degree polynomial
- 7^{th}-degree polynomial *

*) only with FlexProfile

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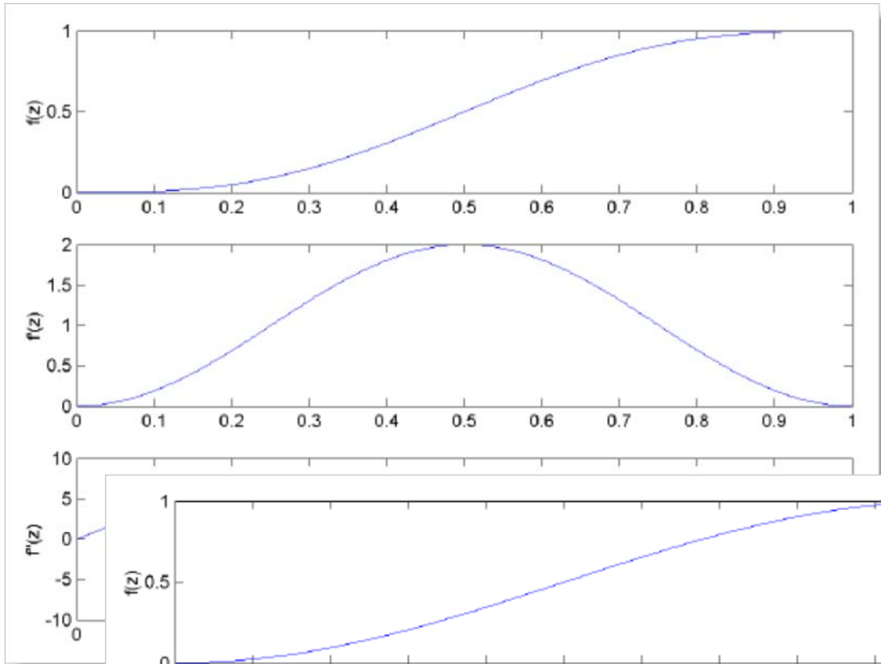
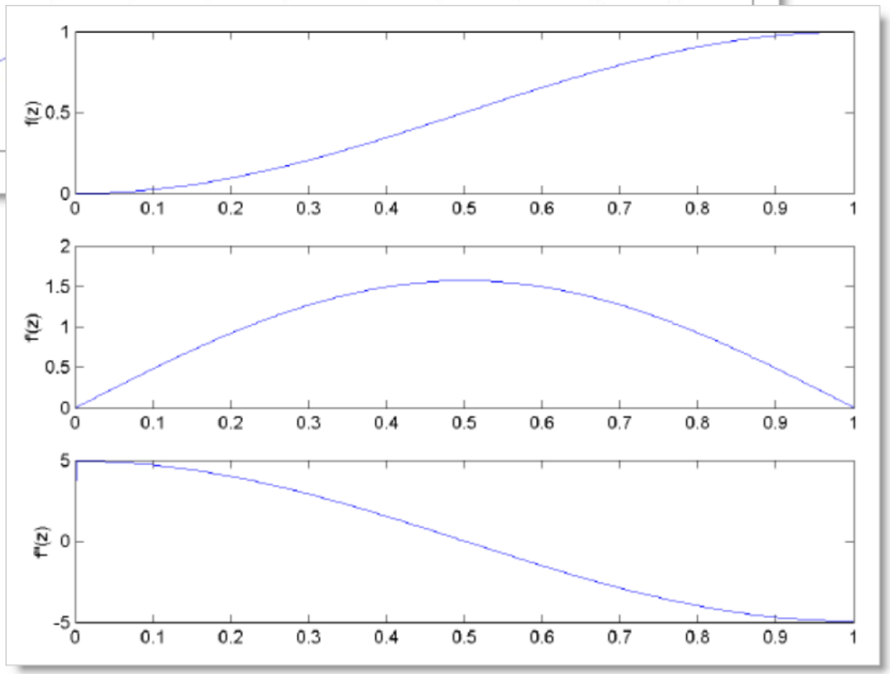
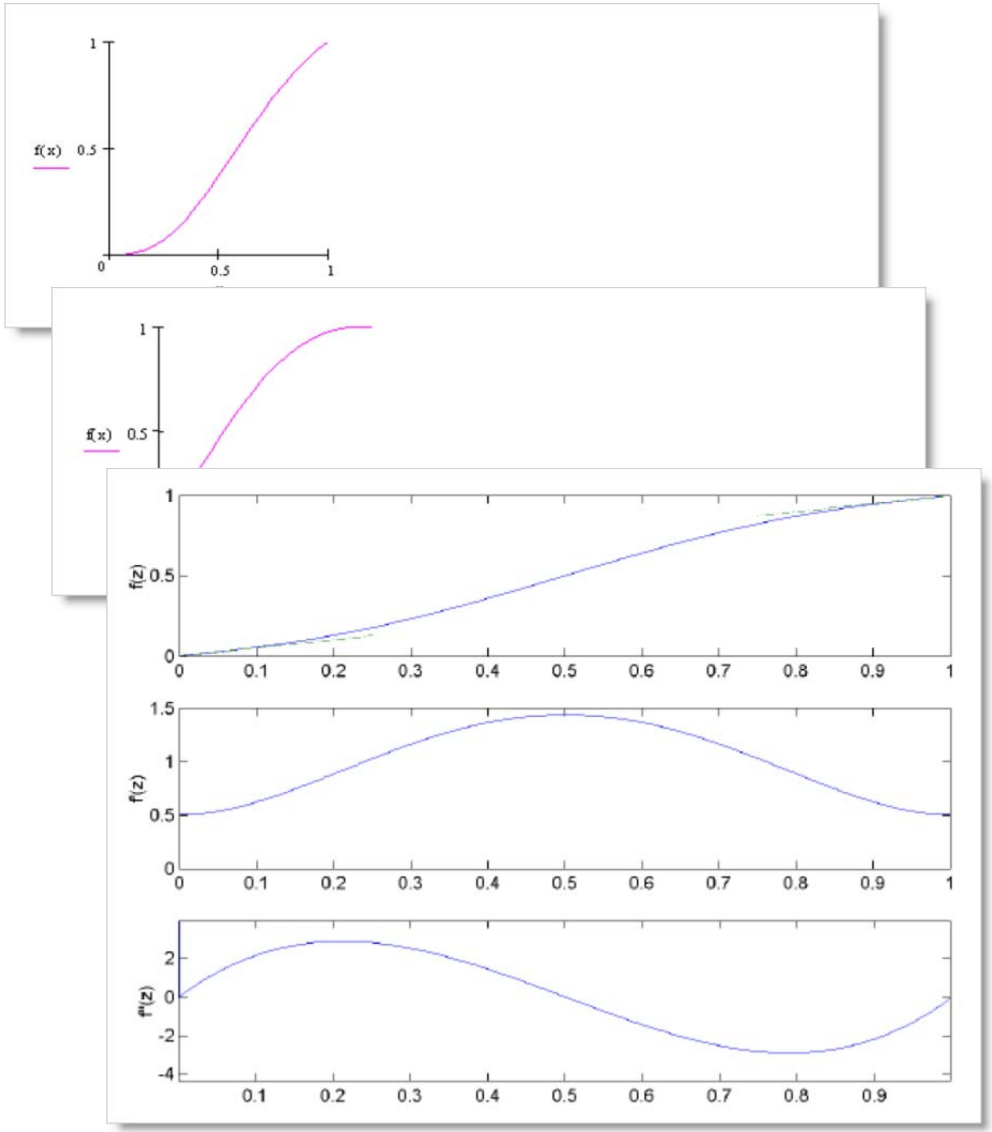
FlexProfile – Motion Rules (examples)





- Rest in Rest 5th-degree polynomial (Position, velocity and acceleration)
- Rest in Rest, simple sinoid (Position, velocity and acceleration)

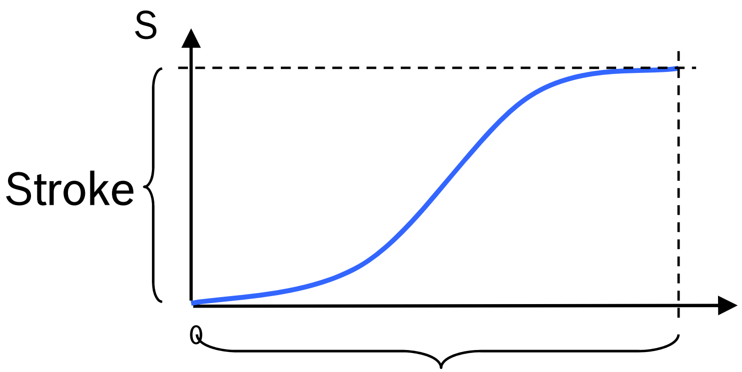
FlexProfile – Motion Rules (examples)



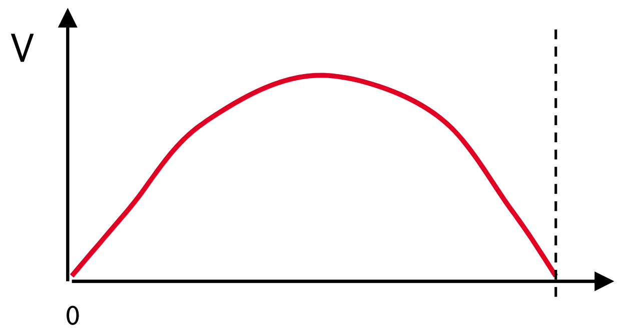
- Rest in Velocity 5th-degree polynomial (position)
- Velocity in Rest 5th-degree polynomial (position)
- Velocity in Velocity 5th-degree polynomial
(Position, velocity and acceleration)

FlexProfile – Supplemental Motion rules

Standardized motion rules
according to VDI 2143

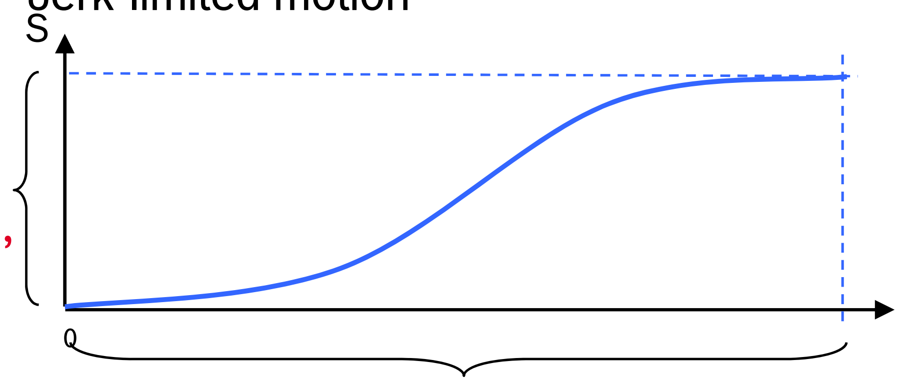


Master axis section

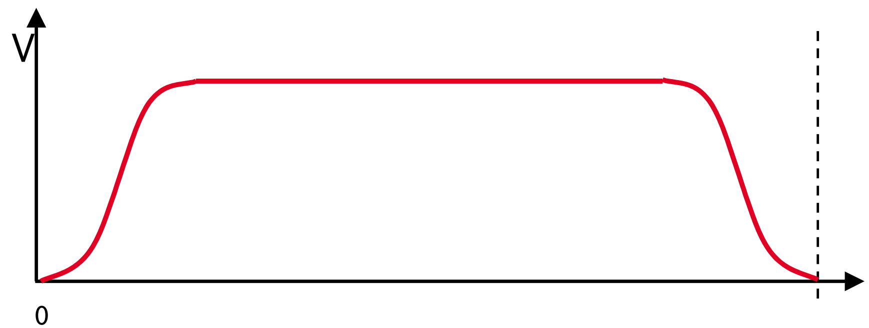


Stroke,
Velocity,
Acceleration,
Jerk

Supplemental motion rules
Acceleration-limited motion (trapezoid profile)
Acceleration-limited sinoid
Jerk-limited motion



Master axis section results



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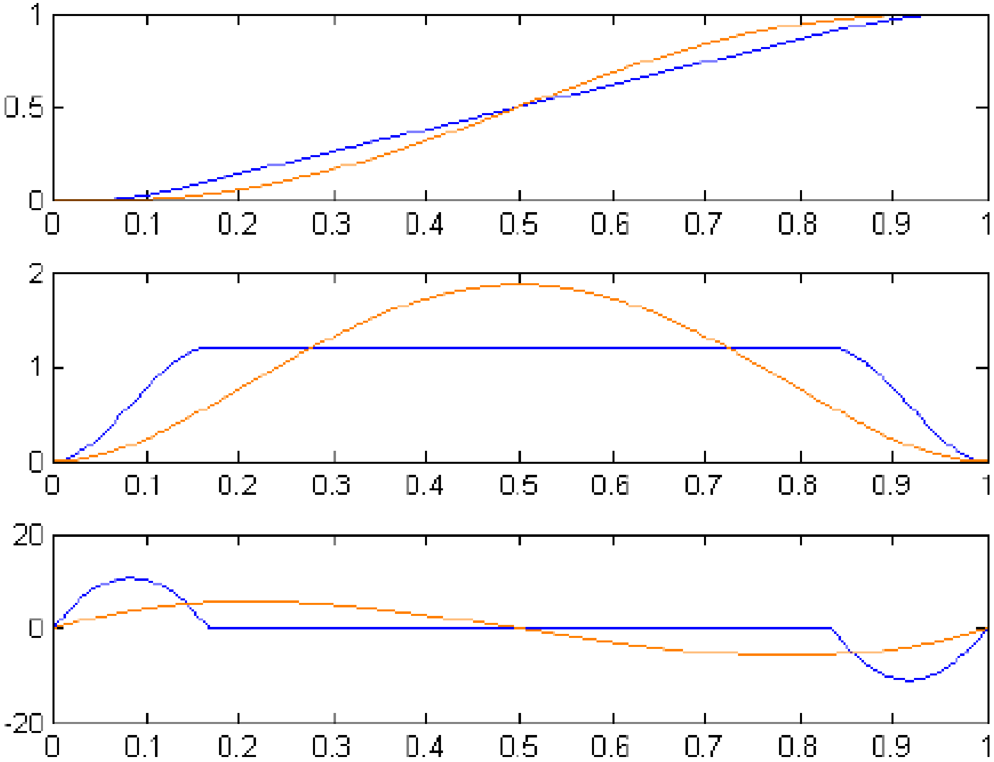
FlexProfile – Supplemental Motion rules

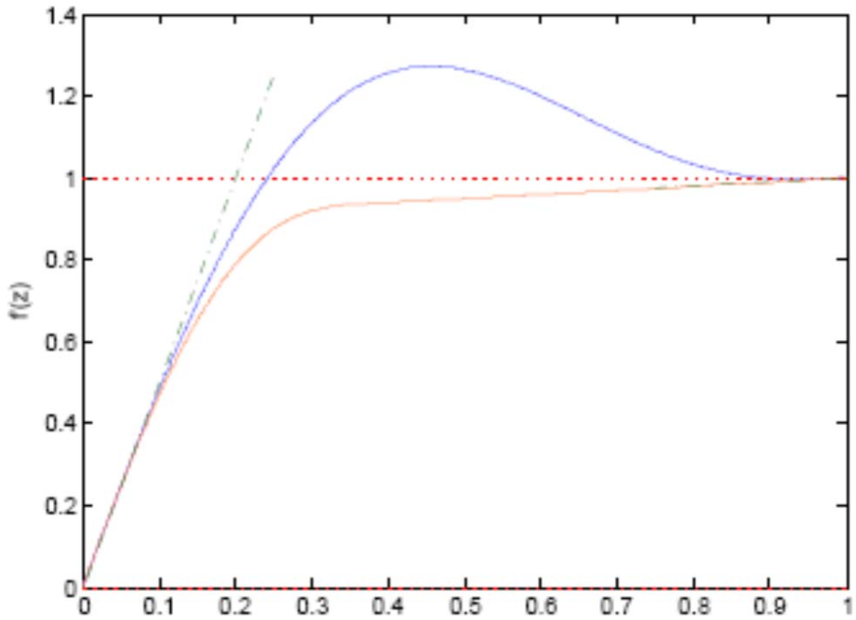
Polynomial 5th order with velocity limitation



Polynomial with velocity limitation

Polynomial 5^{th} order

Polynomial with damped oscillation



Polynomial 5^{th} order

Polynomial with damped oscillation

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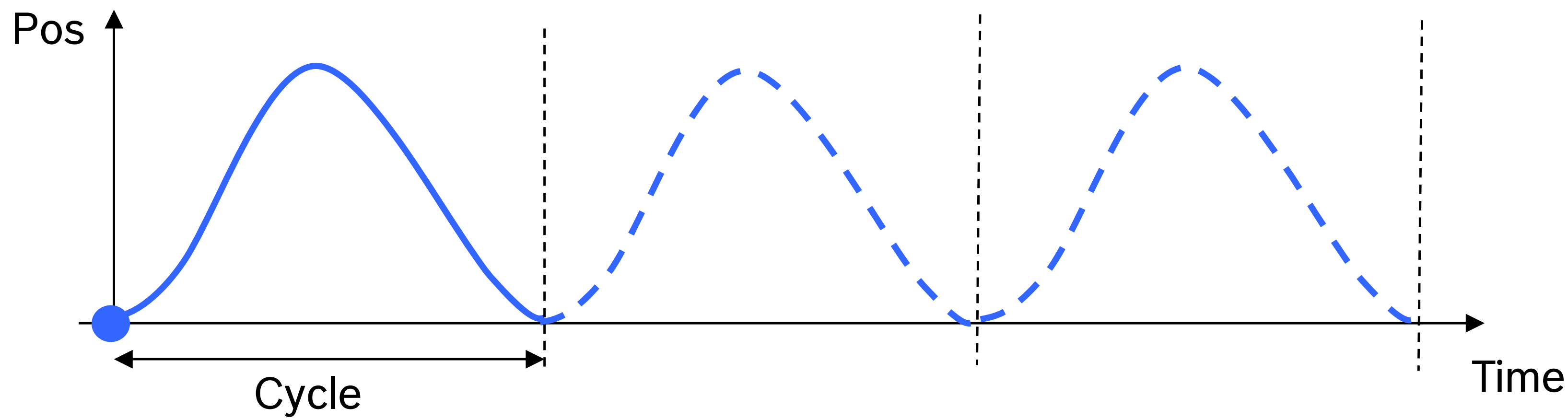
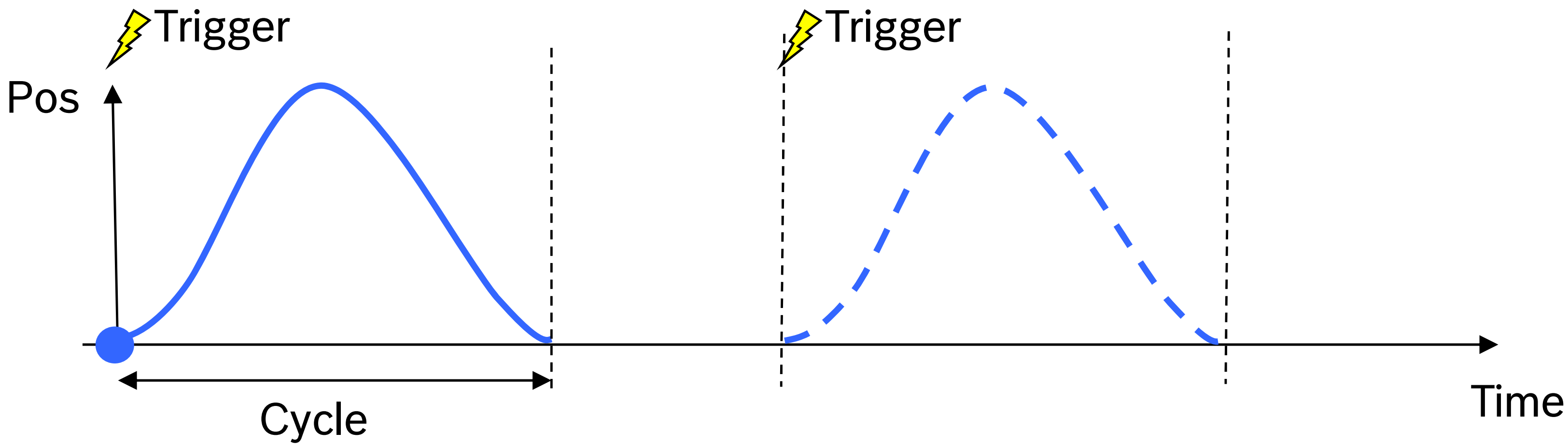
FlexProfile – Execution mode

- **Cyclic execution mode**

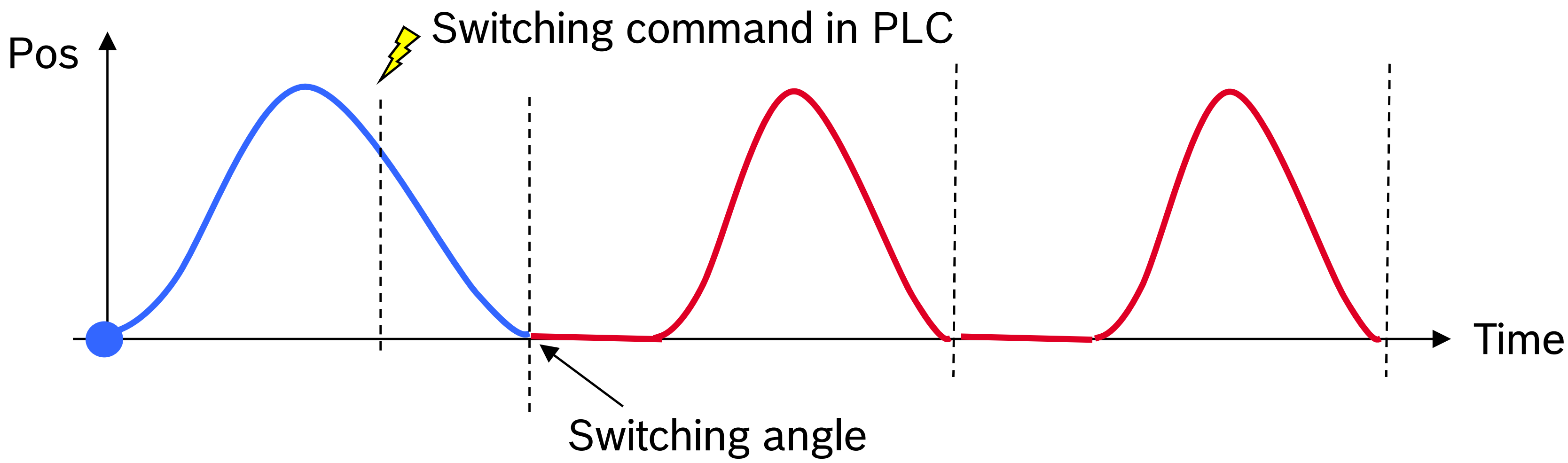


- **Single shot execution mode**

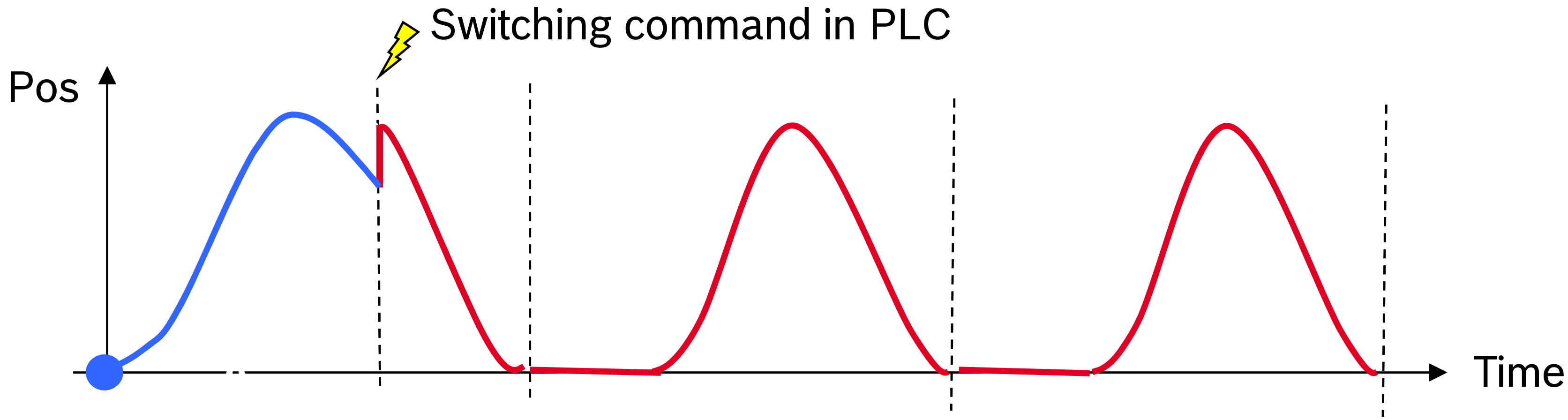


FlexProfile – Switching

- **Switching on angle**

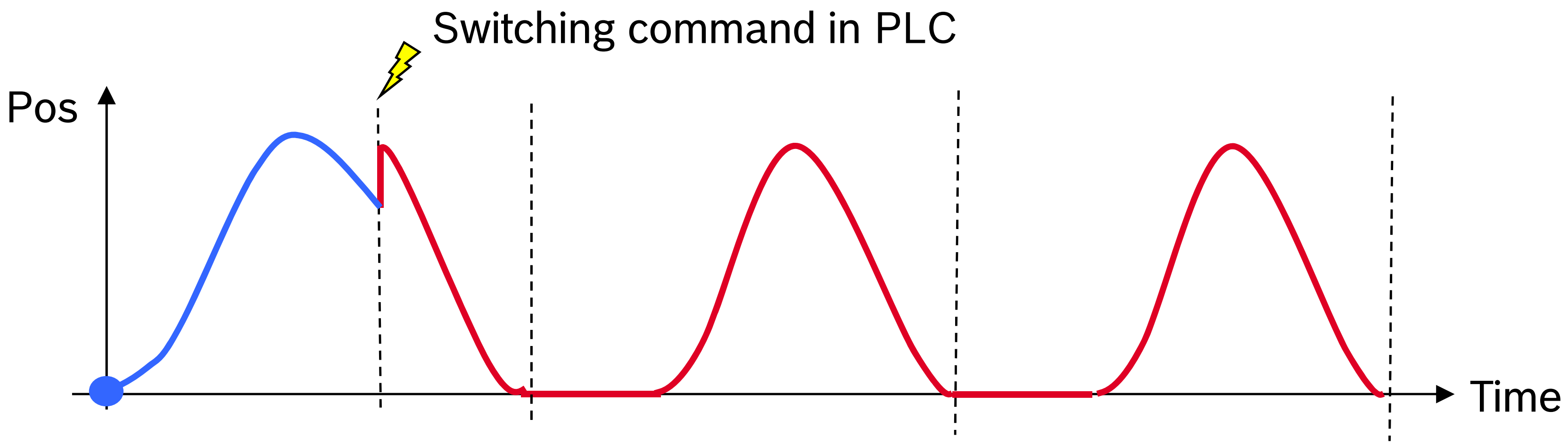


- **Immediate Switching**

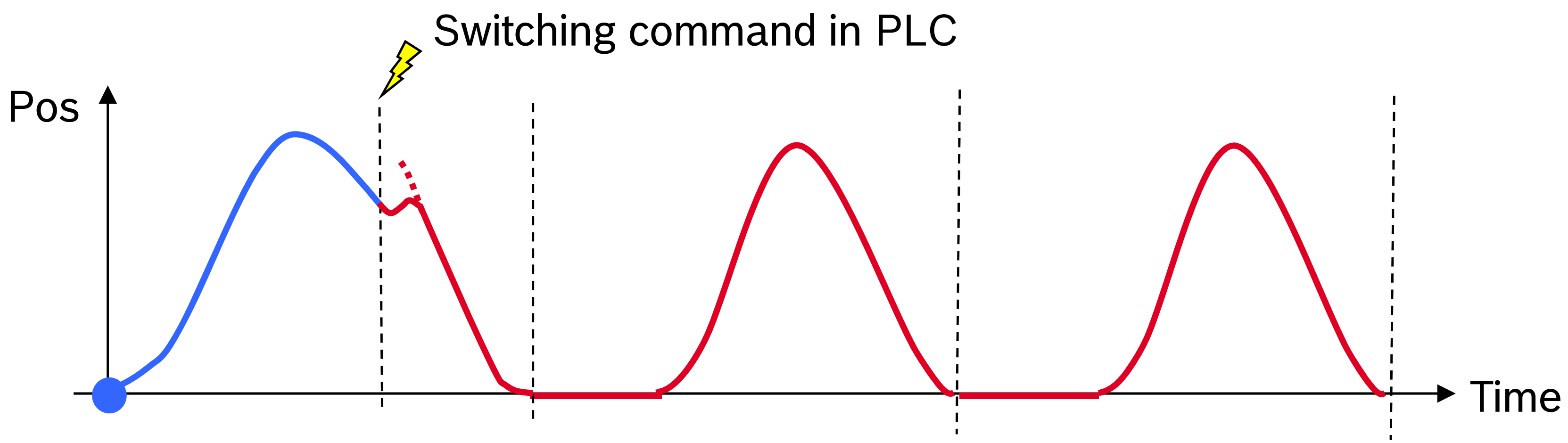


FlexProfile – Switching

- **Hard switching**

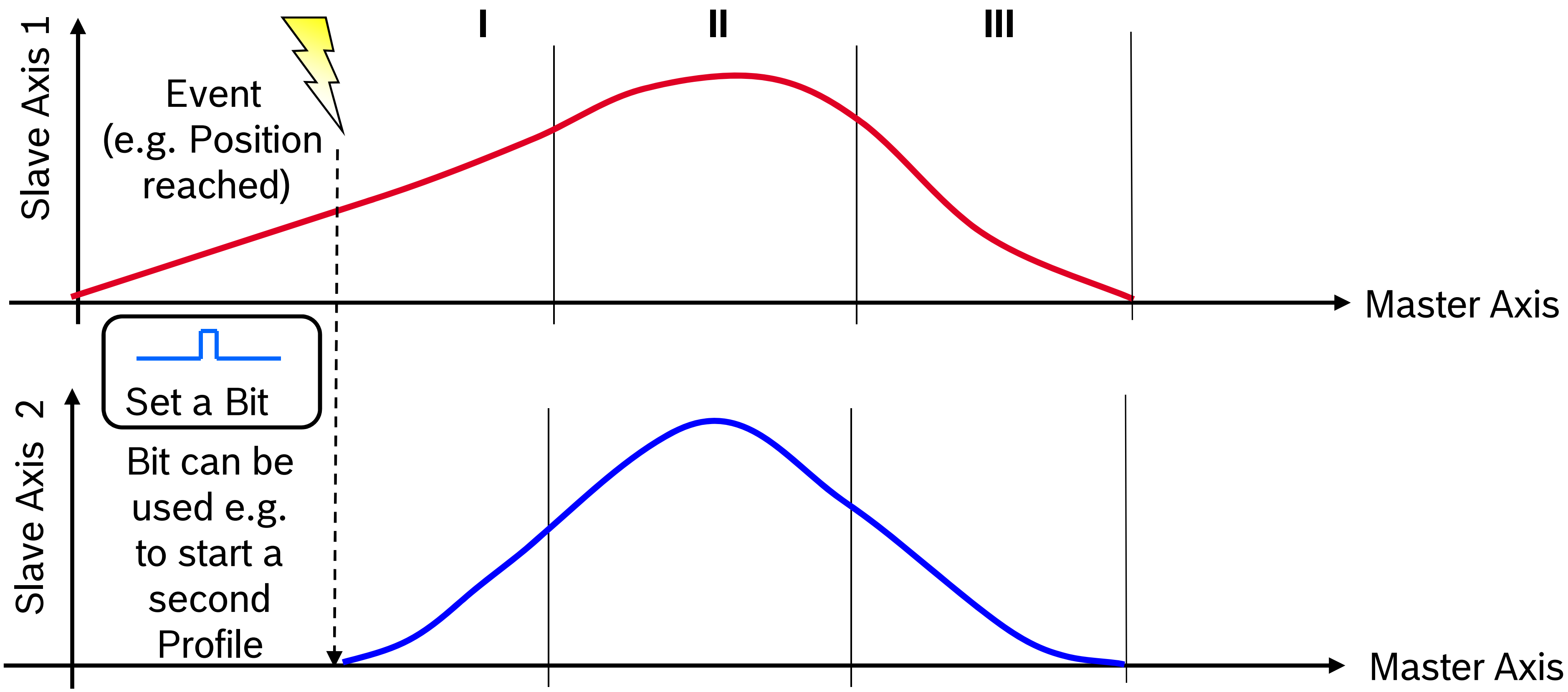


- **Soft switching**



FlexProfile – Events

Events can be attached to sections. Events consist of Trigger and Action



Example: Coordination of motion movements

Trigger:
- Relative / Absolute **master position** respectively **Time** reached
- Relative / Absolute **slave position** reached
- **PLC** Signal

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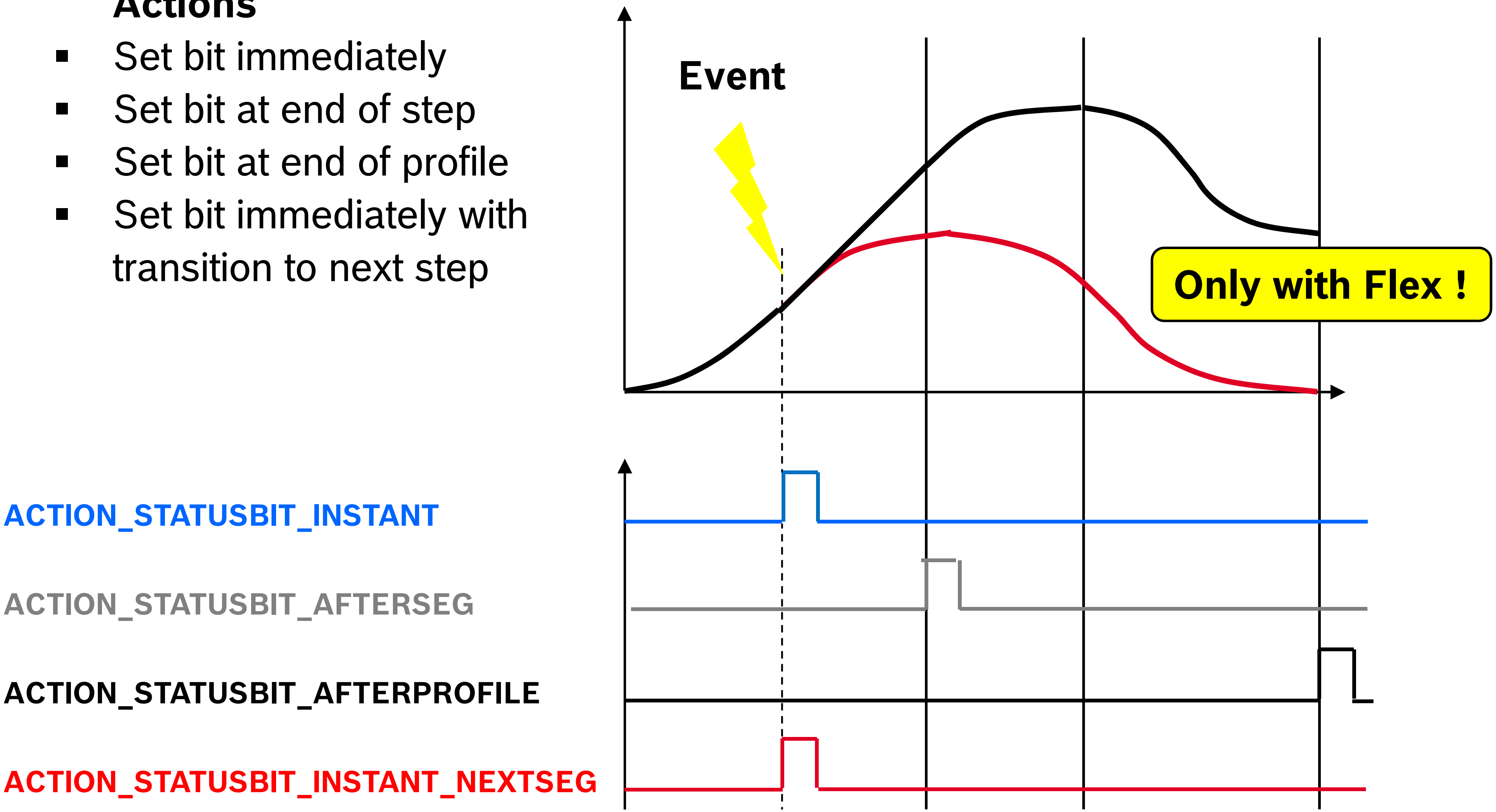
FlexProfile – Events

Actions

- Set bit immediately
- Set bit at end of step
- Set bit at end of profile
- Set bit immediately with transition to next step



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FlexProfile – Events

Event Condition

Absolute master axis position reached
Relative master axis position reached
Absolute slave position reached
Relative slave position reached
Time elapsed
PLC Signal

Event Action

Set Bit immediately
Set Bit at end of step
Set Bit at end of profile
Set Bit immediately and switch to next step

An Event with Event Condition "PLC signal" is fired with a bit in AxisData[ ].**dwFlexEventControlBits_q**

The bit number correlates with the event number! Please note that the event number sometimes is displayed 0-based and on other screens 1-based!

The events can be checked using AxisData[ ].**dwFlexEventStatusBits_i**

Define a high-prior task to evaluate these events (motion driven and priority higher then MotionTask)!

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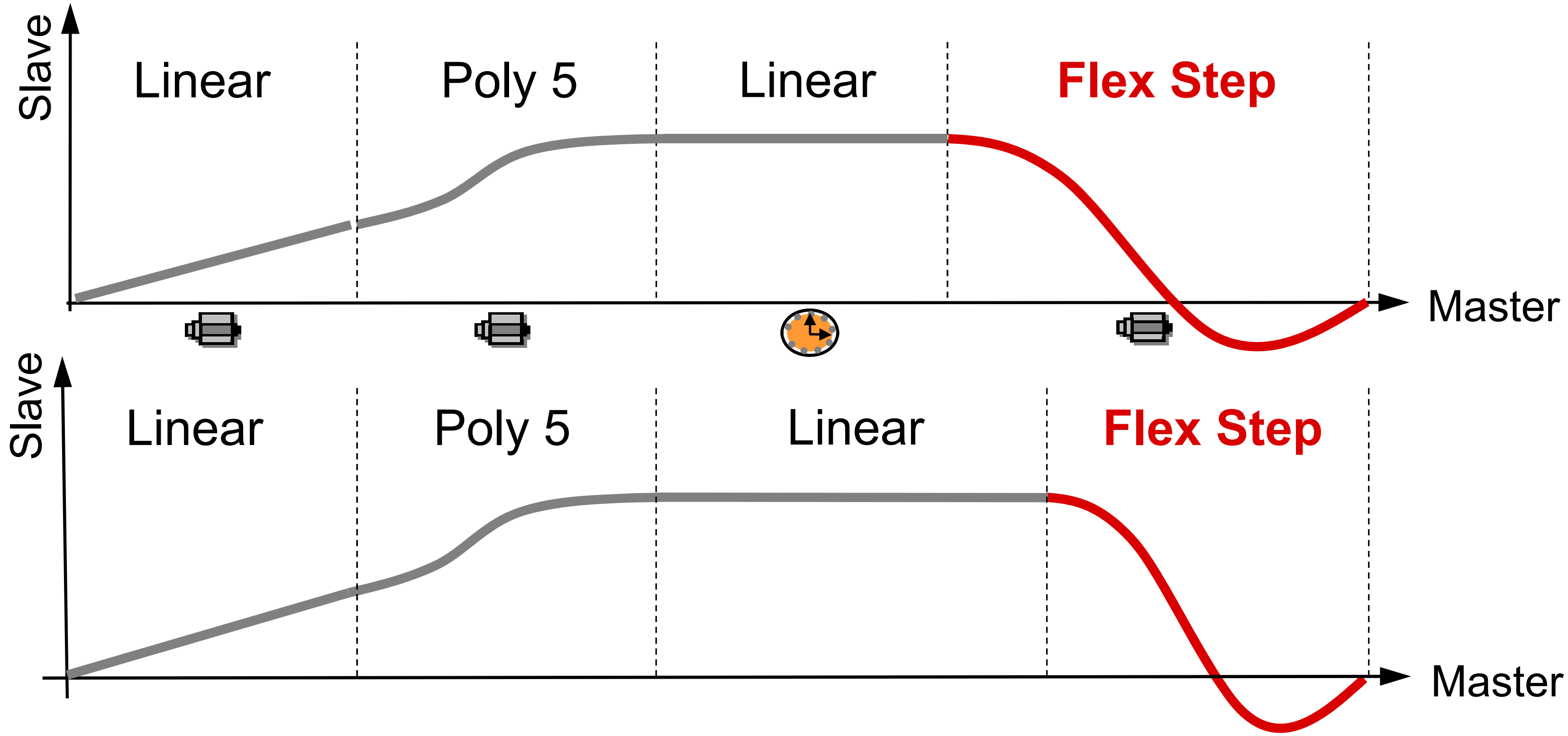
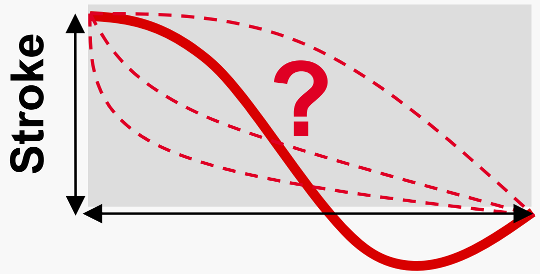
FlexProfile – Time controlled motion



Flex Step
- Dynamic motion step
- Motion curve adapted at runtime
- ... according to the margin values of adjacent motion steps
- Reestablishes relation to master axis position



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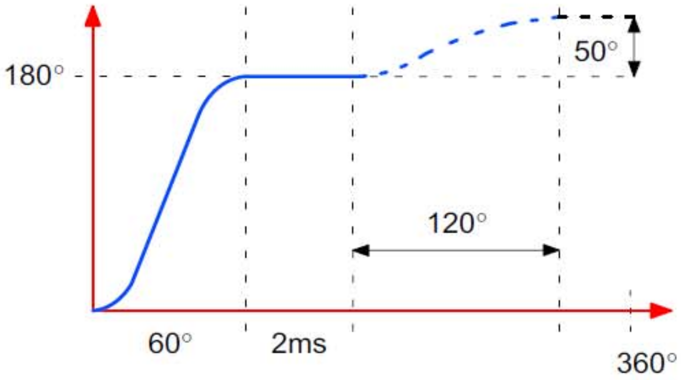
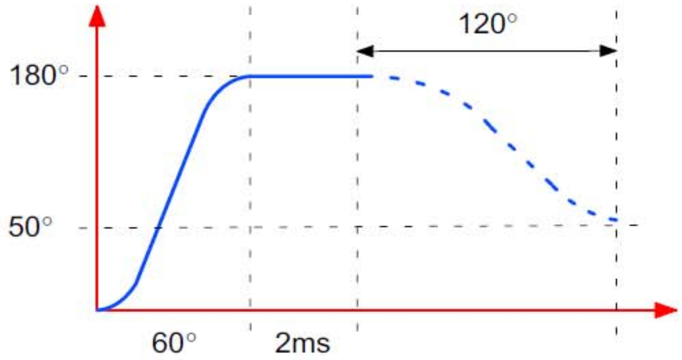
FlexProfile – Flex Step types

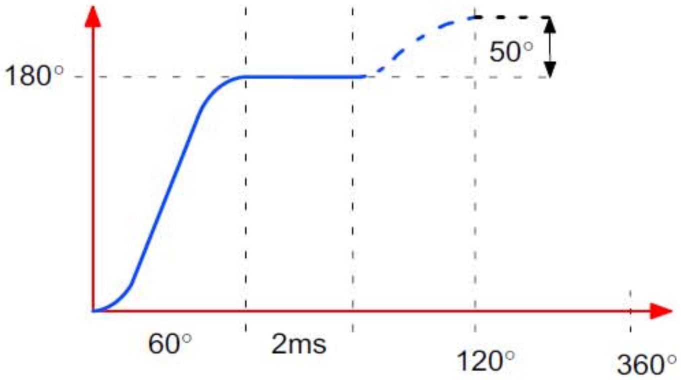
- Flex Step with relative stroke and relative master axis section
- Flex Step with relative stroke and absolute master axis section
- Flex Step with absolute stroke and relative master axis section
- Flex Step with absolute stroke and absolute master axis section



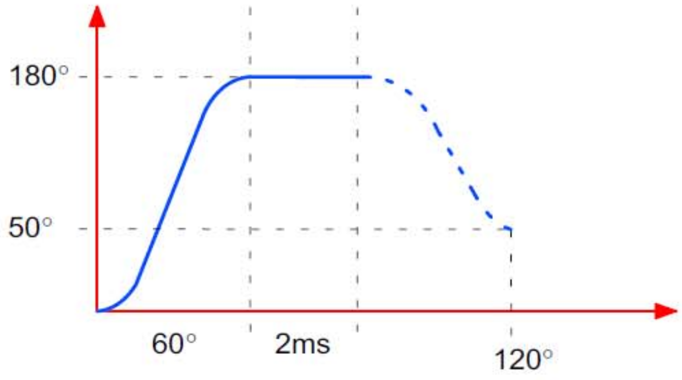
Flex step with a relative stroke and a relative master axis section



Flex step with an absolute stroke and a relative master axis section



Flex step with a relative stroke and an absolute master axis section



Flex step with an absolute stroke and an absolute master axis section

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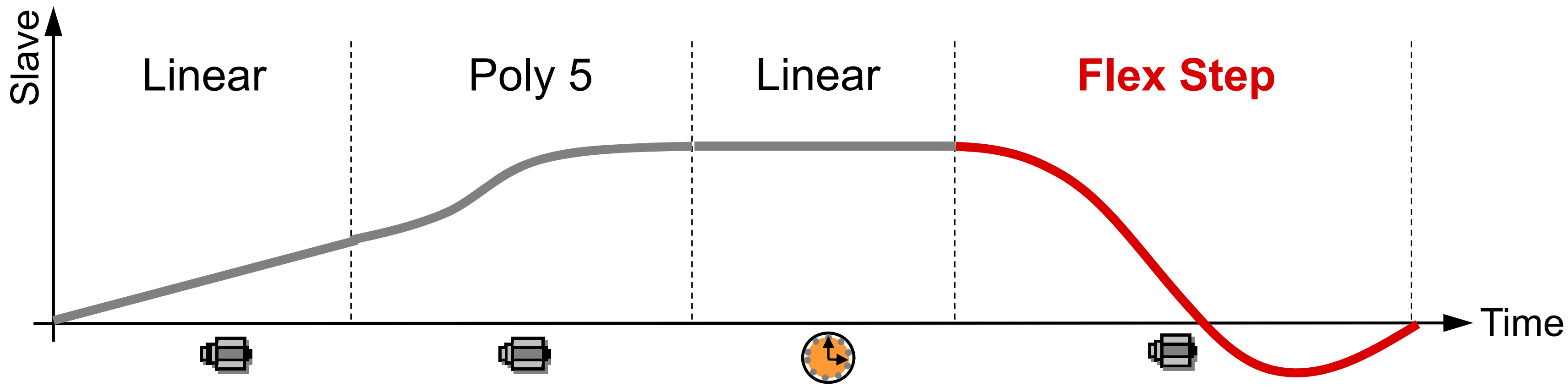
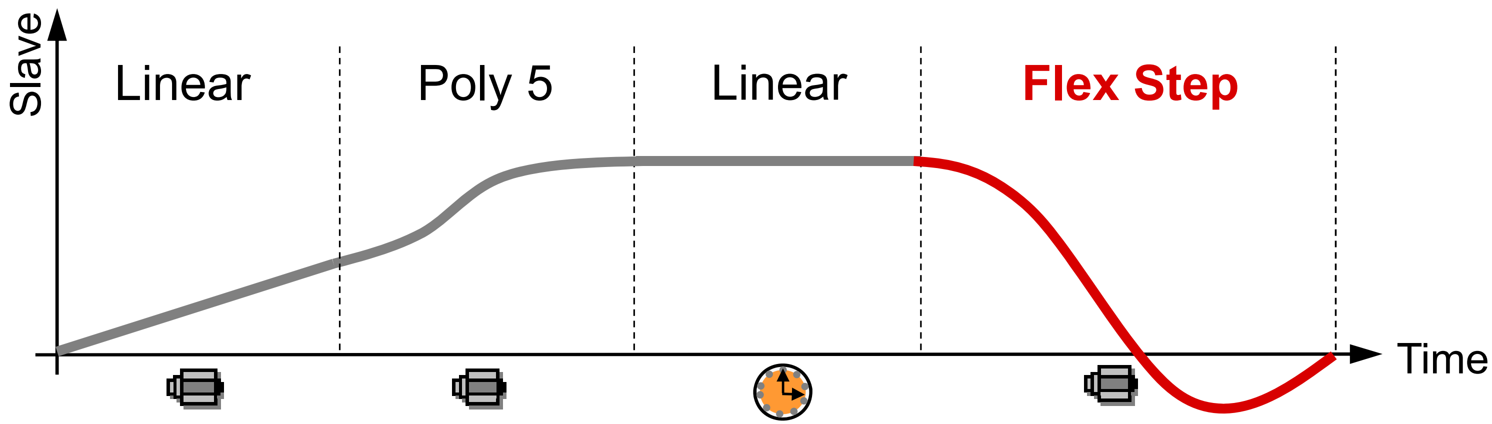
FlexProfile – Time controlled motion



Duration of time controlled step remains constant in spite of modification of master axis velocity!



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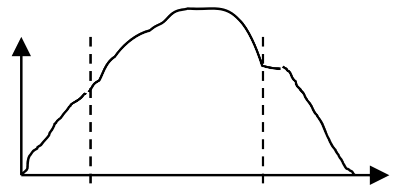
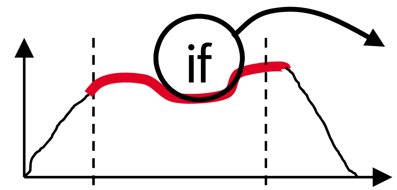
FlexProfile – Overview on Function Blocks

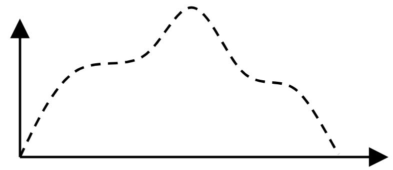
MB_ChangeFlexProfileSet
Setup and write FlexProfile



MB_ChangeFlexEventSet
Define Events and assign them to a step



MB_ChangeCamData
Write Point table in to the MLC



1024 Points

ML_FlexProfile

Activate a FlexProfile

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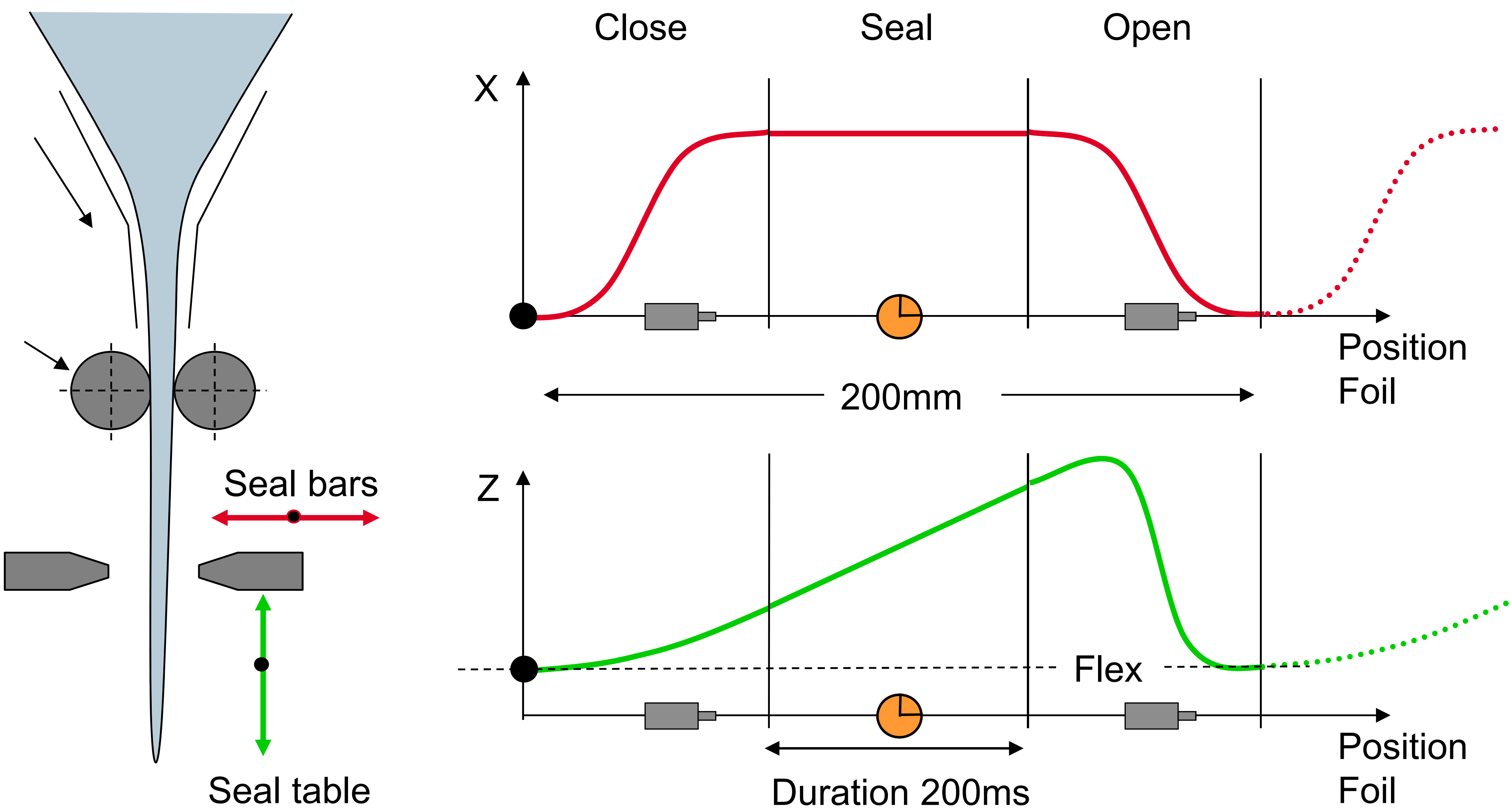
FlexProfile – Example cross sealing



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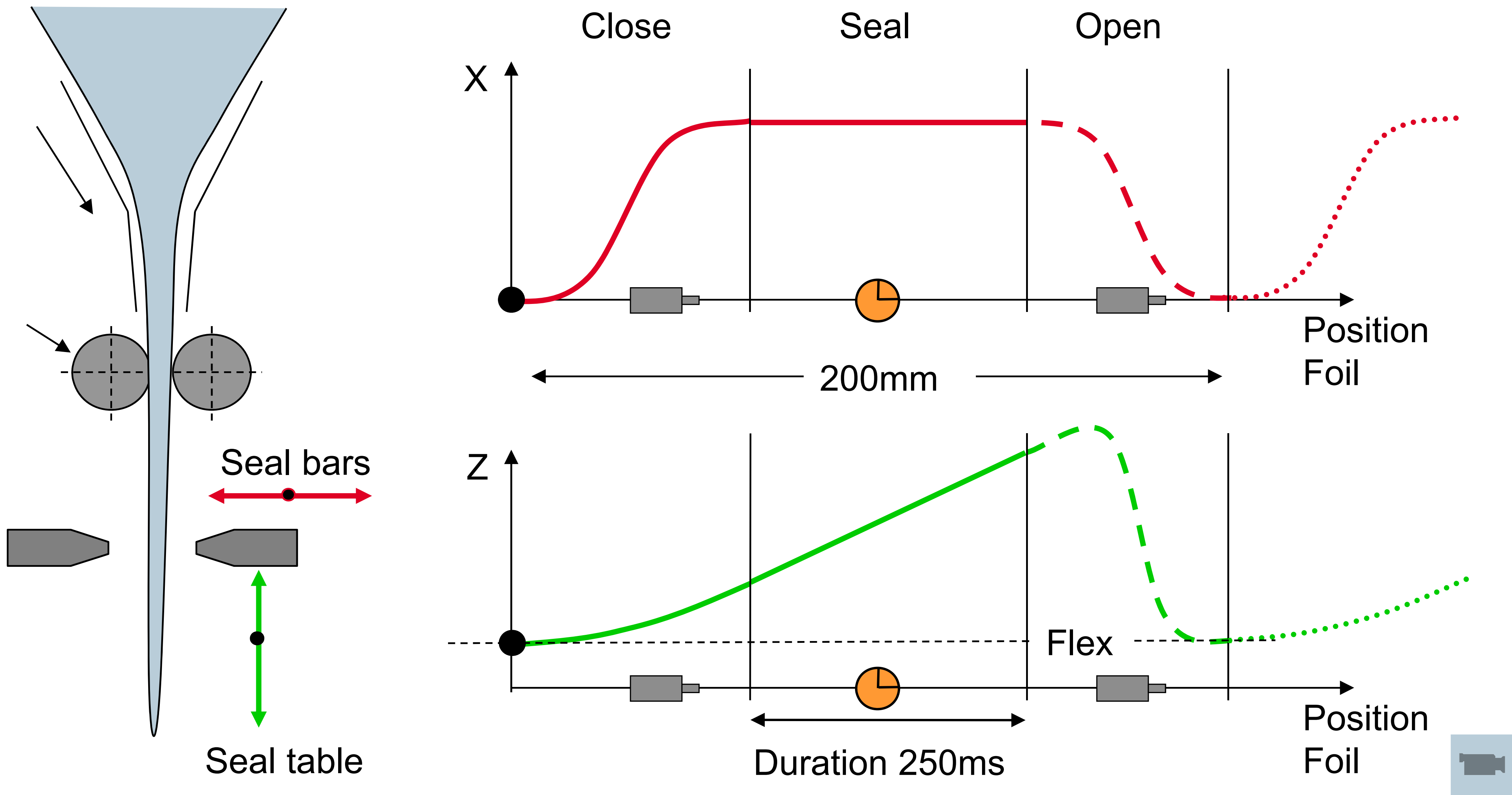
FlexProfile – Example cross sealing



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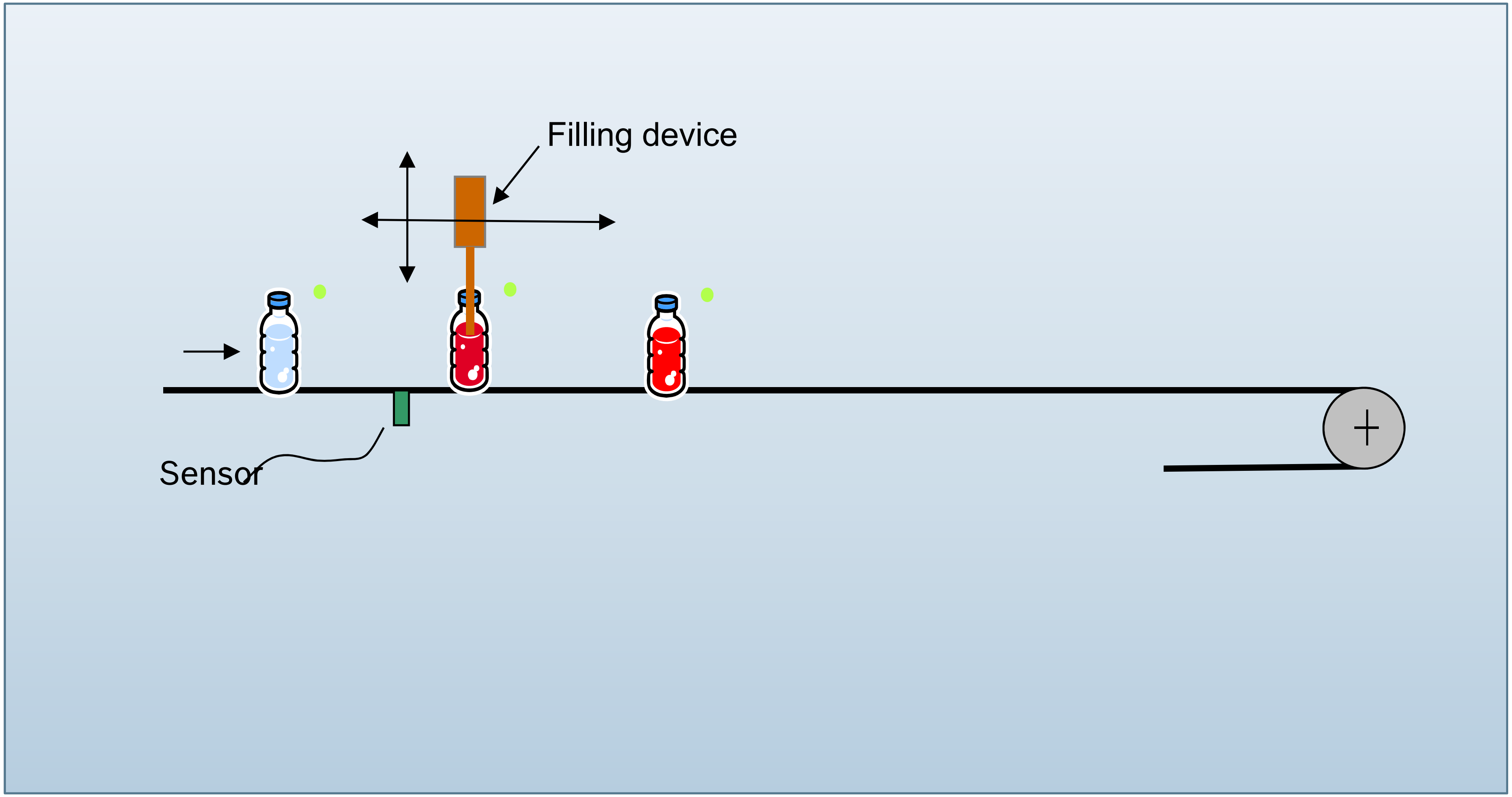
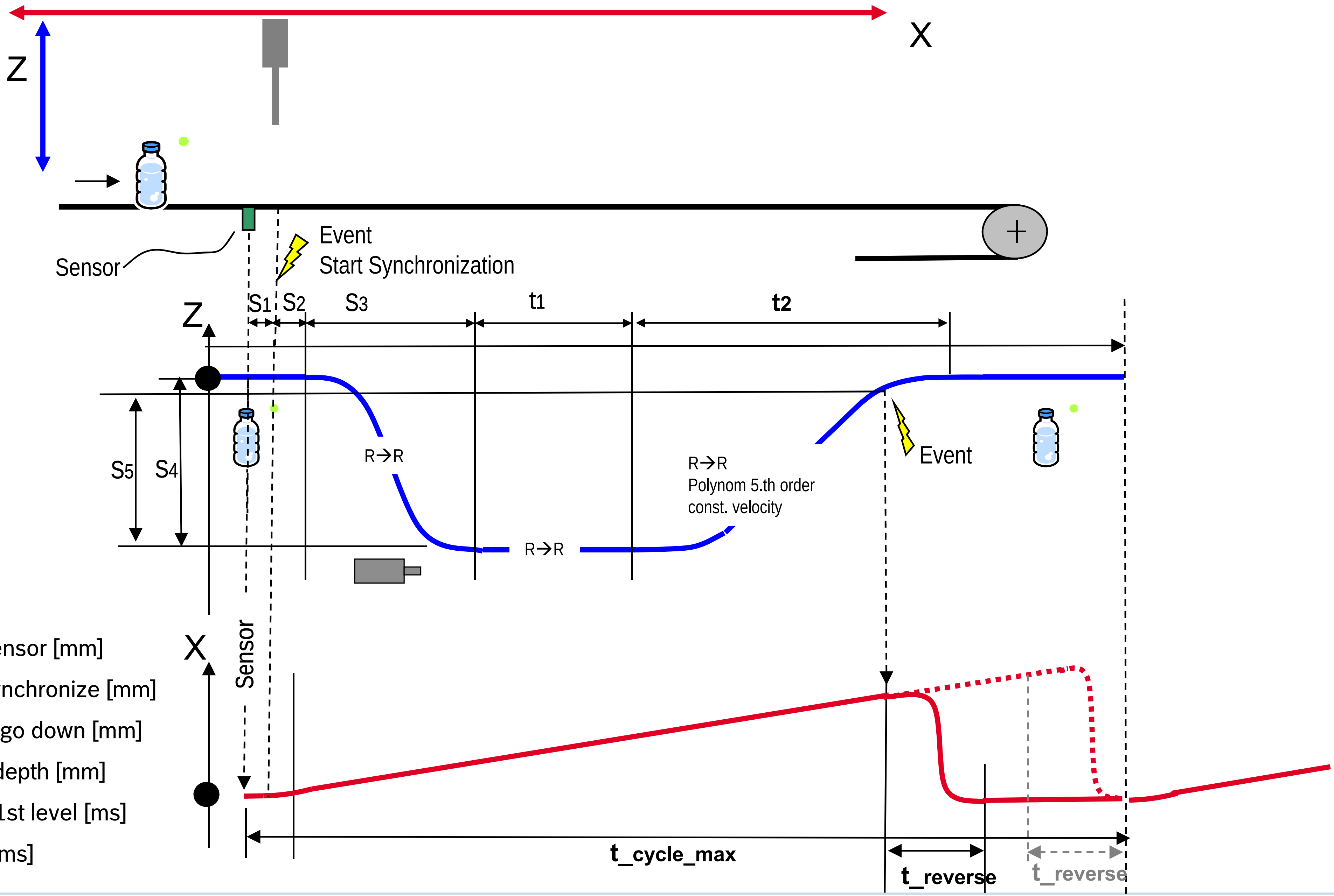
FlexProfile – Example bottling machine





S1: Distance Sensor [mm]

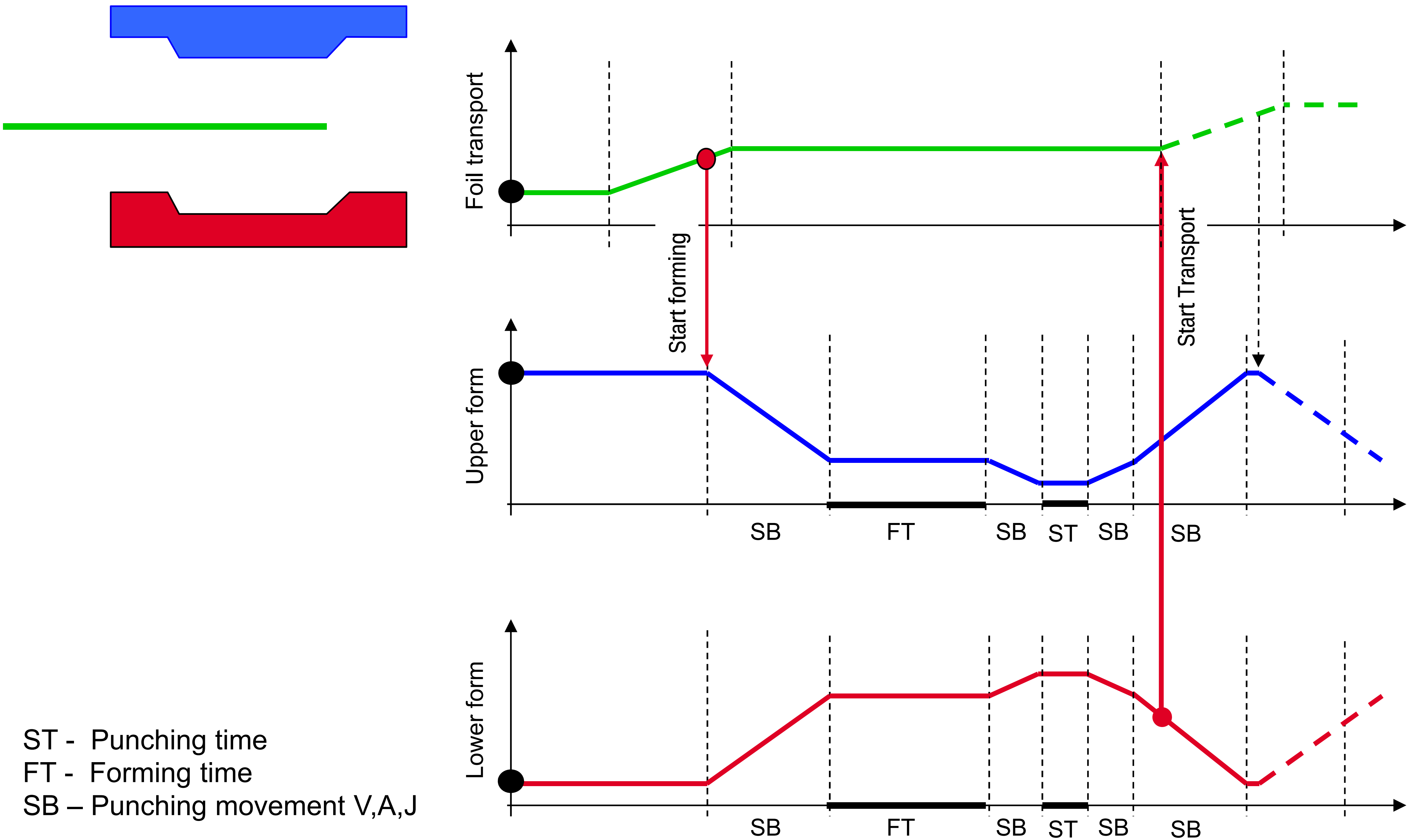
S2: Distance Synchronize [mm]

S3: Distance to go down [mm]

S4: Immersion depth [mm]

t1: Time Filling 1st level [ms]

t2: Time go up [ms]



ST - Punching time
FT - Forming time
SB – Punching movement V,A,J

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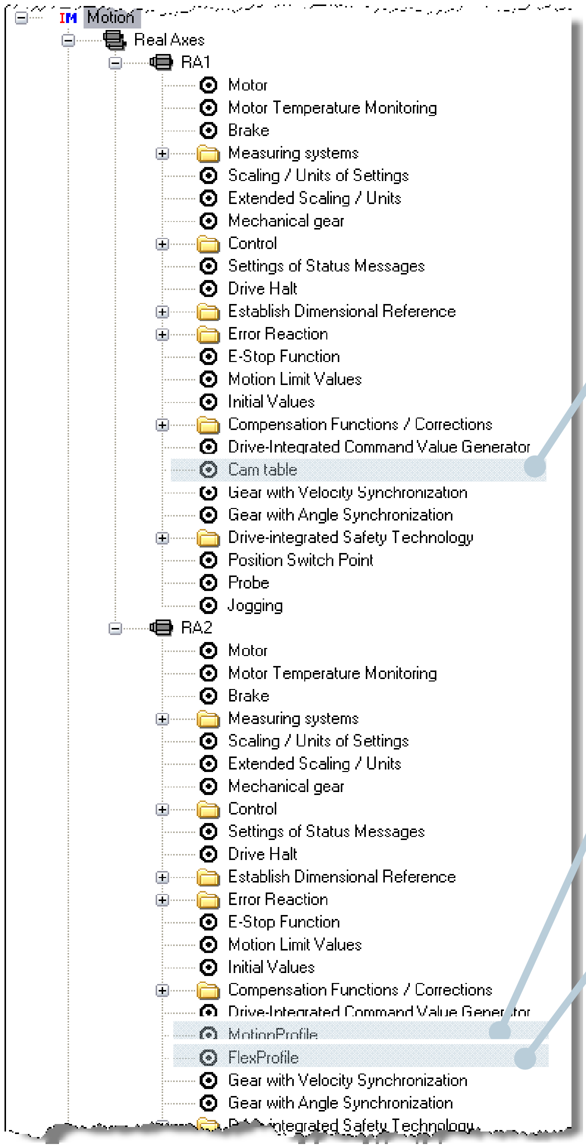
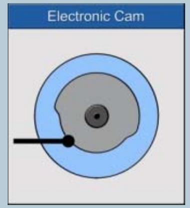
Point table – Motion Profile – FlexProfile

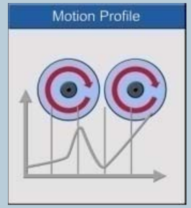


Point tables



- Drive feature
- 4 point tables per axis with max. 1024 points
- Only for real axes

Motion Profile



- Control feature
- 2 motion profiles per axis with up to 16 motion steps
- Always related to master axis position

FlexProfile

- Control feature
- 4 FlexProfiles per axis with up to 16 motion steps
- Related to master axis position or to time
- Relative or absolute reference of master and slave axis
- Variable switching/synchronization of profiles
- Event handling

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Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- Motion Programming Basics
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Synchronized Motion

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CamBuilder – Overview

- CamBuilder is a graphical tool for creation and modification of Cams
- Suitable for
  - IndraDrive
  - IndraMotion MLD and
  - IndraMotion MLC/XLC
- Operations
  - Graphical creation and modification of Cams
  - Download and upload of Cams to/from IndraDrive, IndraMotion MLD and IndraMotion MLC/XLC
  - Import and export of Cams in different formats (Excel, XML, parameter file *.par)
- CAM Builder is integral part of IndraWorks
- A license code has to be entered for activation

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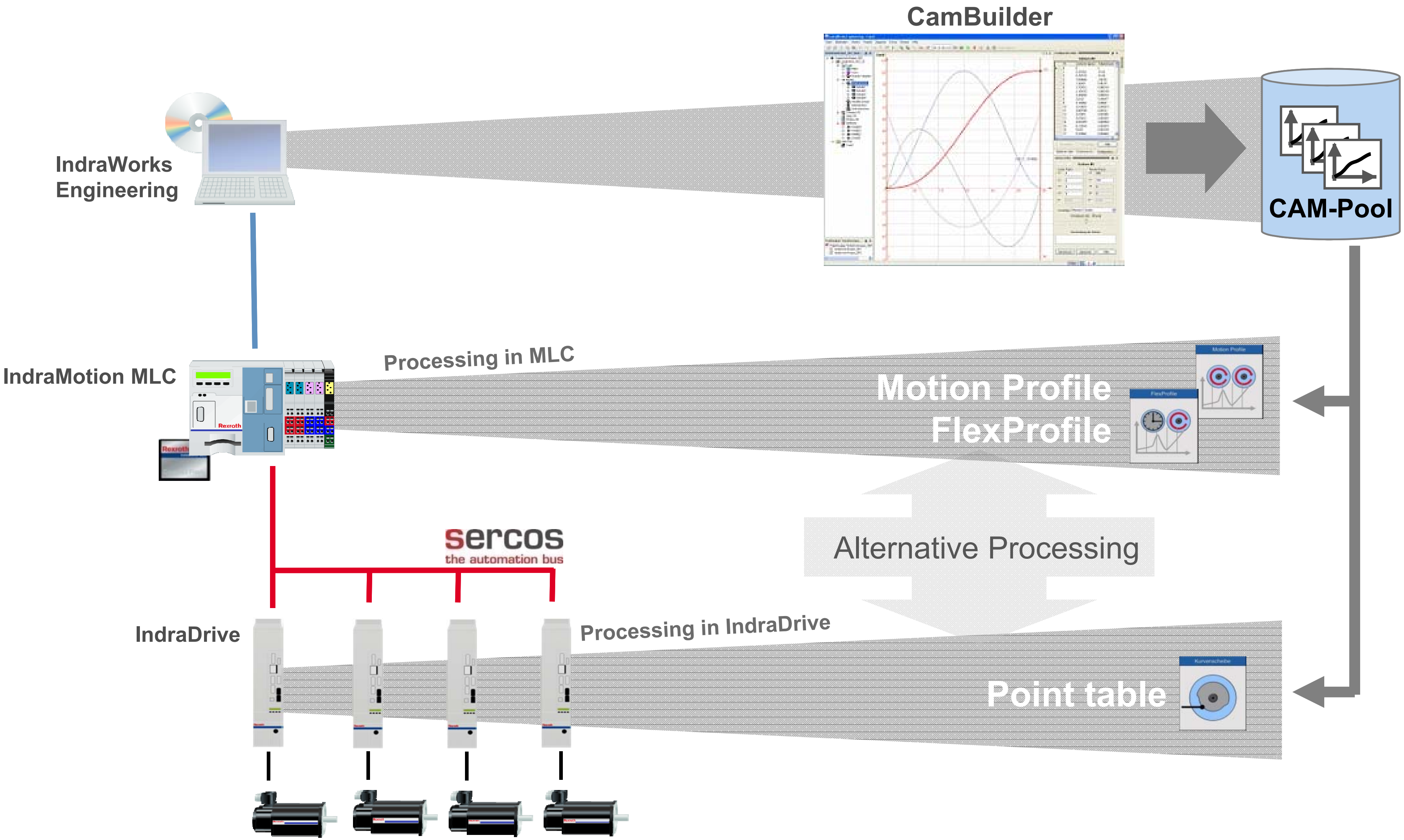
Flexible use of Cams



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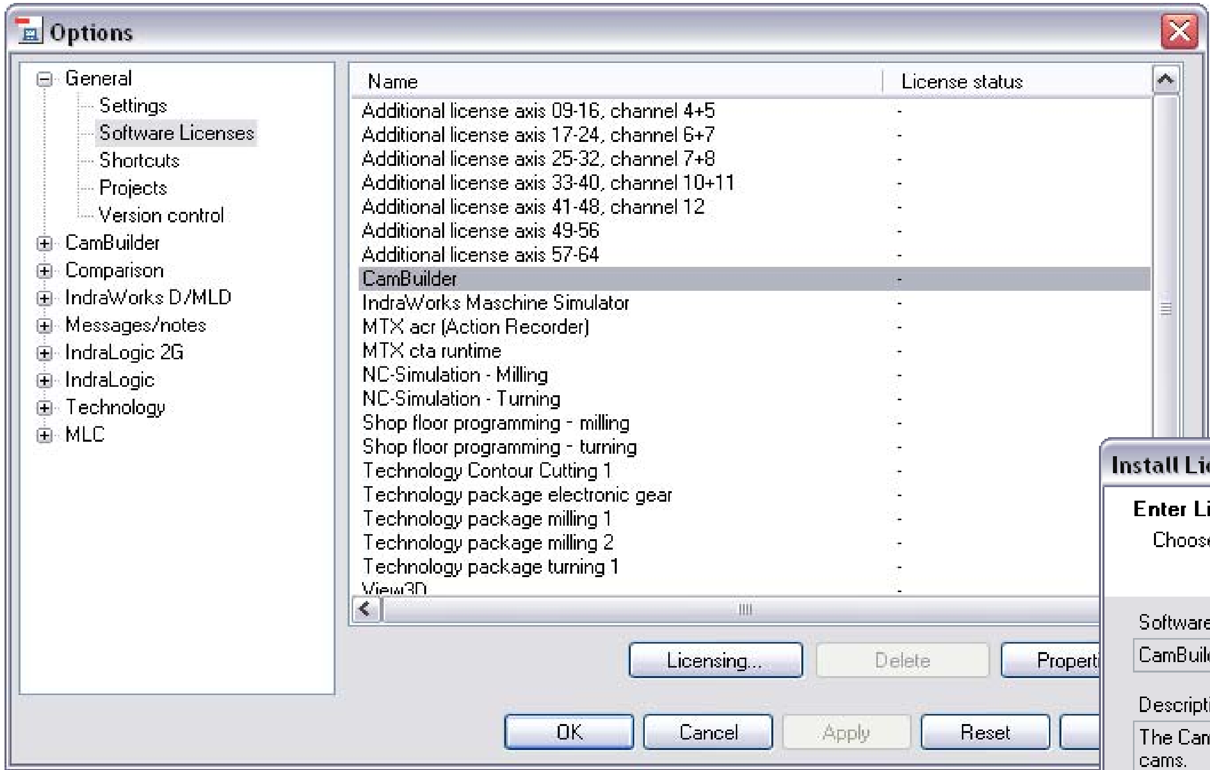
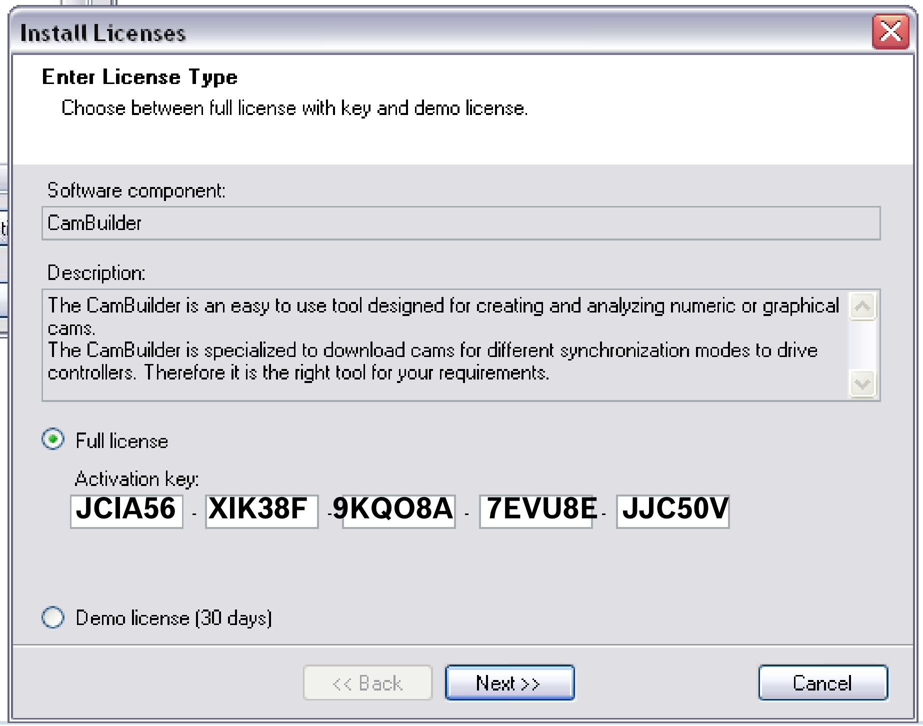
Licensing of CamBuilder



- Select menu entry Tools/Options
- Select General/Software licenses
- The current licensing state is displayed



- You can activate a 30 days demo license
- ... or enter a license key and activate a full license

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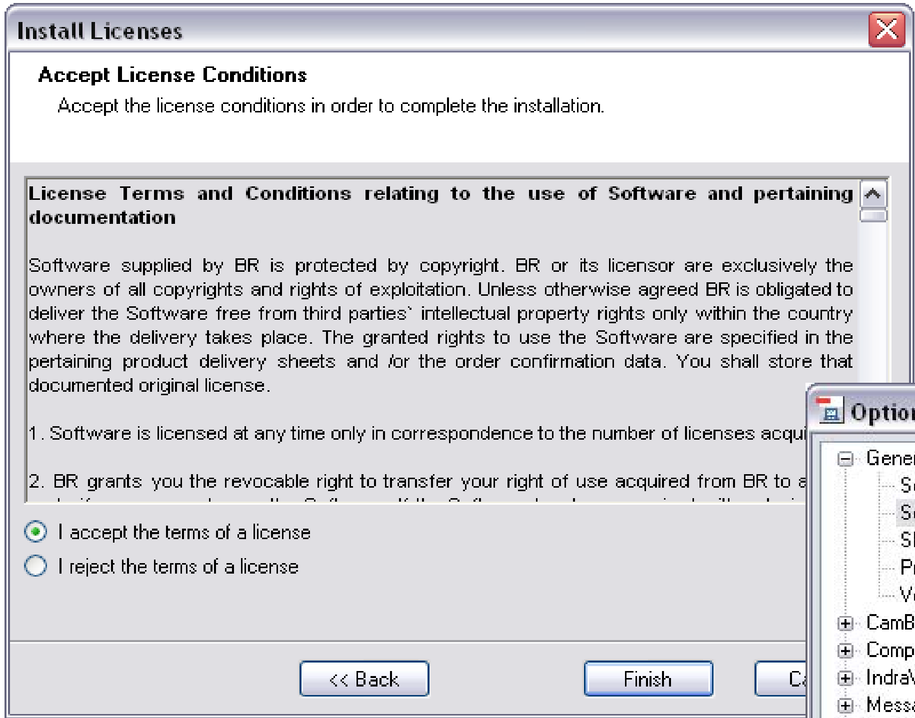
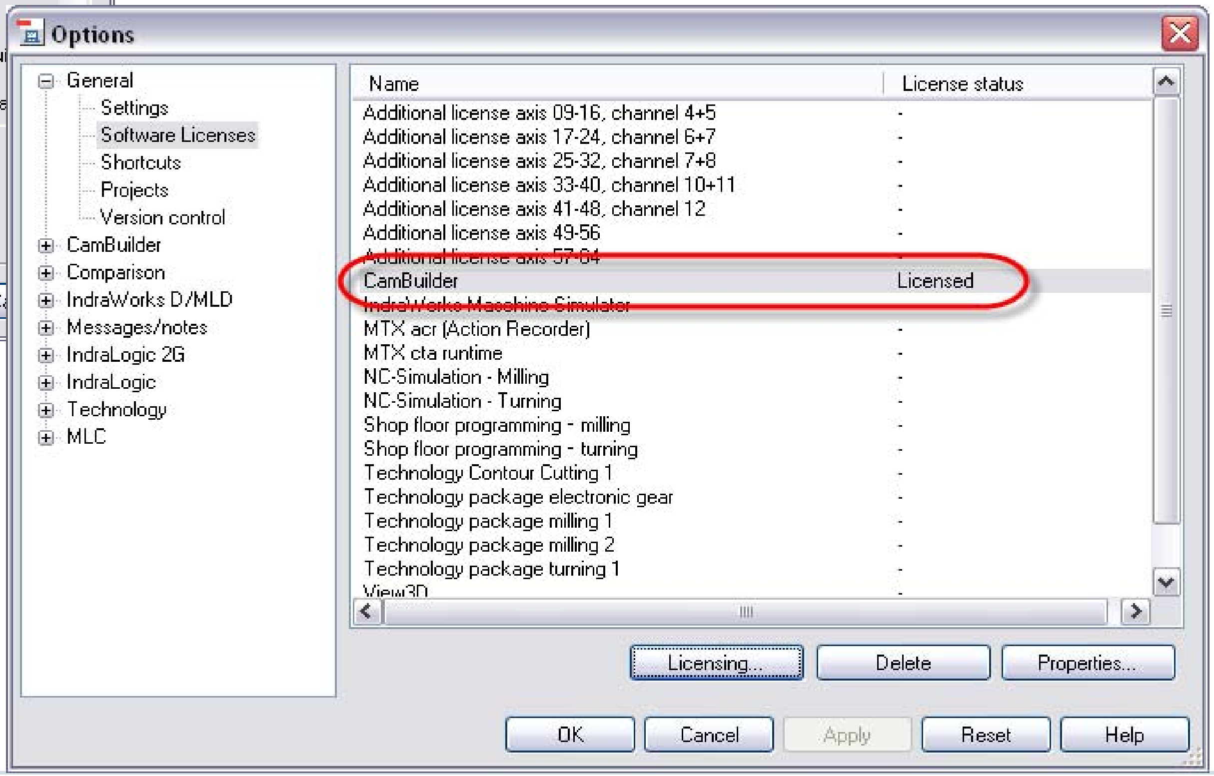
Licensing of CamBuilder



- Accept the terms of license



- The license status of CamBuilder is “Licensed”

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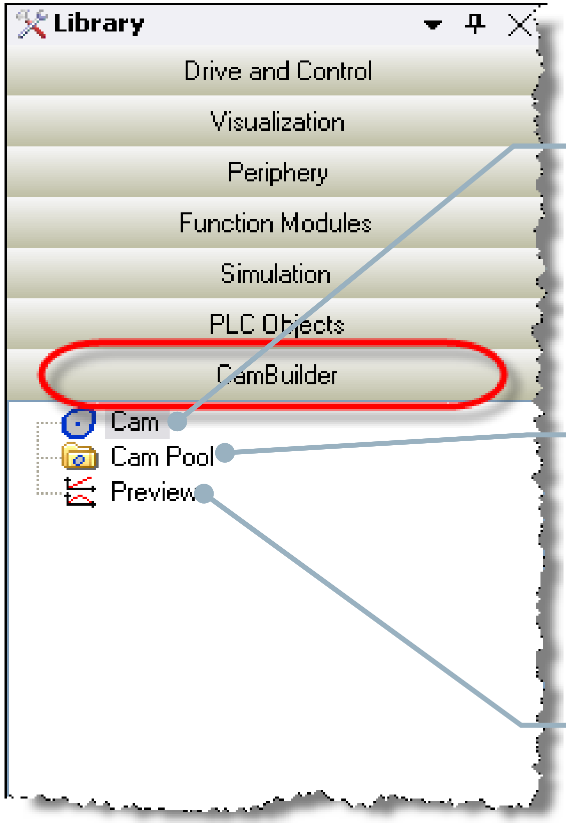
Using the CamBuilder

- After activation of the CamBuilder and opening the group "CamBuilder" in the Device Library you can add the following elements to you project:



Cam
Add a Cam to your CamPool to create your own motion curves. Cam tables, MotionProfiles and FlexProfile are supported

CamPool
CamPool is subfolder in your IndraWorks project which is used as a storage for all your motion curves (Cam tables, Motion profiles and FlexProfiles

Preview
Multiple motion curves can be combined to a Preview to see several axes in correlation to another

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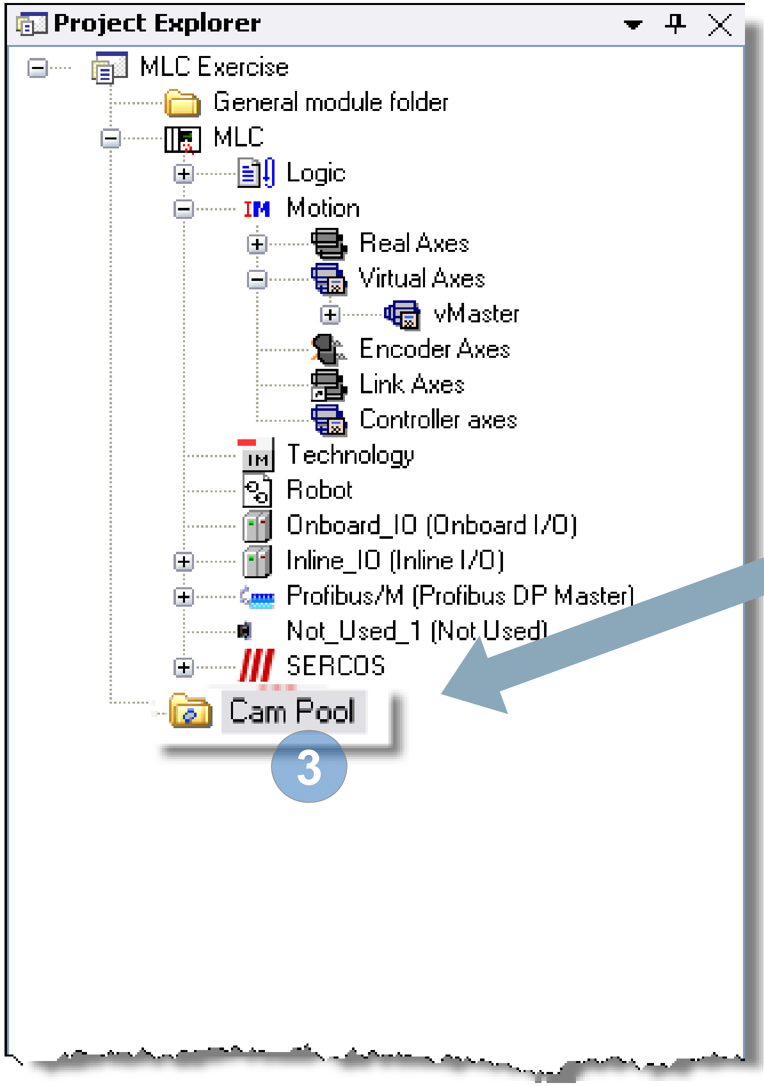
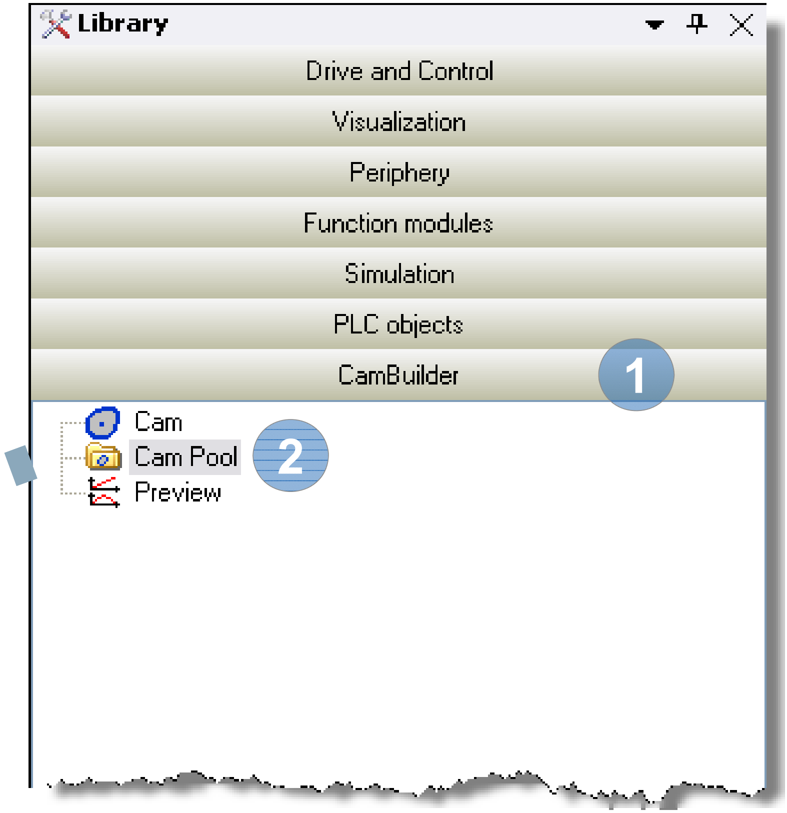
Add CAM Pool to project





1. Open Group CamBuilder in Library
2. Select “CamPool”
3. Add CamPool to IndraWorks Project

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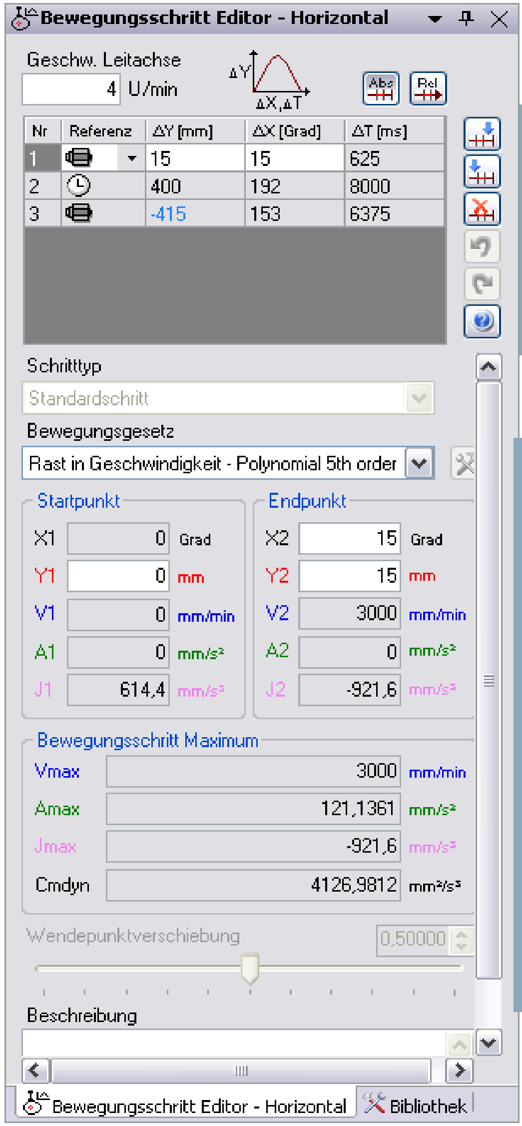
CAM Builder – Motion Step Editor



Sequence of motion steps

Settings for selected motion step

- Easy composition of CAM profile by a sequence of motion steps
- Definition of position-related motion steps, and
- Definition of time-related motion-steps

☞ Graphical creation and modification of FlexProfiles

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How to create a FlexProfile (1)

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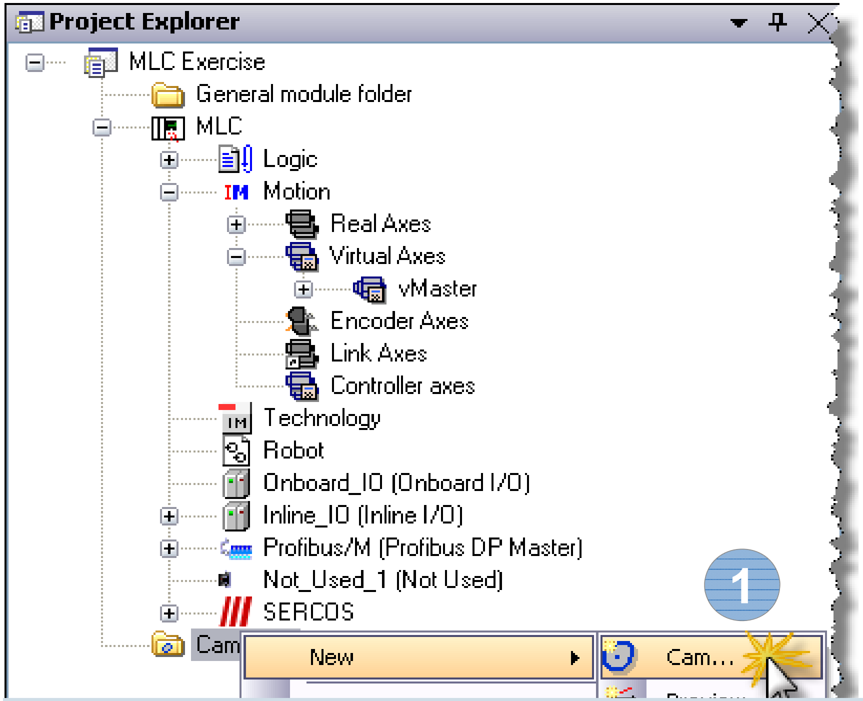
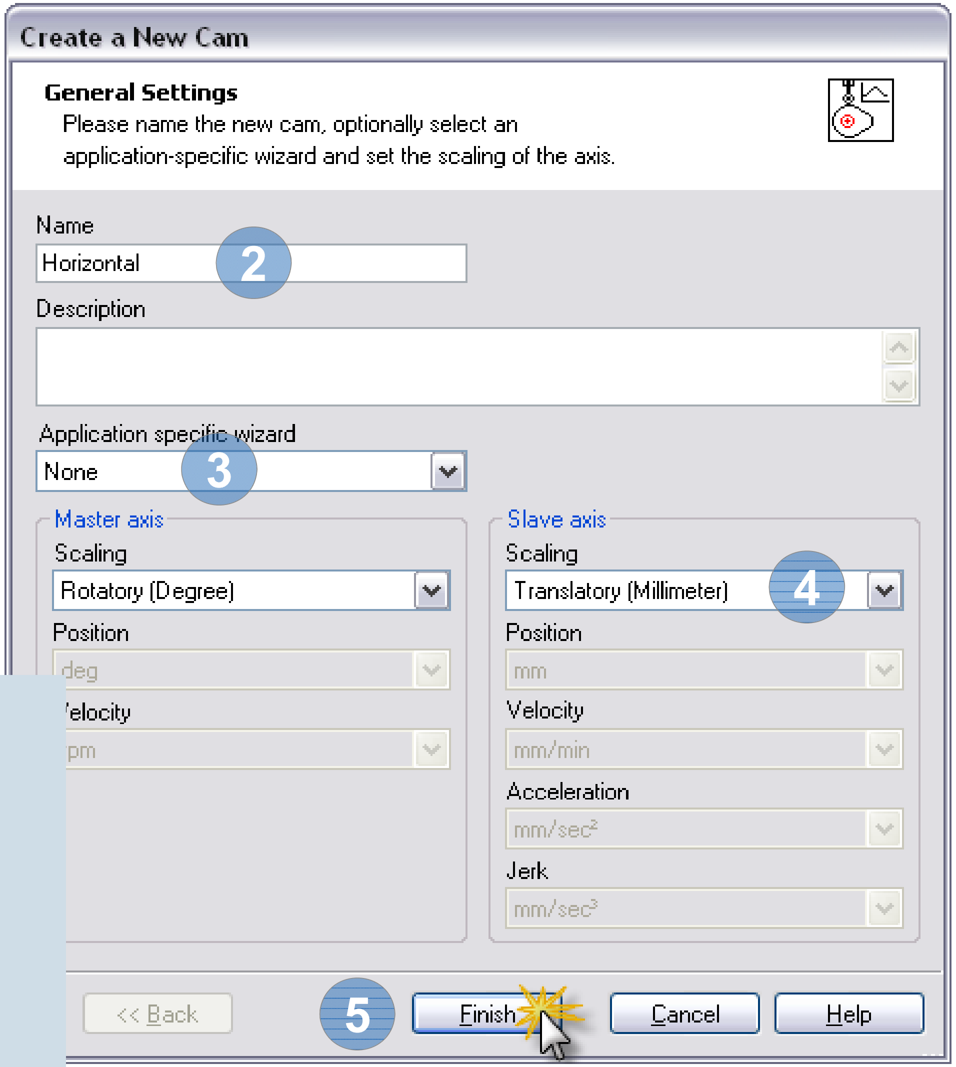
How to create a FlexProfile (2)





1. Create new Cam
2. Enter name “Horizontal”
3. Application specific wizard: None
4. Translatory scaling (mm)
5. Press “Finish” button

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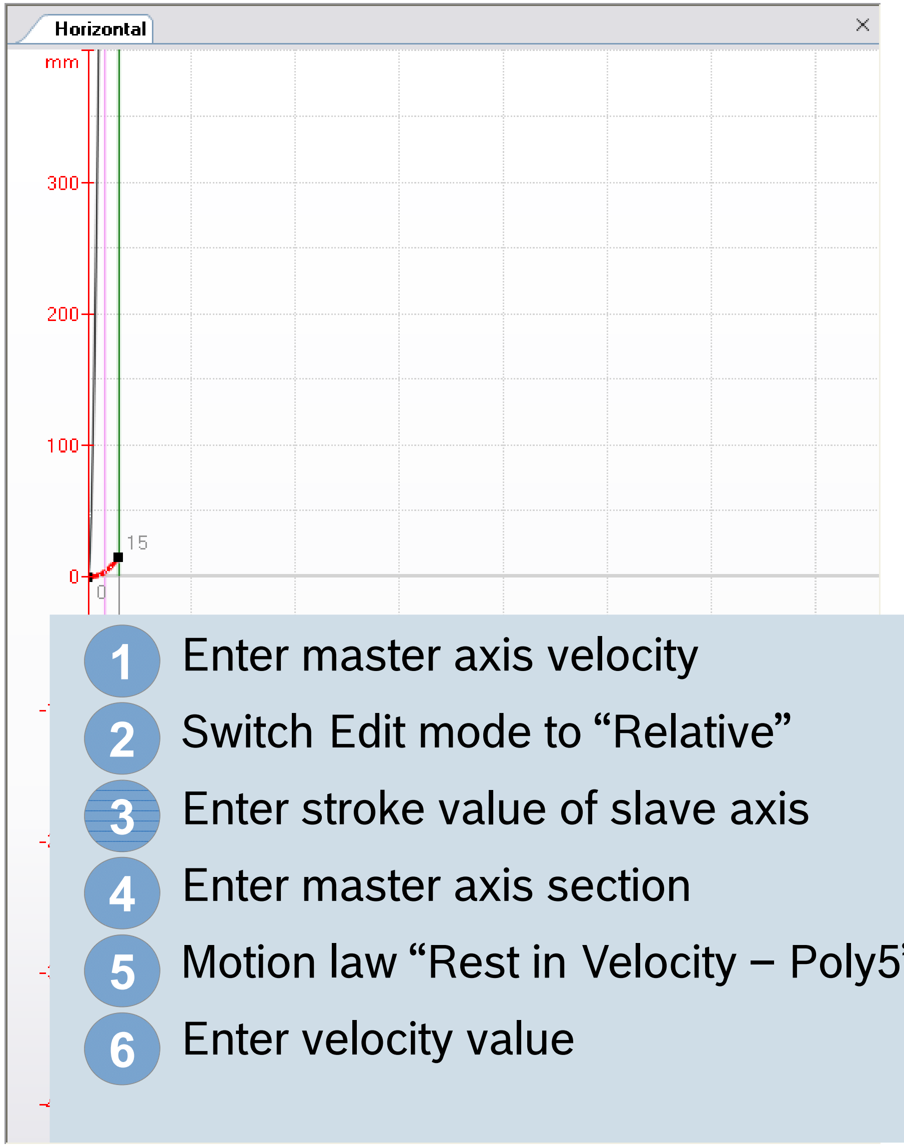
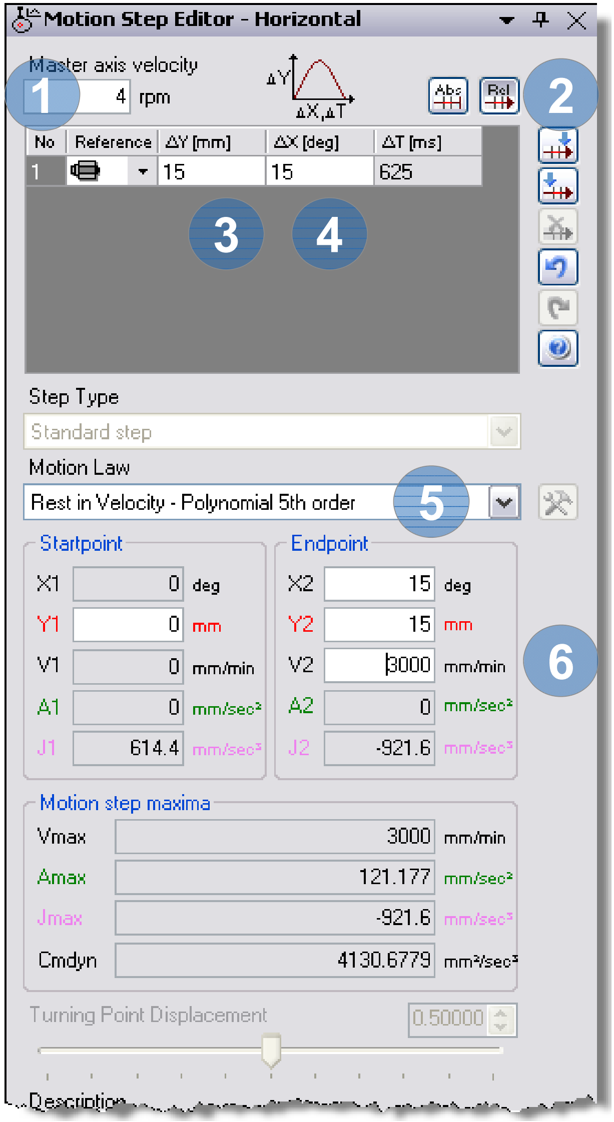
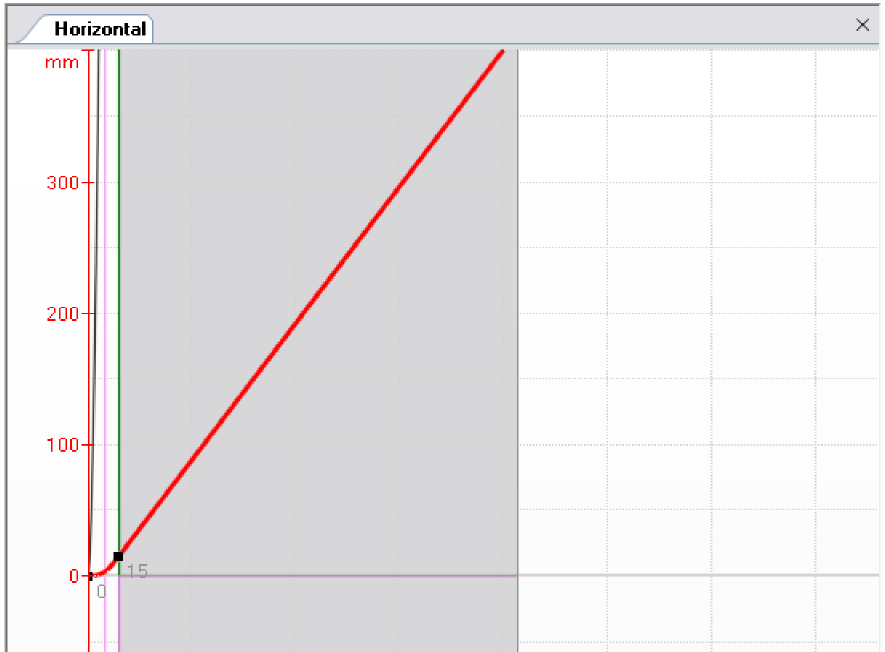
How to create a FlexProfile (3)

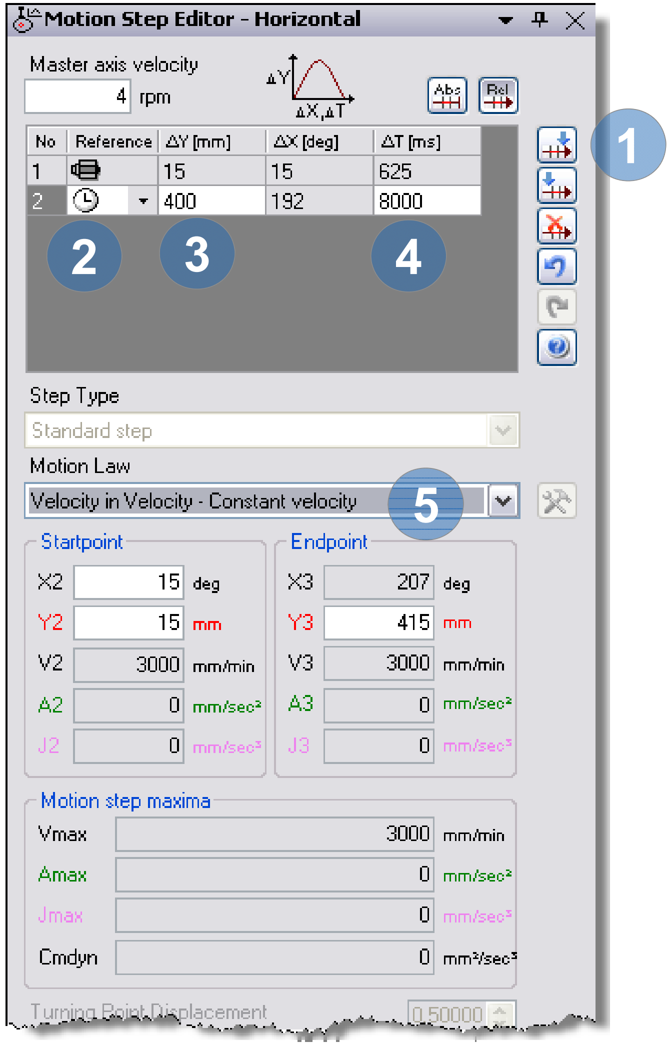


1. Enter master axis velocity
2. Switch Edit mode to “Relative”
3. Enter stroke value of slave axis
4. Enter master axis section
5. Motion law “Rest in Velocity – Poly5”
6. Enter velocity value



How to create a FlexProfile (4)





1. Add motion step after highlighted step
2. Select Reference to Time
3. Enter stroke value for slave axis
4. Enter duration of step
5. Motion law
   “Velocity in Velocity – Constant velocity”

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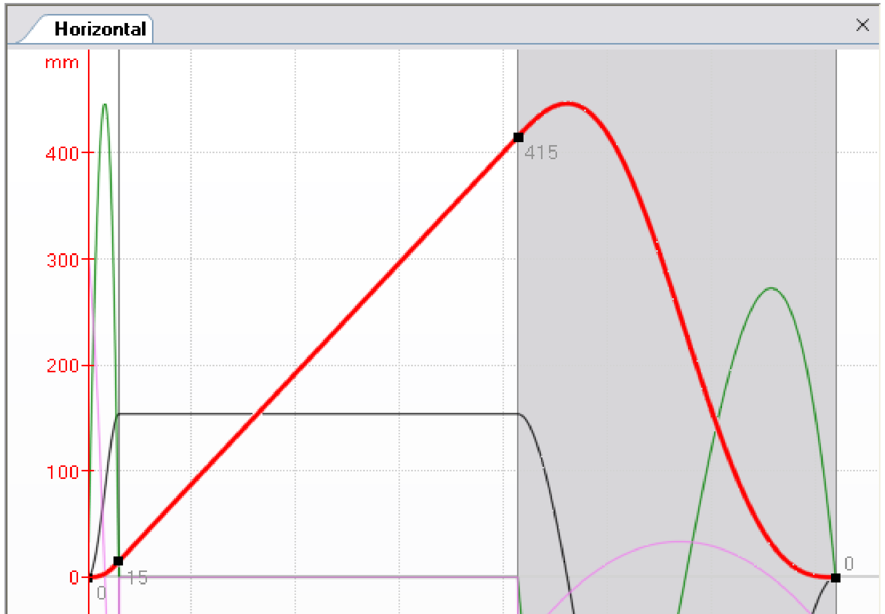
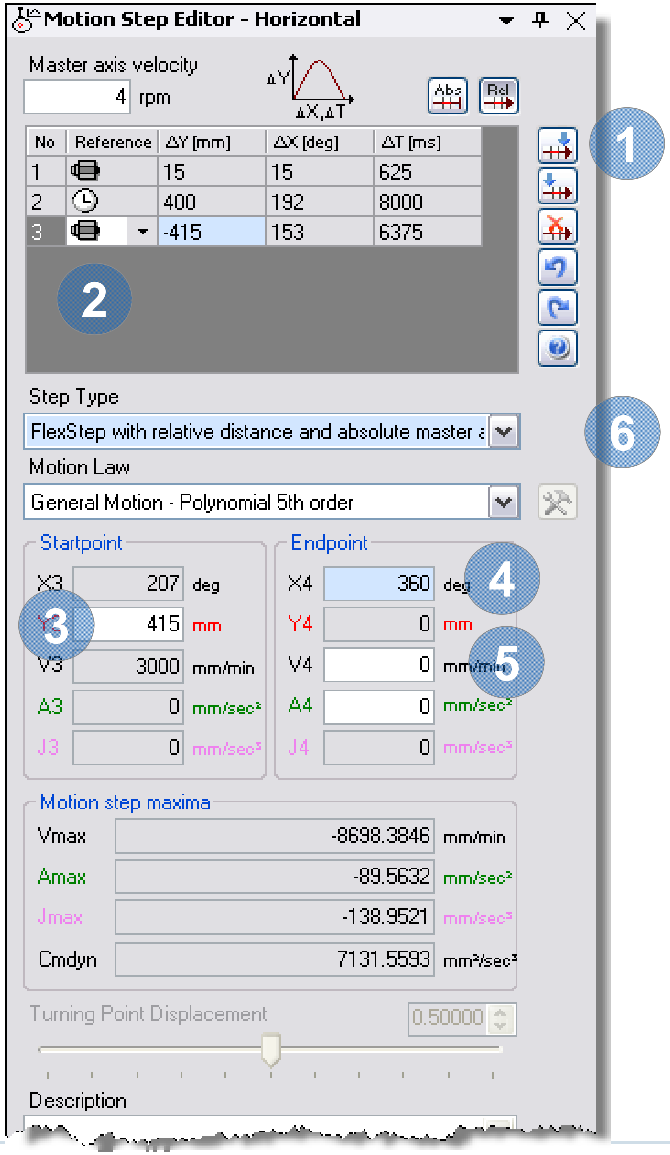
How to create a FlexProfile (5)





1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (-415)
4. Enter Master axis Endpoint (360)
5. Set velocity value to zero
6. Select FlexStep with relative distance and absolute master axis range

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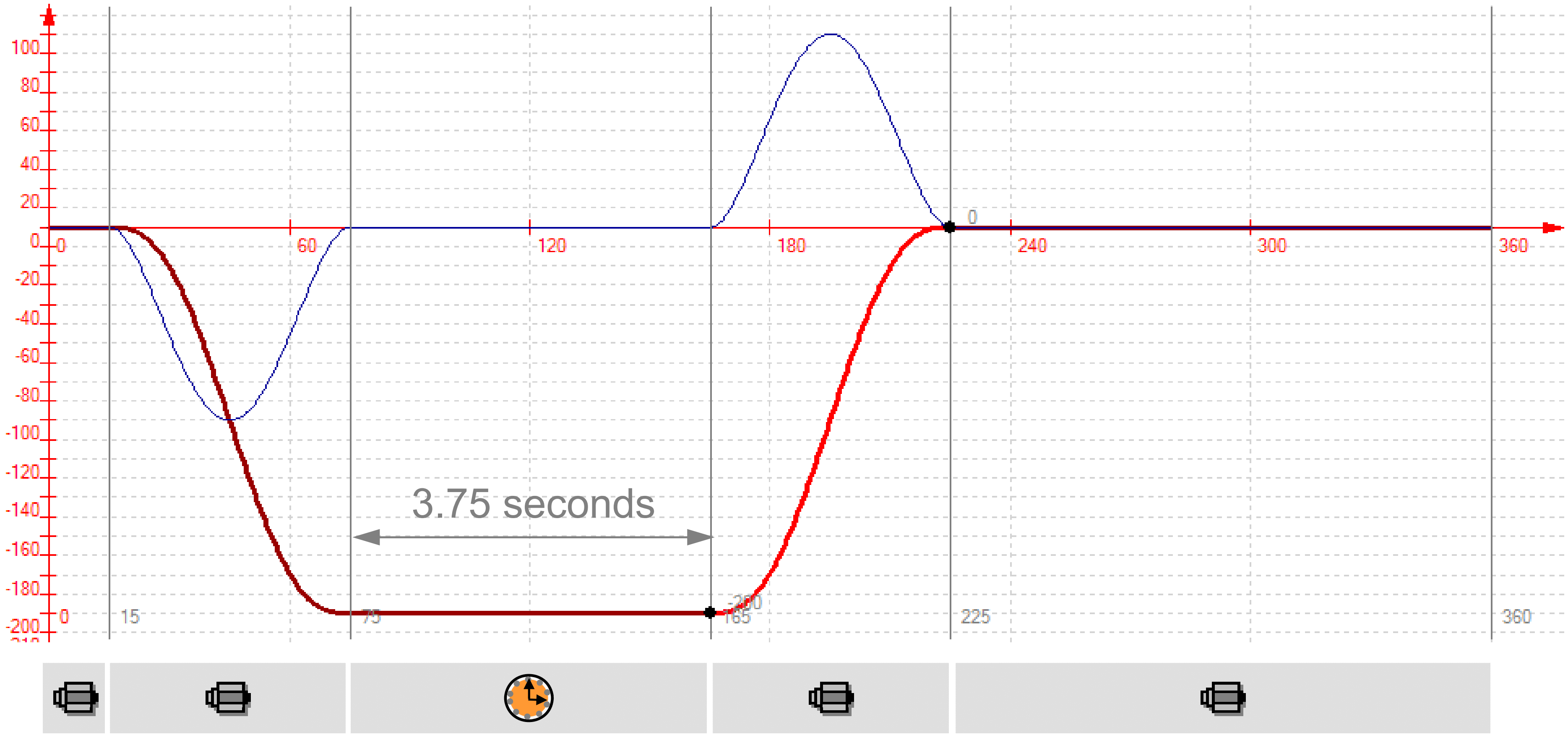
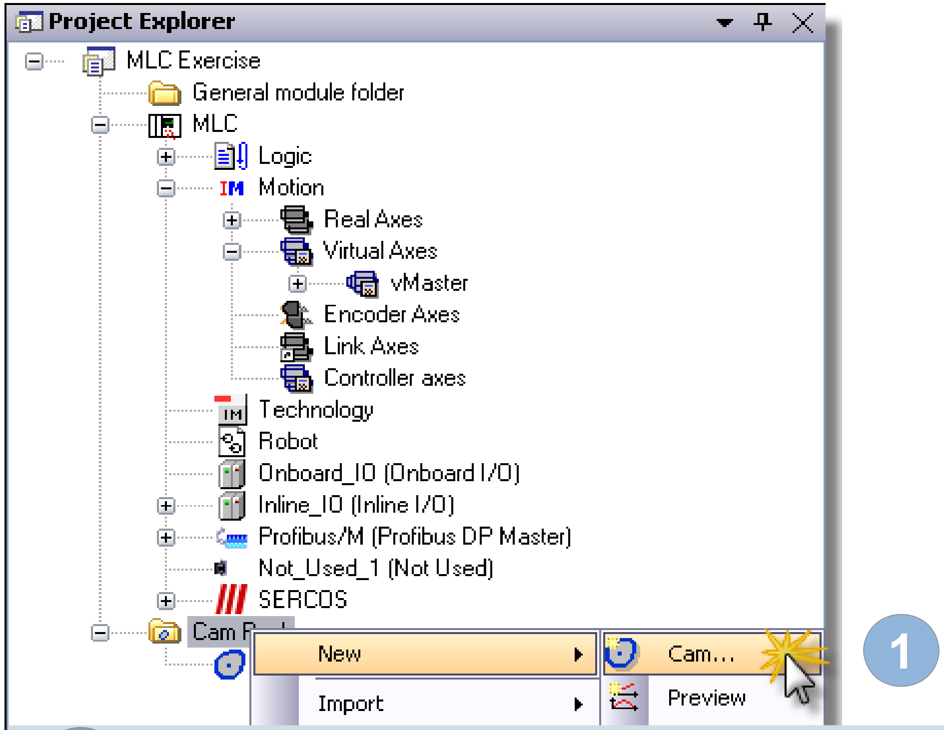
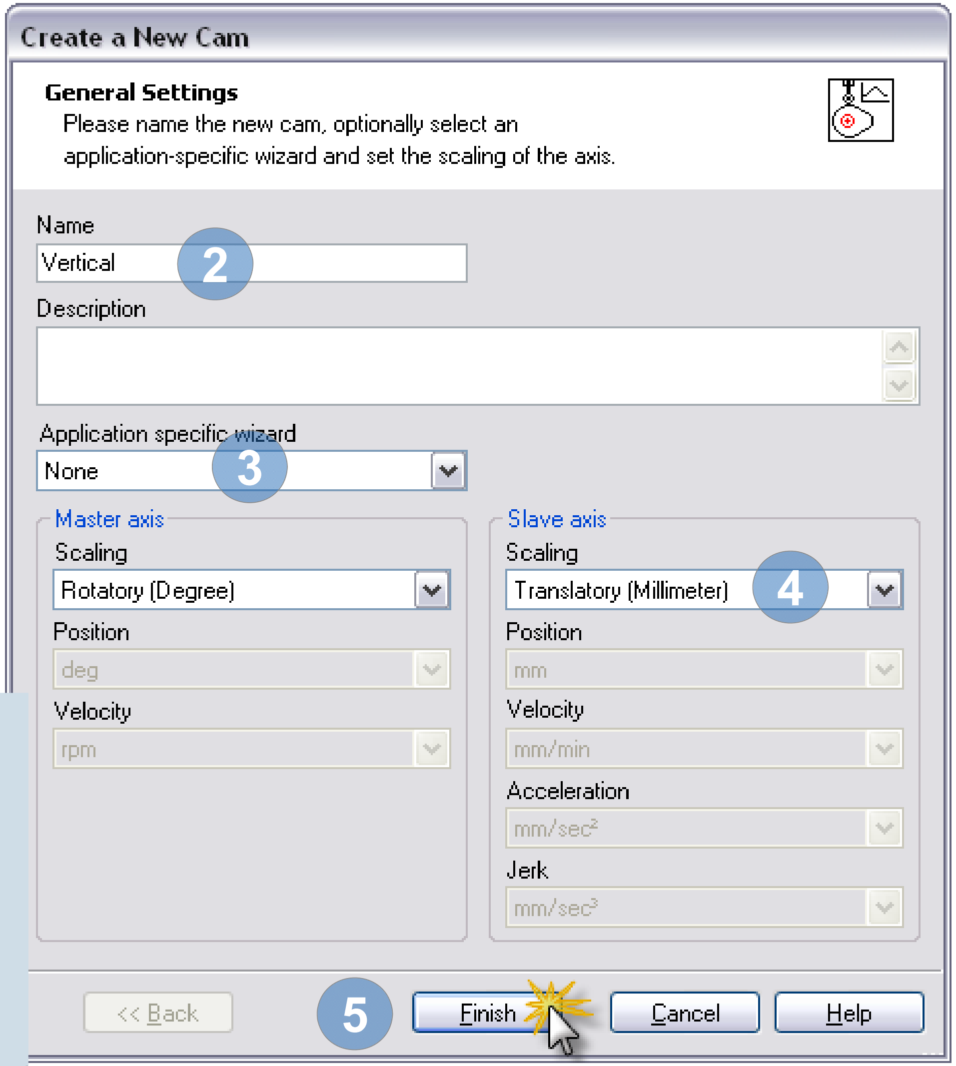
How to create a FlexProfile (6)



How to create a FlexProfile (7)





1. Create new Cam
2. Enter name “Vertical”
3. Application specific wizard: None
4. Translatory scaling (mm)
5. Press “Finish” button

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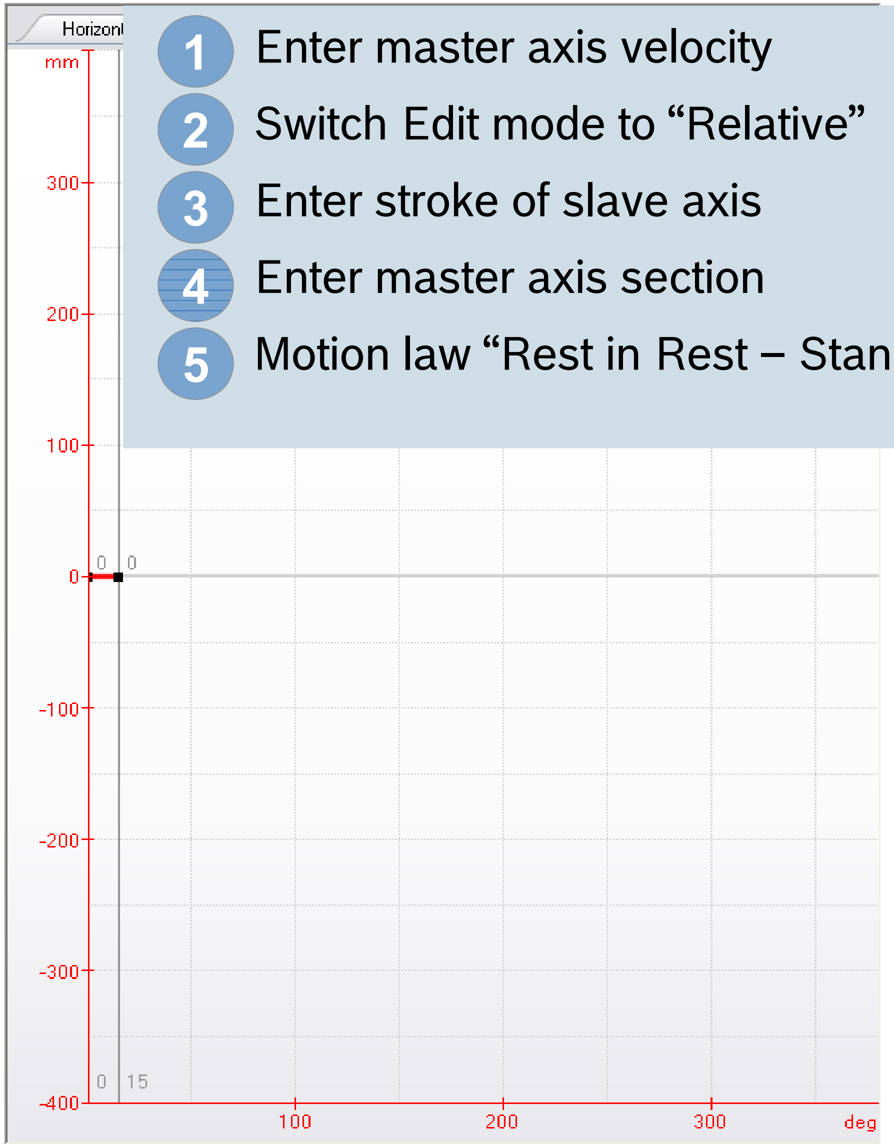
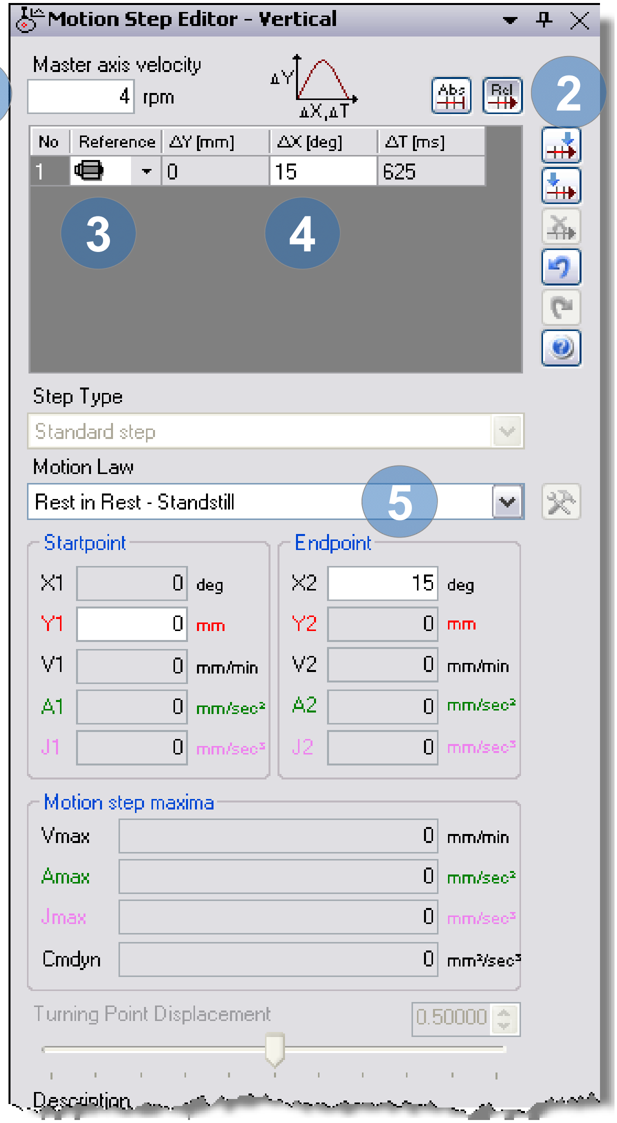
How to create a FlexProfile (8)

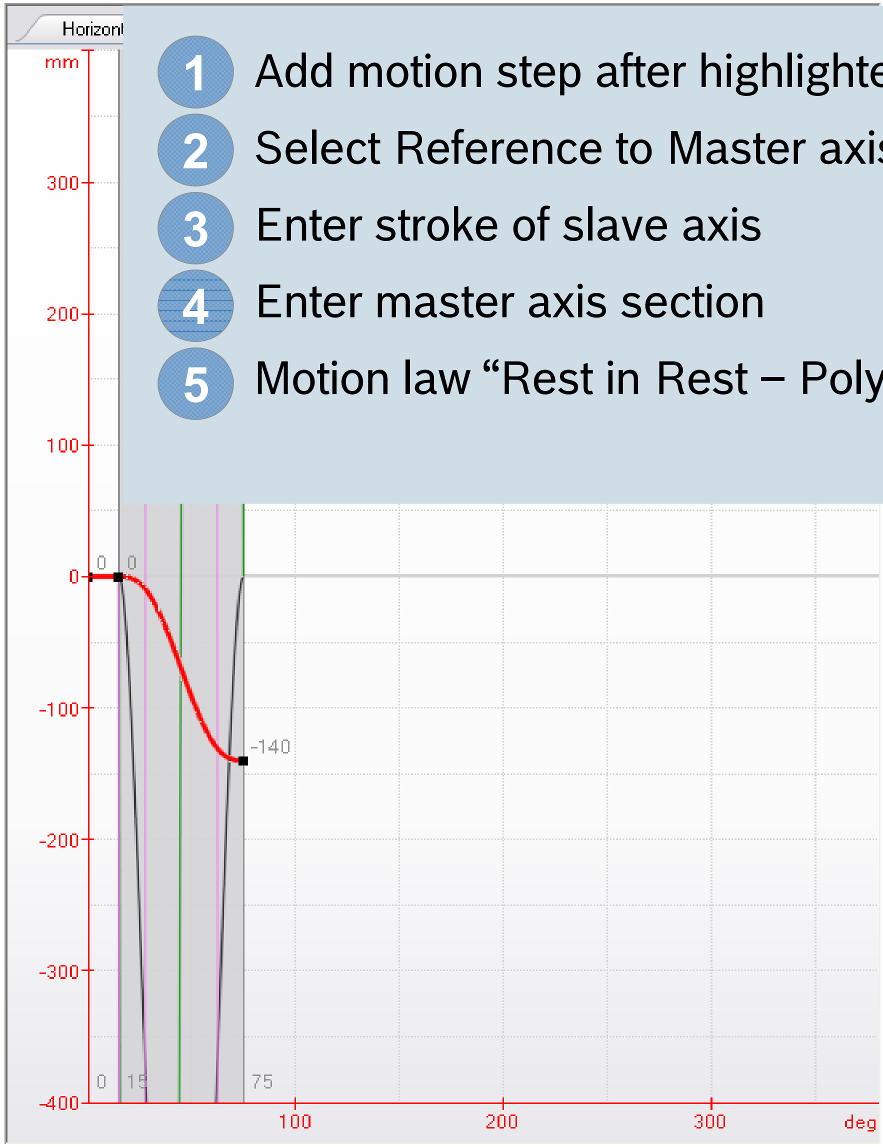
1. Enter master axis velocity
2. Switch Edit mode to “Relative”
3. Enter stroke of slave axis
4. Enter master axis section
5. Motion law “Rest in Rest – Standstill”

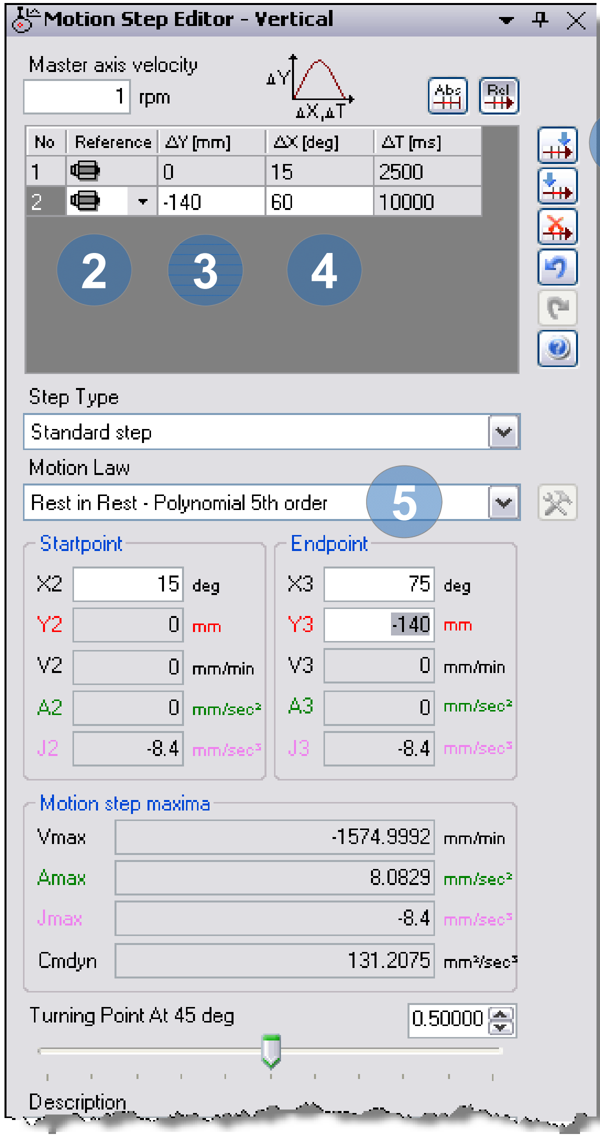




How to create a FlexProfile (9)

1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter stroke of slave axis
4. Enter master axis section
5. Motion law “Rest in Rest – Poly5”





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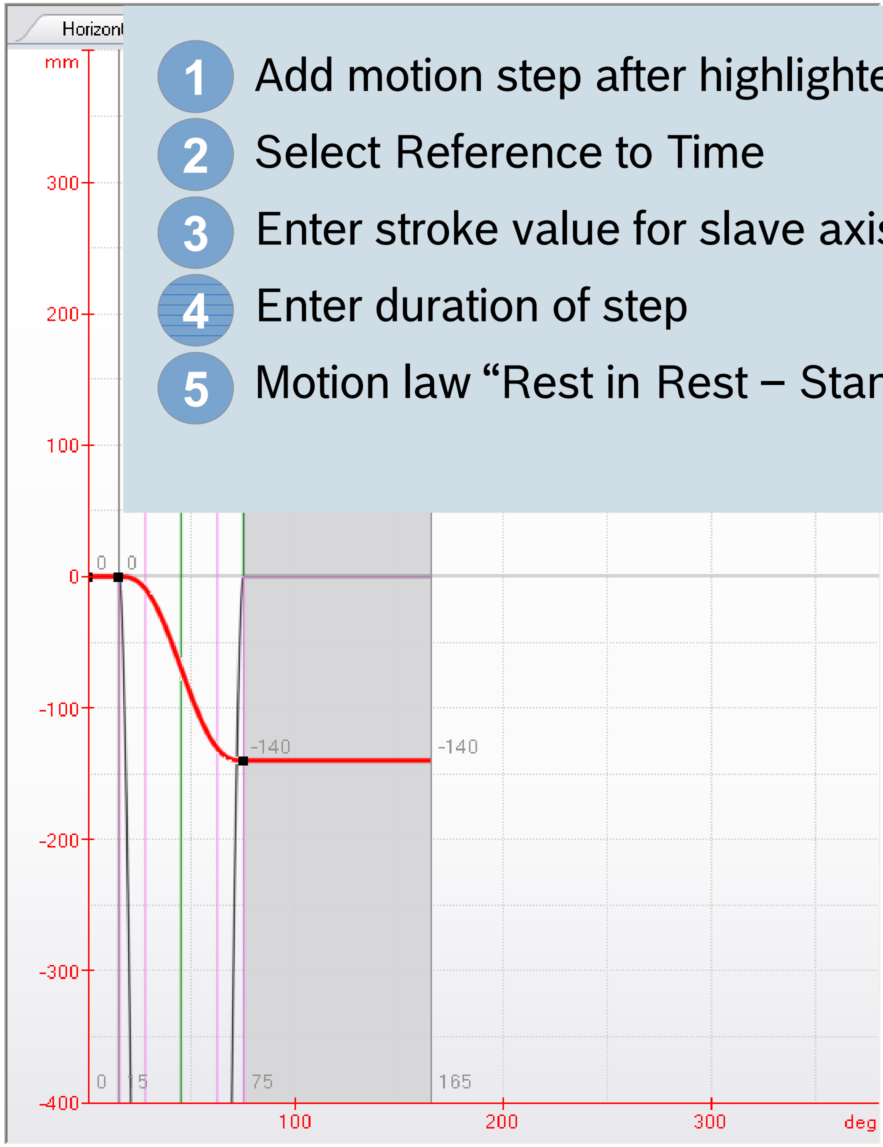
How to create a FlexProfile (10)

1. Add motion step after highlighted step
2. Select Reference to Time
3. Enter stroke value for slave axis
4. Enter duration of step
5. Motion law “Rest in Rest – Standstill”



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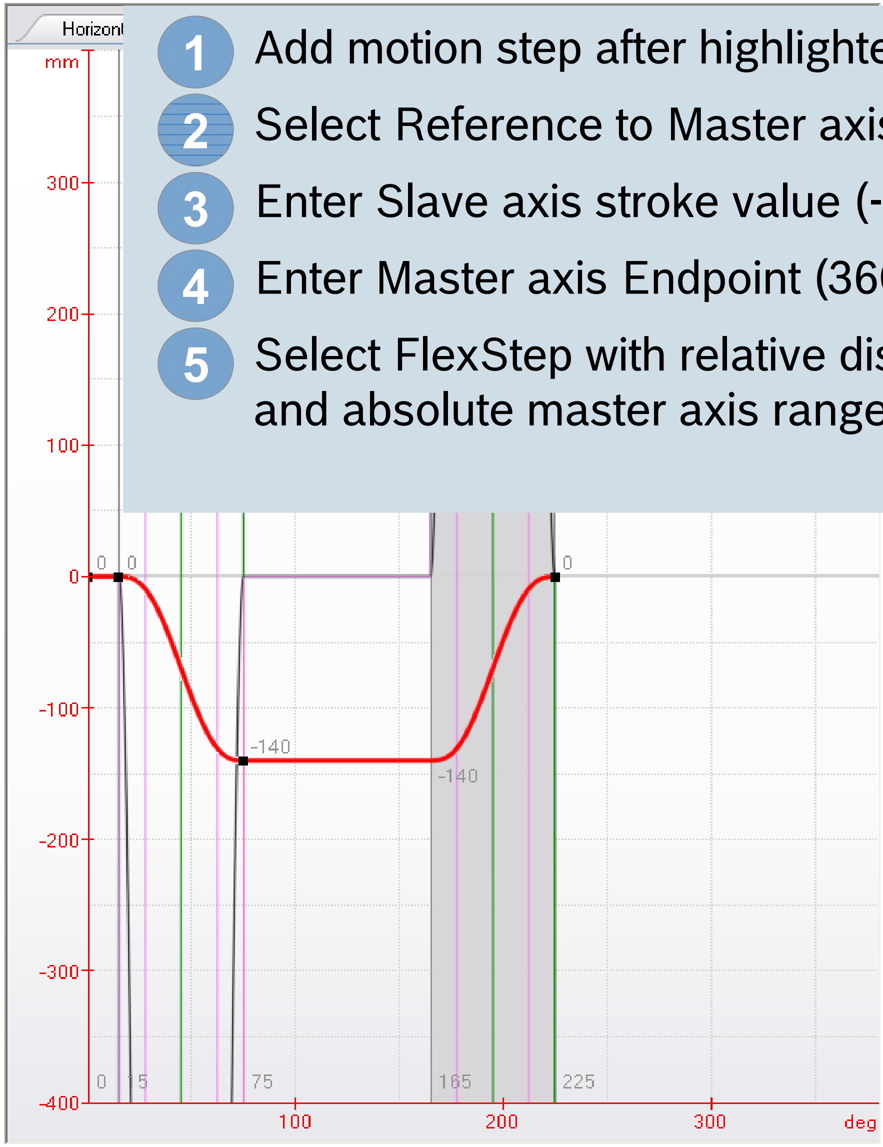
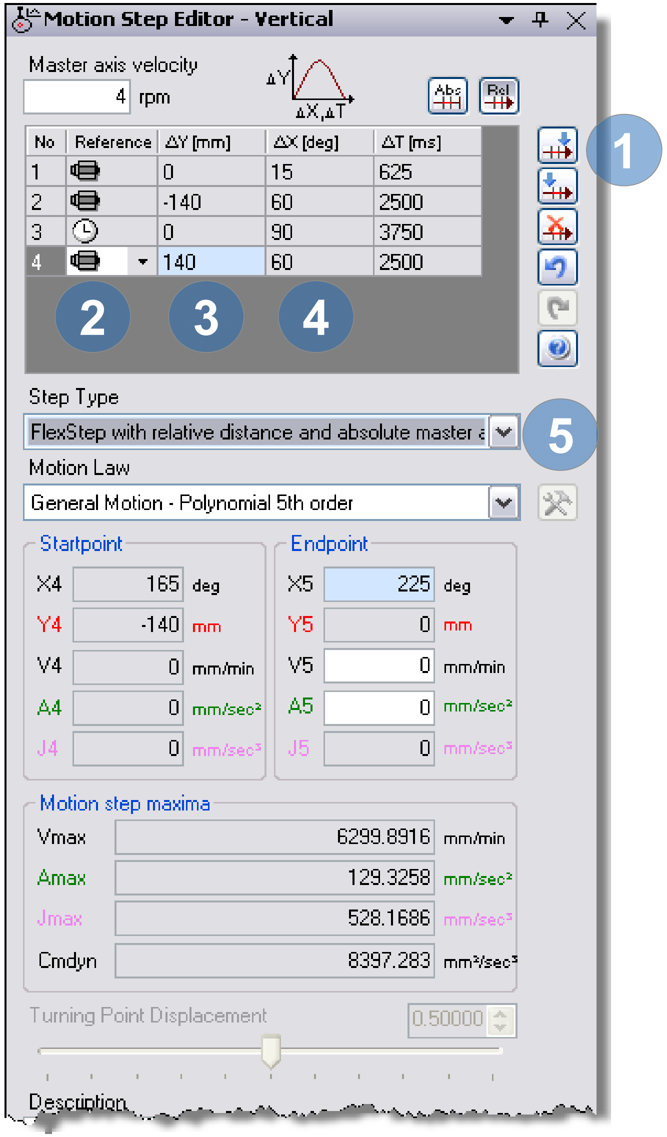
How to create a FlexProfile (11)

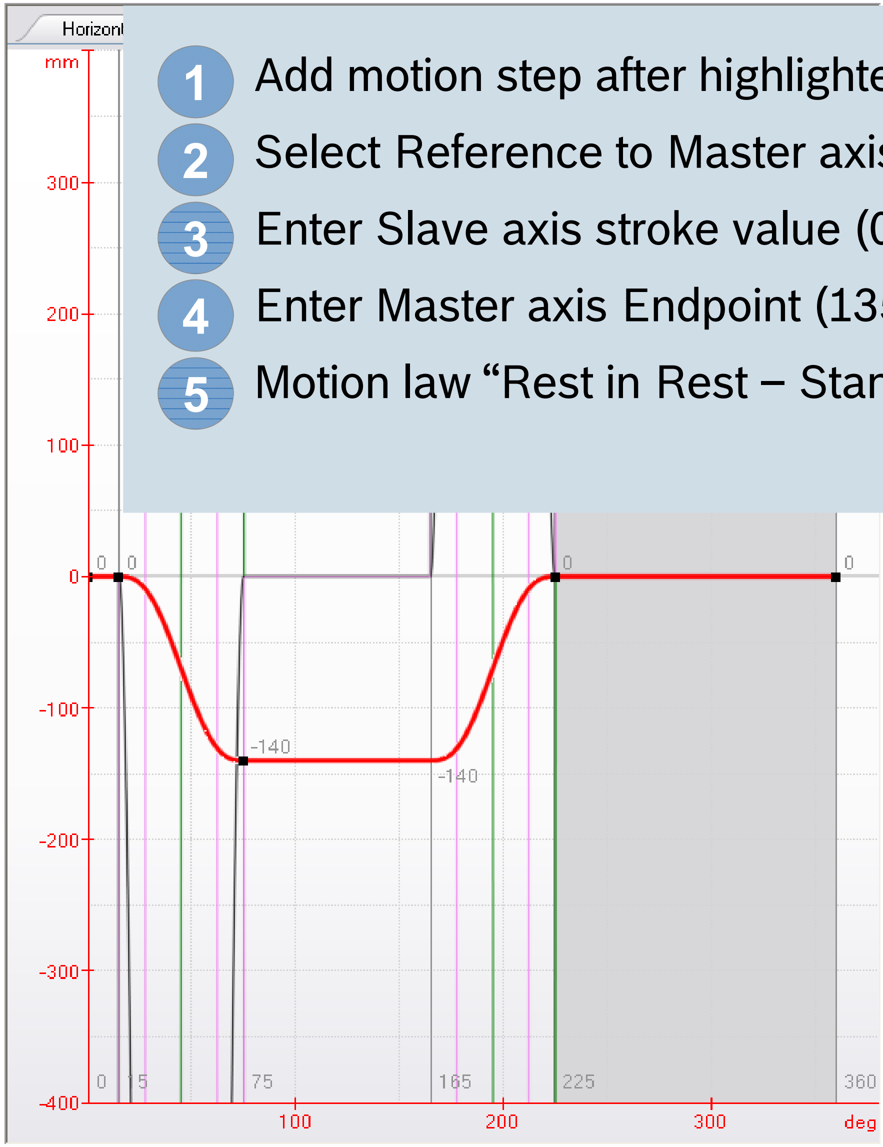
1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (-415)
4. Enter Master axis Endpoint (360)
5. Select FlexStep with relative distance and absolute master axis range





How to create a FlexProfile (12)

1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (0)
4. Enter Master axis Endpoint (135)
5. Motion law "Rest in Rest – Standstill"



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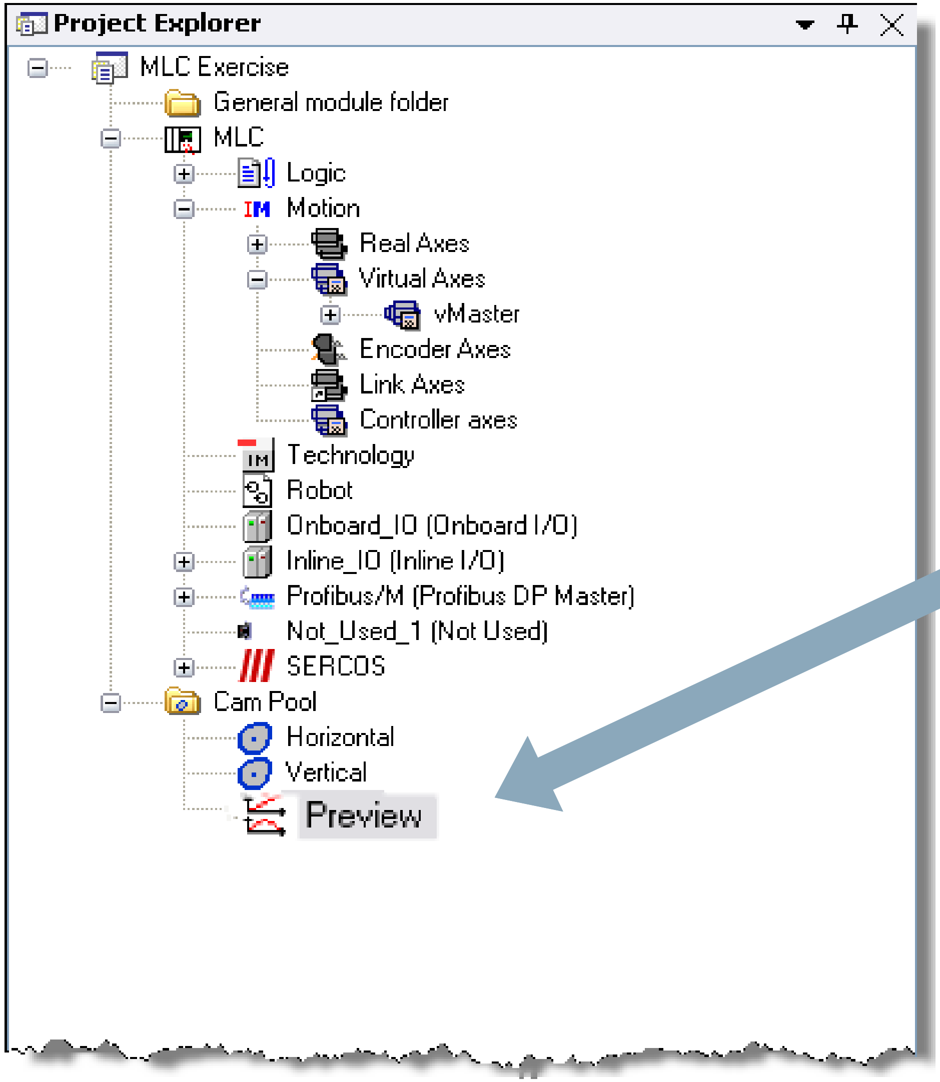
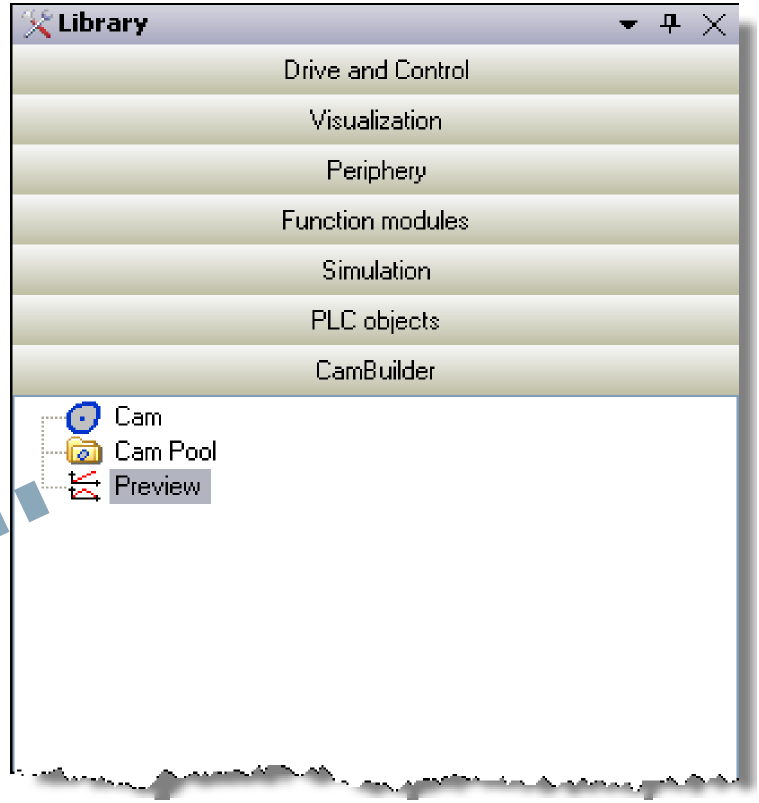
How to create a Preview





1. Open Group CamBuilder in Library
2. Select “Preview”
3. Add Preview to CamPool

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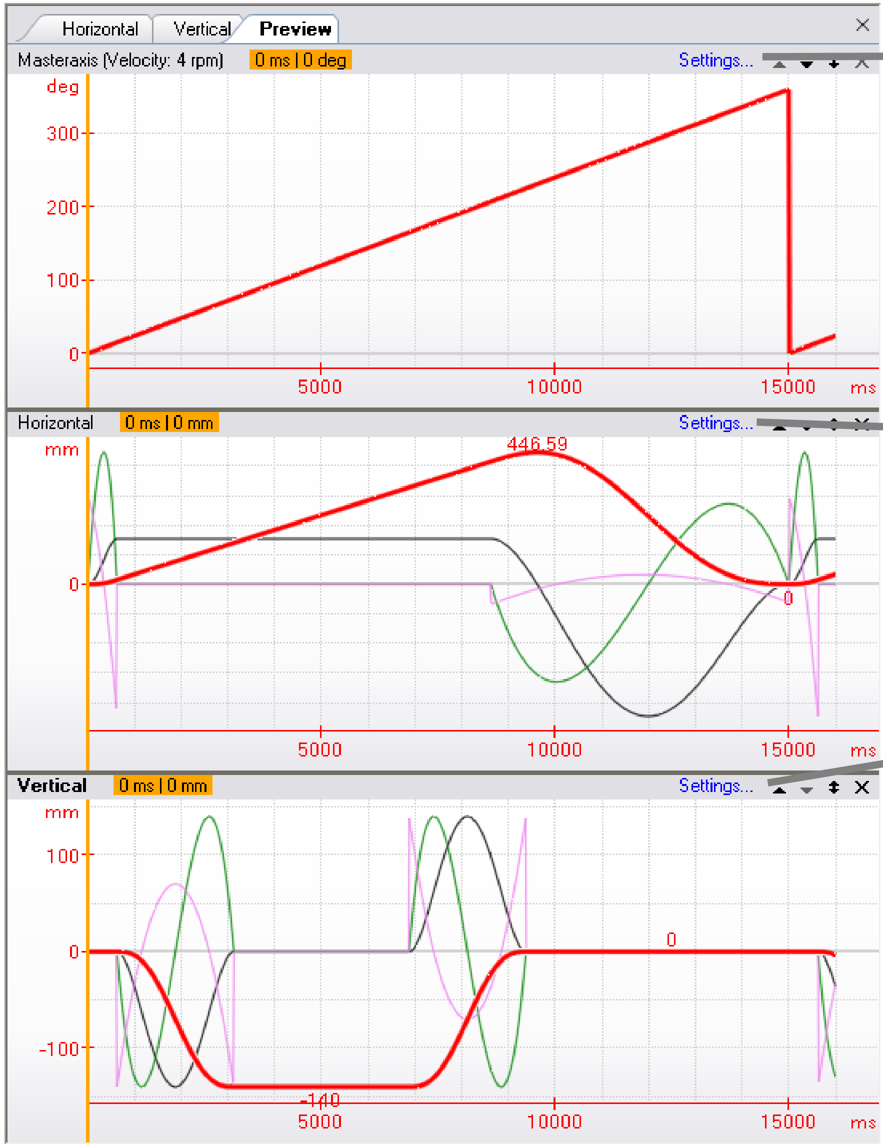
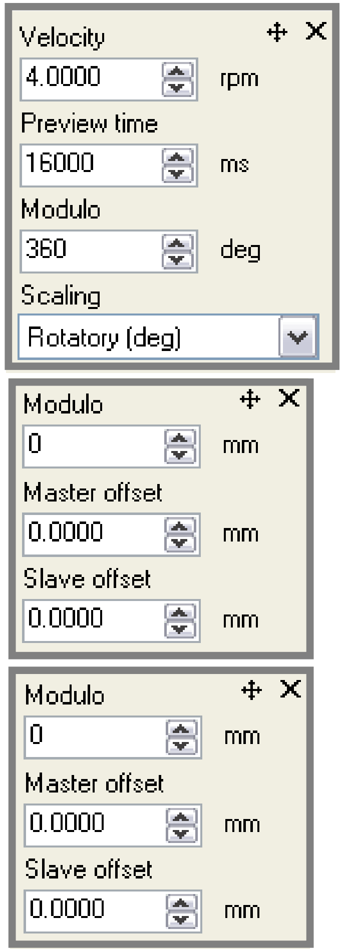
How to create a Preview





Add profiles to preview by
drag & drop and modify settings

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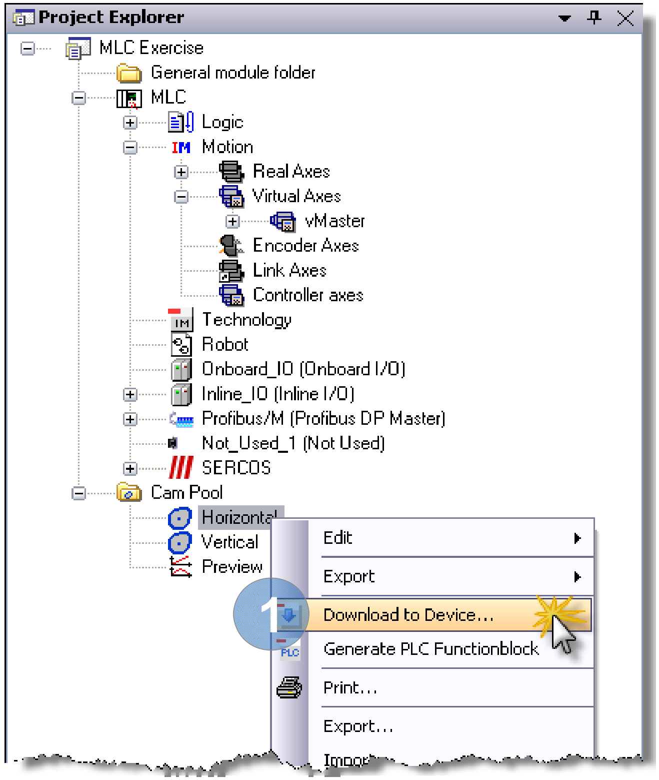
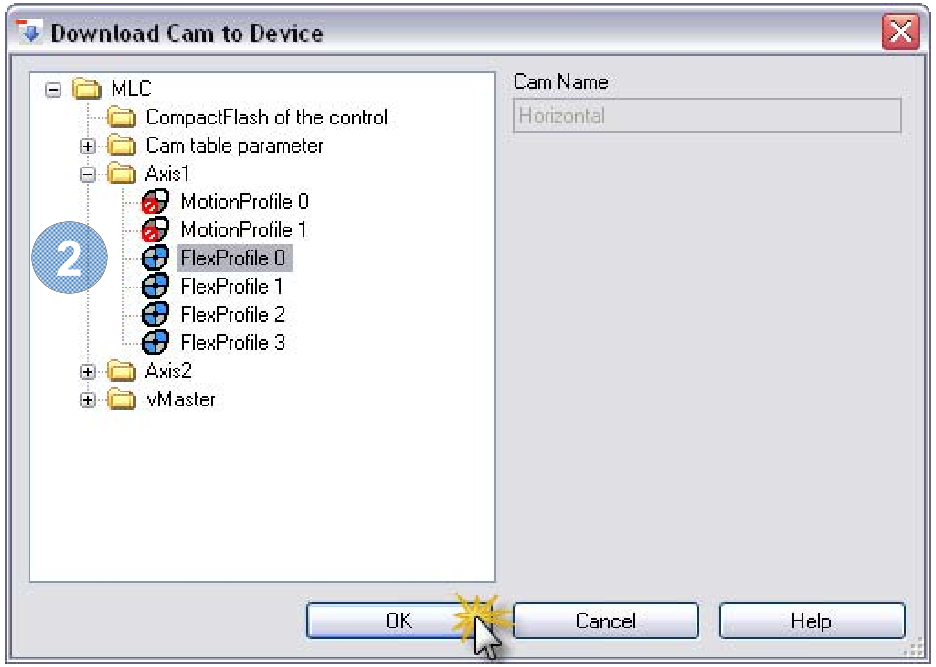
Bosch Group

Download FlexProfile for 1st axis





1. Select “Download to Device ...”
2. Select “FlexProfile 0” for 1st axis
3. After download to MLC an information is displayed

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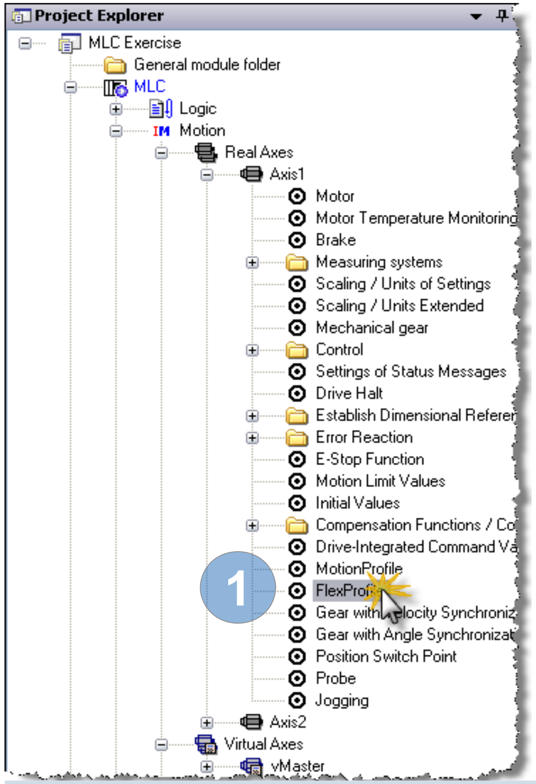
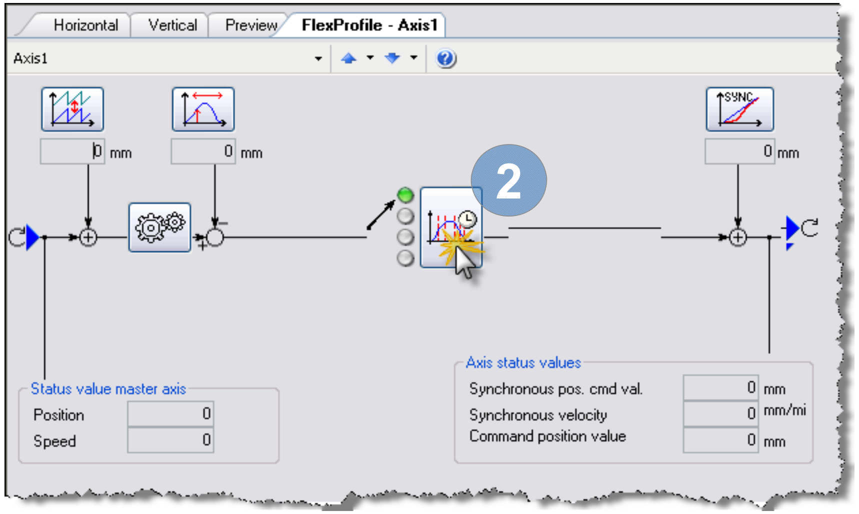
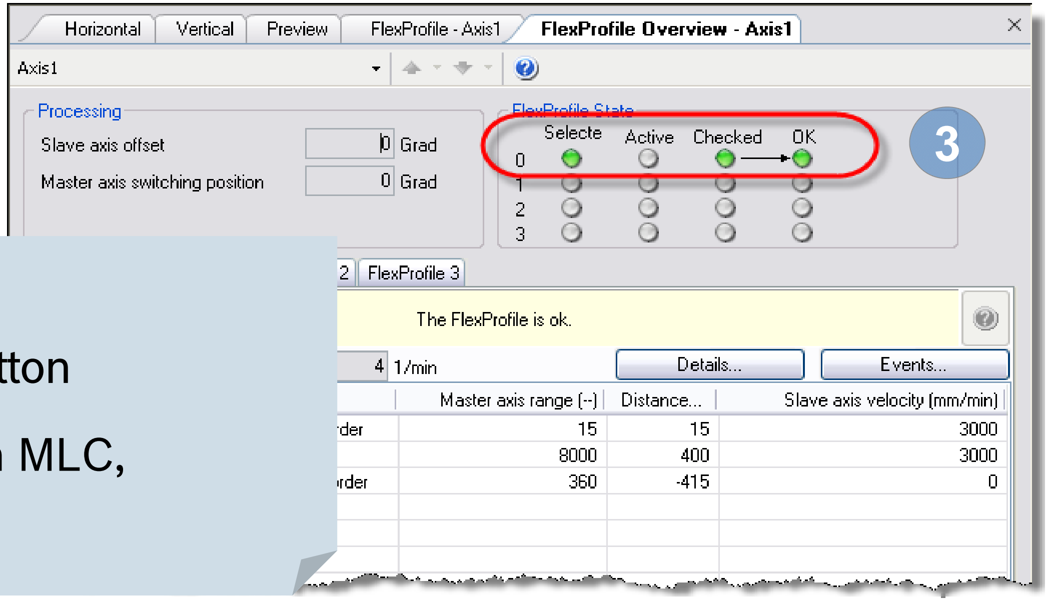
Download FlexProfile for 1st axis

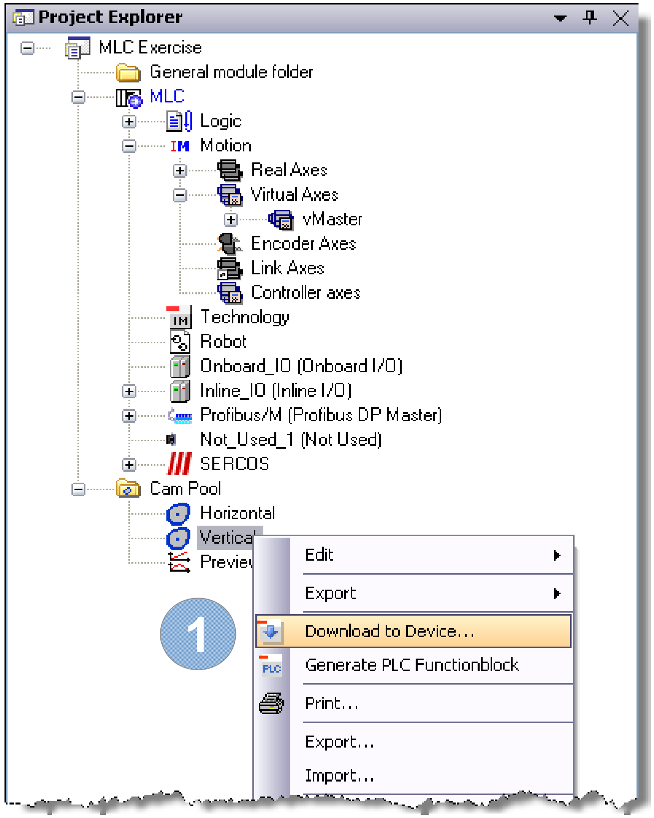


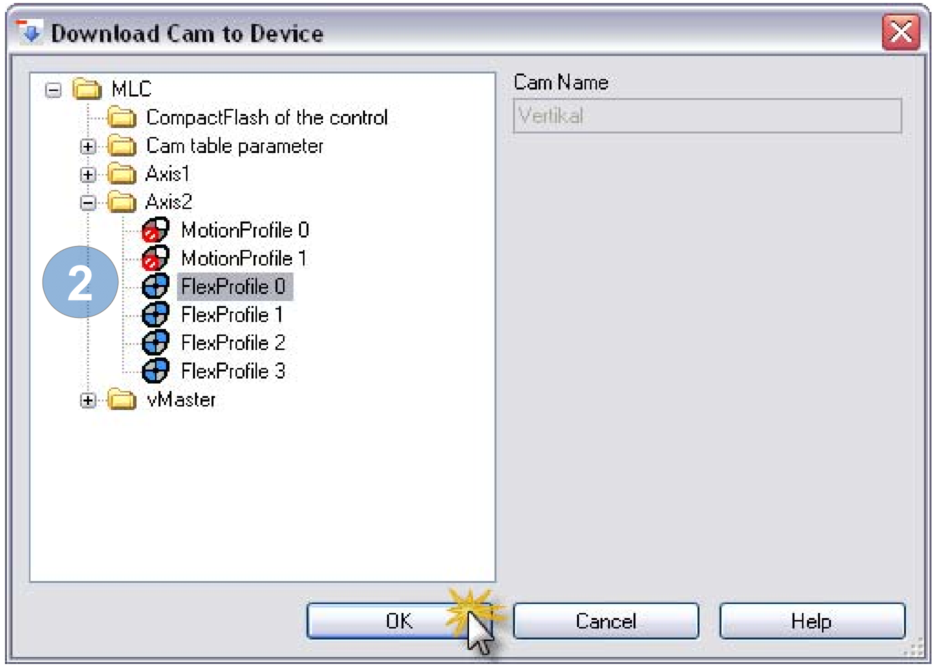




1. Double click on “FlexProfile”
2. Push “FlexProfile overview” button
3. The 1st FlexProfile is stored on MLC, has been checked and is OK!

Download FlexProfile for 2nd axis





1. Select “Download to Device ...”
2. Select “FlexProfile 0” for 2nd axis
3. After download to MLC an information is displayed

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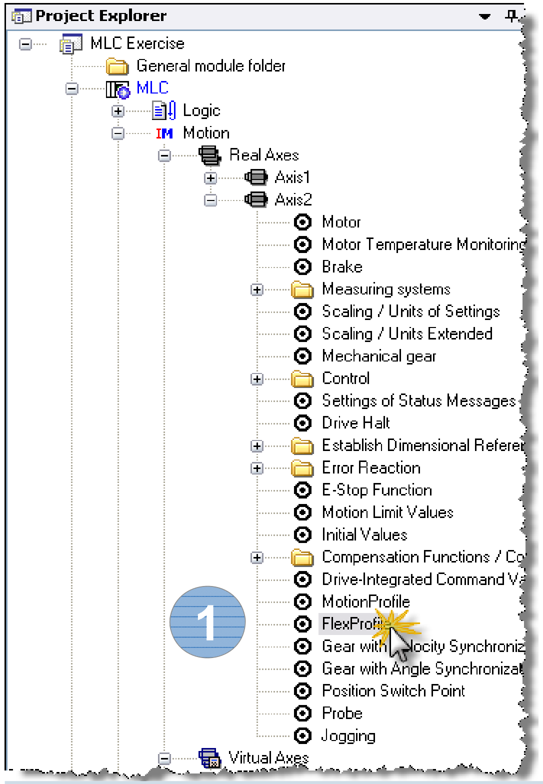
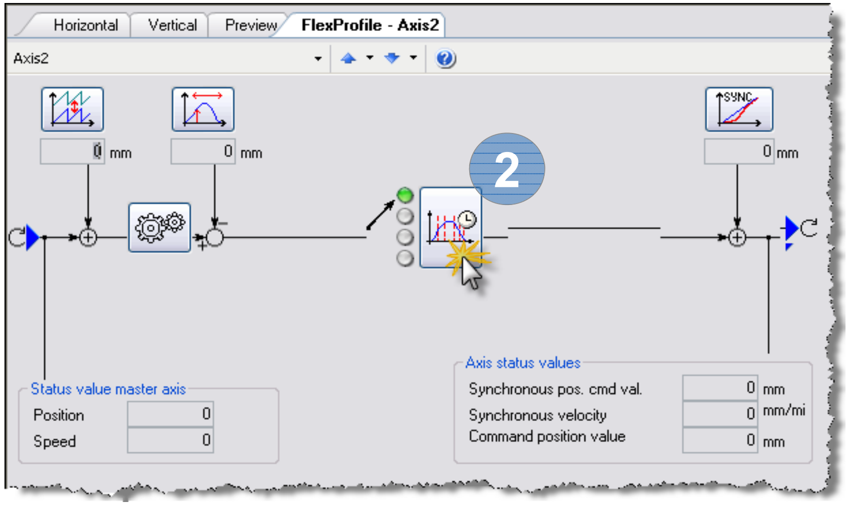
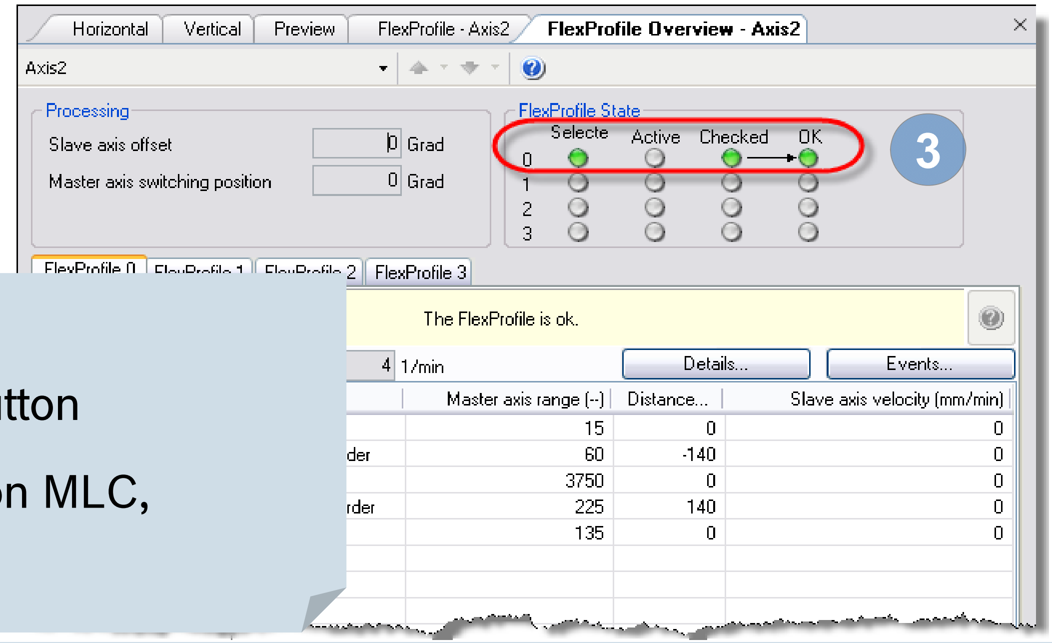
Download FlexProfile for 2nd axis







1. Double click on “FlexProfile”
2. Push “FlexProfile overview” button
3. The 2nd FlexProfile is stored on MLC, has been checked and is OK!

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Agenda

- Introduction to IndraMotion MLC
- System topology and system components
- IndraWorks licenses and installation
- First steps with IndraWorks
- Motion Programming Basics
- MLC Diagnosis system
- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Advanced Topics

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AxisInterface – Documentation

- Complete Documentation of Axis Interface

- ... available on the media directory
- or in print format
- in German and English language

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AxisInterface – Why?

- AxisInterface bundles all functions which are available with PLCopen to a concise and simple data interface
- Bidirectional communication
  - arAxisCtrl_gb – Command interface
  - arAxisStatus_gb – State information
- Less PLCopen knowledge required
- Faster program development (less code required due to efficient command interface)
- Good usability due to “Intellisense”
- Ready-to-use IndraLogic visualization for easy commissioning
- Complete documentation available

AxisInterface considerably simplifies the motion programming!

AxisInterface – Data structure arAxisCtrl_gb

Administration
- Select operating mode (e.g. positioning, velocity, Cam, ...)
- Axis reference ...

Command values in positioning mode
- target position
- velocity
- acceleration ...

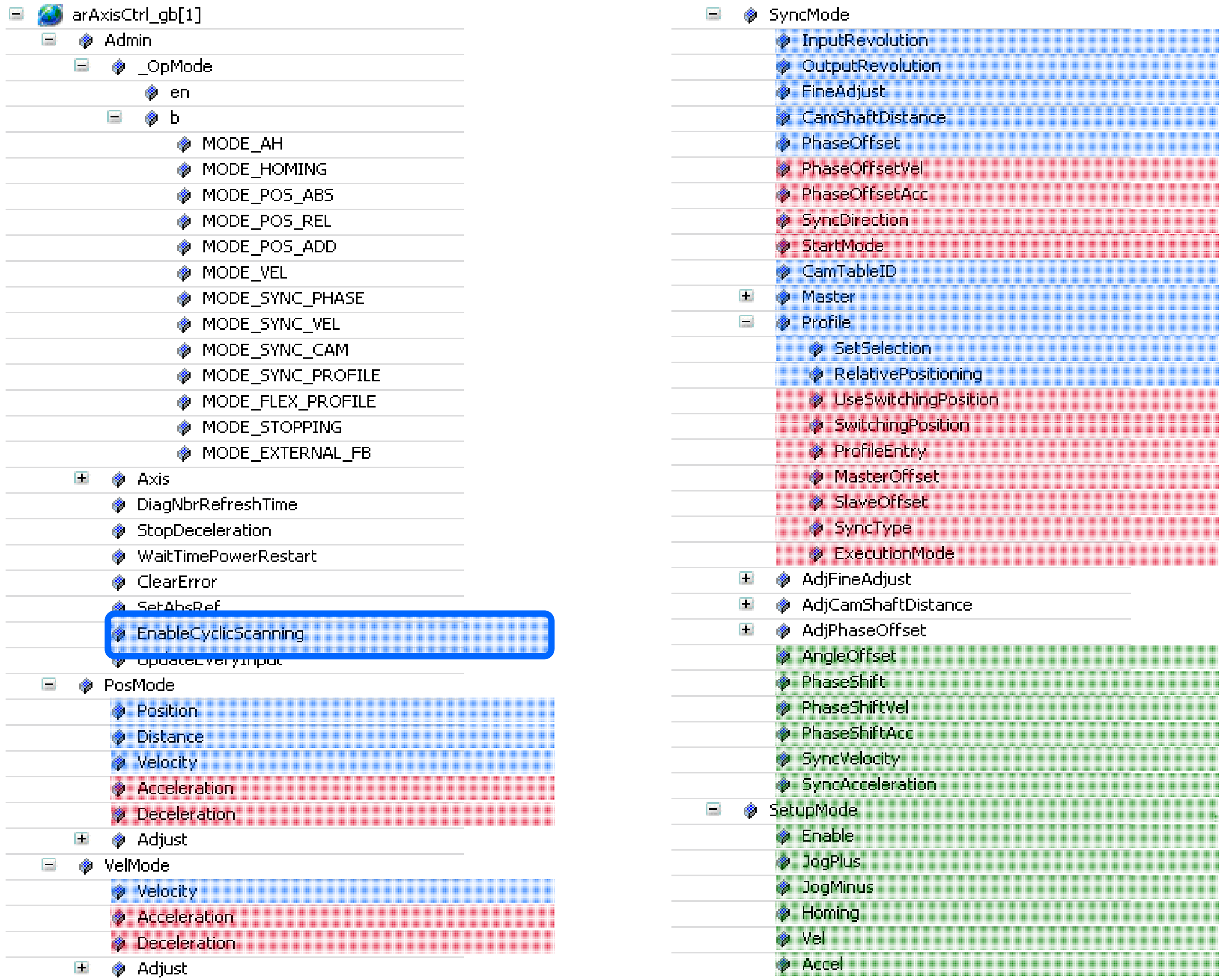
Command values in velocity control mode
- velocity
- acceleration ...

Command values in synchronous mode
- dynamic synchronization
- select master axis
- gear ratio
- select Cam profile ...

Command values for Jogging / Homing
- Jog direction
- velocity
- acceleration ...

Extension by GAT

AxisInterface – Cyclic and non-cyclic scanning



Cyclic scanning

Non-cyclic scanning

GAT assistant extension

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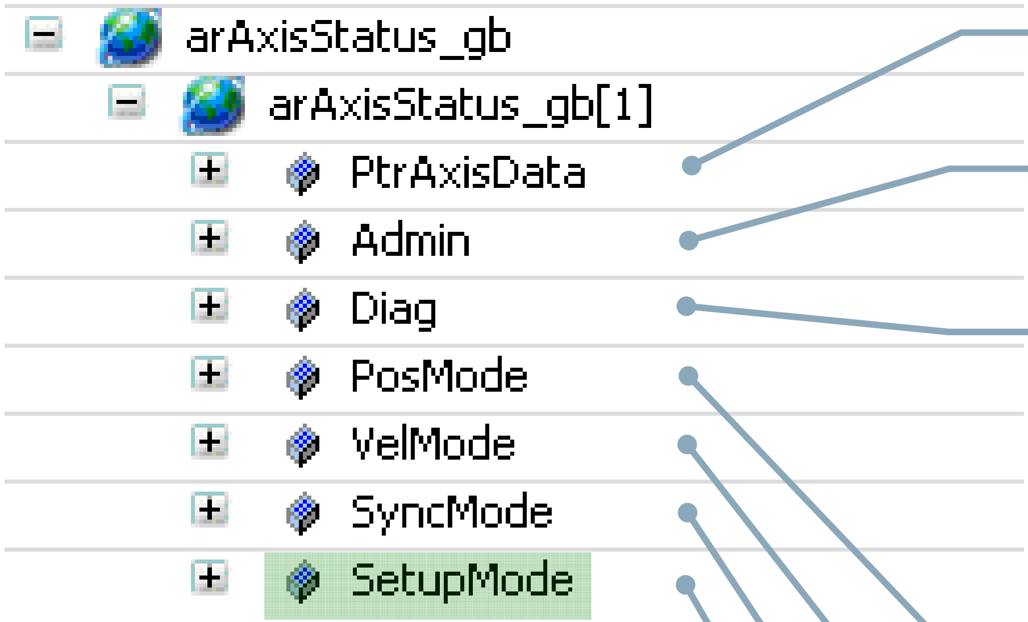
AxisInterface – Data structure arAxisStatus_gb



Pointer to AxisData entry

Administration / Axis state
- Actual operating mode

Axis diagnostics
- Warning
- Error
- Error codes
- ...

Status in positioning mode

Status in velocity control mode

Status for synchronous mode

Status for Jogging / Homing

GAT assistant extension

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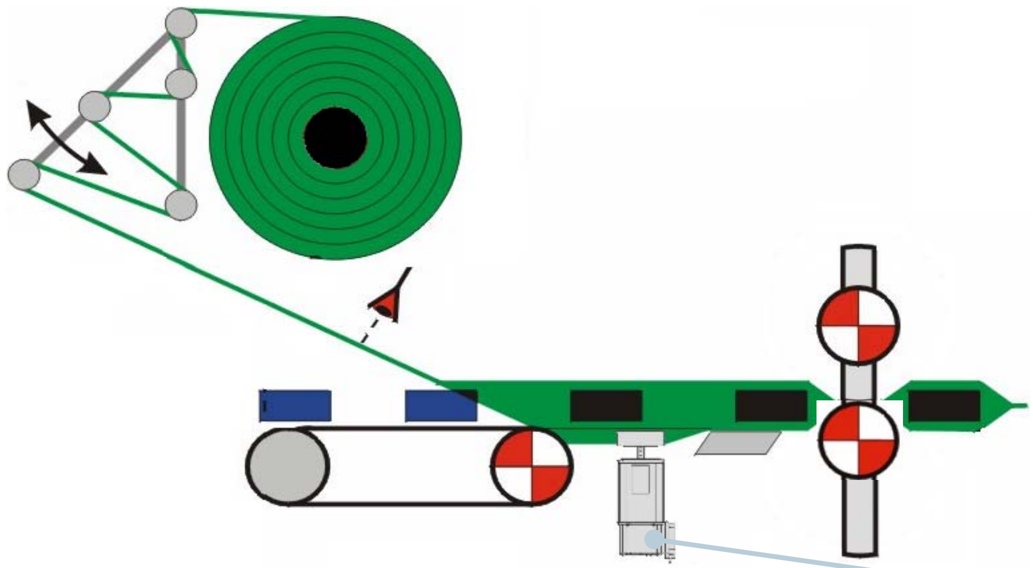
AxisInterface – Application example



- Phase synchronization of the Fin seal axis
  - Master: virtualAxis1
  - Ratio 3:2
  - absolute synchronization
  - dynamic synchronization only in positive direction

Fin seal axis

Sample program:

```
arAxisCtrl_gb[SealAxis.AxisNo].SyncMode.Master := virtualAxis1;
arAxisCtrl_gb[SealAxis.AxisNo].SyncMode.InputRev := 3;
arAxisCtrl_gb[SealAxis.AxisNo].SyncMode.OutputRev := 2;
arAxisCtrl_gb[SealAxis.AxisNo].SyncMode.StartMode := ABSOLUTE;
arAxisCtrl_gb[SealAxis.AxisNo].SyncMode.SyncDirection := POS_DIRECTION;
arAxisCtrl_gb[SealAxis.AxisNo].Admin._OpMode.en := ModePhaseSync;
```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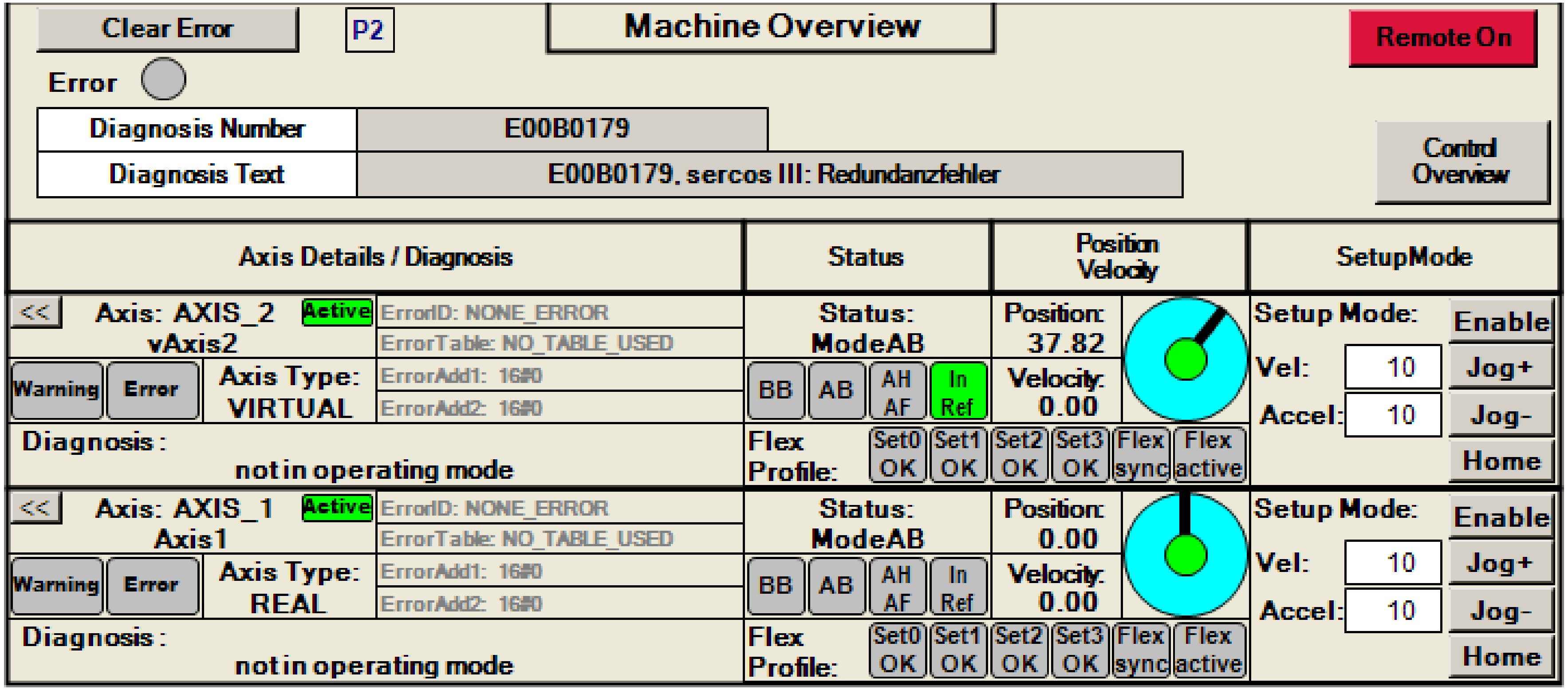
ImcInterface – ImcCtrl and ImcStatus

ImcCtrl
- Admin
  - _OpMode
  - ClearError
  - RetriggerOpMode

ImcStatus
- InOperation
- Admin
  - _OpModeAck
  - ModeStatus_P0
  - ModeStatus_P1
  - ModeStatus_P2
  - ModeStatus_P3
  - ModeStatus_BB
  - PwrUpModeAck
  - PhaseSwitchActive
- Diag
  - Error
  - Warning
  - ErrorID
  - ErrorIdent
  - ClearErrorAck
  - ErrSeverity
  - Number
  - Message
- PassiveMode

AxisInterface – IndraLogic Visualization



- IndraLogic Visualization is part of the AxisInterface
- After adjustment of the visualization to the correct axis configuration the axis can be commanded immediately without any coding!

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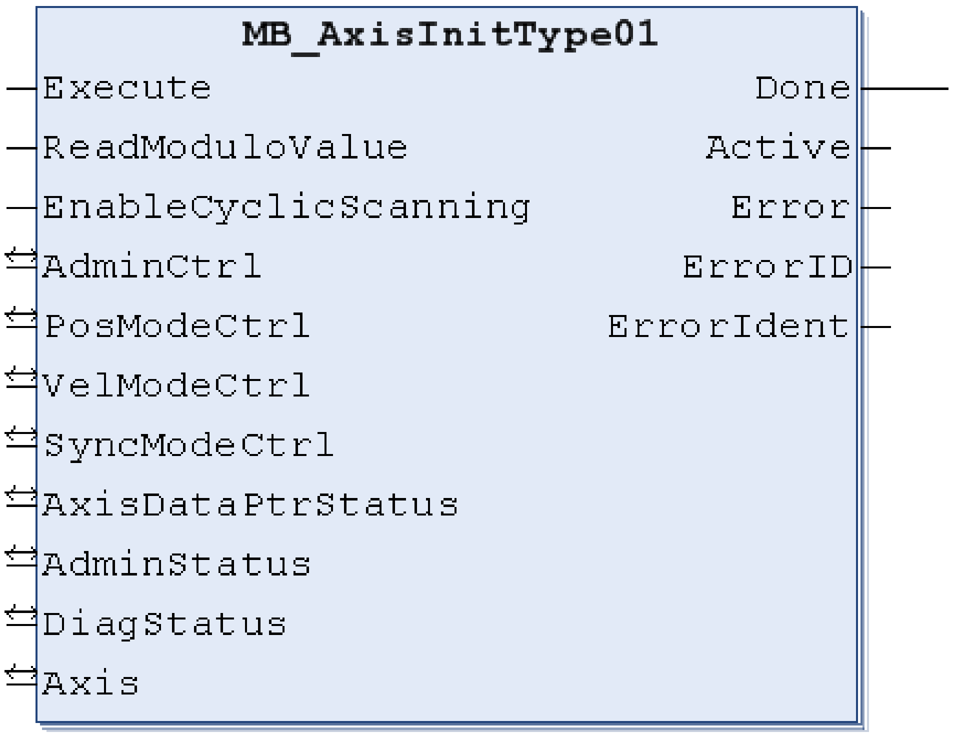
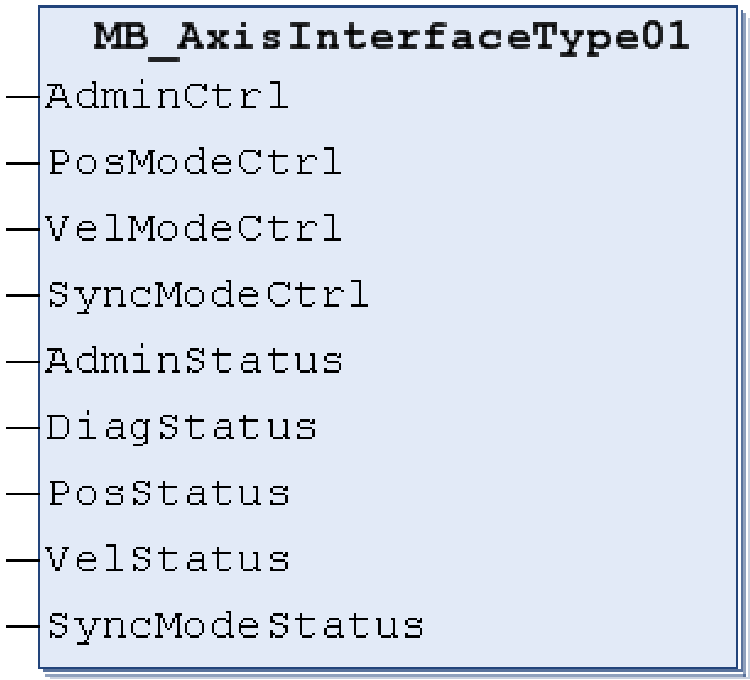
Function blocks for the AxisInterface

- MB_AxisInitType01
  initializes the AxisInterface for one axis
- MB_AxisInterfaceType01
  has to be called cyclically for each axis





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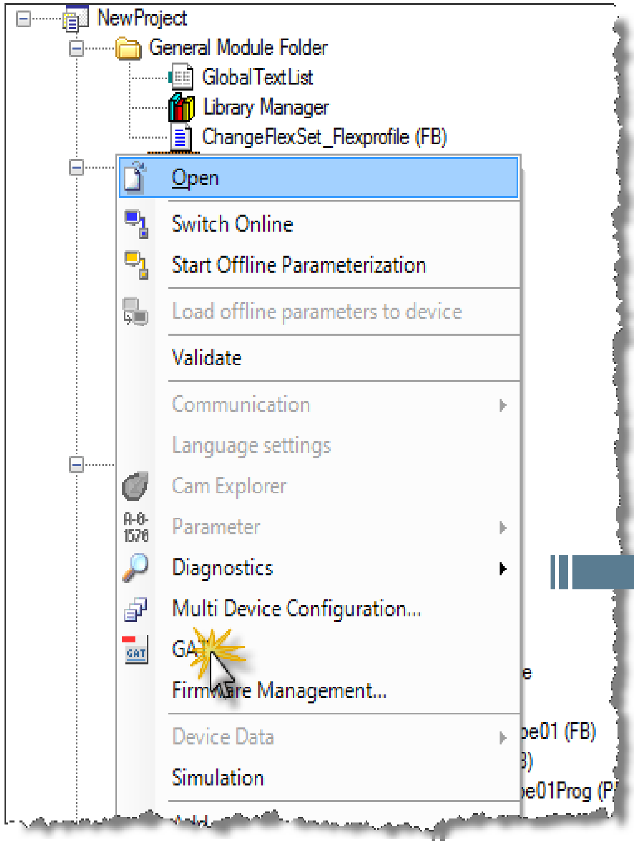
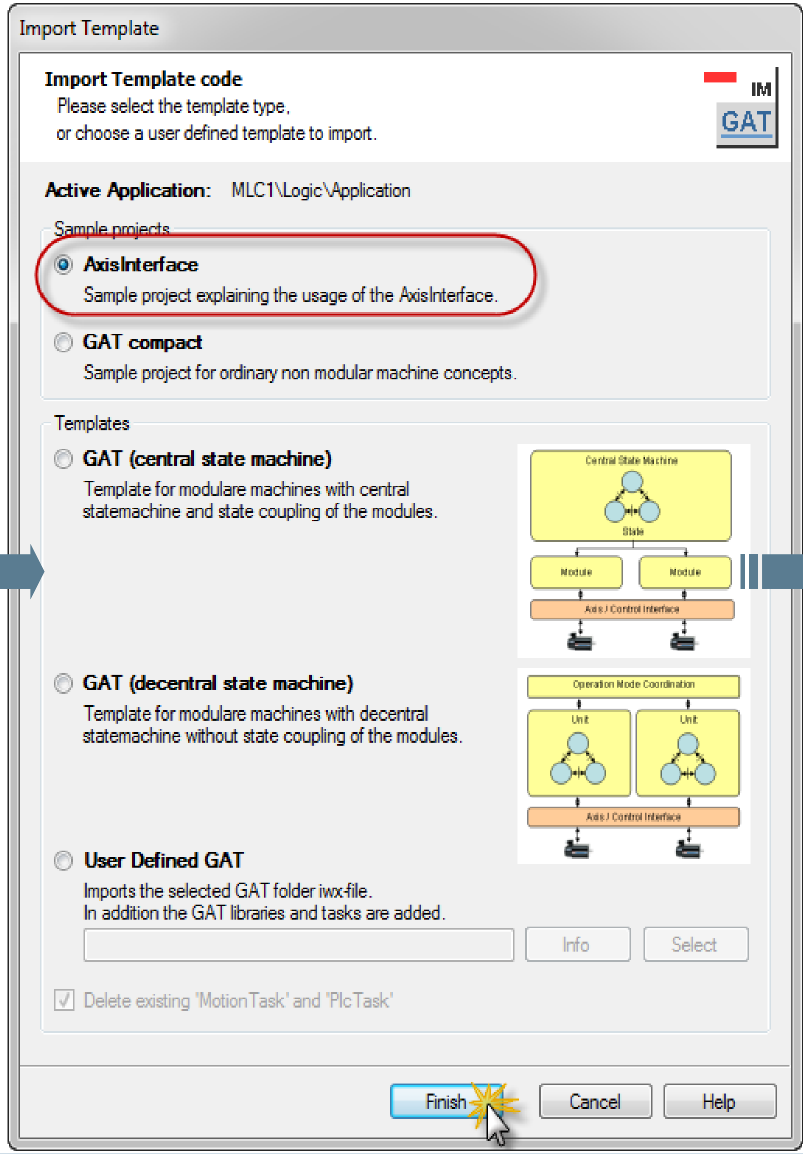
How to get the Axis Interface?





Axis Interface ready-to-use ✓

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Advanced Topics

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Why GAT?

Mark

- Program Frame
- Operation Modes
- Error Handling
- I/O Handling
- Data Organization

Young

- Program Frame
- Operation Modes
- Error Handling
- I/O Handling
- Data Organization

Jerry

- Program Frame
- Operation Modes
- Error Handling
- I/O Handling
- Data Organization

Bob

- Program Frame
- Operation Modes
- Error Handling
- I/O Handling
- DataOrganization

Programmers are like artists, every program is unique!

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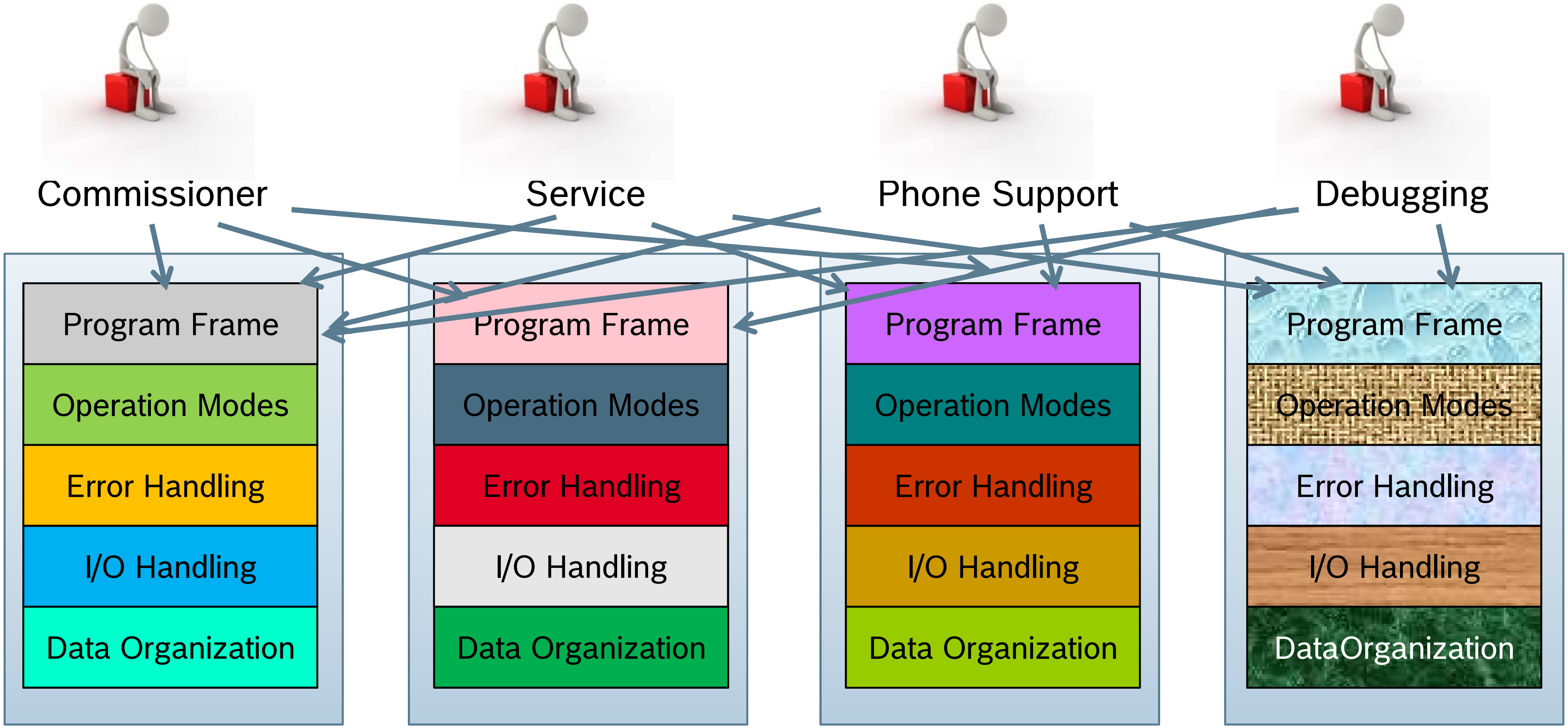
Why GAT?



Much time is lost by understanding the art , the error rate is high!
Artistry is quite expensive!

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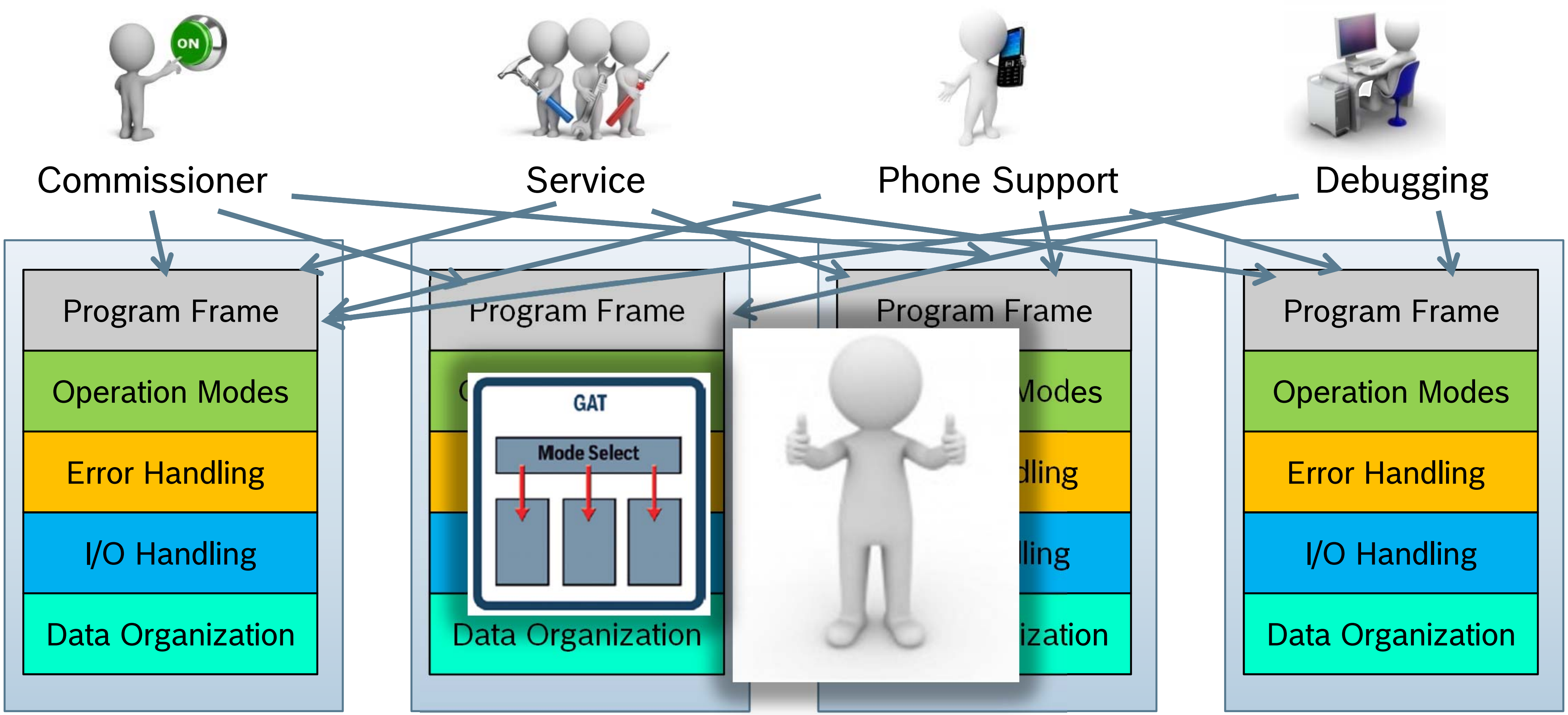
Why GAT?



With GAT all applications have an uniform structure and are easier to understand, to maintain, to extend etc.!

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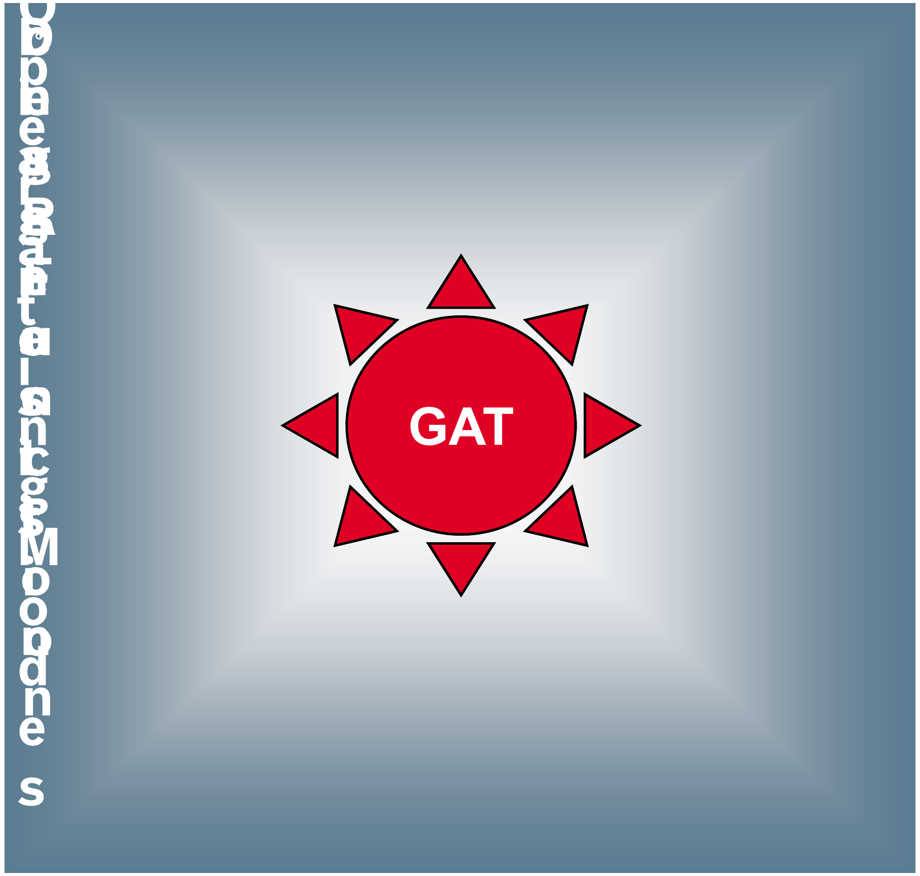
GAT – Generic Application Template



- **Predefined program frame with machine control**
  - Selection of predefined project templates
  - Ready for use operation mode administration
  - Predefined data structure
  - Jog mode
  - Prepared automatic operation
  - Predefined error handling for each machine part
  - Template for WinStudio application (HMI)
- **Wizard supported programming**
  - Adding/removing of machine parts/modules
  - Adding/removing of axis (real, virtual, …)
  - Adding/removing of operation modes
  - Top-Down-programming of operation modes

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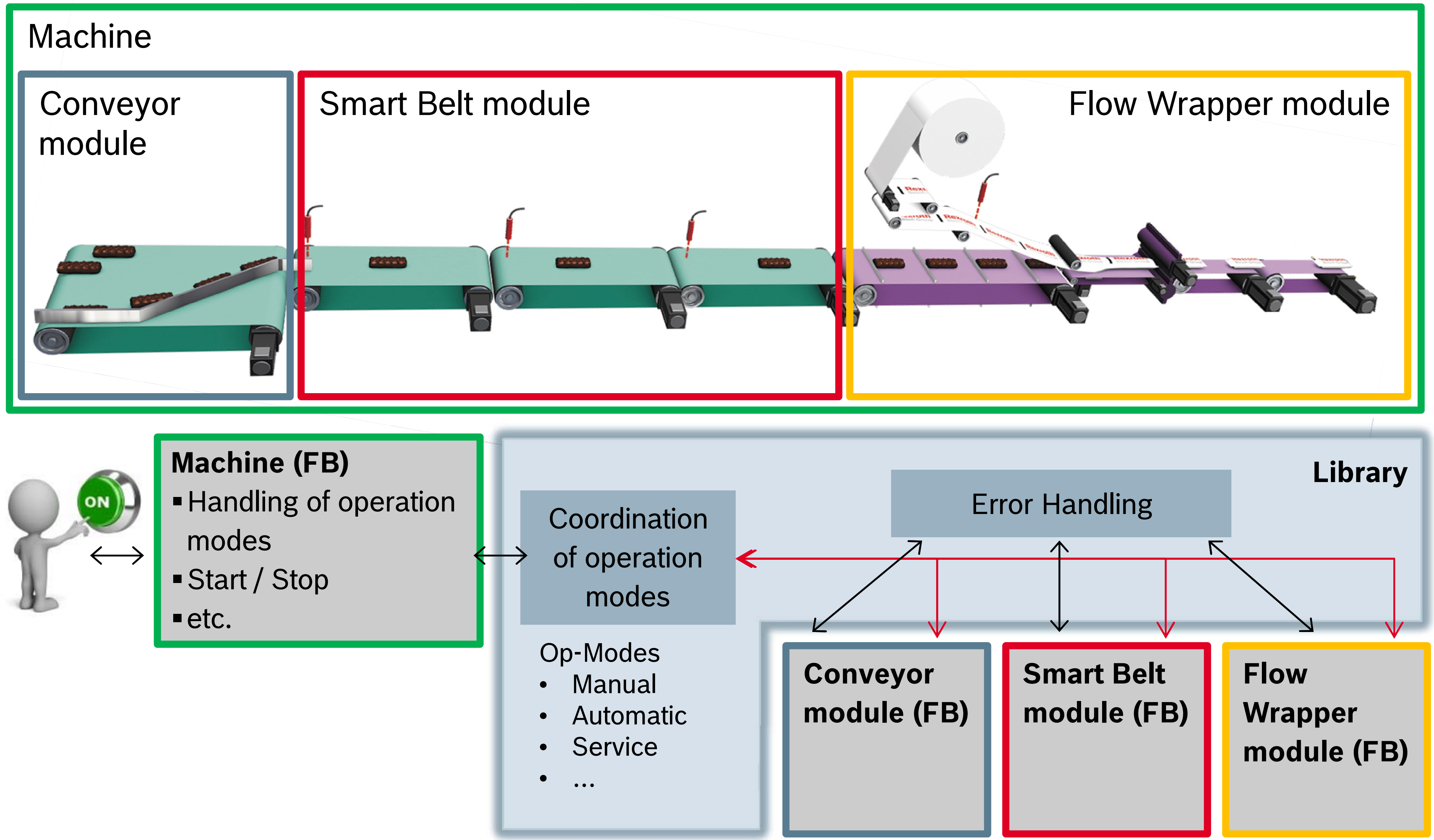
GAT – Generic Application Template



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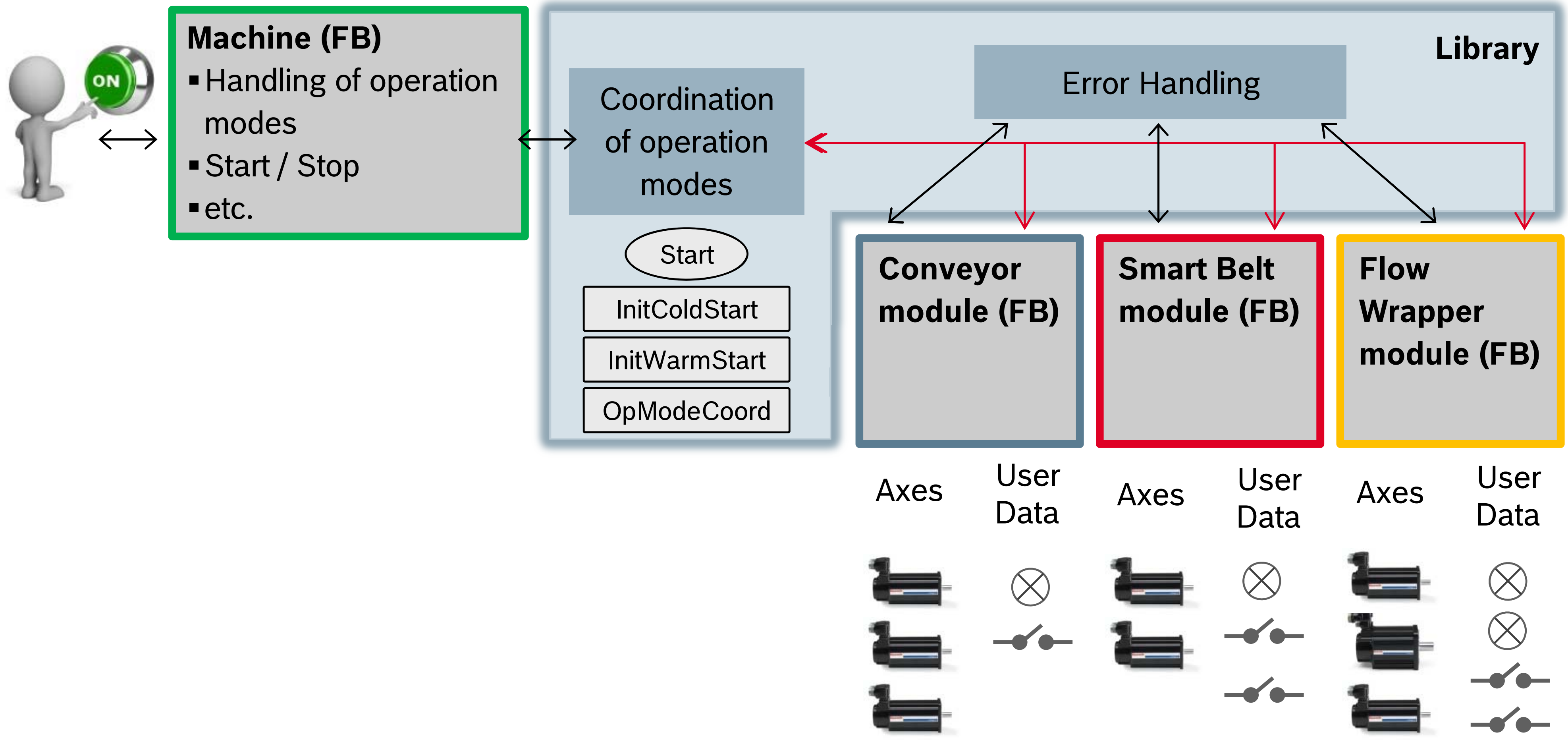
GAT – Generic Application Template



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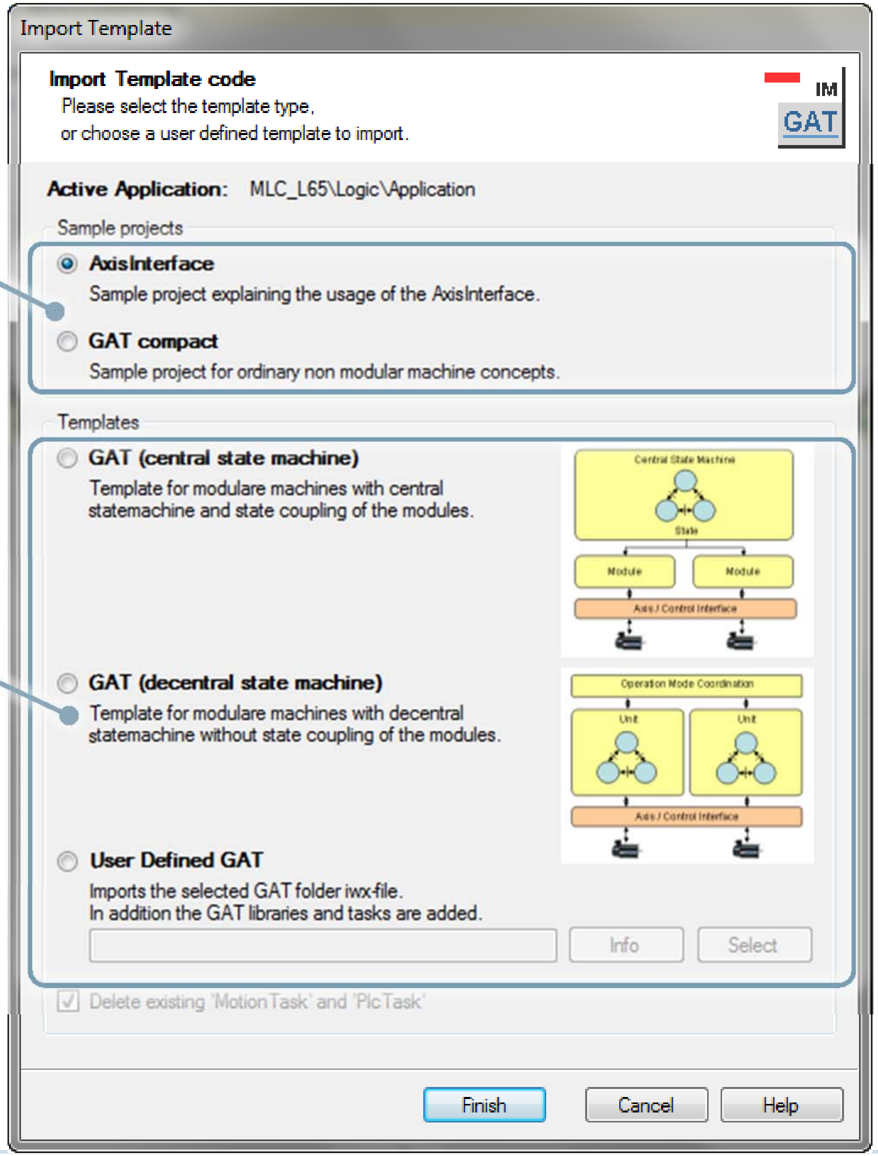
How to get the GAT?

Example projects for XLC and MLC

- AxisInterface
- GAT compact

Templates for MLC

- GAT central
- GAT decentral
- user definded GAT



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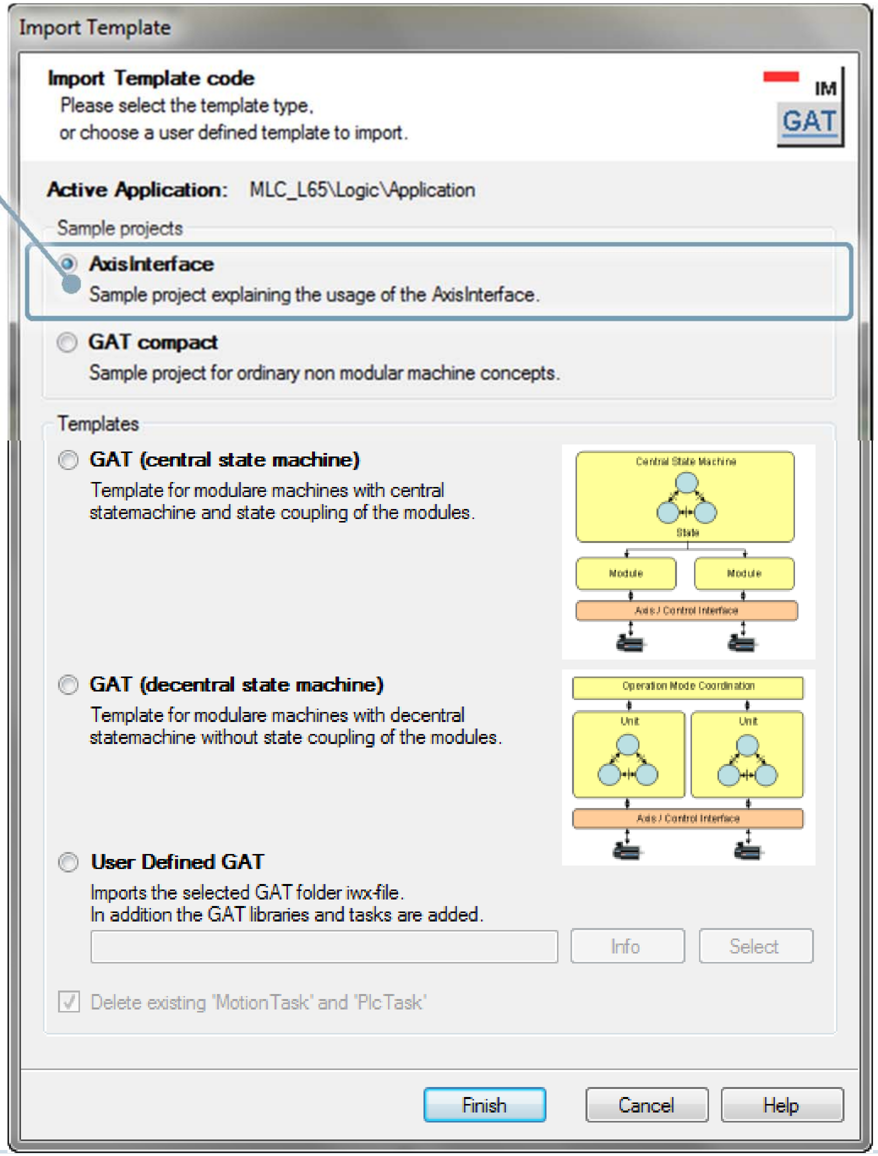
How to get the GAT?

Example project **AxisInterface**

- Example project for the use of AxisInterface and IMC interface
  Open source concept
- Subset of GAT^{compact}
  (without SFC for operation modes)



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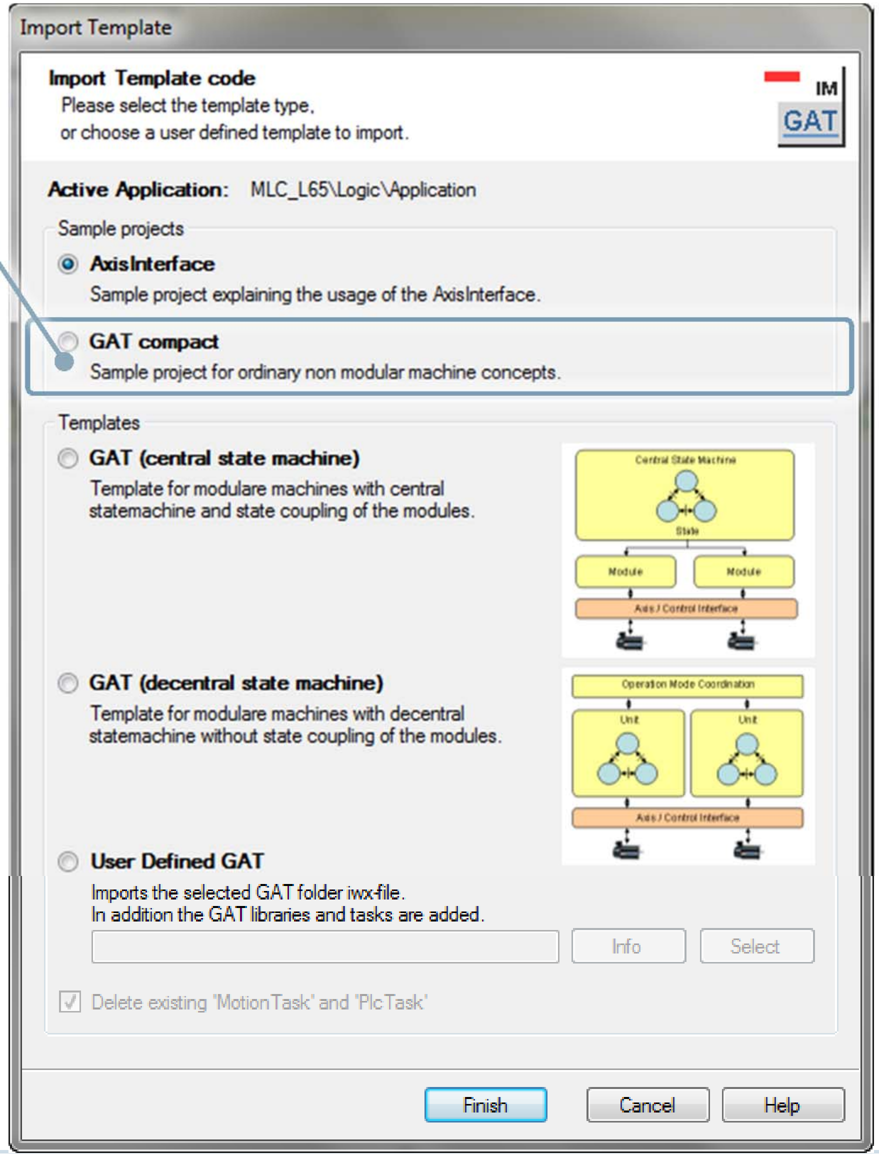
How to get the GAT?

Example project **GAT**[compact]

- Example project for simple and non-modular machine concepts
- Completely open structure based on AxisInterface and IMC interface
- Basis for a fast implementation
- Project includes
  - Complete axes management
  - Ready-for-use operation mode management
  - Jogging of axes
  - Prepared automatic operation
  - Template for WinStudio application



Basic idea:

- Simple machines need operation modes “Automatic” and “Manual”
- Exemplary axes programming

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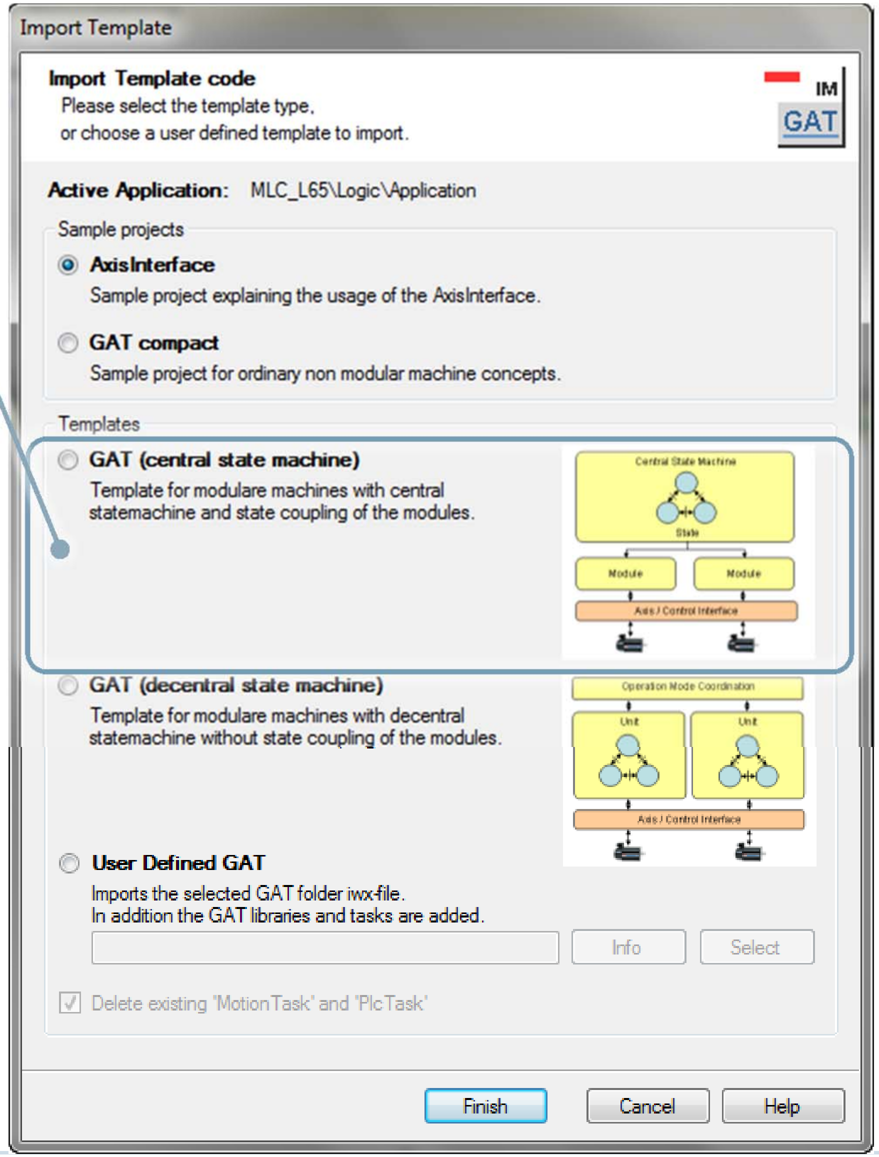
How to get the GAT?

Example project **GAT**[central]

- Template for fast implementation of machines with modular concept
- Wizard based code generation instead of error-prone copy & paste
- Generic code is encapsulated in library
  - Easier version upgrade
  - Easier bug fixing
  - Less code in application
- Template for WinStudio application as basis for own HMI applications



Basic idea:

- Modular machine design
- Common operation modes for all machine modules

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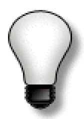
How to get the GAT?

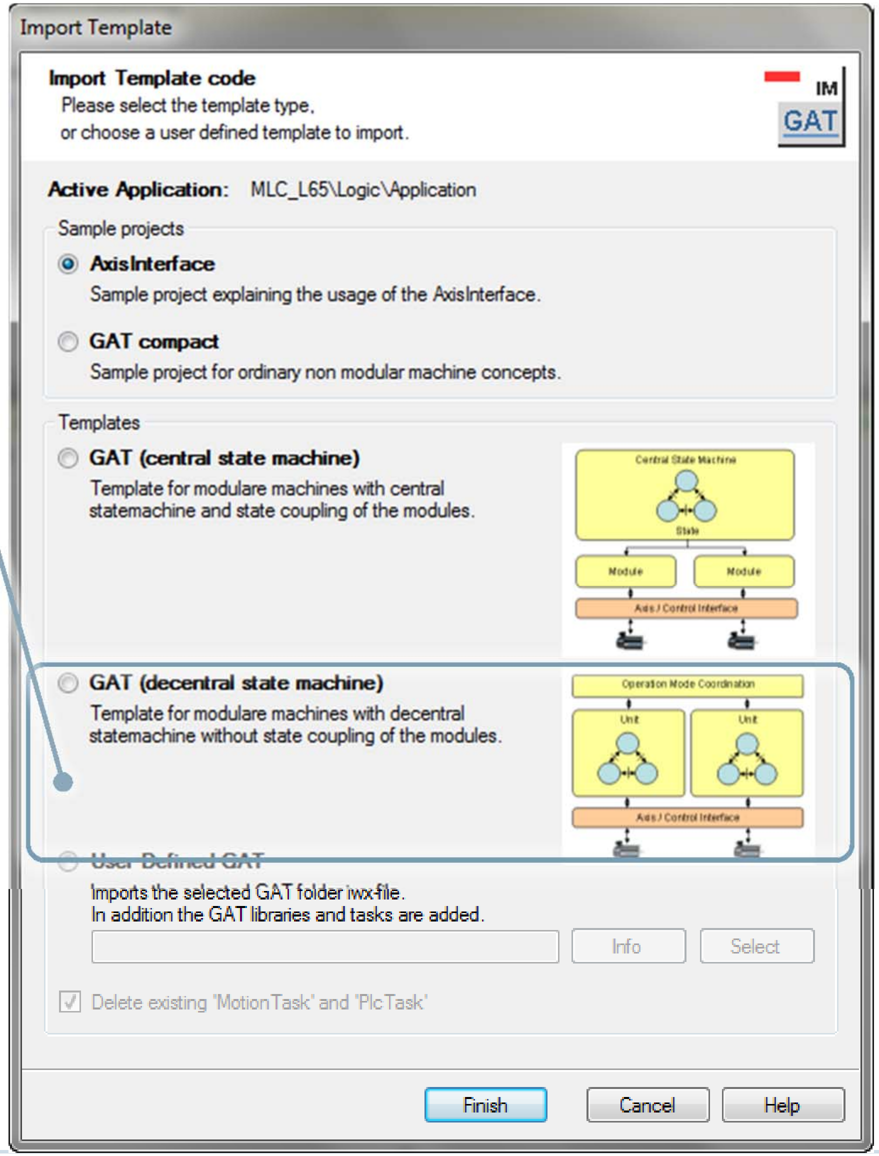
Example project **GAT**decentral

- Template for fast implementation of machines with modular concept
- Wizard based code generation instead of error-prone copy & paste
- Generic code is encapsulated in library
  - Easier version upgrade
  - Easier bug fixing
  - Less code in application
- Template for WinStudio application as basis for own HMI applications



Basic idea:

- Modular machine design
- Part of machine modules are coupled and have a common operation mode
- Others are uncoupled and have their own operation modes



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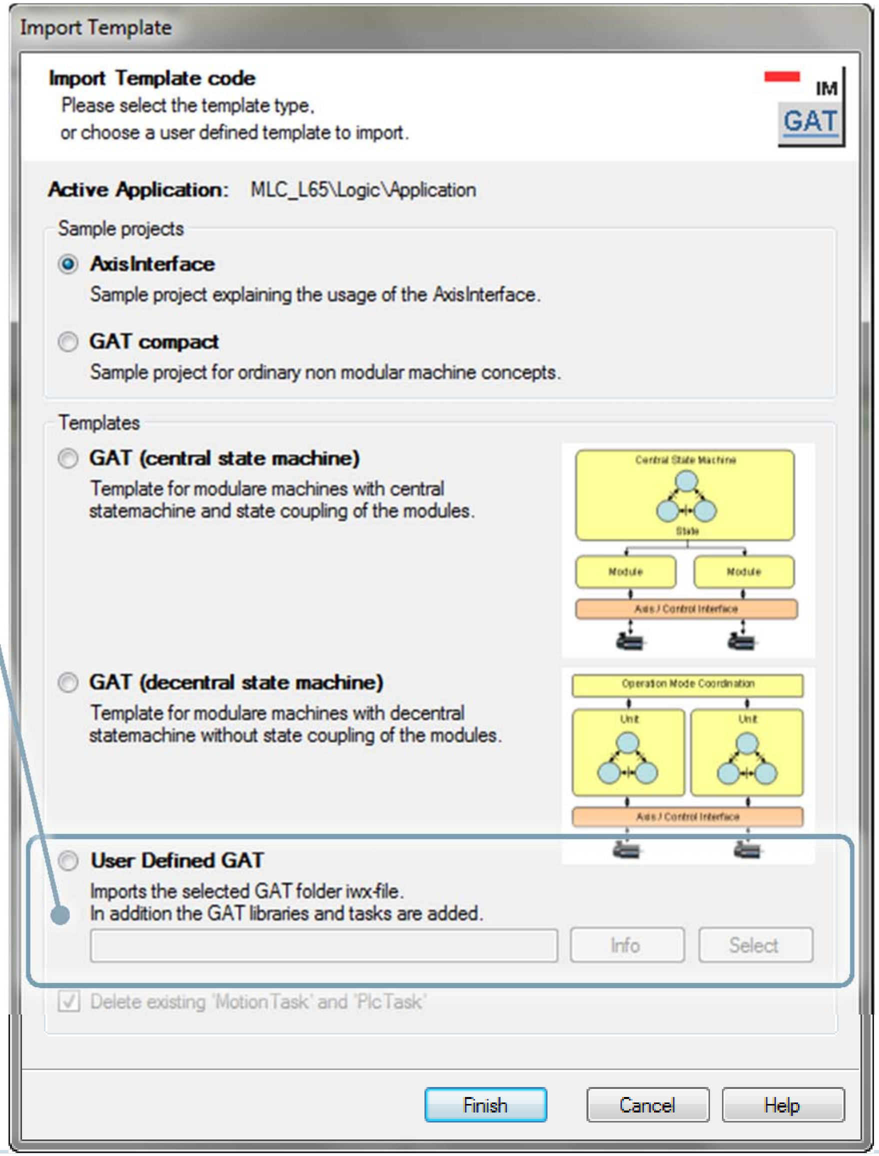
How to get the GAT?

User defined GAT

- For customers using their own template
- OMAC-PackML 3.0 template is installed with IndraWorks
- ... and can be imported also in this way

OMAC PackML



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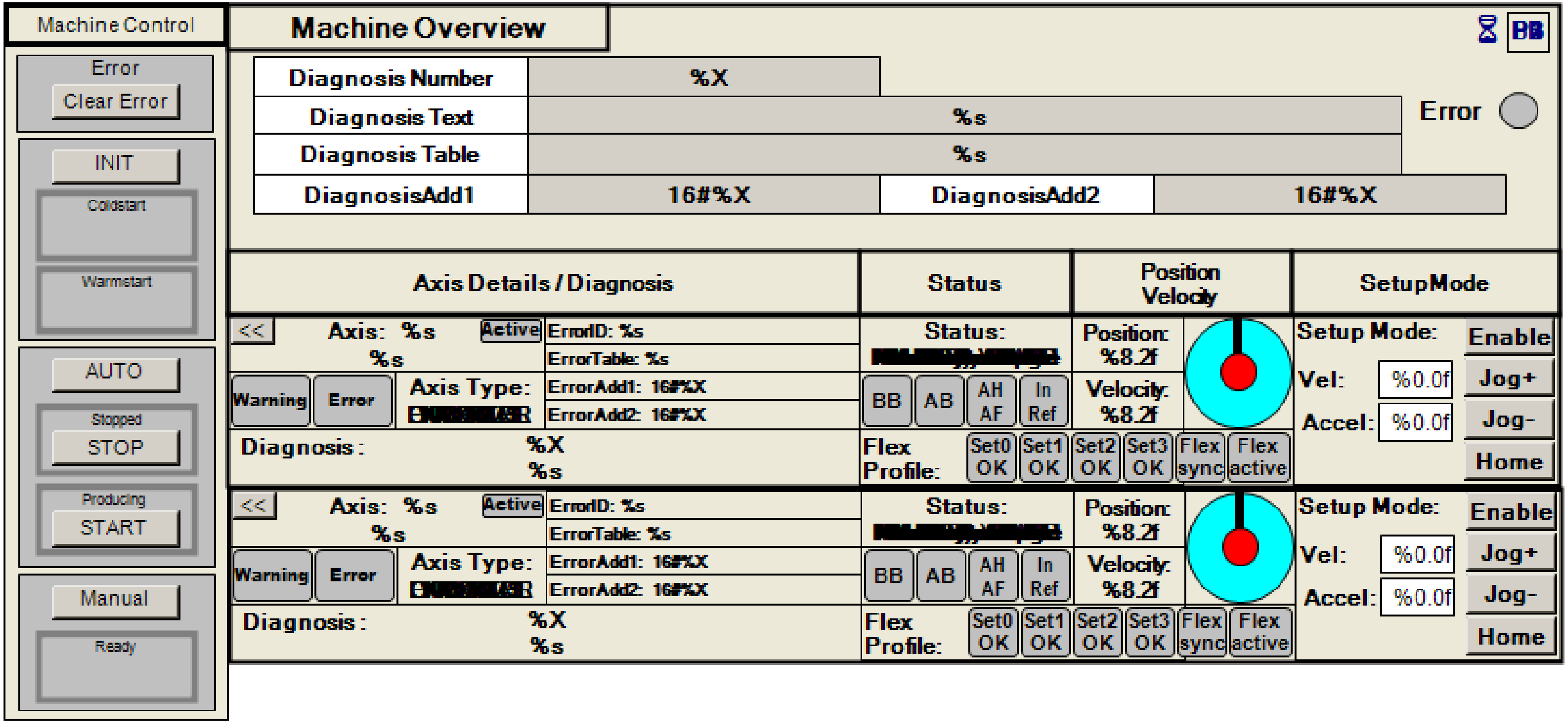
GAT and GAT compact – IndraLogic visualization



Ready-made IndraLogic visualization for easy commissioning

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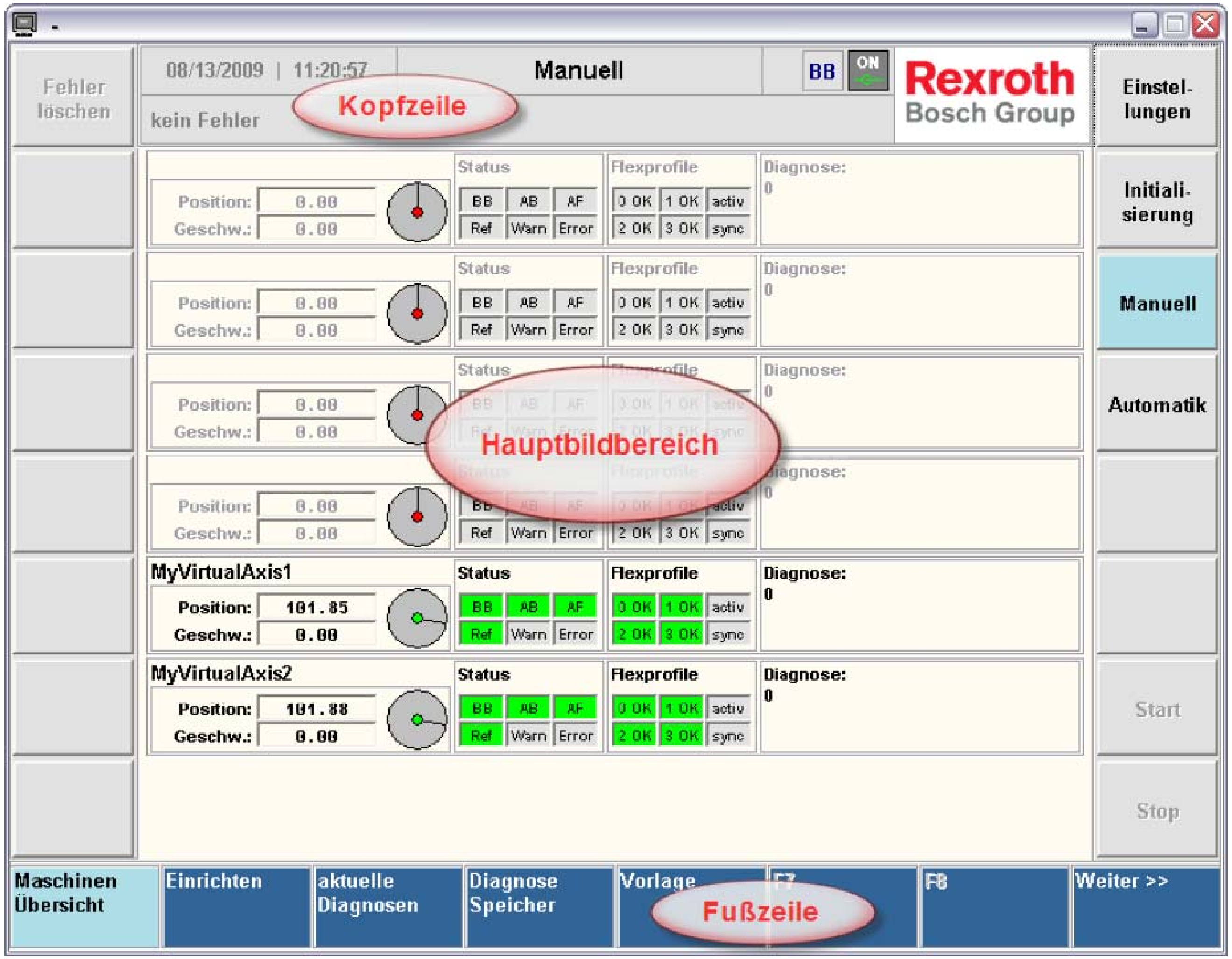
GAT and GAT^{compact} – WinStudio visualization



WinStudio sample project as basis for external visualization

GAT varieties

	AxisInterface & GAT compact	GAT central & GAT decentral
Characteristic	**Ready sample program**	**Program template**
Focus	**Easy** (quick start in case of manageable applications)	**Flexible** (structured implementation of complex applications)
Entry level	higher	
Accessibility of code	**Completely open**	**open** (user code) **encapsulated** (system code) (ease of later update)
Extensibility	**Wizard (axes) / Copy & Paste**	**Wizard based**
Modular design	**no**	**yes**
Maintainability/ Reusability	better	
System support	**MLC, XLC**	**MLC**

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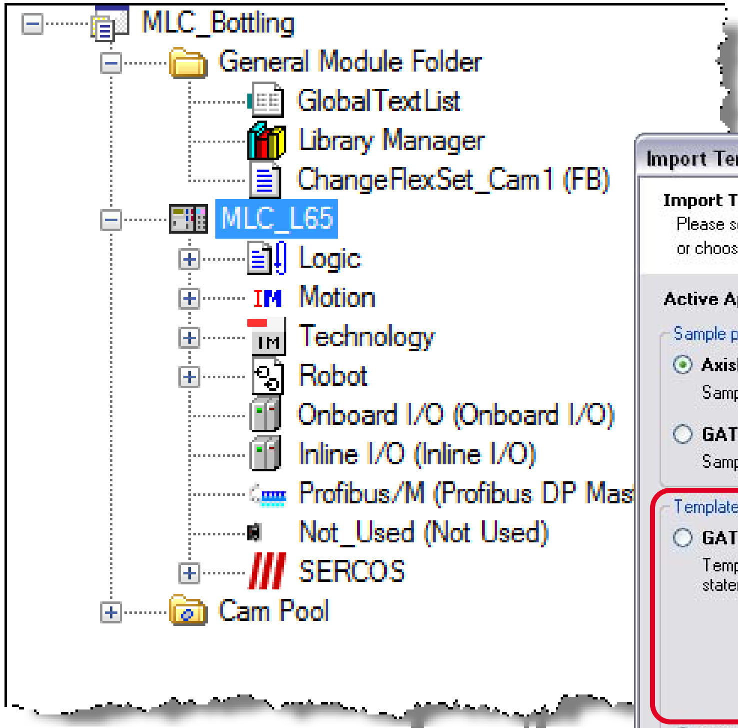
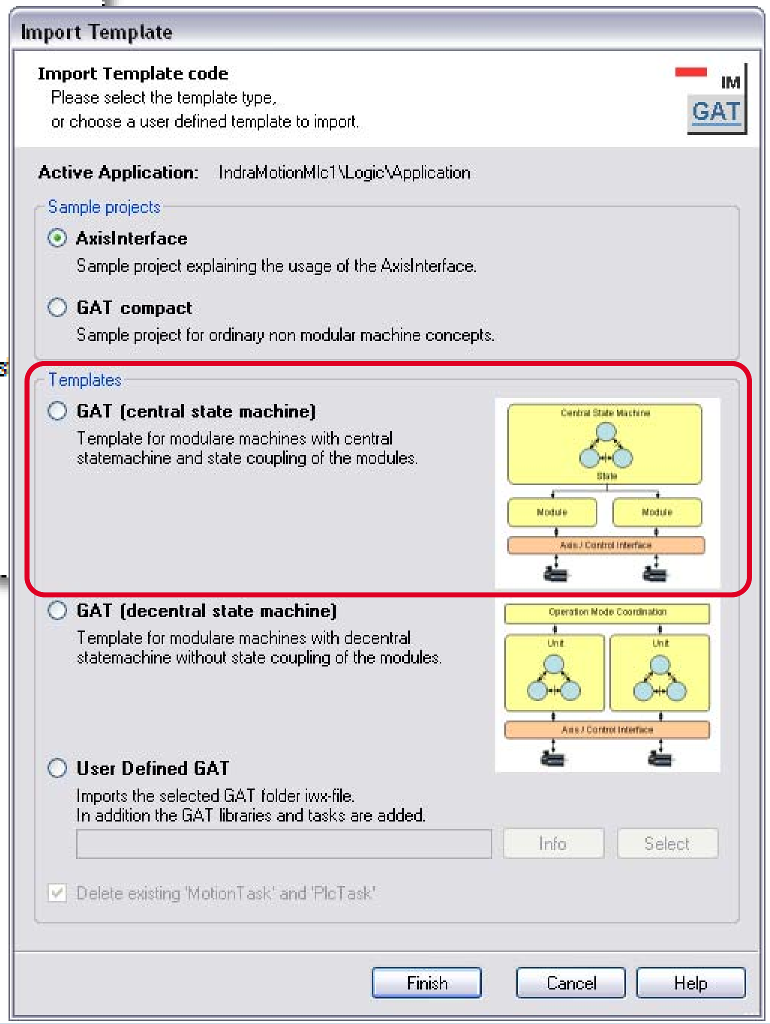
GAT – GAT Wizard

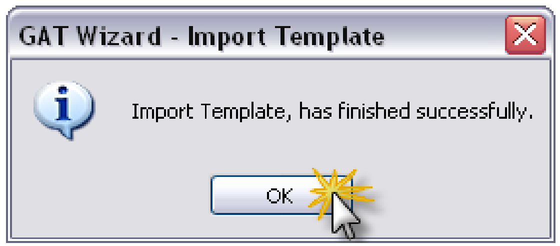


- Call GAT wizard

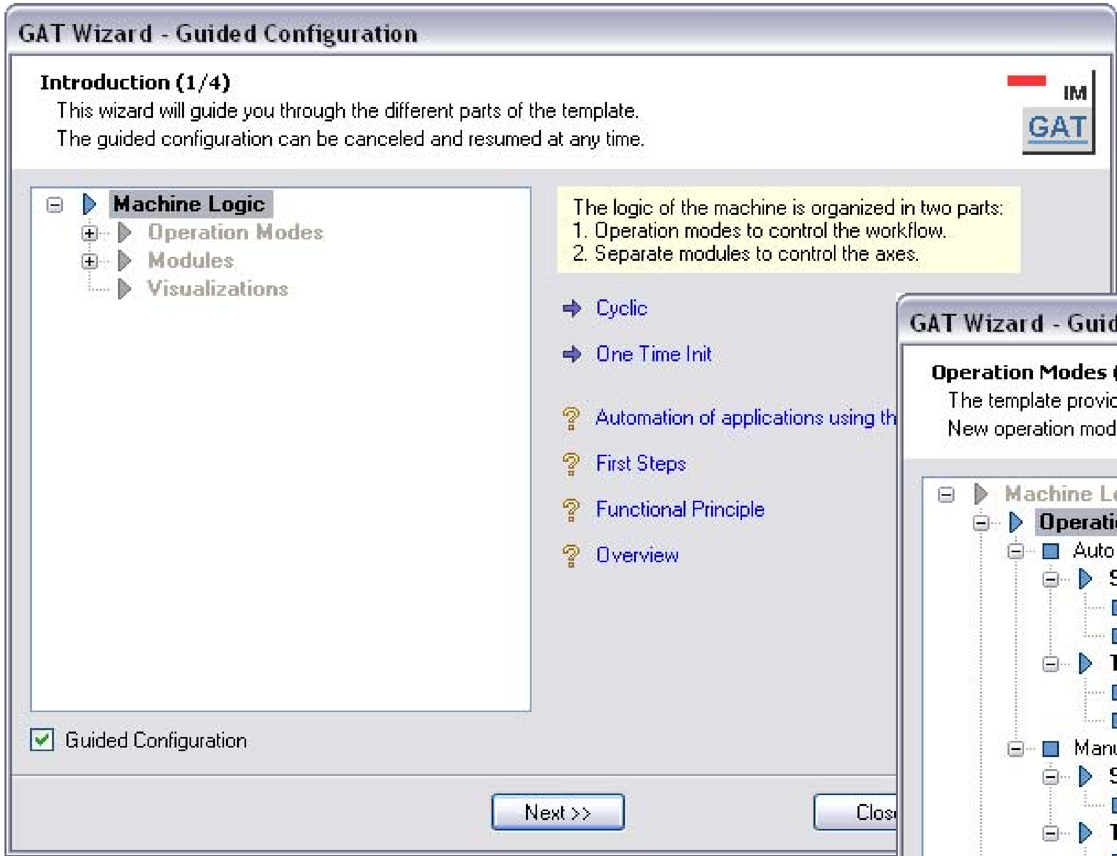


- Select GAT (central state machine)

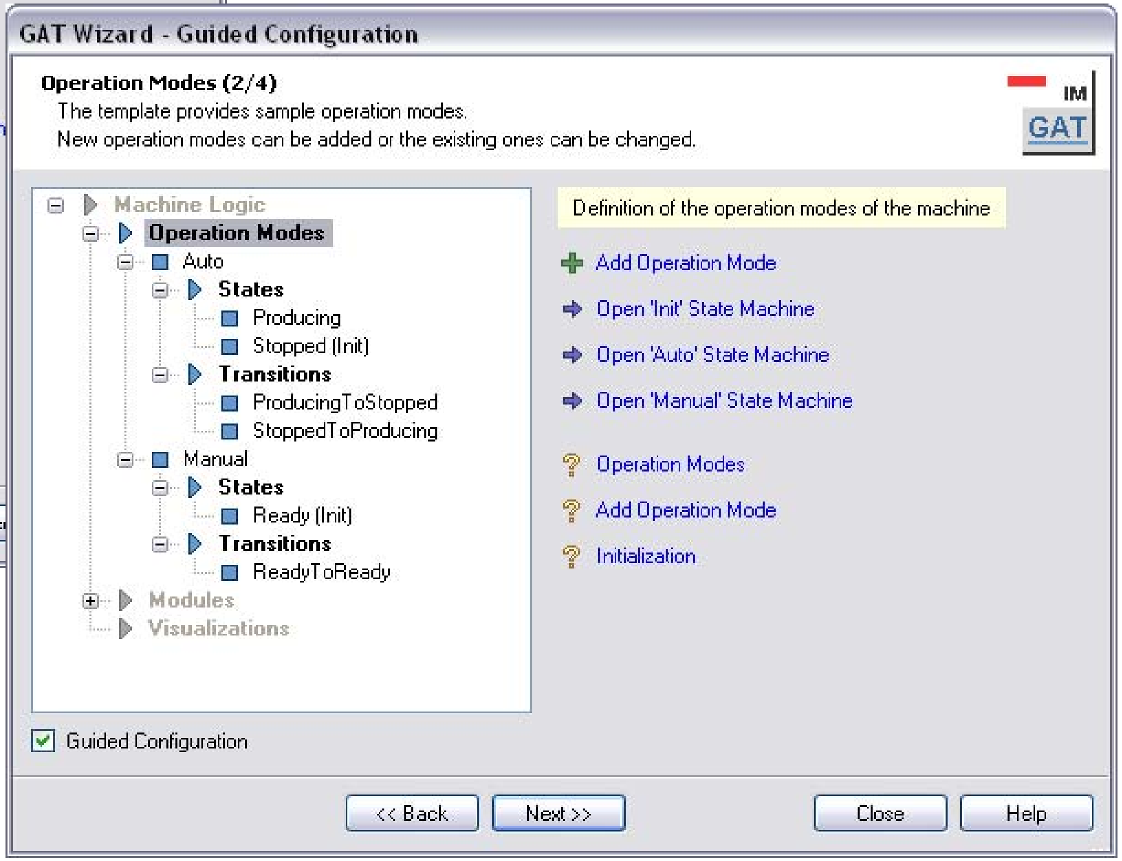
- Import OK!



GAT – GAT Wizard



- The Wizard guides you through the different parts of the template



- 2 operation modes are predefined
- You can add additional operation modes
- ...or modify state machine of predefined modes

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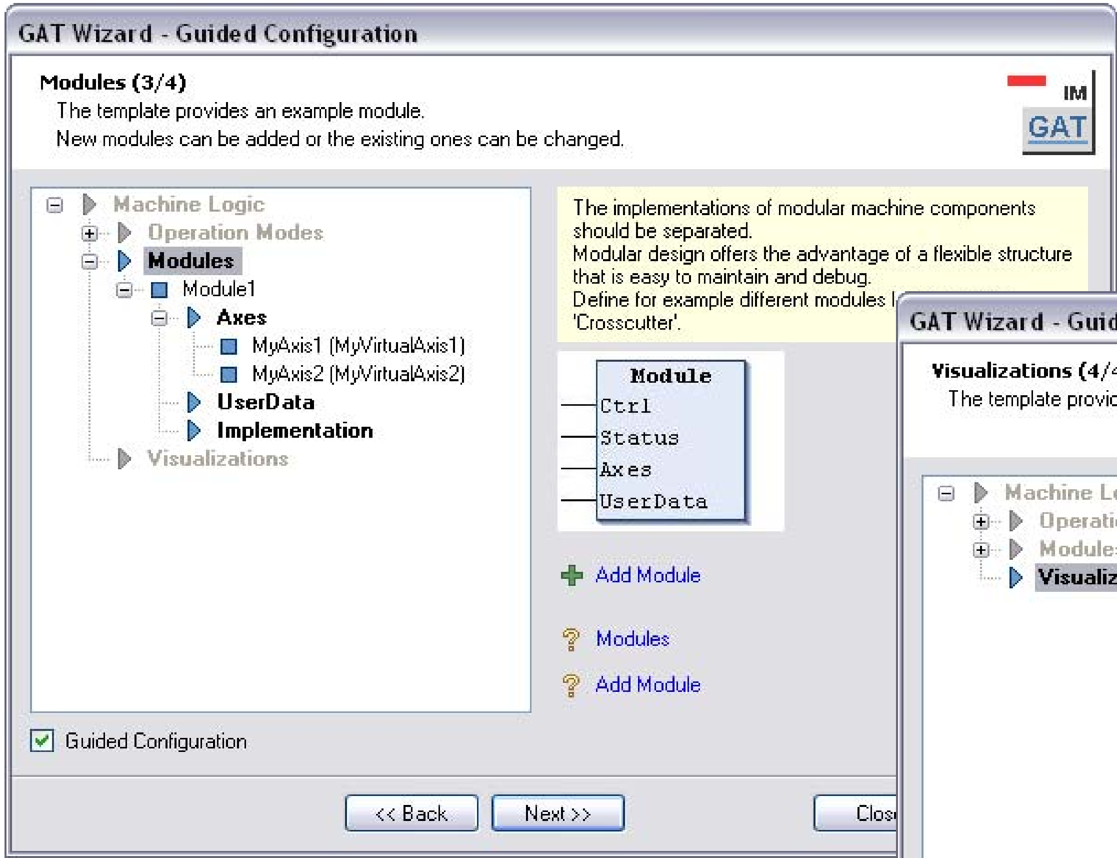
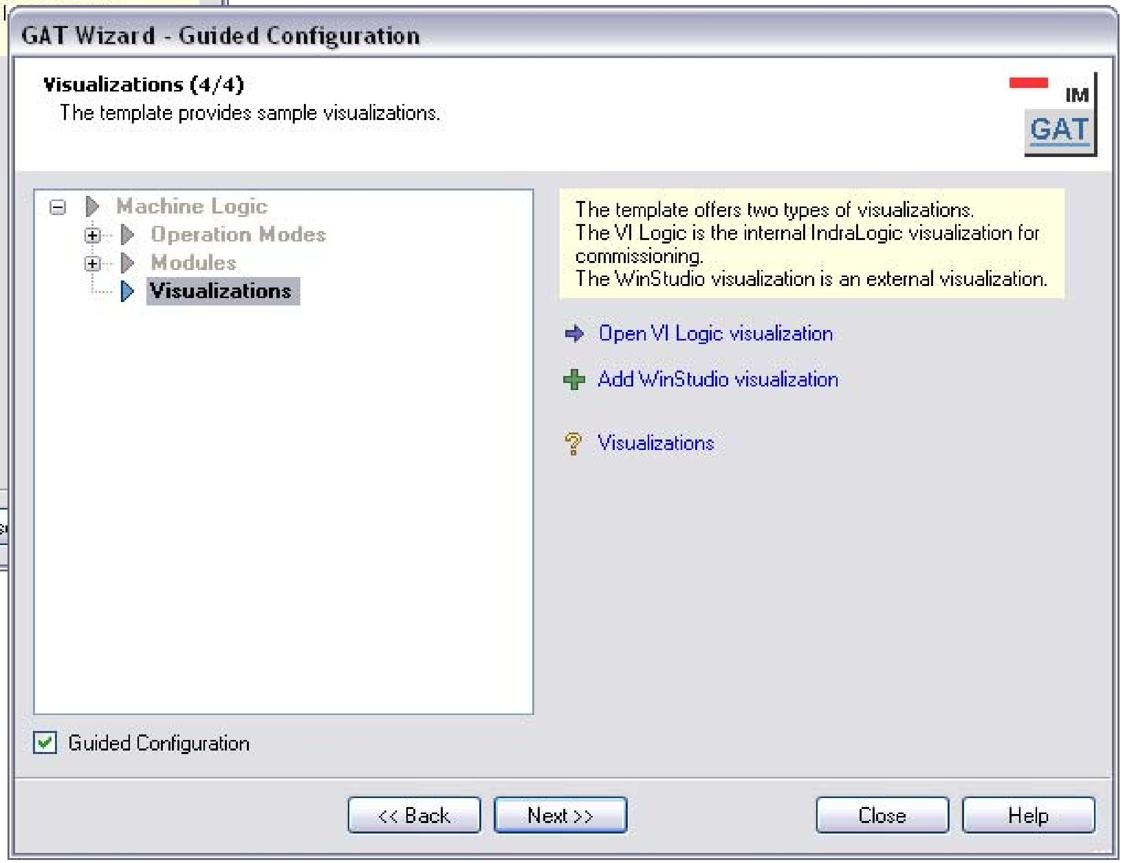
GAT – GAT Wizard





- A module “Module1” is configured by default
- 2 virtual axes are created and assigned to this module

- If a HMI is part of the IndraWorks project
- ... a VI Composer application or
- ... a WinStudio application can be added

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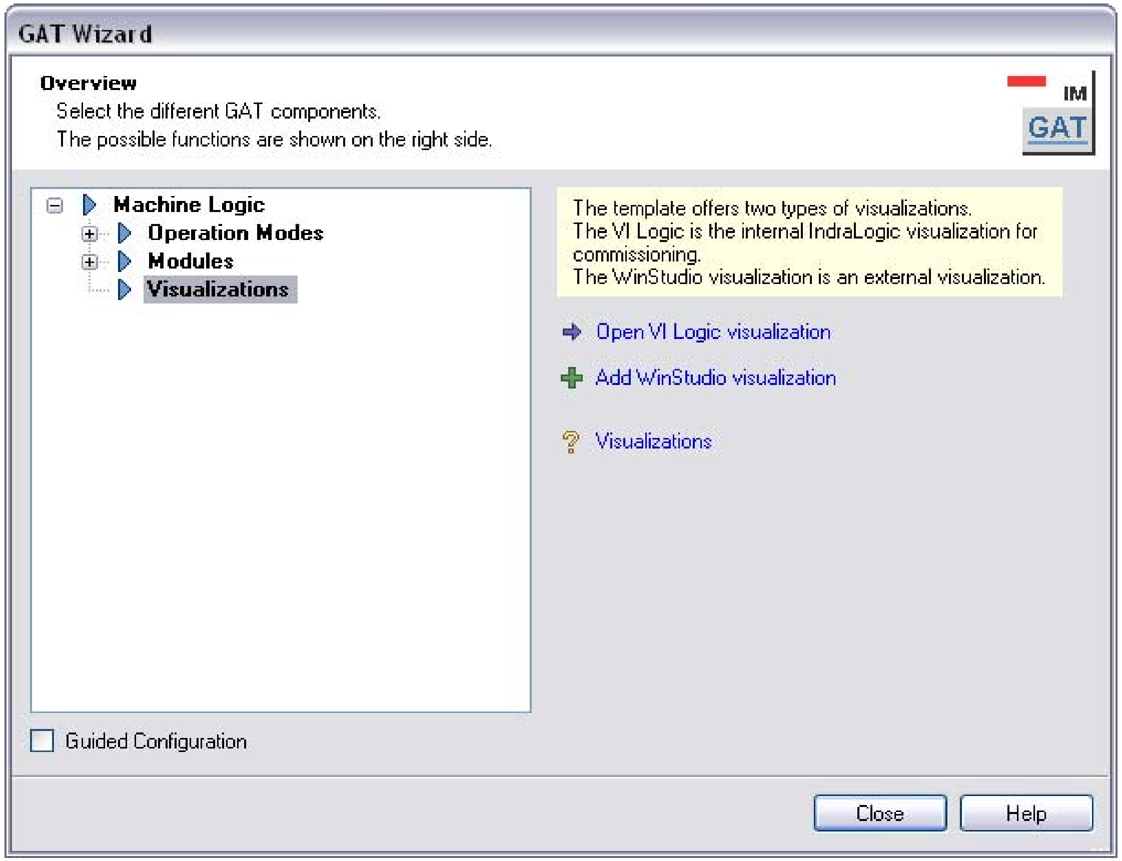
GAT – GAT Wizard



- The configuration is complete
- Navigate in the tree structure for any modifications in your previous settings
- After closing the wizard, the IndraLogic project corresponds to the data you entered in the wizard
- Return to the wizard in order to add or modify
  - operating modes
  - modules
  - or axes
  - etc.
- You can also return to the wizard for assisted coding

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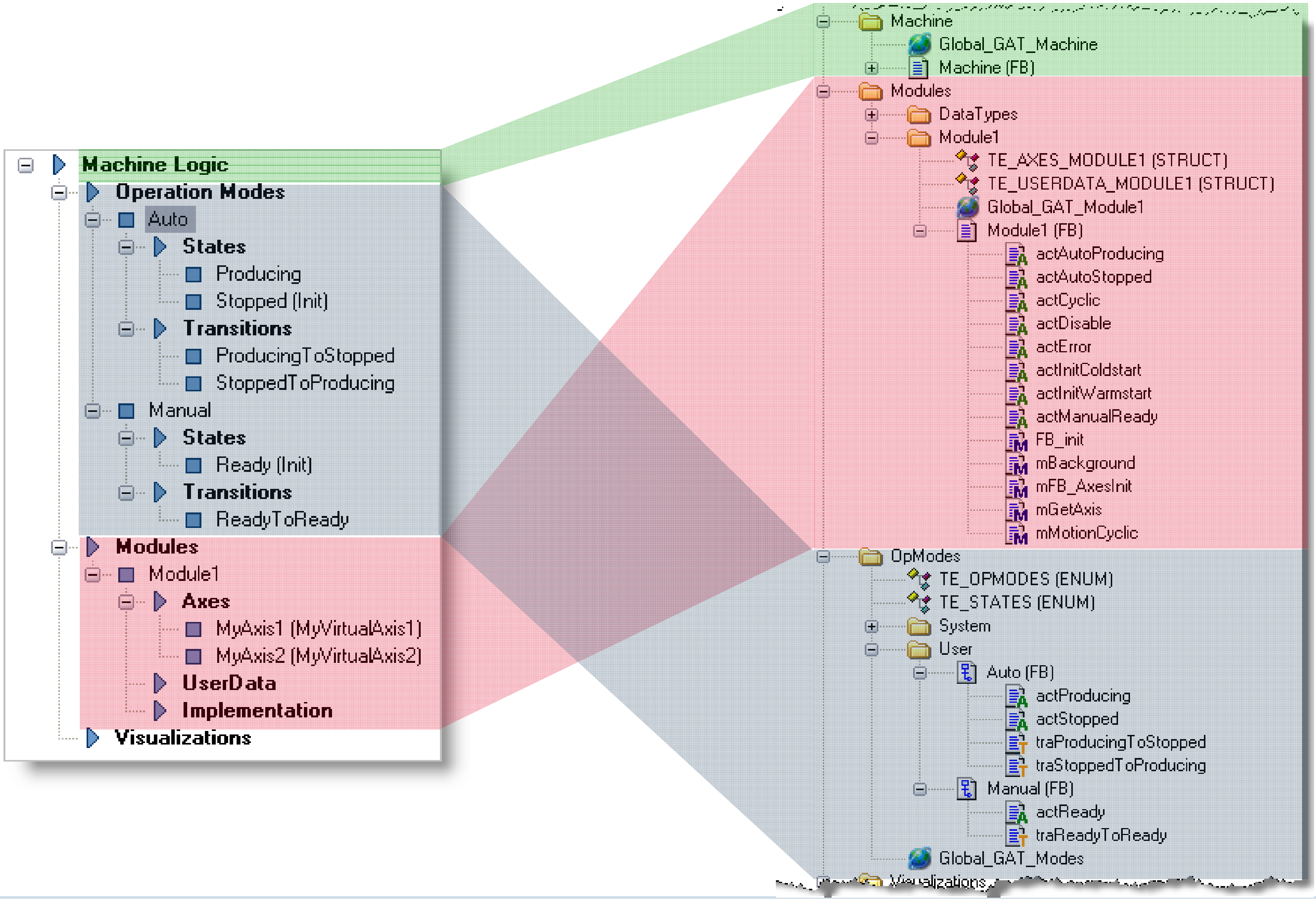
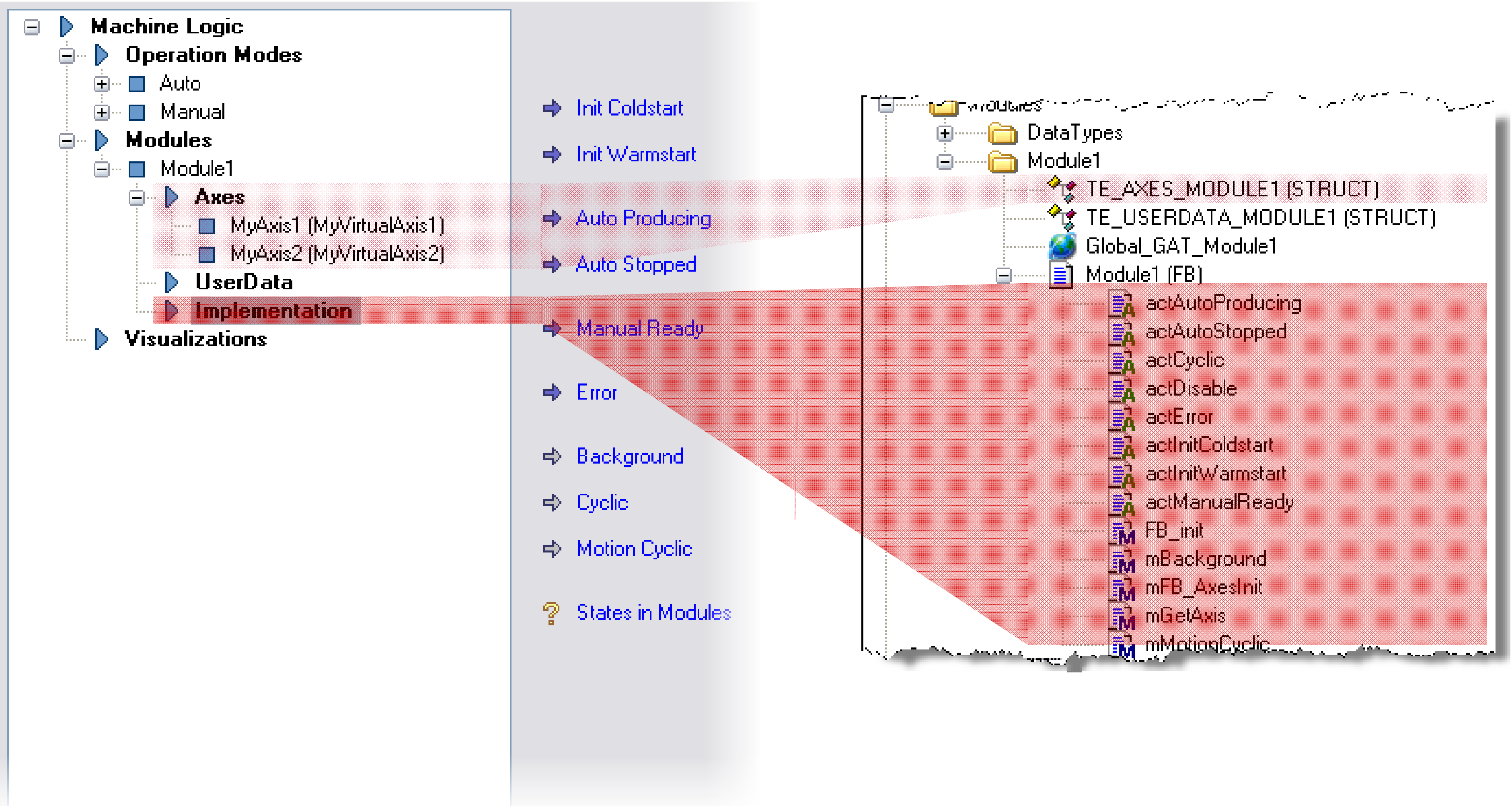
GAT – GAT Wizard & IndraLogic project



GAT – GAT Wizard & IndraLogic project



GAT

Exercise

- Implement Bottling Machine based on GAT

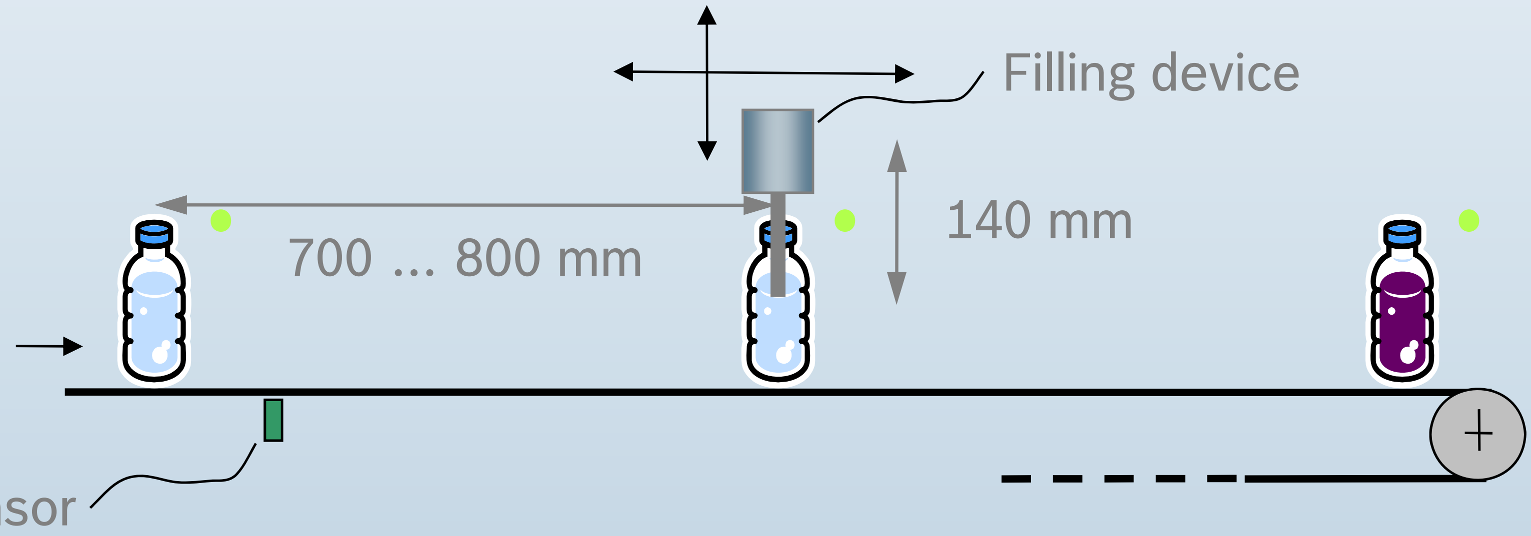
☞ 2 Modules:
  - Master → Controls a virtual master
  - Filling device → Controls the filling device

- Module for conveyor etc. not implemented

GAT

Exercise

ca. 4 bottles / minute → 15 seconds
v ≈ 3000 mm/min
ca. 3.5 sec. filling time
Stroke of vertical axis 140 mm



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GAT – Engineering steps

Exercise

Create new IndraWorks project

add control and axes to project

No.			Drive	Adr.	Name	Axis Type	Firmware
1			Axis1	1	Axis1	IndraDrive	FWA-INDRV*-MPB-05V14-D5-1-SNC
2			Axis2	2	Axis2	IndraDrive	FWA-INDRV*-MPB-05V14-D5-1-SNC
3							
4							
5							
6							

Change names to “Axis1” and “Axis2”!

Select Interpolation on MLC!

Add virtual axis “vAxis3”

configure axes and control

- Scaling of axes
- Maximum travel range
- Travel limits
- Velocity limits
- Acceleration limits

configure IOs

Configure inline IOs and SERCOS III IOs

Configure FlexProfiles

FlexProfile for vertical axis and horizontal axis

Implement motion program

Use GAT wizard to implement motion program

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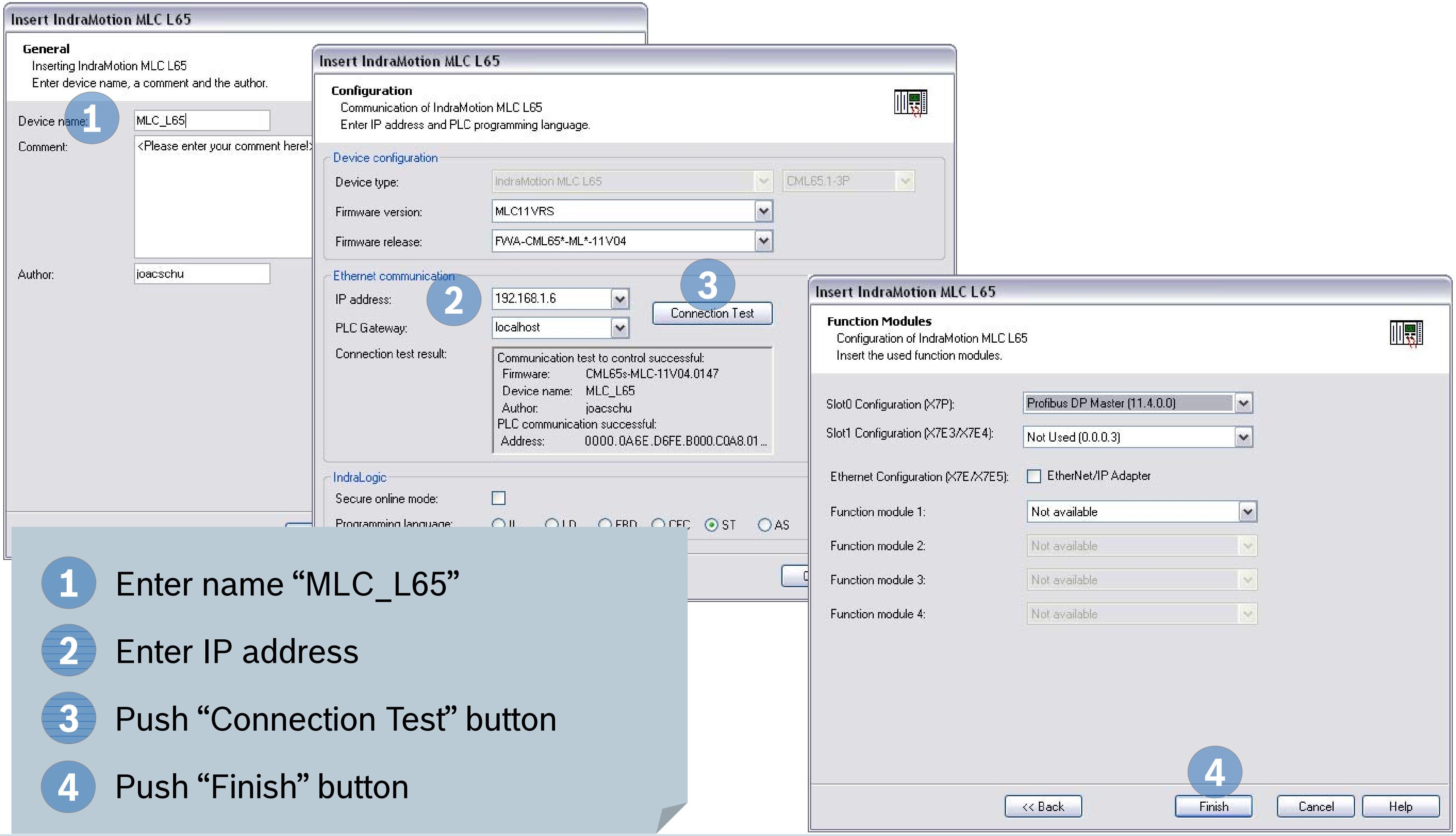
GAT – Create new MLC project

Exercise



1. Enter name “MLC_L65”
2. Enter IP address
3. Push “Connection Test” button
4. Push “Finish” button

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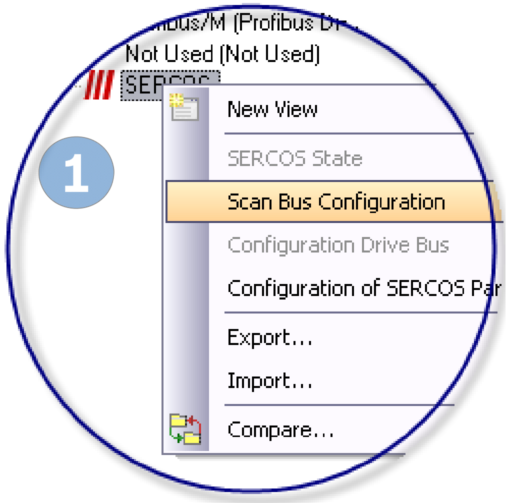
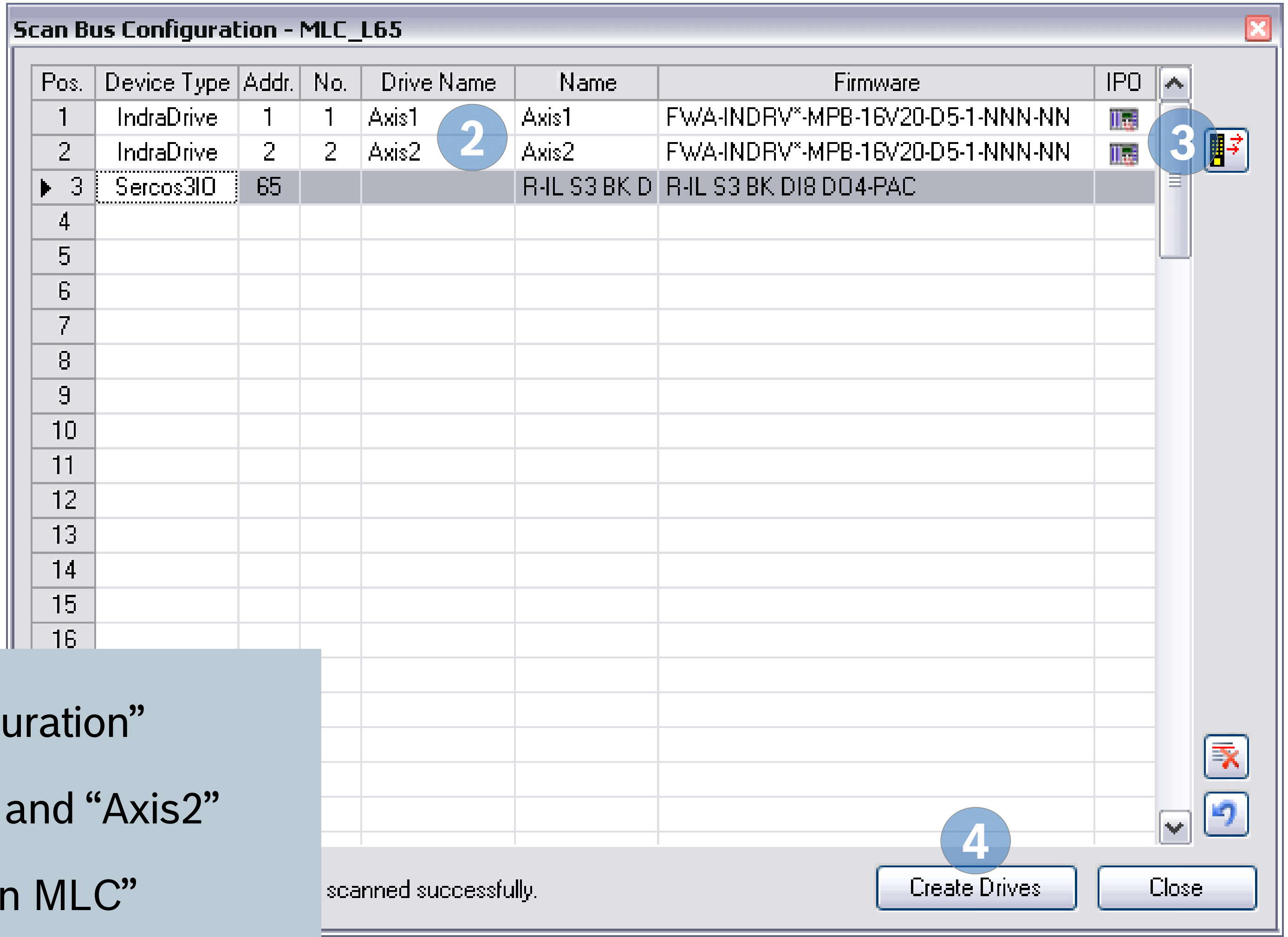
GAT – Scan SERCOS ring

Exercise



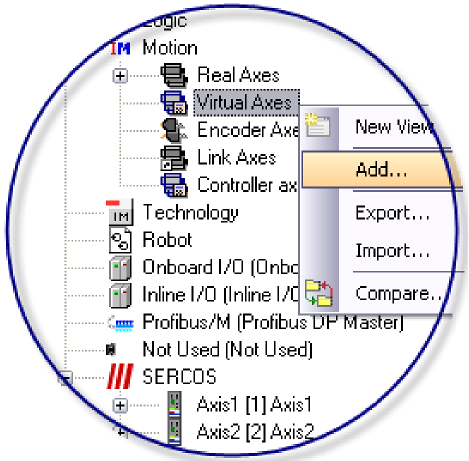


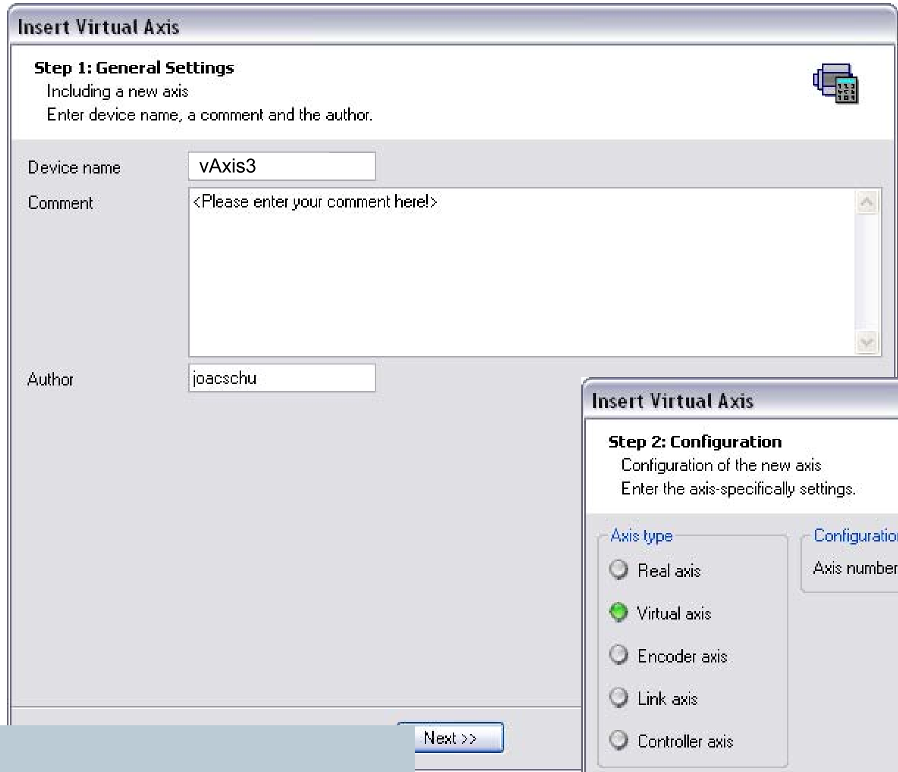
1. Call “Scan Bus Configuration”
2. Modify names “Axis1” and “Axis2”
3. Select “Interpolation on MLC”
4. Push “Create Drives” button

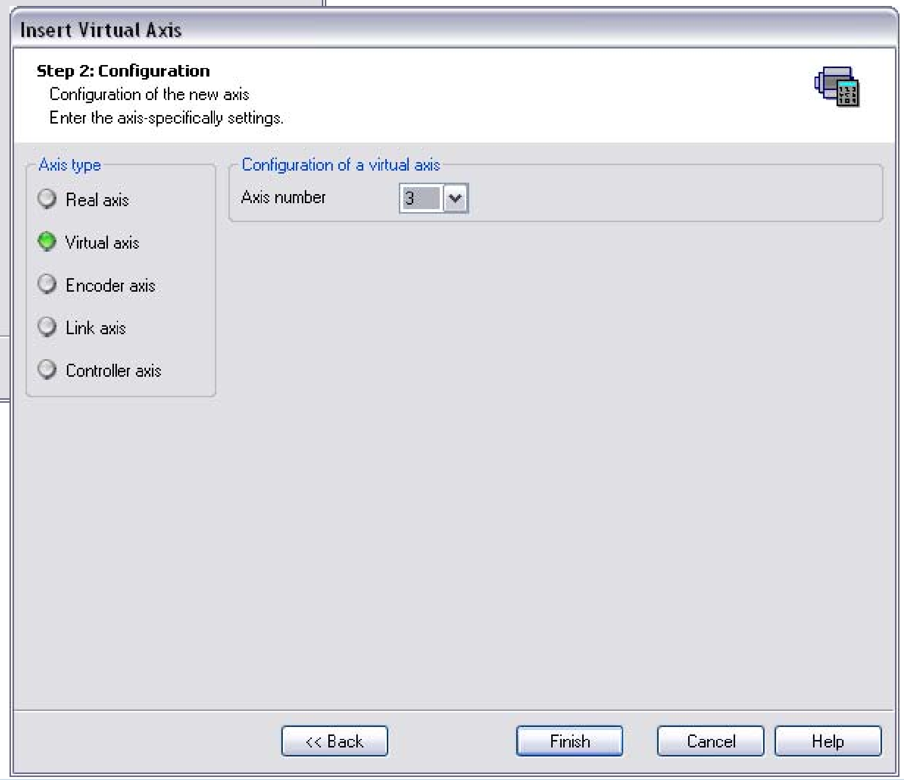
Rexroth Bosch Group

GAT – Add virtual master

Exercise





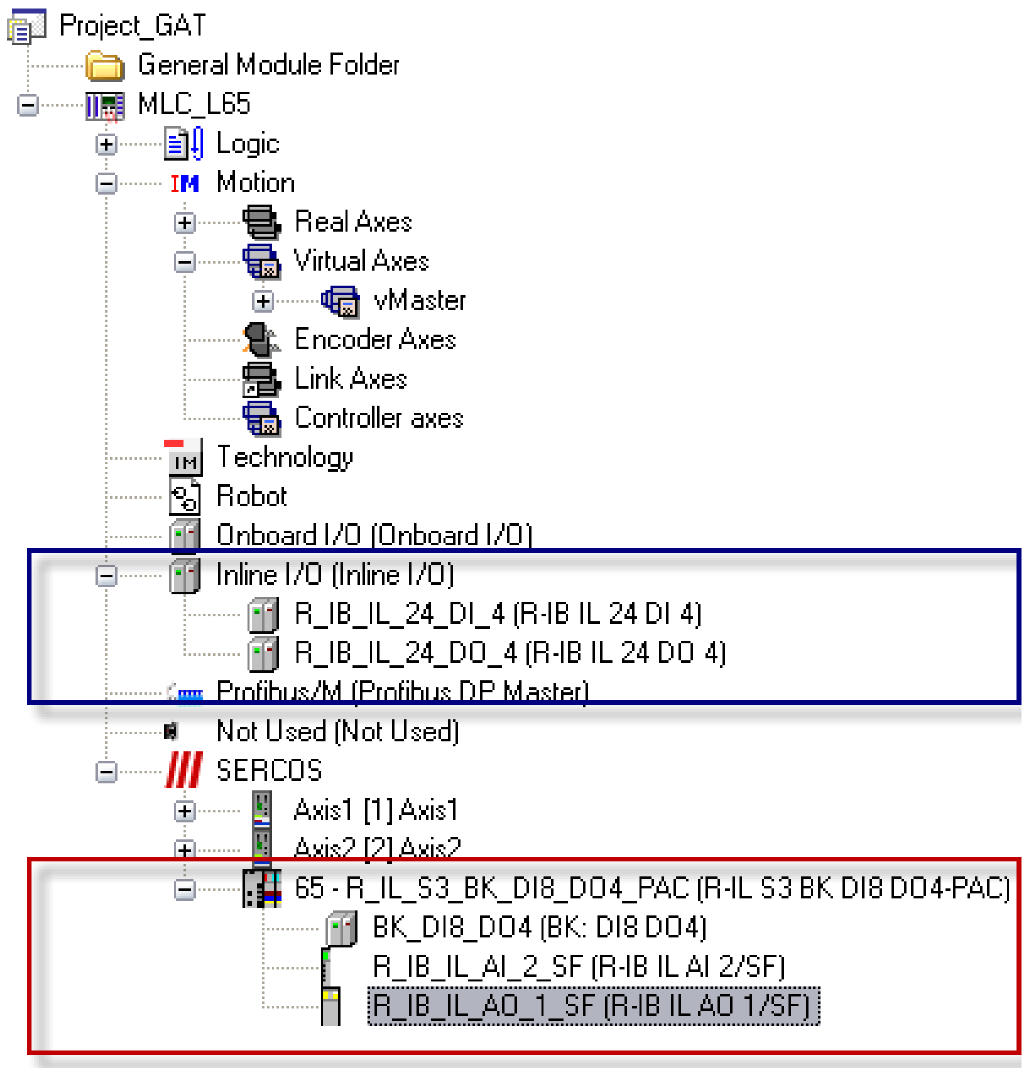


1. Select “Virtual Axes” and call “Add...”
2. Enter name “vAxis3”
3. Push “Next” button
4. Push “Finish” button

Rexroth Bosch Group

GAT – IO Configuration

Exercise



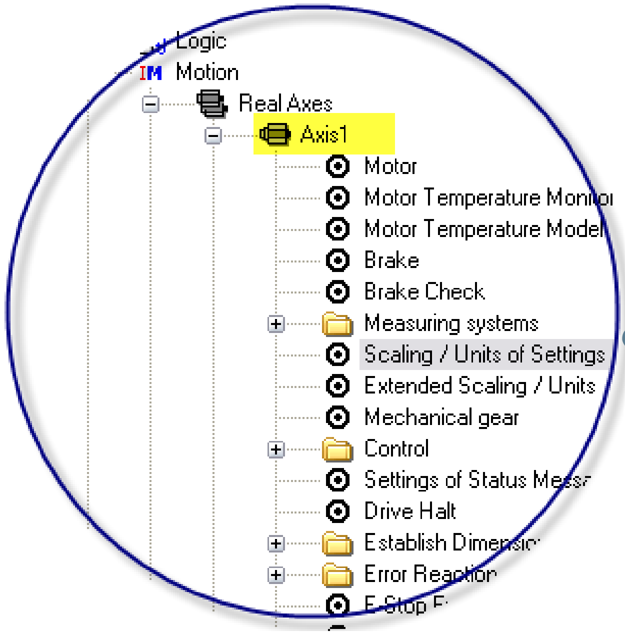
Create

- Inline I/O configuration (if Inline IO present)
- SERCOS III I/O configuration

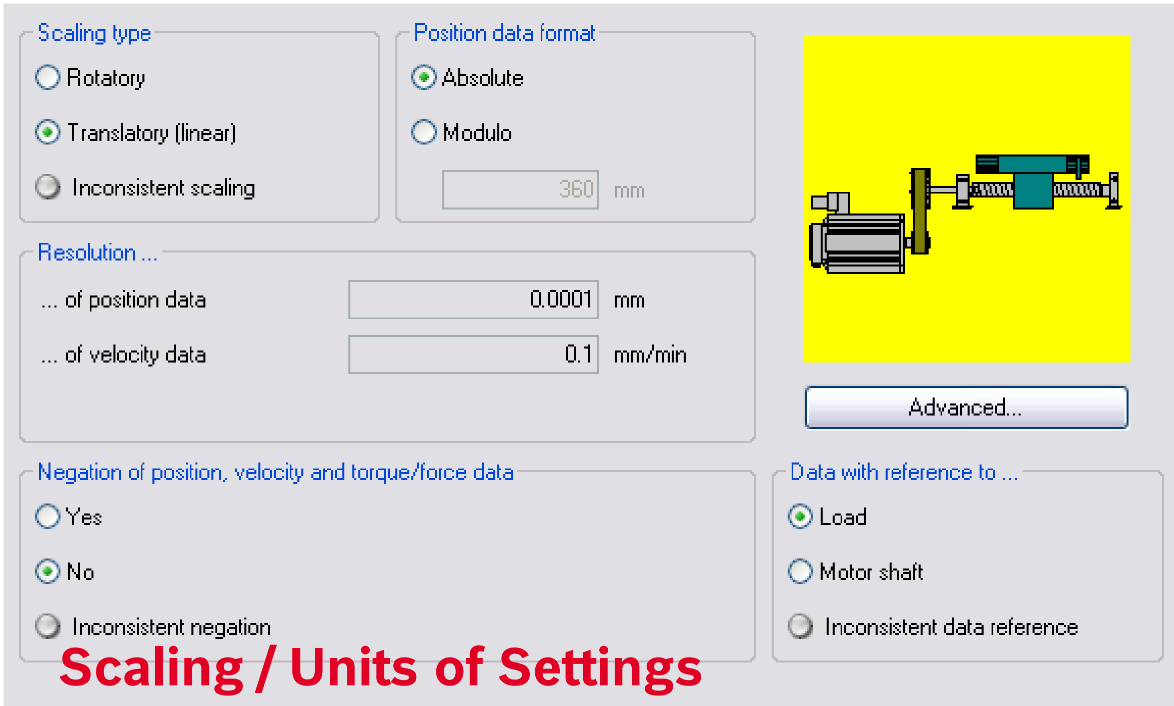
Rexroth Bosch Group

GAT – Axis commissioning (1st axis)

Exercise



After switching to online and loading base parameters ...

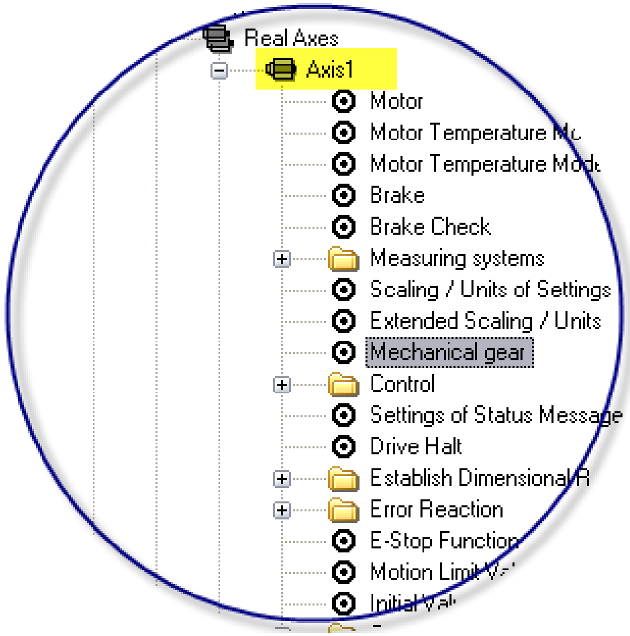


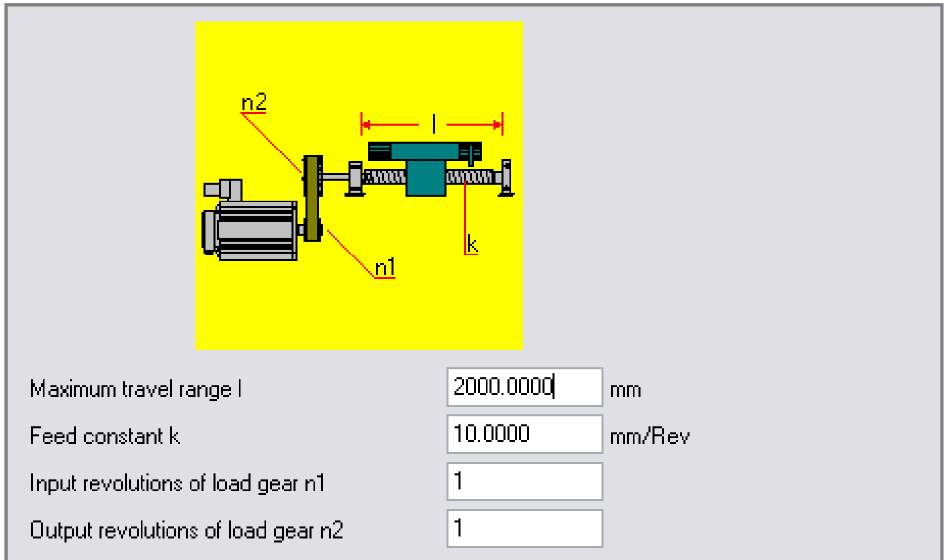
Scaling / Units of Settings

Rexroth Bosch Group

GAT – Axis commissioning (1st axis)

Exercise



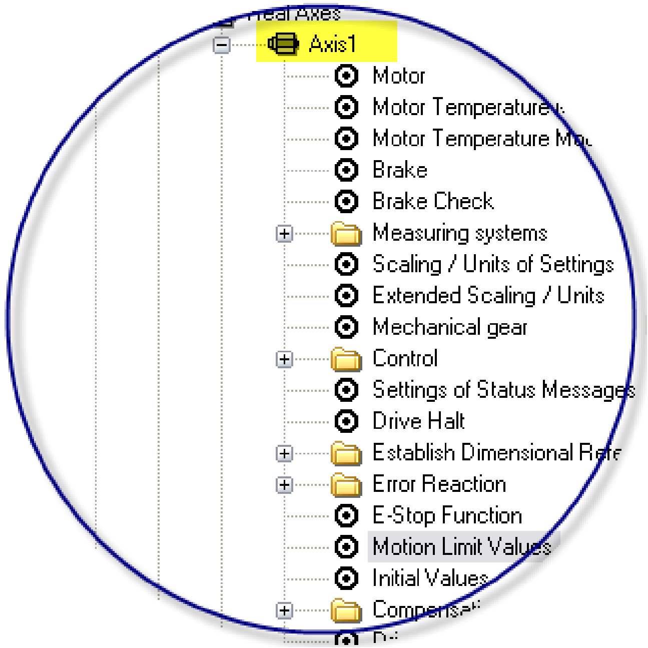


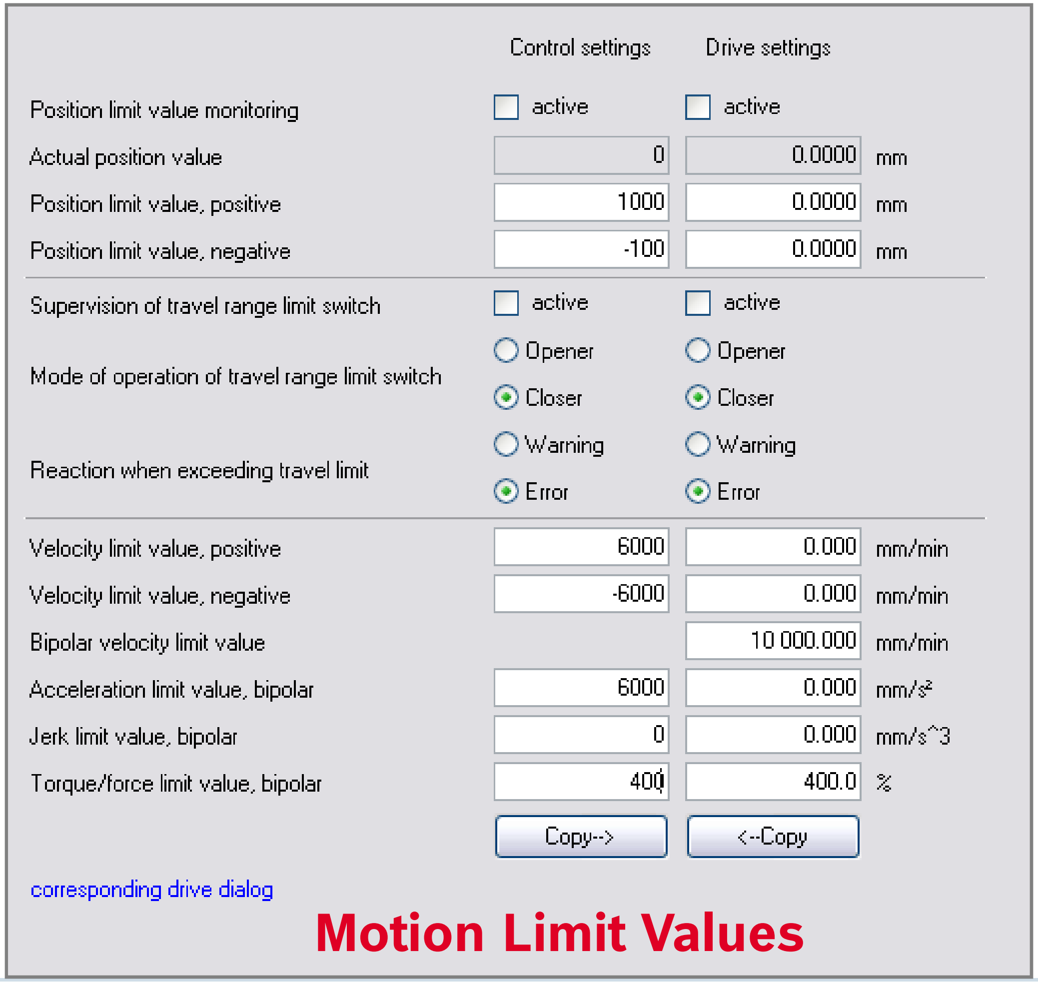
Mechanical Gear

Rexroth
Bosch Group

GAT – Axis commissioning (1st axis)

Exercise





	Control settings	Drive settings	
Position limit value monitoring	☐ active	☐ active	
Actual position value	0	0.0000	mm
Position limit value, positive	1000	0.0000	mm
Position limit value, negative	-100	0.0000	mm
Supervision of travel range limit switch	☐ active	☐ active	
Mode of operation of travel range limit switch	○ Opener ◉ Closer	○ Opener ◉ Closer	
Reaction when exceeding travel limit	○ Warning ◉ Error	○ Warning ◉ Error	
Velocity limit value, positive	6000	0.000	mm/min
Velocity limit value, negative	-6000	0.000	mm/min
Bipolar velocity limit value		10 000.000	mm/min
Acceleration limit value, bipolar	6000	0.000	mm/s²
Jerk limit value, bipolar	0	0.000	mm/s^3
Torque/force limit value, bipolar	400	400.0	%
	Copy-->	<--Copy	

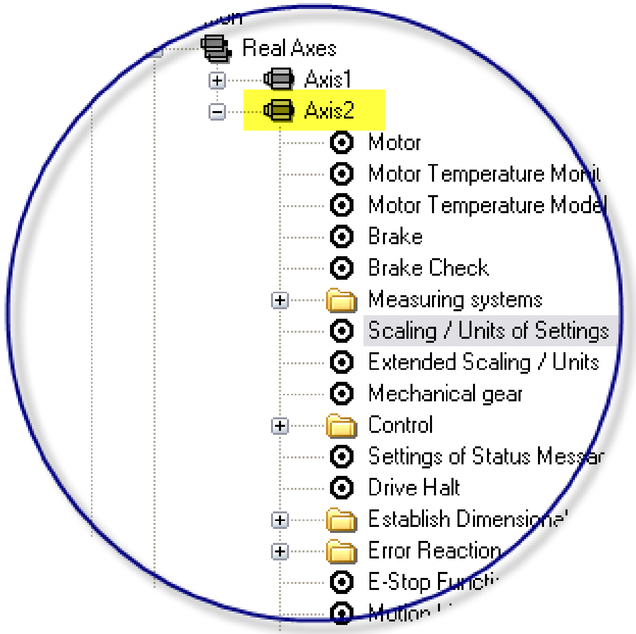
corresponding drive dialog

Motion Limit Values

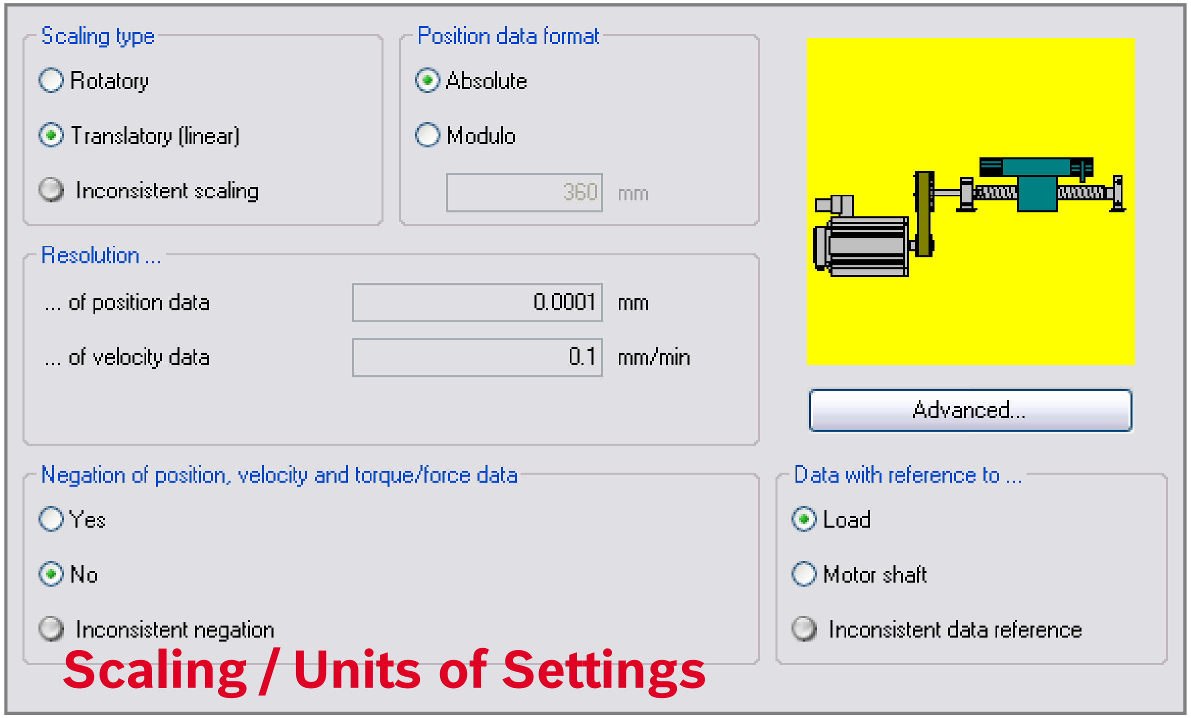
Rexroth
Bosch Group

GAT – Axis commissioning (2^{nd} axis)

Exercise



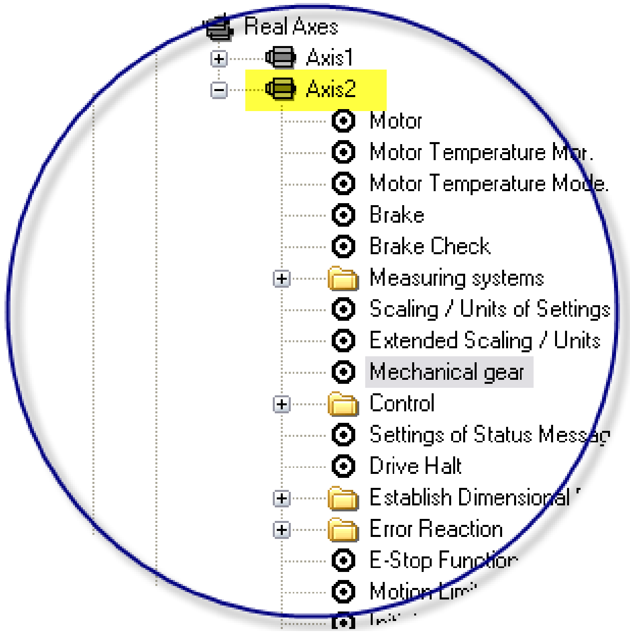
After switching to online and loading base parameters ...

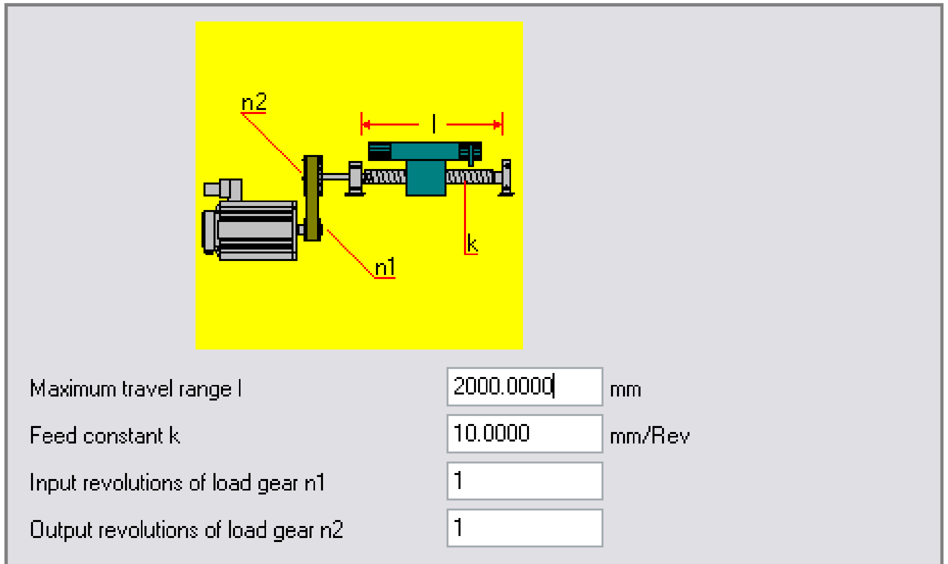


Rexroth Bosch Group

GAT – Axis commissioning (2nd axis)

Exercise



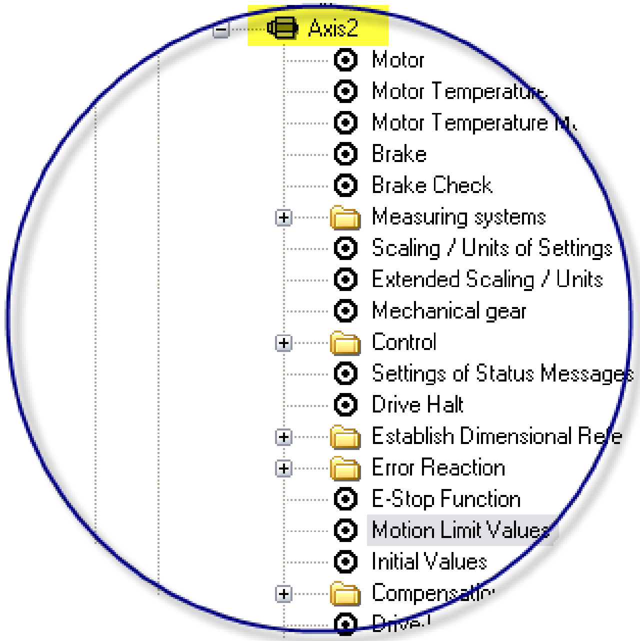


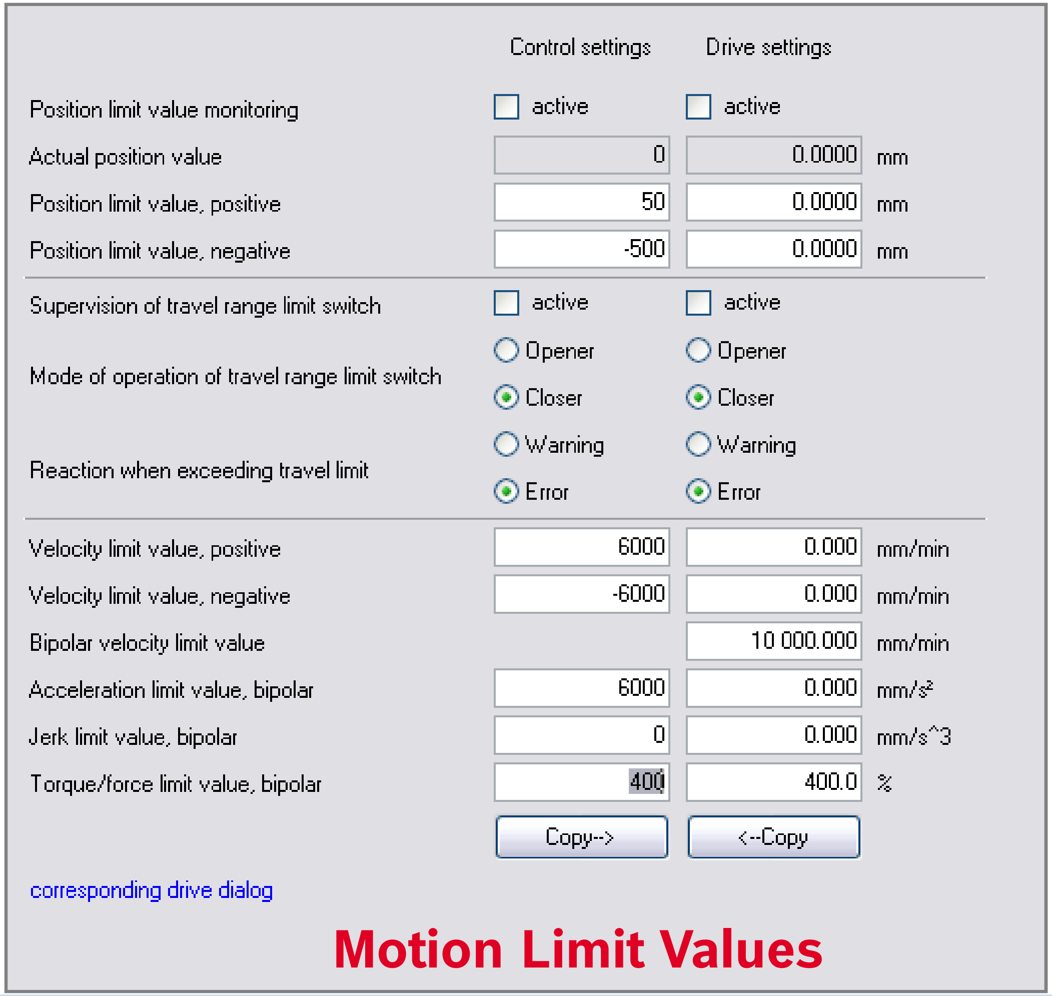
Mechanical Gear

Rexroth Bosch Group

GAT – Axis commissioning (2nd axis)

Exercise





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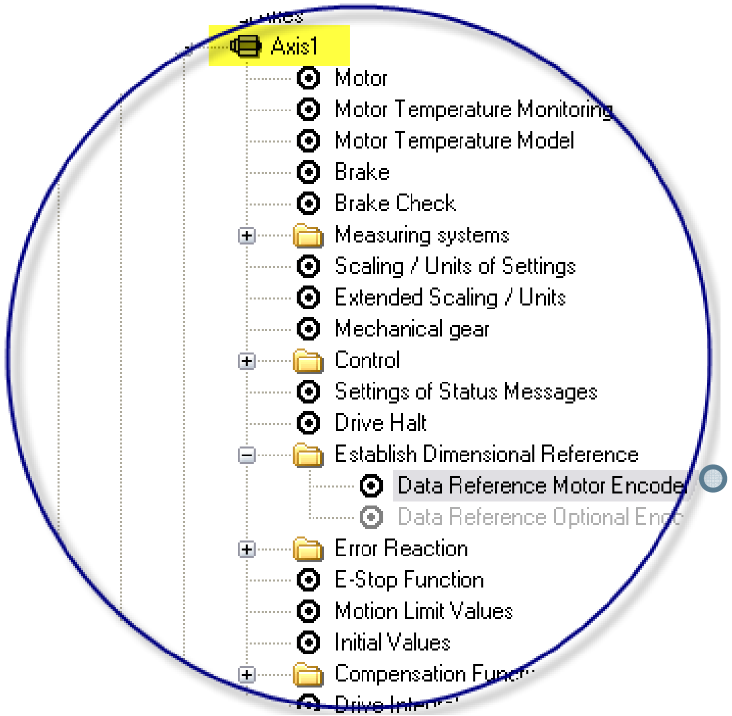
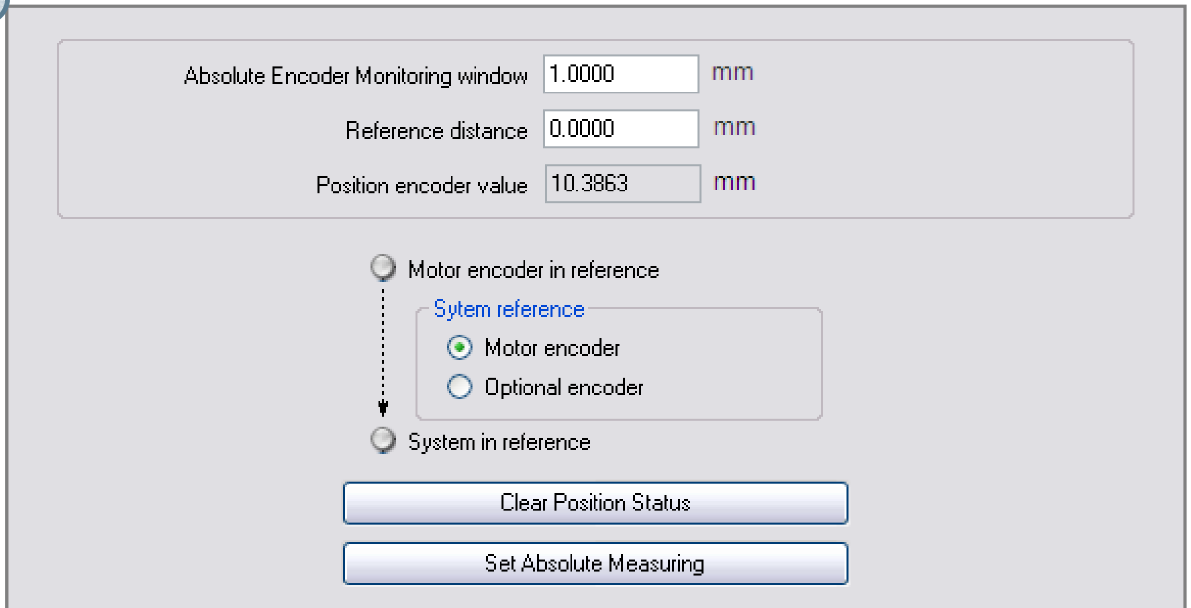
Bosch Group

GAT – Axis commissioning (1st axis)

Exercise



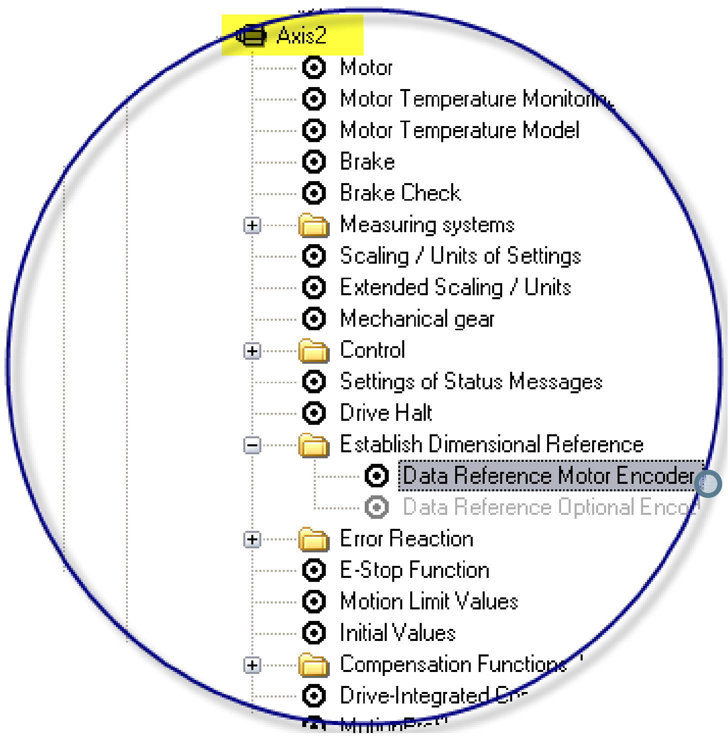
After SERCOS startup and all axes in "AB" ... (only for demo systems with multi turn encoder)



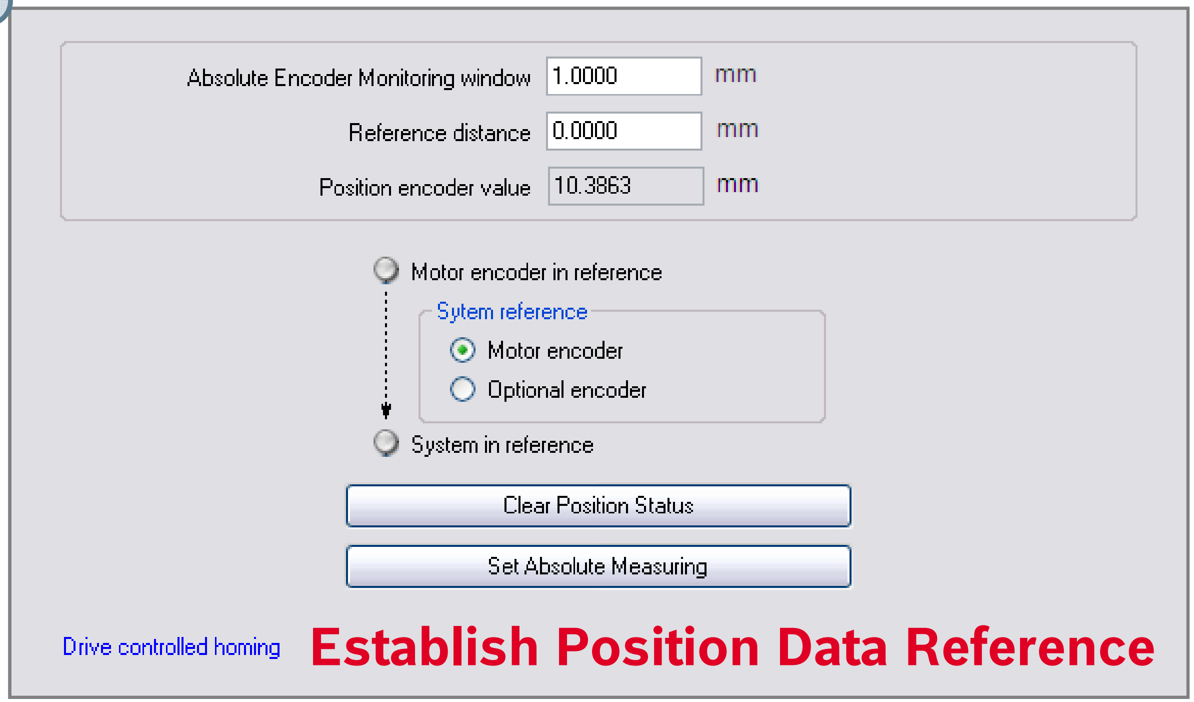
Establish Position Data Reference

GAT – Axis commissioning (2^{nd} axis)

Exercise



After SERCOS startup and all axes in “AB” … (only for demo systems with multi turn encoder)



Establish Position Data Reference

Rexroth Bosch Group

GAT – Import GAT

Exercise

- Use GAT wizard to create the IndraLogic project

- By default 2 operation modes are added to the project
  - Manual
  - Automatic
  - For this exercise we need only these predefined operation modes!

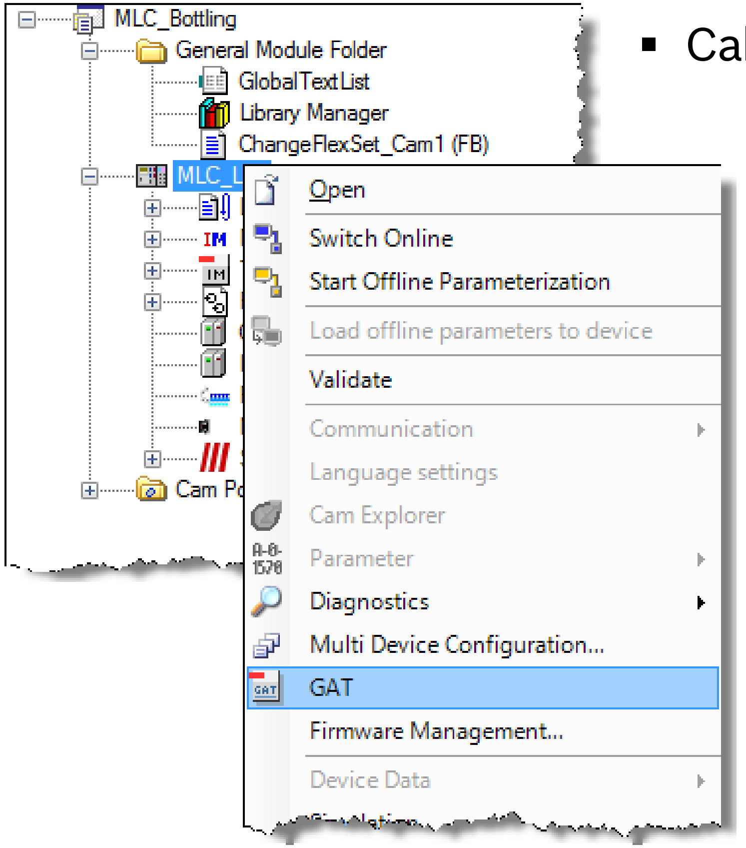
- Modules & axes:
  - Delete axes MyVirtualAxis1 and MyVirtualAxis2!
  - The default module Module1 is not used, rename it!
  - Two modules are required:
    - Module Master with axis vAxis3 (use name vMaster as a name of your GAT axis)
    - Module FillingDevice with axes Axis1, Axis2, vAxis3 (use the names Horizontal (Axis1) and Vertical (Axis2) and vMaster (vAxis3) as names for your GAT axes)

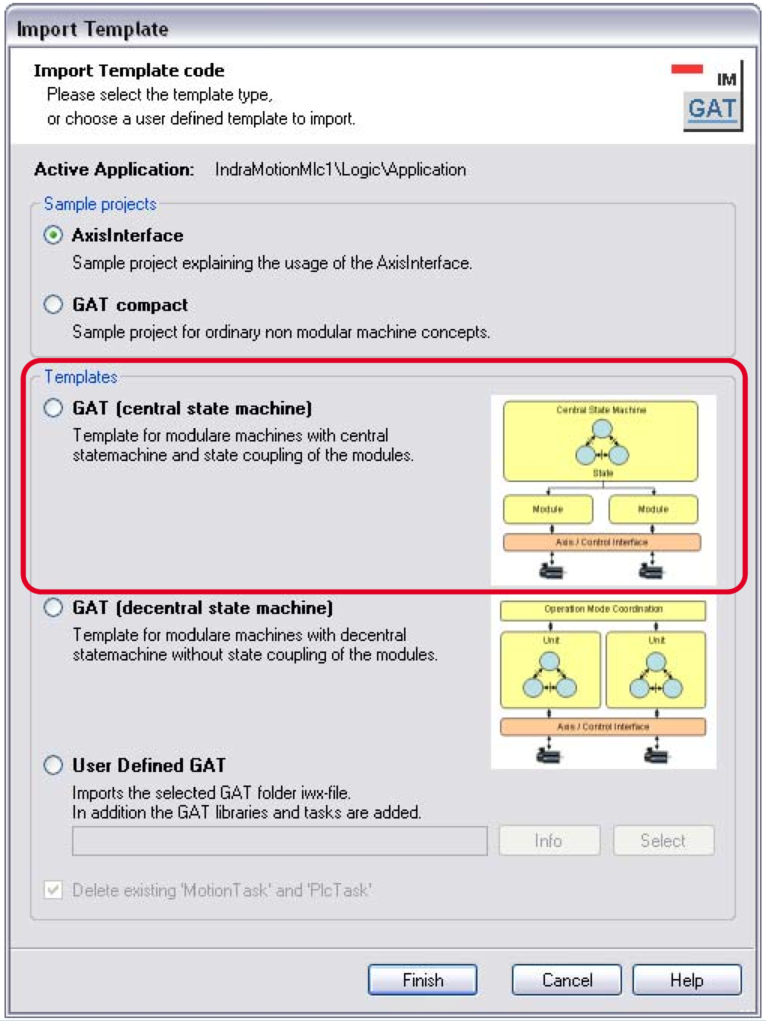
Rexroth Bosch Group

GAT Exercise – Call GAT wizard

Exercise

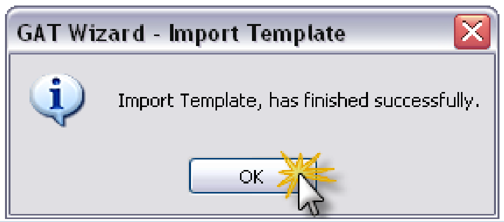
- Call GAT wizard





- Select GAT (central state machine)

- Import OK!



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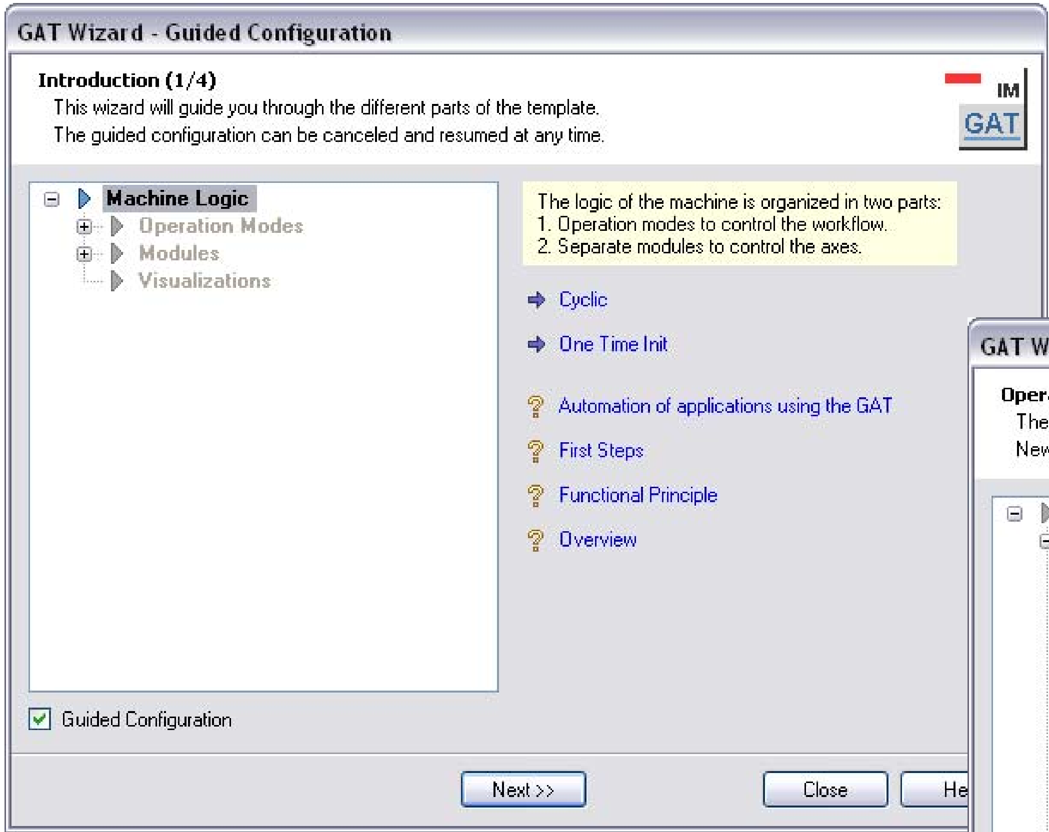
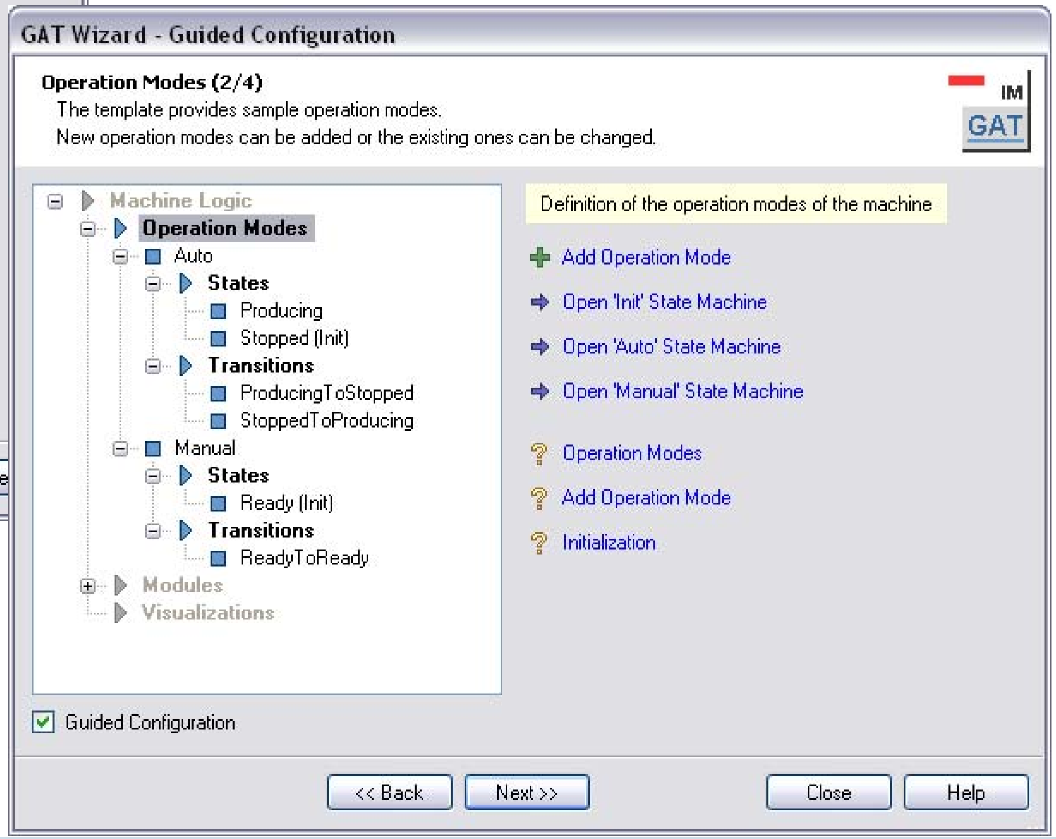
Bosch Group

GAT – Overall Logic & Operation Modes

Exercise



- Don't modify any settings for the machine logic in the 1st step



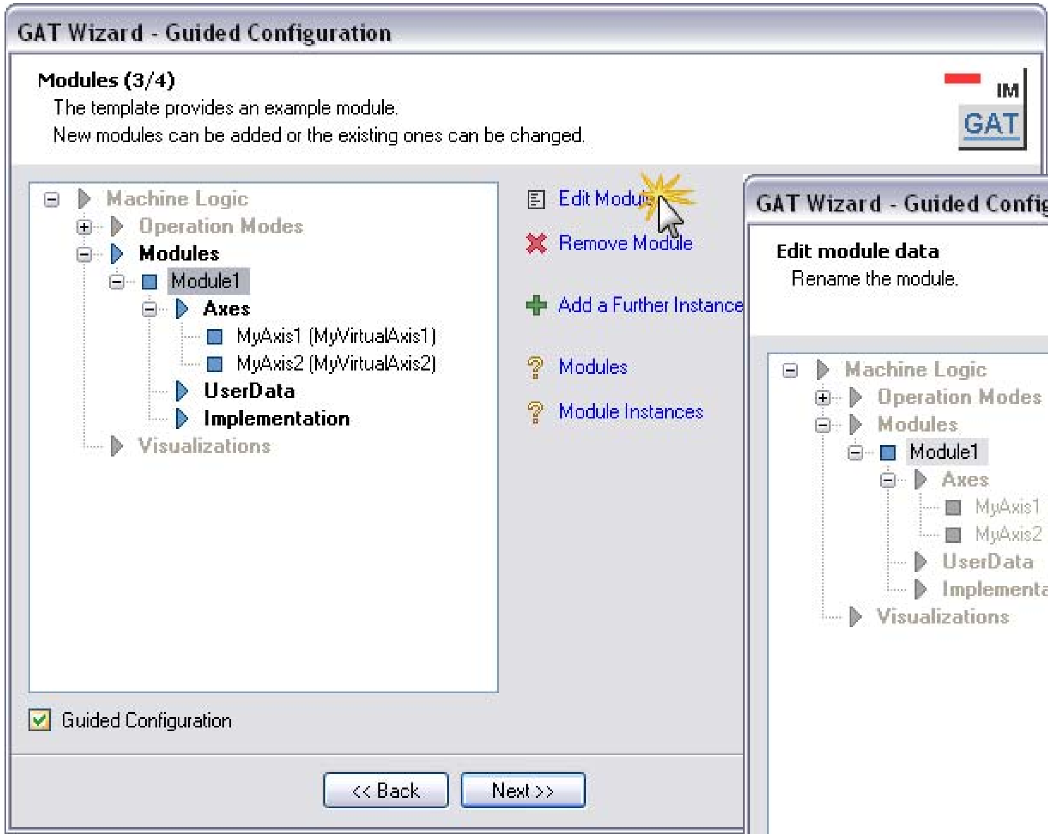
- For this exercise we need only the predefined operation modes

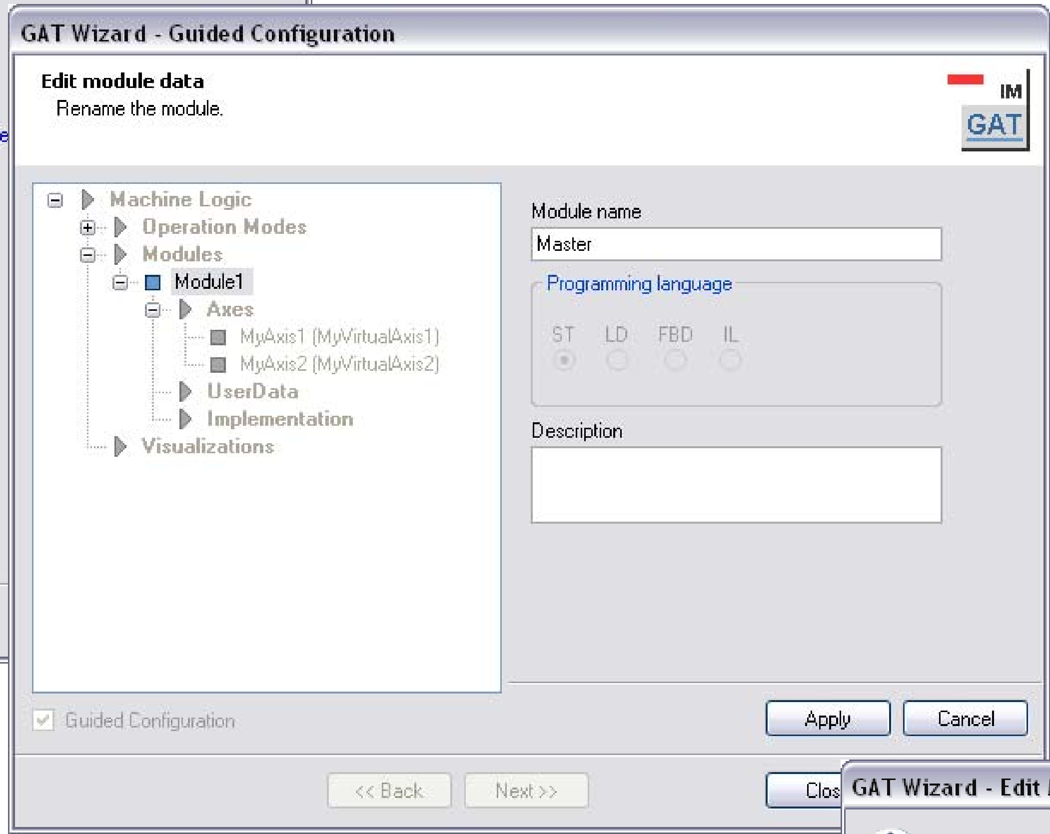
Rexroth Bosch Group

GAT – Rename 1st module

Exercise

- Rename “Module1” to “Master”
- Press “Edit Module”





- Enter the module name “Master” and press “Apply”

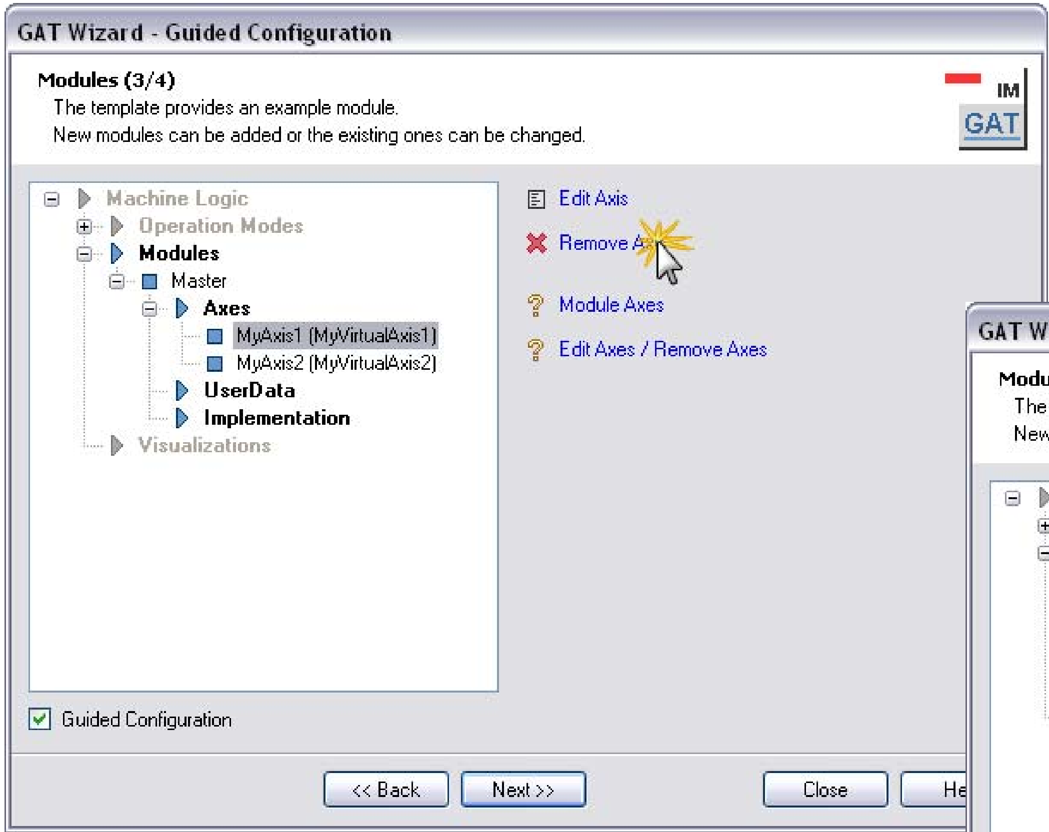
- The module has been renamed successfully

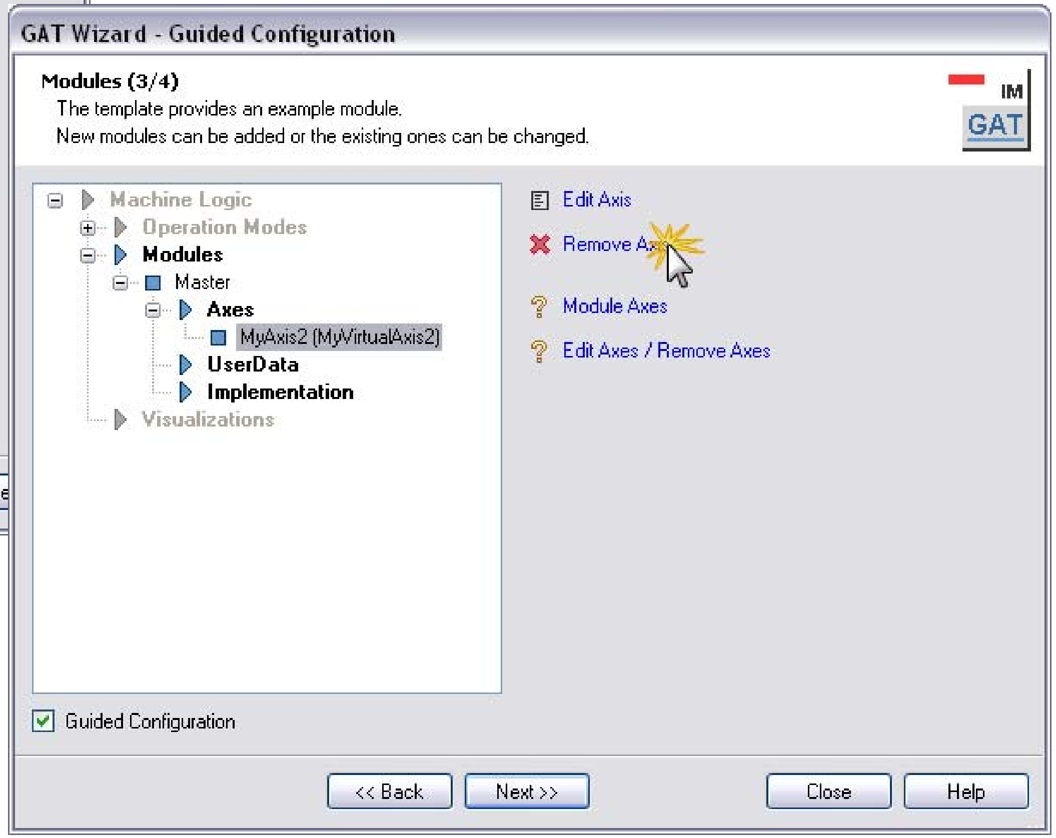
Rexroth
Bosch Group

GAT – Remove axes from module

Exercise

- Delete axis “MyVirtualAxis1” from module Master



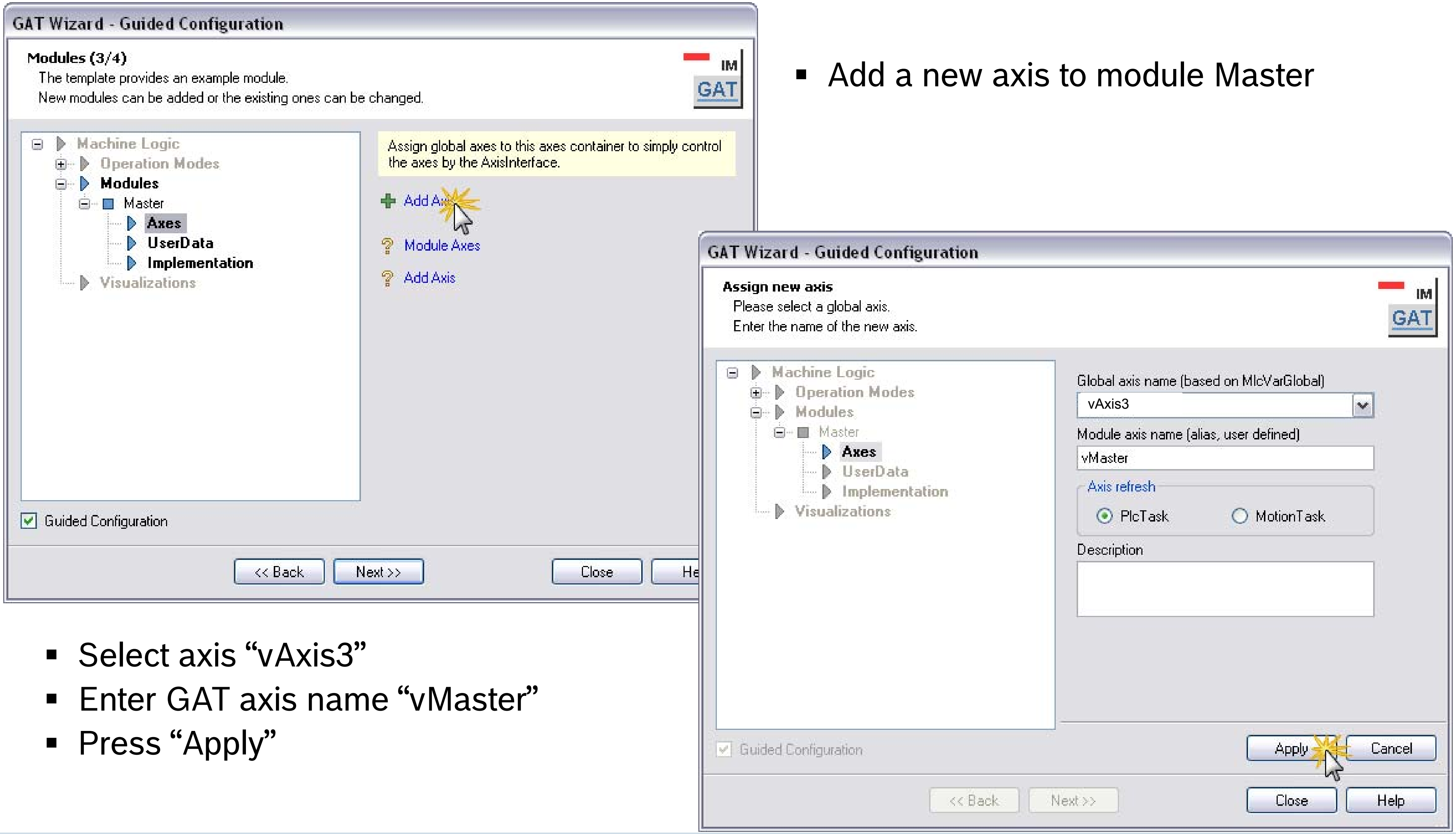


- Delete axis “MyVirtualAxis2” from module Master

Rexroth Bosch Group

GAT – Add axis “vMaster” to module

Exercise



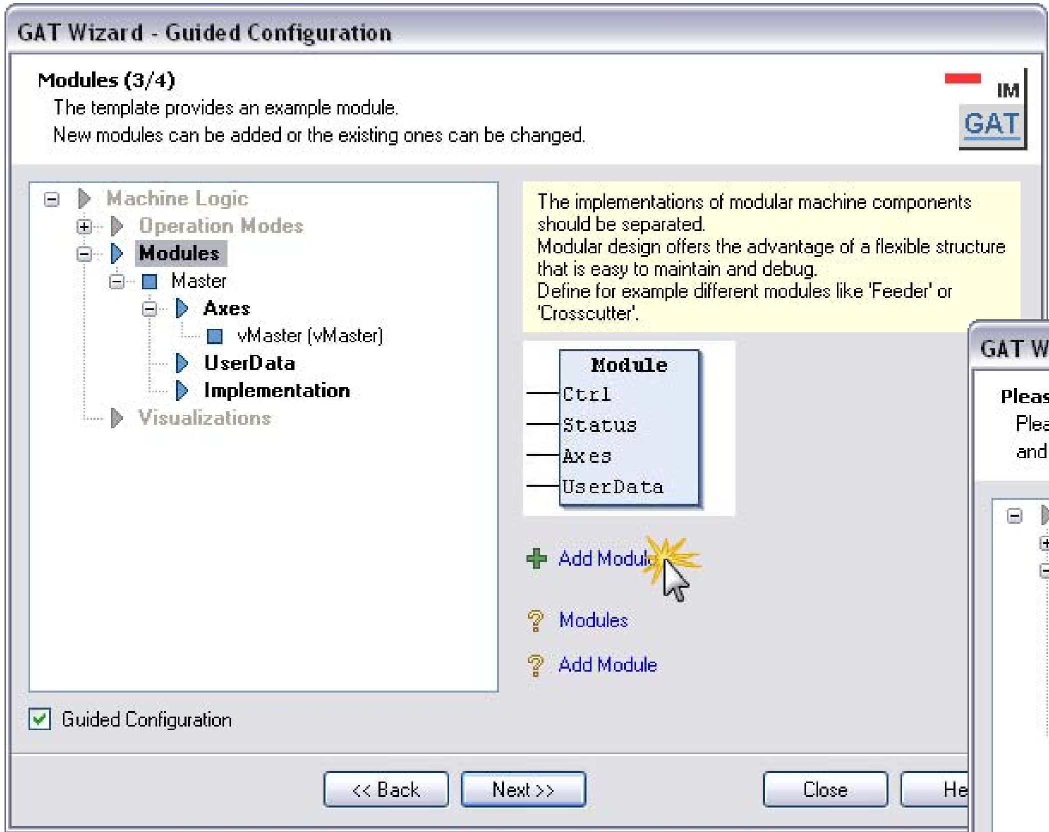
- Add a new axis to module Master

- Select axis “vAxis3”
- Enter GAT axis name “vMaster”
- Press “Apply”

Rexroth Bosch Group

GAT – Add module “FillingDevice”

Exercise



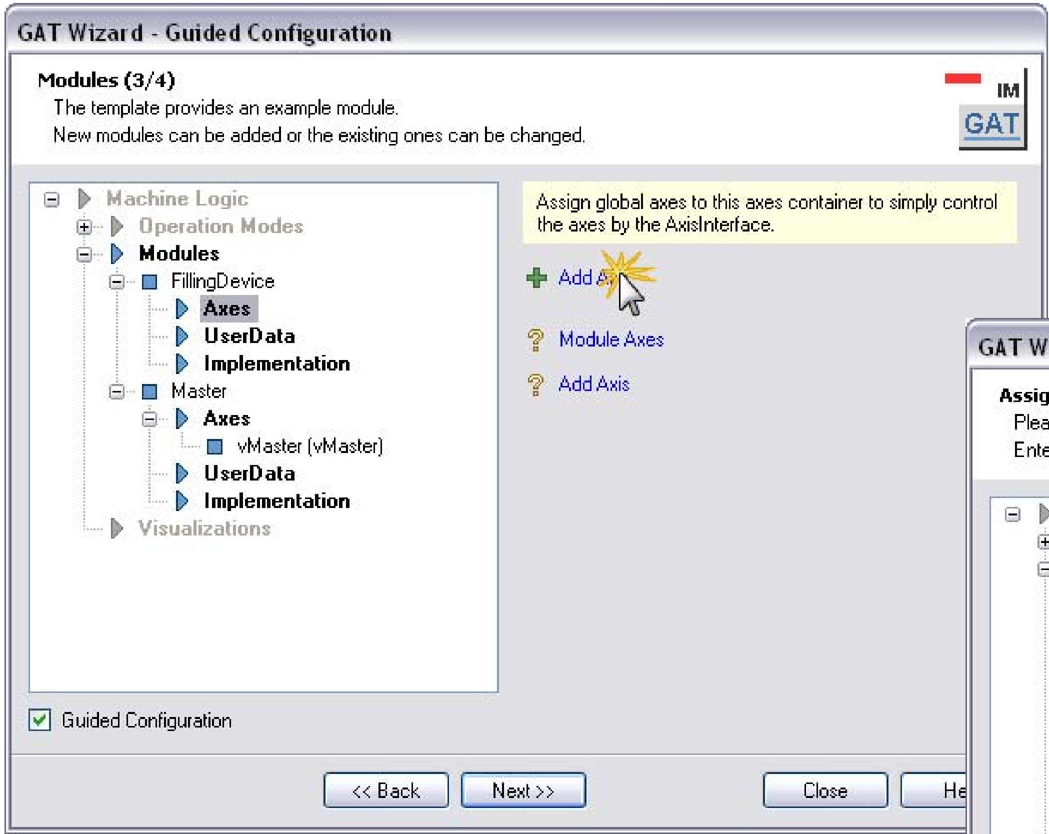
- Add a new module

- Enter module name “FillingDevice”
- Press “Apply”

Rexroth
Bosch Group

GAT – Add axes to “FillingDevice”

Exercise



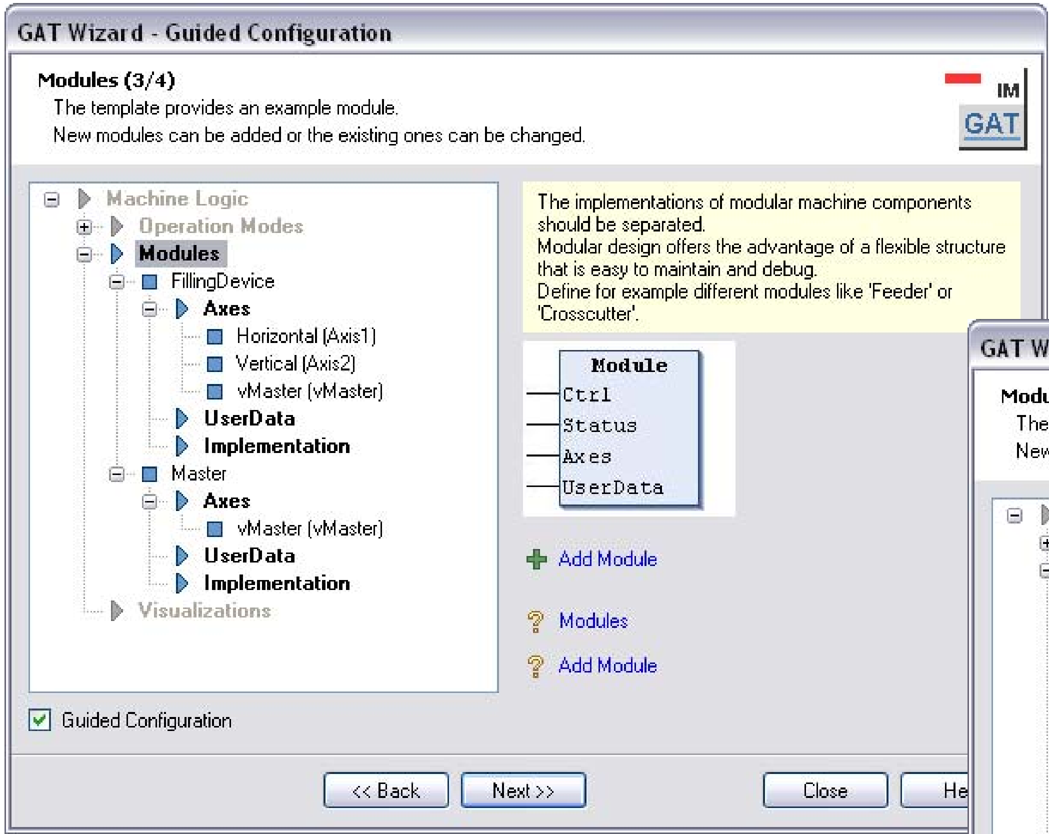
- Assign axes to the module

- Select axis “vAxis3”
- Enter GAT axis name “vMaster”
- Press “Apply”

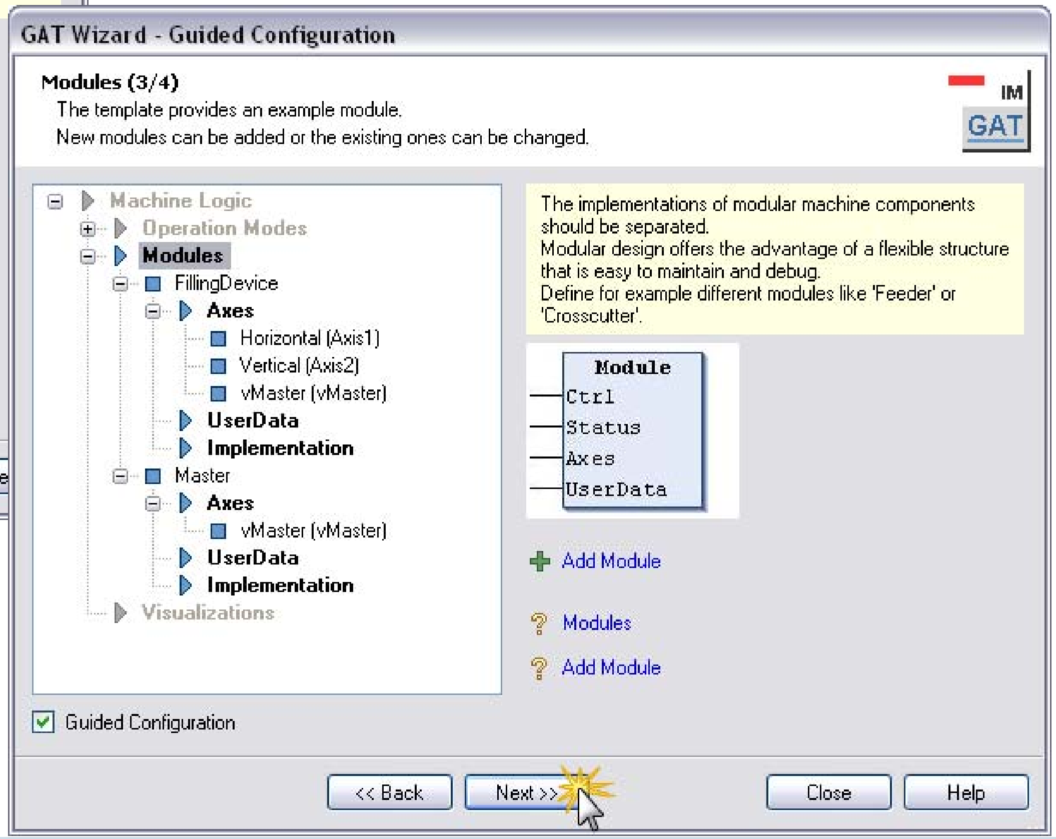
Rexroth Bosch Group

GAT – Add axes to “FillingDevice”

Exercise



- Add 2 additional axes:
  - Horizontal (Axis1) and
  - Vertical (Axis2)

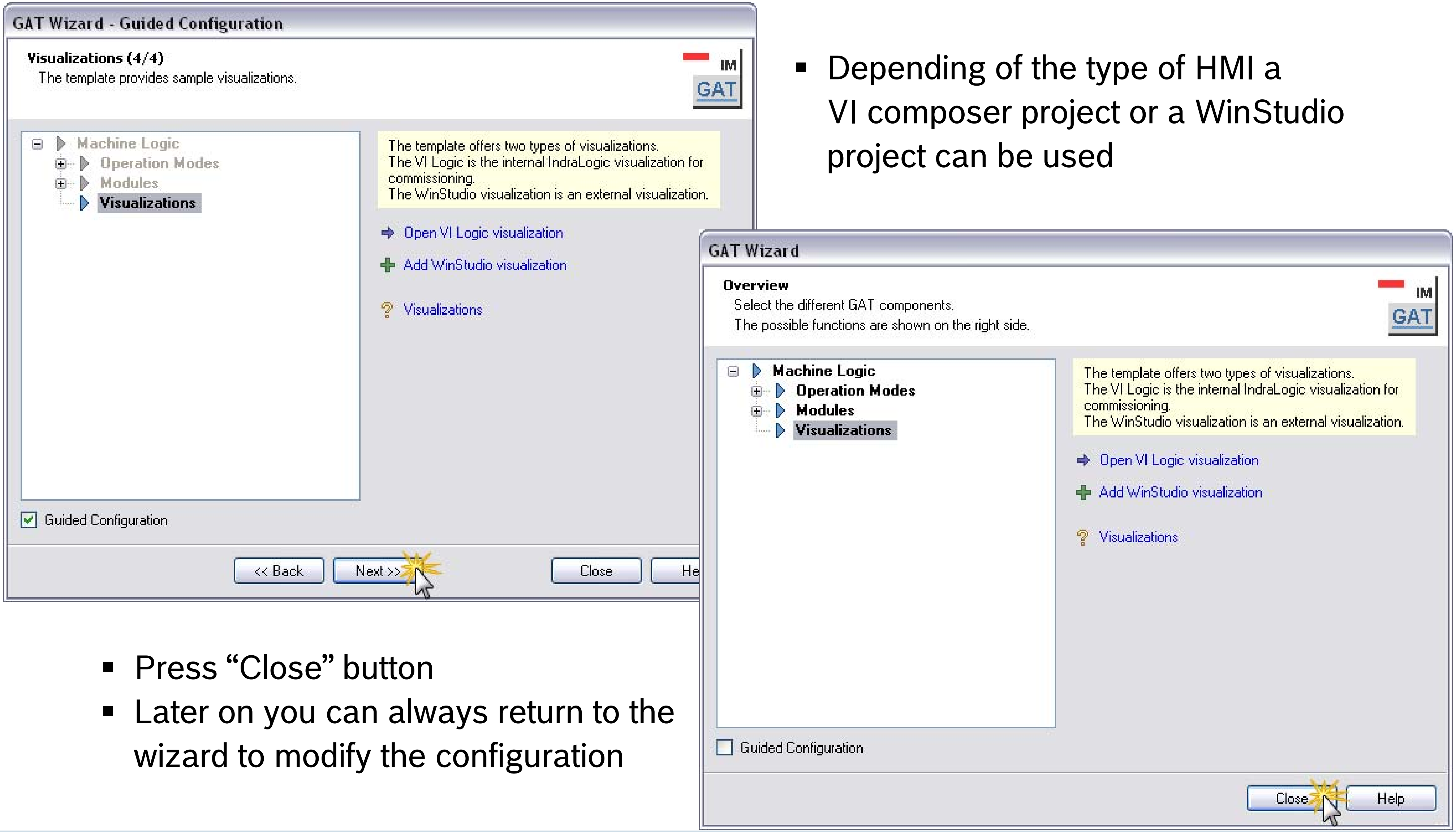


- After completion of the axis assignment
- Press “Next” button

Rexroth
Bosch Group

GAT – Close GAT wizard

Exercise



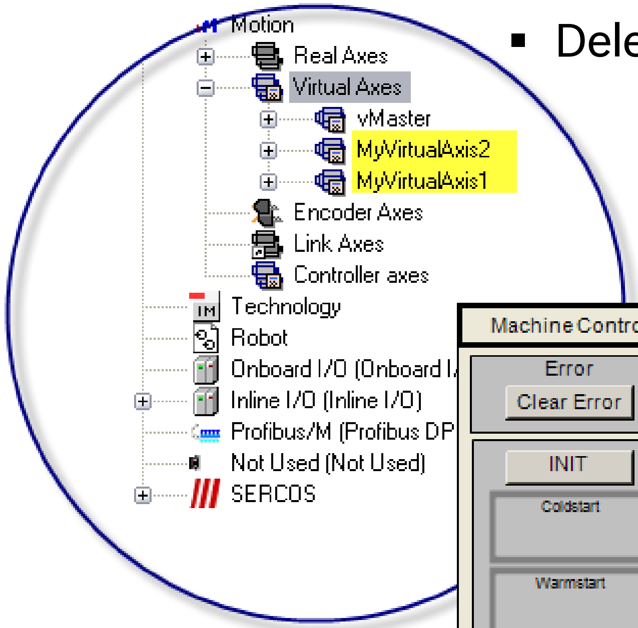
- Depending of the type of HMI a VI composer project or a WinStudio project can be used

- Press “Close” button
- Later on you can always return to the wizard to modify the configuration

Rexroth
Bosch Group

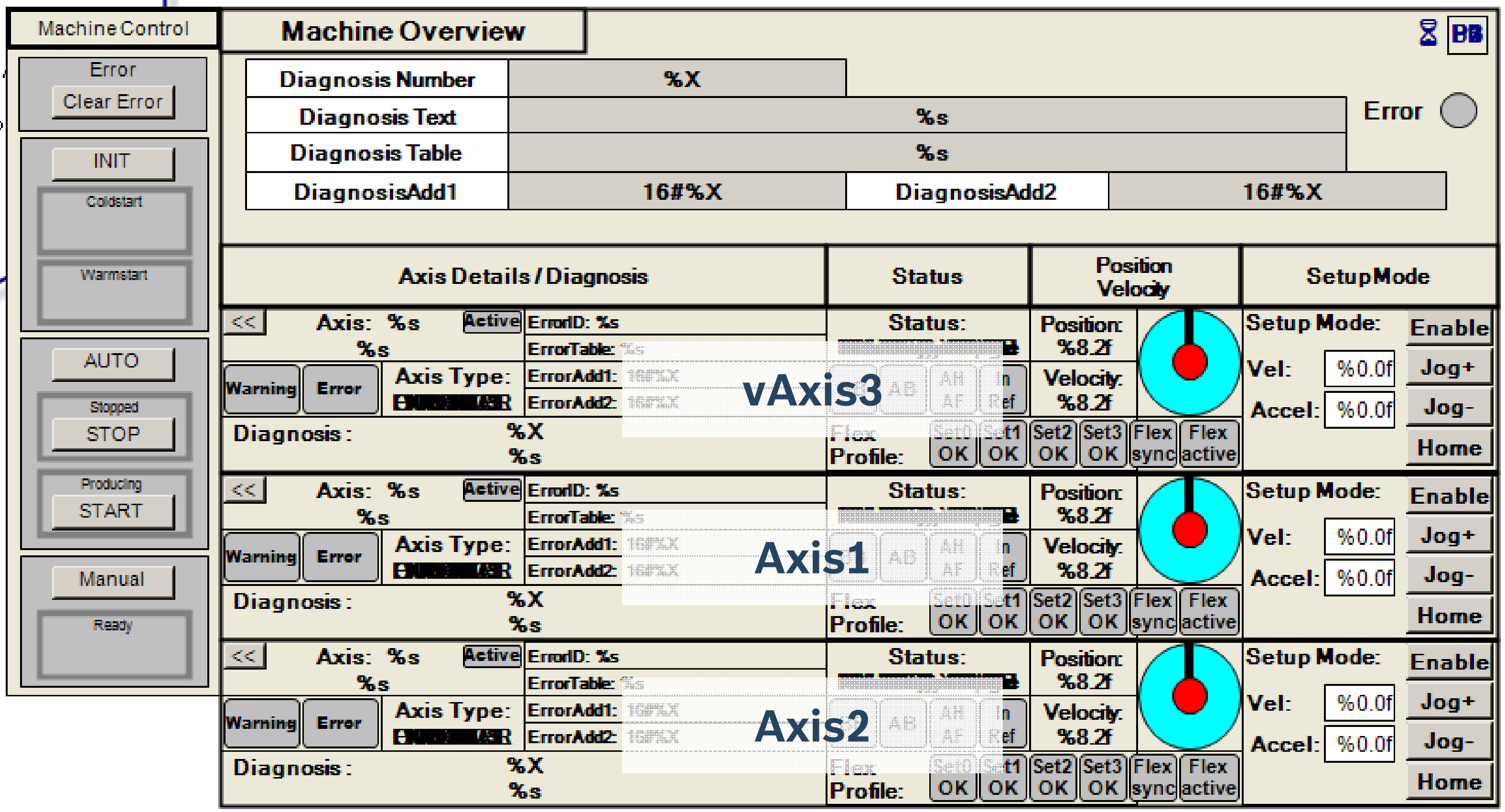
GAT – Remove axes from project

Exercise



- Delete "MyVirtualAxis1" and "MyVirtualAxis2" from IndraWorks project

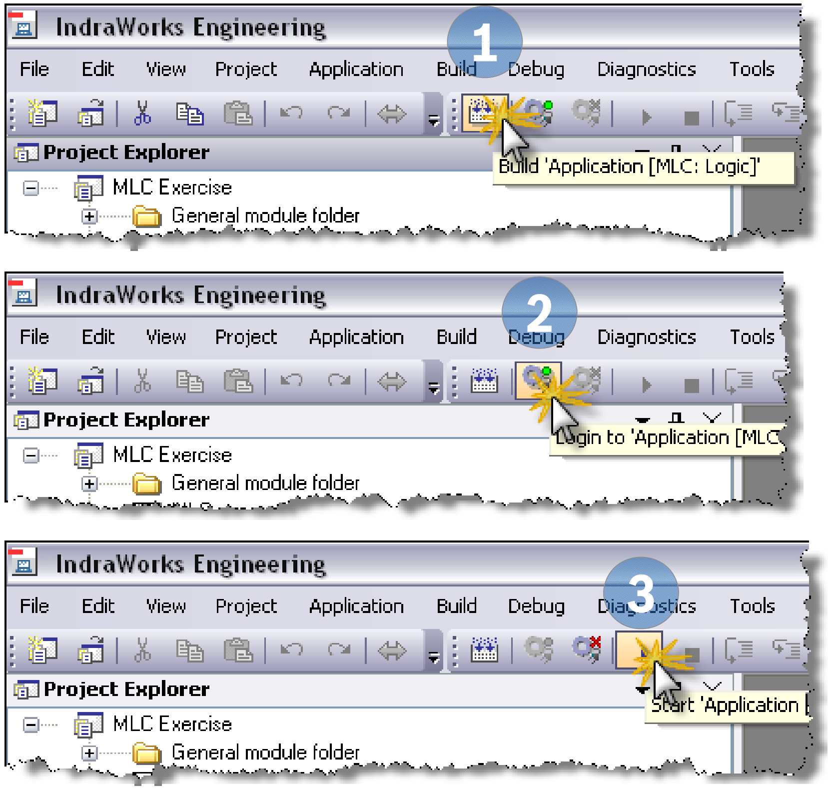
- Modify the IndraLogic visualization "MAIN_VISU" as follows:

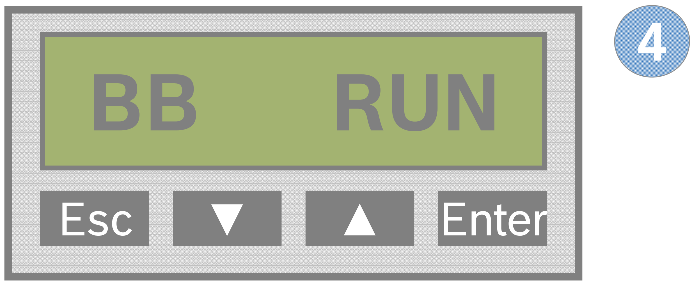


Rexroth Bosch Group

GAT – Build + Download + Start

Exercise

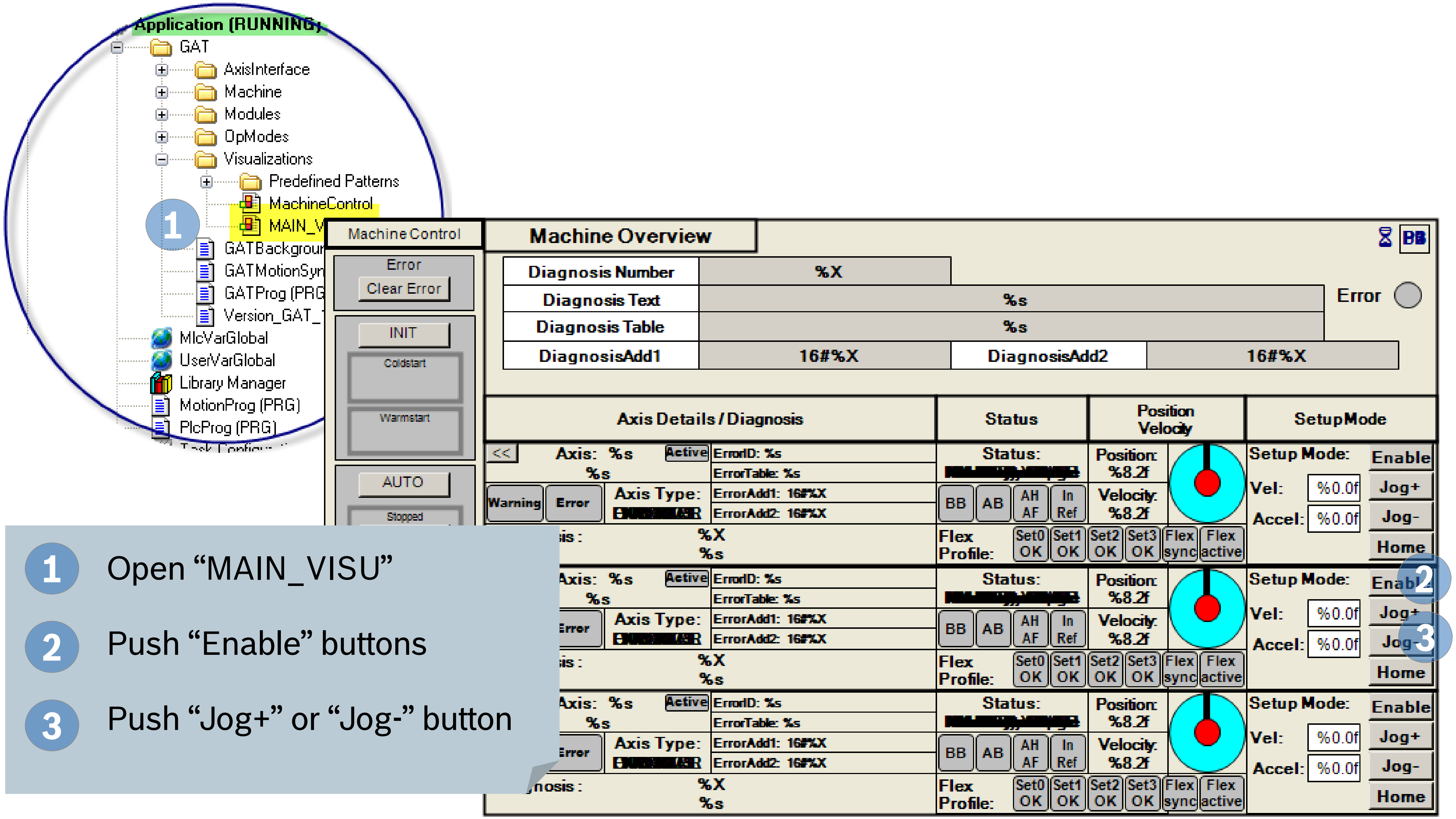




1. Push “Build” button
2. Push “Login” button
3. Push “Start” button
4. “BB RUN” is displayed

Rexroth Bosch Group

GAT – Open Visualization

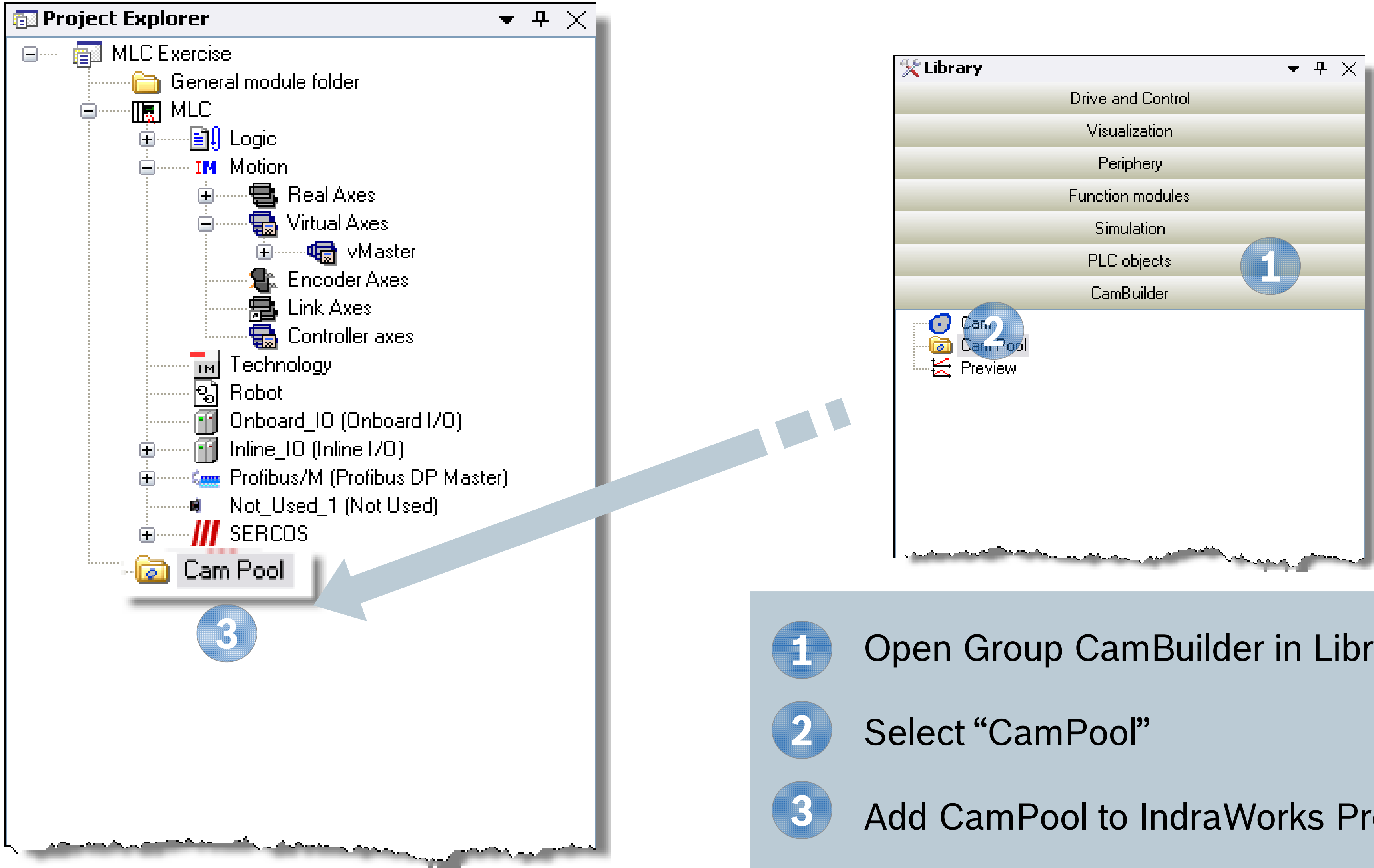


1. Open “MAIN_VISU”
2. Push “Enable” buttons
3. Push “Jog+” or “Jog-” button

Rexroth Bosch Group

GAT – Add CamPool to project

Exercise



1 Open Group CamBuilder in Library

2 Select “CamPool”

3 Add CamPool to IndraWorks Project

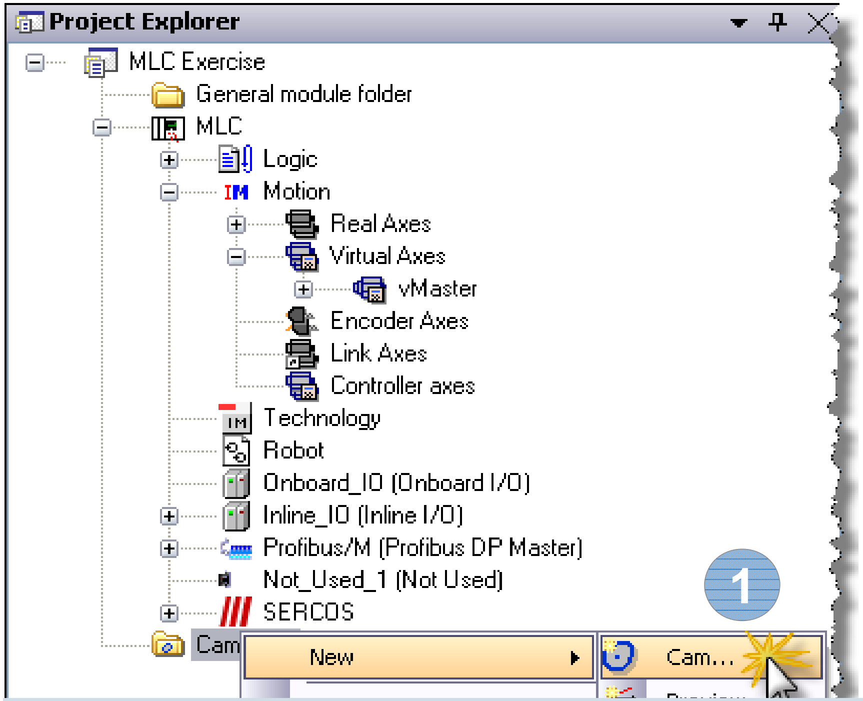
Rexroth Bosch Group

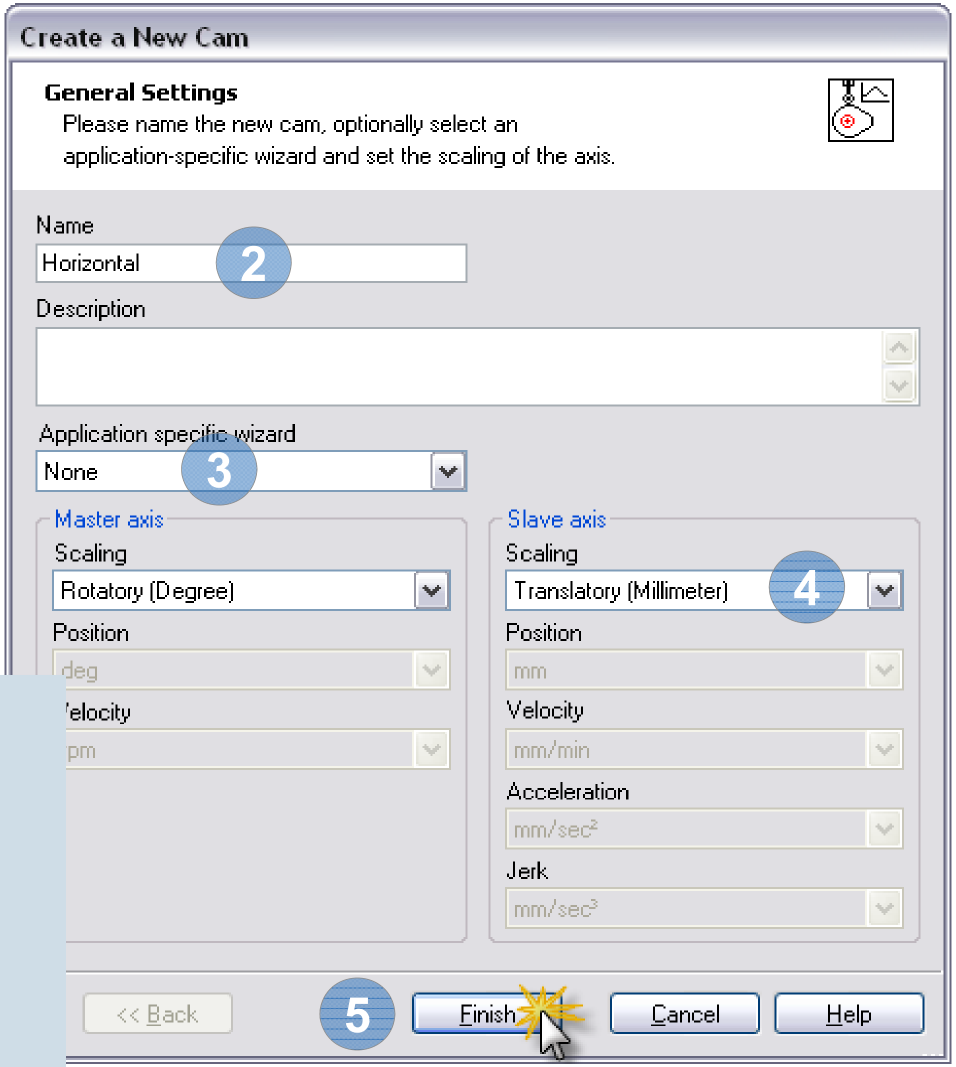
GAT – FlexProfile for horizontal axis (1)

Exercise

GAT – FlexProfile for horizontal axis (2)

Exercise



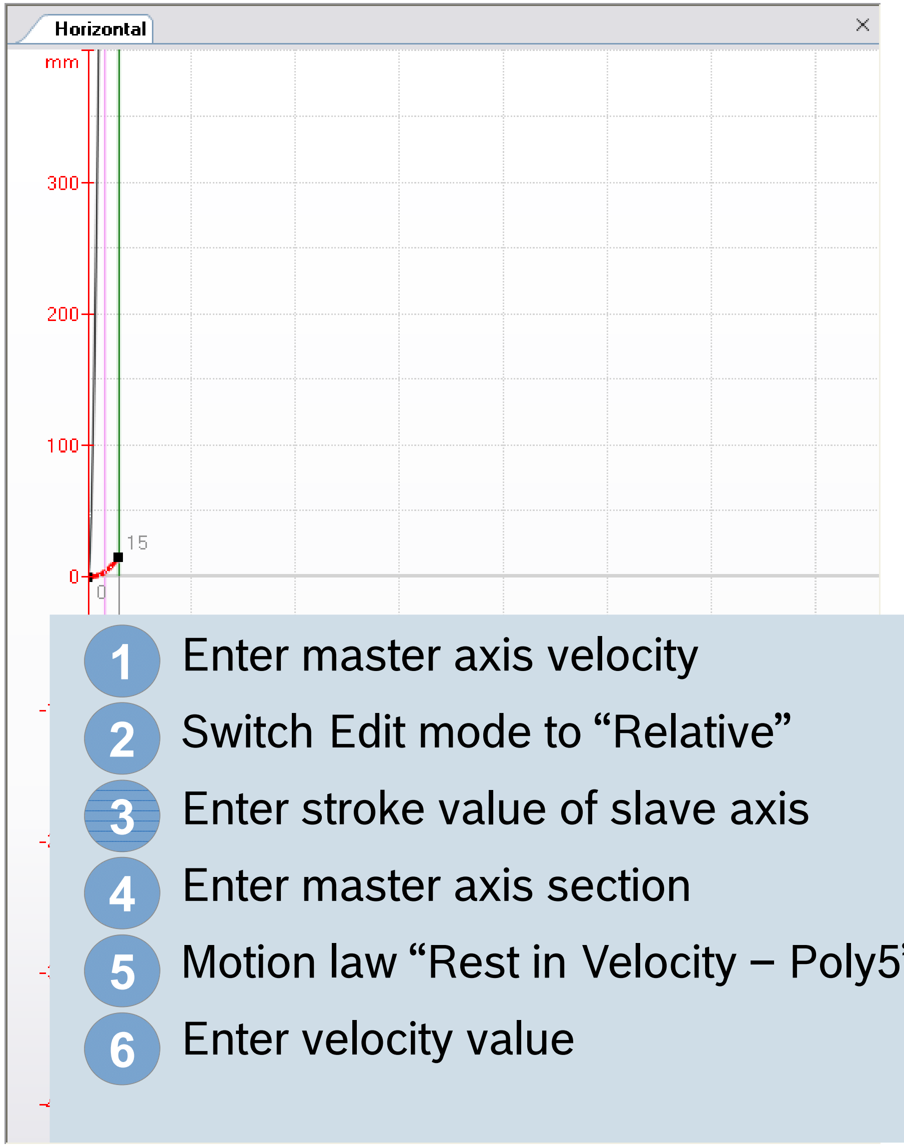


1. Create new Cam
2. Enter name “Horizontal”
3. Application specific wizard: None
4. Translatory scaling (mm)
5. Press “Finish” button

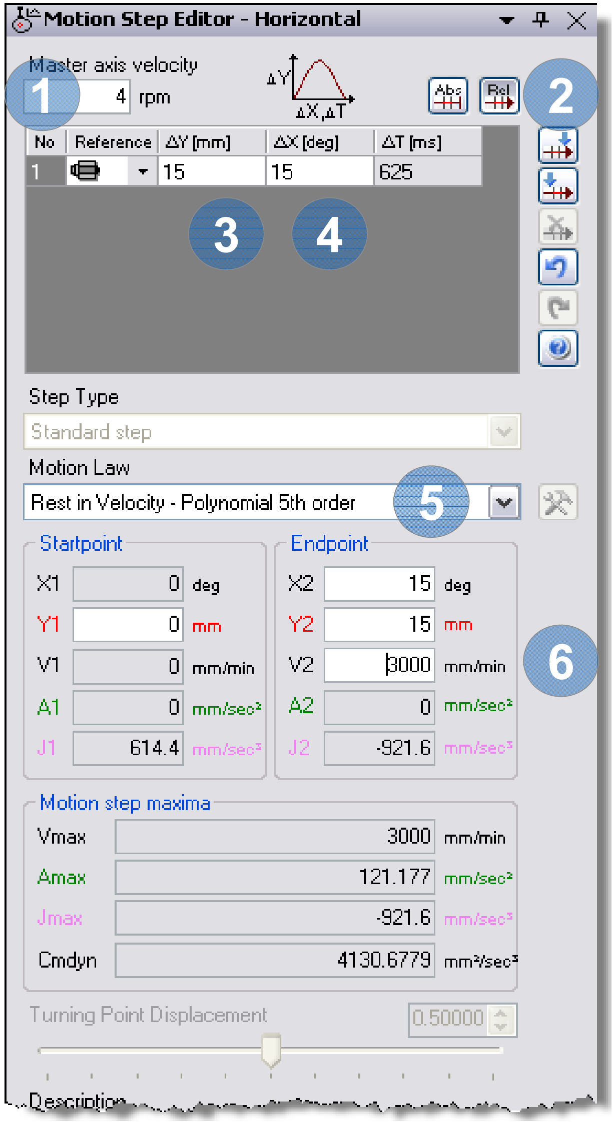
Rexroth Bosch Group

GAT – FlexProfile for horizontal axis (3)

Exercise



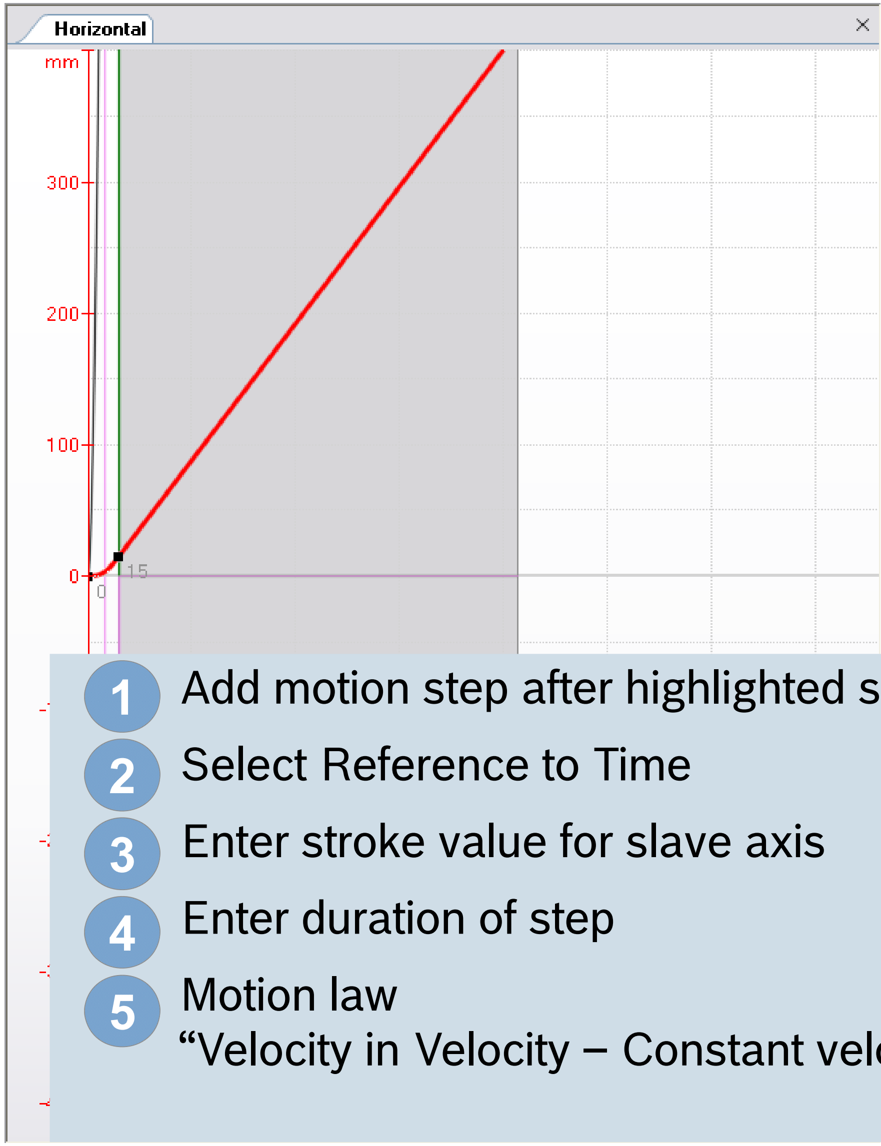
1. Enter master axis velocity
2. Switch Edit mode to “Relative”
3. Enter stroke value of slave axis
4. Enter master axis section
5. Motion law “Rest in Velocity – Poly5”
6. Enter velocity value



Rexroth
Bosch Group

GAT – FlexProfile for horizontal axis (4)

Exercise

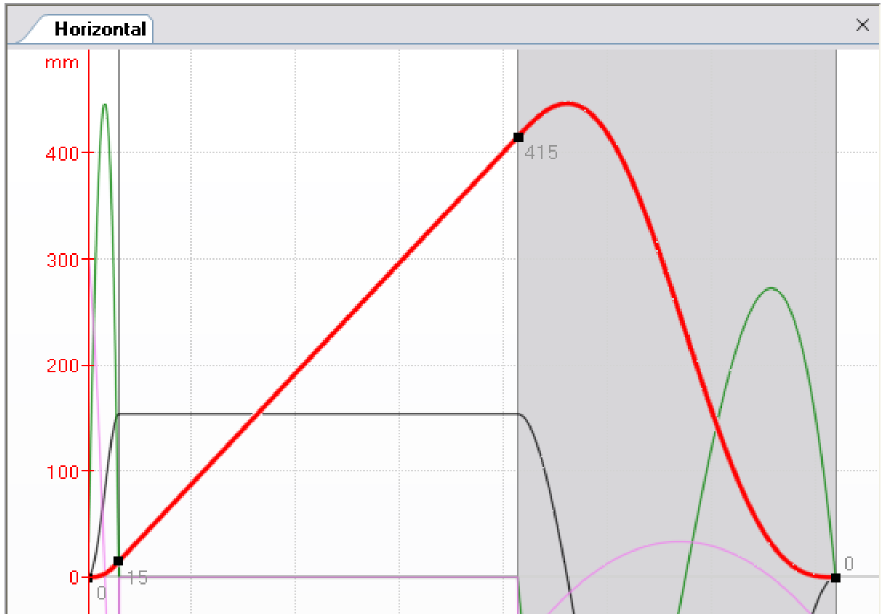


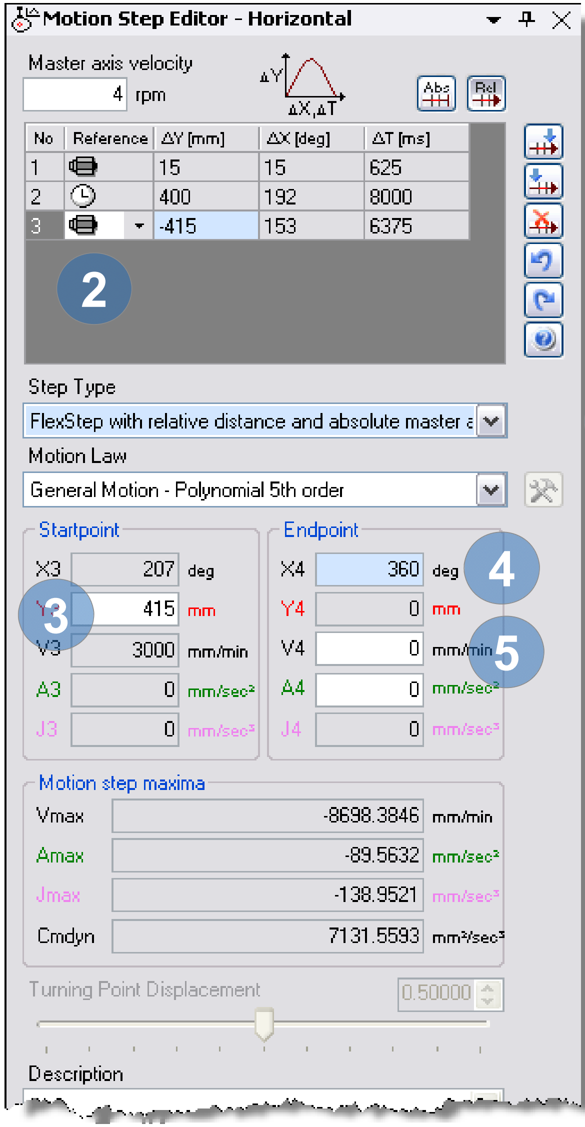
1. Add motion step after highlighted step
2. Select Reference to Time
3. Enter stroke value for slave axis
4. Enter duration of step
5. Motion law
   “Velocity in Velocity – Constant velocity”

Rexroth
Bosch Group

GAT – FlexProfile for horizontal axis (5)

Exercise



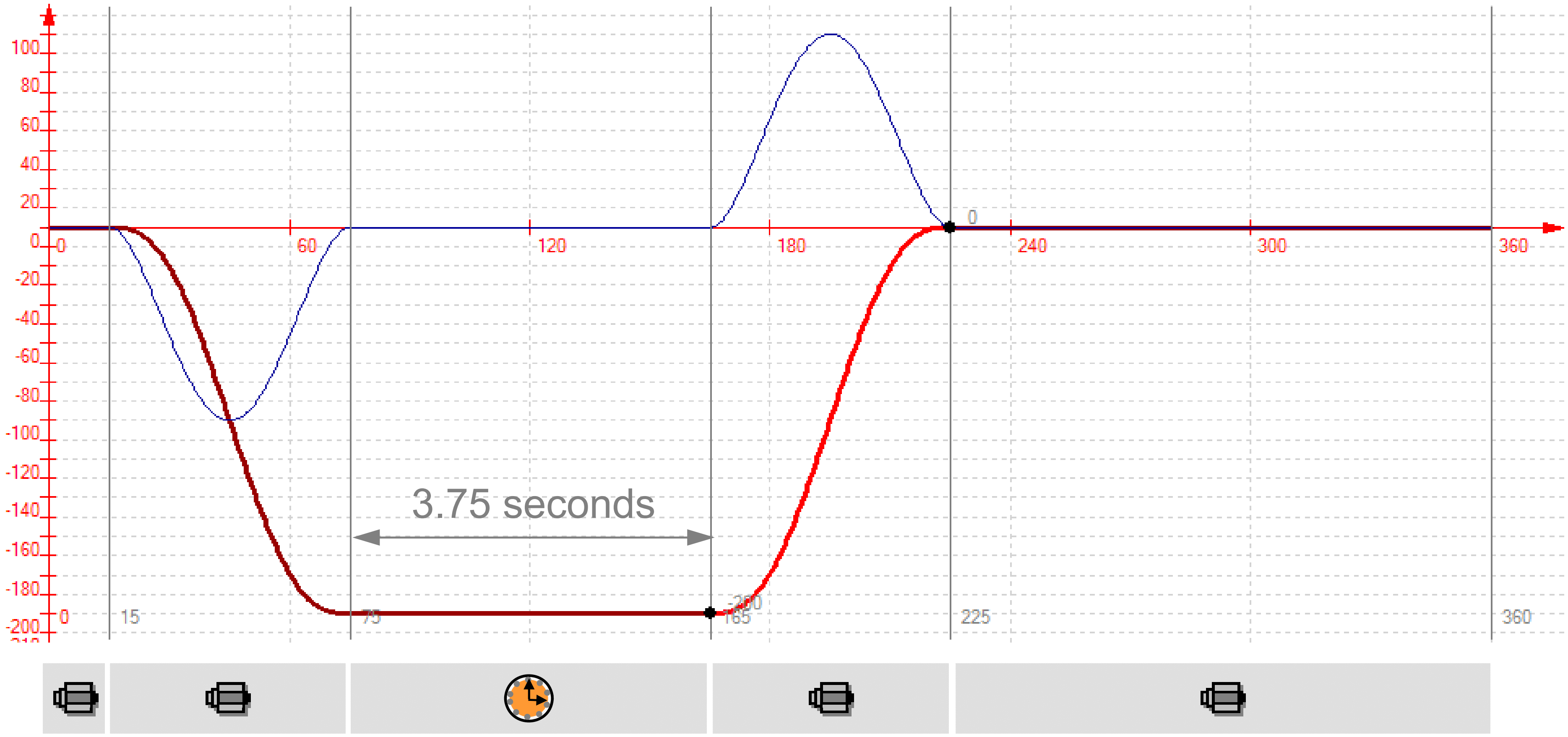


1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (-415)
4. Enter Master axis Endpoint (360)
5. Set velocity value to zero
6. Select FlexStep with relative distance and absolute master axis range

Rexroth Bosch Group

GAT – FlexProfile for vertical axis (1)

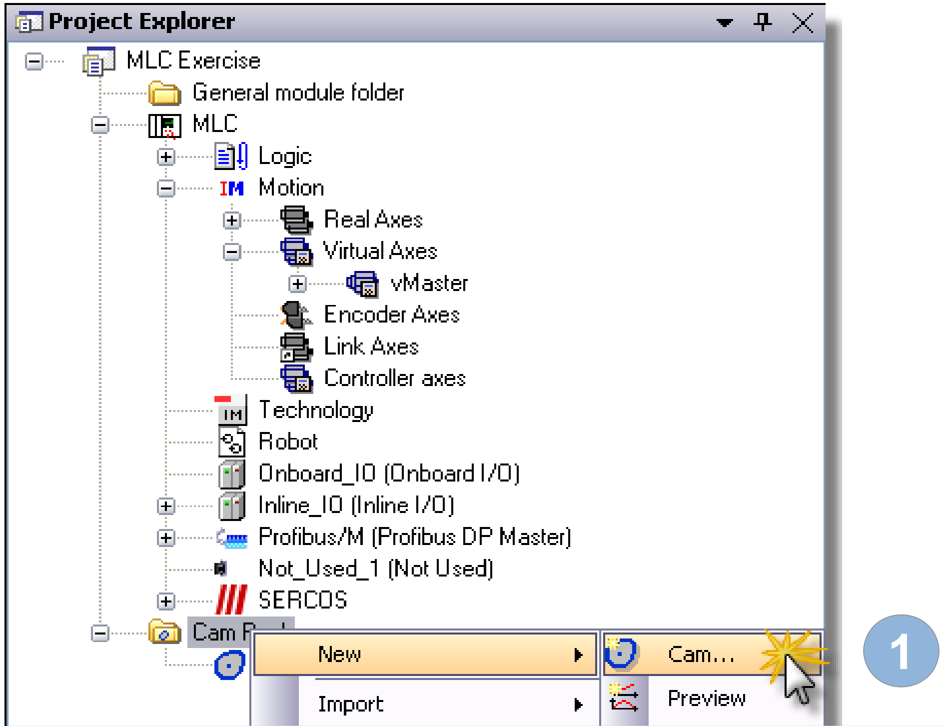
Exercise

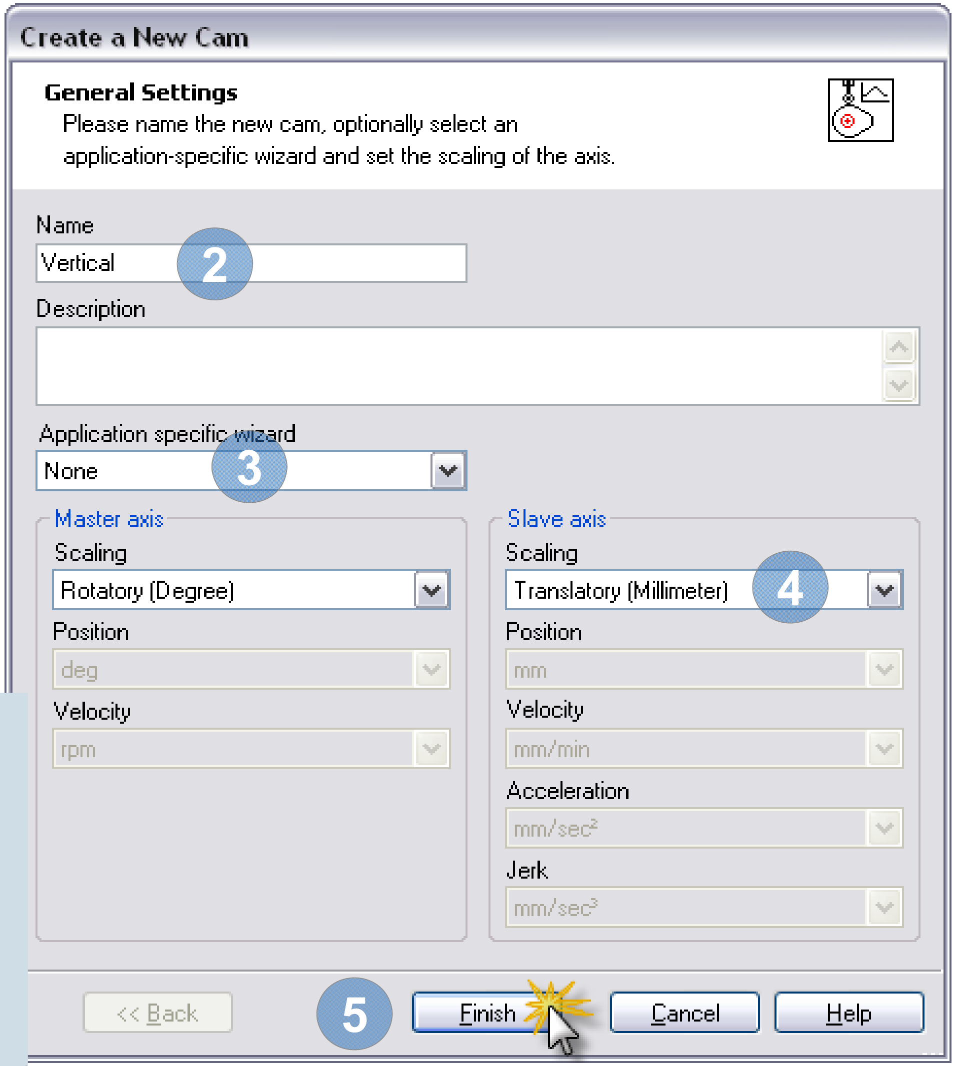


Rexroth Bosch Group

GAT – FlexProfile for vertical axis (2)

Exercise





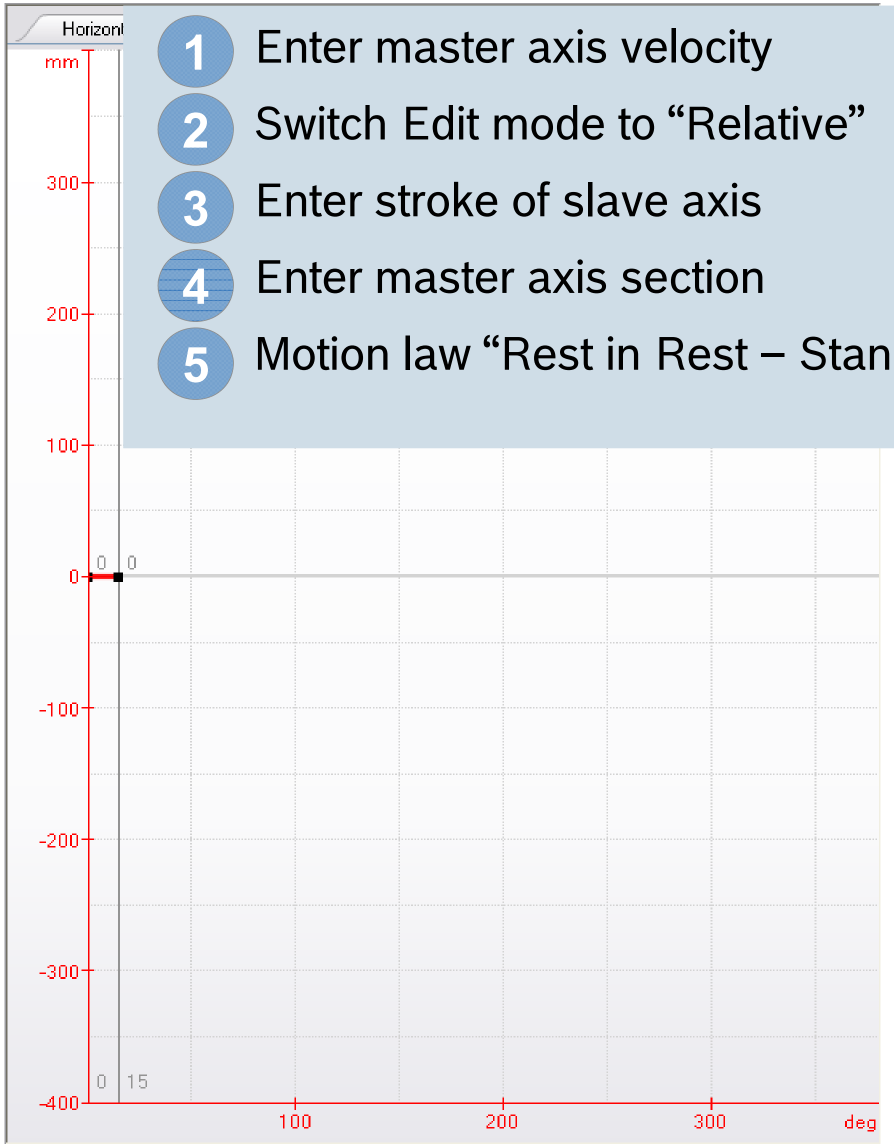
1. Create new Cam
2. Enter name “Vertical”
3. Application specific wizard: None
4. Translatory scaling (mm)
5. Press “Finish” button

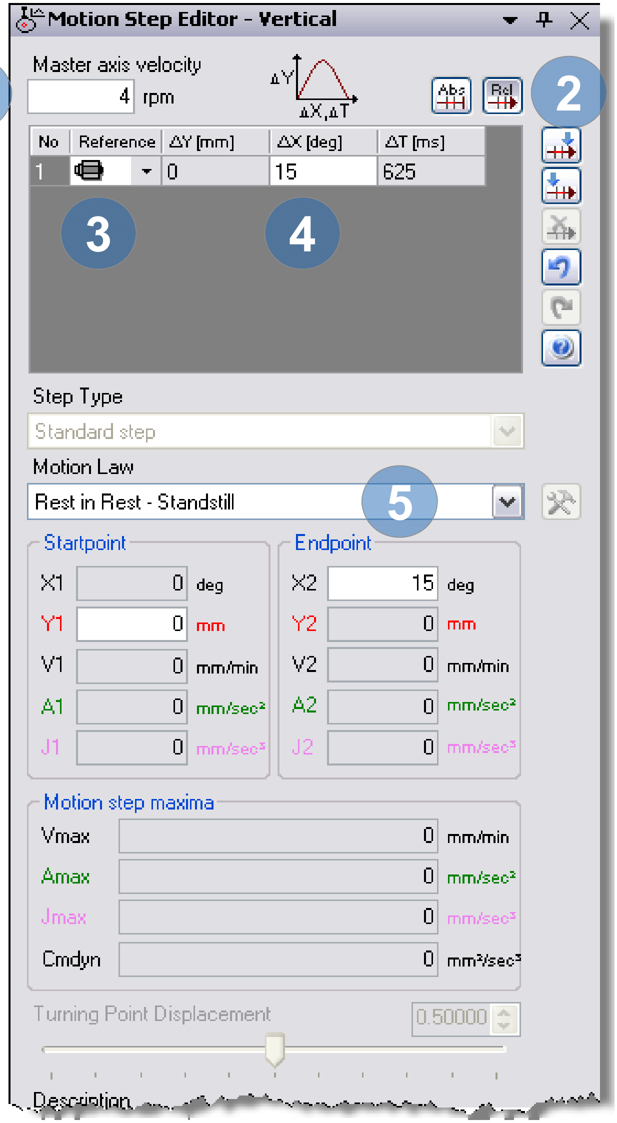
Rexroth
Bosch Group

GAT – FlexProfile for vertical axis (3)

Exercise

1. Enter master axis velocity
2. Switch Edit mode to “Relative”
3. Enter stroke of slave axis
4. Enter master axis section
5. Motion law “Rest in Rest – Standstill”



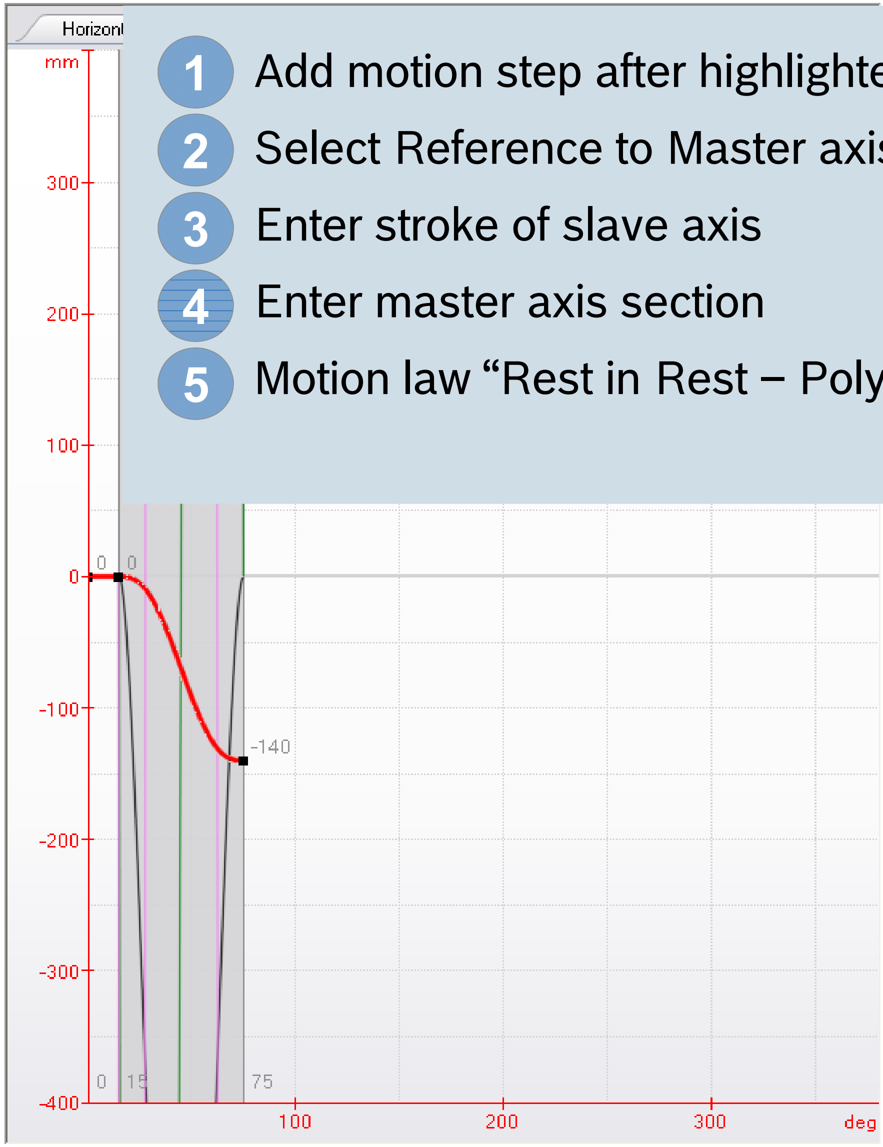


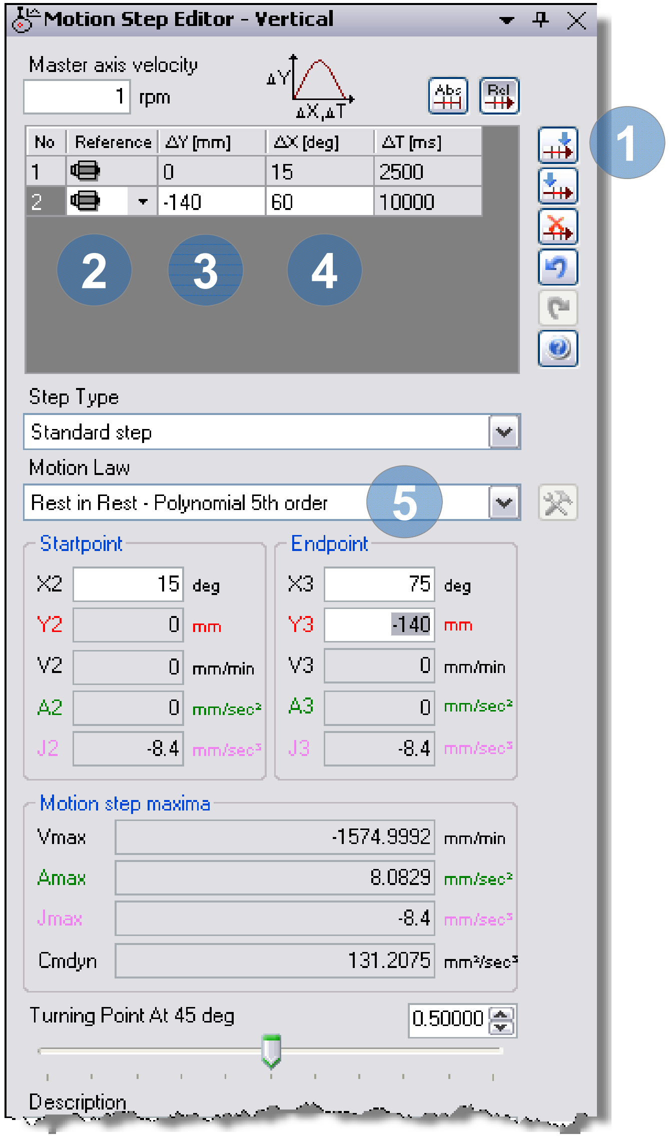
Rexroth Bosch Group

GAT – FlexProfile for vertical axis (4)

Exercise

1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter stroke of slave axis
4. Enter master axis section
5. Motion law “Rest in Rest – Poly5”



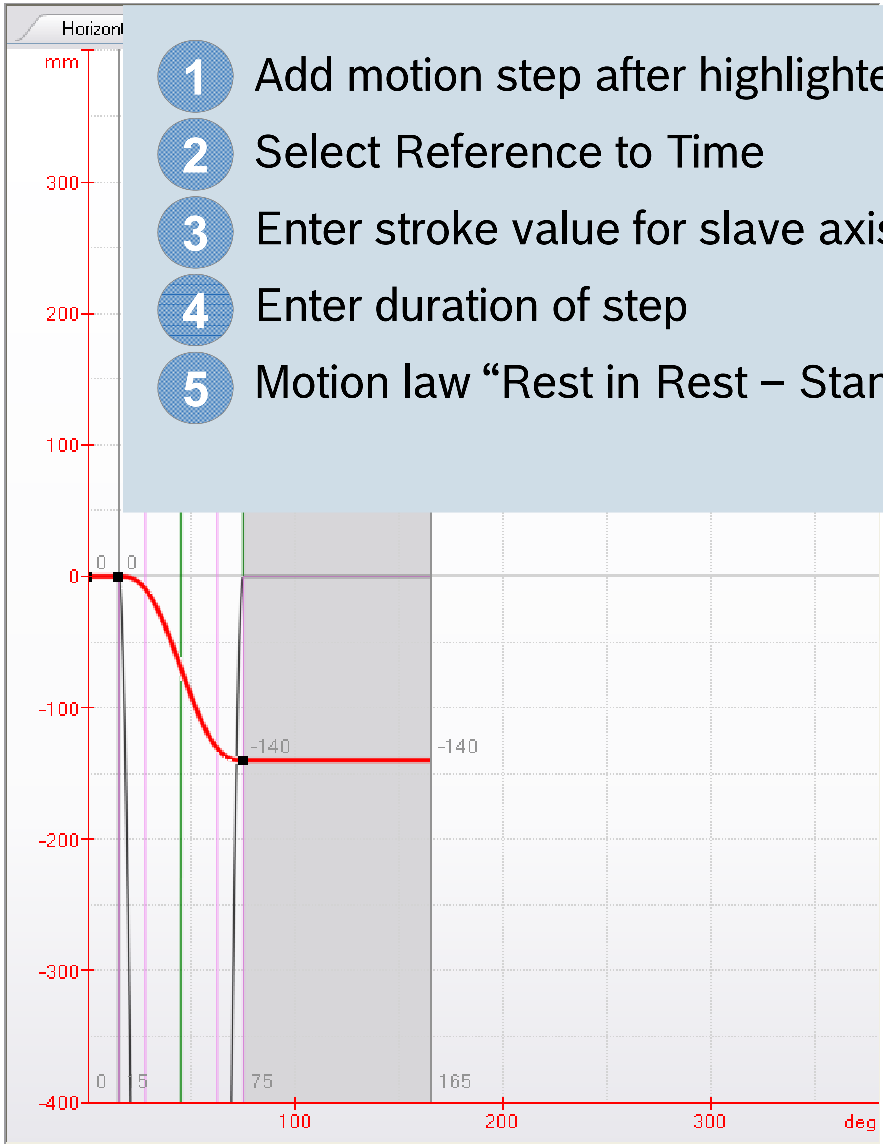


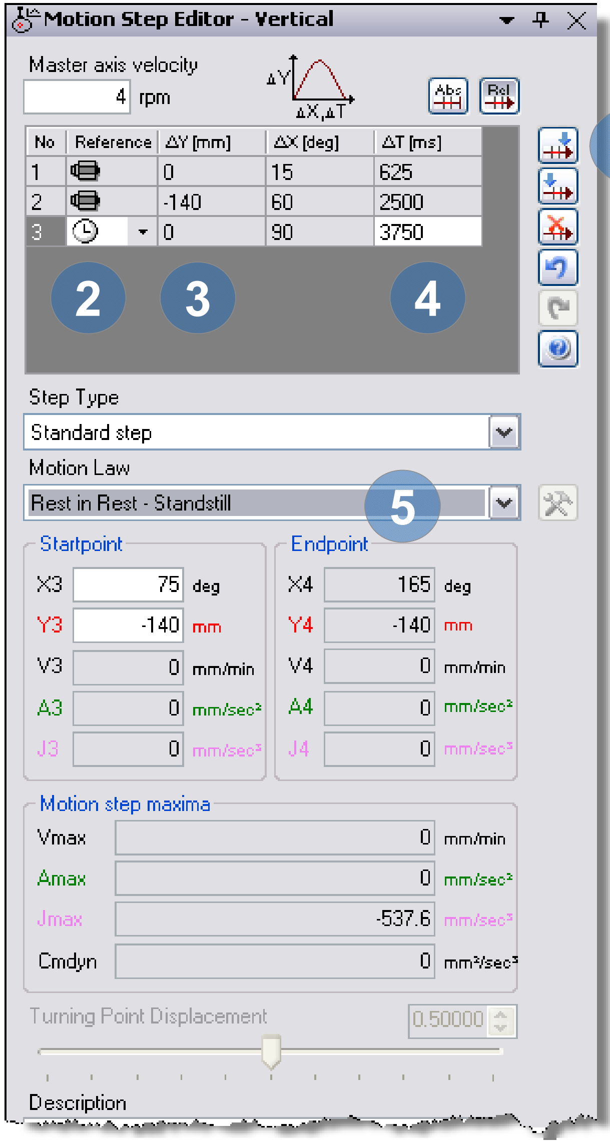
Rexroth Bosch Group

GAT – FlexProfile for vertical axis (5)

Exercise

1. Add motion step after highlighted step
2. Select Reference to Time
3. Enter stroke value for slave axis
4. Enter duration of step
5. Motion law “Rest in Rest – Standstill”





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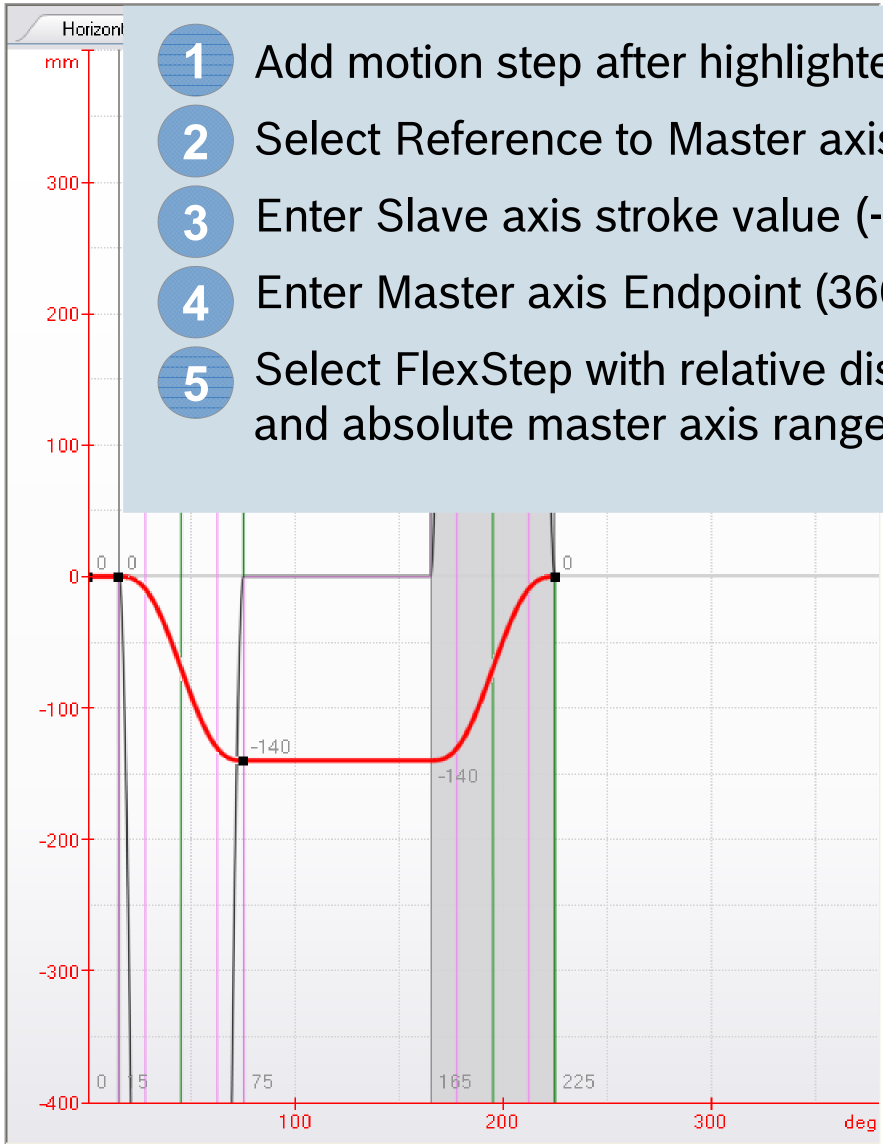
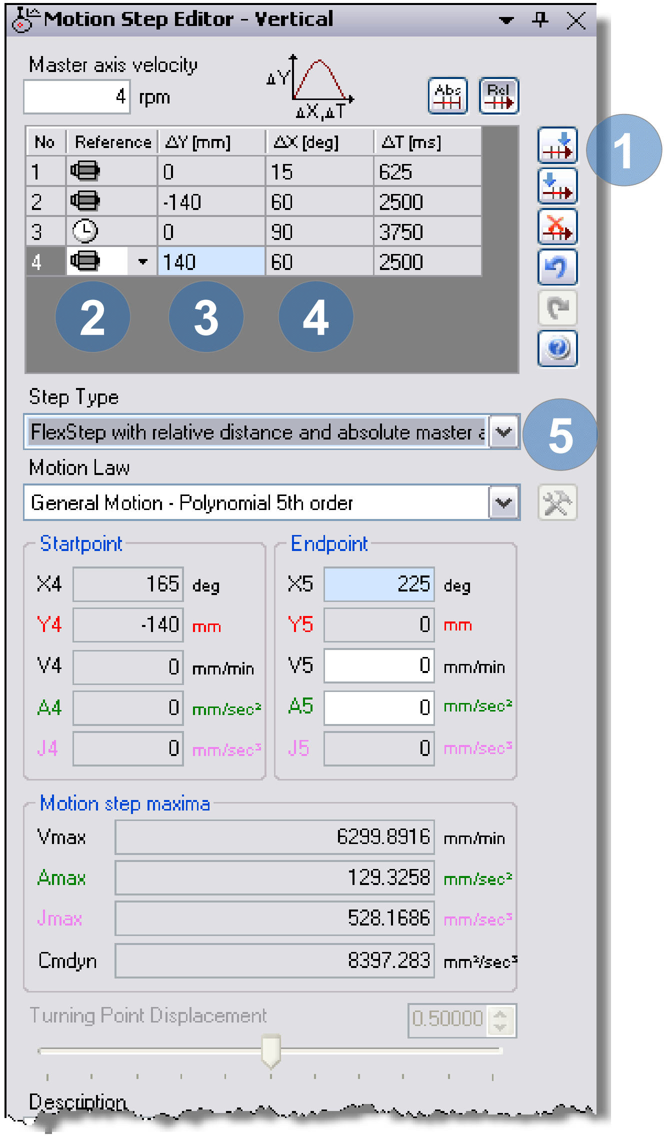
GAT – FlexProfile for vertical axis (6)

Exercise



1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (-415)
4. Enter Master axis Endpoint (360)
5. Select FlexStep with relative distance and absolute master axis range



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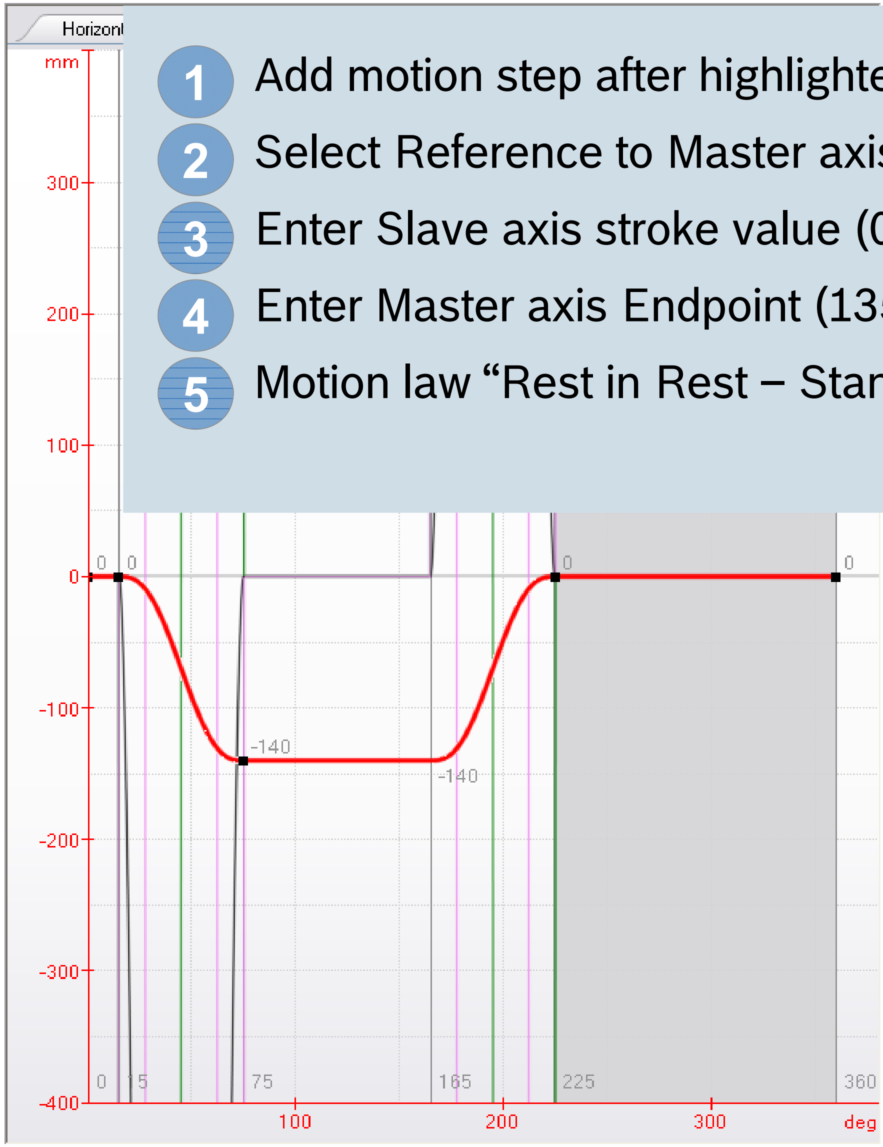
GAT – FlexProfile for vertical axis (7)

Exercise

1. Add motion step after highlighted step
2. Select Reference to Master axis position
3. Enter Slave axis stroke value (0)
4. Enter Master axis Endpoint (135)
5. Motion law “Rest in Rest – Standstill”



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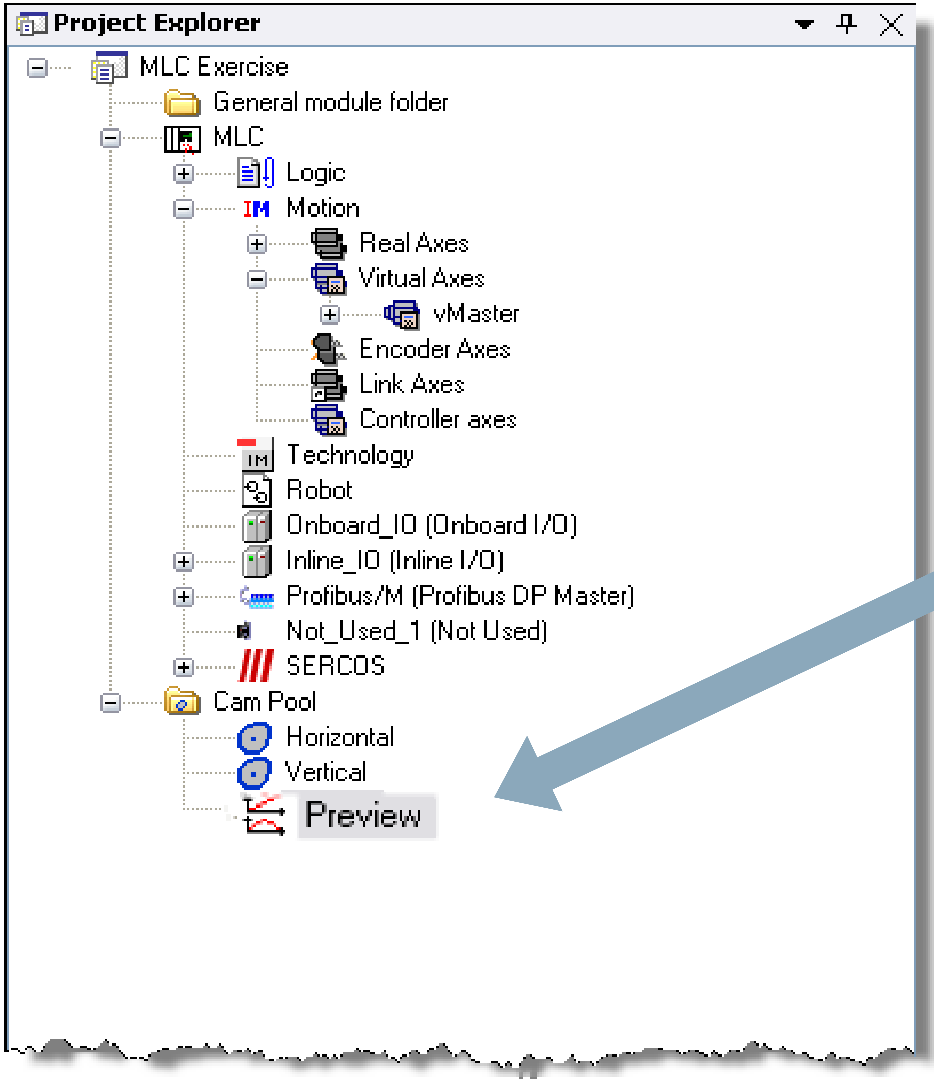
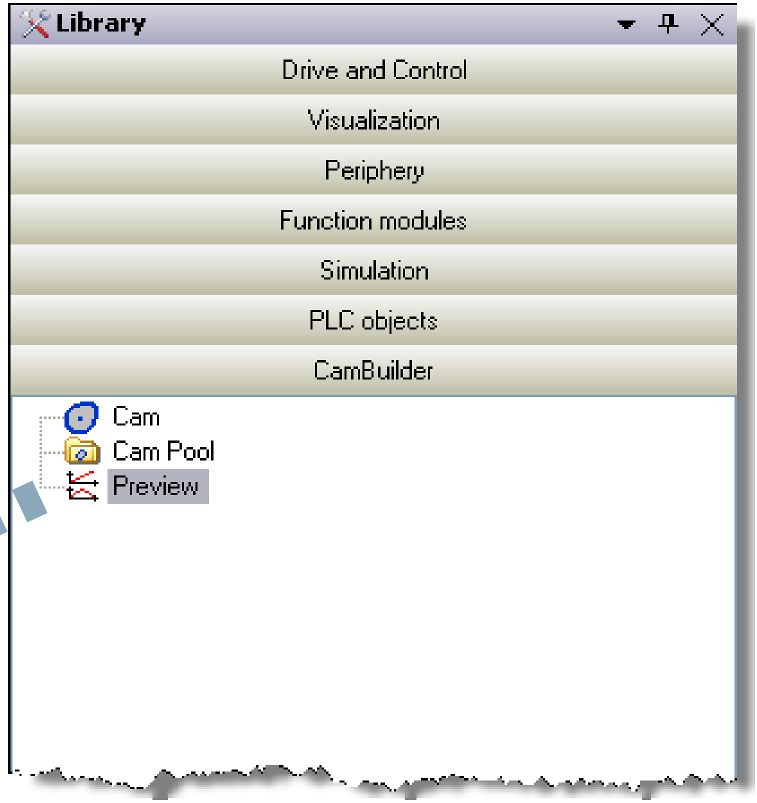
GAT – Create a Preview

Exercise





1. Open Group CamBuilder in Library
2. Select “Preview”
3. Add Preview to CamPool

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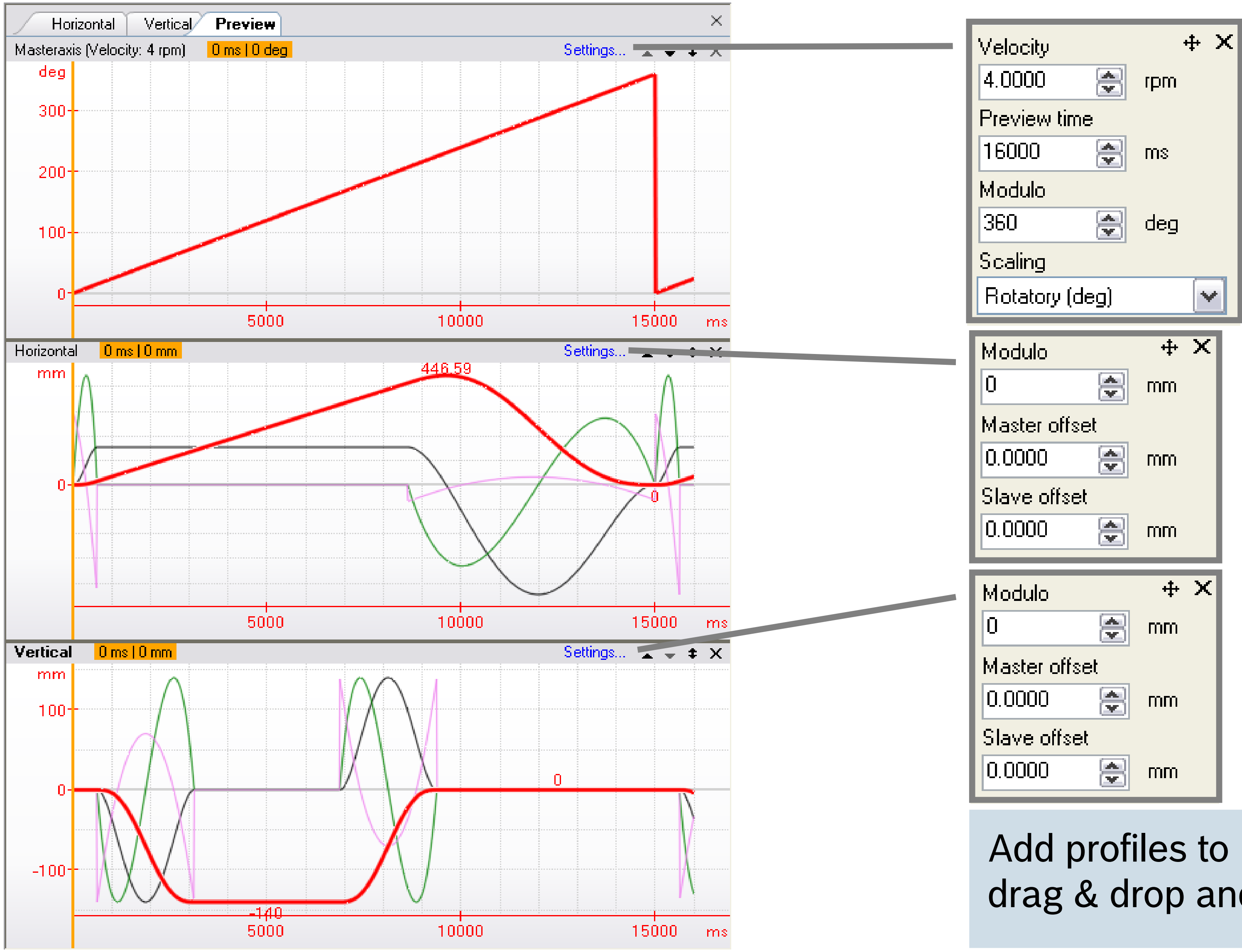
GAT – Create a Preview

Exercise



Add profiles to preview by drag & drop and modify settings

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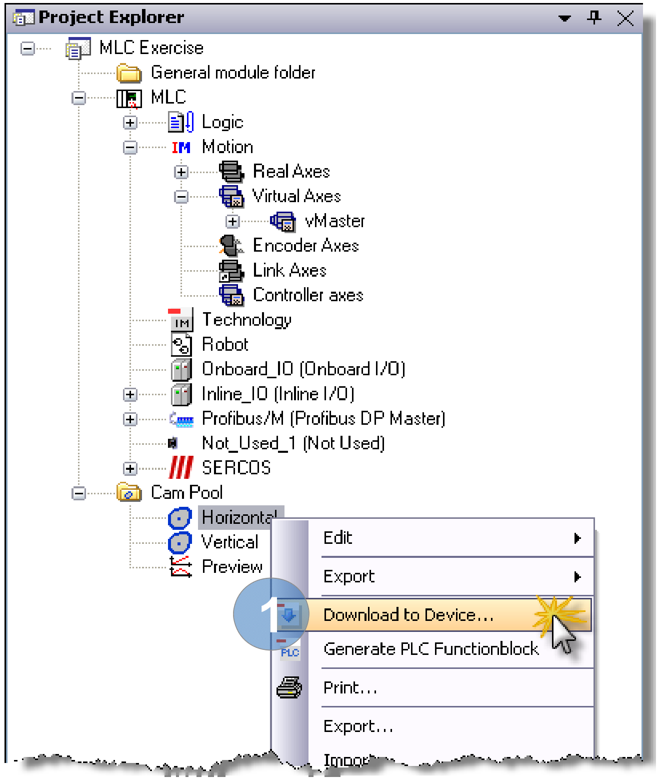
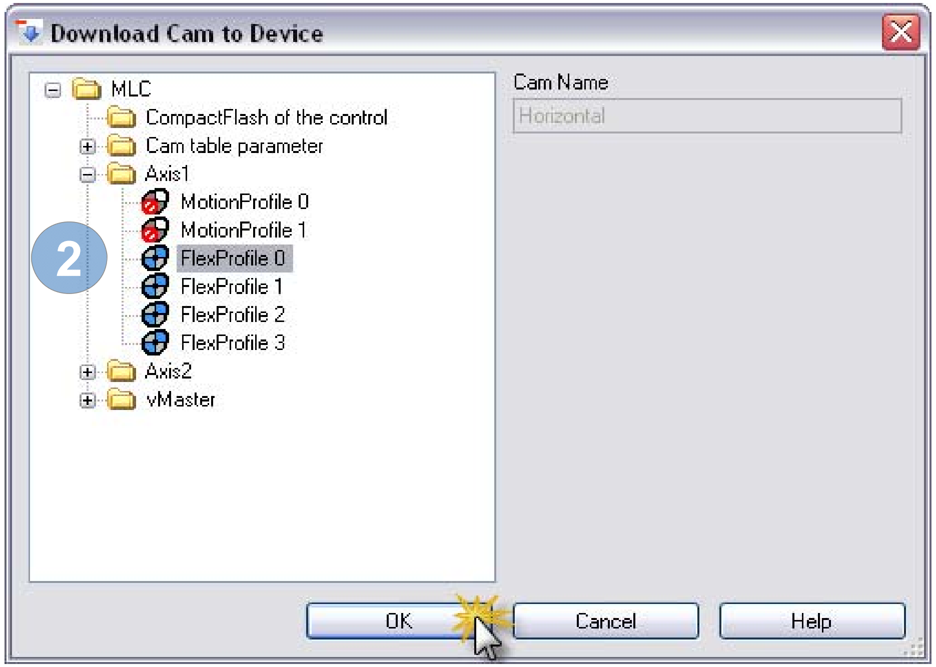
GAT – Download FlexProfile for 1st axis

Exercise





1 Select “Download to Device ...”

2 Select “FlexProfile 0” for 1st axis

3 After download to MLC an information is displayed

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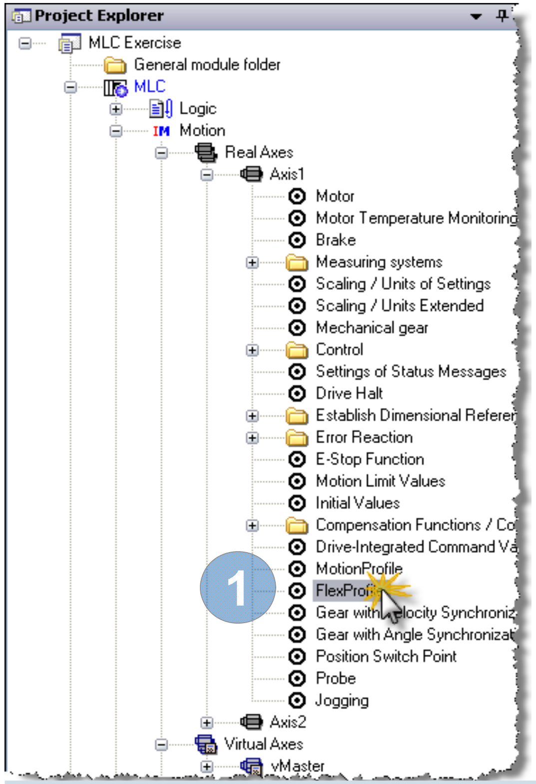
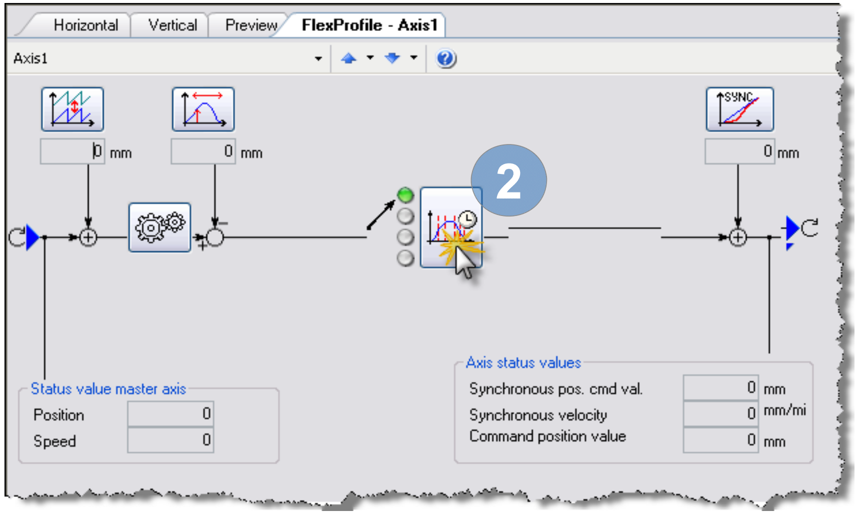
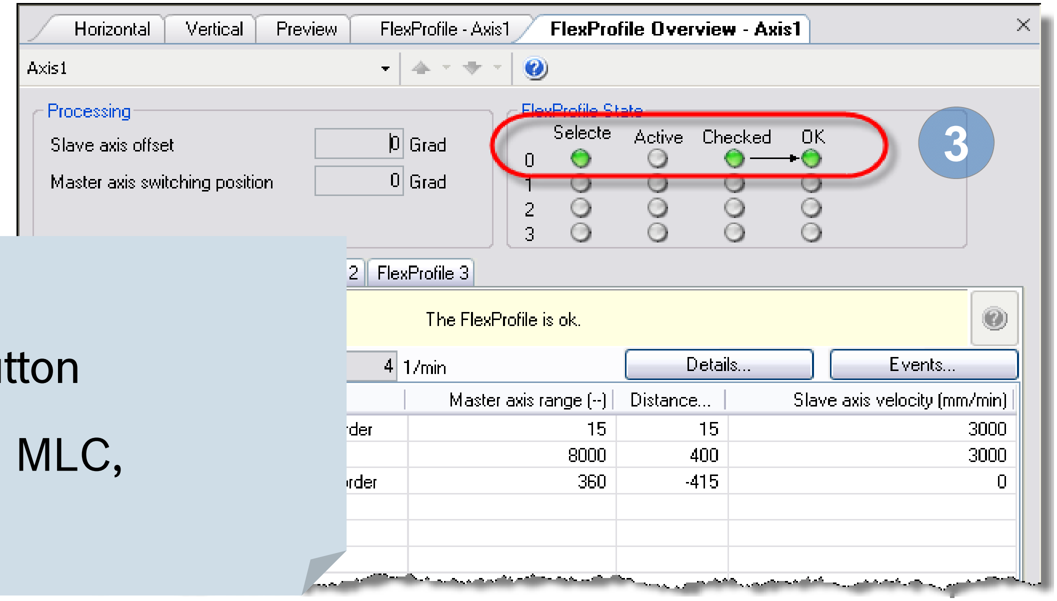
GAT – Download FlexProfile for 1st axis

Exercise







1. Double click on "FlexProfile"
2. Push "FlexProfile overview" button
3. The 1st FlexProfile is stored on MLC, has been checked and is OK!

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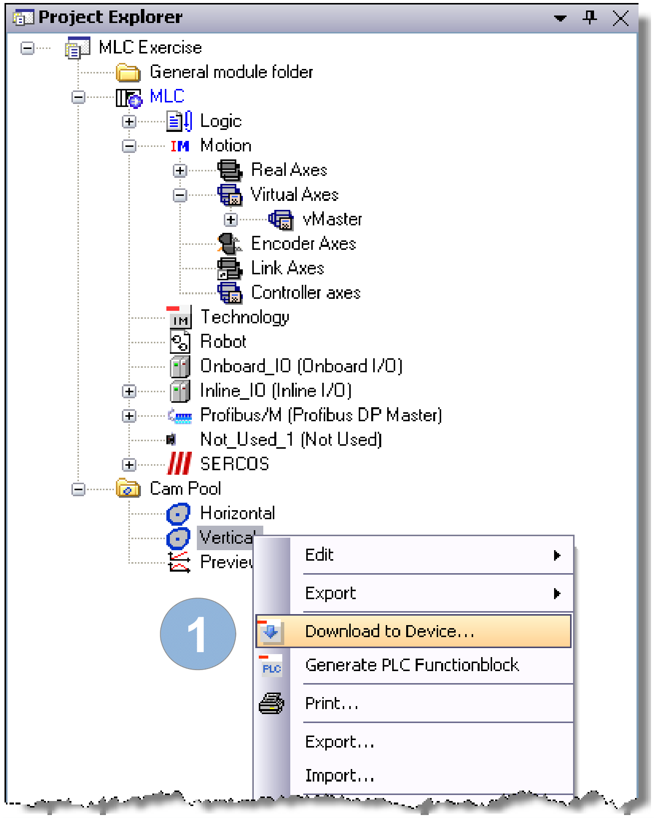
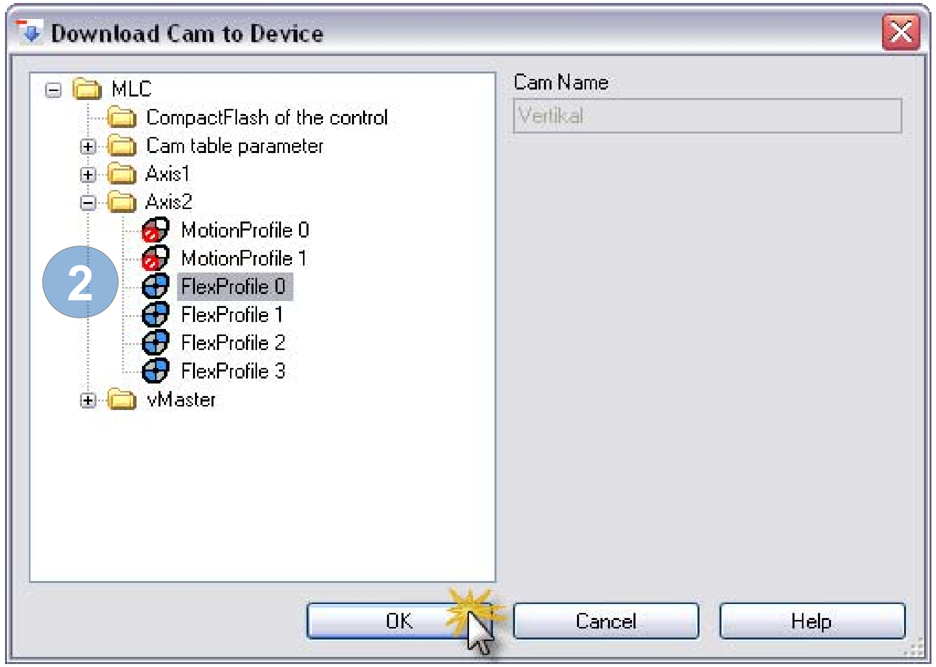
GAT – Download FlexProfile for 2nd axis

Exercise





1. Select “Download to Device ...”
2. Select “FlexProfile 0” for 2nd axis
3. After download to MLC an information is displayed

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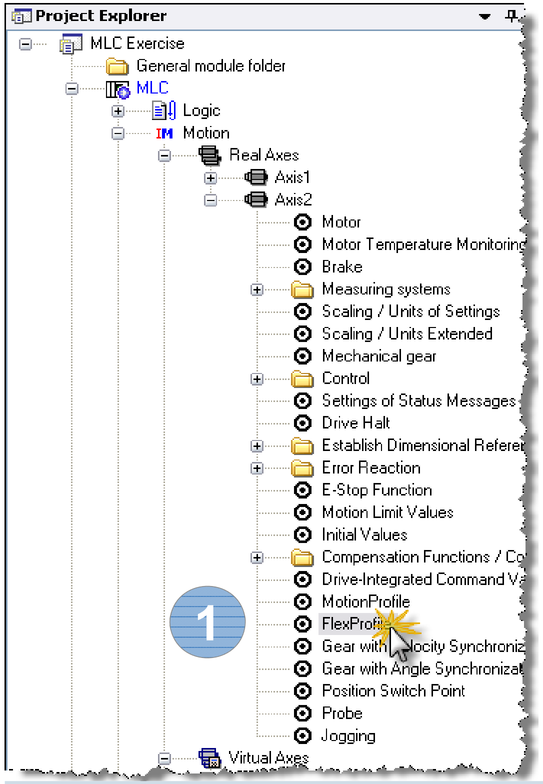
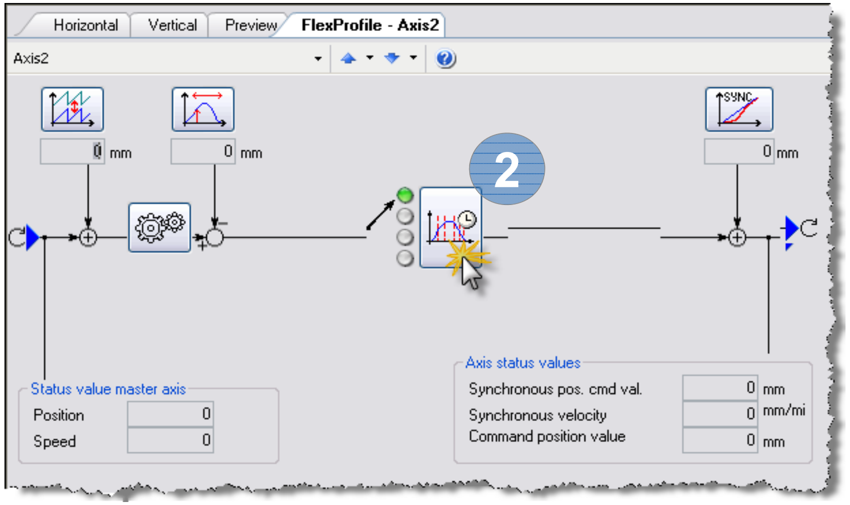
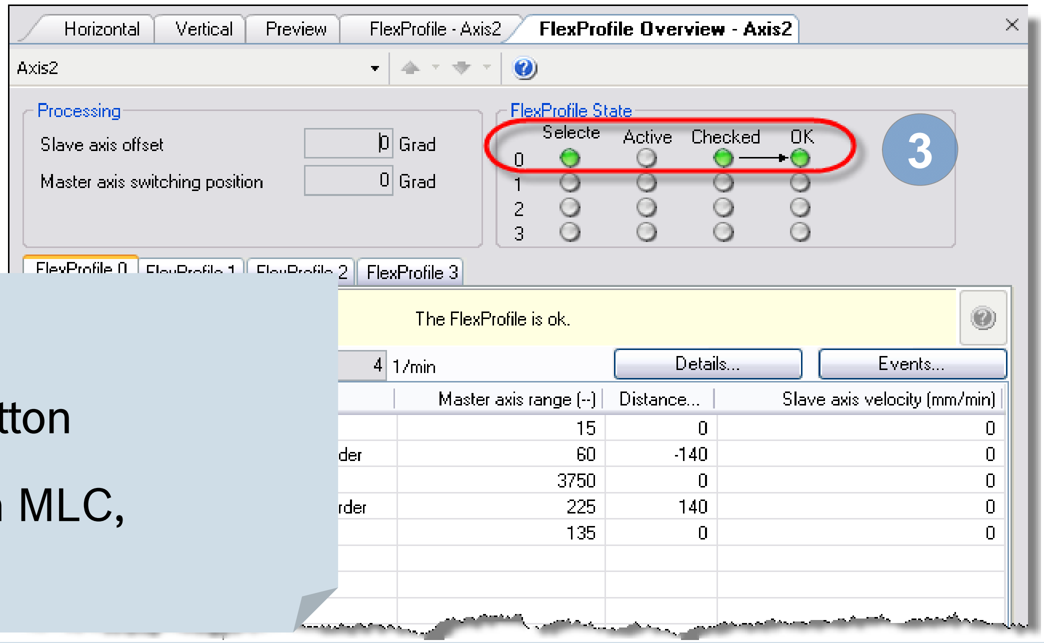
GAT – Download FlexProfile for 2nd axis

Exercise







1 Double click on "FlexProfile"

2 Push "FlexProfile overview" button

3 The 2nd FlexProfile is stored on MLC, has been checked and is OK!

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GAT – Implementing Automatic mode

Exercise

- Machine
  - Initially switch to Manual mode, not to Automatic mode
  - After error also switch to Manual mode

- Module “Master”
  - If Automatic mode is entered nothing has to be done
  - If Automatic mode is started start the virtual master with 4 rpm
  - If Automatic mode is stopped terminate machine cycle

  ☞ Axis setting for “vMaster”: Direction for MC_MoveAbsolute: positive

- Module “Filling Device”:
  - If Automatic mode is entered activate FlexProfiles
  - If Automatic mode is started nothing has to be done
  - If Manual Mode is entered end synchronous operation

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GAT – mOneTimeInit (Machine)

Exercise

```
Ctrl.OpMode_InitSuccessor := OPMODE_MANUAL;
Ctrl.OpMode_ErrorSuccessor := OPMODE_MANUAL;
```

GAT – actAutoStopped (Master)

Exercise

```
IF Ctrl.StateTransient THEN
  // ------------- transient (entry) state part --------------------

  IF Axes.vMaster.Status.Admin._OpModeAck = ModeVel THEN
     Axes.vMaster.Ctrl.Admin._OpMode.en := ModePosAbs;
     Axes.vMaster.Ctrl.PosMode.Position := 0;
     Axes.vMaster.Ctrl.PosMode.Velocity := 4;
  ELSIF Axes.vMaster.Status.Admin._OpModeAck = ModePosAbs AND
        Axes.vMaster.AxisData.Axis_InPosition THEN
     Axes.vMaster.Ctrl.Admin._OpMode.en := ModeAH;
  END_IF

  IF Axes.vMaster.Status.Admin._OpModeAck = ModeAH THEN
     Status.TransientLocked := FALSE;
  ELSE
     Status.TransientLocked := TRUE;
  END_IF
```


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GAT – actAutoProducing (Master)

Exercise

```
IF Ctrl.StateTransient THEN
  // ------------- transient (entry) state part --------------------

  Axes.vMaster.Ctrl.VelMode.Velocity := 4;
  Axes.vMaster.Ctrl.Admin._OpMode.en := ModeVel;
```


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GAT – actManualReady (Master)

Exercise

```
IF Ctrl.StateTransient THEN
  // --------------- transient (entry) state part --------------------

  Axes.vMaster.Ctrl.Admin._OpMode.en := ModeAH;
```


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GAT – actAutoStopped (FillingDevice)

Exercise

```
IF Ctrl.StateTransient THEN
  // --------------- transient (entry) state part ---------------------
  Axes.Horizontal.Ctrl.Admin._OpMode.en := ModeFlexProfile;
  Axes.Horizontal.Ctrl.SyncMode.InputRevolution := 1;
  Axes.Horizontal.Ctrl.SyncMode.OutputRevolution := 1;
  Axes.Horizontal.Ctrl.SyncMode.FineAdjust := 0;
  Axes.Horizontal.Ctrl.SyncMode.Profile.ExecutionMode := EXECUTE_CYCLIC;
  Axes.Horizontal.Ctrl.SyncMode.Profile.MasterOffset := 0;
  Axes.Horizontal.Ctrl.SyncMode.Profile.SlaveOffset := 0;
  Axes.Horizontal.Ctrl.SyncMode.Profile.ProfileEntry :=
   SLAVE_ORIGIN_MASTER_ORIGIN;
  Axes.Horizontal.Ctrl.SyncMode.Profile.RelativePositioning := FALSE;
  Axes.Horizontal.Ctrl.SyncMode.Profile.SyncType :=
   SYNC_RAMPIN_SHORTEST_WAY;
  Axes.Horizontal.Ctrl.SyncMode.Profile.SwitchingPosition := 0;
  Axes.Horizontal.Ctrl.SyncMode.Profile.UseSwitchingPosition := FALSE;
  Axes.Horizontal.Ctrl.SyncMode.Profile.SetSelection := 0;
  Axes.Horizontal.Ctrl.SyncMode.SyncVelocity := 100;
  Axes.Horizontal.Ctrl.SyncMode.SyncAcceleration := 1000;
  Axes.Horizontal.Ctrl.SyncMode.Master := Axes.vMaster.AxisRef;
  …
```


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GAT – actAutoStopped (FillingDevice)

Exercise

```
Axes.Vertical.Ctrl.Admin._OpMode.en := ModeFlexProfile;
Axes.Vertical.Ctrl.SyncMode.InputRevolution := 1;
Axes.Vertical.Ctrl.SyncMode.OutputRevolution := 1;
Axes.Vertical.Ctrl.SyncMode.FineAdjust := 0;
Axes.Vertical.Ctrl.SyncMode.Profile.ExecutionMode := EXECUTE_CYCLIC;
Axes.Vertical.Ctrl.SyncMode.Profile.MasterOffset := 0;
Axes.Vertical.Ctrl.SyncMode.Profile.SlaveOffset := 0;
Axes.Vertical.Ctrl.SyncMode.Profile.ProfileEntry := SLAVE_ORIGIN_MASTER_ORIGIN;
Axes.Vertical.Ctrl.SyncMode.Profile.RelativePositioning := FALSE;
Axes.Vertical.Ctrl.SyncMode.Profile.SyncType := SYNC_RAMPIN_SHORTEST_WAY;
Axes.Vertical.Ctrl.SyncMode.Profile.SwitchingPosition := 0;
Axes.Vertical.Ctrl.SyncMode.Profile.UseSwitchingPosition := FALSE;
Axes.Vertical.Ctrl.SyncMode.Profile.SetSelection := 0;
Axes.Vertical.Ctrl.SyncMode.SyncVelocity := 100;
Axes.Vertical.Ctrl.SyncMode.SyncAcceleration := 1000;
Axes.Vertical.Ctrl.SyncMode.Master := Axes.vMaster.AxisRef;
Status.TransientLocked := NOT Axes.Horizontal.AxisData.Axis_InSynchron OR
   NOT Axes.Vertical.AxisData.Axis_InSynchron;
```


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GAT – actManualReady (FillingDevice)

Exercise

```
IF Ctrl.StateTransient THEN
  // ------------------ transient (entry) state part -----------------
  Axes.Horizontal.Ctrl.Admin._OpMode.en := ModeAH;
  Axes.Vertical.Ctrl.Admin._OpMode.en := ModeAH;

  // --------- NOTE --------
  // It is possible to lock the transition to the quiescent state
  // part using the status element Status.TransientLocked
  // Set "Status.TransientLocked := TRUE" in case the transition to
  // the quiescent state part should be locked
  // Set "Status.TransientLocked := FALSE" in case the transition to
  // the quiescent state part is released (transient activities are
  // done)
```


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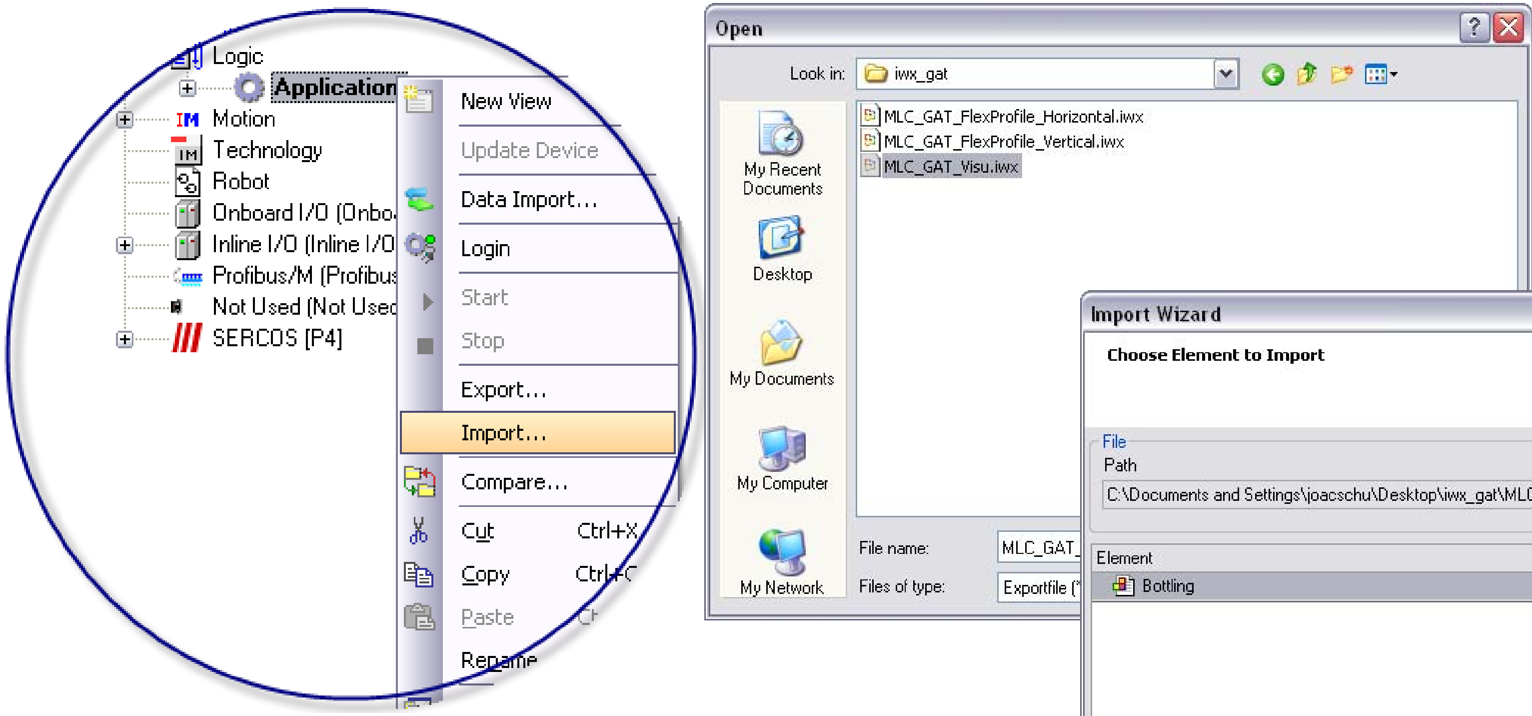
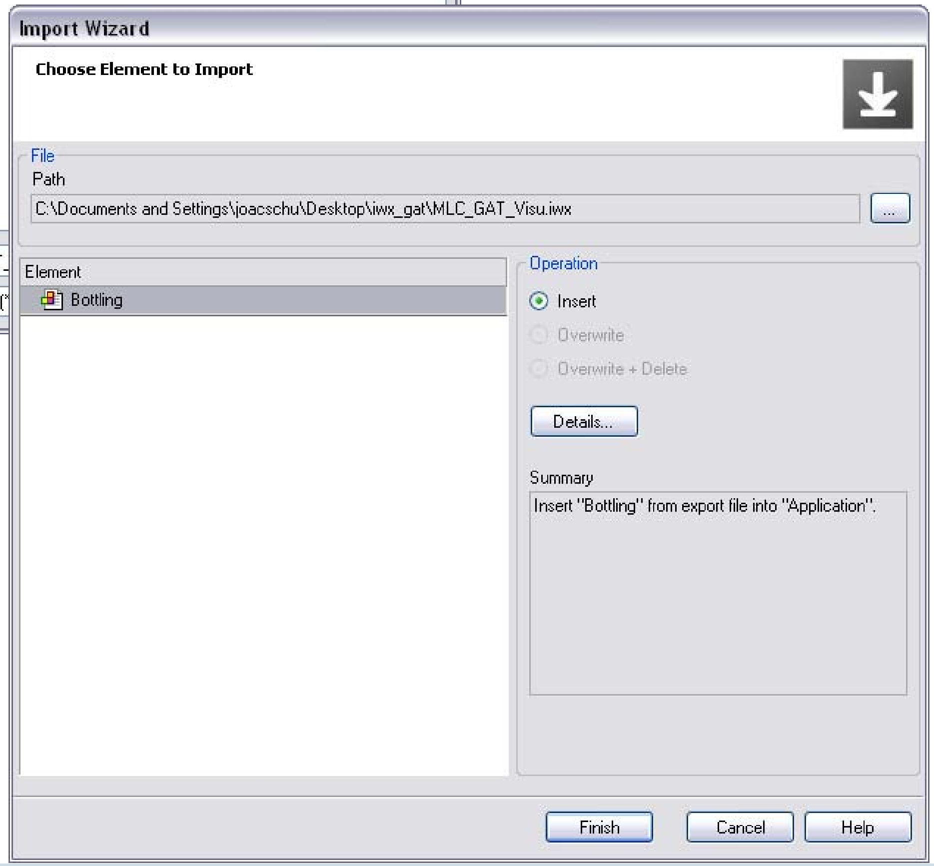
GAT – Import Visualization

Exercise





1. Select node “Application” and call “Import…”
2. Double click file “Visu_Bottling.iwx”
3. Push “Finish” button

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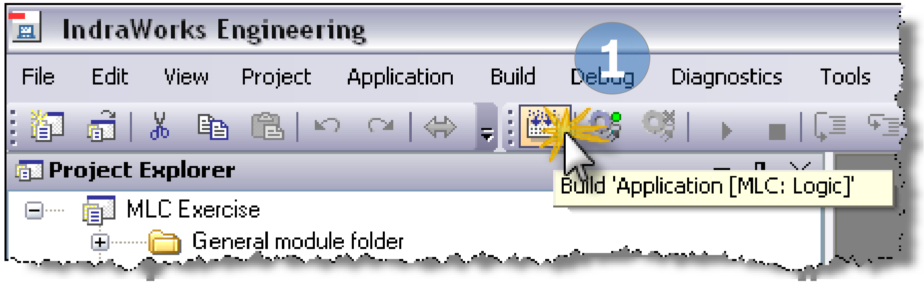
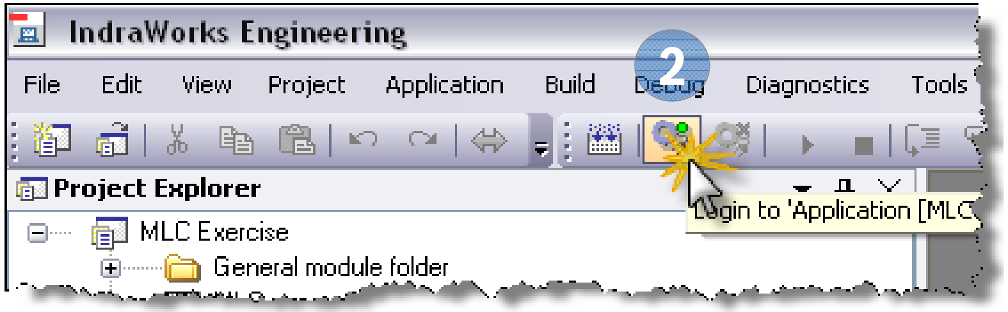
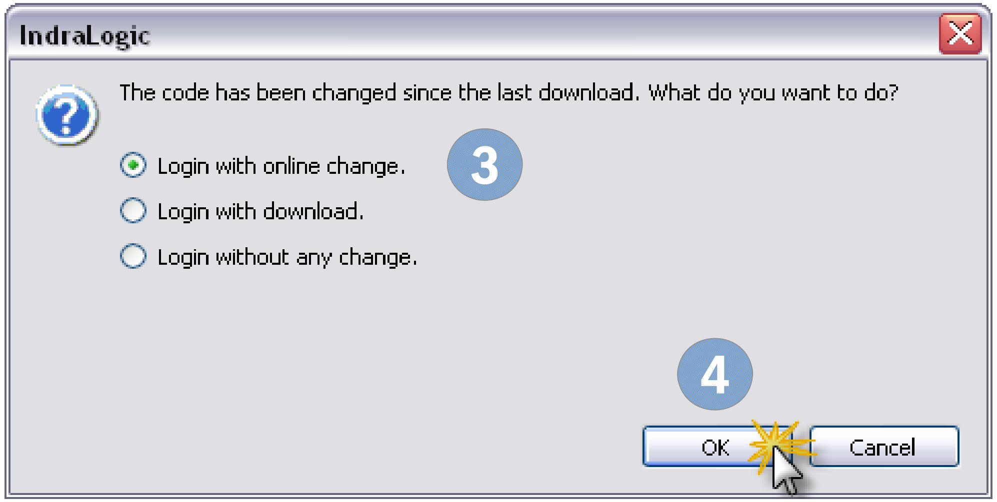
GAT – Build + Login with online change

Exercise







1. Push "Build" button
2. Push "Login" button
3. Select "Login with online change"
4. Push "OK" button

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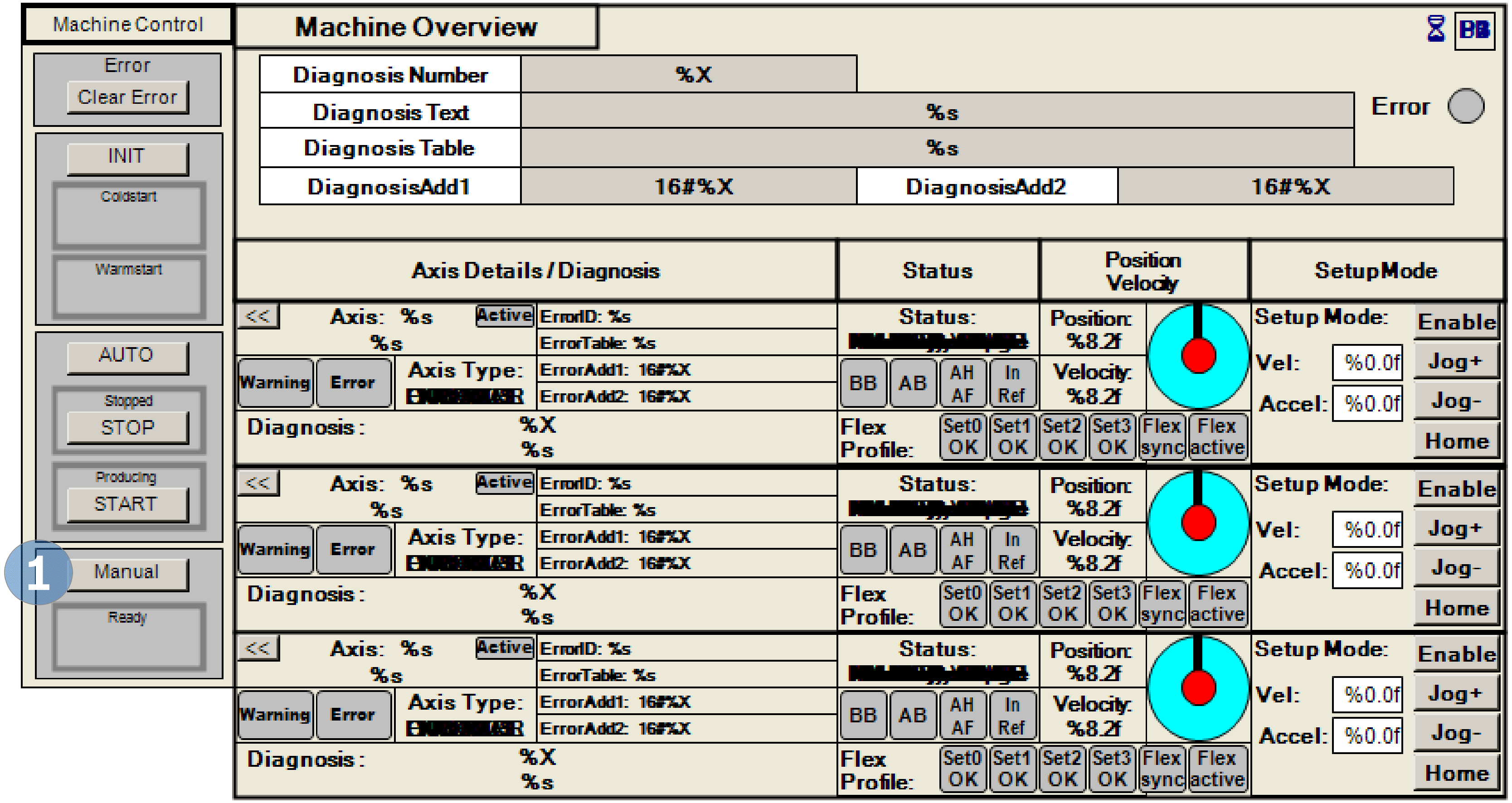
GAT – Home axes

Exercise



1 Open “MAIN_VISU” and select “Manual” mode

2 Move all axes to the zero position

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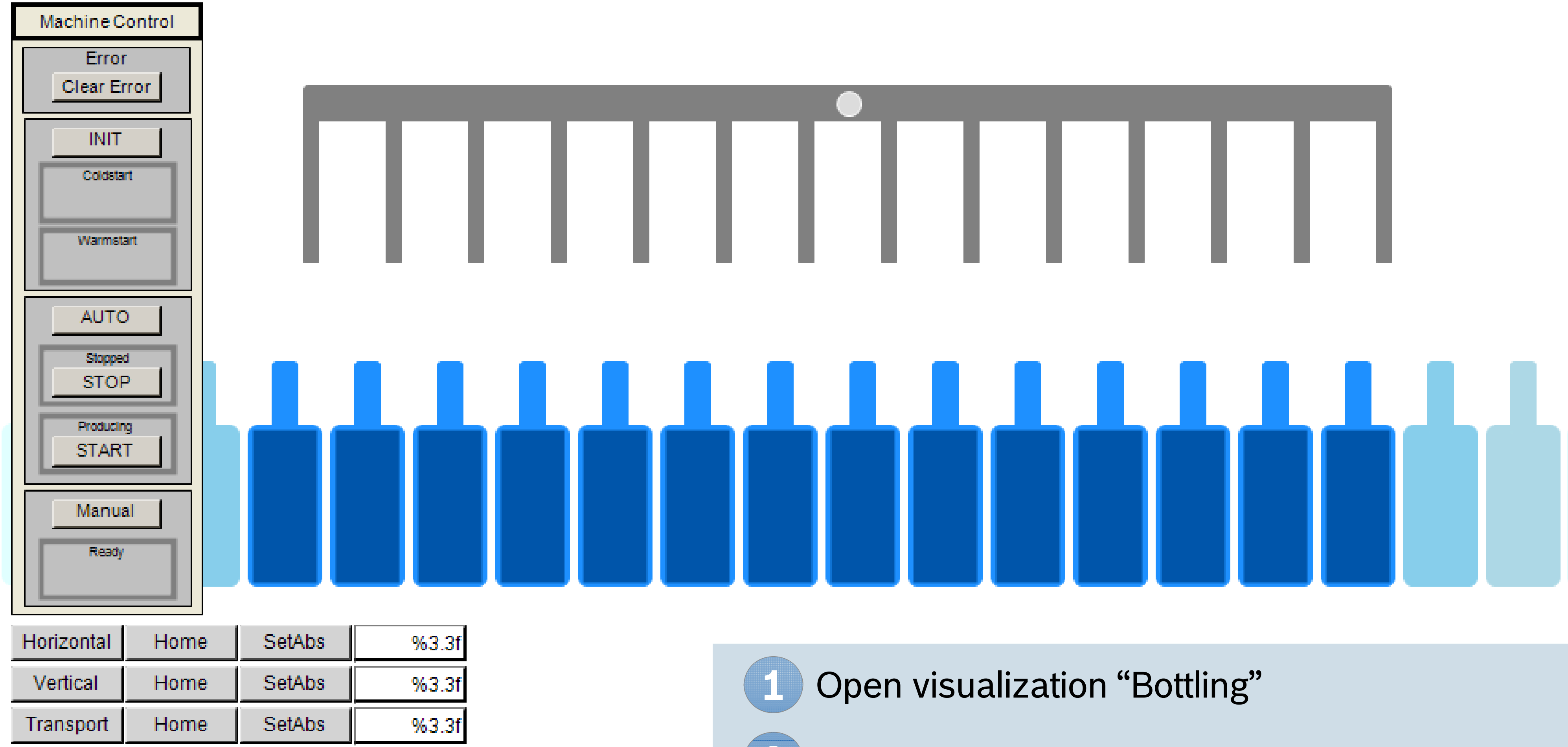
GAT – Start bottling



1. Open visualization "Bottling"
2. Select "Automatic" mode
3. Push "Start" button

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Agenda

- Introduction to IndraMotion MLC
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- Data backup and restore
- Task System
- Synchronized Motion with PLCopen
- Electronic CAMs: Point table – MotionProfile – FlexProfile
- CamBuilder
- AxisInterface and IMC
- Generic Application Template
- IMST
- Additional sources of information

Advanced Topics

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IMST – Documentation

- Complete Documentation of IndraMotion Service Tool
- ... available on the media directory
- or in print format
- in German and English language

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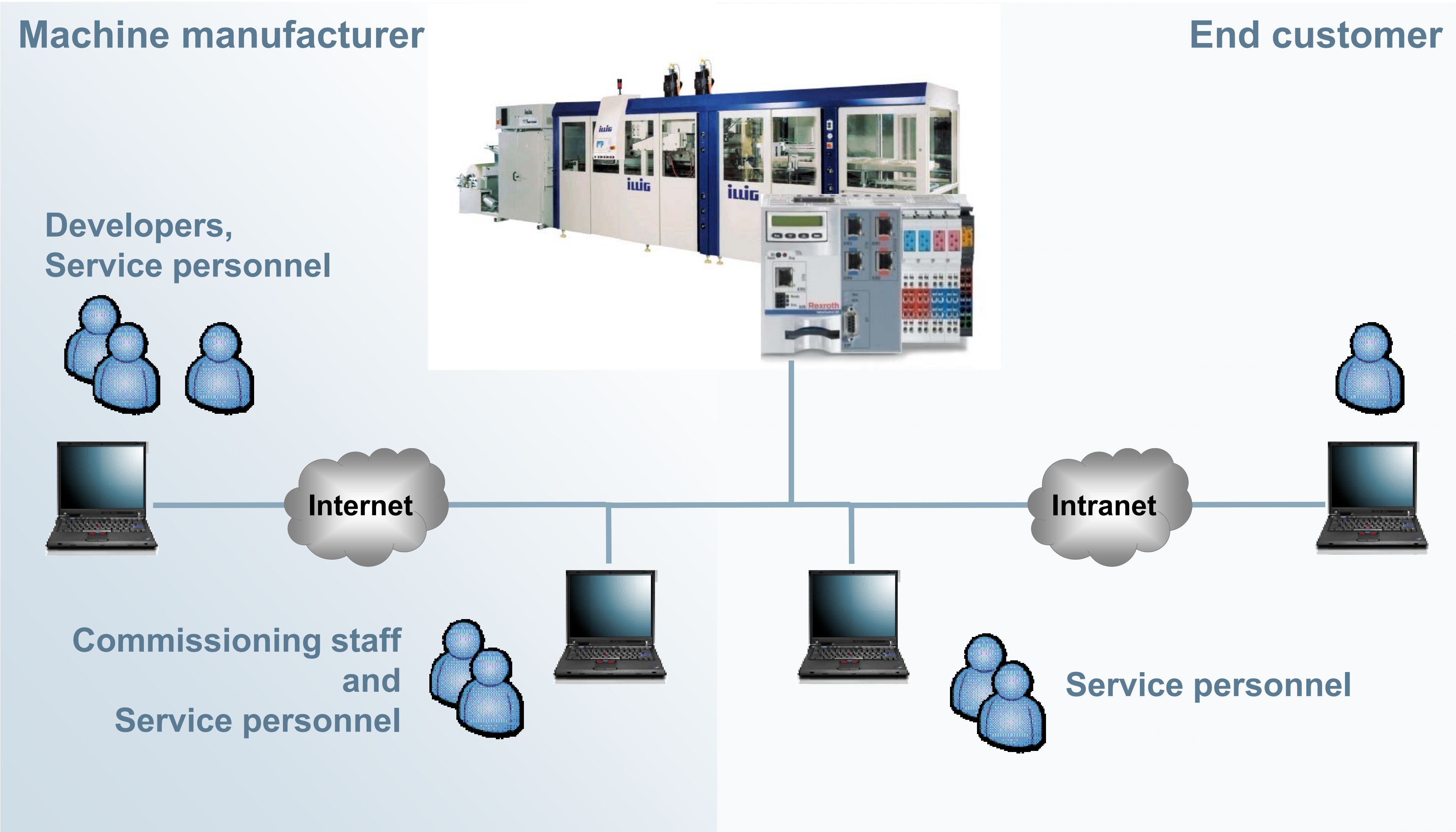
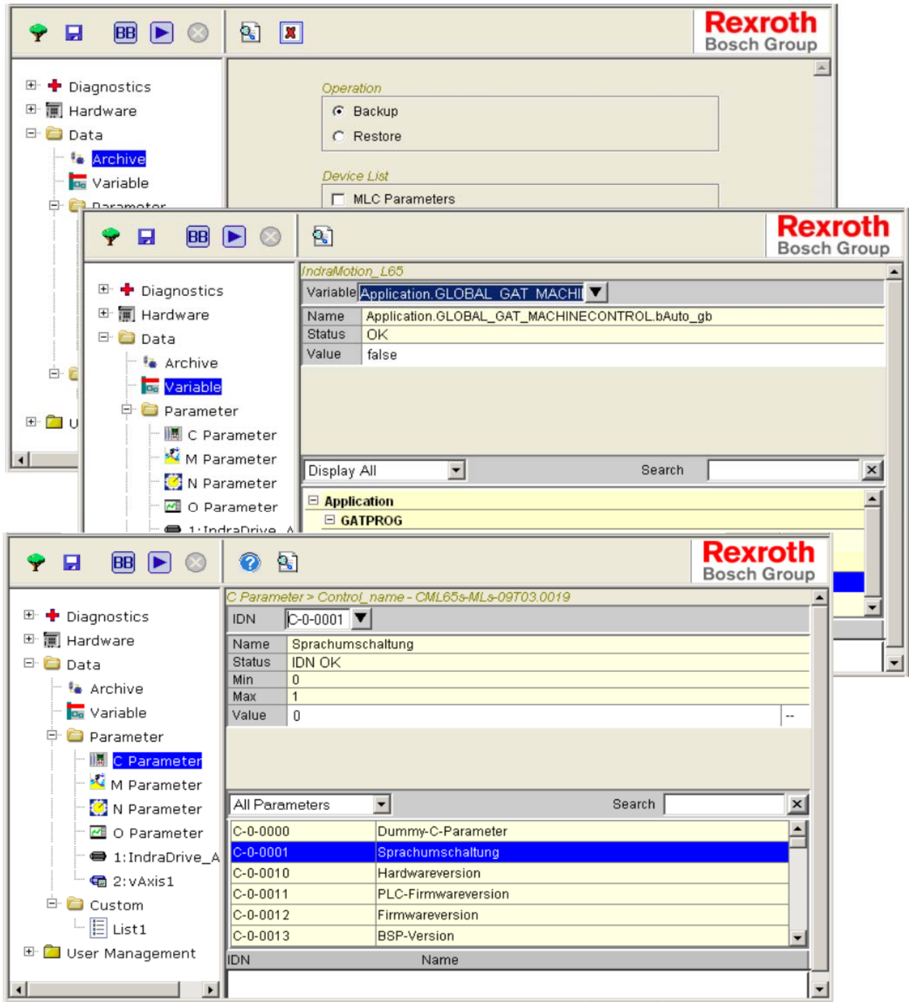
IMST – IndraMotion Service Tool

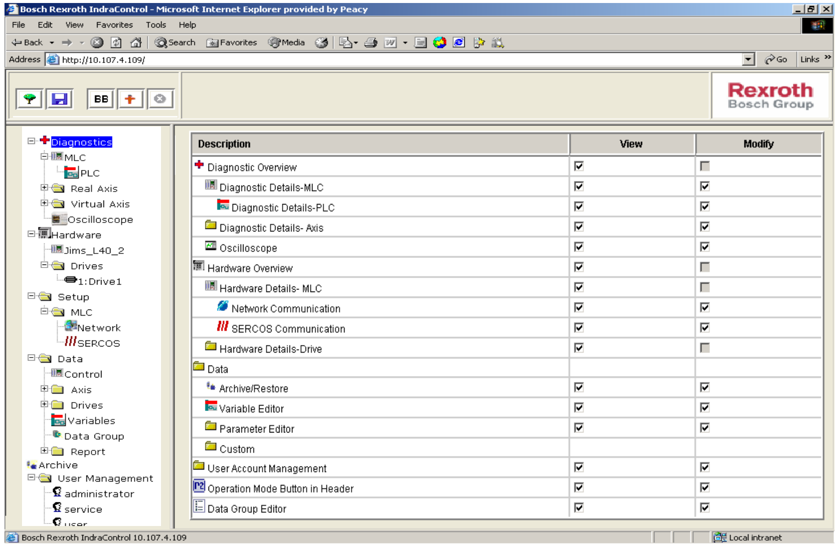


IMST – IndraMotion Service Tool

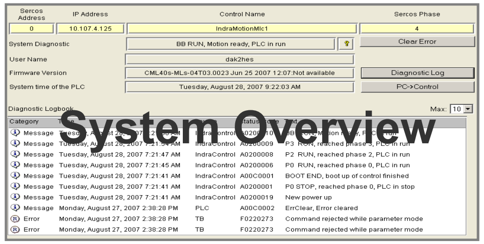
- Easy access using a browser to control and drive data without IndraWorks
- Integrated into runtime system of IndraMotion MLC
- Basic configuration and tool for simplified maintenance and commissioning
- Ease of use by navigation in a tree structure, toolbar and tooltips
- Integrated user management
  - Access privileges depending on current user level
  - Flexible definition of user groups
  - Individual adjustment of privileges

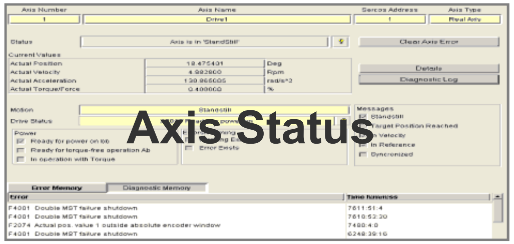


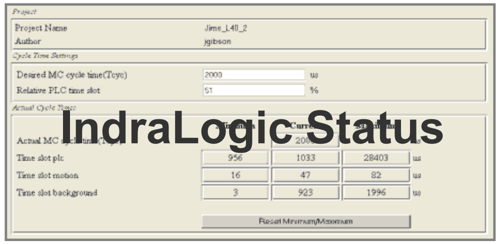
IMST – IndraMotion Service Tool

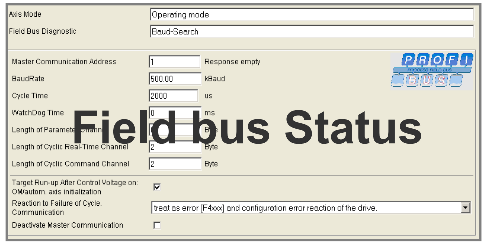


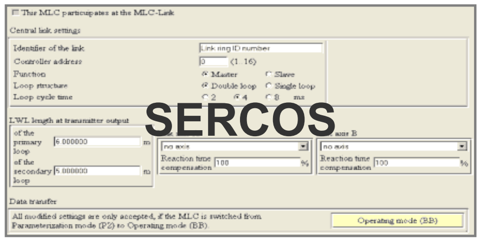
- Status and diagnosis of control hardware, drives and communication interfaces
- Parameter editor
- Access to PLC variables
- User-specific lists
- Electronic type plate (Hardware, Firmware, Serial number)
- Simple Oscilloscope operation

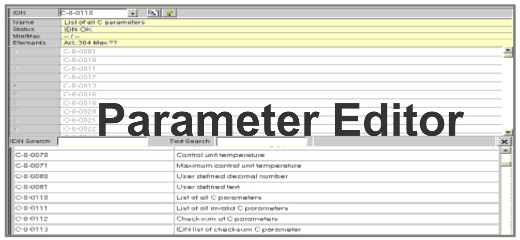


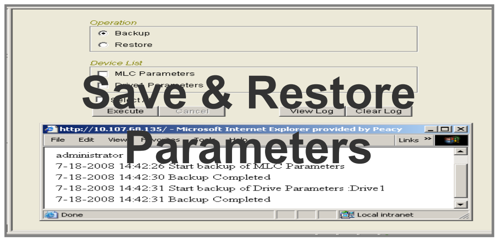


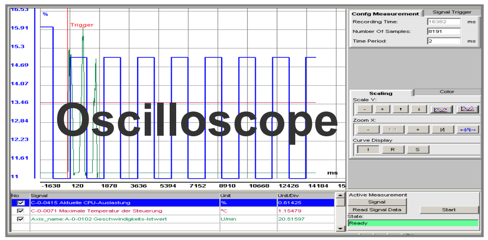












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Advanced Topics

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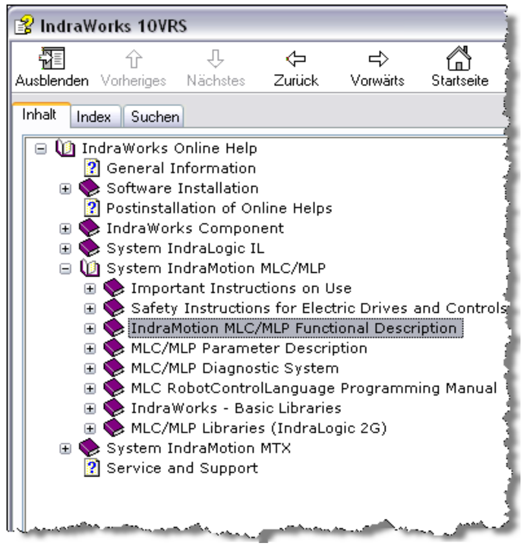
MLC Documentation

- All MLC manuals for released MLC versions are available in the Intranet on the media directory

www.boschrexroth.com/mediadirectory

- MLC manuals for the current version can be found on the installation DVD IndraWorks MLC

- The Online Help system provides the same information



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MLC Documentation – available documents

Title	Document	Edition / Language / Date
Rexroth IndraMotion MLC 12VRS Programming Device for RobotControl Edition 01 PDF 3,68 MB	DOK-IM*ML*-ROCOPHG*V12-AP Instruction Details	R911338802 DOK-IM*ML*-ROCOPHG*V... en-US 08.07.2013
Rexroth IndraMotion MLC 12VRS Winder Function Application PDF 3,12 MB	DOK-MLC***-TF*WIND*V12-AP Instruction Details	R911333870 DOK-MLC***-TF*WIND*V... en-US 22.07.2011
Rexroth IndraMotion MLC 12VRS System Overview PDF 27,00 MB	DOK-MLC***-SYSTEM**V12-PR Instruction Details	R911333860 DOK-MLC***-SYSTEM**V... en-US 05.10.2012
Rexroth IndraMotion MLC 12VRS RCL Programming Instruction PDF 4,75 MB	DOK-MLC***-RCL*PRO*V12-AP Instruction Details	R911333852 DOK-MLC***-RCL*PRO*V... en-US 16.02.2012
Rexroth IndraMotion MLC 12VRS Commissioning PDF 5,79 MB	DOK-MLC***-STARTUP*V12-CO Instruction Details	R911333858 DOK-MLC***-STARTUP*V... en-US 13.05.2011

- MLC Manuals
- ... in English language
- ... for Version 12

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MLC Documentation – available documents

Title	Document	Edition / Language / Date
Rexroth IndraMotion MLC 12VRS Functional Description PDF 12,70 MB	DOK-MLC***-FUNC****V12-AP Instruction Details	R911333848 DOK-MLC***-FUNC****V... en-US 16.10.2012
Rexroth IndraMotion MLC 12VRS Technology Libraries PDF 13,76 MB	DOK-MLC***-TF*LIB**V12-LI Libary Details	R911333868 DOK-MLC***-TF*LIB**V... en-US 31.08.2011
Rexroth IndraMotion MLC 12VRS First Steps PDF 3,81 MB	DOK-MLC***-F*STEP**V12-CO Instruction Details	R911333846 DOK-MLC***-F*STEP**V... en-US 13.05.2011
Rexroth IndraLogic XLC IndraMotion MLC 12VRS Automation Interface PDF 1,11 MB	DOK-XLCMLC-AUT*INT*V12-AP Instruction Details	R911334178 DOK-XLCMLC-AUT*INT*V... en-US 13.05.2011
Rexroth IndraLogic XLC IndraMotion MLC 12VRS Generic Application Template PDF 6,44 MB	DOK-XLCMLC-TF*GAT**V12-AP Instruction Details	R911334191 DOK-XLCMLC-TF*GAT**V... en-US 13.05.2011

- MLC Manuals
- … in English language
- … for Version 12

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MLC Documentation – available documents

Title	Document	Edition / Language / Date
Rexroth IndraLogic XLC IndraMotion MLC 12VRS IndraMotion Service Tool PDF 1,92 MB	DOK-XLCMLC-IMST****V12-RE Reference Book Details	R911334936 DOK-XLCMLC-IMST****V... en-US 13.05.2011
Rexroth IndraLogic XLC IndraMotion MLC 12VRS Project Conversion PDF 2,65 MB	DOK-XLCMLC-PROCONV*V12-AP Instruction Details	R911334187 DOK-XLCMLC-PROCONV*V... en-US 19.11.2012
Rexroth IndraLogic XLC IndraMotion MLC 12VRS PLCopen Libraries PDF 9,95 MB	DOK-XLCMLC-FUNLIB**V12-LI Libary Details	R911334182 DOK-XLCMLC-FUNLIB**V... en-US 23.11.2012
Rexroth IndraWorks 12VRS IndraLogic 2G PLC Programming System PDF 13,18 MB	DOK-IWORKS-IL2GPRO*V12-AP Instruction Details	R911334390 DOK-IWORKS-IL2GPRO*V... en-US 06.09.2011
Rexroth IndraLogic XLC IndraMotion MLC 12VRS Diagnostics PDF 3,40 MB	DOK-XLCMLC-DIAG****V12-RE Reference Book Details	R911334180 DOK-XLCMLC-DIAG****V... en-US 22.10.2012

- MLC Manuals
- ... in English language
- ... for Version 12

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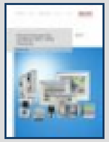
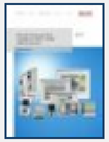
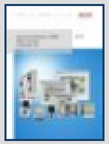
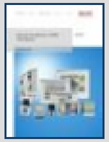
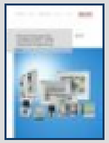
MLC Documentation – available documents

	Title	Document	Edition / Language / Date
	Rexroth IndraLogic XLC IndraMotion MLC 12VRS Parameters PDF 8,50 MB	DOK-XLCMLC-PARAM***V12-RE Reference Book Details	R911334176 DOK-XLCMLC-PARAM***V... en-US 19.10.2012
	Rexroth IndraLogic XLC IndraMotion MLC 12VRS HMI Connection PDF 5,91 MB	DOK-XLCMLC-HMI*****V12-AP Instruction Details	R911334184 DOK-XLCMLC-HMI*****V... en-US 17.11.2011
	Rexroth IndraWorks CamBuilder 12VRS PDF 2,07 MB	DOK-IWORKS-CAMBUIL*V12-AP Instruction Details	R911333842 DOK-IWORKS-CAMBUIL*V... en-US 13.05.2011
	Rexroth IndraWorks 12VRS Software Installation PDF 2,84 MB	DOK-IWORKS-SOFTINS*V12-CO Instruction Details	R911334396 DOK-IWORKS-SOFTINS*V... en-US 16.05.2011
	Rexroth IndraWorks 12VRS Basic Libraries IndraLogic 2G PDF 5,99 MB	DOK-IL*2G*-BASLIB**V12-LI Libary Details	R911333835 DOK-IL*2G*-BASLIB**V... en-US 13.05.2011

- MLC Manuals
- ... in English language
- ... for Version 12

MLC Documentation – available documents

	Title	Document	Edition / Language / Date
	Rexroth IndraWorks 12VRS Engineering PDF 6,37 MB	DOK-IWORKS-ENGINEE*V12-AP Instruction Details	R911334388 DOK-IWORKS-ENGINEE*V... en-US 26.10.2011
	Rexroth IndraWorks 12VRS Field Busses PDF 7,81 MB	DOK-IWORKS-FB******V12-AP Instruction Details	R911334394 DOK-IWORKS-FB******V... en-US 01.03.2012
	Rexroth IndraWorks HMI 12VRS PDF 5,40 MB	DOK-IWORKS-HMI*****V12-AP Instruction Details	R911334392 DOK-IWORKS-HMI*****V... en-US 08.07.2011
	Rexroth IndraLogic XLC IndraMotion MLC 12VRS Sequential Programming PDF 2,02 MB	DOK-XLCML*-SEQPROG*V12-LI Libary Details	R911334189 DOK-XLCML*-SEQPROG*V... en-US 19.11.2012

- MLC Manuals
- ... in English language
- ... for Version 12

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Training IndraMotion MLC

DC-IA/SVF15 – Joachim Schümann

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IndraDrive C/M – Control Panel

- The control panel has an 8-digit display and four keys located underneath it.
- The display shows operating states, drive address, command and error diagnoses, as well as present warnings, if any.
- If a safety operation mode is active, the safety technology code is displayed.
- Using the four keys, the commissioning engineer or service technician can opt to display extended diagnostic messages at the drive controller and to activate simple commands (in addition to master communication using the commissioning tool or NC control unit).

- The control panel is used for visualization and also allows some basic settings
- Firmware and parameter are not stored on the control panel

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IndraDrive C/M – MMC

- The MMC (Multi-Media-Card) is an optional storage medium
- Use is possible with control sections BASIC UNIVERSAL und ADVANCED
- Temporary or permanent use is possible
- Typically used for
  - Firmware update without PC
  - Drive exchange without PC
  - Duplicate axes (Firmware and parameters)
- Can be used in a PC by means of a standard card reader
  (access via FTP over Ethernet is also supported)

Only MMCs from Bosch Rexroth are supported (type code PFM02.1-****-FW)!
If 3rd party MMCs are used a correct functioning can not be guaranteed.

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IndraDrive C/M – MMC

- Directory structure:

Dateiname	Größe	Typ	Geändert
Dokumentation		File Folder	26.11.2004 16:11
Firmware		File Folder	26.11.2004 16:11
Parameter		File Folder	26.11.2004 16:11
Plc		File Folder	26.11.2004 16:11
User		File Folder	26.11.2004 16:11

- Contents:

- User
- Plc
- Parameter
  - parameter#1.... 327.680 12.9.2003 9:51
  - retain#1.rbf 16.384 25.9.2002 17:26
- Firmware
  - FWA-INDRV... 3.278.177 14.2.2006 6:37
- Dokumentation

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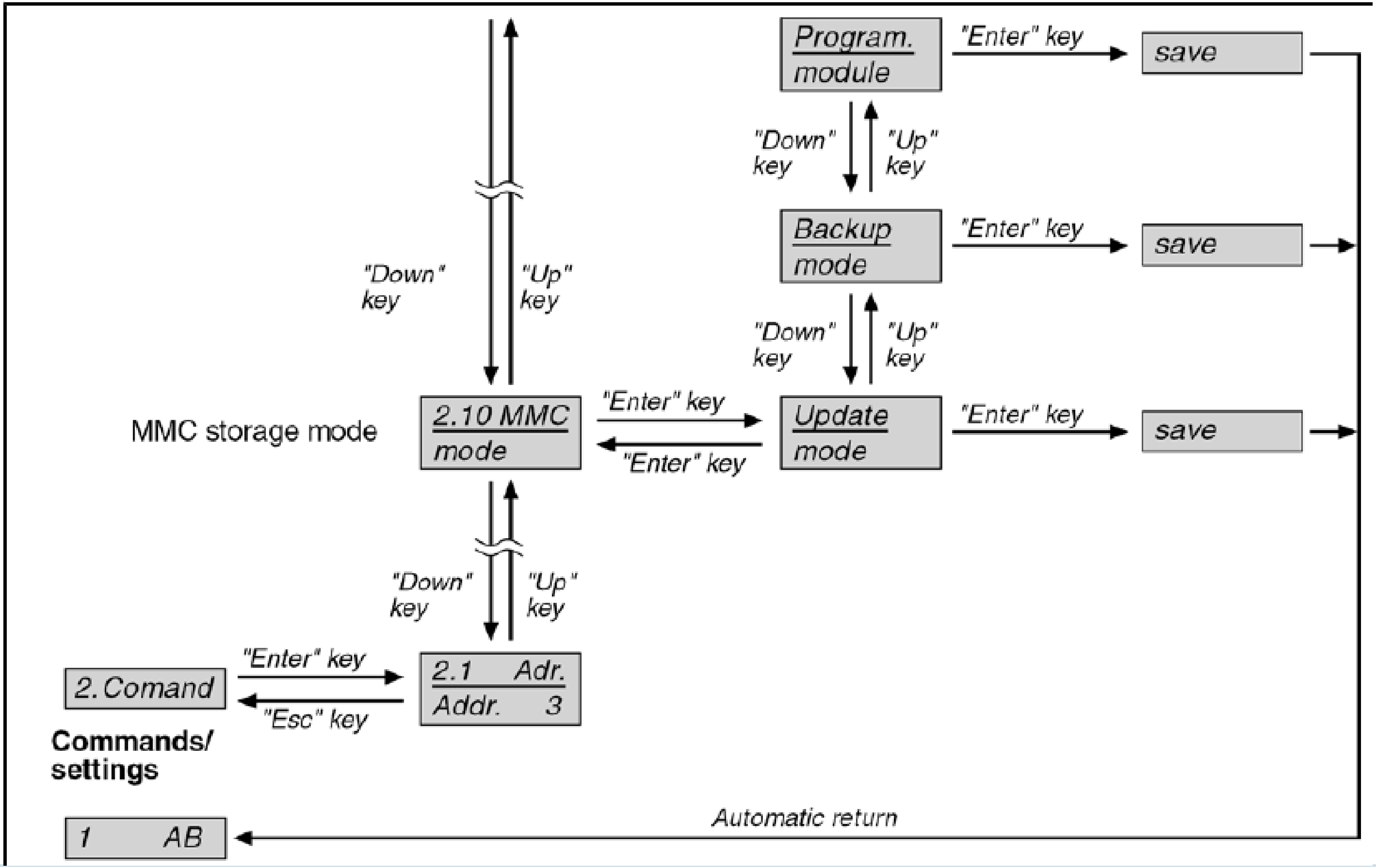
IndraDrive C/M – MMC

P-0-4070 Bit 1/0 Parameter storage configuration

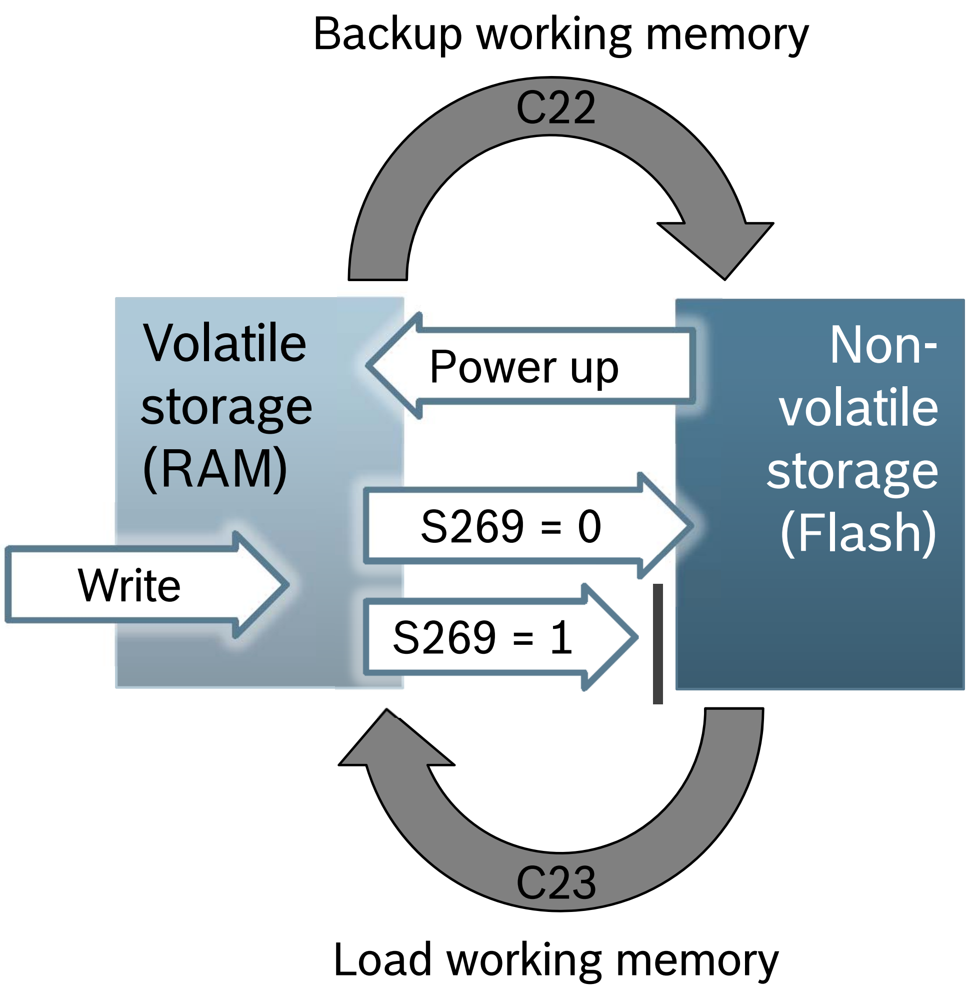
- **00: MMC as init/update medium**
  The MMC can be used for system initialization and as update medium. Boot requests for the firmware (firmware update) and parameters (Load Par from MMC). The active memory is always the onboard flash (see P-0-4065 = 0x02).
- **01: Programming module mode:**
  All parameters and the firmware are stored to the MMC. If the MMC is removed in off state, error message "MMC: Defective or missing, replace" is generated. Operation of the system is not possible. If the MMC is removed during ongoing operation, error message " F2006 MMC was removed" is displayed.
- **10: Backup medium mode**
  The MMC may be available as a storage medium and can be accessed via PLC file services, FTP or drive commands. The MMC is not subject to any checks during booting or ongoing operation.
- **Info:** The changed parameter settings become active at once!

IndraDrive C/M – MMC-Einstellung über Menü

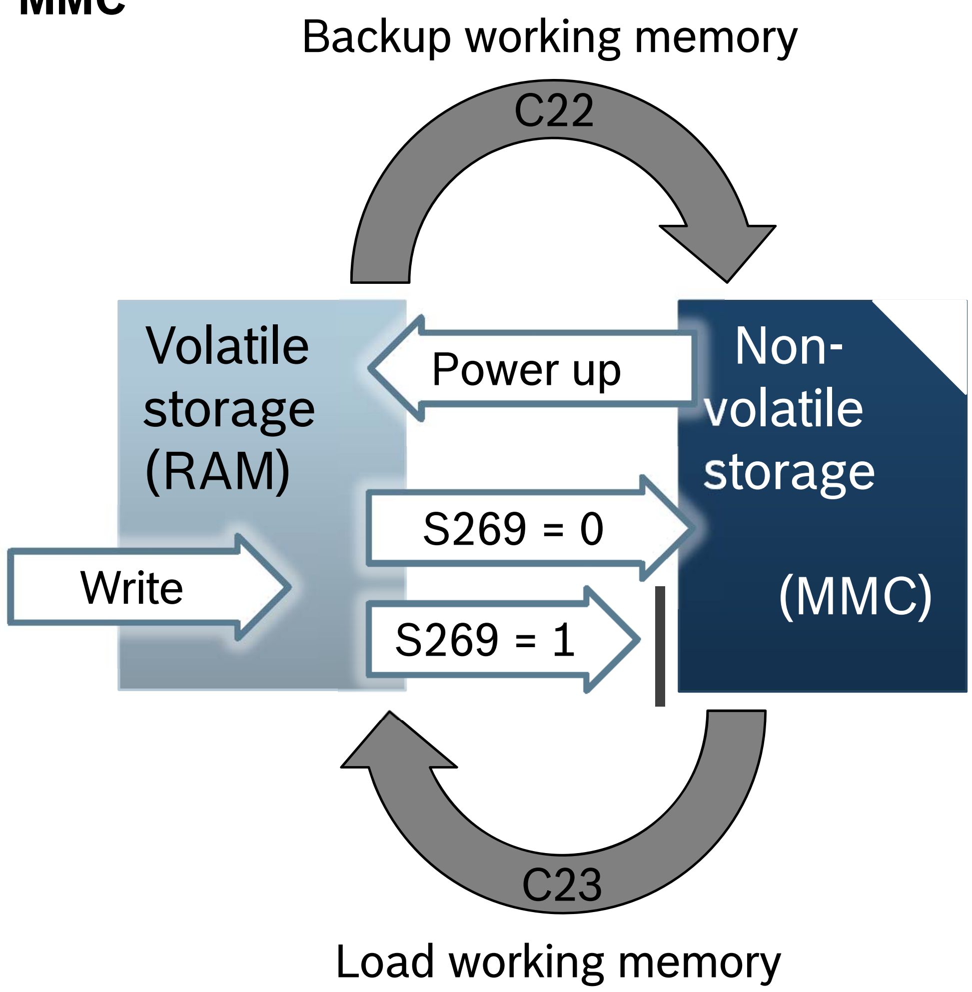


IndraDrive C/M – Storage areas

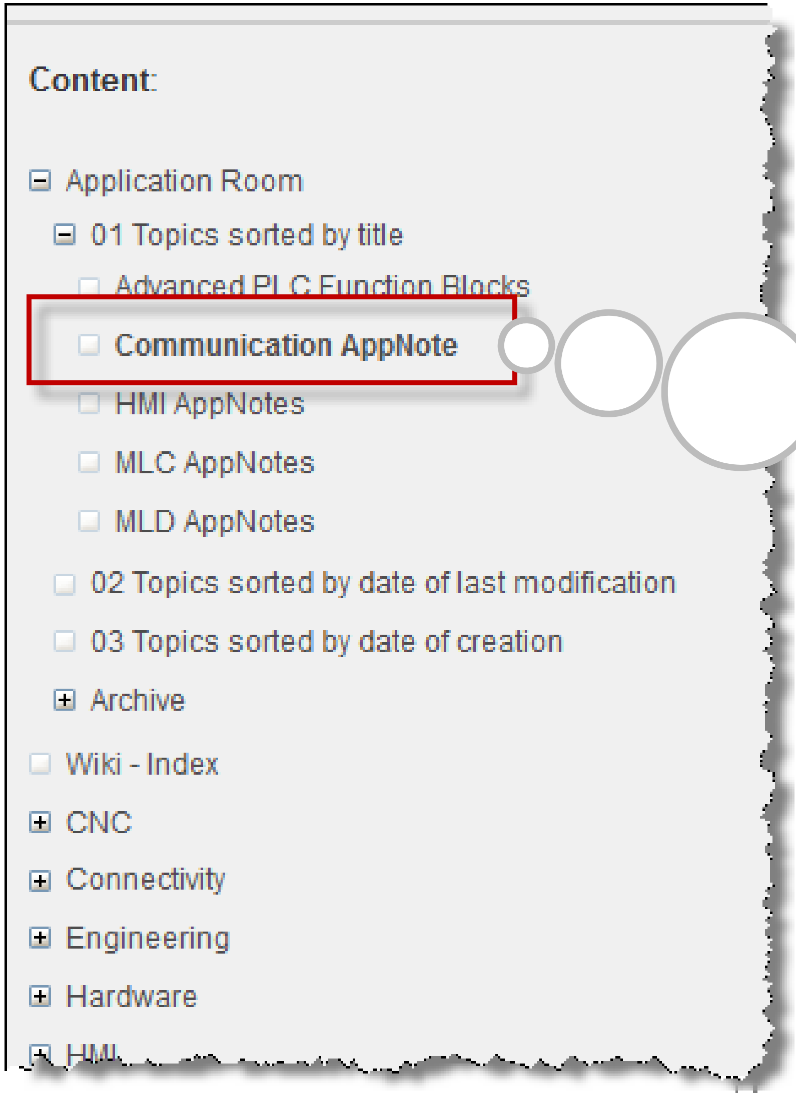
Control section without MMC



Control section with permanently used MMC



socket communication



Examples for Ethernet communication (based on TCP, UDP, or ICMP) are available on the Motion Logic Wiki in the Application Room

socket communication

Motion Logic Wiki

Communication AppNotes

Communicate serial - via RS232 and RS485 Inline modules (Motion-Logic Wiki)
Stichwörter: communication, appnote, faq_inline_com

Communicate TFTP - IndraDrive-Firmware-Download using TFTP FB (Motion-Logic Wiki)
Stichwörter: communication, tftp, firmware, download, indradrive, appnote

Communicate with CS351 (IMpnio) and Profinet (Motion-Logic Wiki)
Stichwörter: faq_profinet, communication, appnote

Communicate with TCP IP - Basic telnet server (Motion-Logic Wiki)
Stichwörter: appnote, communication

Communicate with TCP IP - Exchange big data (Motion-Logic Wiki)
Stichwörter: appnote, communication

Communicate with TCP IP - FB to get product positions from a camera (Motion-Logic Wiki)
Stichwörter: appnote, communication, camera

Communicate with TCP IP - First steps (Motion-Logic Wiki)
Stichwörter: communication, appnote

Communicate with UDP - Simple protocol with cyclic data exchange (Motion-Logic Wiki)
Stichwörter: communication, appnote

Communicate with UDP IP - First steps (Motion-Logic Wiki)
Stichwörter: appnote, communication

FB for Inline terminal R-IB IL SSI-IN-PAC (example) (Motion-Logic Wiki)
Stichwörter: faq_inline_fct, communication, appnote, adv_fb

Profibus connection between S7 and MLC (Motion-Logic Wiki)
Stichwörter: siemens, appnote, s7, profibus, communication

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socket communication

Motion Logic Wiki

Communicate with TCP IP - Basic telnet server

Bearbeiten | Hinzufügen | Extras

4 Hinzugefügt von Steudel Karin (DC-IA/SFP3), zuletzt bearbeitet von Schmidt Bastian (DC-IA/SFP32) am Aug 08, 2012 (Änderung anzeigen) Aktuelle Version V 10 Kommentar einblenden

Short description

Example program of an easy telnet server. The incomming commands are received, executed and the result ist sent back to telnet client.

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Abteilung:	SFP32
Ort:	Lohr am Main / Germany

System components

IL1G
libraries: RIL_SocketComm

Attachments

Name	Größe	Ersteller	Erstellungsdatum	Kommentar	
Telnet_Communication_Appendix_1.doc	259 kB	Bostrom Carl (DCUS/SFP)	Feb 02, 2012 22:12	Notes on enabling Microsoft's telnet client in Windows 7	Eigenschaften | Entfernen
TELNET_SERVER_BASIC_IL2G.iwx	19 kB	Bostrom Carl (DCUS/SFP)	Feb 02, 2012 22:07	Telnet server example ported to IL2G	Eigenschaften | Entfernen
TELNET_SERVER_BASIC.EXP	22 kB	Baron Stefan (DC-IA/SFP32)	Jan 20, 2011 14:07	IL1G Importfile with source code	Eigenschaften | Entfernen
Telnet_Communication.doc	91 kB	Baron Stefan (DC-IA/SFP32)	Jan 20, 2011 13:24	Documentation	Eigenschaften | Entfernen

External Links

Enter your download link here

Please note: If a login box appears, you have to enter your AP Automation Portal login in order to download the files. Typically large files such as IndraWorks projects are stored in the Automation Portal.

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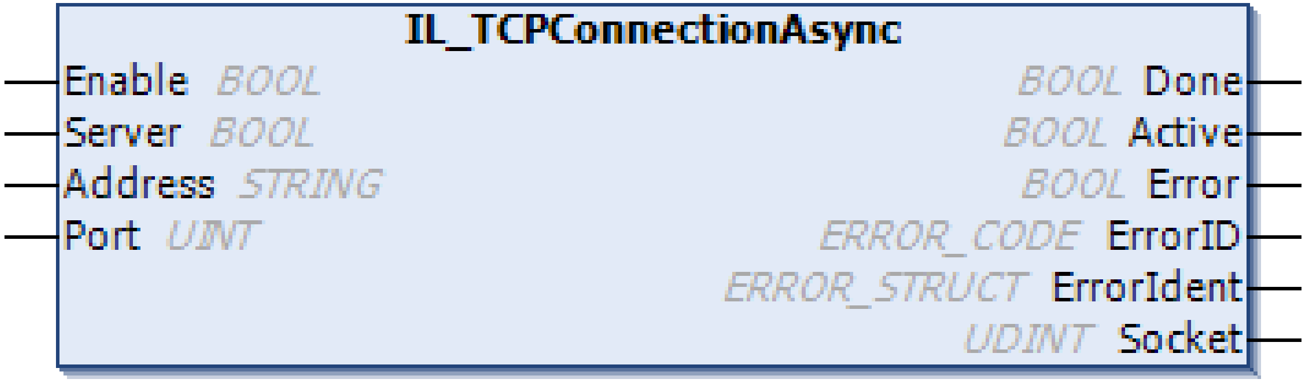
socket communication



	Name	Type	Description
VAR_INPUT	Enable	BOOL	Starts at a rising edge. Binary outputs are reset with falling edge
	Server	BOOL	TRUE: a TCP server is created. FALSE: a TCP client is created.
	Address	STRING	If "Server" is set, this input is not used and the proprietary IP address is used. If "Server" is FALSE, this input contains the IP address of the device to be connected.
	Port	UINT	The TCP port at which incoming connections are expected or to which they are to be connected.

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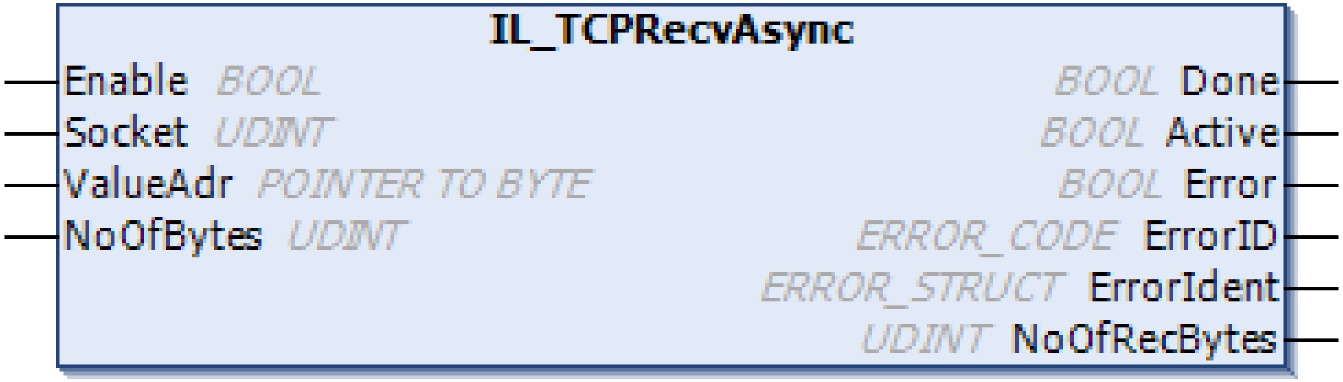
socket communication



	Name	Type	Description
VAR_INPUT	Enable	BOOL	Starts at a rising edge. Binary outputs are reset with falling edge
	Socket	UDINT	IL_TCPConnectionAsync or IL_TCPInitialAsync socket handle
	ValueAdr	POINTER TO BYTE	Initial address of the receive buffer
	NoOfBytes	UDINT	Size of the received data buffer

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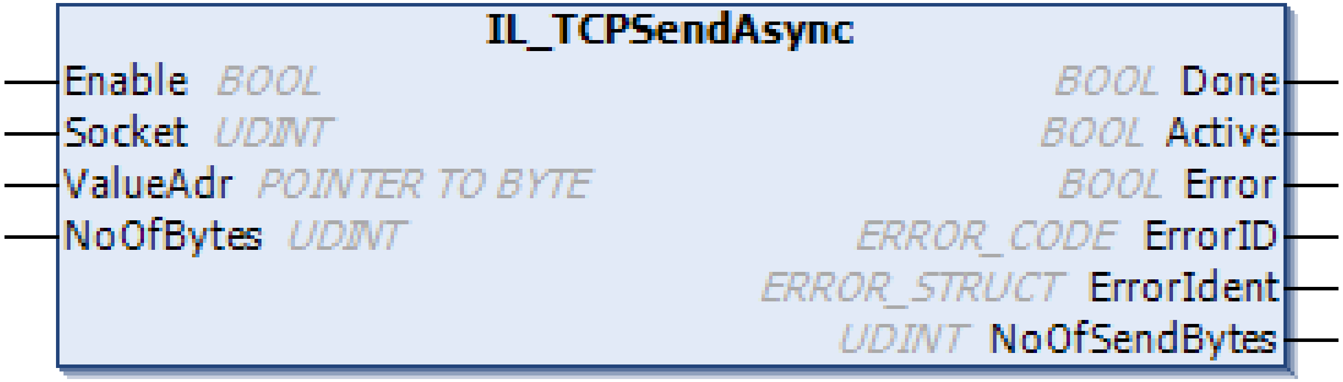
socket communication



	Name	Type	Description
VAR_INPUT	Enable	BOOL	Starts at a rising edge Binary outputs are reset with falling edge
	Socket	UDINT	IL_TCPConnectionAsync or IL_TCPInitialAsync socket handle
	ValueAdr	POINTER TO BYTE	Initial address of the data to be sent
	NoOfBytes	UDINT	Number of characters to be sent

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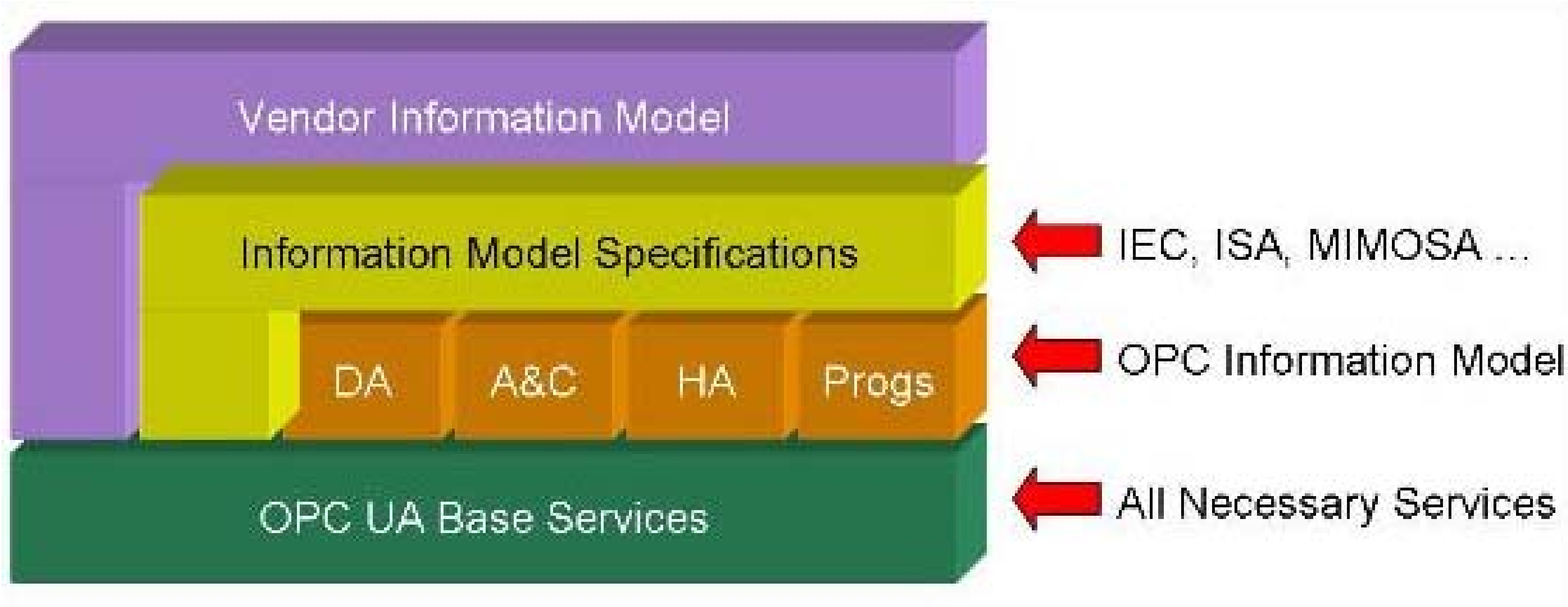
OPC UA

OPC UA – The new communication standard for Industrial Automation



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OPC UA

Classical OPC

- ☑ Today classic OPC is an established standard interface for communication in factory automation
- ☑ It is the essential link for HMI / SCADA systems to subordinate automation components
- ☑ Classic OPC is successfully used in millions of applications worldwide
- ☒ Security aspects are not taken into account
- ☒ Support of simple data types only, no structured data types possible
- ☒ Based on Microsoft COM/DCOM
  - ☒ In 2002 replaced by .NET, no further development for COM technology
  - ☒ Primarily developed for data communication in an office environment (long and non-configurable timeout values, poor error recovery mechanisms)
  - ☒ Firewall pass-through difficult to handle and to configure
  - ☒ Only available for Windows operating systems

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OPC UA

OPC UA – Optimal infrastructure for process automation

- independent → Windows operating system not assumed
- fast → binary data coding
- robust → internal mechanisms for the error recovery
- secure → security mechanisms for access protection and coding
- flexible → UA-TCP (binary) protocol or web-services (e.g. HTTP)
- unlimited → communication trough firewalls (only one port required)

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OPC UA

Support of complex data structures and information models

- Classic OPC supports only basic data types and arrays
- OPC UA supports also compound data types
- Information models
  - Information is not restricted to process data but meta-data can be included (signification of individual elements, etc.)
  - The client can retrieve the type of information (Engineering information)
- By means of “Application-specific information models” standardized objects for certain areas can be specified
- Several organizations are working on such standards

OPC UA – Advantages for Rexroth

Reduction of communication interfaces

- OPC UA will reduce the variety of existing communication interfaces (middle- and long-term), and simplify the communication with Rexroth controls (substitution of IndraLogic OPC server, WinStudio BR_WS driver, IwScp OPC server, communication server within controls

Improved connectivity to 3rd party HMIs

- In spite of using Rexroth control technology some customers want to continue with their existing HMI solutions
- For HMIs based on Windows CE only the IndraLogic OPC server is available with an restricted set of functions (only PLC data)
- Rexroth controls don’t come with a standardized interface for 3rd party HMIs (except Modbus/TCP)
- Most suppliers of HMI solutions don’t offer a driver for Rexroth controls

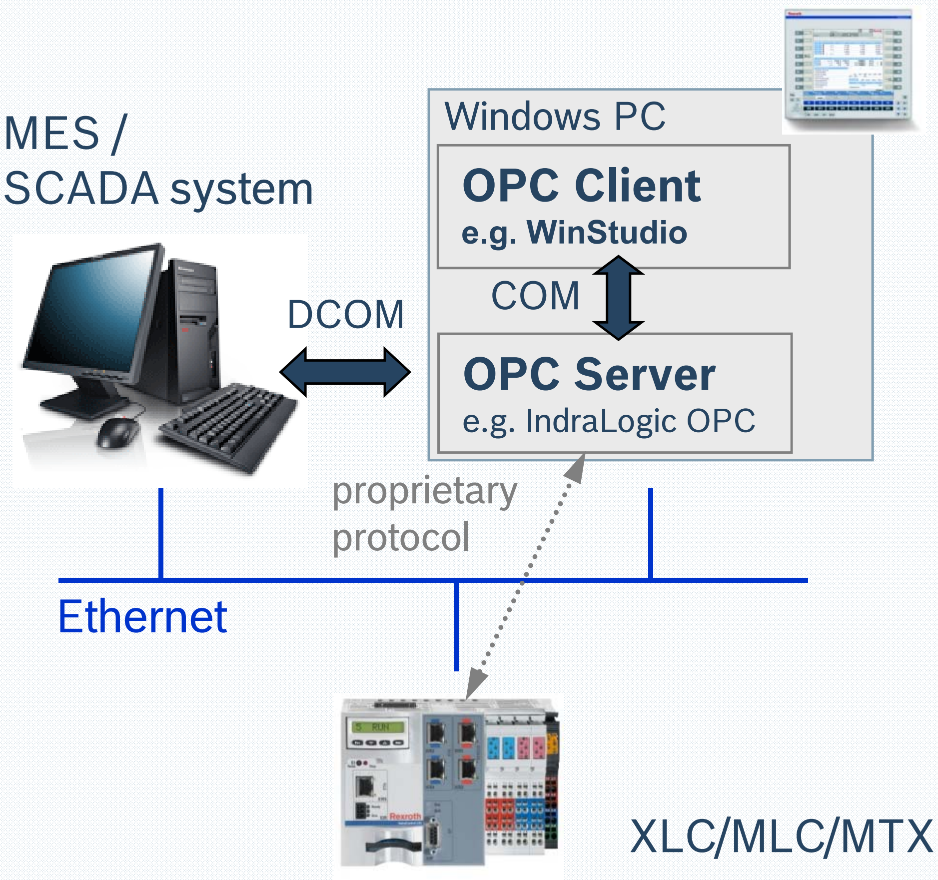
Improved connectivity for SCADA/MES systems

- No direct communication between SCADA systems and Rexroth controls is possible (an intermediate computer with Windows operating system and OPC server is always required)

OPC UA

OPC

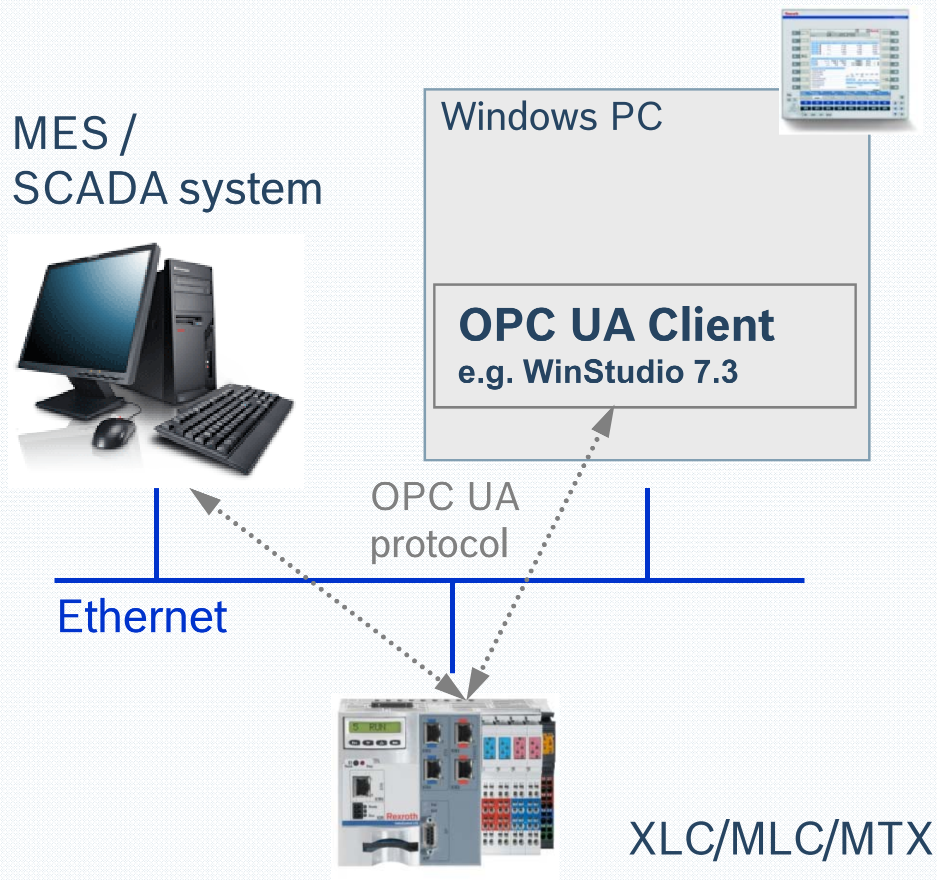
Openess, **P**roductivity and **C**ollaboration



Classic OPC Server installed on HMI device with Windows operating system

OPC UA

OPen **C**onnectivity **U**nified **A**rchitecture



OPC UA server running within the control (direct communication between OPC UA client and control)

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OPC UA

OPC UA Client

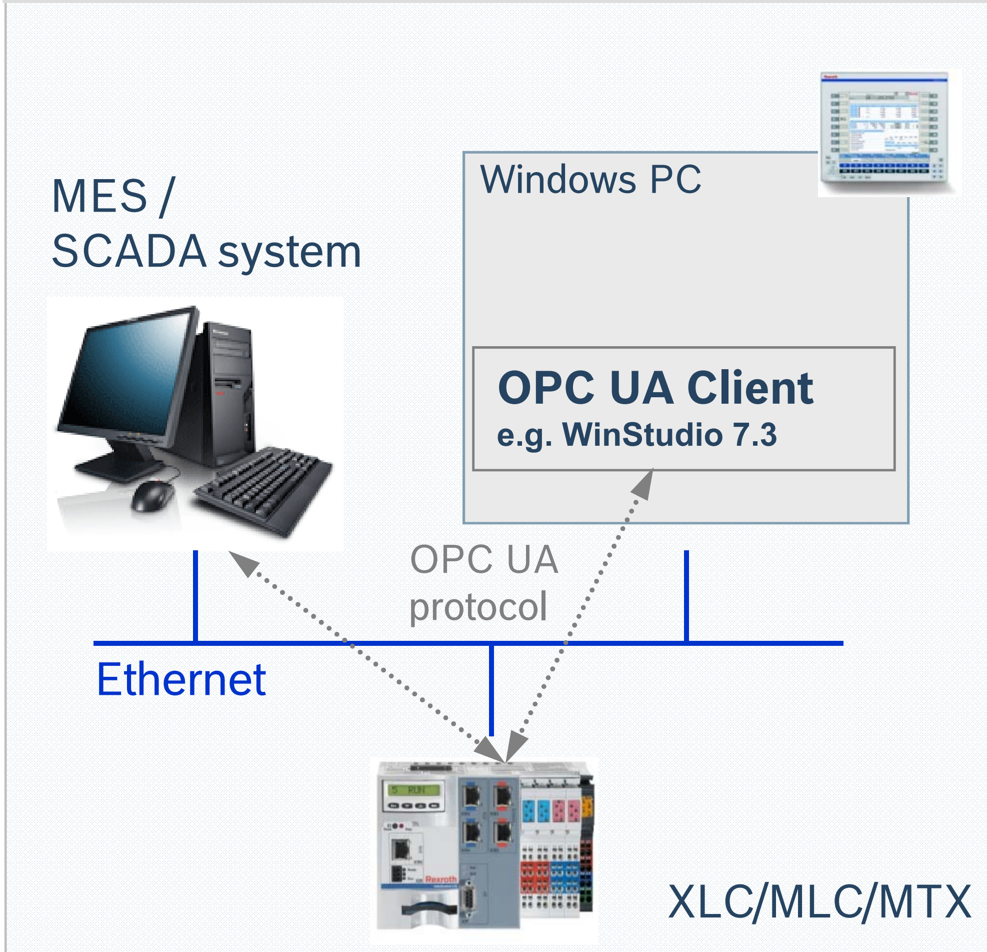
- WinStudio 7.3 – access to standard data types, structures and arrays
- 3rd party OPC UA client (e.g. Zenon / Copa Data)

OPC UA Server

- available for MLC/XLC/MTX
- Supported services:
  Browse, Read, Write, Subscription
- Supported Information models:
  OPC UA Information Model IEC61131-3 Rel. 1.0
- Access only to PLC data
  (Motion or NC data via mapping to PLC variables)

OPC UA

OPen **C**onnectivity **U**nified **A**rchitecture



OPC UA server running within the control (direct communication between OPC UA client and control)

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